

Ya.B. Zeldovich

Higher
Mathematics
for
Beginners

and Its Application to Physics

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Я. Б. ЗЕЛЬДОВИЧ

ВЫСШАЯ МАТЕМАТИКА ДЛЯ НАЧИНАЮЩИХ

и ее приложения к физике

ИЗДАТЕЛЬСТВО «НАУКА» МОСКВА

Ya. B. Zeldovich

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HIGHER MATHEMATICS FOR BEGINNERS

and Its Application to Physics

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by
George Yankovsky



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Preface

to the Fifth Russian Edition

The title of this book gives the clue to our main aim, which is to initiate the reader into the realm of differential and integral calculus and, by applying these methods to the more important divisions of physics, to demonstrate the significance and power of higher mathematics.

The concepts of a derivative and an integral are not so much more involved than the notions of an “unknown quantity” or the “similarity of triangles” that come from the school syllabus of mathematics. The aim to make the concept of the derivative and the integral part and parcel of the education of every single person, no matter what his field of interest, is long since overdue.

The new concepts are introduced in the second chapter as simply and naturally as possible. Then follows a chapter devoted to the computational techniques used in higher mathematics. The fourth chapter and chapters five to eight deal with the application of these techniques to geometry, to processes of nuclear transformations, mechanics, molecular physics, and electricity. The reader who has already forgotten some of his school material will find Chapter 1 on functions and graphs of particular value. The last chapter goes far beyond the scope of any elementary course and is added to give the reader a feeling of what lies ahead. Finally, we conclude with a rough outline of the more complicated problems of mathematical physics.

Many years have passed since the first edition of this book came out in Russian (1960) and was followed by a flush of carping criticism of the author's supposed lack of mathematical rigour (some even went so far as to accuse him of corrupting the youth with a hazy, light-minded approach to a serious subject).

Actually, the matter boils down to two different approaches to the teaching process.

In many textbooks, the exposition is reminiscent of a dispute carried on between two scientists. The student is pictured as an opponent seeking all manner of objections. The instructor puts forth a conse-

cutive and rigorously logical analysis of all objections and proves irrevocably the correctness of his propositions.

In this book, the student is regarded as a friend and ally who puts his faith in the teacher and the textbook and wishes ardently to make use of and apply to nature and technology the mathematical techniques offered to him. Comprehension of the subject expands as the result of analyzing examples and applications. In the strictly logical approach, the question of the significance and usefulness of the theorems studied remains in the background. In the present text, by contrast, we bring to the fore the mathematical ideas and their relationship with the study of nature.

Could it be that this insufficient attention to rigorous proofs stems from a kind of consumers' approach to mathematics on the part of the author who is a physicist? I personally do not believe so. Mathematics advances with the aid of intuition and in terms of general ideas, or to put it more simply still—with the help of inspiration, not merely cold logicity. It is only later on that the work is invested with formulas and chains of rigorous proof. Textbooks often keep hidden the fundamental ideas that inspired the creators of mathematical ideas.

The aged patriarch of modern mathematics, Richard Courant, wrote in 1964 (*Scientific American*, September 1964, p. 43) that for a very long time mathematicians accepted Euclid's geometry as the model of a rigorous axiomatic deduction. Courant then adds: "But emphasis on this aspect of mathematics is totally misleading if it suggests that construction, imaginative induction and combination and the elusive mental process called intuition play a secondary role in productive mathematical activity or genuine understanding. In mathematical education, it is true, the deductive method starting from seemingly dogmatic axioms provides a shortcut for covering a large territory. But the constructive Socratic method that proceeds from the particular to the general and eschews dogmatic compulsion leads the way more surely to independent productive thinking."

Courant places imagination and intuition ahead of all other things!

The notorious pitting of poets against physicists (mathematicians too) is a figment of the imagination of the poet B. Slutsky. In mathematics there is more poetry than any poet ever imagined. The history of science is proof that good mathematics is prophetic: mathematical analysis of the known opens up the path into the realm of the unknown and leads to new physical notions.

In "Higher Mathematics for Beginners" I strove towards a constructive approach, to the eliciting of the meaning and aims of mathematical concepts and attempted, at least in part, to convey the spirit of the heroic period when these notions were born.

Since the first edition of 1960, this text has gone through several Russian editions and has been translated into a number of languages

(Bulgarian, Japanese, French). This fifth edition has been carefully reworked from the pedagogical point of view. I believe that in its present form, the text will be useful to students and to teachers of mathematics, and also to instructors in physics in the senior forms of secondary school and the first year of college.

The last two chapters (Dirac's Remarkable Delta Function and What Next) are entirely different from the remainder of the book. The style too is quite changed. The aim there is to give the reader a feeling (of necessity, very superficial) of what complicated things lie ahead.

In the preparation of this book I have been helped by very many people, and to all of them I am deeply indebted.

Moscow 1973

Academician Ya. B. Zeldovich

Chapter 1

Functions and Graphs

1.1 THE FUNCTIONAL RELATIONSHIP

Nature, technology and mathematics abound in functional relationships. A functional relationship between one quantity (y) and another (x) signifies that to every value of x there corresponds a definite value of y .

The quantity x in this case is called the independent variable; y is the function of this variable. We also sometimes say that x is the argument of the function.

Here are a few examples taken from geometry and physics.

(1) The volume of a sphere V is a function of its radius r ,

$$V = \frac{4\pi}{3} r^3$$

(2) The volume V of a cone with a given altitude h is a function of the radius of the base r :

$$V = \frac{1}{3} \pi r^2 h$$

(3) The distance z traversed by a freely falling body depends on the time t that elapses from the beginning of fall:

$$z = \frac{gt^2}{2}$$

(4) The current i , by Ohm's law, depends on the resistance R of the conductor for a given potential difference u :

$$i = \frac{u}{R} \quad (1.1-4)$$

The list could be extended without end.

It is characteristic that in most cases in nature and technology the quantity of interest (the function) depends on several other quantities. Thus, in the last example, the current depends on two quantities: the potential difference u and the resistance R . The volume of a cone is a function of its altitude h and of the radius r of the base.

Assuming all quantities, except one, to be given and constant, we study the dependence of the function upon a single variable. In this book, we will confine ourselves mainly to functions of one variable.

For example, taking a given storage battery with a definite potential difference u , we will vary the resistance R of the conductor and measure the current i . In this experiment, the current depends only on the resistance, the quantity u in formula (1.1-1) being regarded as a constant coefficient.

In mathematics, functional relationships are ordinarily defined by formulas, for example,

$$y = 2x + 3, \quad y = x^2 + 5, \quad y = 3x^3 - x^2 - x \quad (1.1-2)$$

$$y = \frac{x-1}{x+1}$$

In these formulas it is evident that we have to do with functions of one variable. The formula enables us to compute the values of the function for each given value of the independent variable.

Knowing the formula that states the dependence of y on x , it is easy to form a table of values of y for several arbitrarily given values of x .

By way of an illustration, we will set up a table for the third function in (1.1-2) (see Table 1). The upper row contains the values of x that we choose, the lower row, under each value of x , contains the appropriate value of y .

Table 1

x	-3	-2	-1	0	1	2	3
$y = 3x^3 - x^2 - x$	-87	-26	-3	0	1	18	69

Using this formula it is possible to make a more detailed table specifying, say, the values $x = 0, 0.1, 0.2, \dots$. Thus, the formula is stronger, as it were, than any table. The formula not only contains the information necessary to compile the given table, but also enables one to find the values of the function for values of the independent variable not contained in the table. On the other hand, the table is convenient in that it immediately gives the value of y for any given value of x , provided that the needed x is given in the table since computations via the formula were already carried out in the compilation of the table.

When the law of a phenomenon of nature or of engineering practice has been found, it is expressed by a formula. However, it does happen that the theory of the phenomenon is lacking and the physicist (or chemist, biologist, engineer) is only able to supply experimentally

obtained facts—the dependence of the quantity of interest upon the quantity that was given in the experiment. This is what happens, say, in studies of the relationship between the resistance of a conductor and the temperature of the conductor. Here the functional relationship can only be given in the form of a table containing the experimental data.

Experiments show that for a given conductor (of a given material, a given cross-section and a given length) the electric resistance depends on the temperature of the conductor. For each value of the temperature T , the conductor has a definite resistance R so that we can speak about a relationship in which the resistance R is a function of the temperature T .

Carrying out measurements, we can find the values of R for various T and thus find the dependence (relationship, function) $R(T)$. Here, the results of the experiments are given in Table 2, which gives the values of R for distinct values of T .

Table 2

T (degrees Celsius)	0°	25°	50°	75°	100°
R (ohms)	112.0	118.4	124.6	130.3	135.2

If we are interested in the values of R for other temperatures which do not appear in the table, then additional measurements are required because there is no exact formula defining the function $R(T)$. Practically speaking, we could offer an approximate formula which is in good agreement with experiment at the temperatures for which the measurements were made. Let us take the formula

$$R = 112.0 + 0.272T - 0.0004T^2$$

and compile Table 3.

Table 3

T	0	25	50	75	100
R (by formula)	112.0	118.55	124.6	130.15	135.2

The formula yields values of R that are very close to the experimental values for those temperatures at which the measurements were made, and so one is justified in assuming that for the intermediate temperatures (say for 10° or 80° or 90°) the formula will likewise give a correct description of the functional relationship $R(T)$. How-

ver, if we apply the formula outside the range of the investigated interval (say for -200°C or $+500^{\circ}\text{C}$) this may lead to errors since there are no grounds to expect that $R(T)$ will be expressed as a quadratic trinomial.

Formulas obtained not from theory but experimentally are called empirical formulas.

1.2 COORDINATES

Coordinates are used as a pictorial way of representing functional relationships by means of drawings (graphs). Draw two perpendicular straight lines in a plane. Call the horizontal line the x -axis (also known as the axis of abscissas), the vertical line the y -axis (also known as the axis of ordinates). The point of intersection of the lines is called the origin (of coordinates) (point O in Fig. 1). It is customary to

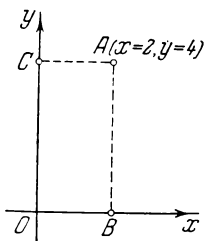


Fig. 1

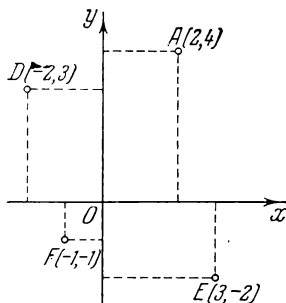


Fig. 2

picture the plane with the x - and y -axes not flat on a table, but vertically in front of the reader. The arrow of the x -axis is from left to right, the arrow of the y -axis is upwards.

A definite pair of values of x and y , say $x = 2$, $y = 4$, is represented on a graph by a single point (point A). The position of the point is determined by two conditions: the perpendicular AB dropped from A on the x -axis cuts off on the axis a line segment OB equal to two units of length; the segment OB from the origin O to the foot B of the perpendicular is taken to be positive and corresponds to a positive value of x when the point B lies to the right of the point O .

The perpendicular AC dropped from point A on the y -axis cuts off on that axis a line segment OC whose length is equal to 4 units. On the y -axis, the positive values of y correspond to the foot C of the perpendicular lying above the origin of coordinates O . It is customary to indicate the positive direction of an axis by an arrow and to place the letter-label (legend) of that axis next to the arrowhead. Picturing the plane as being vertical, we say that "the greater y is, the

higher the point". Points corresponding to negative values of y lie below those corresponding to positive values. For practical work it is convenient to use squared paper (it usually has a grid of millimetre squares) to locate points and plot curves.

An important piece of practical advice: get into the habit of locating any point A (corresponding to specified values of x and y) without drawing the dashed lines (perpendiculars) AB and AC : this should be done mentally so as not to clutter up the drawing with extra lines and designations.

Negative values of x are laid off to the left of O , negative values of y are laid off downwards from O . Fig. 2 shows a few examples of points

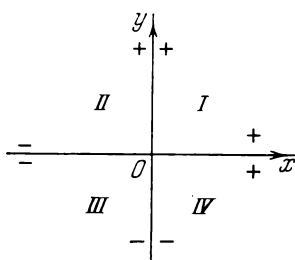


Fig. 3

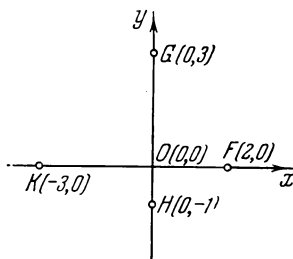


Fig. 4

for which x and y have different signs. The reader should check to see if he agrees with how the points have been indicated. This will test his understanding of the foregoing.

The points given in Fig. 2 are

$$\begin{aligned} A: x = 2, y = 4; & \quad D: x = -2, y = 3; \\ E: x = 3, y = -2; & \quad F: x = -1, y = -1 \end{aligned}$$

The coordinates of points are sometimes given briefly in parentheses after the name of the point, the first number being the value of x (the abscissa of the point) and the second, the value of y (the ordinate of the point). These designations for the four points A , D , E , F are given in Fig. 2.

The coordinate axes divide the plane of the drawing into four parts, or quadrants, which are numbered as in Fig. 3. In each quadrant, x and y have definite signs. In the first quadrant, x and y are positive, $x > 0$, $y > 0$; in the second quadrant, $x < 0$, $y > 0$, or x is negative and y is positive; in the third quadrant, $x < 0$, $y < 0$, or x and y are negative; and in the fourth quadrant $x > 0$, $y < 0$, or x is positive and y is negative. The signs of the four quadrants are shown in Fig. 3. Compare them with the signs of the coordinates of the points A , D , E , F in Fig. 2.

If (see Fig. 4) for point G it is given that $x = 0$, then the point lies on the y -axis. The perpendicular from G to the x -axis then coincides

with the y -axis. This means that the foot of the perpendicular coincides with the origin O so that the distance from the foot of the perpendicular to the origin is zero and we can say that the abscissa of point G lying on the y -axis is zero. If for point F it is given that $y = 0$, then this point lies on the x -axis, and the perpendicular drawn from F to the y -axis coincides with the x -axis. Its foot is the origin O .

Several such points are shown plotted in Fig. 4:

$$G (x = 0, y = 3), \quad F (x = 2, y = 0),$$

$$H (x = 0, y = -1), \quad K (x = -3, y = 0)$$

Finally, the point with $x = 0$ and $y = 0$ is nothing but the origin itself, O .

Our advice has been not to plot the feet of perpendiculars like in Figs. 1 and 2.

In Fig. 1 we wanted to plot point $A (x = 2, y = 4)$. The points B and C merely served as auxiliary points that were used to construct A .

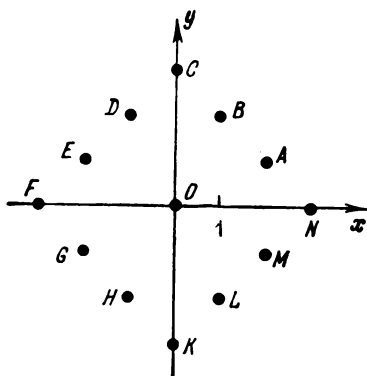


Fig. 5

They proved useful as a first step in learning about the coordinate system. They are no longer needed and we should plot only one point, A . If B and C are also plotted, we might get the idea that they too are needed for some purpose and that we had to construct three points:

$$A (2, 4), B (2, 0) \text{ and } C (0, 4)$$

The reader should take the time to drill himself in plotting points with positive, negative and zero values of x and y . He should also be able, at least in approximate

fashion, to state the values of x and y and their appropriate signs for any points marked in a coordinate system.

Exercise

1. State the coordinates of the points from A to O (Fig. 5).

1.3 GEOMETRIC QUANTITIES EXPRESSED IN TERMS OF COORDINATES

The specification of two numbers—the values of x and y , say—determines the position of a point in a plane. Therefore all geometric quantities referring to this point can be expressed by the coordinates of the point.

Let us find the distance r from the origin to point A with coordinates x and y , that is the length of the line segment r of the straight line OA joining the origin O to the point A (Fig. 6) and also the angle

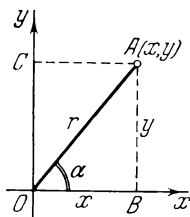


Fig. 6

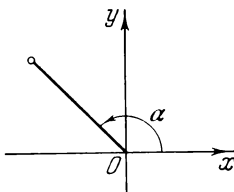


Fig. 7

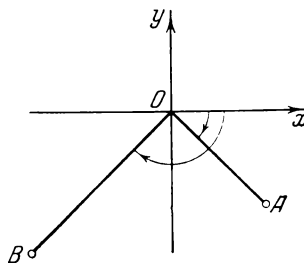


Fig. 8

α (the letter “alpha” of the Greek alphabet, which is given in the Appendix, page 474) between the line OA and the axis of abscissas.

We draw auxiliary lines AB and AC . The length of OB is equal to x , that of AB is equal to OC , or y . From the right triangle OAB , by the Pythagorean theorem, we have

$$(OA)^2 = r^2 = (OB)^2 + (AB)^2 = x^2 + y^2,$$

$$r = +\sqrt{x^2 + y^2}$$

and, by the definition of the tangent, we finally get

$$\tan \alpha = \frac{AB}{OB} = \frac{y}{x}$$

Thus, for example, let $x = 2$, $y = 3$ (Fig. 6). Then $r = \sqrt{13} \approx 3.6$,

$$\alpha = \arctan \frac{3}{2} \approx 56^\circ$$

Note that the angle α is always reckoned from the positive direction of the x -axis; therefore, if $y = 2$ and $x = -2$ (Fig. 7), then the angle α is obtuse, $\tan \alpha = \frac{2}{-2} = -1$, $\alpha = 135^\circ$.

When a point lies below the x -axis, it is customary to reckon the angle α downwards from the axis, taking α negative. In Fig. 8 we have two instances: point A ($x = 2$, $y = -2$) for which $\alpha = -45^\circ$ and point B ($x = -3$, $y = -3$) for which the angle $\alpha = -135^\circ$. Thus, for any point the angle α lies within the range from -180° to $+180^\circ$.

It is easy to solve the inverse problem: suppose we are given point A at a given distance r from the origin O and the line segment OA forming an angle α with the x -axis (the positive direction of the x -axis is assumed as usual). It is required to find the coordinates of the

point A . Looking at Fig. 6 we see that

$$x = r \cos \alpha$$

$$y = r \sin \alpha$$

These formulas are valid, without exception, for arbitrary positive and negative angles α and yield the proper signs of x and y in any quadrant.

Let us now examine problems involving two points A_1 and A_2 . We denote the coordinates of the first point by x_1, y_1 , the coordinates of the second point by x_2, y_2 (Fig. 9). Find the distance r_{12} between these points and the angle α_{12} between line segment A_1A_2 and the x -axis.*

It is convenient to draw through A_1 a straight line parallel to the x -axis and through A_2 a line parallel to the y -axis. In Fig. 9 they are

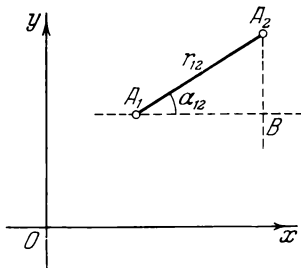


Fig. 9

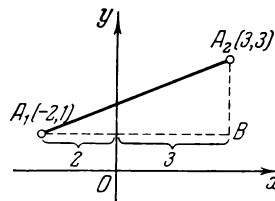


Fig. 10

shown as dashed lines and their point of intersection is B . In the triangle A_1A_2B the line segment A_1B is equal to $x_2 - x_1$ and the segment A_2B is equal to $y_2 - y_1$. The construction of the triangle A_1A_2B is similar to the construction given in Fig. 6.

By the Pythagorean theorem,

$$r_{12} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The angle α_{12} is found from the condition

$$\tan \alpha_{12} = \frac{y_2 - y_1}{x_2 - x_1} \quad (1.3-1)$$

* The subscripts on the letters are known as indices and are not to be confused with exponents (which are superscripts). They are read: x_1 , x sub one, A_2 , A sub two. The same letter with different indices (y_0, y_1, y_2, y_a, y_b) is used in place of a variety of letters to emphasize that we are dealing with similar (yet different) quantities. For instance, x_1 and x_2 are quantities on the x -axis (both are abscissas), but they refer to distinct points. Now quantities denoted by different letters but the same index refer to one and the same point: A_1 denotes a certain point, x_1 denotes the abscissa of that point, y_1 denotes the ordinate of that same point. We sometimes use double-index notation: r_{12} (which is read: " r sub one two" and not " r sub twelve") is the distance between the first point (A_1) and the second (A_2).

The reader should assure himself that the formulas hold true for arbitrary signs of all four quantities x_1, y_1, x_2, y_2 and for any relations: $x_1 > x_2$ or $x_1 < x_2$, $y_1 > y_2$ or $y_1 < y_2$.

For example, in Fig. 10 we have the case $x_1 < 0, x_2 > 0$, the coordinates are $A_1 (x_1 = -2, y_1 = 1)$, $A_2 (x_2 = 3, y_2 = 3)$. Here the length of A_1B is equal to the sum of the absolute values* $|x_1| = 2$, $|x_2| = 3$. But this is strictly in accord with the general formula

$$A_1B = x_2 - x_1 = 3 - (-2) = 3 + 2 = 5$$

Consequently, the expressions for r_{12} and $\tan \alpha_{12}$ are also correct.

We now consider some problems referring to three points: A_1, A_2, A_3 . How can we determine, without construction, merely by computing from the values of the coordinates of the points, whether these

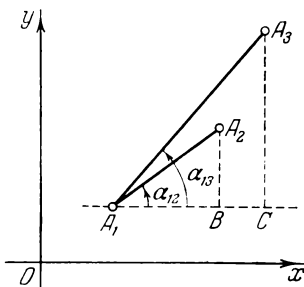


Fig. 11

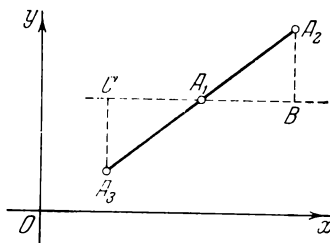


Fig. 12

three points are collinear, that is, lie on a single straight line? It is clear that when the angle α_{12} between A_1A_2 and the x -axis is equal to the angle α_{13} of A_1A_3 with the x -axis, then this means that the line segments A_1A_2 and A_1A_3 are collinear. In Fig. 11 we have a case where $\alpha_{13} > \alpha_{12}$, the point A_3 lies above the extension of A_1A_2 , but the same figure shows us that if α_{13} were equal to α_{12} , then A_3 would be on the straight line A_1A_2 produced.

From the expression of the tangent of an angle (1.3-1) it follows that for $\alpha_{12} = \alpha_{13}$ we have the following relationship between the coordinates of the points:

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_1}{x_3 - x_1} \quad (1.3-2)$$

* The vertical straight lines take the place of the words "absolute value". Thus, for a positive quantity this sign does not change anything, $|3| = 3$, $|0.1| = 0.1$. A negative quantity enclosed between vertical bars is equal to the positive quantity obtained by multiplying the given one by -1 ; thus, for instance, $|-3| = 3$, $|-0.1| = 0.1$. The term "modulus" is a synonym of absolute value. We can say the modulus, or the absolute value, of -3 is $|-3|$ and is equal to plus three.

Without using trigonometry, we can say that condition (1.3-2) is a condition of similarity of two right triangles A_1A_2B and A_1A_3C . The similarity of the triangles indicates that the angles at the vertex A_1 are equal.

The relation (1.3-2) is also applicable in the case where point A_1 lies between A_2 and A_3 (Fig. 12). If the three points are collinear, then from the similarity of the triangles A_1A_2B and A_1A_3C follows the proportion (1.3-2). In the example given in Fig. 12, $x_3 - x_1 < 0$, $y_3 - y_1 < 0$, but their ratio is positive and equal to the ratio of two positive quantities $x_2 - x_1$ and $y_2 - y_1$.

Exercises

1. Plot the points $(1, 1)$, $(-1, 1)$, $(-1, -1)$, $(1, -1)$.
2. Plot the points $(1, 5)$, $(5, 1)$, $(-1, 5)$, $(-5, 1)$, $(-1, -5)$, $(-5, -1)$, $(1, -5)$, $(5, -1)$.
3. Plot the points $(0, 4)$, $(0, -4)$, $(4, 0)$, $(-4, 0)$.
4. Find the angle α and the distance from the origin of the points $(1, 1)$, $(2, -2)$, $(-3, -3)$, $(-4, 4)$.
5. Find the distances between the following pairs of points: $A_1(1, 1)$, $A_2(1, -1)$; $A_1(1, 1)$, $A_2(-1, -1)$; $A_1(2, 4)$, $A_2(4, 2)$; $A_1(-2, -4)$, $A_2(-4, -2)$.
6. Determine whether the point triples lie on a straight line: $A_1(0, 0)$, $A_2(2, 3)$, $A_3(4, 6)$; $A_1(0, 0)$, $A_2(2, 3)$, $A_3(-2, -3)$; $A_1(0, 0)$, $A_2(2, 3)$, $A_3(-2, 3)$.
7. Write out the coordinates of the vertices of a square of side a if the diagonals of the square coincide with the x - and y -axes.
8. Write down the coordinates of the vertices of a regular hexagon with side a if one of the diagonals coincides with the x -axis, and the centre lies at the origin.
9. (a) Write down the coordinates of the vertices of an equilateral triangle with side a , with base on the x -axis, and with the vertex of the subtended angle on the y -axis; (b) the same if the base lies on the x -axis and the vertex of one of the angles lies at the origin.
10. Given a point A_1 with coordinates x_1 , y_1 . Write down the coordinates of point A_2 symmetric to A_1 about the x -axis; the same for A_3 symmetric to A_1 about the y -axis; the same for A_4 symmetric to A_1 with respect to the origin.

1.4 GRAPHICAL REPRESENTATION OF FUNCTIONS. THE EQUATION OF THE STRAIGHT LINE

In Sec. 1.2 it was shown that each pair of values x , y is associated with a definite point in the plane.

If it is given that y is a definite function of x , then this means that to every value of x there corresponds a definite value of y . Therefore, if we are given a range of values of x we can find the various corresponding values of y , and these pairs of values will yield many points in the plane. If we increase the distinct values of x , by taking them closer and closer together, then finally the points will merge into a solid curve. This curve is called the graph of the function. Actually,

only a few points suffice to plot a graph, the intermediate points and the whole graph (curve) itself of the function being obtained by joining the points with a smooth curve. However, in order to avoid crude errors we must have a general picture of the form of curves representing various functions. We begin with a few of the more typical and important functions.

We consider the so-called linear relationship (linear function):

$$y = kx + b$$

Suppose, say,

$$y = 2x + 1$$

We construct a few points for which x and y are given in Table 4. Now plot these points on the graph in Fig. 13. It is immediately seen that these points are collinear (lie on one straight line). In this case, we draw the straight line (whence the term "linear function") and obtain the entire graph of the function; for any x , the corresponding point (x, y) lies on that straight line.

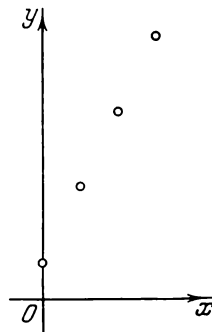


Fig. 13

Table 4

x	0	1	2	3
y	1	3	5	7

How do we prove that, for any function of the form $y = kx + b$ — for arbitrary k and b — all points of the graph lie on a single straight line? To do this, we will verify that the condition derived at the end of Sec. 1.3 remains valid for any triple of points of the graph. Indeed, consider two points $A(x_1, y_1)$ and $B(x_2, y_2)$ whose coordinates satisfy the equation $y = kx + b$. Then

$$y_2 - y_1 = kx_2 + b - (kx_1 + b) = k(x_2 - x_1)$$

whence

$$\frac{y_2 - y_1}{x_2 - x_1} = k$$

This ratio proves to be independent of x_1 and x_2 . Hence, for any other pair of points of the graph, in particular for the points $A(x_1, y_1)$ and $C(x_3, y_3)$, we also get

$$\frac{y_3 - y_1}{x_3 - x_1} = k$$

This means that for any three points of the graph, $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ the relation

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_1}{x_3 - x_1}$$

is valid, which means that any three points are collinear and, hence, *all* points of the graph of the function $y = kx + b$ are collinear. Thus, the graph of the function $y = kx + b$ is a straight line.

The equation $y = kx + b$ is called the equation of a straight line. The coefficient k defines the angle between the straight line and the x -axis. Substituting $x = 0$ into the equation, we get $y = b$, which means that one of the points of the straight line is the point $(0, b)$. This point lies on the y -axis at a distance b above the origin (if $b < 0$, then the point lies below the origin). Thus, b is the ordinate of the point of intersection of the straight line with the y -axis; $|b|$ is the length of the line segment cut off by the straight line on the axis of ordinates (Fig. 13, $b = 1$). It is called the y -intercept.

To construct a straight line corresponding to a given equation, one need not compute the coordinates of a large number of points and plot them on the graph: it is clear that the construction of two points fully determines a straight line passing through these two points.

For instance, we can always take two points: $y = b$ for $x = 0$ and $y = b + k$ for $x = 1$ and draw the line. For the second point we could also take the point of intersection of the straight line with the x -axis ($x = x_0, y = 0$). This is called the x -intercept. From the condition $y = kx_0 + b = 0$ we find $x_0 = -\frac{b}{k}$.

It is useful to do some drilling in the construction of graphs so as to be able to glance at an equation and picture roughly the variation and the position of the curve in question.

This is easy to do when we have a linear function whose graph is a straight line. The line depends only on two quantities, k and b of the equation. Thus, not so many variants have to be examined: k can be positive or negative, k can be large or small in absolute value (greater than 1 or less than 1), b can be positive or negative.

Let us see how to carry out such an investigation.

We start with the case $b = 0$, or the equation $y = kx$. The straight line here will clearly pass through the origin, that is, through the point $x = 0, y = 0$. Fig. 14 depicts several straight lines with different k : $k = 0.1$, $k = 1$, $k = 10$, $k = -0.1$, $k = -1$, $k = -10$. The values of k are indicated at both ends of each line. Check the correctness of each line and you will feel sure of the following conclusions:

- ✓ (1) If $k > 0$, then the line lies in the first and third quadrants, if $k < 0$, it lies in the second and fourth quadrants.

(2) By the foregoing, if $k = 1$, the line lies in the first and third quadrants. Part of a straight line in the first quadrant forms an angle $\alpha = 45^\circ$ with the x -axis, which means it bisects the angle between the x -axis and y -axis. The "angle with the x -axis" here stands for the positive direction of the x -axis (the one with the arrowhead). An extension of the straight line lying in the third quadrant forms with the x -axis an angle $\alpha = -135^\circ$ (the angles are not indicated in Fig. 14).

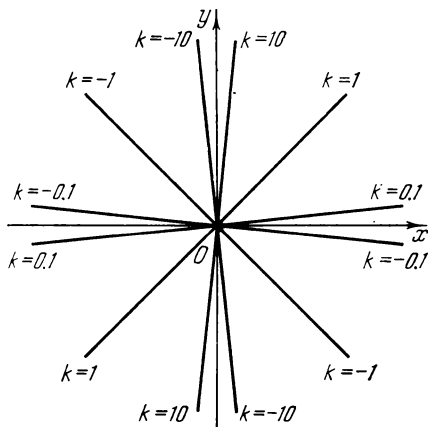


Fig. 14

(3) For $k = -1$, the portion of the line lying in the second quadrant forms an angle $\alpha =$

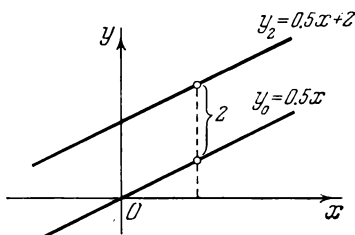


Fig. 15

-135° with the x -axis, while the prolongation of that line in the fourth quadrant forms an angle $\alpha = -45^\circ$.

(4) If $|k| < 1$, the straight line is sloping, i.e., closer to the x -axis than to the y -axis: the smaller $|k|$, the closer the line is to the x -axis. If $|k| > 1$, the straight line is steep, closer to the y -axis than to the x -axis: the greater $|k|$, the closer the line is to the y -axis.

Now that this is clear, let us investigate the general case of a straight line with b different from zero.

In Fig. 15 we have the graph of a straight line with $b = 0$: $y = 0.5x$. How does it differ from the straight lines with $b \neq 0$, but with the same k , say, $y = 0.5x + 2$? For the sake of convenience we denote $y_0 = 0.5x$ and $y_2 = 0.5x + 2$.* For each given x , the quantity y_2 is two units greater than y_0 . To summarize, then, the points of line y_2 are obtained from the points of line y_0 with the same x by an elevation of two units. The straight line y_2 is parallel to y_0 and lies 2

* The subscripts here are used somewhat differently from our earlier practice; y_0 refers to the entire line and not to the ordinate of a point. It is the ordinate of an arbitrary point on a line with given k and $b = 0$; y_2 is the ordinate of an arbitrary point on a line with given k and $b = 2$. That is, y_0 is not a number but a function of x , or $y_0(x)$; accordingly, y_2 is $y_2(x)$, which is another function of x .

units above it. Quite obviously, this rule holds true for any b (if $b < 0$, then the line lies below the origin and below the corresponding straight line $y = kx$).

Now that we see how straight lines with equations $y = kx$ are located for distinct k , we can readily imagine the general positions of straight lines $y = kx + b$ with arbitrary k and b . Exercises that will help you to drill this material are given at the end of the section.

The quantity k in the equation $y = kx + b$ is called the slope of the line. It determines how much the line is slanted. In the particular case of $k = 0$ the equation is $y = b$ (it is assumed that $y = b$ for arbitrary values of x), which is associated with a horizontal straight line with a slant (slope) of zero. We can imagine a man walking from left to right in the direction of increasing values of x . If $k > 0$, then he walks up a positive slope, if $k < 0$, he walks downhill (negative slope). The quantity k indicates the ratio of the variation of the function to the variation of the argument. Indeed,

$$\frac{y(x_2) - y(x_1)}{x_2 - x_1} = \frac{kx_2 + b - (kx_1 + b)}{x_2 - x_1} = k$$

We have already calculated this relation—when we proved that a linear function on a graph is depicted by a straight line.

Henceforth, in the general case of an arbitrary function, we will consider the quantity

$$\frac{y(x_2) - y(x_1)}{x_2 - x_1} = \tan \alpha$$

equal to the tangent of the angle between the line segment joining the two points $x_1, y(x_1)$ and $x_2, y(x_2)$ and the axis of abscissas. A linear function is distinguished by the fact that this quantity is the same for any two points; it is independent of x_2 and of x_1 and for this reason all points of a linear function are collinear.

Exercise

Construct the straight lines: $y = 3x$, $y = 3x + 2$, $y = 3x - 1$, $y = 2 - x$, $y = 2 - 0.5x$, $y = -x - 3$.

1.5 THE PARABOLA

Consider the function

$$y = ax^2$$

with distinct values of a . For the first example we take $a = 1$.

What general properties does this function have?

(1) It is always true that $y > 0$, both for $x > 0$ and for $x < 0$. This means that the entire curve is located above the x -axis and touches the x -axis only at the origin.

(2) y has a minimum (smallest value) at $x = 0$. The minimum of y is equal to 0. On the graph, the minimum is the lowest point of the curve.

(3) Associated with two values of x identical in absolute value but with opposite signs are values of y identical both in sign and absolute value. This means that the curve is symmetric about the y -axis.

The curve is shown in Fig. 16. It is called a *parabola* (for any value of a).

For an arbitrary positive a , the equation $y = ax^2$ has the same properties as indicated above for $y = x^2$.

What will happen if $a < 0$? Consider an example with $a = -2$, $y = -2x^2$. The curve is shown in Fig. 17 (the scale here is smaller than that of Fig. 16). The properties of this curve are:

(1) $y < 0$ for arbitrary x . The whole curve lies below the x -axis and touches the x -axis at the origin,

(2) the function y has a maximum value at $x = 0$. This maximum is equal to $y = 0$. Recalling that negative quantities are smaller

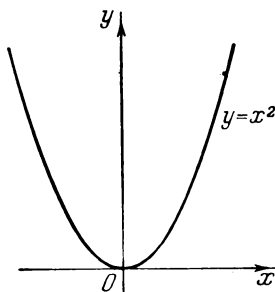


Fig. 16

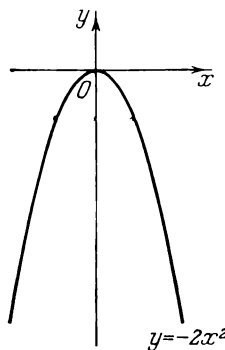


Fig. 17

than zero, we see that the maximum value of y is precisely $y = 0$. On the graph the maximum is the uppermost point of the curve,

(3) the curve is symmetric with respect to the y -axis, as in the case of positive a .

Now let us consider a similar equation:

$$y = a(x - n)^2$$

We take $a = 1$, $n = 3$. The curve is given in Fig. 18. This is the same parabola as shown in Fig. 16 but it has been displaced rightwards 3 units along the x -axis.

This simple fact is not usually realized as readily as it should be.

If a function $y = f(x)$ is given and we compare it with another function $y = f(x - n)$, then the second graph is shifted rightwards

from the first by n units. It is assumed here that in both cases f is one and the same function. In our example, the symbol f denotes a squaring of the argument, i.e., the quantity in brackets under the functional symbol:

$$\begin{aligned} f(g) &= g^2, \quad f(h) = h^2, \\ f(x) &= x^2, \quad f(-x) = x^2, \\ f(x-2) &= (x-2)^2, \\ f(x-n) &= (x-n)^2 \end{aligned}$$

Why is the graph shifted to the right? We will go into this in more detail. Suppose the graph of the function $y_1 = f(x)$ has some kind of characteristic point $x = x_0$ (a kind of notch, so to say). For example,

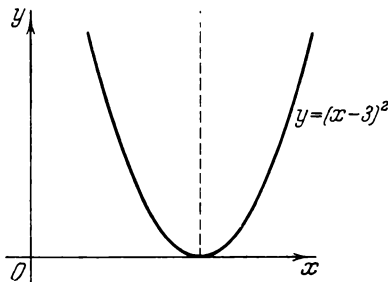


Fig. 18

at this point the function may have, say, a salient point or a maximum, or it may merely assume some definite value y_0 . Then that same value y_0 , or the same salient point, appears on the graph of the new function $y_2 = f(x-n)$ when the argument of the function f is equal to the old value x_0 , i.e., $x-n = x_0$. But this means that now the coordinates of the notch are $x = x_0 + n, y = f(x_0)$. It is

clear then that any notch, as it were, moves together with the whole graph to the right ($x = x_0 + n$ in place of $x = x_0$). Compare the curves in Fig. 16 and Fig. 18. Here, for the notch we can take $x_0 = 0, f(x_0) = (0)^2 = 0$.

All this is very simple and elementary, but it is extremely important and the student should not merely learn it but fully comprehend the meaning of it. The first urge of most students is to say that when we replace $y = x^2$ by $y = (x-3)^2$ the curve is displaced to the left because we subtract 3 from the value of x . It is well worth your time to make a detailed analysis of the examples.

Now we can state the general rules:

(1) The curve $y = a(x-n)^2$ has the vertical line $x = n$ for its axis of symmetry.

(2) This curve, for $a > 0$, lies above the x -axis and has a minimum $y = 0$ for $x = n$. For $a < 0$ the curve lies below the x -axis and has a maximum $y = 0$ for $x = n$.

Finally, there is yet another modification of the equation which does not alter the shape of the curve. Consider the function

$$y = a(x-n)^2 + m$$

This curve clearly differs from the preceding one (without m) solely in the vertical displacement by the quantity m . The position of the axis of symmetry of the curve remains unchanged; for $a > 0$ the function has a minimum at $x = n$ and the value of the function at the minimum is equal to $y = m$ (the minimum, together with the whole curve, was shifted by the amount m). For $a < 0$, the point $x = n$, $y = m$ is a point of maximum. Two examples will suffice:

$$y = (x - 3)^2 + 2 \quad (\text{Fig. 19}),$$

$$y = -(x - 3)^2 + 2 \quad (\text{Fig. 20})$$

The axis of symmetry in both figures is indicated by the dashed line. The minimum point in Fig. 19 and the maximum point in Fig. 20

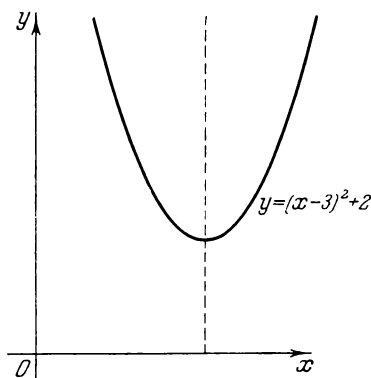


Fig. 19

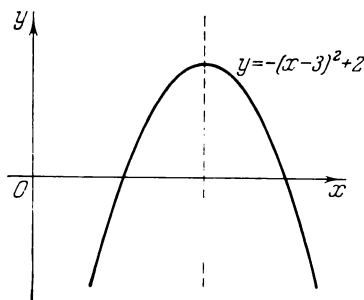


Fig. 20

lie at the intersection of the curve and the dashed axis of symmetry. To summarize, the function

$$y = a(x - n)^2 + m$$

is a parabola with axis of symmetry $x = n$ and minimum y (if $a > 0$) at the point $x = n$, $y = m$. For $a < 0$, that point is the maximum of y .

On the graph, the minimal (smallest) value of y corresponds to the lowermost point of the curve, that is, the point at which y has the smallest value. The maximum of the function $y(x)$ (the greatest value of the function) corresponds, on the graph, to the point located above all other points. Instead of speaking of the point on the graph which corresponds to the maximum or minimum of the function, we simply say the maximum point or the minimum point of the curve.

Removing brackets in the expression $y = a(x - n)^2 + m$, we write

$$y = ax^2 - 2anx + an^2 + m$$

This is a polynomial of degree two, which, in its most general form, has the notation

$$y = a_1x^2 + bx + c$$

Choosing suitable values for a , n , m in the preceding expression, we can make it identical with the latter expression. To do this, compare the corresponding terms with x^2 , with x and without x :

$$a_1x^2 = ax^2, \quad -2anx = bx, \quad an^2 + m = c$$

From the first expression we have $a_1 = a$.

From the second, $-2an = b$, $n = -\frac{b}{2a}$.

From the third, $an^2 + m = c$, $m = c - an^2 = c - \frac{b^2}{4a}$.

Thus, we can write the identity

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right)$$

Using the graph of a parabola, we can investigate the solution of a quadratic equation and the various cases that arise in that connection. We can approach the solution of the quadratic equation

$$ax^2 + bx + c = 0$$

this way: consider the whole curve

$$y = ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right)$$

and find the points of intersection of this curve with the x -axis (x -intercepts). At these points $y = 0$ and so the values of x corresponding to the points of intersection are the roots of the quadratic equation.

But we know that the curve $y = ax^2 + bx + c$ is a parabola. We know that this parabola has an axis of symmetry, the vertical line $x = -\frac{b}{2a}$, and that for $a > 0$ the parabola has a minimum point on the axis of symmetry and the altitude (ordinate) of this minimum is $y = c - \frac{b^2}{4a}$ (we glance at the second part of the last formula; this part has the customary form $a(x - n)^2 + m$). For $a > 0$, the limbs of the parabola point upwards.

It is clear that if the minimum lies above the x -axis, the parabola does not intersect the x -axis at any point (Fig. 21, Curve 1), which means that for

$$a > 0, \quad c - \frac{b^2}{4a} > 0 \quad \checkmark$$

the quadratic equation has no real roots.*

* Only real values of x and y are plotted and so complex and imaginary roots do not correspond to any points of intersection on the graph.

Now, if the minimum lies below the x -axis and the limbs of the parabola point up, the parabola will definitely cut the x -axis in two points; these points will be symmetric with respect to the line $x = -\frac{b}{2a}$ (Curve 2 in Fig. 21).

Then for

$$a > 0, \quad c - \frac{b^2}{4a} < 0 \quad \checkmark$$

the equation has two roots x_1 and x_2 as shown in Fig. 21.

Finally, there may be an intermediate case where the parabola is tangent to the x -axis (Curve 3 in Fig. 21). This case occurs when

$$c - \frac{b^2}{4a} = 0 \quad \checkmark$$

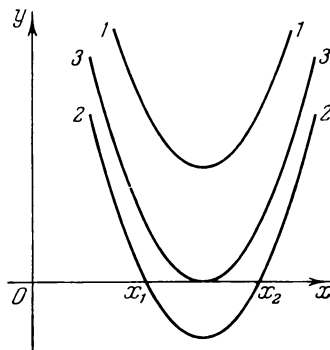


Fig. 21

If we gradually move from Curve 2 to Curve 3 as the parabola moves upwards, then obviously the two roots x_1 and x_2 will come closer together and, ultimately, at the instant of tangency, will merge. That is why, in the case of $c - \frac{b^2}{4a} = 0$, we speak not of one root but of two equal (coincident) roots of the equation.

The case $a < 0$ is considered in a similar manner. Then the curve has a maximum and the limbs point down. The reader is advised to draw the curves himself and to verify that for

$$a < 0, \quad c - \frac{b^2}{4a} < 0, \quad \text{there are no real roots,}$$

$$a < 0, \quad c - \frac{b^2}{4a} > 0, \quad \text{there are two real roots,}$$

$$a < 0, \quad c - \frac{b^2}{4a} = 0, \quad \text{there are two equal roots (tangency).}$$

The ordinary formula for the roots of a quadratic equation is

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The equation has two real roots when we are able to take a square root of $b^2 - 4ac$, that is, when

$$b^2 - 4ac > 0$$

Write this expression as follows:

$$b^2 - 4ac = -4a \left(c - \frac{b^2}{4a} \right)$$

The condition $b^2 - 4ac > 0$ holds true in two cases:

$$(1) \quad a > 0, \quad c - \frac{b^2}{4a} < 0,$$

$$(2) \quad a < 0, \quad c - \frac{b^2}{4a} > 0.$$

These are the two cases of the existence of two roots which were obtained earlier from a consideration of the curves $y = ax^2 + bx + c$.

Observe, finally, that, depending on the sign of the coefficient a of x^2 in the equation of the parabola, the curve is convex down (for $a > 0$) or convex up (for $a < 0$). This property does not depend on the values and the signs of b and c in the equation of the parabola $y = ax^2 + bx + c$.

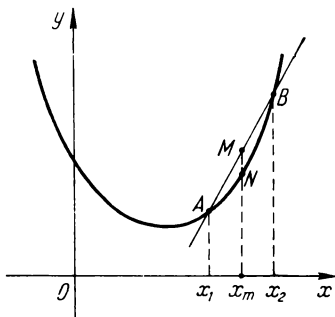


Fig. 22

An exact definition of convexity is this: take two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on a curve and draw a straight line through them. If the portion of the curve between the two points lies **below** the straight line, we say that the curve is **convex down**. If the portion of the curve between the points lies **above** the straight line, we say that the curve is **convex up**.

The convexity of a parabola is readily seen in a drawing, but we can also define it algebraically. Take arbitrary x_1 and x_2 . They are associated with the points on the parabola $A(x_1, y_1 = ax_1^2 + bx_1 + c)$ and $B(x_2, y_2 = ax_2^2 + bx_2 + c)$. We find the coordinates of point M lying at the midpoint of the line segment AB . It may be demonstrated geometrically that if line segments AM and MB are equal (Fig. 22), then the coordinates of the point $M(x_m, y_m)$ are arithmetic means of the coordinates of A and B

$$x_m = \frac{x_1 + x_2}{2} \quad \text{and} \quad y_m = \frac{y_1 + y_2}{2}$$

Now let us find the coordinates of the point $N(x_n, y_n)$ lying on the parabola for $x_n = x_m = \frac{x_1 + x_2}{2}$. Substituting $x_n = \frac{x_1 + x_2}{2}$ into the equation, we find y_n . The reader can assure himself that

$$y_n - y_m = a \left(\frac{x_1 + x_2}{2} \right)^2 - \left(a \frac{x_1^2}{2} + a \frac{x_2^2}{2} \right) = -a \left(\frac{x_1 - x_2}{2} \right)^2$$

The other terms involving b and c cancel out.

The quantity $\left(\frac{x_1 - x_2}{2}\right)^2$ is positive for arbitrary x_1, x_2 . Consequently, for $a > 0$, $y_n < y_m$ is a point on the parabola below the corresponding point (with same x) on the straight line. Thus, the parabola is convex down.

1.6 THE CUBIC PARABOLA, HYPERBOLA AND CIRCLE

We briefly consider a few more examples of curves representing simple functions.

Fig. 23 depicts a curve defined by the formula

$$y = x^3 + x$$

This curve has the distinguishing feature that on any portion of it, y increases as x is increased; the curve constantly rises from left to right. It has neither maximum nor minimum. Quite obviously, such a curve cuts the axis of abscissas only once, when $x = 0$. Fig. 24 shows

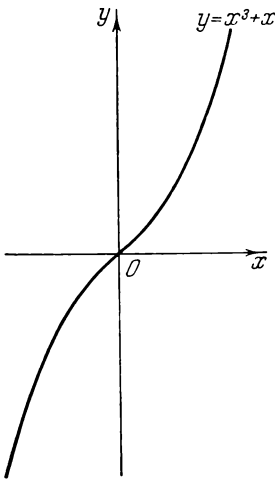


Fig. 23

a curve constructed on the basis of the formula

$$y = x^3 - x \quad (1.6-1)$$

As is evident from the graph, this curve has two portions where y in-

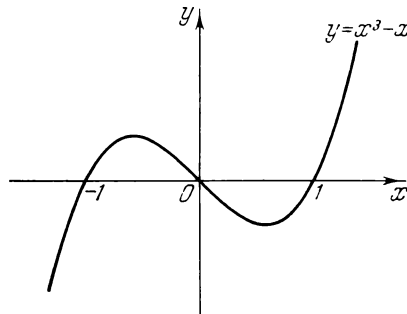


Fig. 24

creases with increasing x : for negative $x < -0.57$ and for positive $x > +0.57$. Between them, on the interval $-0.57 < x < +0.57$, the function is decreasing; y decreases as x grows. The function has a maximum when $x = -0.57$, $y = +0.38$. In this context, the word "maximum" does not mean that in the given case $y = 0.38$ is the greatest possible value of y given by the expression (1.6-1). It is clear that for large positive values of x the quantity y will assume arbitrarily large values. What is conspicuous about the maximum point ($x = -0.57$, $y = +0.38$)?

It will be seen from the graph that at this point y is greater than at adjacent points. The point of maximum separates the portion of the curve where the function is growing (to the left of maximum) from the portion where the function is decreasing (to the right of the maximum point). This is what is called a local (relative) maximum: the value of y at this point is greater than the values of $y(x)$

at other points, but only for x that are not too far away from $x = -0.57$. The same goes for the point $x = +0.57$, $y = -0.38$. Here the function has a local (relative) minimum.

In Fig. 25 we have two more examples of curves describing polynomials of degree three. The cubic equa-

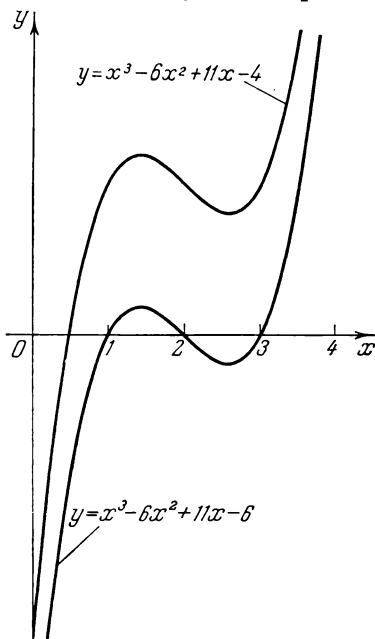


Fig. 25

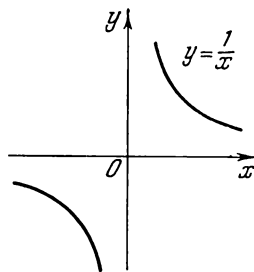


Fig. 26

tion which we get by equating the polynomial to zero has one real solution, $x = 0.48$ in the case of the upper curve, and three roots, $x_1 = 1$, $x_2 = 2$, $x_3 = 3$, in the case of the lower curve. It is easy to see that a cubic equation always has at least one real root: to convince himself, the reader is advised to examine the variation of the curve $y = ax^3 + bx^2 + cx + d$ for very large (in modulus) positive and negative values of x .

The reader is now asked to construct the curves $y = x^3$ and $y = -x^3$.

In Fig. 26 we have the curve

$$y = \frac{1}{x}$$

which is called a hyperbola. This curve has the peculiarity that for x small and negative, y is a negative quantity, which in modulus (absolute value) is very large, but for small positive x , y is a very large positive quantity (see Table 5).

Table 5

x	-1	-0.1	-0.01	-0.001	0.001	0.01	0.1	1
y	-1	-10	-100	-1000	1000	100	10	1

For this reason we say that the value of y for $x = 0$ is $\pm\infty$, i.e., plus infinity or minus infinity depending on the side from which we approach $x = 0$.

From Fig. 26 it is evident that the curve consists of two separate branches for $x < 0$ and for $x > 0$.

Up to now we have given y as a function of x and have constructed the appropriate curve. Using the circle to illustrate our case, we will now solve the converse problem: we specify the curve and desire to find the functional relationship between y and x associated with this curve. We consider a circle of radius r with centre at the origin. Points of the circle are distant r from the origin. By the Pythagorean theorem we have

$$x^2 + y^2 = r^2$$

If we want to express y explicitly as a function of x , we get

$$y = \pm \sqrt{r^2 - x^2}$$

When we consider the circle as a graph of the function $y(x)$, then it is evident from the graph that the function is not single-valued: for every value of x (for $|x| < r$) there are two points on the curve—on the upper and lower semicircles. These two points are associated with the two signs of the square root. The function $y = +\sqrt{r^2 - x^2}$ corresponds to the upper semicircle, the function $y = -\sqrt{r^2 - x^2}$, to the lower semicircle.

Now let us set up the equation of the circle with centre at the point (a, b) . We proceed formally. We know that if x is replaced by $x - a$ in the expression of the function, the graph is then displaced a units to the right. Thus

$$y = \pm \sqrt{r^2 - (x - a)^2}$$

is the equation of a circle shifted rightwards a units, that is, with centre at the point $(x = a, y = 0)$ on the x -axis.

To continue, if we now add one and the same quantity b to all values of y , the entire graph will go upwards b units.*

* We could have said that the graph moves up b units when y is replaced by $y - b$, so that we get

$$y - b = \pm \sqrt{r^2 - (x - a)^2}$$

which is equivalent to the equation in the text.

The desired equation of the circle with centre at the point (a, b) is

$$y = \pm \sqrt{r^2 - (x - a)^2} + b$$

In this particular case it would have been easier to derive the equation directly in geometric fashion from the expression for the distance between the points (x, y) and (a, b) :

$$r^2 = (x - a)^2 + (y - b)^2$$

We purposely use the more formal approach to illustrate once again how the replacement of x by $x - a$ and y by $y - b$ displaces the curve.

1.7 ALTERING THE SCALE OF A CURVE

In the preceding section we learned how to change the equation of a curve so that the curve is displaced (this is called a parallel translation of the curve). When x is replaced by $x + a$ in the expression relating y and x , the corresponding curve is displaced a units leftwards, when y is replaced by $y + b$, the curve is displaced b units downwards. To shift the curve g units to the right, we have to replace x by $x - g$, to raise the curve h units up, we have to replace y by $y - h$.

The equation of a circle of radius r with centre at the origin is $x^2 + y^2 = r^2$. The equation of a circle of the same radius with centre at the point $x_c = g, y_c = h$, that is, displaced g units rightwards and h units up (the initial position was with centre at the origin) is

$$(x - g)^2 + (y - h)^2 = r^2$$

For an arbitrary curve whose equation is written in the form $y = f(x)$, we write, for a displacement g units rightwards and h units up,

$$y - h = f(x - g) \text{ or } y = h + f(x - g)$$

For the same translation for a curve whose equation is written in the form $F(x, y) = 0$, we have to replace the equation by $F(x - g, y - h) = 0$.

Now suppose we want to change the equation of the curve in order to increase C -fold all the vertical dimensions.*

Obviously, in place of the equation $y_0 = f(x)$ we have to take the equation $y_1 = Cf(x)$. Then for the same x the quantity y_1 will be C times greater than before, that is to say, C times y_0 .

As an example, recall the equations of straight lines passing through the origin. The equation of a straight line passing at an angle of 45°

* For the sake of simplicity, we from now on assume that $C > 1$, to increase a quantity two-fold means to multiply it by 2, but to increase it 0.3-fold means to multiply it by 0.3, or actually to reduce it.

in the first quadrant is

$$y_0 = x$$

The equation

$$y_1 = 10x$$

corresponds to a straight line that is more steeply slanted; for a given x the ordinate is 10 times greater (see Fig. 14).

The law of transition from $y_0 = f(x)$ to $y_1 = Cf(x)$ may also be written thus: in the equation of the curve $y_0 = f(x)$ we replace y_0

by $\frac{y_1}{C}$, i.e., we write $\frac{y_1}{C} = f(x)$. Then the dependence of y_1 on x is characterized by the fact that the curve $y_1(x)$ is elongated C -fold vertically as compared with the curve $y_0(x)$.

It would appear at first glance that there is no need to waste time on two different formulations whose identity is obvious:

$$y_1 = Cf(x) \rightarrow \frac{y_1}{C} = f(x)$$

One is as good as the other. But the second formulation (y_0 replaced by $\frac{y_1}{C}$) is convenient for the case where the curve is given by an equation not solved for y , that is, an equation of the form $F(x, y) = 0$.

For example, the equation of a circle of radius 1 is conveniently written as

$$F(x, y_0) = x^2 + y_0^2 - 1 = 0$$

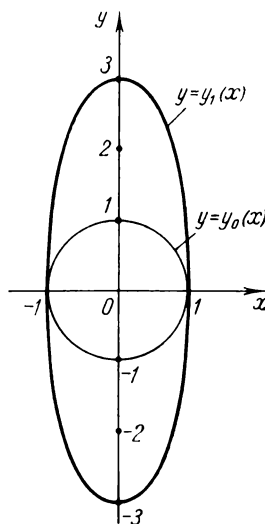


Fig. 27

Now how would you write the equation of a curve elongated three times along the vertical axis [Fig. 27, the curve labelled $y_1(x)$]?

By the rule which we have just stated, in the equation of the circle, replace y_0 by $\frac{y_1}{3}$ to get

$$x^2 + \left(\frac{y_1}{3}\right)^2 - 1 = 0$$

This curve is called an *ellipse*.

The equation is solved in a simple manner:

$$y_0 = \sqrt{1 - x^2}, \quad y_1 = 3\sqrt{1 - x^2}$$

and it is quite evident that $y_1 = 3y_0$ for equal x . But the rule by which replacing y by $\frac{y}{C}$ leads to a C -fold elongation of the curve along the vertical axis holds true also for curves defined by a complicated equa-

tion $F(x, y) = 0$ that cannot be solved algebraically for y , say,

$$x + y \log_{10} y = 0$$

The statement concerning y replaced by $\frac{y}{C}$ is readily extended to the x -coordinate as well. When we replace x_0 by $\frac{x_1}{C}$ in the equation of the curve, the curve stretches C times along the x -axis, which is to say, for equal y the value x_1 is C times x_0 .

We begin with examples instead of a proof:

$$y = x_0 \quad \text{and} \quad y = \frac{x_1}{10} = 0.1x_1$$

(see Fig. 14). The first line slants at an angle of 45° to the x -axis, the second line is less steep.

Another illustration:

$$x_0^2 + y^2 - 1 = 0, \quad \left(\frac{x_1}{2}\right)^2 + y^2 - 1 = 0$$

The first equation corresponds to a circle of radius 1, the second to

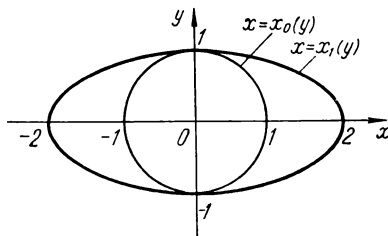


Fig. 28

the equation of a curve stretched 2-fold along the x -axis. It is easy to see that the curve cuts the x -axis at the points

$$y = 0, \quad \left(\frac{x_1}{2}\right)^2 - 1 = 0, \quad x_1 = \pm 2$$

(Fig. 28).

To prove this, we can solve the equation for x : if $y = f(x_0)$, $y = f\left(\frac{x_1}{C}\right)$, then we get

$$x_0 = \varphi(y), \quad \frac{x_1}{C} = \varphi(y)$$

where φ is what is known as the inverse function of f .

The important thing is that f is one and the same function in the formulas involving x_0 and x_1 . Therefore φ is also the same for x_0 and x_1 . Rewriting the second equation

$$\frac{x_1}{C} = \varphi(y) \rightarrow x_1 = C\varphi(y)$$

we get

$$x_1(y) = Cx_0(y) \cdot$$

which fits the formulation: replacing x by $\frac{x}{C}$ stretches the curve C times along the x -axis.

Here is an example:

$$y = 10^{x_0}, \quad y = 10^{\frac{x_1}{2}}$$

The inverse of the power function is the logarithmic function (logarithm)

$$x_0 = \log_{10} y, \quad \frac{x_1}{2} = \log_{10} y, \quad x_1 = 2 \log_{10} y$$

What do we do if in the equation $y = f(x)$, kx is substituted for x ? To take advantage of the above-stated rule, let us recall the rule for dividing by a fraction. Multiplication by k is the same as division by $\frac{1}{k}$:

$$kx = \frac{x}{\frac{1}{k}}$$

Here, $\frac{1}{k}$ plays the role of C in the earlier formulas.

If, for example, $k = \frac{1}{2}$, then $\frac{1}{k} = 2$, i.e., $C = 2$ and the substitution of $0.5x_1$ for x_0 is the same as replacing x_0 by $\frac{x_1}{2}$ and leads to a stretching of the curve along the x -axis by a factor of 2.

If $k = 3$, then $\frac{1}{k} = \frac{1}{3}$, or $C = \frac{1}{3}$, and the replacement of x by $3x$ is the substitution of $\frac{x}{3}$ for x . What does this signify geometrically?

Up to now we have considered only the case of positive values of C , $C > 1$ and we stated the result thus: when, in the equation of the curve, y is replaced by $\frac{y}{C}$, $y \rightarrow \frac{y}{C}$, the curve is stretched vertically by a factor of C ; in the substitution $x \rightarrow \frac{x}{C}$, the curve is stretched horizontally by a factor of C .

If C is positive but less than unity, $0 < C < 1$, which corresponds to $k > 1$, then the substitution $y \rightarrow \frac{y}{C}$ changes the vertical dimensions by a factor of C ; but since $C < 1$, it follows that the C -fold alteration is a compression. For example, for $C = 0.5$, a C -fold change amounts to a multiplication of the height (ordinate) by 0.5, which means an actual reduction in size by one half. The same goes for the substitution $x \rightarrow \frac{x}{C}$: for $0 < C < 1$, this substitution amounts to

a compression of the curve. Here is another case. In Fig. 29 we have two curves

$$y_0 = \sin x, \quad y_1 = \sin 3x$$

The second curve has been compressed horizontally by a factor of three.

The relation $y = \sin x$ is a periodic function: for $x = 2\pi \approx 6.3$ (which corresponds to an angle of 360° in degree measure), the sine has the same value as for $x = 0$. Adding 2π to any angle leaves the value of the sine unchanged. The function $y = \sin 3x$

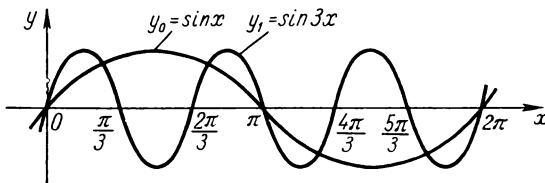


Fig. 29

is also periodic, but the period here is less by a factor of 3. If x varies by $\frac{2\pi}{3} \approx 2.1$ radians, then $3x$ (the angle whose sine is laid off on the axis of ordinates) varies by 2π and $\sin 3x$ returns to the same value

$$\sin 3x = \sin 3 \left(x + \frac{2\pi}{3} \right)$$

Use this example to think over the general assertion that the substitution $x \rightarrow kx$ in the equation of the curve results in multiplying horizontal dimensions by $\frac{1}{k}$. In the given example, $k = 3$, the horizontal dimensions, in particular the distance along the x -axis between the points where $y = 0$, are multiplied by $\frac{1}{3}$, which means a 3-fold reduction in size (compression).

For a periodic function, the substitution $x \rightarrow kx$ reduces the period k times, but increases the frequency (number of periods per unit length) k times.

Despite the simplicity of these arguments, the truly arithmetic nature of the reasoning, beginners (the audience for which this book is designed) frequently make mistakes here.

Finally, let us examine what happens in the substitutions $y \rightarrow \frac{y}{C}$

or $x \rightarrow \frac{x}{C}$ for negative C . A substitution of this kind can be carried out in two stages: we write $C = -1 \cdot b$, where b is positive, and then

perform two substitutions:

$$y_0 \rightarrow \frac{y_1}{b} \rightarrow \frac{y_2}{-1 \cdot b} = -\frac{y_2}{b}$$

The first operation, the substitution $y_0 \rightarrow \frac{y_1}{b}$, where $b > 0$, has already been analyzed. It leads to a b -fold change in the vertical direction. It remains to consider the effect made by the change in sign of y , the replacement of y by $-y$. This was examined for separate points in Sec. 1.1. For curves, we give the answer without proof: a change in the sign of y leads to a reflection of the curve in the x -axis, a change in the sign of x leads to a reflection in the y -axis.

Here is an example:

$$F(x, y) = (x - 3)^2 + (y - 5)^2 - 4 = 0$$

This is the equation of a circle of radius 2 with centre at a point with coordinates $x = 3$, $y = 5$.

The following curves are depicted in Fig. 30:

$$F(x, -y) = (x - 3)^2 + (-y - 5)^2 - 4 = 0,$$

$$F(-x, y) = (-x - 3)^2 + (y - 5)^2 - 4 = 0,$$

$$F(-x, -y) = (-x - 3)^2 + (-y - 5)^2 - 4 = 0$$

As is evident from the formulas, the sign of F in all cases represents one and the same function (follow this through carefully as you regard the first part of the formulas). See what happens to the curve (circle) under the substitution $x \rightarrow -x$, $y \rightarrow -y$, and under the simultaneous substitution $x \rightarrow -x$, $y \rightarrow -y$. A firm grasp of these rules will make it possible, after you have analyzed a curve like

$$y = f(x) \text{ or } F(x, y) = 0$$

to picture the curves of all similar functions

$$\frac{y-b}{c_2} = f\left(\frac{x-a}{c_1}\right) \text{ or } F\left(\frac{x-a}{c_1}, \frac{y-b}{c_2}\right) = 0$$

with arbitrary values of the four constants a , b , c_1 , c_2 .

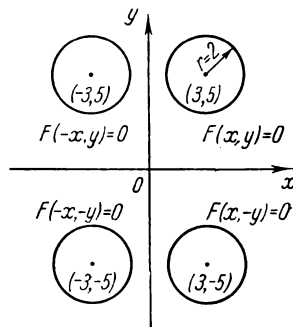


Fig. 30

Exercises

1. Construct the curves $\frac{x^2}{4} + \frac{y^2}{9} - 1 = 0$, $\frac{(x+3)^2}{4} + \frac{(y-5)^2}{9} - 1 = 0$, $\frac{(x-3)^2}{9} + \frac{(y+5)^2}{4} - 1 = 0$, knowing that $x^2 + y^2 - 1 = 0$ is the equation of a circle. In curve sketching, a suggested procedure is the following: mark the

centre, the upper and lower points, the extreme right-hand and extreme left-hand points, and then join them freehand with a smooth curve.

2. Make a detailed sketch of the curve $y = \sin x$, taking x from $-\pi$ to $+\pi$ at 0.25π intervals. It is assumed that x is the angle expressed in radian measure; it is therefore convenient to take definite fractions of π because then the angles (expressed in degrees) will be integral: $0.25\pi = 45^\circ$, $0.5\pi = 90^\circ$ and so on.

You can also take advantage of Table VI of the Appendix, where the sine is given as a function of the angle expressed in radian measure. In all cases, lay off x in radians on the axis of abscissas.

Sketch the following curves.

- (a) $y = 2 \sin x$,
- (b) $y = \sin 0.5x$,
- (c) $y = 3 \sin 3x$,
- (d) $y = \cos x$.

Hint. Take advantage of the trigonometric identity

$$\cos x = \sin \left(x + \frac{\pi}{2} \right)$$

$$(e) \quad y = \cos x + \sin x = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right),$$

$$(f) \quad y = \cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x = \frac{1}{2} + \frac{1}{2} \sin \left(2x + \frac{\pi}{2} \right),$$

$$(g) \quad y = \sin^2 x = \frac{1}{2} - \frac{1}{2} \sin \left(2x + \frac{\pi}{2} \right)$$

Construct all these curves from (a) to (g) by translating, stretching or compressing the curve $y = \sin x$.

3. Plot the following curves.

(a) $y = \pm \sqrt{x^2 - 1}$
or, in symmetric form, $y^2 - x^2 + 1 = 0$, given x from -5 to $+5$ at intervals of 0.5 . If y is imaginary for appropriate x , then there is no curve.

$$(b) \quad y = 2 \pm \sqrt{(x-1)^2 - 1},$$

$$(c) \quad y = \pm \sqrt{x^2 + 1}.$$

Hint. Transforming to $x^2 - y^2 + 1 = 0$, observe that (c) is obtained from (a) by interchanging x and y ;

$$(d) \quad 4y^2 + 4y - x^2 = 0.$$

Hint. Write the equation in the form

$$4 \left(y + \frac{1}{2} \right)^2 - x^2 - 1 = 0$$

and obtain the curve by translating and compressing Curve (c).

1.8 PARAMETRIC REPRESENTATION OF A CURVE

Let each of the quantities x and y be given as a function of time t , i.e., suppose we have two functions $x(t)$ and $y(t)$, say,

$$x = \cos t, \quad y = \sin t$$

These relations can be depicted graphically as two curves by plotting t on the axis of abscissas and x on the axis of ordinates in one drawing, and t on the axis of abscissas and y on the axis of ordinates in the other.

However, the problem may be posed differently: imagine that x and y are the coordinates of a point and each value of t is associated with a specific position of the point. Then we want to see what kind of curve is described by a point in the xy -plane as t varies.

We can eliminate t from the two equations that yield $x = x(t)$ and $y = y(t)$ to get an expression which will involve only y and x , i.e., either $y = y(x)$ or $F(x, y) = 0$. Then we construct the curve in the usual fashion by specifying various x and finding the corresponding y .

Thus, in the example given earlier, we find

$$x^2 + y^2 = \cos^2 t + \sin^2 t = 1,$$

$$y = \pm\sqrt{1 - x^2} \text{ or } x^2 + y^2 - 1 = 0$$

so that the curve is a circle in the xy -plane.

However, it often happens that even comparatively simple expressions for $x(t)$ and $y(t)$ lead to such involved formulas when trying to eliminate t that it makes no sense to tackle them. For instance, if

$$x = a_1 t^4 + b_1 t^3 + c_1 t^2 + d_1 t + e_1,$$

$$y = a_2 t^4 + b_2 t^3 + c_2 t^2 + d_2 t + e_2$$

then to eliminate t we would have to solve a quartic equation, and this leads to extremely unwieldy expressions.

Yet it is possible to construct the curve in the xy -plane without eliminating t : it suffices to specify various values of t and find x and y for each of them. To illustrate, take Table 6.

Table 6

t	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$x = \cos t$	1	0.7	0	-0.7	-1	-0.7	0	0.7	1
$y = \sin t$	0	0.7	1	0.7	0	-0.7	-1	-0.7	0

It is clearly not necessary to take t greater than 2π because the values of x and y repeat. Using this table we can plot the points of the curve. In so doing, we employ only the values of x and y . Those values of t for which the x 's and y 's have been computed are no longer needed for plotting the points. "The Moor has done his work and he may leave," to quote from a familiar play by Schiller.

This method of representing a curve or, what is the same thing, of specifying a functional relationship $y(x)$, is called parametric representation. The quantity t is called the parameter.

Exercises

1. Construct the curve given by the equation

$$x = \cos t, \quad y = \sin 2t$$

The same for

$$x = \cos t, \quad y = \sin 3t$$

Hint. Since $\sin 3t$ varies rapidly, you can take close-lying values of t , say 0, 0.1, 0.2,

2. As a joke, construct the curves

(a) $x = \cos 3t, \quad y = \sin 3t,$

(b) $x = \cos (5t + 1), \quad y = \sin (5t + 1).$

3. This problem is no joke: construct the curve

$$x = \cos t, \quad y = \cos \left(t + \frac{\pi}{4} \right)$$

4. Construct the curve

$$x = \cos t, \quad y = \cos t$$

5. Construct the curve traced out by a moving point A lying on the circumference of a disk of radius 1 cm, the disk is rolling along the x -axis at 1 cm/sec. At the start, the centre of the circle Q lies on the y -axis and the point of interest A lies at the origin. In time t , the coordinates of the centre are $Q_t(t, 1)$, the circle has rotated through an angle of t radians. This curve is called a cycloid.

Chapter 2

The Concepts of a Derivative and an Integral

2.1 MOTION, DISTANCE AND VELOCITY

Let us examine the translational motion of a body along a straight line. Denote the distance of some point of the body to a specific point on the line by z . We will consider the distance in one direction to be positive, in the opposite direction, negative. For example, suppose the line along which our body is moving is vertical. Points above O will correspond to positive z , those below O to negative z .

In the process of the motion, the z -coordinate is dependent on the time (instead of saying "the distance from a specific point of the body to a definite point of the line" we will say "the z -coordinate"). The motion of the body is defined by the dependence of z on the time t , that is to say, by indicating the function $z(t)$. If we know the function $z(t)$, we can find the position of the body at any instant of time.

The function $z(t)$ may be represented graphically by laying off time on the axis of abscissas (t -axis) and the quantity z (which indicates the position of the body) on the axis of ordinates.

In uniform motion with a constant velocity v , the distance covered, s , in time t is equal to $s = vt$.

Denote by z_0 the coordinate of the body at time $t = 0$. The distance covered in time t is equal to the difference $z(t) - z_0$. Thus

$$z(t) = z_0 + vt \quad (2.1-1)$$

Hence, in uniform motion, the dependence of the coordinate on the time is given by a linear function. In the case of uniform motion the graph of the function $z(t)$ is a straight line in the coordinate plane, in which the time t is laid off on the (horizontal) axis of abscissas and the z -coordinate is laid off on the (vertical) axis of ordinates.

In the case of nonuniform motion, the function $z(t)$ is expressed by more involved formulas and the corresponding graph is some kind of curve.

Let us analyze the following problem: given a function $z(t)$, or the dependence of the coordinate of the body on the time, it is required to find the rate of motion v of the body. In the general case of

nonuniform motion, the velocity is not constant, it changes in the course of time. This means that the velocity v is also a function of the time, $v(t)$, and the problem consists in expressing $v(t)$ in terms of the known function $z(t)$.

Everything is simple in the particular case of uniform motion (at a constant velocity). The velocity is defined as the distance covered in unit time. Since the velocity is constant, it is immaterial what

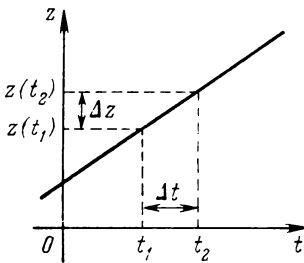


Fig. 31

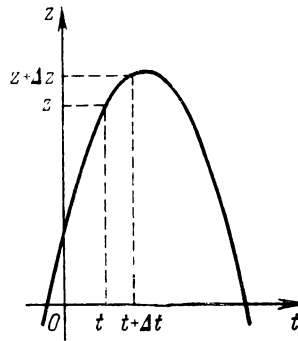


Fig. 32

particular section of the path is taken and what interval of time is chosen to determine the velocity.

Let us find the distance traversed in one second from time t_1 sec to time $t_1 + 1$ sec. This distance is equal to the difference between the coordinates $z(t_1 + 1)$ and $z(t_1)$:

$$z(t_1 + 1) - z(t_1) = [z_0 + v \cdot (t_1 + 1)] - [z_0 + vt_1] = v$$

and, numerically, is equal to the velocity. We can take an arbitrary interval of time between t_1 and t_2 and divide the distance travelled $z_2 - z_1$ by the magnitude of the time interval $t_2 - t_1$:

$$\frac{z_2 - z_1}{t_2 - t_1} = \frac{(z_0 + vt_2) - (z_0 + vt_1)}{t_2 - t_1} = v \quad (2.1-2)$$

It is precisely because the velocity is constant that we were able to take any interval $t_2 - t_1$ to compute the velocity, and the answer is independent both of the time t_1 and of the magnitude of the interval. The situation is different in the case of a variable rate of motion.

Before going over to the more general case, it will be convenient to change the notation. We will write $t_1 = t$, $t_2 = t + \Delta t$ so that the difference $t_2 - t_1$ (the time interval) is denoted by Δt (see Fig. 31). Similarly, we write Δz to denote the difference

$$z(t_2) - z(t_1) = z(t + \Delta t) - z(t) = \Delta z$$

In this notation, the average velocity v_{av} in the interval Δt between t and $t + \Delta t$ is

$$v_{av} = \frac{\Delta z}{\Delta t} \quad * \quad (2.1-3)$$

We speak here of the average velocity because in the general case the velocity itself can change over the interval Δt .

Let us consider a second example in which $z(t)$ is given by the formula

$$z(t) = z_0 + bt + ct^2 \quad (2.1-4)$$

Fig. 32 illustrates a possible graph corresponding to a function of the form (2.1-4). Let us compute the average velocity v_{av} over the interval Δt using formula (2.1-3):

$$\begin{aligned} z(t) &= z_0 + bt + ct^2, \\ z(t + \Delta t) &= z_0 + b(t + \Delta t) + c(t + \Delta t)^2, \\ \Delta z &= z(t + \Delta t) - z(t) = b\Delta t + 2ct\Delta t + c(\Delta t)^2 \end{aligned}$$

From this we get

$$v_{av} = \frac{\Delta z}{\Delta t} = b + 2ct + c\Delta t \quad (2.1-5)$$

Compare the results of (2.1-2) and (2.1-5) for the average velocity when the motion obeys the law (2.1-1) and the law (2.1-4). The second example differs in that here the average velocity depends both on the time t and on the time interval Δt .

How can we find the instantaneous velocity?

The velocity varies gradually, and so the smaller the time interval over which the distance travelled is measured, the smaller will be the change in velocity, and thus the closer will the average velocity be to the instantaneous value.

In formula (2.1-5), v_{av} contains two terms that do not depend on the magnitude of the interval Δt and one term that is proportional to Δt .

For very small Δt we can ignore this term and then v_{av} yields the instantaneous velocity:

$$v_{in} = b + 2ct \quad (2.1-6)$$

The attentive reader will most likely have recognized the expressions (2.1-4) and (2.1-6) from the school course of physics to be the formulas for uniformly accelerated motion:

$$\left. \begin{aligned} z(t) &= z_0 + v_0 t + \frac{at}{2} \\ v(t) &= v_0 + at \end{aligned} \right\} \quad (2.1-7)$$

* Note that Δ is not a factor but a symbol taking the place of the word "increment", and so Δ cannot be cancelled from the numerator and denominator of a fraction. Δ is the capital Greek letter delta. Δt is read "delta t ", Δz , "delta z "; these are also spoken of as the change in time and change in path.

All that is needed is to substitute, for b , the initial velocity v_0 (that is, the velocity at time $t = 0$) and in place of c , to substitute $a/2$, where a is the acceleration.

We have computed the instantaneous velocity at time t on the basis of the average velocity over the interval from t to $t + \Delta t$. Now let us try to compute it by choosing the interval in a somewhat different way. We find the average velocity in the interval from $t_1 = t - 3\Delta t/4$ to $t_2 = t + \Delta t/4$. As before, the duration of the interval is $t_2 - t_1 = \Delta t$. From formula (2.1-4) we get

$$z(t_1) = z_0 + b \left(t - \frac{3\Delta t}{4} \right) + c \left(t - \frac{3\Delta t}{4} \right)^2,$$

$$z(t_2) = z_0 + b \left(t + \frac{\Delta t}{4} \right) + c \left(t + \frac{\Delta t}{4} \right)^2,$$

$$z(t_2) - z(t_1) = b\Delta t + 2ct\Delta t - \frac{1}{2}c(\Delta t)^2$$

Whence it follows that

$$v_{av} = \frac{z(t_2) - z(t_1)}{\Delta t} = b + 2ct - \frac{1}{2}c\Delta t \quad (2.1-8)$$

Comparing (2.1-5) and (2.1-8), we see that the average velocities over the interval from t to $t + \Delta t$ and over the interval from $t - 3\Delta t/4$ to $t + \Delta t/4$ differ by the quantity $c\Delta t [1 - (-1/2)] = 3c\Delta t/2$. But if we want to find the instantaneous velocity, we have to take a very small time interval Δt . Then the difference will vanish and we again obtain for the instantaneous velocity $v_{in} = b + 2ct$.

We have considered the concept of instantaneous velocity for two specific cases: uniform and uniformly accelerated motion. In the next section we give a more exact definition of instantaneous velocity for an arbitrary law of motion.

2.2 THE DERIVATIVE OF A FUNCTION AS THE LIMIT OF A RATIO OF INCREMENTS

In the preceding section we considered the problem of instantaneous velocity and examined ratios of the form

$$\frac{z(t_2) - z(t_1)}{t_2 - t_1}$$

for very close-lying values of t_2 and t_1 .

The expression "close-lying" is not exact, it is not rigorous. The exact formulation is this. It is necessary to find the *limit* to which the following ratio tends:

$$\frac{z(t_2) - z(t_1)}{t_2 - t_1} \quad (2.2-1)$$

as t_2 approaches t_1 . Using the designations Δt and Δz , we can rewrite this ratio as

$$v_{av} = \frac{\Delta z}{\Delta t} \quad (2.2-2)$$

In (2.2-2), the quantities Δt and Δz are related: any time interval Δt may be chosen, but after the denominator Δt has been selected, it is assumed that Δz (the numerator) is not just any distance interval but precisely that distance which corresponds to the time interval Δt . This was obvious in (2.2-1) from the way the arguments of the function $z(t_2)$, $z(t_1)$ were written in the numerator. Formula (2.2-2) is simply a different way of writing (2.2-1).

The quantity that interests us, the instantaneous velocity $v(t)$ at time t , is the limit of the ratio $\frac{\Delta z}{\Delta t}$ as Δt tends to zero. It is obvious that the approach of Δt to zero is equivalent to the approach of t_2 to t_1 , since $\Delta t = t_2 - t_1$. This statement can then be written

$$v(t) = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta z}{\Delta t} \right)$$

where $\lim$ stands for "limit". The particular kind of limit we have in mind is indicated underneath $\lim$ —when Δt approaches zero; and the arrow stands for "approaches". The quantity $\frac{\Delta z}{\Delta t}$ in brackets is the one whose limit is being sought.

What meaning do we attribute to the terms "limit", "approaching the limit"? The calculations carried out in Sec. 2.1 served as an illustration of these notions. We saw that for small intervals Δt the value of v_{av} in the second example differed from the value of v_{in} by a small quantity proportional to Δt . Although the constant of proportionality of Δt could differ for various choices of the interval, for small values of Δt in the expression for v_{av} we could always neglect the term involving Δt .

Thus, the ratio

$$\frac{\Delta z}{\Delta t} = \frac{z(t_2) - z(t_1)}{t_2 - t_1}$$

tends to a definite limit when $\Delta t = t_2 - t_1$ tends to zero. When Δt tends to zero, t_2 and t_1 approach each other without bound, and we denote their common value (as $\Delta t \rightarrow 0$) $t_2 = t_1 = t$.

The limit of the ratio, that is, the instantaneous velocity v , is a definite function of t ,

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = v(t)$$

Why is it that, when computing the velocity from the given formula $z(t)$ we have to carry out so many calculations and find Δz for distinct Δt and only then find the limit $\lim_{\Delta t} \frac{\Delta z}{\Delta t}$? Couldn't we

simply take the value $\Delta t = 0$ from the very start? We would then have $\Delta z = 0$ since $\Delta t = t_2 - t_1$ and if $t_2 = t_1$, then also $z(t_2) = z(t_1)$ and $\Delta z = z(t_2) - z(t_1) = 0$. By this thoughtless mode of operations we would get $\frac{\Delta z}{\Delta t} = \frac{0}{0}$, which means we would get nothing definite.

When computing velocity, the whole idea is to take small Δt and small Δz which correspond to Δt . In this way, we obtain a very definite ratio $\frac{\Delta z}{\Delta t}$ each time. When Δt is reduced, tends to zero, then Δz diminishes in approximate proportion to the quantity Δt , and so the ratio remains approximately constant.

The ratio $\frac{\Delta z}{\Delta t}$ approaches a definite limit when Δt tends to zero.

This limit—the *instantaneous velocity* $v(t)$ in the case of motion or, in the general case, the *derivative* of the function $z(t)$ —depends on the type of function $z(t)$ and on the value of the variable t . In the next section we will carry out the algebraic computations of the derivatives of several elementary functions and will find the exact value of the limit, that is to say, of the derivative.

2.3 NOTATION OF DERIVATIVES. THE DERIVATIVE OF A POWER FUNCTION

The limit of the ratio of the increment of the function to the increment of the independent variable as the increment of the independent variable tends to zero is of prime importance in higher mathematics and its applications. We have already seen, for example, that such an important concept as the instantaneous velocity of motion is found with the aid of the limit of such a ratio. That is why the limit of this ratio has a special name: the *derivative of the function* or, simply, the *derivative*. The first name is due to the fact that if z is a function of t , $z(t)$, then the limit of the ratio $\frac{\Delta z}{\Delta t}$, $\lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = v$, is also a function (a different function) $v(t)$ of the variable t . It depends on the value of t approached by t_1 and t_2 or, to put it differently, v depends on the value of t at which the derivative z is taken.

We have special notations for the derivative.

One notation (differential notation) is

$$\frac{dz}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t}$$

Here, the quantity $\frac{dz}{dt}$ (it is read “d z d t”) is not a fraction but is simply an abbreviated way of writing the limit on the right. The

quantity $\frac{dz}{dt}$ is written in the form of a fraction to remind us that it is obtained from the fraction $\frac{\Delta z}{\Delta t}$ by a passage to the limit.

A different notation for derivatives is the so-called prime notation, $v = z'(t)$, or, for example, for the function $y(x)$,

$$y' = y'(x) = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

In mechanics, the dot notation, $\frac{dx}{dt} = \dot{x}$, is sometimes used for time derivatives, but we will not do so here.

Occasionally, in place of the function symbol, one gives the expression of the function: if $z = at^2 + b$, then instead of $\frac{dz}{dt}$ we can write $\frac{d(at^2 + b)}{dt}$ or $(at^2 + b)'$.

Let us find the derivative of the function

$$z = t^2$$

algebraically (from first principles, that is). To do this, we form the ratio

$$\frac{\Delta z}{\Delta t} = \frac{(t + \Delta t)^2 - t^2}{\Delta t}$$

Removing the brackets in the numerator, we have

$$(t + \Delta t)^2 - t^2 = t^2 + 2t \cdot \Delta t + (\Delta t)^2 - t^2 = 2t \cdot \Delta t + (\Delta t)^2$$

Now form the ratio

$$\frac{\Delta z}{\Delta t} = \frac{2t \cdot \Delta t + (\Delta t)^2}{\Delta t} = 2t + \Delta t$$

It is now easy to find the limit: quite obviously, if the quantity is the sum consisting of a term that does not depend on Δt (in this case $2t$) and of Δt itself, then, as Δt tends to zero, all we have left is a summand that does not depend on Δt :

$$\frac{dz}{dt} = \frac{d(t^2)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = \lim_{\Delta t \rightarrow 0} (2t + \Delta t) = 2t$$

Let us consider another example:

$$z = t^3,$$

$$\Delta z = (t + \Delta t)^3 - t^3 = t^3 + 3t^2 \Delta t + 3t(\Delta t)^2 + (\Delta t)^3 - t^3,$$

$$\frac{\Delta z}{\Delta t} = 3t^2 + 3t \cdot \Delta t + (\Delta t)^2,$$

$$\frac{dz}{dt} = \frac{d(t^3)}{dt} = \lim_{\Delta t \rightarrow 0} [3t^2 + 3t \cdot \Delta t + (\Delta t)^2] = 3t^2$$

The limit was readily found in these examples since Δt cancelled out when we computed the ratio $\frac{\Delta z}{\Delta t}$. Let us consider a more complicated example:

$$z = \frac{1}{t}, \quad \frac{\Delta z}{\Delta t} = \frac{\frac{1}{t+\Delta t} - \frac{1}{t}}{\Delta t}$$

Can we disregard the quantity Δt in the first fraction, in the expression $\frac{1}{t+\Delta t}$, when we pass to the limit? No, we cannot because we have not yet cancelled out the quantity Δt in the denominator. By substituting $\frac{1}{t}$ for $\frac{1}{t+\Delta t}$ when Δt is small, we commit a small error in one of the summands of the numerator of the fraction $\frac{\Delta z}{\Delta t}$. But in this fraction both numerator and denominator are small if Δt is small. For this reason we cannot allow for a small error in the numerator.

Here is the proper way to do this:

$$\begin{aligned} \Delta z &= \frac{1}{t+\Delta t} - \frac{1}{t} = \frac{t-(t+\Delta t)}{t(t+\Delta t)} = -\frac{\Delta t}{t(t+\Delta t)}, \\ \frac{\Delta z}{\Delta t} &= -\frac{1}{t(t+\Delta t)} \end{aligned}$$

Now we can find the limit (the derivative) by dropping Δt in the denominator:

$$\frac{dz}{dt} = \frac{d\left(\frac{1}{t}\right)}{dt} = \lim_{\Delta t \rightarrow 0} \left[-\frac{1}{t(t+\Delta t)} \right] = -\frac{1}{t^2}$$

In these examples we have a very important property, the *fundamental property of limits*. As Δt is made smaller and smaller, the difference between the value of the ratio $\frac{\Delta z}{\Delta t}$ and the limit of this ratio

(this limit is equal to the derivative) $\lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = \frac{dz}{dt}$ may be made as small as we please, which is to say, less than any given number.

An example will serve to illustrate this point. For $z = \frac{1}{t}$,

$$\frac{dz}{dt} = -\frac{1}{t^2}, \quad \frac{\Delta z}{\Delta t} = -\frac{1}{t(t+\Delta t)}$$

Let us take, say, $t = 2$, $\frac{dz}{dt} = -0.25$. Can we choose Δt so that $\frac{\Delta z}{\Delta t}$ differs from the limit by less than 0.0025? What this means is that Δt will have to be chosen so that $\frac{\Delta z}{\Delta t}$ lies in the range between

$-0.25 + 0.0025 = -0.2475$ and $-0.25 - 0.0025 = -0.2525$. Substituting the expression $\frac{\Delta z}{\Delta t}$ for $t = 2$, we find that Δt must be less than 0.02 in absolute value.

The same goes for other functions as well: the approach to a limit as $\Delta t \rightarrow 0$ signifies the opportunity of choosing Δt so that any degree of closeness to the limit is attainable.

Finding the derivative in the special case of $z = t$ is particularly simple: quite obviously $\Delta z = \Delta t$, $\frac{\Delta z}{\Delta t} = 1$. The ratio is equal to 1 for arbitrary (large and small) Δt and hence in the limit as well. Thus

$$z = t, \quad \frac{dz}{dt} = \frac{dt}{dt} = 1$$

Finally, the constant $z = C$ can also be regarded as a special case of the function, but in this case clearly $\Delta z = 0$ for any Δt , and so we have

$$z = C, \quad \frac{dz}{dt} = \frac{dC}{dt} = 0$$

If we multiply the function by a constant factor, then the derivative is multiplied by the same factor. For instance,

$$z = 3t^2, \quad \frac{dz}{dt} = \frac{d(3t^2)}{dt} = 3 \cdot \frac{dt^2}{dt} = 3 \cdot 2t = 6t$$

In the general case, if

$$z(t) = a \cdot y(t)$$

then

$$\frac{dz}{dt} = a \cdot \frac{dy}{dt}$$

It is also obvious that the derivative of a sum of two functions is equal to the sum of the derivatives of the two functions:

$$z(t) = x(t) + y(t), \quad \frac{dz}{dt} = \frac{dx}{dt} + \frac{dy}{dt}$$

Using these two rules, we find that the derivative of a sum of several functions taken with constant (but, generally speaking, different) coefficients is equal to the sum of the derivatives of these functions with the same coefficients:

$$z(t) = a \cdot x(t) + b \cdot y(t) + c \cdot u(t),$$

$$\frac{dz}{dt} = a \frac{dx}{dt} + b \frac{dy}{dt} + c \frac{du}{dt}$$

Each of these rules is readily proved by forming $\Delta z = z(t + \Delta t) - z(t)$. These rules, which hold true for $\frac{\Delta z}{\Delta t}$, given arbitrary Δt , are also valid for the limit, that is, for $\frac{dz}{dt}$.

It is now easy to find the derivative of a polynomial. First let us write down all the derivatives we have found so far:

$$\frac{dC}{dt} = 0, \quad \frac{dt}{dt} = 1, \quad \frac{d(t^2)}{dt} = 2t, \quad \frac{d(t^3)}{dt} = 3t^2$$

In Sec. 2.1 we considered

$$z(t) = z_0 + bt + ct^2$$

We find

$$v(t) = \frac{dz}{dt} = \frac{dz_0}{dt} + b \frac{dt}{dt} + c \frac{dt^2}{dt} = 0 + b \cdot 1 + c \cdot 2t = b + 2ct$$

This is the formula for the instantaneous velocity which we obtained there.

The technique for finding derivatives (also called the *differentiation of functions*) is given in detail at the beginning of the next chapter.

Running ahead a bit, we may point out that differentiating functions given by formulas is a relatively simple job, much easier, say, than the solution of algebraic equations. The formulas for the derivatives of functions are never more complicated than the formulas defining the functions. For instance, if the function is a polynomial,

$$y = a + bx + cx^2 + lx^3 + fx^4$$

then its derivative is also a polynomial,

$$\frac{dy}{dx} = b + 2cx + 3lx^2 + 4fx^3$$

(this is true of polynomials of any degree). If the function is an algebraic fraction, then the derivative is also a fraction. If the function contains roots or fractional powers, then the derivative also contains them. The derivatives of trigonometric functions are also trigonometric functions, and in some cases (the logarithmic function for instance) the derivative turns out to be a still simpler function (in the given case, an algebraic fraction).

Finding derivatives does not require any kind of special ingenuity or imagination. The problem is always solved in a neat fashion through the use of simple rules which are given in Chapter 3 (also see Table I of the Appendix).

The derivatives of more involved functions are considered in the next chapter.

So far, all the functions we have considered are defined by formulas. This is not however absolutely necessary for the existence of a derivative. For example, we can regard the dependence of distance covered upon time as having been found from experiments, in the form of very extensive tables. It is clearly possible, using these

tables, to compute the instantaneous velocity (that is, the derivative) using the very same rules we applied to the functions defined by formulas.

In view of the importance of this section, let us review the basic conclusions.

(1) A derivative function is defined as the limit of the ratio of the increment of the function to the increment of the independent variable as the increment of the independent variable tends to zero:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right)$$

(2) The instantaneous velocity of a body is equal to the derivative of the coordinate of the body with respect to time. By analogy, when x does not represent time and y is not a coordinate one says that the derivative $\frac{dy}{dx}$ yields the rate of change of the function y as the variable (argument) x changes.

Exercises

1. Find the derivatives of the following functions algebraically (from first principles): (a) $z = t^2$ and (b) $z = t^3$, given $t_1 = t - \frac{\Delta t}{2}$ and $t_2 = t + \frac{\Delta t}{2}$. Here, the time t for which the derivative is sought lies, for arbitrary t , at the midpoint of the time interval from t_1 to t_2 .

Find the derivatives of the following functions.

2. $y = x^4$. 3. $y = (x+1)^2$. 4. $y = \frac{1}{x^2}$. 5. $y = a + \frac{b}{x^2}$. 6. $y = \sqrt{x}$.

Hint. In Problem 6, multiply the numerator and the denominator of the expression $\frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x}$ by the sum $\sqrt{x+\Delta x} + \sqrt{x}$.

2.4 APPROXIMATING THE VALUES OF A FUNCTION BY MEANS OF A DERIVATIVE

The derivative $\frac{dz}{dt}$ is defined as the limit of the ratio of the increments $\frac{\Delta z}{\Delta t}$ as $\Delta t \rightarrow 0$. When Δt is not equal to zero, the ratio of the increments $\frac{\Delta z}{\Delta t}$ is not equal to the derivative $\frac{dz}{dt}$, but this ratio is approximately equal to $\frac{dz}{dt}$ and the approximation is the better, the smaller Δt is.

Therefore, let us write, approximately,*

$$\frac{\Delta z}{\Delta t} \approx \frac{dz}{dt} = z'(t), \quad \Delta z \approx \frac{dz}{dt} \cdot \Delta t = z'(t) \Delta t \quad (2.4-4)$$

From this we can find the approximate value of the function $z(t + \Delta t)$:

$$z(t + \Delta t) = z(t) + \Delta z \approx z(t) + \frac{dz}{dt} \cdot \Delta t = z(t) + z'(t) \Delta t \quad (2.4-2)$$

Note that in (2.4-2) the first equality sign is exact in accord with the definition of Δz , while the second one denotes approximate equality.

Let us now return to the designations $t_2 = t + \Delta t$, $t_1 = t$, which we used earlier. We have

$$z(t_2) \approx z(t_1) + z'(t_1) \cdot (t_2 - t_1) \quad (2.4-3)$$

Thus, given a small difference $t_2 - t_1$, that is, when t_2 is close to t_1 , the function $z(t_2)$ can be expressed by an approximate formula involving the value of the function $z(t)$ and its derivative $z'(t)$ for $t = t_1$. Note that in this formula t_2 is linear (to the first power).

Let us take an example. Suppose $z = t^3$ and we are interested in the values of z when t is close to 1. We take $t_1 = 1$ and then $z(t_1) = t_1^3 = 1$, $z'(t_1) = 3t_1^2 = 3$ and the approximate formula has the form

$$z(t_2) = t_2^3 \approx 1 + 3(t_2 - 1) = 3t_2 - 2$$

We compare the exact and approximate expressions in Table 1.

Table 1

t_2	1	1.01	1.02	1.05	1.1	1.5	2
t_2^3	1	1.0303	1.0612	1.1576	1.3310	3.375	8
$3t_2 - 2$	1	1.03	1.06	1.15	1.30	2.50	4.0

* The assertion that the approximate equality

$$\Delta z = z(x + \Delta x) - z(x) \approx z'(x) \Delta x$$

becomes exact in the limit as $\Delta x \rightarrow 0$ requires some explanation. It is clear that $\Delta z \rightarrow 0$ as $\Delta x \rightarrow 0$, and so the approximate equality $\Delta z \approx a \cdot \Delta x$ becomes exact for any a as $\Delta x \rightarrow 0$, since this equation yields $0 = 0$. But we assert still more: given a finite Δx , it follows from $\Delta z \approx z'(x) \Delta x$ that $\frac{\Delta z}{\Delta x} \approx z'(x)$.

We assert that this consequence of the approximate equality $\Delta z \approx z'(x) \Delta x$ also becomes exact in the limit as $\Delta x \rightarrow 0$. This fact follows from the definition of the derivative $z'(x)$.

Another example is $z = \sqrt{t}$. We find the values of the function for t close to 4. Then $z(4) = \sqrt{4} = 2$. The derivative $z'(t) = \frac{1}{2\sqrt{t}}$ (see Exercise 6 in Sec. 2.3). Therefore $z'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$ and the approximate formula is of the form

$$z(t_2) = \sqrt{t_2} \approx 2 + 0.25(t_2 - 4) = 1 + 0.25t_2$$

We again compare the approximate and exact expressions (see Table 2).

Table 2

t_2	4	5	6	7	8	9
$\sqrt{t_2}$	2	2.24	2.45	2.65	2.83	3
$1 + 0.25 t_2$	2	2.25	2.50	2.75	3.0	3.25

Suppose that Δt is the time interval, $z'(t)$ is the instantaneous velocity, Δz is the increment in distance, that is, the distance covered during time Δt . The formula

$$\Delta z = z'(t) \Delta t \quad (2.4-4)$$

then signifies that the distance covered is equal to the product of the instantaneous velocity and the time interval. But the instantaneous velocity itself varies with time. Therefore, (2.4-4) is true only when the instantaneous velocity does not perceptibly change during the time Δt . Hence, the faster $z'(t)$ varies, the smaller Δt can be taken in (2.4-4), and conversely, the slower $z'(t)$ varies, the larger Δt may be taken. That is to say, the magnitude of the increment Δt for which formula (2.4-4) still yields a small error depends on the rate of change of the derivative over the interval Δt .

The cases we have examined confirm this conclusion. In the first example, when t varies from 1 to 2 ($\Delta t = 1$) the derivative $z'(t) = 3t^2$ varies from 3 to 12 (which is to say, by a factor of 4). In the second example, when t varies from 4 to 9, the derivative $z'(t) = \frac{1}{2\sqrt{t}}$ varies from 0.25 to 0.167 (or roughly by 30%). Therefore, in the latter instance the formula yields a good result for larger values of Δt . A detailed discussion of the range of application of the formula (for a given requisite accuracy) and the possibility of making it more precise is given in the last sections of Chapter 3.

All this applies in equal measure to positive and negative increments; an example involving negative increments is given in the exercises.

Exercises

1. Find $(1.2)^2$, $(1.1)^2$, $(1.05)^2$, $(1.01)^2$, using formula (2.4-3). Compare the results with the exact values.

2. Using the derivative of the function $z(t) = 2 + 20t - 5t^2$ find $z(1.1)$, $z(1.05)$, $z(0.98)$. Compare them with the exact values.

Hint. In the last case, take $t = 1$, $\Delta t = -0.02$.

2.5 A TANGENT TO A CURVE

Using a derivative, we can solve an important problem in analytical geometry: to find the tangent line to a curve given by the equation $y = f(x)$. The coordinates of the point A of tangency are given: $x = x_0$, $y = y_0 = f(x_0)$.

To find the tangent line means to find the equation of the line. It is clear that the equation of the tangent line is the equation of

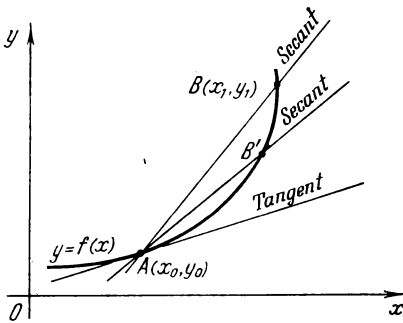


Fig. 33

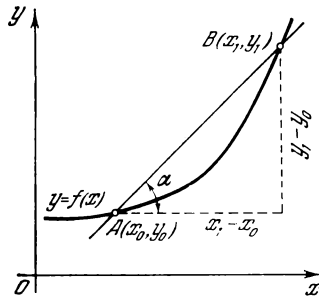


Fig. 34

the straight line passing through the point of tangency. The equation of any straight line passing through a given point $A(x_0, y_0)$ can be written as

$$y - y_0 = k(x - x_0)$$

In order to find the equation of the tangent line, it remains to determine the quantity k , the slope of the tangent line. To do this, we first find the slope of the line passing through two given points A and B of the curve at hand (Fig. 33). We call this line a secant line. When these two points of the curve approach each other, the line approaches the tangent line. In Fig. 33 we see two secant lines through points A and B and through A and B' , B' lying closer to A than B .

The closer the second point is to A , the closer the secant line is to the tangent line. Therefore the slope of the tangent line is equal to the limit approached by the slope of the secant line as the distance between the two points of intersection of the secant and the curve tends to zero.

The slope of the secant line can readily be expressed in terms of the values of the function at the points of intersection.

For one of the points of intersection of the secant and the curve we take the point $A(x_0, y_0)$ at which we desire to draw a tangent line to the curve; we denote by x_1, y_1 the coordinates of the second point of intersection, B .

Since these points lie on the curve whose equation is $y = f(x)$, it follows that $y_0 = f(x_0)$ and $y_1 = f(x_1)$. As may be seen from Fig. 34, the slope of the secant line k_c is

$$k_c = \tan \alpha = \frac{y_1 - y_0}{x_1 - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

The expression of the slope of a straight line passing through two given points is considered in Secs. 1.3 and 1.4.

In order to obtain the slope of the *tangent line* at the point $x = x_0$, we have to take point B closer and closer to A , which means that x_1 must approach x_0 . Consequently, the slope k of the tangent line is equal to the limit of k_c as x_1 tends to x_0 :

$$k = \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

We denote by Δx the difference $x_1 - x_0$, $x_1 = x_0 + \Delta x$ and accordingly

$$\Delta f = f(x_1) - f(x_0) = f(x_0 + \Delta x) - f(x_0)$$

In this notation, the slopes k_c and k of the secant and tangent, respectively, are given by the formulas

$$k_c = \frac{\Delta f}{\Delta x}, \quad k = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

Thus, the slope of the tangent line is the *derivative* of the function $f(x)$:

$$k = \frac{df}{dx} = f'(x_0)$$

We know that the derivative of a function $f(x)$ is itself a function of x . Since we sought the slope of the tangent at the point $A(x_0, y_0)$, we assumed, in computing the limit of $\frac{\Delta f}{\Delta x}$, that $x = x_0$ is fixed. That is why in the final formula we have $f'(x_0)$, which is the value of the derivative at $x = x_0$.

Let us consider the example of a parabola $y = x^2$, i.e., $f(x) = x^2$. Set up the equation of the tangent line at the point $x_0 = 2$, $y_0 = 4 = f(x_0)$.

We know the derivative

$$f'(x) = \frac{df}{dx} = \frac{dx^2}{dx} = 2x$$

Consequently, at the point of interest the slope of the tangent line is

$$k = f'(x_0) = 2x_0 = 4$$

The equation of the tangent line is

$$y - y_0 = k(x - x_0), \quad y - 4 = 4(x - 2), \quad y = 4x - 4$$

Without the aid of derivatives, it is rather difficult to draw a tangent to a curve given by an equation $y = f(x)$: you have to compute a large number of points of the curve, then, using a French curve, draw the curve through these points and, by eye, apply a ruler to the curve at the given point and pay special attention to see that you do not intersect the curve near the point of tangency. Using derivatives,

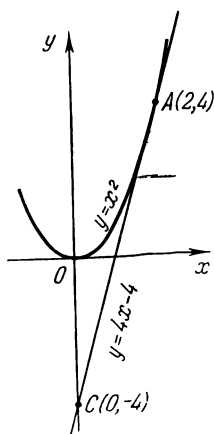


Fig. 35

we find the equation of the tangent line, then from this equation we find two points lying on the straight line given by this equation, and then we draw the straight line (tangent line) with a ruler (through the two points). For one of the two points it is natural to take the point of tangency itself, $A(x_0, y_0)$. The second point C may be taken on the straight line a good distance from A ; we can then more accurately determine the slope and the position of the tangent as a straight line passing through the two points A and C .

For example, above we found the equation of a straight line tangent to the parabola $y = x^2$ at the point $x_0 = 2, y_0 = 4$. This is an equation of the form $y = 4x - 4$. Let us find the coordinates of two points on this line: at $x = 2$ we find $y = 4 \cdot 2 - 4 = 4$. This is the point of tangency

$A(2, 4)$. The coordinates need not have been computed since the tangent must pass through it. For the second point (C) we choose the point of intersection of the tangent line and the y -axis. Putting $x = 0$, we find $y = -4$, so that $C(0, -4)$ (Fig. 35).

Note the curious fact that for $x = 0, y = -y_0$ the point C of intersection of the tangent with the y -axis lies below the x -axis just as much as the point of tangency itself lies above the x -axis. This is not accidental. The rule holds true for all tangents to quadratic parabolas with equation $y = ax^2$. Indeed, if the tangent is drawn to the point $A(x_0, y_0 = ax_0^2)$, then its equation is

$$y - y_0 = 2ax_0(x - x_0)$$

and for $x = 0$ we get

$$y - y_0 = -2ax_0^2, \quad y = y_0 - 2ax_0^2 = y_0 - 2y_0 = -y_0$$

Thus, the tangent passes through the points $A(x_0, y_0 = ax_0^2)$ and $C(0, y = -y_0 = -ax_0^2)$.

When plotting a curve by points it is hard to construct the curve if there are few points. Using derivatives, one can draw the tangents

to the curve at these points beforehand, and then the curve itself can be drawn with greater ease and accuracy.

Pictorially, it is clear that the tangent is horizontal at the points of maximum and minimum. The equation of a horizontal straight line is $y = \text{const}$, and the slope of the horizontal straight line is $k = 0$. Hence, the derivative of the function $y = f(x)$ (the graph of which is a curve—for details see Sec. 2.6) is zero at the points of maximum and minimum of the curve. Thus, using the condition $f'(x) = 0$ one can find the x -coordinates of the points of maximum and minimum of the curve. The y -coordinate is then easily found

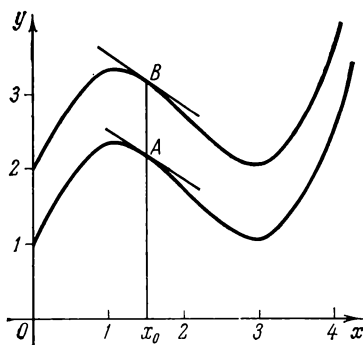


Fig. 36

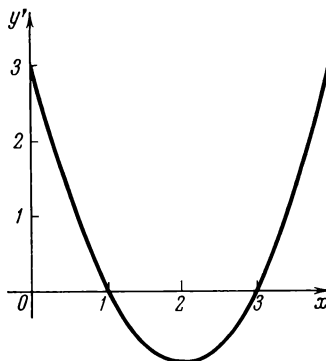


Fig. 37

by substituting x into the equation of the curve. It is also obvious that knowing the coordinates of the points of maximum and minimum one can draw the curve itself more accurately.

It is a useful exercise to draw a curve $y(x)$ freehand and then rapidly draw the curve $y'(x)$, noting the sign of $y'(x)$ and the points where $y'(x)$ vanishes. This is illustrated in Fig. 36 [the graph of $y(x)$] and Fig. 37 [the graph of the derivative $y'(x)$].

For the derivative $y'(x)$, the vanishing points of the function $y(x)$ are of no interest. If the curve $y(x)$ is raised parallel to itself (the upper curve in Fig. 36), then the curve $y'(x)$ does not change in any way because in parallel translation all the slopes remain the same; for instance, when $x = x_0$ the tangents to the curve $y(x)$ (point A) and to the displaced curve (point B) are parallel, and the angles are the same. This result is in accord with the property of derivatives: the addition of a constant to a function (this corresponds to a vertical parallel translation of the graph) does not change the derivative.

Another mathematical game is this: draw freehand the graph of the derivative and then give a rough construction of the graph

of the function. Here you have to specify (in arbitrary fashion) one point $(x_0, y(x_0))$ and then draw the curve up or down (in accordance with the sign of the derivative).

Note in conclusion that up till now we assumed the scales on the x -axis and y -axis to be the same, that is, one unit of x and one unit of y are expressed on the graph by line segments of equal length. Then we indeed have $\tan \alpha = \frac{dy}{dx}$.

In constructing graphs one often uses different scales, particularly if y and x are quantities with different dimensions. For instance, let y be the distance traveled and x the time. We construct the graph of the position of a body depending on the time, $y(x)$. On the axis of ordinates (y -axis) we lay off y using the scale 1 metre of distance = 1 cm in the drawing. On the axis of abscissas (x -axis) we lay off time using the scale 1 sec of time = 1 cm in the drawing. Then the velocity v expressed in metres per second and equal to the derivative $\frac{dy}{dx}$ will indeed be equal to $\tan \alpha$, the tangent of the angle of the tangent line in the drawing. But if we choose a different scale for x , say 1 sec = l cm = 5 cm in the drawing, we then get

$$\tan \alpha = \frac{dy}{l dx} = \frac{1}{l} \frac{dy}{dx} = \frac{1}{5} \frac{dy}{dx} = \frac{1}{5} v \text{ m/sec}$$

In the general case, if one x unit in the drawing is laid off to a scale of l cm and one y unit is laid off to a scale of n cm, then

$$\tan \alpha = \frac{n}{l} \frac{dy}{dx}$$

When y and x are denominate quantities (i.e., quantities having dimensions), say y metres, x seconds or y kilograms and x months (the weight of a baby as a function of time), the derivative $\frac{dy}{dx}$ also has dimensions: in the former case, $\frac{dy}{dx} = v$ is the velocity with dimensions m/sec, in the latter case, $\frac{dy}{dx}$ is the rate of increase in weight, kg/month.

The trigonometric function $\tan \alpha$ is dimensionless (being equal to the ratio of the lengths of two line segments). Therefore we cannot have the simple equation $\tan \alpha = \frac{dy}{dx}$ since the left and right members have different dimensions. It is precisely the scale factors l and n in the formula $\tan \alpha = \frac{n}{l} \cdot \frac{dy}{dx}$ that make the equation proper from the standpoint of dimensions. Thus, in the latter example, l has the dimensions of cm/month (1 cm in the drawing per month of age), n has the dimensions of cm/kg (1 cm on the graph per kilogram of

weight), so that $\frac{n}{l} \cdot \frac{dy}{dx}$ is dimensionless. In the formula

$$\tan \alpha = \frac{n \left(\frac{\text{cm}}{\text{kg}} \right)}{l \left(\frac{\text{cm}}{\text{month}} \right)} \frac{dy}{dx} \left(\frac{\text{kg}}{\text{month}} \right)$$

all the dimensions cancel out.

This should be borne in mind when comparing the derivative and the slope of a curve.

Exercises

1. Construct the graph of the function $y = x^2 + 1$ within the range from $x = -1.5$ to $x = 2.5$ and draw tangent lines at the points $x = -1$, $x = 0$, $x = 1$, $x = 2$.

2. The same for the function $y = x^3 - 3x^2$, $-1 < x < 3.5$; tangent lines at $x = -1, 0, 3$. Find the points with horizontal tangent lines.

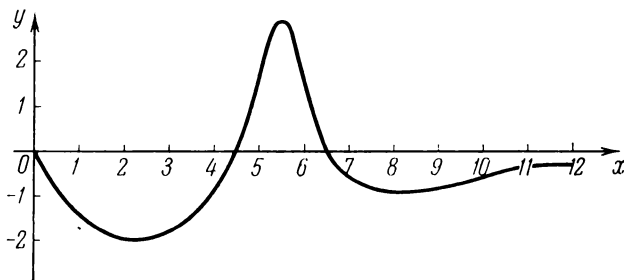


Fig. 38

3. On the curve $y = x - x + 1$, find points having horizontal tangents. Construct the curve for $-2 < x < 2$.

Hint. In Exercises 1 to 3 it is advisable to use graph paper and a large scale.

4. Construct the curve $y'(x)$ for the function $y(x)$ given in Fig. 38.

Suggestion. First copy Fig. 38 on a fresh sheet of paper and then construct $y'(x)$ there (so as not to deprive the next reader of the pleasure of drawing this curve).

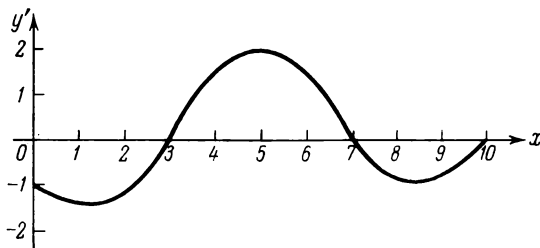


Fig. 39

5. Draw the curve $y(x)$ through the point $x = 5$, $y = 0$ for the curve $y'(x)$ given in Fig. 39. At what angle will $y(x)$ intersect the y -axis? At what angle will $y(x)$ cut the x -axis for $x = 5$?

Suggestion. The same as for Exercise 4.

6. Set up the equations of the tangent lines to the curve $y = x^3$ at the points $x = 0.5$ and $x = 1$. Find the points of intersection of the tangent lines with the x - and y -axes.

7. Find the general rule for points of intersection with the axes of tangent lines to the curves $y = ax^2$, $y = bx^3$.

2.6 INCREASE AND DECREASE OF FUNCTIONS. MAXIMUM AND MINIMUM

Suppose we have a relationship between some physical quantity (say temperature) and time.

We have z for temperature, t for time and the formula for the function $z(t)$ is given. How can we determine whether the temperature is rising or falling at a given time t ? How can we determine at what time the temperature reaches a maximum or a minimum value?

Without a knowledge of derivatives, we have to seek the answer to the first question numerically: take the temperature at a given time t and then take it at some following time t_1 and see if it has increased or decreased. This is clearly not a reliable approach: if $z(t_1)$ is greater than $z(t)$, it still might be that at time t the temperature fell, then soon afterwards (after t but prior to t_1) it reached a minimum and only then began to increase and by t_1 had risen above $z(t)$.

Using derivatives we get an exact solution: we have to find the derivative $\frac{dz}{dt}$. If $\frac{dz}{dt} = z'(t)$ is a positive quantity for a given t , then $z(t)$ is an increasing function: if t increases by a small amount Δt , the temperature increases by a small amount $\Delta z = z'(t) \cdot \Delta t$ (as was clarified earlier, the smaller Δt , the more exact the equation). We consider $\Delta t > 0$, time increases. If $z'(t) > 0$, $\Delta t > 0$, then also $\Delta z > 0$, i.e., the temperature rises with time. If $z'(t) < 0$, $\Delta t > 0$, then $\Delta z < 0$, i.e., the temperature at the next instant of time $z(t + \Delta t)$ will be below the temperature $z(t)$ at the given time.

Thus, a positive derivative indicates that the function is increasing, a negative derivative, that it is a falling, decreasing function.

The expressions "increasing function" and "decreasing function" are applied to any function $y(x)$ and not only to those that depend on time (functions of time). An increasing function is one in which y increases as the independent variable x increases.

The derivative $\frac{dy}{dx}$ is what indicates the rate of growth, that is, the ratio of the variation of y to the variation of x . A negative rate of growth means a falling, a decrease in y as x increases, and if $\frac{dy}{dx} < 0$, then $\left(-\frac{dy}{dx}\right)$ is the rate of fall (decrease).

The expression "the quantity y has a large negative derivative with respect to x " means that y is falling fast as x increases. A positive derivative $\frac{dy}{dx}$ means that y grows with the growth of x .

Physicists and mathematicians, especially those in the making who have just learned what a derivative is, frequently put it to use in everyday life like this: "the derivative of my mood with respect to time is positive" in place of "my mood is definitely improving".

Solve this joke problem: what sign does the derivative of my mood have with respect to the distance from the dentist's chair? My mood deteriorates, "decreases", becomes "negative" as the distance *decreases*; hence the derivative is positive.

The serious editor may complain about abuse of the English language, but actually this free-style use of mathematical concepts is good practice for future serious applications.

There are functions that have the same sign of the derivative for any values of the variable: such is the property of the linear function $y = kx + b$, whose derivative $\frac{dy}{dx} = k$ is a constant. Later on we

will see that in the case of the exponential function $y = a^x$ the derivative has a constant sign (although it is not constant in magnitude) for arbitrary x . However, a derivative need not have a constant sign; the sign of the derivative of a given function may be different for different values of the independent variable.

Let us imagine a function $y(x)$ whose derivative $y'(x)$ is positive for $x < x_0$ and negative for $x > x_0$: in short, $y'(x) > 0$, $x < x_0$; $y'(x) < 0$, $x > x_0$.

What can we say about such a function? We begin with $x < x_0$. As x increases to x_0 , y will increase; as x continues to increase, y falls. The conclusion is that for $x = x_0$ the function $y(x)$ has a maximum.

Consider the contrary case:

$$y'(x) < 0, \quad x < x_0, \quad y'(x) > 0, \quad x > x_0$$

Reasoning as before, we conclude that in this case $y(x)$ has a minimum value when $x = x_0$.

If a function $y(x)$ is defined by a formula associated with a smooth curve, so that $y'(x)$ also varies smoothly as x varies, then the different sign of $y'(x)$ for $x < x_0$ and $x > x_0$ in both cases signifies that for $x = x_0$, $y'(x_0) = 0$. Thus, by equating the derivative to zero, we can find those values of the independent variable for which the function has a maximum value or a minimum value. We will discuss the exceptions to this rule for nonsmooth curves in Chapter 4.

Let us take a numerical example. In Sec. 1.1 we compiled a table for the function $y = 3x^3 - x^2 - x$ (see page 14). Judging by this

table, one might think that the function is increasing for all values of x since every increase in x by unity caused an increase in y .

Take the derivative:

$$y' = 9x^2 - 2x - 1$$

Taking $x = 0$, we get $y'(0) = -1 < 0$. Hence, when $x = 0$, the function is decreasing. This refutes the supposition (obtained from a glance at the table) that the function is an everywhere increasing function.

We equate $y'(x)$ to zero. Solving the equation

$$9x^2 - 2x - 1 = 0$$

we find two roots:

$$x_1 = -0.24, \quad x_2 = +0.46$$

Now form a detailed table (Table 3) including the maximum and minimum points just found.

Table 3

x	-2	-1	-0.30	-0.24	-0.18	
y	-26	-3	+0.129	+0.140	+0.131	
x	0	0.40	0.46	0.52	1	2
y	0	-0.372	-0.381	-0.370	+1	+18

We see that, true enough, on the portion from $x = -0.24$ to $x = +0.46$ the function y falls from $+0.14$ to -0.38 .

A comparison of the values $y(-0.24)$ with the adjacent values $y(-0.30)$ and $y(-0.18)$ confirms the fact that when $x = -0.24$, y reaches a maximum, the adjacent values of y being smaller. The graph of the function $y = 3x^3 - x^2 - x$ is shown in Fig. 40.

We see here that the word *maximum* should not be understood as meaning the largest of all possible values of y . Indeed, at the maximum point $y(-0.24) = +0.14$ and for $x = 1$, $y = 1$, for $x = 2$, $y = +18$, for $x = 10$, $y = 269$ and so on, y increasing without bound as x increases without bound. In what way does the maximum point $x_{\max} = -0.24$, $y = 0.14$ that we found differ?

The difference is that for close values of x , both larger than $x_{\max}$ and less than $x_{\max}$, the quantity y is less than $y_{\max} = y(x_{\max})$. This peculiarity of $x_{\max}$ is clearly seen in the table [compare $y(-0.30)$, $y(-0.24)$ and $y(-0.18)$]. The same arguments can be applied to the minimum: when $x_{\min} = 0.46$, $y_{\min} = -0.381$; for

large (in absolute value) negative x , y decreases without bound and becomes less than $y_{\min}$, but $x_{\min}$, $y_{\min}$ differs in that $y_{\min}$ is less than the values of y for x close to $x_{\min}$. The condition of a vanishing derivative enables us to find just such maxima and minima.

The determination of maxima and minima arithmetically (by computing and comparing the values of the function for different values of the argument) is many times more arduous and less exact. Higher mathematics is not only a remarkable achievement of the mind. Practical computational problems are resolved much more easily by the methods of higher mathematics.

To conclude this section, let us investigate the problem of how to distinguish a maximum from a minimum when we use the condition $y'(x) = 0$. This condition holds true both at maximum and at minimum, the difference being in the sign of $y'(x)$ for $x < x_0$ and for $x > x_0$.

How is it possible to determine the sign of $y'(x)$ for x close to x_0 without computing y' directly for other values of x ? In the first case we saw that the function $y(x)$ has a maximum when $y'(x) > 0$ for $x < x_0$ and $y'(x) < 0$ for $x > x_0$.

Thus, in this case the derivative $y'(x)$ is itself a decreasing function: as x increases, the derivative, which was at first positive (for $x < x_0$), vanishes (when $x = x_0$) and, continuing to fall, becomes negative when $x > x_0$. But we already know how to distinguish a decreasing function: its derivative is negative. Hence, in the first case, for the value $x = x_{\max}$ for which y has a maximum, $y'(x_0) = 0$, and the derivative of a derivative is negative. This quantity—the derivative of a derivative—which by the ordinary rules can be written as a “double-decker” fraction,

$$\frac{dy'}{dx} = \frac{d\left(\frac{dy}{dx}\right)}{dx},$$

is called the second derivative and has the notation $y''(x)$ or $\frac{d^2y}{dx^2}$.

To summarize, the condition of a maximum is

$$y'(x) = 0, \quad y''(x) < 0$$

In the same way we can verify that for the x for which

$$y'(x) = 0, \quad y''(x) > 0$$

the function $y(x)$ has a minimum.

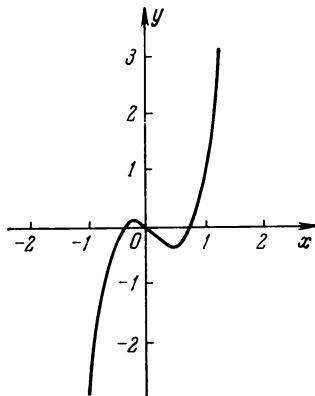


Fig. 40

Let us revert to the example given above:

$$y = 3x^3 - x^2 - x, \quad y' = 9x^2 - 2x - 1$$

Taking the derivative of y' , we get

$$y'' = 18x - 2$$

For $x = -0.24$, $y' = 0$, $y'' = -6.3 < 0$ and true enough $x = -0.24$, $y = 0.14$ is a maximum. For $x = +0.46$, $y' = 0$, $y'' = +6.3 > 0$; for $x = 0.46$, $y = -0.38$, y has a minimum.

Exercises

Find the values of x for which the following functions have a maximum or a minimum. In each case determine whether the minimum or maximum is involved. For functions involving a constant (given by the letter a) give the answer for $a > 0$ and for $a < 0$.

1. $y = ax^2$. 2. $y = x + \frac{1}{x}$. 3. $y = x + \frac{a}{x}$

4. $y = x^3 - x$. 5. $y = x^4 + ax^2 + b$.

2.7 THE AREA UNDER A CURVE AND DETERMINING DISTANCE FROM THE RATE OF MOTION

The problem of determining the instantaneous rate of motion $v(t)$ from a given dependence of the position of a body upon the time $z(t)$ led us to the concept of the derivative:

$$v(t) = \frac{dz}{dt}$$

The inverse problem consists in determining the position of a body and the distance covered by the body in a given interval of time when we know the instantaneous velocity $v(t)$ as a function of the time. This problem brings us to the second most important concept of higher mathematics—that of the *integral*.

Let us agree on some convenient notation. We consider the distance traveled during time from t_1 to t_2 . So as to avoid subscripts, let us call the beginning of the time interval n , $t_1 = n$, and the terminal point of the interval, k , $t_2 = k$. We denote the distance covered by $z(n, k)$. Remember that when the two quantities n and k stand under the function symbol z in parentheses, then $z(n, k)$ is the distance covered during the interval of time from n to k , whereas $z(t)$ with the single quantity in parentheses is the position (coordinate) of the body at a specified time t . These quantities are related in a simple manner:

$$\left. \begin{aligned} z(k) &= z(n) + z(n, k), \\ z(n, k) &= z(k) - z(n) \end{aligned} \right\} \quad (2.7-1)$$

The distance covered during time from n to k is equal to the difference between the coordinate at the end of the time interval under consideration $z(k)$ and that at the beginning of the interval, $z(n)$.

Now let us compute $z(n, k)$.

In the most elementary case, if the velocity is constant,

$$v(t) = \text{const} = v_0 \quad (2.7-2)$$

the distance covered will obviously be equal simply to the product of the time of motion into the velocity:

$$z(n, k) = (k - n) v_0 \quad (2.7-3)$$

Taking advantage of the graph of velocity versus time, we find that a constant velocity is associated with a horizontal straight line

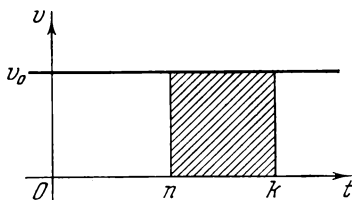


Fig. 41

(Fig. 41). The distance covered is clearly equal to the hatched area because the area of a rectangle is equal to the product of the base $(k - n)$ by the altitude (v_0) .

What do we do in the general case when the instantaneous velocity is not a constant quantity?

Let us make a detailed study of a numerical example. Suppose the velocity is given by the formula $v = t^2$.* We seek the distance covered during the time interval from $t = n = 1$ to $t = k = 2$.

We partition the entire interval from n to k into ten subintervals and set up a table of velocity (Table 4). We call Δt the subintervals of time (0.1 second each) into which we split up the large interval from $t = n$ to $t = k$.

Table 4

t	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
v	1.0	1.21	1.44	1.69	1.96	2.25	2.56	2.89	3.24	3.61	4.0

* The velocity v is expressed in cm/sec, t in seconds. In order to maintain the requirements of dimensionality, we write $v = at^2$, where a has the dimensions of cm/sec³. We consider the special case when the coefficient a is numerically equal to 1 cm/sec³.

Why is it difficult to compute the distance traveled at a velocity of $v(t)$ as given by the formula? Clearly because the velocity is variable (for a constant velocity, the answer is trivial). In the case at hand, the velocity changes 4 times over the time interval from $t = 1$ to $t = 2$. However, after this interval is partitioned into 10 parts (subintervals), the velocity varies less over each subinterval of duration 0.1 second (only 10 to 20%). Therefore, in the subintervals the velocity can roughly be taken to be constant and we can compute the distance covered during such a subinterval of time as the product of that subinterval by the velocity.

To compute the distance covered during each subinterval Δt , equal to 0.1 sec, we utilize the initial velocity in the given subinterval Δt : 1 cm/sec in Δt from 1 to 1.1 sec, 1.21 cm/sec in Δt from 1.1 to 1.2 sec, and so forth. Finally, 3.61 cm/sec in the last subinterval Δt extending from 1.9 to 2.0 seconds. The total distance covered during the time interval from $t = 1$ to $t = 2$ is then computed to be

$$z(1, 2) = 0.1 + 0.121 + 0.144 + \dots + 0.361 = 2.185 \text{ cm}$$

It is quite clear that we have reduced the actual distance covered because the velocity here increases with time and so the velocity at the beginning of each subinterval Δt is less than the average velocity. Each of the ten terms into which the entire distance was partitioned is slightly less than the actual value, and so the result has a deficit too.

Let us now compute the distance somewhat differently, namely, in each subinterval Δt we will take the value of velocity at the end of the subinterval. For the first subinterval Δt from 1 to 1.1 sec, this velocity is equal to 1.21 cm/sec, for the last one from 1.9 to 2 sec, it is 4 cm/sec. We then get

$$z(1, 2) = 0.121 + 0.144 + \dots + 0.400 = 2.485 \text{ cm}$$

for the distance.

This calculation clearly yields the distance $z(1, 2)$ with an excess. Hence the true value lies between 2.185 and 2.485 cm. The difference between these numbers amounts to about 15%. Rounding off the boundary values for z , we get

$$2.18 < z(1, 2) < 2.49$$

These computations can be illustrated by means of a graph. We construct the graph (Fig. 42) with time laid off on the axis of abscissas and velocity on the axis of ordinates. In the figure we divide the time interval into five parts instead of ten, as we did in the table. This makes each part (step) stand out more clearly. Each term in the first sum is the area of a narrow rectangle with the corresponding subinterval Δt as the base and the velocity at the beginning of the subinterval as the altitude. Thus, the sum is the area under the poly-

gonal (step-like) line (hatched in Fig. 42). The second sum, in which the velocity in each subinterval is taken at the end of the subinterval, corresponds to the hatched area in Fig. 43.

How can we make a more accurate computation of the distance covered during a given time from $t = n = 1$ sec to $t = k = 2$ sec?

The difference between the lower and upper estimate, that is, the difference between 2.18 and 2.49, depends on the variation of velocity within the limits of each subinterval Δt .

To find a more exact value of $z(1, 2)$, we have to partition the time interval between 1 and 2 seconds into a larger number of subintervals, each of smaller duration. For instance, if we split up the

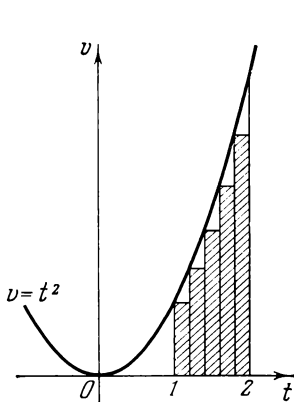


Fig. 42

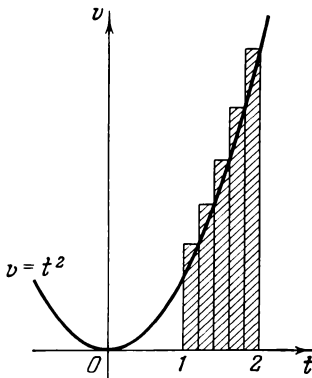


Fig. 43

1-to-2-sec interval into 20 subintervals Δt of 0.05 sec duration each, then, using the initial velocities in each Δt , we compute the distance to be

$$z(1, 2) = 0.05 + 0.05 \cdot 1.1025 + \dots + 0.05 \cdot 3.8025 = 2.25875$$

using the terminal velocities in each subinterval, we get the distance

$$z(1, 2) = 0.05 \cdot 1.1025 + 0.05 \cdot 1.21 + \dots + 0.05 \cdot 4 = 2.40875$$

The difference between 2.25875 and 2.40875 now amounts to about 7%. The range within which $z(1, 2)$ lies has narrowed down. Rounding off these figures, we have

$$2.26 < z(1, 2) < 2.41$$

As we reduce the subinterval Δt , the result approaches the true value of the distance covered. It will be computed later on and proves to be equal to

$$z(1, 2) = 2 \frac{1}{3} = 2.333 \dots$$

As we reduce Δt , the difference between the initial and terminal velocities in each subinterval Δt decreases, and so also does the relative error in each summand. That is why the whole sum of the distances for all subintervals Δt , that is, the quantity $z(1, 2)$, is more exactly determined if we take smaller and smaller subintervals Δt (then the number of subintervals, which is equal to $\frac{k-n}{\Delta t}$, increases).

Geometrically, it is obvious that as we increase the number of subintervals Δt and reduce the length of each one, the dimensions of each rectangle in figures like 42 and 43 become smaller and, consequently, the step-like line approaches closer and closer to the curve $v(t)$.

We thus conclude that the distance covered during time $t = n$ to $t = k$, given an arbitrary dependence of the instantaneous velocity on the time, $v(t)$, is equal to the area bounded by the curve $v(t)$, the vertical lines $t = n$ and $t = k$ and the t -axis (Fig. 44).

This conclusion yields a method for practical computation of the distance: we can construct the graph on graph paper and determine the hatched area either by counting the squares or, for instance, by cutting out the area of paper, weighing the sheet and

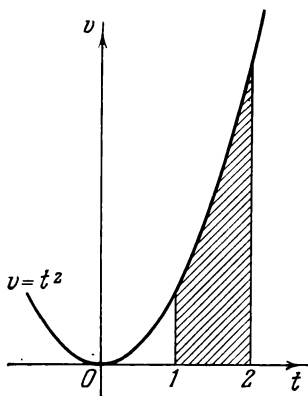


Fig. 44

comparing its weight with the weight of a rectangular or square piece of the same paper of known area.

This method is convenient and quite justified when the velocity is not known exactly or is given in the form of a table or graph obtained empirically (in an experiment). But we will not dwell on these approximate methods and will attempt to express the distance by a formula when the velocity is given by a formula.

We can also make more precise the numerical method used above to determine the distance: to do this, we determine the distance in each subinterval on the basis of the arithmetic mean (half-sum) of the initial and terminal velocities in the given subinterval. In this approach, with a partition into ten subintervals, the velocity in the first subinterval from 1 to 1.1 sec is taken equal to $\frac{1+1.21}{2} = 1.105$ cm/sec and the distance covered during this subinterval of time is 0.1105 cm, the distance covered during the second subinterval is $0.1 \cdot \frac{1.21+1.44}{2} = 0.1325$ cm, and so on. Adding them, we get

the distance covered during the time interval between $n = 1$ sec and $k = 2$ sec to be

$$z(1, 2) = 0.1105 + 0.1325 + \dots = 2.335 \text{ cm}$$

If we divide the interval into 20 subintervals, we get (using the same half-sum of velocities)

$$z(1, 2) = 2.33375 \text{ cm}$$

These values are much closer to the true value of 2.3333 cm than those computed on the basis of the initial and terminal values of velocity for the same number of subintervals: for ten subintervals, the error is equal to 0.07% instead of 15% in the earlier method, and for 20 subintervals the error is only 0.02% instead of 7%.

This method can also be displayed vividly on a graph. The product of the half-sum of the velocities at the beginning and end of an interval by the magnitude of the interval of time is equal to the area of the trapezoid $ABCD$ (Fig. 45). With bases AB and DC and altitude AD , the area is

$$\frac{AB+DC}{2} AD = \frac{v(t_1)+v(t_2)}{2} (t_2-t_1)$$

For this reason, the determination of distance on the basis of the half-sum of the velocities is called the "trapezoid method". For the shape of the curve shown in Fig. 45, the area of the trapezoid is somewhat greater than the area bounded by the straight lines BA , AD , DC

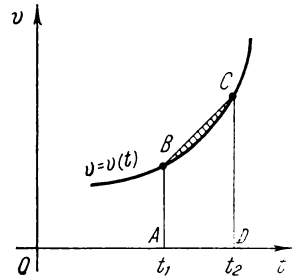


Fig. 45

and the portion BC of the curve. The difference between the area of the trapezoid and the area bounded by the arc of the curve is equal to the area of the crescent-like figure formed by the chord BC and the portion BC of the curve (shown hatched in Fig. 45). This area yields the error, that is, the difference between the true value of the distance and the value computed by the method of trapezoids. A comparison with Figs. 42 and 43 shows vividly that the error in the trapezoid method should be less than that in the method of rectangles.

When one compares the distance and the area on a graph, it is important to take into account the scale used. Suppose 1 cm along the axis of abscissas on the graph corresponds to a time interval of T seconds and 1 cm on the axis of ordinates corresponds to a velocity of V cm/sec. Then if the motion is at a constant velocity v_0 during a time from n to k , the distance covered is equal to $v_0(k-n)$, and the area of the rectangle on the graph (Fig. 41) is equal to

$$S = \frac{v_0}{V} \frac{(k-n)}{T} \text{ cm}^2$$

Thus

$$z(n, k) = SVT$$

This relationship between the distance covered and the area on the graph of velocity bounded by the curve $v(t)$, the axis of abscissas and the vertical lines is preserved in the case of a variable velocity and an arbitrary function $v(t)$.

We have thus examined in detail some methods for an approximate numerical and graphical determination of distance on the basis of a given velocity as a function of time.

2.8 THE DEFINITE INTEGRAL

In the preceding section, two problems—finding the distance traversed by a body and the equivalent problem of finding the area under a curve—led to a consideration of sums of a special type with a large number of small terms (summands).

These problems lead to the concept of the integral.

The distance $z(n, k)$ found from a given velocity $v(t)$ is called “the definite integral of the function $v(t)$ (velocity) with respect to the variable t (time) taken from n to k ”.

We now give a mathematical definition of the integral that corresponds to the ideas which were illustrated by the numerical example

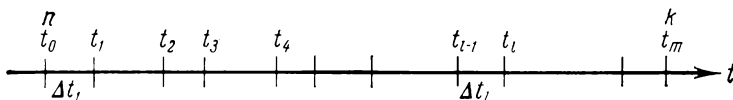


Fig. 46

of Sec. 2.7. This definition will remain valid when we consider physical or mathematical quantities of a nature different from velocity and distance.

Suppose we have a function $v(t)$. To find its integral from n to k we partition the interval from n to k into a large number m of subintervals. We denote the values of the argument t at the endpoints of the subintervals by $t_0, t_1, t_2, \dots, t_{m-1}, t_m$. Here, obviously, $t_0 = n$ and the last value $t_m = k$ (Fig. 46).

The lengths of the small subintervals of time Δt are equal to the difference between adjacent values of t .*

* If the interval (n, k) is specially partitioned into m equal parts, then each subinterval $\Delta t = \frac{k-n}{m}$. In what follows, however, it is not obligatory that the subintervals be equal, the only thing we require is that the subintervals be small. The reader will see the truth of this if he thinks through the distance-velocity example of Sec. 2.7.

The number label of each subinterval corresponds to that of the argument at the end of the subinterval (Fig. 46). Thus, for an arbitrary l ,

$$\Delta t_l = t_l - t_{l-1}$$

The subscripts on the quantities t and Δt are not factors but merely number labels, or indices, as they are sometimes called (see footnote on page 20).

The approximate value of the integral $z(n, k)$ is given by the formula

$$z(n, k) \approx \sum_{l=1}^{l=m} v(t_{l-1}) \Delta t_l \quad (2.8-1)$$

The symbol $\sum$ is the capital Greek letter sigma. It corresponds to S in the Latin alphabet, the first letter of the term *sum*. $\sum_{l=1}^{l=m}$ signifies that the expression which stands to the right of the symbol and depends on the index l is to be taken for all values of l from 1 to m and then all these expressions are to be added together. For example, if $m = 10$, then

$$\sum_{l=1}^{l=10} v(t_{l-1}) \Delta t_l = v(t_0) \Delta t_1 + v(t_1) \Delta t_2 + \dots + v(t_9) \Delta t_{10}$$

In the example of Sec. 2.7, in Table 4, $t_0 = 1$, $t_1 = 1.1$, $t_2 = 1.2$, . . . ,

$$z(1, 2) = z(n, k) \approx \sum_{l=1}^{l=10} t_{l-1}^2 \Delta t_l = 2.185$$

In the approximate expression (2.8-1), the value of the function $v(t)$ in each subinterval was taken at the beginning of the subinterval, at the point t_{l-1} . A different approximate expression is obtained if we take the value of the function at the endpoint of each subinterval:

$$z(n, k) \approx \sum_{l=1}^{l=m} v(t_l) \Delta t_l \quad (2.8-2)$$

In the example of Sec. 2.7, this sum for $m = 10$ was equal to 2.485.

The *definite integral* of a function $v(t)$, taken from n to k , is the limit approached by the sums (2.8-1) and (2.8-2) as all subintervals Δt tend to zero.

The integral is written

$$z(n, k) = \int_n^k v(t) dt \quad (2.8-3)$$

(read: $z(n, k)$ equals the definite integral of $v(t)$ from n to k , dt). The integral sign $\int$ is merely an elongated S (the first letter of the word *summa* which is the Latin for sum).

Unlike Δt , the symbol dt signifies that in order to obtain the exact value of the integral it is necessary to pass to the limit as all subintervals Δt tend to zero. The formulas (2.8-1) and (2.8-2) with finite subintervals Δt only yield approximate values of the integral. Recall that in Sec. 2.2 when we considered the derivative we also replaced the finite line-segments Δz and Δt with the differentials dz and dt .

When the subintervals Δt become smaller and smaller, it is immaterial whether we take the value of the function v at the beginning, at the end, or in the middle of the subinterval, which is to say that it is immaterial whether we proceed from (2.8-1) or from (2.8-2); and in formula (2.8-3) we simply have $v(t)$, which is the value of the function in the subinterval dt without any indication of the value of $v(t)$ being taken at the beginning or at the end of the subinterval.

Another way in which the integral (2.8-3) differs from the sums (2.8-1) and (2.8-2), which yield approximate values of the integral, lies in the fact that as the quantities Δt become smaller and smaller and the number of subintervals increases, we no longer label them. For this reason, we indicate on the integral only the limits of the variation of t (range of t) from n to k .

The quantity n is placed at the bottom of the integral sign and is termed the lower limit of integration, k is placed at the top and is called the upper limit.*

The range of t from n to k is called the interval of integration. The function $v(t)$ in the expression of the integral is called the integrand, t being the variable of integration.

Thus, the integral is defined as the limit approached by the sum of products of the values of the function multiplied by the difference of the values of the arguments when all differences of the arguments tend to zero:

$$\lim_{\Delta t_i \rightarrow 0} \sum_{l=1}^{l=m} v(t_l) \Delta t_l = \lim_{\Delta t_i \rightarrow 0} \sum_{l=1}^{l=m} v(t_{l-1}) \Delta t_l = \int_n^k v(t) dt \quad (2.8-4)$$

Although the first and second sums in (2.8-4) are different for a finite number of small intervals (subintervals), their limits are the same when all subintervals Δt decrease without bound.

As Δt tends to zero, each separate summand tends to zero, but on the other hand the number of terms in the sum increases and ap-

* In this section, the word "limit" is used in two meanings: the integral is the limit of a sum in the same sense that the derivative is the limit of a ratio. Here, limit corresponds to the sign $\lim$. We also speak of the limits of variation of t from n to k , the limits of integration n and k . The meaning here is different. The attentive reader will readily see which of the two meanings is used in any given case.

proaches infinity. The sum itself tends to a very definite limit, which is the solution of the problem and is termed the integral. This limit, that is, the integral of the function, is equal to the distance covered if the function represents instantaneous velocity.

However, not just any sum of a large number m of small terms tends to a definite limit as $m \rightarrow \infty$. We shall try to explain why in our case the limit must exist. Let us partition the interval $k - n$ into m equal subintervals, the length of each one being $\Delta t_i = (k - n)/m$. If for the sake of simplicity we take the velocity v as being constant, then we get a sum of m terms each equal to $v\Delta t = v(k - n)/m$. The total sum (i.e., the distance covered) is equal to $mv\Delta t = mv(k - n)/m = v(k - n)$, which means that it is independent of m . Here it is very important to point out that each separate term diminishes in exactly the same proportion (in proportion to $1/m$) as does the number of terms m increase. It is also clear that for the case of a variable velocity the result will not depend on m for a very large number m of subintervals, $\Delta t = (k - n)/m$. The reader will see this to be true if he does the exercises at the end of the section.

Since the variable of integration can assume the values n and k it is clear that the limits of integration have dimensions and their dimensions are equal to the dimensions of the variable of integration (in the distance-velocity example, the limits of integration have the dimensions of time). The dimensions of the integral are readily obtained from (2.8-1). Indeed, the dimensions of a sum are equal to the dimensions of the separate terms (summands).

The separate summands of (2.8-1) have dimensions equal to the product of the dimensions of the variable of integration into the dimensions of the integrand. In the distance-velocity example, the dimensions of the integral are $\text{sec} \cdot \text{cm}/\text{sec} = \text{cm}$.

Note that the value of a definite integral depends on the values of the function under the integral sign solely within the interval of integration. Values of the function outside the interval of integration have no effect on the magnitude of the integral. This is made abundantly clear when we consider the distance-velocity example.

The distance covered of course depends on the velocity $v(t)$, but only for values of the function inside the interval of integration. The distance $z(n, k)$ is in no way dependent on the velocity *prior* to time $t = n$ (which is when we began to consider the motion) and *subsequent* to time $t = k$ (end of motion).

It was pointed out in Sec. 2.7 that the distance can be determined by computing the area on the graph in which the velocity is a function of time. The problem of finding the area S bounded above by a curve with a specified equation $y(x)$, below by the axis of abscissas (x -axis), on the sides by the lines $x = a$ and $x = b$ (Fig. 47) also

reduces to computing the integral

$$S = \int_a^b y(x) dx$$

To explain this, recall Figs. 42 and 43. Imagine that the values of some function $y(x)$ are laid off on the axis of ordinates, the independent variable x on the axis of abscissas, and $y(x)$ has no connection with motion and velocity. In place of n and k we substitute the letters a and b . The sum of the areas of the hatched rectangles in Fig. 42 is equal

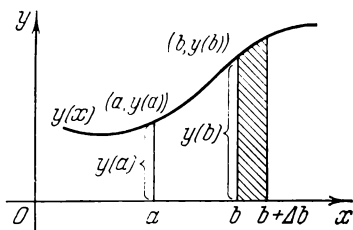


Fig. 47

to $\sum_{l=1}^{l=m} y(x_{l-1}) \Delta x_l$; the same sum in

Fig. 43 is equal to $\sum_{l=1}^{l=m} y(x_l) \Delta x_l$. In the limit, as $\Delta x_l \rightarrow 0$, these sums are, by definition, equal to the integral, and the sum of the areas of the rectangles tends to the area bounded by the curve $y(x)$, since the smaller the subintervals Δx_l , the closer to the curve is the polygonal (step-like) line bounding the rectangles.

In conclusion we note that the definite integral depends on the integrand and the limits of integration but is independent of the designation of the variable of integration. Judge for yourself. Suppose we have the integrand function

$$v(t) = 3t^2 + 5$$

Substitute $t = x$ to get

$$v(x) = 3x^2 + 5$$

When we compute the integral, it is immaterial how the variable of integration is designated, the only important thing being over what range it varies and what are the values of the function. For this reason

$$z(n, k) = \int_n^k v(t) dt = \int_n^k v(x) dx$$

Any letter will do for the variable of integration.

A variable which (like the variable of integration) does not appear in the final result is called a dummy variable. The variable of integration under the integral sign can be changed to any letter without disrupting the validity of the formulas. An ordinary (not dummy) variable can be replaced by a different letter only in **all** parts of a formula: for instance, in the formula $(x+1)^2 = x^2 + 2x + 1$

one cannot write $(t + 1)^2 = x^2 + 2x + 1$, but in integrals we can write

$$\int_n^k v(t) dt = \int_n^k v(x) dx$$

Exercises

1. Consider the case $v = at + b$ (uniformly accelerated motion). Find the distance during time n to k by dividing this time into m equal subintervals; take advantage of the fact that the terms of the sum form an arithmetic progression. Find the limit of the sum as $m \rightarrow \infty$. Compare the resulting expression with the area, equal to the distance covered, of a trapezoid on the vt -plane.

2. Consider the case $v = t^2$ and for this case find the distance covered in the time interval from $t = 1$ to $t = 2$; in other words, find the integral

$$\int_1^2 t^2 dt$$

To do this, partition the interval from 1 to 2 into m equal parts and compute the sum $\sum t_{i-1}^2 \cdot \frac{1}{m}$ or the sum $\sum t_i^2 \cdot \frac{1}{m}$. Compare these two sums.

2.9 THE RELATIONSHIP BETWEEN THE INTEGRAL AND THE DERIVATIVE (NEWTON-LEIBNIZ THEOREM)

In the preceding sections we considered, separately, the concepts of the derivative and the integral. In implicit fashion, these notions were utilized by mathematicians even before Newton and Leibniz. The great achievement of these two mathematicians lay precisely in establishing the relationship between these concepts that so greatly speeded the development of mathematics from then on. In the present section we examine this relationship using the distance-velocity example.

We assume as given and known the instantaneous velocity as a function of time, $v(t)$. We regard as constant the time $t_1 = n$ at the beginning of the distance. We consider the distance covered in the time interval from $t_1 = n$ to $t_2 = k$ as a function of the terminal instant k . We know that

$$z(k, n) = z(k) - z(n)$$

Let us take the derivative of the left and right members, regarding n as a constant. Thus, $z(n)$ is also a constant quantity. This yields

$$\frac{dz(k, n)}{dk} = \frac{dz(k)}{dk}$$

But we know that the derivative of the coordinate of a body with respect to time is nothing but the instantaneous velocity of that body,

$$\frac{dz(k)}{dk} = v(k)$$

and so we also have

$$\frac{dz(n, k)}{dk} = v(k)$$

Substituting here the expression of $z(n, k)$ in the form of an integral, we get

$$\frac{d}{dk} \left(\int_n^k v(t) dt \right) = v(k) \quad (2.9-1)$$

This equation is the most important general property of the definite integral. In the given form, this equation is a general mathematical theorem. Its validity is independent of whether $v(t)$ is velocity (and the integral is distance) or $v(t)$ is some other quite different quantity. For any function, say $y(x)$, we have

$$\frac{d}{db} \left(\int_a^b y(x) dx \right) = y(b) \quad (2.9-2)$$

The theorem is stated thus: *the derivative of a definite integral with respect to the upper limit is equal to the value of the integrand at the upper limit.*

Because this theorem is so important, we give a different derivation of it based on a consideration of area. We will compute the derivative by first principles, that is as the limit of the ratio of the increment of the function to the increment of the independent variable.

We consider

$$I(a, b) = \int_a^b y(x) dx$$

This integral is the area bounded from above by the curve $y(x)$, from below by the x -axis, on the left by the vertical line $x = a$, and on the right by the vertical line $x = b$ (see Fig. 47).

How do we find the increment of the integral? By first principles, we have $\Delta I = I(a, b + \Delta b) - I(a, b)$. The area equal to the integral $I(a, b + \Delta b)$ differs from the area $I(a, b)$ in that the right vertical is displaced rightwards by Δb (see Fig. 47).

Consequently, the increment ΔI is the difference between two areas: that with the base from a to $b + \Delta b$ and that with the base from a to b . ΔI is then clearly the area of the strip that is hatched in Fig. 47. The base of this strip on the x -axis is a line segment of length Δb .

The desired derivative is equal to the limit

$$\frac{dI(a, b)}{db} = \lim_{\Delta b \rightarrow 0} \frac{\Delta I}{\Delta b}$$

Quite obviously, as Δb tends to zero, the area of the strip approaches $y(b) \Delta b$, and the ratio $\frac{\Delta I}{\Delta b}$ approaches the quantity $y(b)$. We have thus once again given a pictorial proof of the theorem

$$\frac{d}{db} \left(\int_a^b y(x) dx \right) = y(b) \quad (2.9-2)$$

The definite integral of a known function $y(x)$ or $v(t)$ is a function of the limits of integration a , b or n , k .

The definition of an integral as the limit of a sum, which was given in the preceding section, explains to us the role the integral concept plays in the solution of physical problems: in computing the distance traveled when a body is moving with a variable velocity $v(t)$, in determining the area bounded by a curve given by an equation $y = y(x)$. But this definition does not yield a convenient general method for computing an integral and it does not yield a convenient general method for finding an integral as a formula, as a function of the limits of integration.*

A method for finding such a formula follows from the theorem, proved above, concerning the derivative of an integral. Here, besides the property of the derivative of an integral, we make use of yet a second property of the definite integral: the definite integral is equal to zero when the upper and lower limits of the integral coincide:

$$z(n, k=n) = \int_n^n v(t) dt = 0$$

This property is obvious because the distance is equal to zero if the time in transit is $k - n = n - n = 0$.

The formula itself which yields the value of the integral as a function of the limits of integration will be derived in this fashion in Sec. 2.12. First we give a simpler derivation in Sec. 2.11.

2.10 THE INTEGRAL OF A DERIVATIVE

Let the integrand $v(t)$ be equal to the derivative of a known function $f(t)$,

$$v(t) = f'(t) = \frac{df}{dt} \quad (2.10-1)$$

* Only in rare cases and with great difficulty is one able to sum an arbitrary number of small summands.

In this case we can find the exact value of the integral in the following manner. Recall the approximate expression for the increment of a function f (Sec. 2.4):

$$\Delta f \approx f'(t) \Delta t = v(t) \Delta t \quad (2.10-2)$$

The quantity in the right member of the equation is precisely one of those summands whose sum is equal to the integral. And so we can write, approximately,

$$\Delta f = f(t_{l+1}) - f(t_l) \approx v(t_{l+1})(t_{l+1} - t_l) \approx v(t_l)(t_{l+1} - t_l) \quad (2.10-3)$$

As already mentioned, (2.10-2) is approximate and its accuracy increases as the increment Δt becomes smaller, that is to say, the smaller the difference $t_{l+1} - t_l$. But as the difference $t_{l+1} - t_l$ decreases, that is, as t_{l+1} approaches t_l , the difference between

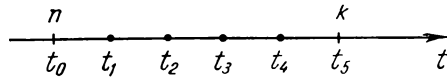


Fig. 48

$v(t_{l+1})$ and $v(t_l)$ also diminishes. For this reason, we have just as much right (the degree of accuracy is the same) to put both $v(t_{l+1})$ and $v(t_l)$ in the right member of (2.10-3), as was done above. Let us write formulas like (2.10-3) for *all* subintervals into which the domain of integration (that is, the interval from n to k) is partitioned. By way of illustration, let the interval be split up into five subintervals (Fig. 48) so that $t_0 = n$, $t_5 = k$. Take the pains to write out all five equations:

$$\begin{aligned} f(t_1) - f(t_0) &\approx v(t_1)(t_1 - t_0) \approx v(t_0)(t_1 - t_0), \\ f(t_2) - f(t_1) &\approx v(t_2)(t_2 - t_1) \approx v(t_1)(t_2 - t_1), \\ f(t_3) - f(t_2) &\approx v(t_3)(t_3 - t_2) \approx v(t_2)(t_3 - t_2), \\ f(t_4) - f(t_3) &\approx v(t_4)(t_4 - t_3) \approx v(t_3)(t_4 - t_3), \\ f(t_5) - f(t_4) &\approx v(t_5)(t_5 - t_4) \approx v(t_4)(t_5 - t_4) \end{aligned}$$

Now add them together. In the left members, all values of the function f for intermediate values of t cancel out leaving only

$$f(t_5) - f(t_0) = f(k) - f(n)$$

On the right we have precisely those sums with the aid of which we approximately expressed the integral in the preceding section, expressing the distance $z(n, k)$ for a given velocity $v(t)$. Thus

$$\begin{aligned} f(k) - f(n) &\approx \sum v(t_{l+1})(t_{l+1} - t_l) \approx \sum v(t_l)(t_{l+1} - t_l) \approx z(n, k) \\ &= \int_n^k v(t) dt \quad \text{for } v(t) = \frac{df}{dt} \end{aligned}$$

The smaller each increment Δt , i.e., the quantity $t_{i+1} - t_i$, the more exact the expression (2.10-3) of the increment of f ; but as the differences $t_{i+1} - t_i$ decrease, the sums tend to the integral.

Therefore, the equation

$$f(k) - f(n) = \int_n^k v(t) dt \quad \text{for } v(t) = \frac{df}{dt} \quad (2.10-4)$$

is indeed **exact**.

Formula (2.10-4) establishes a relationship between the integral and the derivative. From this formula it follows that if we are able to find a function f whose derivative is equal to the integrand function v , then the problem of evaluating the integral is solved, for all that remains is to substitute the values $f(k)$ and $f(n)$ and find the difference $f(k) - f(n)$.

Since this formula is so important, in the sections which follow we will give a different derivation of (2.10-4) on the basis of a more detailed consideration of the properties of the integral (see the end of Sec. 2.9) and the function f .

2.11 THE INDEFINITE INTEGRAL

In the preceding sections we introduced the concept of a definite integral as the limit of a sum of a large number of small terms. In Sec. 2.9 we elucidated the principal property of the definite integral: the derivative of a definite integral with respect to the upper limit is equal to the integrand:

$$z(n, k) = \int_n^k v(t) dt, \quad \frac{dz(n, k)}{dk} = v(k) \quad (2.11-1)$$

We now wish to take advantage of this property to compute a definite integral.

We seek a function of k whose derivative is the known function $v(k)$ and denote this function by $f(k)$. Then, by definition,

$$\frac{df(k)}{dk} = v(k) \quad (2.11-2)$$

This equation does not fully define the function $f(k)$. We know that the addition of any constant to $f(k)$ does not alter the derivative of the function. Hence, if $f(k)$ satisfies (2.11-2), then the function $g(k) = f(k) + C$ will also satisfy this equation.

The function $f(k)$ which satisfies equation (2.11-2) is called the "indefinite integral of the function $v(k)$ ". This term reflects two properties of $f(k)$: the derivative of $f(k)$ is the same as that of the definite integral $z(n, k)$,* and so $f(k)$ is called an integral. To the function

* Compare formulas (2.11-1) and (2.11-2).

$f(k)$ that satisfies (2.11-2) we can add any constant quantity, whence the modifying adjective "indefinite".

Any solution of (2.11-2) can differ from some solution $f(k)$ solely by a constant. Indeed, if there is another solution of (2.11-2), which we denote as $g(k)$, then for their difference we get

$$\frac{d}{dk} [f(k) - g(k)] = v(k) - v(k) = 0$$

But only the derivative of a constant is equal to zero for arbitrary values of the argument.

According to (2.11-1), the definite integral $z(n, k)$ is also one of the solutions of (2.11-2). Hence, $z(n, k)$ can likewise be represented as

$$z(n, k) = f(k) + B \quad (2.11-3)$$

where $f(k)$ is a solution of (2.11-2) and B is a constant, and it only remains to determine the constant. To do this, we take advantage of the second property of a definite integral: it is equal to zero when the upper limit coincides with the lower limit,

$$z(n, k = n) = z(n, n) = 0 \quad (2.11-4)$$

Putting $k = n$ into (2.11-3), we get, using (2.11-4),

$$0 = f(n) + B, \quad B = -f(n)$$

And from this we finally have

$$z(n, k) = f(k) - f(n) \quad (2.11-5)$$

It is to be noted that the "indeterminacy" of the function $f(k)$ does not in any way hamper computation, with its aid, of a definite integral from formula (2.11-5). Indeed, in place of $f(k)$ take some other solution of (2.11-2), say $g(k)$, which differs from $f(k)$ by a constant

$$g(k) = f(k) + C$$

We will evaluate the definite integral using the formula (2.11-5), taking g instead of f :

$$z(n, k) = g(k) - g(n) = f(k) + C - [f(n) + C] = f(k) - f(n)$$

The result is the same as (2.11-5).

It is convenient to denote the indefinite integral by the same letter z as we used for the definite integral.

For a given integrand $v(t)$, the definite integral depends on the upper and lower limits, which is to say, it is a function of two variables, $z(n, k)$. The indefinite integral is a function of one variable. Let us denote it by t . Thus the indefinite integral $z(t)$ is a function which satisfies the equation

$$z'(t) = \frac{dz(t)}{dt} = v(t) \quad (2.11-6)$$

Using this function, the definite integral $z(n, k)$ of the function $v(t)$ can be found from the formula

$$z(n, k) = \int_n^k v(t) dt = z(k) - z(n) \quad (2.11-7)$$

The following compact notation is used to state the difference between the values of one and the same function for two different values of the variable:

$$z(t) \Big|_n^k \equiv z(k) - z(n) \quad (2.11-8)$$

Here, on the left is the function of a dummy variable t , this is followed by a vertical line at the top of which is the value of the variable at which we desire to take the function with a plus sign, and below, the value for which we want to take the function with a minus sign.

Substituting, under the integral sign in (2.11-7), $v(t)$ expressed in terms of $z(t)$ in accord with (2.11-6) and putting expression (2.11-8) into the right side, we get the identity

$$\int_n^k z'(t) dt = z(t) \Big|_n^k \quad (2.11-9)$$

It will be seen that the positions of n and k on the left and on the right are the same. This is a good mnemonic device.

It is now time to examine some examples.

Let us consider the problem of the distance covered during the time interval from n to k at a velocity equal to $v(t) = t^2$. This distance is equal to the definite integral

$$z(n, k) = \int_n^k t^2 dt$$

In this problem, the indefinite integral $z(t)$ is obtained by solving the equation

$$\frac{dz(t)}{dt} = v(t) = t^2$$

But we know that $\frac{d(t^3)}{dt} = 3t^2$, hence $\frac{d\left(\frac{t^3}{3}\right)}{dt} = \frac{1}{3}(3t^2) = t^2$. Consequently, the equation is satisfied by

$$z(t) = \frac{t^3}{3}$$

Substituting this solution into (2.11-9), we get

$$\int_n^k t^2 dt = \frac{t^3}{3} \Big|_n^k = \frac{k^3}{3} - \frac{n^3}{3}$$

The special case of $n = 1$, $k = 2$ yields

$$\int_1^2 t^2 dt = \frac{8}{3} - \frac{1}{3} = \frac{7}{3} = 2,333 \dots$$

Thus, using the indefinite integral, we obtain in a few lines the *exact* result which we laboriously approached by numerical computations in Sec. 2.9.

The definite integral is the limit of a sum of the form

$$v(t_0)(t_1 - t_0) + v(t_1)(t_2 - t_1) + \dots$$

as each term tends to zero and the number of terms increases correspondingly. In an approximate computation, one has to partition the interval of integration into a number of subintervals, find the approximate value of the distance $v\Delta t$ in each subinterval and then add. To obtain good accuracy requires numerous arithmetic operations. But if we know the indefinite integral $z(t)$, that is, if we know the function whose derivative is equal to the integrand $v(t)$, then

any definite integral $\int_n^k v(t) dt$ is obtained immediately from formula (2.11-9). The possibility of finding functions with a given derivative (indefinite integrals) has "suddenly" put at our disposal a powerful method for computing sums (definite integrals).

The indefinite integral is sometimes called a primitive function (antiderivative). This term is used in textbooks in cases where the problem of finding a function from the known derivative of the function is solved before definite integrals are considered. We do not use this term here.

An indefinite integral can always be expressed in terms of a definite integral:

$$z(t) = C + \int_a^t v(x) dx \quad (2.11-10)$$

Applying the rule concerning the derivative of a definite integral with respect to the upper limit, it is easy to verify that $z(t)$ given by equation (2.11-10) satisfies (2.11-6) for arbitrary constants C and a .

In all problems, the answer always involves the difference between the values $z(k) - z(n)$, and this is independent of C and a . Therefore (2.11-10) may be written more compactly:

$$z(t) = \int_n^t v(x) dx$$

This is frequently abridged still further to

$$z(t) = \int v(t) dt \quad (2.11-11)$$

This notation is widely used and we will employ it, but one should bear in mind that, properly speaking, it is not correct. It may be compared with those grammatically loose expressions that occur in everyday speech and are clear to all (except pedants), such as, say, "Who did you see there?"

The notation of (2.11-11) violates the rule by which the dummy variable of integration does not appear in the result. Therefore, when using the abridged notation (2.11-11) one should always bear in mind that this is a conventional contraction of the exact expression (2.11-10).

The familiar formulas for derivatives yield a table of indefinite integrals:

$$\int dt = t, \quad \int t \, dt = \frac{t^2}{2}, \quad \int t^2 \, dt = \frac{t^3}{3}, \\ \int \frac{dt}{t^2} = -\frac{1}{t}, \quad \int \frac{dt}{\sqrt{t}} = 2\sqrt{t}$$

(See Exercise 6 of Sec. 2.3 concerning the last integral.) To verify any of these formulas, it suffices to find the derivative of the right-hand side. If in doing so, we get the function under the integral sign, then the formula is correct. Methods for finding indefinite integrals of a variety of functions are considered in detail in Chapter 3. Thanks to the relationship which exists between an integral and a derivative, we are able to find the integrals of a large number of functions.

The problem of integration is technically a much more complicated job than the problem of finding derivatives. The complexity is due, for one thing, to the fact that in the integration of rational (i.e., not containing radicals) algebraic expressions we get answers involving logarithms and inverse trigonometric functions. In the integration of algebraic expressions with radicals the result is sometimes expressible only with the aid of new, nonelementary, functions that cannot be expressed in terms of a finite number of operations involving algebraic, power and trigonometric functions.

However, the difficulties of expressing integrals by formulas should not eclipse the fundamental simplicity and clarity of the integral concept. If it is impossible (or difficult) to evaluate an integral by formula (2.11-9), it is always possible to approximate it by means of cumbersome, yet fundamentally very simple, computations.

Exercises

Evaluate the following integrals.

$$1. \int_0^1 t^2 \, dt. \quad 2. \int_1^{1.1} t^2 \, dt. \quad 3. \int_1^2 \frac{dt}{t^2}. \quad 4. \int_1^3 \frac{dt}{\sqrt{t}}.$$

5. Using an integral, find the area of a right triangle having base b and altitude h . Put the origin of coordinates at the vertex of an acute angle and the

right angle on the x -axis at point $x = b, y = 0$ (Fig. 49). Find the equation of the hypotenuse in this system of coordinates and find the area as an integral.

In integrating, take advantage of the formula $\int x \, dx = \frac{x^2}{2}$.

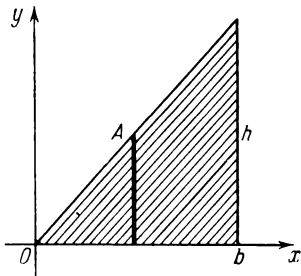


Fig. 49

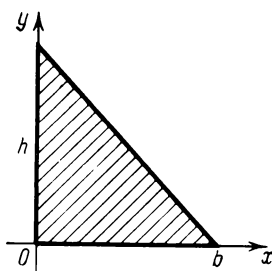


Fig. 50

Remark. Do not be indignant that a lot of effort is put into finding the familiar answer $S = \frac{1}{2}bh$ because the method of integration will be used later on in cases where elementary methods do not suffice.

6. Find the area of the same triangle by placing the right angle at the origin and an acute angle at the point $x = b, y = 0$ (Fig. 50). In integrating, make use of the obvious property of the integral of a sum of two terms $\int (f + g) \, dx = \int f \, dx + \int g \, dx$ for arbitrary f and g which are constant or functions of x (positive or negative).

Remark. The same as in Exercise 5.

7. Find the area under the parabola $y = Ax^2$ passing through the point

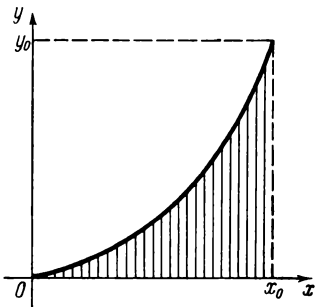


Fig. 51

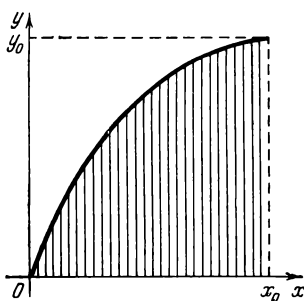


Fig. 52

$x = x_0, y = y_0$ bounded by the vertical line $x = x_0$ and the x -axis. Express the area in terms of x_0, y_0 (Fig. 51).

8. The same for a parabola passing through the origin with horizontal tangent at the point (x_0, y_0) (Fig. 52).

Hint 1. The answer may be obtained at once by using the result of the preceding exercise.

Hint 2. Still and all, take the hard way and do all the operations in their requisite order. Seek the equation of the parabola in the form $y = kx^2 + nx + m$, where k, n, m are found from the conditions of passage through the points

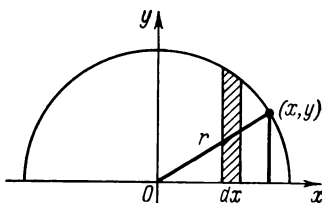


Fig. 53

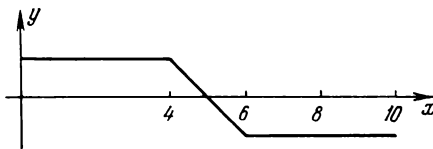


Fig. 54

(x_0, y_0) and the origin and from the condition that the tangent at the point $x = x_0, y = y_0$ is horizontal. Express the area in terms of x_0, y_0 .

Hint 3. If you are not able to follow Hint 1, then first try Hint 2 and the result will suggest how to carry out Hint 1.

9. Write the expression for the area of a semicircle of radius r (Fig. 53) in the form of a definite integral.

Hint. From the drawing, using the Pythagorean theorem,

$$x^2 + y^2 = r^2$$

This is the equation of a circle (see Sec. 1.6).

10. Evaluate the integral $\int_0^1 \frac{dx}{1+x^2}$ using the trapezoidal formula and taking $m = 5$ and $m = 10$. Carry the computations to the fourth decimal place.

Remark. The exact value of this integral is $\frac{\pi}{4}$. An approximate evaluation

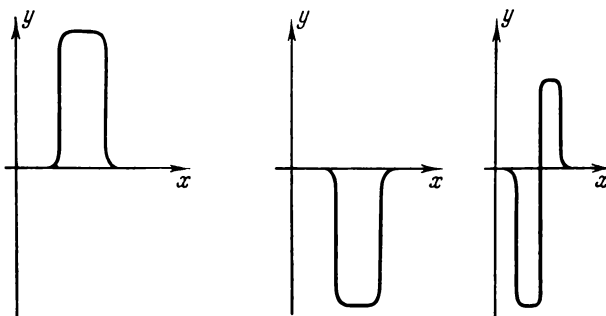


Fig. 55

of the integral enables us to obtain an approximate value for the number π .

11. Construct the graph of the function

$$F(x) = \int_a^x y(x) dx$$

The function $y(x)$ is given graphically (Fig. 54). For values of a take $a = 0$, $a = 4$, $a = 8$.

12. Construct the graph of the function

$$F(x) = \int_0^x y(x) dx$$

The function $y(x)$ is given by the graphs in Fig. 55.

13. Construct the curves

$$F(x) = \int_0^x \varphi(x) dx$$

where the functions $\varphi(x)$ are given by the curves which appear as answers to Problems 4 and 5 of Sec. 2.5. Compare $F(x)$ with the curves $y(x)$ given in Figs. 38 and 39 on page 63.

2.12 PROPERTIES OF INTEGRALS

Above we considered the simplest case of a definite integral having a positive integrand and with upper limit greater than the lower limit:

$$z(n, k) = \int_n^k v(t) dt, \quad v > 0, \quad k > n$$

The integral here is clearly positive since it is equal to the limit of a sum of positive terms. The integral has the simple physical meaning of a distance covered [$v(t)$ is the velocity] or an area [$v = v(t)$ is the equation of the curve].

What is the sign of the integral of a negative function, that is, in the case of $v(t) < 0$?

For the time being, we leave the condition $k > n$. In the expression of the sum (which in the case of passage to the limit becomes an integral), the factor Δt in each term is positive, the factor $v(t)$ is negative, each summand is negative, the sum is negative, and the integral is also negative. To summarize, if $v(t) < 0$ for $n < t < k$ so that $k > n$, then

$$\int_n^k v(t) dt < 0$$

In the case of motion, the meaning of the answer is simple: a negative value of v signifies that the motion occurs in a direction opposite to the positive direction or the direction of increasing z -coordinate). The distance traversed in the negative direction will always be regarded as negative. In this case, z decreases, $z(k) < z(n)$. Since

$\int_n^k v \, dt$ is negative here, the general formula

$$z(k) = z(n) + z(k, n) = z(n) + \int_n^k v \, dt$$

remains valid.

In the case of a velocity that changes sign, it may happen that, in particular $\int_n^k v \, dt = 0$ although $k > n$, $k \neq n$; this occurs if part of the time (between n and k) the body moves in one direction, and during another time interval it moves in the opposite direction

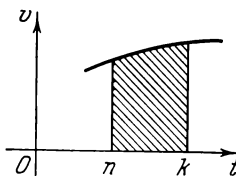


Fig. 56

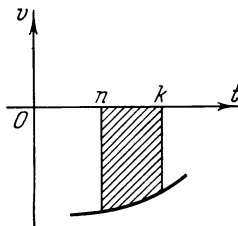


Fig. 57

with the result that by time k it returns to the position of time n .

Let us examine the problem of the area under a curve. For $k > n$ and $v(t) > 0$, the integral is equal to the area bounded by the curve $v(t)$, the t -axis and the vertical lines $t = n$, $t = k$ (Fig. 56). For $v < 0$, $k > n$, $\int_n^k v \, dt < 0$. In this case the curve lies below the axis of abscissas (Fig. 57).

Thus, in order to preserve the law by which the area is equal to an integral, it is necessary to regard the area as *negative* if the curve lies below the axis of abscissas.

If we take a function which changes sign, say, $v(t) = \sin t$, then the area under such a curve over an interval equal to a period from $t = 0$ to $t = 2\pi$ will be equal (by our definition) to zero (Fig. 58). This means that the area under the first arch (which we take to be positive) and the negative area of the second arch cancel each other exactly.

If our problem is to find out how much paint is required to paint over the hatched portions in Fig. 58, then such a definition of area is no good. In this case we have to partition the entire integral into parts over each of which v does not change sign (in our case there will be two parts, from 0 to π and from π to 2π), then compute the inte-

gral over each portion separately and finally add the absolute values of the integrals of the separate parts.

The definite integral also generalizes to the case when the upper limit is less than the lower limit. In this case we will no longer speak

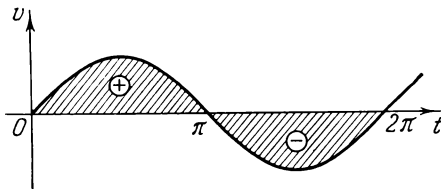


Fig. 58

of the distance, time and velocity (Sec. 2.7) and will regard the definition of the integral as a sum (see Sec. 2.8). Referring to Fig. 59, we again partition the interval between n and k by the intermediate

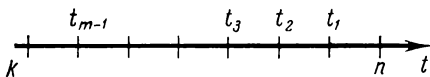


Fig. 59

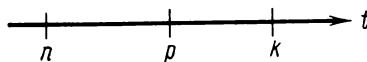


Fig. 60

values $t_1, t_2, \dots, t_{m-1}$ and convince ourselves that all Δt are now negative. It is now easy to see that

$$\int_n^k v(t) dt = - \int_k^n v(t) dt \quad (2.12-1)$$

since in any partition of the interval $[n, k]$ the corresponding sums will differ as to the signs of the subintervals Δt in all terms.

An essential property of the integral consists in the fact that the domain of integration may be divided into parts: the distance covered in the time interval between n (beginning) and k (end) may be represented as the sum of the distances traversed between time n and p (an intermediate point of time) and between p and k (Fig. 60):

$$\int_n^k v(t) dt = \int_n^p v(t) dt + \int_p^k v(t) dt \quad (2.12-2)$$

With the aid of (2.12-1) we can extend the formula (2.12-2) to the case where p lies outside the interval $[n, k]$.

Suppose $p > k > n$ (Fig. 61). Then obviously

$$\int_n^p v(t) dt = \int_n^k v(t) dt + \int_k^p v(t) dt \quad (2.12-3)$$

Transpose the last term to the left and take advantage of (2.12-1):

$$\int_n^p v dt - \int_k^p v dt = \int_n^p v(t) dt + \int_p^k v(t) dt = \int_n^k v(t) dt \quad (2.12-4)$$

In this way we get (2.12-4) which coincides exactly with (2.12-2).

We can likewise consider different arrangements of numbers n , p , k (there are six such variants in all). The reader will find no difficulty in considering them and in convincing himself that formula

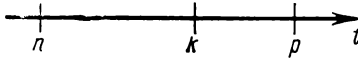


Fig.61

(2.12-2) proves to be valid in all these cases, that is to say, irrespective of the mutual arrangement of the numbers n , p , k .

Actually, we have derived all these properties of definite integrals from the definition of an integral as the limit of a sum.

These properties likewise follow from the expression of a definite integral in terms of an indefinite integral. Indeed, suppose the indefinite integral

$$\int v(t) dt = z(t)$$

Then

$$\begin{aligned} \int_n^k v(t) dt &= z(k) - z(n) \\ \int_k^n v(t) dt &= z(n) - z(k) = - \int_n^k v(t) dt \end{aligned}$$

The fundamental law by which the derivative of an integral is equal to the integrand refers to the derivative with respect to the *upper* limit.

If the definite integral is regarded as a function of the lower limit with the upper limit held constant, then we get the answer with opposite sign:

$$\frac{dz(n, k)}{dn} = \frac{d}{dn} \left(\int_n^k v(t) dt \right) = -v(n) \quad (2.12-5)$$

The minus sign in this formula is easy to understand if we regard the integral as an area: the increment of n clearly *diminishes* the area

(Fig. 62).^{*} We can formally obtain the same result by interchanging the limits (this will introduce a minus sign) and by using the familiar law on the derivative with respect to the upper limit:

$$\frac{d}{dn} \left(\int_n^k v(t) dt \right) = \frac{d}{dn} \left(- \int_k^n v(t) dt \right) = -v(n)$$

In connection with the question of the sign of the integral, we note an example that frequently disconcerts the beginner. Let us consider

$$\int \frac{dx}{x^2} = -\frac{1}{x} \quad (2.12-6)$$

This equation follows from the earlier found value of the derivative

$$\frac{d \left(\frac{1}{x} \right)}{dx} = -\frac{1}{x^2}$$

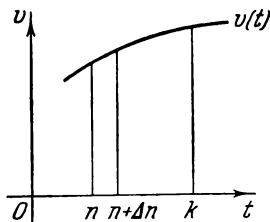


Fig. 62

Is the sign of the integral correct here? Can the integral of a positive function,

$\frac{1}{x^2}$, be negative? Does not this sign contradict the assertions made above?

Any dismay is due to the fact that formula (2.12-6) is not written in proper fashion. If we write it as

$$\int \frac{dx}{x^2} = -\frac{1}{x} + C$$

then we cannot say that the sign of the integral is always negative since this also depends on the sign and value of the quantity C .

Actually, all statements concerning the sign of the integral have referred to the definite integral. Let us take

$$\int_a^b \frac{1}{x^2} dx = -\frac{1}{x} \Big|_a^b = \left(-\frac{1}{b} \right) - \left(-\frac{1}{a} \right) = \frac{1}{a} - \frac{1}{b} = \frac{b-a}{ab}$$

When $b > a$, the integral is positive, as it should be, that is, formula (2.12-6) is correct and leads to a correct result for the definite integral. Looking ahead, we may note that the integral $\int \frac{dx}{x^2}$ involves other, no longer fictitious but real, difficulties which will be considered in Sec. 3.16.

^{*} The area bounded by the vertical lines $t = n + \Delta n$, $t = k$, the curve, and the t -axis is less than the area bounded by the vertical lines $t = n$, $t = k$, the curve, and the t -axis.

2.13 MEAN VALUES

Using an integral we can give an exact definition of the mean of a quantity which is a function of some variable.

If we have a quantity that assumes a sequence of separate values, say m values,

$$v_1, v_2, v_3, \dots, v_m$$

then the mean value is naturally determined by the formula

$$\frac{v_1 + v_2 + v_3 + \dots + v_m}{m}$$

How do we determine the mean value of a function $v(t)$ of the variable t which assumes all values in a given interval from n to k ($n < t < k$)?

Suppose that $v(t)$ is the instantaneous velocity. How do we determine the mean value $\bar{v}(n, k)$ that is, the mean (average) velocity over a time interval from n to k ? The average velocity is determined as the ratio of the distance traversed to the time spent in transit:

$$\bar{v}(n, k) = \frac{z(n, k)}{k - n} = \frac{\int_n^k v(t) dt}{k - n}$$

This definition of the mean value of a function is reasonable also in those cases when the function is not velocity but some other quantity. For instance, let $y = y(x)$ be the equation of a curve in the xy -plane (Fig. 63).

Then $\int_a^b y(x) dx$ is the area under the curve. The formula

$$\bar{y} = \frac{\int_a^b y(x) dx}{b - a}, \quad (b - a) \bar{y} = \int_a^b y(x) dx$$

signifies that $\bar{y}$ is the altitude of a rectangle with base $b - a$, the area of which is equal to the area under the curve. This means that in Fig. 63 the hatched area above the line $y = \bar{y}$ labelled plus (+) is exactly equal to the area labelled minus (−) on the portion where the curve is below the line $y = \bar{y}$. The graph of the function $y(x)$ (if it is not a straight line parallel to the x -axis) must definitely pass partly above and partly below the mean value $\bar{y}$ defined with the aid of an integral. Consequently, $\bar{y}$ is greater than the smallest value of $y(x)$ and is less than the greatest value of $y(x)$ on the averaging interval $n < x < k$.

Let us consider some examples.

Let $y(x)$ be a linear function,

$$y = kx + m$$

Here the integral is the area of a trapezoid (Fig. 64) with altitude $b - a$, bases $y(a)$ and $y(b)$ and midline $y\left(\frac{a+b}{2}\right)$. Hence

$$I(a, b) = \frac{y(a) + y(b)}{2} (b - a) = y\left(\frac{a+b}{2}\right) (b - a)$$

This expression can readily be obtained without geometrical considerations:

$$\begin{aligned} I(a, b) &= \int_a^b (kx + m) dx = \left(\frac{kx^2}{2} + mx \right) \Big|_a^b \\ &= \frac{kb^2}{2} + mb - \left(\frac{ka^2}{2} + ma \right) = (b - a) \left(\frac{kb}{2} + \frac{ka}{2} + m \right), \\ y(b) &= kb + m, \quad y(a) = ka + m, \\ y\left(\frac{a+b}{2}\right) &= k\left(\frac{a+b}{2}\right) + m \end{aligned}$$

whence quite obviously follow the expressions of the preceding formula.

Thus, for a linear function,

$$\bar{y} = \frac{y(a) + y(b)}{2} = y\left(\frac{a+b}{2}\right) \quad (2.13-1)$$

To summarize, then, for a linear function, the mean value of the function over a given interval from a to b is exactly equal to the

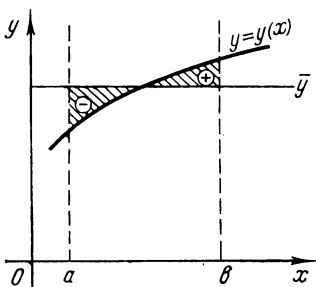


Fig. 63

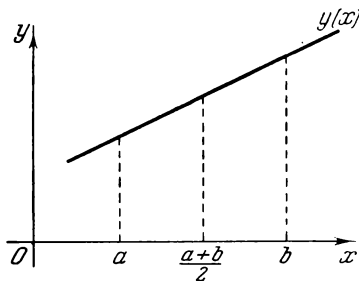


Fig. 64

arithmetic mean of the values of the function at the endpoints of the interval, $y(a)$ and $y(b)$. This may be stated differently: the mean value of a linear function is equal to the value of the function at the midpoint of the interval, that is, for $x = \frac{a+b}{2}$.

An important example of a linear relationship is the dependence of velocity on time in the case of uniformly accelerated or uniformly decelerated motion, that is to say, when a body is moving under the action of a constant force, for instance, under the action of the force of gravity when

$$v = gt + v_0$$

In computing distance we use the properties of the mean value of a linear function,

$$z(n, k) = (k - n) \left(\frac{v(k) + v(n)}{2} \right) = (k - n) \left(\frac{gk + gn}{2} + v_0 \right)$$

It must be kept in mind, however, that in the case of a nonlinear relationship, the expressions for the mean value (2.13-1) no longer hold true.

Let us consider the quadratic function (parabola) $y = rx^2 + px + q$ + q . For the sake of definiteness, take $r > 0$ and consider some portion of the parabola $a < x < b$. From the drawing in Fig. 65 it is first of all evident that

$$y\left(\frac{a+b}{2}\right) < \frac{y(a) + y(b)}{2}$$

Indeed, $y\left(\frac{a+b}{2}\right)$ is the ordinate of the point C lying on the curve, while the half-sum $\frac{y(a) + y(b)}{2}$ is the ordinate of

the point D lying at the midpoint of the chord connecting points A and B of the curve. And it is clear from the figure that C lies below D .*

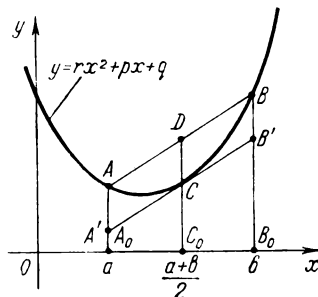


Fig. 65

Now let us turn to the integral $\int_a^b y(x) dx$, i.e., to a computation of the area under the curve. It is clear that this area is less than the area of a trapezoid with bases A_0A and B_0B . On the other hand, if we draw a tangent to the curve at point C , then this tangent will intersect the vertical lines at the points A' and B' and will form a trapezoid with midline C_0C . The area of this trapezoid is obviously less than the area under the curve. Thus, in the case of a parabola with $r > 0$,

$$(b-a) y\left(\frac{a+b}{2}\right) < \int_a^b y(x) dx < (b-a) \frac{y(a) + y(b)}{2}$$

* We note that the parabola $y = rx^2$, $r > 0$, is convex downwards, and the parabola $y = rx^2 + px + q$ with arbitrary p and q is obtained from the parabola $y = rx^2$ by a parallel translation (see Sec. 1.5).

We accordingly obtain the inequalities for the mean value $\bar{y}$ in the interval between a and b :

$$y\left(\frac{a+b}{2}\right) < \bar{y} < \frac{y(a)+y(b)}{2}$$

For the quadratic function, there is an exact formula (given without derivation, see Exercise 4) which holds true for any sign of r :

$$\bar{y} = \frac{2}{3} y\left(\frac{a+b}{2}\right) + \frac{1}{3} \left(\frac{y(a)+y(b)}{2}\right) = \frac{1}{6} y(a) + \frac{2}{3} y\left(\frac{a+b}{2}\right) + \frac{1}{6} y(b) \quad (2.13-2)$$

This expression is a good approximate formula for computing the area under any smooth curve (see Exercises 6 and 7). The employment of mean values is very convenient in a practical sense, even frequently more so than the use of integrals.

These quantities are essentially of an equal status: if we know the integral $I = \int_a^b y \, dx$, we find the mean as $\bar{y} = \frac{I}{b-a}$; having com-

puted the mean, it is just as easy to find the integral $I = (b-a)\bar{y}$.

The convenience of the mean consists in the fact that this quantity, $\bar{y}$, has the same dimensions as y and, obviously, is of the same order of magnitude as the values of y on the interval under study. It is therefore more difficult to miss an error that is ten-fold the value of $\bar{y}$ than the same error in the value of the integral.

It is commonly held that students of higher mathematics have a perfect knowledge of arithmetic and algebra and never make a mistake by a factor of 10 or err in sign. Experience shows that this is hardly the case! Computations should therefore always be carried out so as to reduce the probability of an unnoticed mistake.

Exercises

1. Find the mean value of $y = x^2$ on the interval from 0 to 2.
2. Compare this mean value with the arithmetic mean of the values of the function at the endpoints and with the value at the midpoint of the interval.
3. Check the formula (2.13-2) for the mean value of the data given in Exercise 1.
4. Verify the formula for the mean (2.13-2) in the general form for the parabola $y = rx^2 + px + q$.
5. The force of gravity decreases with distance from the centre of the earth as $F = \frac{A}{r^2}$. Using an integral, find the mean value of the force of gravity over the interval from the earth's surface (the earth's radius is R) out to a distance of R from the earth's surface (that is, $2R$ from the centre).
6. Compare the exact value of the mean in the preceding exercise with the arithmetic mean at the endpoints of the interval.
7. Compare the exact value of the mean in Exercise 5 with the mean from formula (2.13-2) referring to a parabola.

2.14 EXAMPLES OF DERIVATIVES AND INTEGRALS

In the preceding sections we examined the relationship between distance and velocity, the relationship between the equation of a curve and the area under the curve. These relations are concrete problems on the basis of which the differential and integral calculus took shape in the history of mathematics. But the concepts of the integral and the derivative are of course applicable not only to the foregoing problems but to an extraordinarily broad range of phenomena in the most diverse fields of science, technology and everyday life. Actually, the integral and the derivative constitute a definite language that is extremely suitable for a description of nature.

The student beginning the study of a foreign language has to repeat all manner of simple phrases just to get used to the new medium: "there is a table in the room", "a glass is on the table", "there is a cat on the floor", "there is a mouse near the cat". The same goes for the student of higher mathematics. The student should repeat the relationships between the derivative and the integral in a multitude of similar examples. One learns a language before attempting to express ideas in it. So that is our task now; we must learn to express familiar relationships and to formulate problems in the language of higher mathematics before solving problems and obtaining new results.*

Here are a few typical examples:

A. Derivatives with respect to time

1. Picture a vessel of arbitrary shape with water flowing out (Fig. 66). The mass of liquid in the vessel at a given instant of time is equal to M . This quantity is a function of the time, $M(t)$. The liquid collects in another vessel; the quantity of liquid here is $m(t)$. We denote the amount of liquid flowing out of the vessel in unit time by $W(t)$. This quantity has the dimensions of g/sec. The quantities m , M and W are connected by the relations

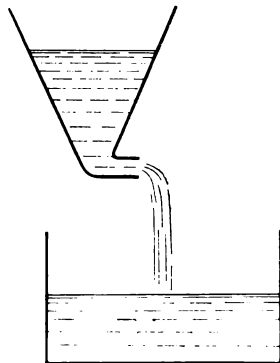


Fig. 66

$$\frac{dM}{dt} = -W(t), \quad \frac{dm}{dt} = +W(t) \quad (2.14-1)$$

* Goethe put it this way, "Mathematicians are a kind of Frenchman. You tell him something, he then translates it into his language and it turns out to be something quite different from what you had in mind."

These same relations may be written as integrals. We do so and we state that at a certain initial time t_0 the amount of liquid in the first vessel is $M(t_0) = M_0$ and the second vessel is empty: $m(t_0) = 0$. Then

$$\begin{aligned} m(t_1) &= \int_{t_0}^{t_1} W(t) dt, \\ M(t_1) &= M(t_0) - \int_{t_0}^{t_1} W(t) dt \end{aligned} \quad (2.14-2)$$

We note the fact that if we are interested in the quantity of liquid at a definite time t_1 , it is expressed in terms of an integral in which the variable of integration t runs through all values from t_0 to t_1 .

If we want to write expressions for $m(t)$ and $M(t)$, then for greater clarity it will be convenient to rename the variable of integration (using the fact that it is a dummy variable) and call it, say, tau (τ is the Greek letter tau which corresponds to the Latin t). We then have

$$\begin{aligned} m(t) &= \int_{t_0}^t W(\tau) d\tau \\ M(t) &= M(t_0) - \int_{t_0}^t W(\tau) d\tau \end{aligned} \quad (2.14-3)$$

Ordinarily, this is simply written as

$$\begin{aligned} m(t) &= \int_{t_0}^t W(t) dt, \\ M(t) &= M(t_0) - \int_{t_0}^t W(t) dt \end{aligned} \quad (2.14-4)$$

but remember that the t under the integral sign has a different meaning from the argument t in $M(t)$ and $m(t)$, which coincides with the t in the upper limit. In this respect, the notation (2.14-2) and (2.14-3) is more exact than (2.14-4).

The formulas given above correspond to an experiment in which M and the flow of liquid W are measured at distinct instants of time.

The problem is frequently stated thus: W —outflow of liquid—depends in some known fashion on the pressure, that is, the height of a column h of the liquid. In turn, given a definite shape of vessel, the quantity h is a function of M . Thus, we know the outflow W as a function of the quantity of liquid in the vessel,

$$W = W(M)$$

Then (2.14-1) takes the form

$$\frac{dM}{dt} = -W(M)$$

This is a differential equation. The solution of such equations will be considered in Chapter 5. Formulas like (2.14-2) to (2.14-4) cannot be used in this case since W is not given as a function of time.

2. Consider a capacitor (Fig. 67). We denote the charge accumulated in it (the quantity of electricity) by q . q may be measured in coulombs (absolute practical system of units). The electric current j flowing in the wire is the quantity of electricity flowing in unit time and is measured in amperes. One ampere is a current of one coulomb per second.

The charge on the capacitor* and the current are connected by the equation

$$\frac{dq}{dt} = j \quad (2.14-5)$$

(the positive direction of the current is indicated by the arrow in Fig. 67). If the variation of current flow with time is given or has been found via experiment, then we can write the integral relation

$$q(t) = q(t_0) + \int_{t_0}^t j(t) dt$$

If the capacitance of the capacitor C is given, then the voltage drop on the capacitor may be expressed in terms of q : $\varphi_C = \frac{q}{C}$. The voltage drop on the resistance R is

$$\varphi_R = E_0 - \varphi_C = E_0 - \frac{q}{C}$$

where E_0 is the battery voltage. By Ohm's law, the current flowing through a resistance is $j = \frac{1}{R} \left(E_0 - \frac{q}{C} \right)$. Using (2.14-5) we get the differential equation

$$\frac{dq}{dt} = \frac{1}{R} \left(E_0 - \frac{q}{C} \right)$$

Problems involving capacitors are discussed in detail in Chapter 8.

3. **Acceleration.** Earlier, we considered the velocity of motion as a derivative of the coordinate with respect to time. But after the

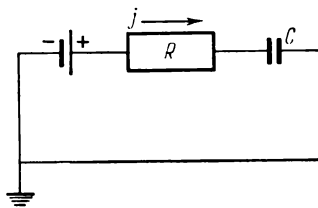


Fig. 67

* We use the term charge on the capacitor to mean the quantity of positive electricity on the left-hand plate of the capacitor C in Fig. 67 expressed in coulombs.

instantaneous velocity v has been found and we know the relationship between the instantaneous velocity and time, $v(t)$, we can ask how the velocity varies with time.

The derivative of velocity with respect to time is called the acceleration and is ordinarily denoted by a :

$$\frac{dv}{dt} = a \quad (2.14-6)$$

Since the dimensions of velocity are cm/sec or m/sec, the dimensions of acceleration are cm/sec² or m/sec².

If we know the acceleration as a function of time, then the instantaneous value of velocity may be written in the form of an integral:

$$v(t) = v(t_0) + \int_{t_0}^t a(t) dt \quad (2.14-7)$$

For example, in the case of motion under the action of gravity, $a = -g$, where $g = 9.8$ m/sec² (the minus sign being due to the fact that the upward direction is taken to be positive). Assuming $a = -g$ in (2.14-7), we get

$$v(t) = v(t_0) - \int_{t_0}^t g dt = v(t_0) - (t - t_0)g$$

We write down the velocity as the derivative of distance with respect to time, $v = \frac{dz}{dt}$, and substitute it into (2.14-6) to get

$$a = \frac{d}{dt} \left(\frac{dz}{dt} \right)$$

Such a quantity—the derivative of a derivative—is called a second derivative and is denoted

$$a = \frac{d^2z}{dt^2}$$

and is read “d two z with respect to d t square”.

Note how sensibly the superscripts (twos) are placed in the expression $\frac{d^2z}{dt^2}$. The dimensions of acceleration are just $\frac{z}{t^2}$; dropping the dimensionless symbols d , we get the proper dimensions of the second derivative.

B. Derivatives with respect to a coordinate

4. Imagine a vertical column of air with constant cross-section S cm². The density of the air is ρ g/cm³ and depends on the altitude h above the earth's surface. The volume of a thin layer contained

between h and $h + dh$ (Fig. 68) is equal to Sdh . Inside this thin layer, the density $\rho(h)$ may be regarded as constant, which is precisely why we took a thin layer. In the given instance, dh may be pictured as 1 metre or 10 metres or even (with a slightly lower accuracy) as 100 metres, since air density varies roughly by 12 to 14% per kilometre of altitude.

The mass of air in the layer dh is equal to $dm = \rho Sdh$. The mass of air in a column extending from h_1 to h_2 is defined by the integral

$$m(h_1, h_2) = S \int_{h_1}^{h_2} \rho(h) dh$$

The mass of air in a column from the earth's surface ($h = 0$) to an altitude h is

$$m(0, h) = S \int_0^h \rho(h) dh$$

The mass of air above a given altitude h is

$$m = S \int_h^{\infty} \rho(h) dh^*$$

The pressure P at some altitude h multiplied by the area S is equal to the force with which the entire column of air above h is attracted to the earth. The force of gravity is equal to the mass multiplied by the acceleration of gravity g , whence

$$P(h) = \int_h^{\infty} g\rho(h) dh$$

Using formula (2.12-5), we get

$$\frac{dP}{dh} = -g\rho(h)$$

This formula could have been written straight off by considering the equilibrium of a thin layer dh acted upon from below by the pressure $P(h)$ and from above by the pressure $P(h + dh)$; the resultant of these two forces balances the attraction to earth of the mass of air in the layer.

5. We wish to express the volume of a solid in the form of an integral (Fig. 69). Slice the solid by planes $x = \text{const}$ into thin layers. The volume dV of a thin layer is equal to the product of the cross-sectional

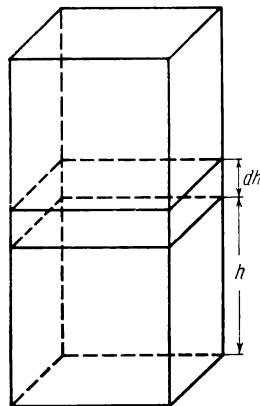


Fig. 68

* The symbol ∞ in the upper limit takes the place of a very large number h such that any subsequent increase in this quantity does not substantially change the integral.

area S by the thickness dx of the layer. Thus, if we know the area of a cross-section of the solid cut by a vertical plane as a function of the coordinate x of the cross-section, then the volume of the solid may be computed by the formula

$$V = \int_{x_0}^{x_h} S(x) dx \quad (2.14-8)$$

Let us apply this formula to a regular quadrangular pyramid with vertex at the origin of coordinates and with axis of symmetry direct-

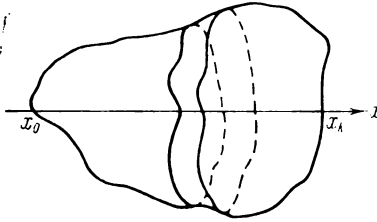


Fig. 69

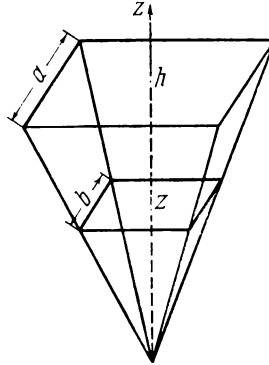


Fig. 70

ed along the z -axis (Fig. 70). Let the altitude of the pyramid be h and the base (at the top of the figure), a square with side a . From geometry we recall that a cross-section of a pyramid cut by a horizontal plane at an altitude z is a square, the side b of which is to a as z is to h :

$$b = b(z) = a \frac{z}{h}$$

Hence, the area of the cross-section is $S(z) = b^2 = \frac{a^2}{h^2} z^2$. The volume of the pyramid is

$$V = \int_0^h \frac{a^2}{h^2} z^2 dz = \frac{a^2}{h^2} \int_0^h z^2 dz$$

Let us take advantage of the result of Sec. 2.11:

$$\int z^2 dz = \frac{1}{3} z^3, \quad \int_0^h z^2 dz = \frac{1}{3} h^3$$

We then get an expression for the volume of a pyramid:

$$V = \frac{a^3}{h^2} \frac{1}{3} h^3 = \frac{1}{3} a^2 h$$

The volume of a pyramid is equal to one third the product of the area of the base by the altitude of the pyramid. To derive this formula in solid geometry without the aid of integrals is a rather complicated job.

Exercise

Derive the formula for the volume of an arbitrary pyramid using the properties of parallel sections.

SUMMARY

In Chapter 2 we considered the concepts of the derivative and the integral, some of their simpler properties, and the relationship between the integral and the derivative. The techniques involved in practical computation of derivatives and integrals of various functions will be examined in Chapter 3. Only the simplest examples were illustrated in Chapter 2.

A word of warning to the reader: do not get into the habit of measuring the difficulty and significance of any section of a mathematics course by the number of formulas, their complexity and unwieldiness. Actually, the most important thing and the most difficult thing is the mathematical statement (formulation) of a problem in the form of an algebraic equation or an integral or a differential equation. That is where our attention should be focused.

If the last three sections of this chapter have appeared to be difficult, the reader will do well to reread the whole of Chapter 2.

From his own experience, the author knows that those pieces of research which he did not succeed in completing (and which other workers did complete) were left undone because he confined himself to a general reflection and did not find the courage to write down the equations and formulate the problem mathematically. The computational difficulties in a properly posed problem with a clear-cut physical content are always surmountable, at least via approximate procedures if not by precise methods.

Chapter 3

Computation of Derivatives and Integrals

3.1 THE DIFFERENTIAL SIGN. THE DERIVATIVE OF A SUM OF FUNCTIONS

Convenient and pictorial modes of notation and simple rules that permit carrying out computations mechanically without errors is of great significance both for teaching and for the development of mathematics as such.

In Chapter 2 we analyzed the meaning of the derivative concept. In Chapter 3 we present the rules for finding the derivatives of various functions: polynomials, rational functions involving ratios of polynomials, radicals and, generally, fractional powers, the exponential function, trigonometric functions, and so forth. We have to find general rules for the derivatives of various combinations of functions: a sum of functions, a product of functions, a composite function. A table of derivatives of a number of functions that summarizes the work carried out in Secs. 3.1 to 3.12 is given in the Appendix.

From the definition of a derivative we have the following four-step rule: in each case specify value of x and its increment Δx ; find $f(x)$ and $f(x + \Delta x)$, find the increment Δf ; form the ratio $\frac{\Delta f}{\Delta x}$ and then pass to the limit as $\Delta x \rightarrow 0$.

However, the formula which yields the general expression $\frac{\Delta f}{\Delta x}$ for arbitrary Δx (not tending to zero) is, as a rule, more complicated than the formula for the limit, $\lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \frac{df}{dx}$, that is, for the derivative. For this reason, we will frequently write formulas which are only valid in the limit, when the increment tends to zero, and in this case we will write dy , dx instead of Δy , Δx . We have to work out rules for handling the quantities dy , dx so that the basic equation,

$$\frac{dy}{dx} = y'(x)$$

holds true, that is, so that the ratio of differentials is identically equal to the derivative. Before, we wrote an approximate expression for

the increment of the function:

$$y(x + \Delta x) - y(x) = \Delta y \approx y'(x) \cdot \Delta x$$

This expression becomes exact in the limit as $\Delta x \rightarrow 0$ (see footnote on page 56). For differentials we will write the exact equation

$$dy = y'(x) dx$$

The words about the limit as $\Delta x \rightarrow 0$ that had to be added to the approximate equation $\Delta y \approx y'(x) \cdot \Delta x$ are no longer needed when we write the second formula. They are taken for granted when we use the differentials dy , dx .

The rules for using differentials must be such that the ratio of differentials is equal to the derivative. To achieve this, in formulas we have to drop all terms proportional to $(dx)^2$ and higher powers of dx .

Let us consider the most elementary example, $y = x^2$. We will compare the increment technique (the Δ -process) here with the technique of differentials. Originally, we did as follows:

$$\begin{aligned} \Delta y &= (x + \Delta x)^2 - x^2 = 2x \cdot \Delta x + (\Delta x)^2, \\ \frac{\Delta y}{\Delta x} &= 2x + \Delta x, \quad y' = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = 2x \end{aligned}$$

Using differentials we write

$$dy = (x + dx)^2 - x^2 = 2x dx$$

We immediately dropped the term $(dx)^2$ in the right member!

$$y'(x) = \frac{dy}{dx} = 2x$$

As another illustration, we consider the sum of two functions taken with constant coefficients:

$$y = Cf(x) + Eg(x)$$

Using differentials, we write

$$\begin{aligned} dy &= y(x + dx) - y(x) \\ &= Cf(x + dx) + Eg(x + dx) - Cf(x) - Eg(x) \\ &= C[f(x + dx) - f(x)] + E[g(x + dx) - g(x)] \\ &= C df + E dg = Cf' dx + Eg' dx, \\ y' &= \frac{dy}{dx} = Cf' + Eg' \end{aligned}$$

The reader can easily obtain the same formula using increments and limits.

The symbols dx , dy (read: differential x and differential y) take the place of the words pertaining to limit and simplify the aspect

of formulas. The general rule is: when writing formulas involving differentials drop (this is necessary) all terms proportional to $(dx)^2$, $(dx)^3$, etc. In all other respects, we can treat differentials as ordinary algebraic quantities.

We will use the differential of a variable, dx , dt , etc., the differential of a function, df , the differential of some involved expression consisting of several functions, for instance $d\left(\frac{f^2}{g}\right)$. By definition, the ratio $\frac{df}{dx}$ is the derivative function of $f(x)$.

3.2 THE DERIVATIVE OF AN INVERSE FUNCTION

Specifying y as a function of x signifies that to each x there corresponds a definite value of y . Hence, conversely, we can say that with each definite y there is associated an x . Thus, the specification of $y(x)$ also yields the functional relation $x(y)$. This relationship is called the inverse function.

Here are some examples: on the left is the direct function $y(x)$, on the right the inverse function x of y :

$$\begin{array}{ll} y = x + a & x = y - a \\ y = x^2 & x = \sqrt{y} \\ y = x^3 + 1 & x = \sqrt[3]{y-1} \end{array}$$

In many cases, the inverse function is of a simpler form than the direct function: for example, whereas the direct function $y = \sqrt[3]{x-1}$ contains a cubic root, the inverse function $x = y^3 + 1$ is a power function, which is simpler than the direct function. In this case, it is simpler and easier to find the derivative of the inverse function, $\frac{dx}{dy}$, than it is to find the derivative of the direct function, $\frac{dy}{dx}$. The problem now is to express the derivative of a direct function in terms of the derivative of the inverse function.

For the direct function $y(x)$ we have

$$dy = \frac{dy}{dx} dx = y'(x) dx$$

From this we get the answer for the derivative $x'(y)$ of the inverse function $x(y)$:

$$x'(y) = \frac{dx}{dy} = \frac{1}{y'(x)} \quad (3.2-1)$$

On the right is an expression in the form of a function of x , namely, written with the aid of $y'(x)$. But if the inverse function $x(y)$ is known, then this expression can be represented as a function of y .

A few examples will serve as illustrations of the foregoing. The first example (a linear function) is too simple. We start with the second example:

$$y = x^2, \quad \frac{dy}{dx} = y' (x) = 2x, \quad \frac{dx}{dy} = \frac{1}{y' (x)} = \frac{1}{2x} \quad (3.2-2)$$

Substituting into (3.2-2) the inverse function $x = \sqrt{y}$, we get

$$\frac{dx}{dy} = \frac{d(\sqrt{y})}{dy} = \frac{1}{2x} = \frac{1}{2\sqrt{y}} \quad (3.2-3)$$

In Chapter 2 we obtained the same result in a more roundabout fashion.

Here is a third example:

$$y = x^3 + 1, \quad \frac{dy}{dx} = y' (x) = 3x^2, \quad \frac{dx}{dy} = \frac{1}{y' (x)} = \frac{1}{3x^2},$$

$$\frac{dx}{dy} = \frac{d(\sqrt[3]{y-1})}{dy} = \frac{1}{3x^2} = \frac{1}{3\sqrt[3]{(y-1)^2}} = \frac{1}{3} (y-1)^{-\frac{2}{3}}$$

This method will come in handy later on. When we study the exponential function a^x , we will be able to regard the logarithmic function as the inverse of the exponential function. Studying the derivatives of the trigonometric functions, $\sin x$, $\cos x$, $\tan x$, we will find the derivatives of the inverse trigonometric functions $\arcsin x$, $\arccos x$, $\arctan x$.

3.3 THE COMPOSITE FUNCTION

Let z be given as a function of y , say $z = \frac{1}{y}$, and y as a function of x , say $y = x^2 + 5$. It is clear that to each x there corresponds a definite y and since to each y there corresponds a definite z , then finally, each x is associated with a definite z , and z is a function of x . It is always possible, by substituting the expression of y in terms of x , to write directly $z(x)$; in the given example, $z = \frac{1}{x^2 + 5}$.

But for our purposes it is more advantageous to reduce all functions to combinations of the simplest possible functions: separately, each of the functions $z = \frac{1}{y}$ and $y = x^2 + 5$ is simpler than $z = \frac{1}{x^2 + 5}$.

By reducing all functions to the simplest ones, we will be able to get by with the rules for finding the derivatives of these simple functions.

Let us find the differential of the composite function $z[y(x)]$. Regarding z as a function of y , we write

$$dz = \frac{dz}{dy} \cdot dy$$

But y is a function of x , and so

$$dy = \frac{dy}{dx} \cdot dx$$

Substituting, we obtain

$$dz = \frac{dz}{dy} \cdot \frac{dy}{dx} \cdot dx \quad (3.3-1)$$

Dividing both sides of (3.3-1) by dx , we get a rule for determining the derivative of a composite function *

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} \quad (3.3-2)$$

The form of the formula is in full accord with what was stated about the possibility of handling differentials as ordinary algebraic quantities: we can cancel out dy in the product $\frac{dz}{dy} \cdot \frac{dy}{dx}$.

Recall that z is given as a function of y , and so $\frac{dz}{dy}$ is also a function of y . But since y itself is a function of x , it follows that by substituting $y = y(x)$ into the expression $\frac{dz}{dy}$, we get $\frac{dz}{dy}$ as a function of x and, hence, also $\frac{dz}{dx}$ as a function of x .

Let us carry out the computations for a case that will be needed later on. Suppose

$$z = \frac{1}{y(x)}$$

We know that for $z = \frac{1}{y}$, $\frac{dz}{dy} = -\frac{1}{y^2}$ and so

$$dz = -\frac{1}{y^2} dy = -\frac{1}{y^2} \cdot \frac{dy}{dx} dx$$

and

$$\frac{dz}{dx} = -\frac{1}{y^2} \cdot \frac{dy}{dx} \quad (3.3-3)$$

For example, if $y = x^2 + 5$, $z = \frac{1}{x^2 + 5}$, then

$$\frac{dz}{dx} = -\frac{1}{(x^2 + 5)^2} \cdot \frac{d(x^2 + 5)}{dx} = -\frac{2x}{(x^2 + 5)^2}$$

This rule for forming the derivative of a composite function holds true for a more complicated relationship as well:

$$\left. \begin{aligned} z &= z(y), \quad y = y(x), \quad x = x(t), \quad t = t(w), \\ \frac{dz}{dw} &= \frac{dz}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{dt} \cdot \frac{dt}{dw} \end{aligned} \right\} \quad (3.3-4)$$

* For derivatives we have used the notations $\frac{dz}{dx}$ and z' . But the prime notation may lead to confusion. When we write z' it is not clear whether we mean $\frac{dz}{dx}$ or $\frac{dz}{dy}$. For this reason, in case of doubt we will not use the prime notation.

If a function is represented parametrically (see Sec. 1.8), then this representation may be regarded as a special case of a composite function. Indeed, if it is given that

$$x = f(t), \quad y = g(t)$$

then the first of these equations may be regarded as an equation whose solution yields $t(x)$, substituting this $t(x)$ into the second equation, we get

$$y = g(t) = g(t(x))$$

Hence

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$

But to use this formula one need not express t as a function of x (if we did so we would get rid of the parameter, but this is not always possible). It suffices to know $x = f(t)$. This is the inverse of the function $t(x)$. Thus

$$\frac{dt}{dx} = \frac{1}{\frac{df}{dt}} = \frac{1}{\frac{dx}{dt}}$$

and so

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

This formula is yet another instance showing that we can handle differentials like ordinary algebraic quantities: the quantity dt in the right member is cancelled out.

Here is an example:

$$x = t^2 - t, \quad y = t^2 + t,$$

$$\frac{dx}{dt} = 2t - 1, \quad \frac{dy}{dt} = 2t + 1, \quad \frac{dy}{dx} = \frac{2t + 1}{2t - 1}$$

Thus, when we set up a table for construction of the graph and (given t) find x and y , we can find at any point the value of the derivative $\frac{dy}{dx}$, which value yields the slope of the tangent at that point.

Exercises

1. Find the derivative of $z = (ax + b)^2$ as a composite function of $y = ax + b$. Remove the brackets and find the same derivative.
2. Find the derivative

$$z = \frac{1}{ax + b}, \quad z = \frac{1}{(ax + b)^2}, \quad z = \frac{1}{1 + \frac{1}{x}}$$

3.4 THE DERIVATIVE OF A PRODUCT OF FUNCTIONS

Let us find the derivative of a product of two functions: $g(x)$ and $h(x)$. Put

$$f(x) = g(x) h(x),$$

But $df = f(x + dx) - f(x) = g(x + dx) h(x + dx) - g(x) h(x)$

$$g(x + dx) = g(x) + dg$$

$$h(x + dx) = h(x) + dh$$

Therefore

$$\begin{aligned} df &= [g(x) + dg][h(x) + dh] - g(x) h(x) \\ &= g(x) h(x) + g(x) dh + h(x) dg + dg dh - g(x) h(x) \\ &= g(x) dh + h(x) dg + dg dh \end{aligned}$$

Note that

$$dg = g'(x) dx,$$

$$dh = h'(x) dx$$

whence

$$dh dg = g'(x) h'(x) (dx)^2$$

The quantity $dh dg$ is proportional to $(dx)^2$ and so we ignore the product $dh dg$ in the expression for df . Finally we get

$$df = g(x) dh + h(x) dg \quad (3.4-1)$$

Dividing all the terms of (3.4-1) by dx , we get

$$\frac{df}{dx} = \frac{d(gh)}{dx} = g \frac{dh}{dx} + h \frac{dg}{dx} \quad (3.4-2)$$

This expression is remembered in the following manner: the derivative of the product gh is equal to the sum of the derivative taken on the assumption that only h is a function of x while g is held constant (the term $g \frac{dh}{dx}$) and the derivative taken on the assumption that h is held constant and only g depends on x (the term $h \frac{dg}{dx}$). Here, naturally, the constant value g in the term $g \frac{dh}{dx}$ is taken for the x for which the value of the derivative is being sought. The same goes for h in the second term.

How would we have handled this in the old way? Simple algebra yields the exact equation

$$\Delta f = g(x) \cdot \Delta h + h(x) \cdot \Delta g + \Delta h \cdot \Delta g$$

Dividing both members by Δx we get,

$$\frac{\Delta f}{\Delta x} = g(x) \cdot \frac{\Delta h}{\Delta x} + h(x) \cdot \frac{\Delta g}{\Delta x} + \frac{\Delta h}{\Delta x} \cdot \frac{\Delta g}{\Delta x} \cdot \Delta x \quad (3.4-3)$$

Note that in the last term for convenience sake we multiplied and divided by Δx . Up to now all the equations are exact and hold true for all values of Δx . Now pass to the limit as $\Delta x \rightarrow 0$. Then

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = f', \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta h}{\Delta x} = h', \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta g}{\Delta x} = g'$$

and by virtue of (3.4-3)

$$f' = gh' + hg'$$

In passing to the limit, the last term in (3.4-3) vanished since the first two factors yield in the limit the product $h' \cdot f'$ and we allowed Δx to vanish.

Using increments and passage to the limit, we obtain the same result as that obtained with the aid of differentials, but it takes more time. This is not surprising since in the case of differentials we dropped $df \cdot dg$ mechanically, on the basis of an earlier acquired rule according to which we have to reject terms involving $(dx)^2$, $(dx)^3$, etc., hence, any products of two, three or more differentials.

When carrying out the computation with the aid of increments, we actually, in the very process, *proved* this rule once again for the case of a product of functions.

The succession of operations using increments is needed to justify the rules and to understand them. But once these are understood, the use of differentials is faster and more efficient. It would be silly every time to start from rock bottom in solving a specific problem and to write out in full that the derivative is the limit of a ratio, etc.

Example. $f = (2x^2 + 5)(3x + 4)$.

Find $f'(x)$ and, in particular, $f'(2)$. Here,

$$g = 2x^2 + 5, \quad \frac{dg}{dx} = 4x,$$

$$h = 3x + 4, \quad \frac{dh}{dx} = 3,$$

$$\frac{df}{dx} = (2x^2 + 5) \cdot 3 + (3x + 4) \cdot 4x,$$

$$f'(2) = \left. \frac{df}{dx} \right|_{x=2} = (2 \cdot 4 + 5) \cdot 3 + (3 \cdot 2 + 4) \cdot 4 \cdot 2 = 39 + 80 = 119$$

The rule for finding the derivative of a product generalizes to the case of many factors. For example, for a product of four functions $f(x)$, $g(x)$, $h(x)$, $k(x)$ we get

$$\frac{d(fghk)}{dx} = fgh \frac{dk}{dx} + fgk \frac{dh}{dx} + fhk \frac{dg}{dx} + ghk \frac{df}{dx} \quad (3.4-4)$$

The derivative of a quotient (ratio) of two functions is found by writing $f = \frac{h}{g}$ in the form of a product:

$$f = h \frac{1}{g}$$

Then

$$f' = h \left(\frac{1}{g} \right)' + h' \frac{1}{g} \quad (3.4-5)$$

We find the derivative of the function $\frac{1}{g}$ by using (3.3-3),

$$\left(\frac{1}{g} \right)' = -\frac{1}{g^2} g'$$

Substituting this into (3.4-5), we get

$$f' = \left(\frac{h}{g} \right)' = -\frac{h}{g^2} g' + h' \frac{1}{g}$$

or

$$\left(\frac{h}{g} \right)' = \frac{h'g - hg'}{g^2} \quad (3.4-6)$$

The rule by which the derivative of a product of several functions can be found as the sum of the derivatives computed on the assumption that each time only one function varies is actually applicable not only to a product of functions but to other combinations as well. It is easy to see that the formula for the derivative of a sum of functions also agrees with this statement.

Later on we will see that the same formulation is applicable also to cases, like, say, $g(x)^{h(x)}$, where the function g is raised to the power of h , which is a function of x .

Exercises

1. Find the derivative of the function $y = x^4$ by writing $x^4 = x^2 x^2$.

Find the derivatives of the following functions.

2. $y = (2x^2 + x) \sqrt{x}$. 3. $y = \frac{x^3 + 5x^2}{x + 1}$. 4. $y = \frac{x - 1}{x^2 + 2}$.

3.5 THE POWER FUNCTION

Let us consider the derivative of the power function

$$y = x^n$$

where n is a constant. For n a positive integer, x^n is the product of n identical factors:

$$y = \overbrace{x \cdot x \cdot x \cdots x}^{n \text{ times}},$$

$$\frac{dy}{dx} = \overbrace{1 \cdot x^{n-1} + 1 \cdot x^{n-1} + \cdots + 1 \cdot x^{n-1}}^{n \text{ times}}$$

[by a formula of the type (3.4-4)] whence*

$$\frac{dy}{dx} = nx^{n-1} \quad .$$

We will show that this formula holds true for arbitrary n , whether fractional or negative.

For fractional n , we write $n = \frac{m}{p}$, where the numbers m and p are integers. We get $y = x^{\frac{m}{p}}$ or

$$y^p = x^m \quad (3.5-1)$$

The expression y^p in the left member of (3.5-1) is a composite function of x since y is a function of x . Therefore, computing the derivative of both members of (3.5-1), we get

$$\frac{d}{dx} (y^p) = p y^{p-1} \frac{dy}{dx} = m x^{m-1}$$

whence

$$\frac{dy}{dx} = \frac{m}{p} \frac{x^{m-1}}{y^{p-1}} = \frac{m}{p} \frac{x^{m-1}}{\left(\frac{m}{x^{\frac{m}{p}}}\right)^{p-1}} = \frac{m}{p} \frac{x^{m-1}}{x^{\frac{m-p}{p}}} = \frac{m}{p} x^{\frac{m}{p}-1}$$

Noting that $\frac{m}{p} = n$, we finally have

$$\frac{dy}{dx} = nx^{n-1}$$

For a negative exponent, we write $n = -k$, where k is a positive number:

$$y = x^n = x^{-k} = \frac{1}{x^k}$$

By the rule for finding the derivative of the composite function $y = \frac{1}{f}$, $f = x^k$, we get

$$\frac{dy}{dx} = -\frac{1}{f^2} \frac{df}{dx} = -\frac{1}{x^{2k}} k x^{k-1} = -k x^{-k-1}$$

Substituting $k = -n$ back again, we also get for a negative n

$$\frac{dy}{dx} = \frac{d(x^n)}{dx} = nx^{n-1}$$

Thus, the formula of the derivative of a power is applicable to any rational exponent n . It can also be extended to the case of an irrational exponent.

This formula is of the greatest importance. Simple as it is, this formula can also be written usefully as

$$y = cx^n, \quad \frac{dy}{dx} = n \frac{y}{x} \quad (3.5-2)$$

* For n an integer, we can obtain this formula with the aid of the binomial theorem, but the derivative is easily found without resorting to this theorem.

One has to get the feeling of this result. For positive n , the power function has the obvious property that for $x = 0$, y is also equal to 0. For a given $n > 0$, the curve $y = cx^n$ can be drawn through any point (x_0, y_0) : it suffices to choose $c = y_0/x_0^n$. Let the curve pass through the origin and through the point (x_0, y_0) . We find the mean value of the derivative on the section of the curve between the origin and the point (x_0, y_0) . According to the definition of the mean (see Sec. 2.13),

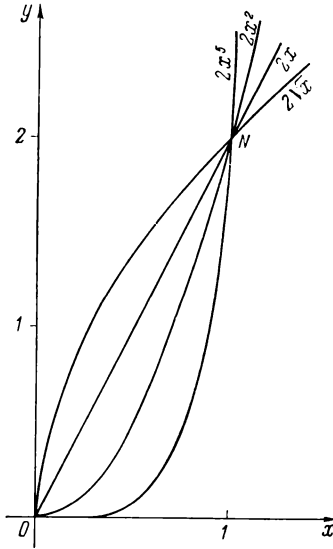


Fig. 71

$$\bar{y}' = \frac{\int_0^{x_0} y'(x) dx}{x_0 - 0}$$

whence, using (2.11-9), we get

$$\bar{y}' = \frac{y(x_0) - y(0)}{x_0} = \frac{y(x_0)}{x_0} = \frac{y_0}{x_0}$$

Indeed, as x varies from 0 to x_0 , y grows from 0 to y_0 . Hence, the average rate of growth of y (that is, the mean value of the derivative) is equal to y_0/x_0 , this is obvious without integrals!

As is evident from (3.5-2), the value of the derivative at the point (x_0, y_0) differs by a factor of n (n is the exponent) from the mean value of the derivative. Fig. 71 depicts a number of curves with different n : $n = \frac{1}{2}, 1, 2, 5$, passing through one and the same point $N(x_0, y_0)$ and, thus, having the same mean derivative on the interval $0 - x_0$. It is clearly evident that the larger n , the greater the derivative at the point N (the more steeply the curve rises). Let us return to formula (3.5-2):

$$\frac{dy}{dx} = n \frac{y}{x}$$

whence $dy = n \frac{y}{x} dx$ and so for small increments

$$\Delta y = n \frac{y}{x} \Delta x \quad (3.5-3)$$

We will assume that the relation (3.5-3) is sufficiently exact for $\Delta x = 0.01x$, that is, for a 1-percent variation of the argument. Then from (3.5-3) we get

$$\Delta y = n \frac{y}{x} 0.01x$$

or

$$\Delta y = n \cdot 0.01y$$

When the argument varies by 1 %, the power function with exponent n varies by n %.

Exercises

Find the derivatives of the following functions.

1. $y = x^5 - 3x^4 + x^3 + 7x^2 - 2x + 5$. 2. $y = (x^3 + x + 1)^2$. 3. $y = (x^2 - x + 1)^4$. 4. $y = (3x^2 - 1)^{10}$. 5. $y = \sqrt{x^2 - 1}$.

6. $y = \sqrt[5]{x^2}$.

7. Find the values of $y(9)$ and $y(11)$ if it is given that $y(10) = 5$ in the case: (a) $y \sim \sqrt{x}$, (b) $y \sim \frac{1}{x}$, (c) $y \sim x^2$ (the sign $\sim$ stands for proportionality). Solve the problem mentally without any computations. Compare the answers with the exact values.

3.6 THE DERIVATIVES OF ALGEBRAIC FUNCTIONS WITH CONSTANT EXPONENTS

The collection of rules in Secs. 3.1 to 3.5 enables one to find the derivative of any function involving addition, subtraction, multiplication, division, and raising to a (constant) power, including fractional powers (radicals).

An example will illustrate how to do this in the best practical fashion. Find the derivative of the function

$$f(x) = x \sqrt[3]{x^2 - 1}$$

It is best to write the answer at once, that is, without introducing any new designations (such as $\sqrt[3]{x^2 - 1} = y$). The derivative is taken, as it were, separately with respect to each site involving x , and we say roughly the following (the letters "a", "b", "c", . . . show to what sites of the expression of the derivative that is written underneath the words refer): the derivative of (a) with respect to x under the radical sign, plus (b) the derivative with respect to $\sqrt[3]{x^2 - 1}$ multiplied by (c) the derivative of $\sqrt[3]{x^2 - 1}$ with respect to $x^2 - 1$ multiplied by (d) the derivative of $x^2 - 1$:

$$\frac{df}{dx} = \overset{(a)}{\sqrt[3]{x^2 - 1}} + x \cdot \overset{(b)}{\frac{1}{3}} \cdot \overset{(c)}{\frac{\sqrt[3]{x^2 - 1}}{x^2 - 1}} \cdot \overset{(d)}{2x}$$

It is well to get used to this efficient approach (without indulging in a lot of writing) by applying the following principles:

(a) the rule for differentiating a composite function [formulas (3.3-2), (3.3-4)];

(b) if the expression is made up of several functions, then its derivative is equal to the sum of the derivatives computed on the assumption that each time only one of the functions is taken to be variable and the rest are held constant [see formulas (3.4-2), (3.4-4), (3.4-6)].

The formula for the derivative of a power is conveniently used in the form

$$y = cx^n, \quad \frac{dy}{dx} = n \frac{y}{x}$$

as was done above in the example; see expression (c).

To acquire the necessary facility in handling these rules, a good 10 to 20 exercises devoted to the pure techniques (without regard for the physical content of the problems) is definitely needed.

Exercises

Find the derivatives of the following functions.

1. $y = x^3 (x^2 - 1)^2$. 2. $y = x^3 \sqrt{x^2 + x}$.
3. $y = x^5 \sqrt[3]{x^2 - 1} (x^3 - 2x)^{\frac{1}{5}}$. 4. $y = \left(x + \frac{1}{\sqrt{x}}\right) \sqrt{x^3 - 2}$.
5. $y = x^2 \sqrt{\sqrt{x} + x}$. 6. $y = \left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{3}}\right)^5 x$. 7. $y = \frac{x}{1 - x^2}$. 8. $y = \frac{x^2 + x + 1}{x^2 - x + 1}$.
9. $y = \frac{(x-1)(x+3)}{x+1}$. 10. $y = \frac{3x-1}{x^5} \sqrt{x^3 + 2}$. 11. $y = \frac{x}{\sqrt{x^2 - 1}}$.
12. $y = \frac{x}{\sqrt[3]{x+1}}$. 13. $y = \sqrt{x^2 + x} \sqrt{x}$. 14. $y = x \sqrt{x^2 + \sqrt[3]{x}}$.
15. $y = \frac{x^2 - 2}{(1 + x^2) \sqrt{1 + x^2}}$. 16. $y = x \sqrt[3]{(2x+3)^2}$.
17. $y = (x^3 - 1) \sqrt{x-1} + x \sqrt[3]{x^2 - 1}$. 18. $y = \frac{x \sqrt[3]{(2x-3)^2}}{(x-1)^2}$. 19. $y = \sqrt{\frac{x-1}{x+1}}$.
20. $y = \frac{x^2 + x + 1}{x-2} \sqrt[3]{x+1}$. 21. $y = \frac{x \sqrt{x^2 - 1}}{x^3 + 1}$. 22. $y = \sqrt[3]{\frac{x^2 + x + 1}{x+1}}$.
23. $y = x \sqrt{x^2 - 1} \sqrt[3]{x + \sqrt{x}}$. 24. $y = \left(x + \frac{1}{\sqrt{x}}\right)^{\frac{1}{7}} x^2$. 25. $y = \frac{\sqrt[3]{x-2x}}{(x+1)^{\frac{1}{3}}}$.

3.7 THE EXPONENTIAL FUNCTION

Consider the function

$$y = a^x$$

where the number a exceeds 1. The graph of the function y is shown in Fig. 72. When $x = 0$, $y = 1$ for arbitrary a .

For all x , the function y is positive and grows with increasing x so that the derivative is everywhere positive as well. When x is increased by a constant quantity c , we get $y(x+c) = a^{x+c} = a^c \cdot a^x = b \cdot a^x = b \cdot y(x)$, where $b = a^c$. The quantity y is multiplied by the constant. Thus, if x is varied in successive fashion, by identical steps (in arithmetic progression),

$$x = x_0, x_0 + c,$$

$$x_0 + 2c, \dots, x_0 + nc$$

then y will assume the values

$$y_0, by_0, b^2y_0, \dots, b^ny_0$$

It will be recalled that such a law of growth is called a geometric progression.

Let us find the derivative of the exponential function for $a = 10$ ($a = 10$ is taken simply to facilitate computations):

$$\frac{d(10^x)}{dx} = \frac{10^{x+\Delta x} - 10^x}{\Delta x} = 10^x \frac{10^{\Delta x} - 1}{\Delta x}$$

What is the quantity

$$\frac{10^{\Delta x} - 1}{\Delta x} ?$$

It is the limit of the ratio $\frac{10^{\Delta x} - 1}{\Delta x}$

as $\Delta x \rightarrow 0$. Let us find this limit numerically, arithmetically. Using a four-place table of logarithms, we get

$$10^{0.1} = 1.2586, \quad \frac{10^{0.1} - 1}{0.1} = 2.586,$$

$$10^{0.01} = 1.0233, \quad \frac{10^{0.01} - 1}{0.01} = 2.33,$$

$$10^{0.001} = 1.0023, \quad \frac{10^{0.001} - 1}{0.001} = 2.3$$

Thus we have

$$\frac{10^{\Delta x} - 1}{\Delta x} = 2.3$$

Hence the derivative

$$\frac{d}{dx}(10^x) = 10^x \cdot 2.3$$

We found the derivative of 10^x in experimental fashion, so to speak, by means of tables. For any other exponential function it is now easy to reduce the problem to the preceding one: using the concept

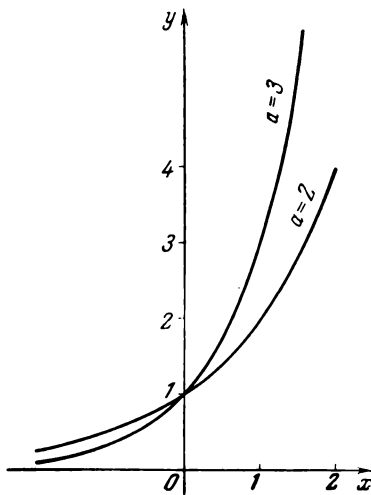


Fig. 72

of a logarithm, we write

$$a = 10^{\log_{10} a}, \quad a^x = 10^{x \log_{10} a}$$

By the rule for finding a derivative of a composite function, we get

$$\frac{da^x}{dx} = 10^{x \log_{10} a} \cdot 2.3 \cdot \log_{10} a = a^x \cdot 2.3 \cdot \log_{10} a \quad (3.7-1)$$

The remarkable peculiarity of the exponential function is that its derivative is directly proportional to the function itself. Therein

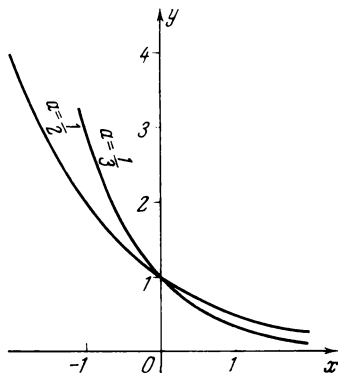


Fig. 73

lies the chief property of a geometric progression: the greater the quantity, the faster it grows. The properties of geometric progressions, their exceedingly fast rate of build-up are a favourite topic in Perelman's interesting popular-science books "Recreational Algebra" and "Figures for Fun".

If $0 < a < 1$, the graph of an exponential function will have the form shown in Fig. 73. When x increases in arithmetic progression, y diminishes in geometric progression. Formula (3.7-1) is still applicable. In this formula, $\log_{10} a$ is negative

for $a < 1$ and hence the derivative, which is proportional to the function, is of opposite sign.

In Chapter 5 we will give some instances of a quantity diminishing with time in such a manner that the rate of decrease is proportional to the quantity at a given instant:

$$\frac{dy}{dt} = -ky$$

From the foregoing, it is evident that in this case the solution of the problem is the exponential function

$$y = y_0 a^t \quad (a < 1)$$

These problems will be discussed in detail in Chapter 5.

Exercises

Find the derivatives of the following functions.

$$1. \ y = 10^{\sqrt{x}}. \quad 2. \ y = 2x. \quad 3. \ y = 5^{x+1}. \quad 4. \ y = \left(\frac{1}{2}\right)^x.$$

3.8 THE NUMBER e

Let us find a base for which the derivative of an exponential function is of the simplest form, namely such that the coefficient in the expression of the derivative is equal to unity, so that it need not be written at all. We denote this number by e . Thus

$$\frac{de^x}{dx} = e^x \quad (3.8-1)$$

This number is easily found by using formula (3.7-1):

$$2.3 \cdot \log_{10} e = 1, \quad \log_{10} e = \frac{1}{2.3} = 0.4343$$

whence, referring to a table of logarithms, we get

$$e = 2.718$$

This practical approach does not follow the historical development of mathematics and is fundamentally unsatisfactory. We made use of numbers taken from logarithmic tables and did not stop to think how they were computed.*

Let us find the number e on the basis of formula (3.8-1). By the general property of exponential functions, $e^0 = 1$. Let us consider the function $y = e^x$. Then $y(0) = 1$, and from (3.8-1) $y'(0) = 1$.

Let us take a small $\Delta x = r$ and compute the increment of the function $y = e^x$ when passing from $x = 0$ to $x = r$: $\Delta y = y' \Delta x$, and therefore $\Delta y = 1 \cdot \Delta x = r$, $y(x) = y(0) + \Delta y$, whence

$$e^r = 1 + r \quad (3.8-2)$$

We write the small number r as a fraction with a large denominator:

$r = \frac{1}{n}$; if $r \ll 1$, then $n \gg 1$.** Then from (3.8-2) we get $e^{\frac{1}{n}} = 1 + \frac{1}{n}$, whence $e = \left(1 + \frac{1}{n}\right)^n$. This expression is the more exact, the larger n , so that a rigorous definition of the number e is written thus:

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

[and is read: e is the limit of the expression $\left(1 + \frac{1}{n}\right)^n$ as n tends to infinity].

* The accuracy of e here is also greater than that obtainable when determining the derivative of 10^x by means of a four-place table of logarithms.

** The notation $r \ll 1$ means that the number r is very much less than 1.

But one should not fear such words as "limit" or "infinity". Actually $\left(1 + \frac{1}{100}\right)^{100} = 2.705$, which is only slightly different from the exact value. I advise the reader to find $\left(1 + \frac{1}{8}\right)^8$ for himself.

We have found that for r small,

$$e^r = 1 + r$$

and this is the more exact, the smaller r is (detailed tables have been compiled for the function $y = e^x$). Let us check this using numbers.

From Table 1 (the values of e^x are taken from a four-place table) we see that even when $r = \pm 0.3$, the error does not exceed 6%. For computational purposes it is worth remembering not only $e = 2.718$, but also the approximate values $e^2 = 7.4$, $e^3 = 20$, $e^4 = 55$, $e^5 = 150$. Values of e^x and e^{-x} are given in Table IV of the Appendix.

Table 1

r	$1 + r$	e^r	r	$1 + r$	e^r
-0.5	0.5	0.6065	0	1	1
-0.4	0.6	0.6703	+0.01	1.01	1.0101
-0.3	0.7	0.7408	0.1	1.1	1.1052
-0.2	0.8	0.8187	0.2	1.2	1.2214
-0.1	0.9	0.9048	0.3	1.3	1.3499
-0.01	0.99	0.9900	0.4	1.4	1.4918
			0.5	1.5	1.6487

The number e greatly simplifies the solution of problems involving geometric progressions and compound interest. Consider the following example. How many times will production increase over a period of 50 years if the annual growth rate is 2%? We have to compute 1.02^{50} . The use of the number e consists in our setting, approximately, $1.02 = e^{0.02}$, whence $1.02^{50} = e^{0.02 \times 50} = e = 2.72$. The general formula is

$$(1 + r)^m = e^{mr} \quad (r \ll 1) \quad (3.8-3)$$

This formula can be used if r is small: m and mr need not be small. If mr is also small, then $e^{mr} = 1 + mr$ and we obtain the earlier formula $(1 + r)^m = 1 + mr$; but for large values of mr this formula is not applicable, whereas expression (3.8-3) remains valid.

Thus, for the example given above, the exact value of 1.02^{50} is 2.693, approximately $1.02^{50} = e^1 = 2.718$, and by the formula $(1 + r)^m = 1 + mr$ we get $1 + 50 \times 0.02 = 2$. Computations using e yielded an error of about 1% whereas the formula $(1 + r)^m = 1 +$

+ mr was in error in this case by about 25%. The error estimate of the formula is given in general form in Exercise 5 of Sec. 3.17.

In accordance with the original definition of the number e by formula (3.8-1), the derivatives of exponential functions are especially simple when taking powers of e . These derivatives are conveniently expressed in terms of the function itself. Here are a number of formulas.

$$\begin{aligned} y &= e^x, & \frac{dy}{dx} &= e^x = y, \\ y &= Ce^x, & \frac{dy}{dx} &= C \frac{de^x}{dx} = Ce^x = y, \\ y &= Ce^{kx}, & \frac{dy}{dx} &= C \frac{de^{kx}}{dx} = Ce^{kx} \frac{d(kx)}{dx} = ky, \\ y &= e^{m(x)}, & \frac{dy}{dx} &= \frac{de^{m(x)}}{dx} = e^{m(x)} \frac{dm(x)}{dx} = y \frac{dm(x)}{dx}, \\ y &= f(x) e^{m(x)}, & \frac{dy}{dx} &= f'(x) e^{m(x)} + f(x) e^{m(x)} m'(x) \\ & & &= y \left(\frac{f'(x)}{f(x)} + m'(x) \right) \end{aligned}$$

The exponential function of x to the base e is written e^x . But if x turns out to be a complicated and unwieldy expression, this is not a convenient notation. For example, for $x = \left(\frac{7t^2+24t}{t^3+5} \right)^3$ in the expression

$$e^{\left(\frac{7t^2+24t}{t^3+5} \right)^3}$$

one might simply not notice the e and would fail to realize what the whole matter was about. There is another designation for the function e^x :

$$e^x \equiv \exp(x)$$

(read: the exponential of x). The law e^x is called an exponential law and the function e^x is termed the exponential function.

In the new notation, our example takes the form

$$e^x = \exp \left[\left(\frac{7t^2+24t}{t^3+5} \right)^3 \right]$$

Conclusion. To summarize, we can give three distinct definitions of the number e : (1) from the condition $(e^x)' = e^x$, (2) from the condition $e^r = 1 + r$ for $r \ll 1$, and (3) as the limit $\left(1 + \frac{1}{n}\right)^n$ as $n \rightarrow \infty$. To fix this important material in his mind, the reader is advised to put aside the book and demonstrate for himself how from each definition follow the other two regarded as properties of e .

The number e has still other remarkable definitions and properties. In particular, the series which is convenient to use in computing e is given on page 157, formula (3.18-2). Then there is the formula $e^{i\varphi} = \cos\varphi + i \sin\varphi$, where $i = \sqrt{-1}$ is the imaginary unit. For small φ , the validity of this formula is confirmed by the fact that from the second definition of e follows $e^{i\varphi} \approx 1 + i\varphi$, and $\cos\varphi \approx 1$, $\sin\varphi \approx \varphi$ for $\varphi \ll 1$. Since the functions $\sin(\omega t)$ and $\cos(\omega t)$ describe harmonic oscillations, the function $e^{i\omega t}$ is very frequently used in the theory of oscillations.

Exercises

Find the derivatives of the following functions.

1. $y = e^{-x}$. 2. $y = e^{x^2}$. 3. $y = e^{x^3-3x+1}$. 4. $y = e^{\sqrt{x}}$. 5. $y = 5e^x - e^{3x}$.

3.9 LOGARITHMS

By definition, the logarithm of a quantity f to a base a is the exponent g of the power to which a (the logarithmic base) must be raised in order to obtain the given number f :

$$f = a^g, \quad g = \log_a f$$

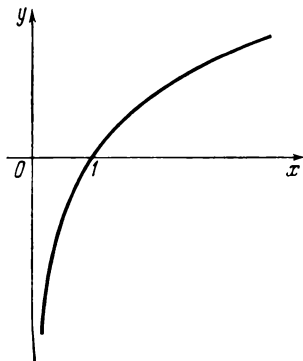


Fig. 74

The curve representing the relationship $y = \log_a x$ (for the case $a > 1$) is shown in Fig. 74. Note that $y = 0$ for $x = 1$; $y > 0$ for $x > 1$, and $y < 0$ for $x < 1$. The entire curve is located to the right of the axis of ordinates. Since a positive number a raised to any power yields a positive number, there are no logarithms of negative numbers. The reader will also note that in the equations $y = \log_a x$, $x = a^y$ the quantities x , y and a are dimensionless.

As will be seen from Fig. 74, the derivative of the function $y = \log_a x$ is positive for all values of x . The derivative decreases as x increases. Logarithms to the base e (see Sec. 3.8) are called natural logarithms. They are denoted by $\ln x$.

Let us find the derivative of a natural logarithm. We consider $d \ln x = \ln(x + dx) - \ln x$. Take advantage of the familiar formula $\ln a - \ln b = \ln \frac{a}{b}$. Then

$$d \ln x = \ln \frac{x+dx}{x} = \ln \left(1 + \frac{dx}{x} \right) \quad (3.9-1)$$

We already know (see Sec. 3.8) that for small r ,

$$e^r = 1 + r$$

Take the logarithms of both sides:

$$\ln e^r = r = \ln(1 + r) \quad (3.9-2)$$

Using (3.9-2) we obtain from (3.9-1)

$$d \ln x = \ln \left(1 + \frac{dx}{x} \right) = \frac{dx}{x}$$

and so

$$\frac{d \ln x}{dx} = \frac{1}{x} \quad (3.9-3)$$

The derivative of a natural logarithm can also be found by using the fact that the logarithmic function and the exponential function are inverse functions. We can write

$$y = \ln x, \quad x = e^y, \quad x' = \frac{dx}{dy} = \frac{d(e^y)}{dy} = e^y, \quad \frac{dy}{dx} = \frac{1}{x'} = \frac{1}{e^y} = \frac{1}{x}$$

When x varies in geometric progression, $\ln x$ varies in arithmetic progression:

$$x = ab^m, \quad \ln x = \ln a + m \ln b$$

For this reason, the larger x is, the slower $\ln x$ grows and the smaller is the derivative.

Let us derive a formula connecting the logarithms of one and the same number to different bases. Suppose

$$f = \log_a h, \quad a^f = h \quad (3.9-4)$$

Taking the logarithms of both sides of (3.9-4) to the base b , we have $f \log_b a = \log_b h$, whence $f = \frac{\log_b h}{\log_b a}$. Taking (3.9-4) into account, we get

$$\log_a h = \frac{\log_b h}{\log_b a} \quad (3.9-5)$$

Using (3.9-5) we can obtain the derivative of the logarithm to any base. Let $y = \log_a x$. Then

$$y = \frac{\ln x}{\ln a}, \quad \frac{dy}{dx} = \frac{1}{\ln a} \frac{d \ln x}{dx} = \frac{1}{\ln a} \frac{1}{x} \quad (3.9-6)$$

In formula (3.9-5) put $b = e$ and $h = e$ to get $\log_a e = \frac{1}{\ln a}$ and then rewrite (3.9-6) as

$$\frac{d \log_a x}{dx} = \frac{\log_a e}{x} \quad (3.9-7)$$

The simplest one of the formulas (3.9-3), (3.9-6) and (3.9-7) is (3.9-3). It is obtained if the logarithms are taken to the base e .

That is why they are called natural logarithms. For rough mental calculations, it is advisable to memorize: $\ln 2 = 0.69$, $\ln 3 = 1.1$, $\ln 10 = 2.3 = \frac{1}{0.434}$. A short table of natural logarithms is given in the Appendix, Table V.

If some function $f(x)$ is under the sign of the logarithm, the derivative is found by the rule for differentiating composite functions (Sec. 3.3):

$$\frac{d \ln f(x)}{dx} = \frac{1}{f(x)} \cdot \frac{df(x)}{dx} \quad (3.9-8)$$

Note that by using the concept of a logarithm, it is easy to find the derivative of the function $y = a^x$ for arbitrary a . Indeed, $\ln y = x \ln a$ and so $y = e^{x \ln a}$, whence

$$y' = e^{x \ln a} \ln a = a^x \ln a = y \ln a$$

Formula (3.9-8) enables one to find the derivatives of expressions of the form $f(x)^{h(x)}$, that is, such that contain the variable both in the base and in the exponent. Let

$$y = f(x)^{h(x)} \quad (3.9-9)$$

Taking logs (the logarithms can be taken to any base, we choose natural logarithms), we have

$$\ln y = h(x) \ln f(x) \quad (3.9-10)$$

Let us take the derivatives of both sides of (3.9-10) and have regard for the fact that $\ln y$ is a composite function of x [just as $\ln f(x)$]:

$$\frac{1}{y} y' = h'(x) \ln f(x) + h(x) \frac{f'(x)}{f(x)}$$

whence

$$y' = y \left[h'(x) \ln f(x) + h(x) \frac{f'(x)}{f(x)} \right]$$

or, using (3.9-9),

$$y' = f(x)^{h(x)} h'(x) \ln f(x) + h(x) f(x)^{h(x)-1} f'(x) \quad (3.9-11)$$

Consider the formula (3.9-11). On the right we have a sum of two terms: the first term, $f(x)^{h(x)} h'(x) \ln f(x)$, is the derivative of the expression f^h computed under the assumption that only h is variable while f is held constant, the second term, $h(x) f(x)^{h(x)-1} f'(x)$, is the derivative of the expression f^h computed under the assumption that f is variable and h is constant. This confirms the general principle expressed at the end of Sec. 3.4.

Exercises

1. Recalling $\ln 10$, find $\ln 100$.
2. Use formula (3.9-5) to find $\log_5 15$.
3. Using the fact that $\ln(u \cdot v) = \ln u + \ln v$ and differentiating both sides, obtain the formula for the derivative of a product.

4. Starting with the relation $\ln \frac{u}{v} = \ln u - \ln v$, obtain the formula for the derivative of a quotient.

Find the derivatives of the following functions.

5. $y = \ln 2x$. 6. $y = \ln (x + 3)$. 7. $y = \ln 3x$. 8. $y = \ln (x^2 + 1)$.
 9. $y = \ln (3x^2 - x + 1)$.
 10. $y = \ln \frac{x-1}{x+1}$. 11. $y = \ln \frac{\sqrt{x}}{x+1}$.
 12. $y = x \ln x$. 13. $y = x^3 \ln (x + 1)$. 14. $y = x^x$.
 15. $y = x \sqrt{x^2 - 1}$.

3.10 TRIGONOMETRIC FUNCTIONS

In this section we will find the derivatives of trigonometric functions.

The trigonometric functions are defined as ratios of line segments and, consequently, are dimensionless. They depend on dimensionless quantities (angles).

For angles from zero to a right angle, the trigonometric functions may be defined as ratios of line segments in a right triangle (the sine of an angle is equal to the ratio of the side opposite the angle to the hypotenuse, and so forth). But we are interested in defining the functions of arbitrary angles (greater than right angles and also negative angles), and so we will consider the trigonometric functions in a circle.

The sole measure of angles used in higher mathematics is the radian. Short tables of the trigonometric functions depending on angles expressed in radians are given in the Appendix, Table VI.

So as to avoid speaking all the time of the ratio of the length of the line of sines to the radius of the circle or of the angle as a ratio of the arc length to the radius, we will consider a circle of radius unity. Then we will briefly say that the sine is equal to the length of the line of sines in that circle, the angle is equal to the arc length, etc.

The reader must bear in mind, however, that both the trigonometric functions and the angles are dimensionless and are not measured by any units of length (centimetres, inches or metres). The sine is equal to the length of the line of sines (in centimetres) divided by the length of the radius (in centimetres) and for $r = 1$ cm is numerically equal to the length of the line of sines. The lines of sines and cosines are shown in Fig. 75.

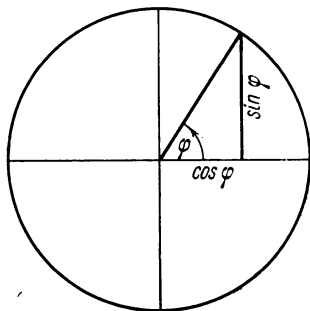


Fig. 75

Recall the form of the graphs of sine and cosine as functions of the angle (see Fig. 76). The period of the sine, like that of the cosine, is equal to $2\pi = 6.28$ and corresponds to a complete revolution of the radius of the circle.

Let us find the derivatives of the sine and cosine geometrically. In Fig. 77, the endpoint of the radius drawn at an angle φ is A ; the

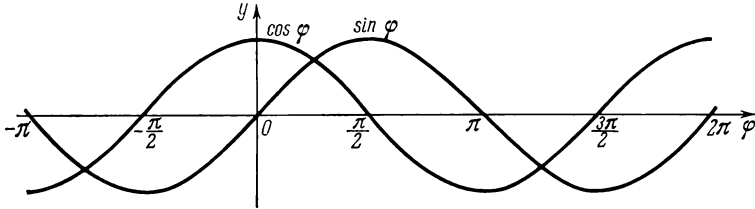


Fig. 76

endpoint of the radius drawn at the angle $\varphi + d\varphi$ is B . Thus, the length of arc AB is equal to $d\varphi$. Draw from A a perpendicular AC to the line of sines BB' of the angle $\varphi + d\varphi$. As can be seen from Fig. 77,

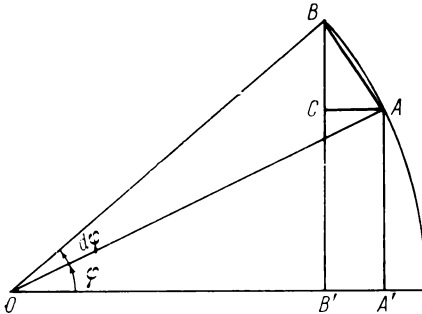


Fig. 77

$$AA' = \sin \varphi, \quad BB' = \sin (\varphi + d\varphi)$$

and

$$BC = \sin (\varphi + d\varphi) - \sin \varphi = d(\sin \varphi)$$

Furthermore

$$OA' = \cos \varphi, \quad OB' = \cos (\varphi + d\varphi)$$

and

$$A'B' = AC = \cos \varphi - \cos (\varphi + d\varphi) = -d(\cos \varphi)$$

Since the angle $d\varphi$ is small, the arc length AB does not differ from the length of the chord AB and the angle ABC formed by the chord AB and the vertical line BCB' is equal to φ .*

From a consideration of triangle ABC , we find $BC = AB \cos \varphi$, $AC = AB \sin \varphi$. Thus

$$d(\sin \varphi) = \cos \varphi d\varphi, \quad d(\cos \varphi) = -\sin \varphi d\varphi$$

* The exact value of the angle is equal to $\varphi + \frac{d\varphi}{2}$ but the triangle ABC is small ($AB = d\varphi$) and so, neglecting $d\varphi$ in the expression of the angle ABC , we have errors, in the quantities BC and AC , that are proportional to $(d\varphi)^2$.

and consequently

$$\frac{d(\sin \varphi)}{d\varphi} = \cos \varphi, \quad \frac{d(\cos \varphi)}{d\varphi} = -\sin \varphi \quad (3.10-1)$$

Here is another way of computing the derivatives of $\sin \varphi$ and $\cos \varphi$ without using a drawing. According to the general formulas, $\Delta \sin \varphi = \sin(\varphi + \Delta\varphi) - \sin \varphi$. Recall the formula for the sine of the sum of two angles:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

and apply it to $\sin(\varphi + \Delta\varphi)$ to get

$$\sin(\varphi + \Delta\varphi) = \sin \varphi \cdot \cos \Delta\varphi + \cos \varphi \cdot \sin \Delta\varphi$$

whence

$$\Delta \sin \varphi = \sin \varphi \cdot \cos \Delta\varphi + \cos \varphi \cdot \sin \Delta\varphi - \sin \varphi$$

Let us form the ratio of the increments as follows:

$$\frac{\Delta \sin \varphi}{\Delta\varphi} = \cos \varphi \cdot \frac{\sin \Delta\varphi}{\Delta\varphi} - \sin \varphi \cdot \frac{1 - \cos \Delta\varphi}{\Delta\varphi}$$

Now we have to pass to the limit as $\Delta\varphi \rightarrow 0$. We know that for angles α or $\Delta\varphi$ tending to zero, the sine is equal to the arc: $\sin \alpha = \alpha$, $\sin \Delta\varphi = \Delta\varphi$. In other words,

$$\lim_{\Delta\varphi \rightarrow 0} \frac{\sin \Delta\varphi}{\Delta\varphi} = 1$$

The second term must first of all be transformed by using the familiar formula

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha, \quad 1 - \cos 2\alpha = 2 \sin^2 \alpha,$$

$$1 - \cos \Delta\varphi = 2 \sin^2 \left(\frac{\Delta\varphi}{2} \right)$$

In this formula, we make the substitution $\sin \left(\frac{\Delta\varphi}{2} \right) = \frac{\Delta\varphi}{2}$ for small $\Delta\varphi$ to get

$$\frac{1 - \cos \Delta\varphi}{\Delta\varphi} = \frac{2 \left(\frac{\Delta\varphi}{2} \right)^2}{\Delta\varphi} = \frac{\Delta\varphi}{2}$$

Hence, in the limit, as $\Delta\varphi \rightarrow 0$, the second term vanishes:

$$\lim_{\Delta\varphi \rightarrow 0} \frac{1 - \cos \Delta\varphi}{\Delta\varphi} = 0 \quad \text{whence}$$

$$\lim_{\Delta\varphi \rightarrow 0} \frac{\Delta \sin \varphi}{\Delta\varphi} = \frac{d \sin \varphi}{d\varphi} = \cos \varphi$$

The relations (3.10-1) are valid for arbitrary angles and not only those in the first quadrant. It is also useful to verify, glancing at the graphs of the functions $\sin x$ and $\cos x$, that the formulas (3.10-1)

give the proper signs of the derivatives for any x and not only in the first quadrant.

Let us check the formulas (3.10-1) for small angles. For small φ , it is obvious geometrically that

$$\sin \varphi \approx \varphi, \quad \cos \varphi \approx 1$$

For small φ the first formula $\frac{d \sin \varphi}{d\varphi} = \cos \varphi$ yields $\frac{d \sin \varphi}{d\varphi} = 1$.

The second formula yields $\frac{d \cos \varphi}{d\varphi} = -\varphi$, $\frac{d \cos \varphi}{d\varphi} = 0$ for $\varphi = 0$.

The fact that the derivative is zero signifies that the cosine has a maximum at $\varphi = 0$.

If we know the derivatives of the functions $y = \sin x$ and $y = \cos x$, it is easy to find the derivatives of all other trigonometric functions by using the formulas interrelating them.

Thus, for example, we know that $\tan x = \frac{\sin x}{\cos x}$ and so, by the formula for differentiating a fraction, we get

$$\frac{d \tan x}{dx} = \frac{\cos^2 x - \sin x (-\sin x)}{\cos^2 x}$$

whence

$$\frac{d \tan x}{dx} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} \quad (3.10-2)$$

From Fig. 78 (the graph of $\tan x$) we can see that the function $y = \tan x$ has a positive derivative for arbitrary x . Near the points of

discontinuity ($x = \frac{\pi}{2}, x = \frac{3\pi}{2}, \dots$) the derivative increases without bound. Both of these conclusions are in full agreement with formula (3.10-2).

By a completely analogous device we find

$$\frac{d (\cot x)}{dx} = -\frac{1}{\sin^2 x}$$

The derivatives of a tangent and a cotangent can also be found directly. Note that

$$\tan \alpha - \tan \beta = \frac{\sin \alpha}{\cos \alpha} - \frac{\sin \beta}{\cos \beta} = \frac{\sin \alpha \cos \beta - \sin \beta \cos \alpha}{\cos \alpha \cos \beta} = \frac{\sin (\alpha - \beta)}{\cos \alpha \cdot \cos \beta}$$

whence

$$\Delta \tan \varphi = \tan (\varphi + \Delta \varphi) - \tan \varphi = \frac{\sin \Delta \varphi}{\cos (\varphi + \Delta \varphi) \cos \varphi} \quad (3.10-3)$$

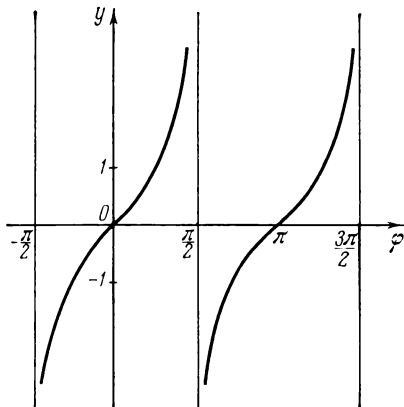


Fig. 78

Bearing in mind (see page 129) that

$$\lim_{\Delta\varphi \rightarrow 0} \frac{\sin \Delta\varphi}{\Delta\varphi} = 1$$

we get, from (3.10-3),

$$\frac{d(\tan \varphi)}{d\varphi} = \lim_{\Delta\varphi \rightarrow 0} \frac{\Delta \tan \varphi}{\Delta\varphi} = \lim_{\Delta\varphi \rightarrow 0} \frac{\sin \Delta\varphi}{\Delta\varphi} \cdot \lim_{\Delta\varphi \rightarrow 0} \frac{1}{\cos(\varphi + \Delta\varphi) \cdot \cos \varphi} = \frac{1}{\cos^2 \varphi}$$

Exercises

Find the derivatives of the following functions.

1. $y = \sin(2x + 3)$. 2. $y = \cos(x - 1)$. 3. $y = \cos(x^2 - x + 1)$. 4. $y = \sin^2 x$. 5. $y = \sin 3x \cos^2 x$. 6. $y = (\sin 2x)^x$.

7. $y = x \tan x$. 8. $y = e^{\tan 2x}$. 9. $y = \cot \frac{x}{2}$.

3.11 INVERSE TRIGONOMETRIC FUNCTIONS

New and very interesting results are obtained when we consider the inverse trigonometric functions. We remind the reader of how these functions are defined. The function

$$y = \text{Arcsin } x \quad (3.11-1)$$

is an angle such that

$$\sin y = x \quad (3.11-2)$$

These two equations denote the same thing. Similarly, the function

$$y = \text{Arctan } x$$

denotes an angle y such that

$$\tan y = x$$

The definitions are similar for the functions $y = \text{Arccos } x$ ($x = \cos y$) and $y = \text{Arccot } x$ ($x = \cot y$). Note that the function $y = \text{Arcsin } x$ is meaningful only for values of x that satisfy the inequality $-1 \leq x \leq 1$, as is seen from (3.11-2). The function $y = \text{Arctan } x$ is meaningful for all values of x .

Let us consider in more detail the function $y = \text{Arcsin } x$. For instance, let $x = \frac{1}{2}$, $y = \text{Arcsin } \frac{1}{2}$. We can take $y = \frac{\pi}{6}$ since $\sin \frac{\pi}{6} = \frac{1}{2}$, however we can also take $y = \frac{5\pi}{6}$, since $\sin \frac{5\pi}{6}$ is also equal to $\frac{1}{2}$. We can likewise take $y = \frac{13\pi}{6}$, $y = \frac{17\pi}{6}$ and so on. We see that one value of x is associated with an infinitude of values of y . All

these properties of the function $y = \text{Arcsin } x$ are seen in the graph of Fig. 79.

Take the portion of the curve for which $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. This part of the curve is called the principal value of the function $y = \text{Arcsin } x$ and is denoted by $y = \arcsin x$ (the "a" in arcsin is lower-case). If we confine ourselves to a consideration of $y = \arcsin x$, then to each x there corresponds only one value of y . The principal value of the arctangent function is defined in similar fashion:

$$-\frac{\pi}{2} \leq \arctan x \leq \frac{\pi}{2}$$

Find the derivative of the function $y = \arcsin x$. We take advantage of the fact that the arcsine is the inverse function of the sine:

$$y = \arcsin x, \quad x = \sin y,$$

$$x'(y) = \frac{dx}{dy} = \cos y,$$

$$y'(x) = \frac{dy}{dx} = \frac{1}{x'(y)} = \frac{1}{\cos y} \quad (3.11-3)$$

But we consider x the argument and so $\frac{dy}{dx}$ should be expressed in terms of x and not in terms of y , as in (3.11-3). We take advantage of the familiar formula $\sin^2 y + \cos^2 y = 1$, whence $\cos y = \pm \sqrt{1 - \sin^2 y}$. Since we are considering the principal value of the arcsine function, it follows that $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, $\cos y \geq 0$ and so we take

the plus sign in front of the radical: $\cos y = \sqrt{1 - \sin^2 y}$. Since $\sin y = x$ by (3.11-2), it follows that $\cos y = \sqrt{1 - x^2}$. Substituting this into (3.11-3) we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}$$

or

$$\frac{d(\arcsin x)}{dx} = \frac{1}{\sqrt{1 - x^2}} \quad (3.11-4)$$

Formula (3.11-4) may be used not only for the principal value but also for other portions of the curve if we choose the appropriate sign in front of the radical. Indeed, for one and the same value of x , the derivative has different signs on various portions of the curve:

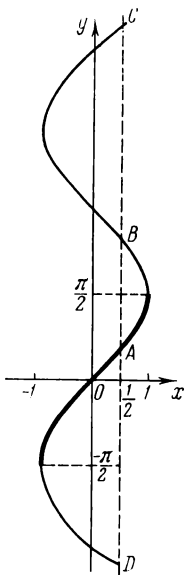


Fig. 79

at points A and C (Fig. 79) the derivative is positive and at points B and D it is negative.

Let us now find the derivative $\frac{d(\arctan x)}{dx}$. If $y = \arctan x$, then $x = \tan y$, whence, by the foregoing, we find

$$\begin{aligned} x'(y) &= \frac{dx}{dy} = \frac{1}{\cos^2 y}, \\ y'(x) &= \frac{dy}{dx} = \frac{1}{x'(y)} = \cos^2 y \end{aligned} \quad (3.11-5)$$

From trigonometry we have

$$\tan^2 y + 1 = \frac{1}{\cos^2 y}$$

therefore

$$\frac{1}{\cos^2 y} = 1 + x^2$$

Using the relation (3.11-5) we finally get

$$\frac{dy}{dx} = \frac{d(\arctan x)}{dx} = \frac{1}{1+x^2} \quad (3.11-6)$$

The formula (3.11-6) holds true for any other branch of the arctangent (Fig. 80) since any other branch is obtained from the principal one by a parallel translation, and this does not affect the magnitude of the derivative.

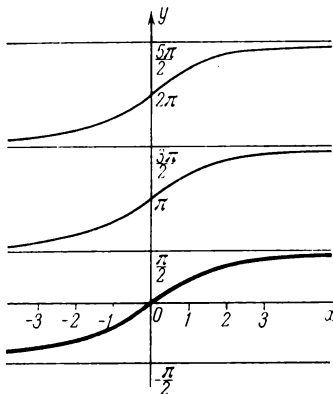


Fig. 80

Exercises

1. Find the derivatives of the functions $y = \arccos x$ and $y = \operatorname{arccot} x$.
2. Knowing that $\frac{d(e^x)}{dx} = e^x$ find $\frac{d(\ln x)}{dx}$ using the fact that from the equation $y = \ln x$ follows $x = e^y$.
- Find the derivatives of the following functions.
3. $y = \arcsin 2x$.
4. $y = \arctan(3x + 1)$.
5. $y = \arctan(x^2 - x)$.
6. $y = e^{\arctan \sqrt{x}}$.

3.12 THE DERIVATIVE OF AN IMPLICIT FUNCTION

To define a function implicitly means to define it by means of an expression of the form

$$F(x, y) = 0 \quad (3.12-1)$$

If the equation can be solved for x or y , then we revert to the ordinary representation of the function. But sometimes such a solution leads to complicated formulas and at other times it cannot be found at all. For instance, the equation of a circle in the form

$$x^2 + y^2 - 1 = 0 \quad (3.12-2)$$

is simpler than the following expression derived from it:

$$y = \pm \sqrt{1 - x^2} \quad (3.12-3)$$

If the left-hand member of (3.12-1) is an arbitrary polynomial involving x and y to a power exceeding the fourth, then in the general case this equation cannot be solved for x or for y . Also, for example, unsolvable is the simple-looking equation

$$F(x, y) = x \sin x + y \sin y - \pi = 0 \quad (3.12-4)$$

However, even in those cases where there is no solution in the form of a formula that specifies directly a procedure for computing y for a given x , it still remains a fact that y is a definite function of x . For every x it is possible, by solving the equation numerically, to find a corresponding y and to construct a curve in the xy -plane. It may be that the curve will not exist for all x (in the case of a circle, for example, it exists only for x between $-r$ and $+r$, where r is the radius of the circle) and for a given x there may be more than one value of y (in the case of the circle, for instance, there are two values to accord with the $\pm$ sign in front of the square root sign). However, these complications do not detract from the basic fact, which is that the equation $F(x, y) = 0$ defines y as a function of x .

How can we find the derivative $\frac{dy}{dx}$? And can this be done without solving the equation, that is, without expressing $y(x)$ in explicit fashion?

This was done by Newton. Let x, y satisfy the equation

$$F(x, y) = 0 \quad (3.12-1)$$

Let us take adjacent values $x + \Delta x, y + \Delta y$, which also satisfy the equation:

$$F(x + \Delta x, y + \Delta y) = 0 \quad (3.12-5)$$

Then, using (3.12-1), we write

$$F(x + \Delta x, y + \Delta y) = F(x + \Delta x, y + \Delta y) - F(x + \Delta x, y) + F(x + \Delta x, y) - F(x, y) \quad (3.12-6)$$

The difference $F(x + \Delta x, y) - F(x, y)$ is the increment of the function $F(x, y)$ regarded as a function of the variable x alone, with y held constant. This increment, as we know, can, in the limit,* be expressed as follows:

$$F(x + \Delta x, y) - F(x, y) = \frac{dF(x, y)}{dx} \Big|_{y=\text{const}} \cdot \Delta x$$

* The expression "is equal in the limit" for small Δx or Δy is explained in detail in Sec. 2.4, where we considered the expression for the increment of a function with the aid of a derivative.

We note here that when computing the derivative with respect to x of a function of two variables x and y , we consider y to be constant. The derivative thus evaluated is called the partial derivative and in place of the letter d we write the 'mirror 6' ∂ :

$$F(x + \Delta x, y) - F(x, y) = \frac{\partial F(x, y)}{\partial x} \Delta x,$$

$$\frac{\partial F(x, y)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{F(x + \Delta x, y) - F(x, y)}{\Delta x}$$

Similarly, for the first difference in (3.12-6) we can write

$$F(x + \Delta x, y + \Delta y) - F(x + \Delta x, y) = \frac{\partial F(x + \Delta x, y)}{\partial y} \Delta y$$

The condition (3.12-5) yields

$$\frac{\partial F(x, y)}{\partial x} \cdot \Delta x + \frac{\partial F(x + \Delta x, y)}{\partial y} \cdot \Delta y = 0$$

or

$$\frac{\Delta y}{\Delta x} = - \frac{\frac{\partial F(x, y)}{\partial x}}{\frac{\partial F(x + \Delta x, y)}{\partial y}}$$

Passing to the limit as $\Delta x \rightarrow 0$, we get the derivative on the left; on the right we can discard Δx . Finally we have

$$\frac{dy}{dx} = - \frac{\frac{\partial F(x, y)}{\partial x}}{\frac{\partial F(x, y)}{\partial y}} \quad (3.12-7)$$

Note the minus sign in (3.12-7) and also the fact that we cannot simply cancel the $\partial F(x, y)$ in the numerator and the denominator.

We will demonstrate the application of (3.12-7) using as an example the equation (3.12-2). We have $F(x, y) = x^2 + y^2 - 1$;

$$\frac{\partial F(x, y)}{\partial x} = 2x, \quad \frac{\partial F(x, y)}{\partial y} = 2y,$$

$$\frac{dy}{dx} = - \frac{2x}{2y} = - \frac{x}{y} \quad (3.12-8)$$

It is easy to see that this result coincides with that obtained if we compute the derivative of (3.12-3).

Let us find the derivative in the case of (3.12-4):

$$\frac{\partial F(x, y)}{\partial x} = \sin x + x \cos x, \quad \frac{\partial F(x, y)}{\partial y} = \sin y + y \cos y,$$

$$\frac{dy}{dx} = - \frac{\sin x + x \cos x}{\sin y + y \cos y}$$

Thus, the expression of the derivative of an implicit function involves both quantities, x and y . To find it numerically, we have to find y numerically for a given x . But if we did not have formula (3.12-7), then to find the derivative would require finding, numerically, two values y_2 and y_1 for two adjacent values x_2 and x_1 and finding the ratio $\frac{y_2 - y_1}{x_2 - x_1}$. Here, the closer x_2 and x_1 are, the more exactly we would need to compute y_2 and y_1 , but this is often very difficult to do.

Finally note that if $F(x, y) = 0$ leads to an ambiguous curve, that is, when there are two or more values of y for one value of x (several branches of the curve), then (3.12-7) yields the values of the derivative at appropriate points for a given x when different y are substituted. The reader is advised to verify this using the equation of the circle (3.12-2) for which the derivative is given by formula (3.12-8).

In finding the derivative of an implicit function we had to introduce a new concept, that of the partial derivative. This notion is of great importance and is necessary when considering functions of several variables (we do not study them in this book). Actually, we have already, latently, made use of the concept of a partial derivative even in such elementary questions as the derivative of a product of many functions $y = h(x) g(x)$ or, say, the derivative of a power, $y = h(x)^{g(x)}$ (see pages 112 and 126) when we said that y' is composed of a term obtained when taking the derivative with respect to x in $h(x)$ and with respect to x in the expression $g(x)$. Using partial derivatives, we can write this rule as follows: if

$$y = F[g(x), h(x)]$$

then

$$y' = \frac{dy}{dx} = \frac{\partial F}{\partial g} \cdot \frac{dg}{dx} + \frac{\partial F}{\partial h} \cdot \frac{dh}{dx}$$

Exercises

1. Find the derivative $\frac{dy}{dx}$ at the point $x = \frac{\pi}{2}$, $y = \frac{\pi}{2}$ of a function defined by the equation (3.12-4). The same at the point $x = -\frac{\pi}{2}$, $y = \frac{\pi}{2}$.
2. Find the derivative $\frac{dy}{dx}$ at the point $x = y = 1$ of a function defined by the equation $x^3 + 3x + y^3 + 3y - 8 = 0$.

3.13 INTEGRALS. STATEMENT OF THE PROBLEM

In Chapter 2 we introduced the concept of the integral and noted the close connection between two different (at first glance) problems. These problems are:

(1) finding the sum of a large number of small summands when the terms can be represented as $v(t) dt$;

(2) finding the function $z(t)$, the derivative of which is equal to the given function $v(t)$:

$$\frac{dz}{dt} = v(t)$$

Before going on, the reader is advised to reread Secs. 2.7 to 2.12.

Most of the problems that arise in physics, mathematics and chemistry are problems involving the computation of a sum. This statement of the problem is more pictorial. The problem itself suggests a simple, though approximate, way to compute the quantity of interest. This approach does not yield any general formulas however.

The second statement of the problem is more artificial, but it has its advantages. The finding of derivatives proved to be a very simple matter that reduced to four or five formulas (the derivatives of x^n , e^x , $\ln x$, $\sin x$, $\cos x$) and to two or three rules. It is therefore easy to find the derivatives of a large number of functions. Every time the derivative $\frac{dz}{dt} = v$ of some function is found, we can record the fact that for this v the integral z is known (see Sec. 3.14). In this way we can build up a range of particular cases in which it is possible to solve the problem of finding the integral. For certain simple types of functions v , it has been possible, with the aid of identity transformations, to find the rules for evaluating integrals (see Sec. 3.15).

However, this is not possible to do for all the elementary functions so that integration is more difficult than differentiation (finding derivatives). Nevertheless, the formulas found for certain integrals in the second statement of the problem are very important. If for a given v it is possible to find the integral (the indefinite integral or the antiderivative), then all problems in the first statement of the

problem, all sums, that is, all definite integrals $\int_a^b v(t) dt$ are then

expressed by simple formulas via a function z : $\int_a^b v(t) dt = z(b) - z(a)$. Such a result is more complete, more exact and more valuable than the result of every separate numerical computation of a sum,

that is, of the definite integral $\int_a^b v(t) dt$ between definite limits a and b . For this reason we aim primarily to solve the problem in its second statement.

3.14 ELEMENTARY INTEGRALS

Let us write down the formulas for the derivatives that have been found in the preceding sections and their corresponding integrals:

$$\begin{array}{ll}
 \frac{d}{dx} (x^n) = nx^{n-1}, & n \int x^{n-1} dx = x^n + C; \\
 \frac{d}{dx} (e^{kx}) = ke^{kx}, & k \int e^{kx} dx = e^{kx} + C; \\
 \frac{d}{dx} (\ln x) = \frac{1}{x}, & \int \frac{1}{x} dx = \ln x + C; \\
 \frac{d}{dx} (\sin kx) = k \cos kx, & k \int \cos kx dx = \sin kx + C; \\
 \frac{d}{dx} (\cos kx) = -k \sin kx, & -k \int \sin kx dx = \cos kx + C; \\
 \frac{d}{dx} (\tan x) = \frac{1}{\cos^2 x}, & \int \frac{1}{\cos^2 x} dx = \tan x + C; \\
 \frac{d}{dx} (\cot x) = -\frac{1}{\sin^2 x}, & -\int \frac{1}{\sin^2 x} dx = \cot x + C; \\
 \frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}, & \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C; \\
 \frac{d}{dx} (\arctan x) = \frac{1}{1+x^2}, & \int \frac{1}{1+x^2} dx = \arctan x + C
 \end{array}$$

Let us perform a few manipulations. In the first integral, we denote $n - 1 = m$ (then $n = m + 1$), and we can rewrite it as

$$\int x^m dx = \frac{1}{m+1} x^{m+1} + C$$

It is clear that the formula is valid for all m except $m = -1$; for $m = -1$ the denominator vanishes, $x^{m+1} = x^0 = 1$, and we have an expression unsuitable for computations, $\frac{1}{0} + C$. However, it is precisely in the case where $m = -1$, i.e., for $\int \frac{1}{x} dx$, that we have the formula

$$\int \frac{1}{x} dx = \ln x$$

This formula holds true only for positive values of x , since $\ln x$ is meaningful only for $x > 0$. For $x < 0$, $\ln x$ is meaningless, but $\ln (-x)$ is meaningful. Since

$$\frac{d \ln (-x)}{dx} = \frac{1}{-x} (-1) = \frac{1}{x}$$

it follows that $\int \frac{dx}{x} = \ln (-x) + C$ if $x < 0$. Both formulas for

$\int \frac{dx}{x}$ can be combined into one:

$$\int \frac{dx}{x} = \ln |x| + C \quad (3.14-1)$$

This formula may be used for any interval of integration that does not contain $x = 0$.

The integral of the exponential function looks like this:

$$\int e^{kx} dx = \frac{1}{k} e^{kx} + C$$

Similarly, for the sine and the cosine, we get

$$\begin{aligned} \int \sin kx dx &= -\frac{1}{k} \cos kx + C, \\ \int \cos kx dx &= \frac{1}{k} \sin kx + C \end{aligned}$$

3.15 GENERAL PROPERTIES OF INTEGRALS

In Secs. 3.1 to 3.3 we established the properties of the derivative of a sum of functions, the derivative of a composite function and the derivative of a product of functions. To each of these properties corresponds a definite property referring to integrals.

For integrals we have the equation

$$\int [Cf(x) + Eg(x)] dx = C \int f(x) dx + E \int g(x) dx \quad (3.15-1)$$

To prove this we have to take the derivative of the expression on the right. If the equation is a true statement, then we get the integrand function. Differentiating we have

$$\begin{aligned} \left[C \int f(x) dx + E \int g(x) dx \right]' &= C \left[\int f(x) dx \right]' \\ &+ E \left[\int g(x) dx \right]' = Cf(x) + Eg(x) \end{aligned}$$

and the proof of equation (3.15-1) is complete. It shows that the integral of a sum of several terms splits up into the sum of the integrals of the separate summands, and any constant factors can be taken outside the integral sign.

It is possible, under the integral sign, to make a change of variable and pass to a new and more convenient variable. Let us examine a number of simple examples.

1. Find $\int (ax + b)^n dx$ ($n \neq -1$).

For the new variable, call it z , we take the expression in brackets:

$$ax + b = z \quad (3.15-2)$$

In so doing, we also have to pass from the differential dx to the differential dz .

From (3.15-2) we get

$$dz = a \, dx, \quad dx = \frac{dz}{a}$$

Thus

$$\int (ax + b)^n \, dx = \int z^n \frac{dz}{a} = \frac{1}{a} \int z^n \, dz = \frac{z^{n+1}}{a(n+1)} + C = \frac{(ax + b)^{n+1}}{a(n+1)} + C$$

The correctness of this result is easily seen if we compute the derivative of the right member:

$$\begin{aligned} \frac{d}{dx} \left[\frac{(ax + b)^{n+1}}{a(n+1)} + C \right] &= \frac{d}{dx} \left[\frac{(ax + b)^{n+1}}{a(n+1)} \right] \\ &= \frac{n+1}{a(n+1)} (ax + b)^n \frac{d}{dx} (ax + b) = \frac{(ax + b)^n}{a} a = (ax + b)^n \end{aligned}$$

2. In similar fashion, in the integral $\int \frac{dx}{ax+b}$ we can make the change of variable

$$z = ax + b, \quad dz = a \, dx, \quad dx = \frac{dz}{a},$$

$$\int \frac{dx}{ax+b} = \int \frac{dz}{a \cdot z} = \frac{1}{a} \int \frac{dz}{z} = \frac{1}{a} \ln z + C = \frac{\ln(ax+b)}{a} + C$$

When dealing with such simple examples in practical situations, the transformations are ordinarily carried out without introducing separate designations for the new intermediate variables. For example one writes

$$\int (ax + b)^n \, dx = \int (ax + b)^n \frac{1}{a} d(ax + b) = \frac{1}{(n+1)a} (ax + b)^{n+1} + C$$

Let $f(x)$ and $g(x)$ be two distinct functions of the variable x . The rule for finding the derivative of a product yields

$$\frac{d}{dx} (fg) = g \frac{df}{dx} + f \frac{dg}{dx} \quad (3.15-3)$$

The equation (3.15-3) enables us to write

$$fg = \int f \frac{dg}{dx} \, dx + \int g \frac{df}{dx} \, dx \quad (3.15-4)$$

We can see the validity of (3.15-4) by taking the derivative of the left and right members to get the true equation (3.15-3).

Let us rewrite (3.15-4) as

$$\int f \frac{dg}{dx} \, dx = fg - \int g \frac{df}{dx} \, dx$$

This is usually compactly written as

$$\int f \, dg = fg - \int g \, df \quad (3.15-5)$$

What is the meaning of formula (3.15-5)? When evaluating an integral, there is no rule that expresses the integral of a product of two functions in terms of the integrals of each of the factors. However, if in the product of two functions fw the integral of one of the factors is known,

$$\int w \, dx = g, \quad w = \frac{dg}{dx}$$

then it becomes possible to express the integral $\int fw \, dx$ in terms of the integral involving the derivative $\frac{df}{dx}$. Using w we rewrite (3.15-5) as

$$\int fw \, dx = f \left(\int w \, dx \right) - \int \left(\int w \, dx \right) \frac{df}{dx} \, dx \quad (3.15-6)$$

Since $\int w \, dx = g$, it follows that the last integral in (3.15-6) is $\int g \frac{df}{dx} \, dx$. Sometimes it is simpler than the original integral $\int fw \, dx$ or reduces to a known integral. In particular, if f is a power function, then $\frac{df}{dx}$ has the power of f minus unity. The formula (3.15-5) or (3.15-6) is called the formula for *integration by parts*.

Here are some examples.

1. Find $\int xe^x \, dx$.

Put $f = x$; then $w = \frac{dg}{dx} = e^x$, $e^x \, dx = dg$, $g = \int e^x \, dx = e^x$, $df = dx$. By formula (3.15-5)

$$\int xe^x \, dx = xe^x - \int e^x \, dx = xe^x - e^x = e^x (x - 1) + C$$

2. Find $\int x^2 e^x \, dx$.

Set $f = x^2$, then $w = \frac{dg}{dx} = e^x$, $e^x \, dx = dg$, $g = \int e^x \, dx = e^x$, $df = 2x \, dx$. Using (3.15-5) we get $\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx$. Using the result of the first example, we obtain

$$\int x^2 e^x \, dx = x^2 e^x - 2xe^x + 2e^x + C = (x^2 - 2x + 2)e^x + C$$

To find $\int P_n(x) e^{hx} \, dx$, where $P_n(x)$ is a polynomial of degree n , we have to perform integration by parts n times. We then get $Q_n(x) e^{hx}$, where $Q_n(x)$ is a polynomial of degree n . Knowing this, we need not perform integration by parts n times, but can write down directly the coefficients of the polynomial $Q_n(x)$.

Let us take the same example. Find $\int x^2 e^x dx$. We write the equation with the (still) unknown coefficients of the polynomial $Q_n(x)$:

$$\int x^2 e^x dx = (a_2 x^2 + a_1 x + a_0) e^x + C \quad (3.15-7)$$

Taking the derivatives of both sides of (3.15-7), we get

$$x^2 e^x = (2a_2 x + a_1) e^x + (a_2 x^2 + a_1 x + a_0) e^x,$$

$$x^2 e^x = [x^2 a_2 + x(2a_2 + a_1) + (a_1 + a_0)] e^x$$

Equate the coefficients of identical powers of x in the polynomials on the right and left to get

$$a_2 = 1,$$

$$2a_2 + a_1 = 0 \text{ whence } a_1 = -2,$$

$$a_1 + a_0 = 0 \text{ whence } a_0 = 2$$

Finally, as before, we get

$$\int x^2 e^x dx = (x^2 - 2x + 2) e^x + C$$

By a similar technique we can find the integrals of the functions $P_n(x) \cos kx$ and $P_n(x) \sin kx$, where $P_n(x)$ is a polynomial. In both cases the answer is of the form

$$Q_n(x) \cos kx + R_n(x) \sin kx$$

where $Q_n(x)$ and $R_n(x)$ are polynomials of degree n (or less than n). Examples of this kind are given in the exercises.

The following is an instance of an integral that can be reduced to familiar integrals by means of algebraic manipulations.

We consider the integral $\int \frac{dx}{(x-a)(x-b)}$. Note that the identity

$$\frac{1}{x-a} - \frac{1}{x-b} = \frac{a-b}{(x-a)(x-b)}$$

holds true. Using this identity we get

$$\frac{1}{(x-a)(x-b)} = \frac{1}{a-b} \left[\frac{1}{x-a} - \frac{1}{x-b} \right]$$

Therefore

$$\begin{aligned} \int \frac{dx}{(x-a)(x-b)} &= \frac{1}{a-b} \int \left(\frac{1}{x-a} - \frac{1}{x-b} \right) dx \\ &= \frac{1}{a-b} [\ln(x-a) - \ln(x-b)] + C = \frac{1}{a-b} \ln \frac{x-a}{x-b} + C \end{aligned}$$

There are techniques which permit one to express in terms of elementary functions the integral of any algebraic fraction with integral powers of the variable. However, the results involve not only

algebraic functions but also logarithms and inverse trigonometric functions (arctangents). The general theory for finding such integrals is much too complicated for this book.

The integration of many functions involving radicals and trigonometric functions may be reduced, by means of an appropriate change of variable, to the integration of polynomials or algebraic fractions with integral powers. Let us consider an example.

Find $\int x \sqrt{x+1} dx$. We make a change of variable: $z = \sqrt{x+1}$, $x+1 = z^2$, whence $2z dz = dx$. Passing to the new variable in the integral, we obtain

$$\begin{aligned} \int x \sqrt{x+1} dx &= \int (z^2 - 1) z 2z dz = 2 \int (z^4 - z^2) dz \\ &= 2 \frac{z^5}{5} - 2 \frac{z^3}{3} + C = \frac{2}{5} \sqrt{(x+1)^5} - \frac{2}{3} \sqrt{(x+1)^3} + C \end{aligned}$$

A few more examples of this kind are given in the exercises.

Finally, we give an example of an integral which *cannot* be represented in terms of a finite number of elementary functions:

$$f(x) = \int e^{-x^2} dx$$

The proof that it cannot be expressed in terms of a finite number of elementary functions is extremely complicated and we will not give it here.

This integral is a function whose properties may be studied. From the definition of $f(x)$ it follows that

$$\frac{df(x)}{dx} = e^{-x^2}$$

Since $e^{-x^2} > 0$ for arbitrary x , it follows that $f(x)$ is an increasing function. The derivative has a maximum at $x = 0$; hence, at $x = 0$, $f(x)$ has a maximum angle of the tangent line with the x -axis. For large absolute values of x (positive or negative), the derivative $\frac{df}{dx}$ is very small. This means that the function is almost constant. The

graph of the function $f(x) = \int_0^x e^{-x^2} dx$ is shown in Fig. 81 (for the

sake of definiteness, the lower limit has been chosen equal to zero).

Extensive tables have been compiled for this function and so computations involving this integral are no more complicated than, say, those involving trigonometric functions.

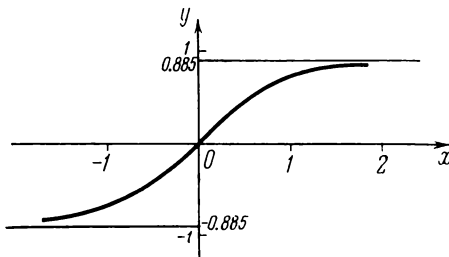


Fig. 81

Exercises

Find the following integrals.

1. $\int x(x-1)^2 dx$. 2. $\int \frac{x^2+2x-3}{x} dx$. 3. $\int \cos(3x-5) dx$. 4. $\int \sin(2x+1) dx$.
 5. $\int \sqrt{3x-2} dx$.

Hint. In Examples 3, 4, 5 make a change of variable.

6. $\int x \cos x dx$. 7. $\int \ln x dx$.

Hint. In Examples 6 and 7 take advantage of the formula for integration by parts.

8. $\int x^2 \sin 2x dx$. 9. $\int x^3 e^{-x} dx$. 10. $\int (x^2 + x + 1) \cos x dx$.
 11. $\int (2x^2 + 1) \cos 3x dx$.

Hint. Example 11 is considered in detail in the "Answers and Solutions". The other three are handled similarly.

12. $\int \frac{x dx}{(x-2)(x-3)}$

Hint. Take advantage of the identity

$$\frac{x}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$$

the numbers A and B are found by equating the coefficients of identical powers of x after clearing of fractions.

13. $\int \frac{x+1}{x^2-3x+2} dx$. 14. $\int \frac{dx}{(x+1)(x-2)}$. 15. $\int \frac{x dx}{x + \sqrt{x}}$

Hint. Make the substitution $\sqrt{x} = z$.

16. $\int \frac{x dx}{\sqrt{x^2-5}}$.

Hint. Make the change of variable $x^2 - 5 = z$.

17. $\int \sin^3 x \cos x dx$.

Hint. Make the change of variable $\cos x = z$.

18. $\int \frac{\cos^3 x}{\sin^4 x} dx$.

Hint. Make the change of variable $\sin x = z$.

19. $\int \tan x dx$. 20. $\int \frac{dx}{x^2+a^2}$.

Hint. Make the change of variable $x = at$.

21. $\int \frac{dx}{\sqrt{a^2-x^2}}$. 22. $\int \arcsin x dx$. 23. $\int \arctan x dx$.

24. $\int e^{2x} \sin 3x dx$. 25. $\int e^x \cos 2x dx$.

Hint. In Examples 22 to 25 use integration by parts.

General remark. When using various techniques, one sometimes obtains distinct expressions for one and the same integral. This should not dismay the reader. If the computations are correct, such expressions should differ by a constant only. The results are identical when evaluating a definite integral.

Verify this remark using Example 17 and making the change of variable $\sin x = z$.

3.16 CHANGE OF THE VARIABLE IN A DEFINITE INTEGRAL

We consider an example. Let it be required to calculate

$$\int_n^k (ax + b)^2 dx$$

We can do as follows: first evaluate the indefinite integral $\int (ax + b)^2 dx$ and then form the difference of its values for $x = k$ and for $x = n$.

To compute $\int (ax + b)^2 dx$ we make a change of variable using the formula $z = ax + b$. Then $dz = a dx$ and

$$\int (ax + b)^2 dx = \frac{1}{a} \int z^2 dz = \frac{z^3}{3a} = \frac{(ax + b)^3}{3a}$$

Therefore

$$\int_n^k (ax + b)^2 dx = \left. \frac{(ax + b)^3}{3a} \right|_n^k = \frac{(ak + b)^3 - (an + b)^3}{3a}$$

However, it is possible to do otherwise. Let us determine how z will vary when x varies from n to k . Since z and x are connected by the formula $z = ax + b$, it follows that as x varies from n to k , z will vary from $an + b$ to $ak + b$. Hence

$$\int_n^k (ax + b)^2 dx = \frac{1}{a} \int_{an+b}^{ak+b} z^2 dz = \frac{z^3}{3a} \Big|_{an+b}^{ak+b} = \frac{(ak + b)^3 - (an + b)^3}{3a}$$

When evaluating integrals it is convenient to do just that, i.e., when making a change of variable, to find the new limits of integration at the same time. This will obviate a return to the old variable in the expression of the indefinite integral.

Let us consider some examples.

1. Calculate $\int_0^1 \frac{dx}{(2-x)^3}$. Note from the start that the function $\frac{1}{(2-x)^3}$ assumes positive values as x varies from 0 to 1, therefore $\int_0^1 \frac{dx}{(2-x)^3} > 0$. At the same time, the denominator in this interval

does not vanish, so that the integrand is finite throughout the interval. Make the change of variable $2 - x = y$, $dx = -dy$. Then for $x = 0$, $y = 2$ and for $x = 1$, $y = 1$ and

$$\int_0^1 \frac{dx}{(2-x)^3} = - \int_2^1 \frac{dy}{y^3} \quad (3.16-1)$$

In the right member of (3.16-1) the limits of integration are given for y . The reader may wonder about the minus sign in the last equation. Indeed, on the right and left we have integrals of positive functions so why is the right side of (3.16-1) positive? The point is that the lower limit of the integral on the right is greater than the upper limit. Since an integral changes sign upon interchanging the limits of integration, (3.16-1) may be rewritten as follows:

$$\int_0^1 \frac{dx}{(2-x)^3} = \int_1^2 \frac{dy}{y^3}$$

Now, in the right-hand integral, the upper limit exceeds the lower one and it is clear that the integral on the right is positive. The computations can now be completed with ease:

$$\int_1^2 \frac{dy}{y^3} = -\frac{1}{2y^2} \Big|_1^2 = -\frac{1}{8} + \frac{1}{2} = \frac{3}{8}$$

2. In Sec. 3.15 we considered the function $f(x) = \int_0^x e^{-x^2} dx$. One

often has to deal with the function $\varphi(a) = \int_0^a e^{-kx^2} dx$, where k is a constant. We will show that there is a simple relationship between the functions φ and f .

In the expression for $\varphi(a)$ make the change of variable $kx^2 = t^2$. From this we find $\sqrt{k}x = t$, $x = \frac{t}{\sqrt{k}}$, $dx = \frac{1}{\sqrt{k}} dt$. For $x = 0$, $t = 0$, for $x = a$, $t = a\sqrt{k}$. And so we get

$$\varphi(a) = \int_0^{a\sqrt{k}} e^{-t^2} \frac{dt}{\sqrt{k}} = \frac{1}{\sqrt{k}} \int_0^{a\sqrt{k}} e^{-t^2} dt = \frac{1}{\sqrt{k}} f(a\sqrt{k})$$

Thus $\varphi(a) = \frac{1}{\sqrt{k}} f(a\sqrt{k})$. Consequently, for an arbitrary value of the independent variable x ,

$$\varphi(x) = \frac{1}{\sqrt{k}} f(x\sqrt{k})$$

If we have a table of the function $f(x)$, it is possible to find the integral $\varphi(x)$ for any value of k .

3. In Chapter 2 we saw that the definite integral has dimensions if the integrand and the limits of integration have. It is often convenient however to reduce the integral to a dimensionless form by taking all factors having dimensions outside the integral sign. This can be done in the following manner.

Suppose we have $\int_a^b f(x) dx$. Denote by f_{max} the greatest value of the function $f(x)$ on the interval of integration:

$$\int_a^b f(x) dx = \int_a^b \frac{f(x)}{f_{max}} f_{max} dx = f_{max} \int_a^b \frac{f(x)}{f_{max}} dx \quad (3.16-2)$$

It is clear that in the last integral the integrand $\frac{f(x)}{f_{max}}$ is dimensionless since $f(x)$ and f_{max} have the same dimensions. Let us pass to dimensionless limits of integration. To do this, make the change of variable

$$z = \frac{x-a}{b-a} \quad \text{or} \quad x = a + z(b-a) \quad (3.16-3)$$

From (3.16-3) it is clear that z is a dimensionless quantity. Since $dx = (b-a) dz$ and $z = 0$ for $x = a$, $z = 1$ for $x = b$, it follows that the integral in (3.16-2) takes the form

$$\int_a^b \frac{f(x)}{f_{max}} dx = (b-a) \int_0^1 \frac{f[a+z(b-a)]}{f_{max}} dz \quad (3.16-4)$$

Set

$$\frac{f[a+z(b-a)]}{f_{max}} = \varphi(z)$$

then from (3.16-4) we have

$$\int_a^b \frac{f(x)}{f_{max}} dx = (b-a) \int_0^1 \varphi(z) dz$$

and, finally,

$$\int_a^b f(x) dx = f_{max} \cdot (b-a) \int_0^1 \varphi(z) dz \quad (3.16-5)$$

In the formula (3.16-5), $\int_0^1 \varphi(z) dz$ is a dimensionless quantity.

If $f(x)$ varies but slightly on the interval of integration, then $\frac{f(x)}{f_{\max}} \approx 1$, and so

$$\varphi(z) \approx 1 \quad \text{and} \quad \int_0^1 \varphi(z) dz \approx 1 \cdot \int_0^1 dz = 1$$

Thus, in this case, the dimensionless factor $\int_0^1 \varphi(z) dz$ is a number of the order of 1 and the value of the integral is mainly determined by the product

$$f_{\max} \cdot (b - a)$$

Let us consider a simple example: the free fall of a body during time t_0 . Let $\int_0^{t_0} v(t) dt$ exist. The rate of fall of the body is $v = gt$ and the maximum velocity attained at time t_0 is $v_{\max} = gt_0$.

It will be noted that the maximum here is not due to any decrease in velocity after $t = t_0$, but simply to the fact that times exceeding t_0 lie outside the interval over which the integration is taken, $0 < t < t_0$. We introduce

$$z = \frac{t}{t_0}, \quad \varphi(z) = \frac{v}{v_{\max}} = \frac{gt}{gt_0} = \frac{t}{t_0} = z,$$

$$\int_0^{t_0} v(t) dt = v_{\max} t_0 \int_0^1 z dz = gt_0^2 \int_0^1 z dz = \frac{gt_0^2}{2}$$

We conclude with an example that shows the necessity of a careful examination of the function and the danger of a purely formal approach. We will evaluate $I = \int_a^b \frac{dx}{x^2}$. The indefinite integral

$$\int \frac{dx}{x^2} = -\frac{1}{x} + C \quad \text{and so}$$

$$I = \int_a^b \frac{dx}{x^2} = -\frac{1}{x} \Big|_a^b = -\frac{1}{b} + \frac{1}{a} = \frac{b-a}{ab} \quad (3.16-6)$$

Since the integrand is positive, the result must be positive if $b > a$. The answer by formula (3.16-6) is indeed positive for $b > a$, if a and b have the same sign. However, for the integral $\int_{-1}^{+1} \frac{dx}{x^2}$ formula (3.16-6) yields the obviously absurd result $I = -2$. This is because the integrand becomes infinite inside the interval of integra-

tion when $x = 0$ and, at this very same place, the function $\left(-\frac{1}{x}\right)$, which is the indefinite integral of the function $\frac{1}{x^2}$, has an infinite discontinuity.

In order to get at the crux of the problem, we have to eliminate from the whole interval $-1 < x < +1$ a small region about the singular point $x = 0$: $-\varepsilon_1 < x < \varepsilon_2$ (ε_1 and ε_2 are small positive numbers) and consider

$$K = \int_{-1}^{-\varepsilon_1} \frac{dx}{x^2} + \int_{\varepsilon_2}^1 \frac{dx}{x^2}$$

From (3.16-6) we get

$$K = \frac{1-\varepsilon_1}{\varepsilon_1} + \frac{1-\varepsilon_2}{\varepsilon_2} = -2 + \frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2}$$

It is clear that for ε_1 and ε_2 tending to zero, $K \rightarrow \infty$.

In other cases, an integral with the integrand function becoming infinite on the interval of integration can yield a very definite

finite result. For instance, $\int_0^1 \frac{dx}{\sqrt{x}} = 2$. To prove this, evaluate $\int_\varepsilon^1 \frac{dx}{\sqrt{x}} = 2 - 2\sqrt{\varepsilon}$. The integral tends to 2 as $\varepsilon \rightarrow 0$.

An analysis of this kind is always necessary when the integrand becomes infinite.

3.17 SERIES

We pose the problem of constructing a simple and convenient approximate expression of a function $y(x)$ (defined exactly by some formula) over a small range of the argument x , say for values of x close to a .

The definition of a derivative given in Chapter 2 may be written as follows:

$$y'(a) = \lim_{x \rightarrow a} \frac{y(x) - y(a)}{x - a}$$

From this definition, it follows that in the limit, that is, the smaller the difference $(x - a)$ the greater the accuracy, we can write

$$y(x) = y(a) + (x - a) y'(a) \quad (3.17-1)$$

This formula fits the meaning of the derivative as the rate of change of the function. If we know the value of the function at a given point $y(a)$ and the rate of change of the function at that point, $\left.\frac{dy}{dx}\right|_{x=a} = y'(a)$, then for x close to a , when the argument has changed by $(x - a)$ as compared to the initial value a , the function changes by $(x - a) y'(a)$. The expression (3.17-1) is an approximate

expression and its accuracy decreases with increasing interval $(x - a)$. Indeed, when computing the variation of the function by the formula $(x - a) y' (a)$, we used the value of the rate of change of the function $y' (a)$ at the beginning of the interval between a and x , yet the rate y' itself changes in this interval. The exact formula is

$$y (x) = y (a) + \int_a^x y' (t) dt \quad (3.17-2)$$

Applying (3.17-1) to the derivative $y' (x)$, we get

$$y' (x) = y' (a) + (x - a) y'' (a) \quad (3.17-3)$$

Before proceeding, let us recall that $y'' (x)$ is the second derivative function of y with respect to x , denoted $\frac{d^2 y}{dx^2}$, which is to say, the derivative of $y' (x)$ with respect to x :

$$y'' (x) = \frac{dy'}{dx}$$

so that y'' is connected with y' in the same way that y' is connected with y . The third derivative, y''' , or $y^{(3)}$, is defined in similar fashion:

$$y''' = \frac{dy''}{dx}$$

The fourth derivative is denoted y^{IV} or $y^{(4)}$, the fifth y^V or $y^{(5)}$, and so on. The derivative of order n , or n th derivative, which is obtained by taking the derivative of the function $y (x)$ n times in succession is denoted by $y^{(n)} (x)$ or $\frac{d^n y}{dx^n}$. In the notation $y^{(n)}$, the n is enclosed in parentheses to distinguish it from an exponent.

Now let us return to the problem of the approximate expression of the function. Formula (3.17-3) for the derivative is nothing but formula (3.17-1) in which the function $y' (x)$ is substituted for $y (x)$. Now substitute the expression of the derivative (3.17-3) into (3.17-2) to get

$$\begin{aligned} y (x) &= y (a) + \int_a^x [y' (a) + (t - a) y'' (a)] dt \\ &= y (a) + (x - a) y' (a) + \frac{(x - a)^2}{2} y'' (a) \end{aligned} \quad (3.17-4)$$

This formula is more exact than (3.17-1). In deriving (3.17-4) we assumed (to a first approximation) that the rate of change of the function y , i.e., its derivative y' , is constant and equal to the value of the derivative at $x = a$. The result was a linear dependence of y on

x .* In deriving (3.17-4) we took into account that the derivative $y'(x)$ is not constant, but the variation of $y'(x)$ was considered only approximately: the formula (3.17-3) which we made use of when deriving (3.17-4) assumes that $y''(x)$ is constant, which is what gives us the linear dependence of y' on x . For $y(x)$ the relationship is quadratic.

Let us make formula (3.17-4) still more precise. To do this, we take into consideration that y'' is not constant. We take advantage of the formula

$$y'(x) = y'(a) + \int_a^x y''(t) dt \quad (3.17-5)$$

which is obtained from (3.17-2) by substituting y' for y . Also note that [like (3.17-2)] this formula can readily be verified by evaluating the integral. Now write $y''(x)$ using a formula of the type (3.17-1)

$$y''(x) = y''(a) + (x-a)y'''(a) \quad (3.17-6)$$

Then from (3.17-5) and (3.17-6) we obtain

$$y'(x) = y'(a) + \int_a^x [y''(a) + (t-a)y'''(a)] dt$$

or

$$y'(x) = y'(a) + y''(a)(x-a) + \frac{(x-a)^2}{2} y'''(a) \quad (3.17-7)$$

Note that formula (3.17-7) is a formula of the type (3.17-4) written for $y'(x)$.

Substitute the expression for $y'(x)$ from (3.17-7) into (3.17-2):

$$\begin{aligned} y(x) &= y(a) + \int_a^x \left[y'(a) + y''(a)(t-a) + y'''(a) \frac{(t-a)^2}{2} \right] dt \\ &= y(a) + y'(a)(x-a) + \frac{y''(a)}{2} (x-a)^2 + \frac{y'''(a)}{2 \cdot 3} (x-a)^3 \end{aligned} \quad (3.17-8)$$

It is now easy to imagine the aspect of the formulas for $y(x)$ if the approximation process is continued: if we take into account that y''' is not constant, then $y^{(4)}(a)$ will be involved, the expression for $y(x)$ will contain $(x-a)^4$. Each subsequent step in approximating $y(x)$ yields an additional term with a higher power of $(x-a)$.

This law becomes all the more obvious if we compare the expressions we have already obtained. To the roughest approximation, if $x-a$ is small, we take $y(x)$ to be equal to $y(a)$. This does not require any higher mathematics. We call this equation the *zeroth approxi-*

* In the expression for y , (3.17-1), x appears only to the first power. In other words, y is a polynomial of the first degree in x . This relationship is termed linear because its graph is a straight line (see Sec. 1.4).

tion. Then expression (3.17-1) is called the *first approximation*, expression (3.17-4), the *second approximation*, and expression (3.17-8), the *third approximation*. Listed, they are

$$\begin{aligned} y(x) &= y(a) && \text{(zeroth approximation),} \\ y(x) &= y(a) + (x-a)y'(a) && \text{(first approximation),} \\ y(x) &= y(a) + (x-a)y'(a) + \frac{(x-a)^2}{2}y''(a) && \text{(second approximation),} \\ y(x) &= y(a) + (x-a)y'(a) \\ &\quad + \frac{(x-a)^2}{2}y''(a) + \frac{(x-a)^3}{2 \cdot 3}y'''(a) && \text{(third approximation)} \end{aligned}$$

It will readily be seen that if we continue to improve the formulas, each subsequent approximation will contain one more term than the preceding one. This means that the more powers of $(x-a)$ that enter into the formula, the more exact the formula is.

A formula of this type can also be obtained in a somewhat different manner.

Let us take the exact equation (3.17-2) and integrate it by parts after first replacing* dt under the integral sign by $d(t-x)$:

$$\begin{aligned} y(x) &= y(a) + \int_a^x y'(t) dt = y(a) + \int_a^x y'(t) d(t-x) \\ &= y(a) + y'(t)(t-x) \Big|_a^x - \int_a^x (t-x)y''(t) dt \\ &= y(a) + (x-a)y'(a) + \int_a^x (x-t)y''(t) dt \quad (3.17-9) \end{aligned}$$

Performing the integration by parts n times, we get an exact expression for $y(x)$ consisting of $n+2$ terms. The first $n+1$ terms coincide with the n th approximation of the preceding derivation. The last term yields the expression of the remainder in the form of an integral of the $(n+1)$ th derivative of the function $y(x)$:

$$\begin{aligned} y(x) &= y(a) + (x-a)y'(a) + \frac{(x-a)^2}{2}y''(a) + \dots + \frac{(x-a)^n}{2 \cdot 3 \dots n}y^{(n)}(a) \\ &\quad + \frac{1}{2 \cdot 3 \dots n} \int_a^x (x-t)^n y^{(n+1)}(t) dt \quad (3.17-10) \end{aligned}$$

* When integrating, t is a dummy variable and x is regarded as a constant, therefore $dt = d(t-x)$ and the substitution is admissible.

If the last term involving the integral is absent, the formula is approximate. In the general case of an arbitrary function $y(x)$, no finite number of powers of $(x - a)$ can yield an absolutely exact formula.* This can only be given by an expression consisting of an infinite number of powers of $(x - a)$:

$$y(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots + c_n(x-a)^n \dots \quad (3.17-11)$$

An expression of this kind is called an infinite series. Ordinarily, we drop the word “infinite” and simply say “series”.

The coefficients $c_0, c_1, \dots, c_n, \dots$ are distinct for different functions. They are also dependent on the value of a . These coefficients can be found more quickly. Write down (3.17-11) and compute the first, second, and ... n th derivatives of both sides:

$$\begin{aligned} y(x) &= c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \dots + c_n(x-a)^n + \dots \\ y'(x) &= c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \dots + nc_n(x-a)^{n-1} + \dots \\ y''(x) &= 2c_2 + 3 \cdot 2c_3(x-a) + \dots + n(n-1)c_n(x-a)^{n-2} + \dots \\ &\vdots \\ y^{(n)}(x) &= n(n-1) \dots 3 \cdot 2c_n + (n+1)n(n-1) \dots 3 \cdot 2c_{n+1}(x-a) + \dots \end{aligned}$$

Each of the foregoing equations enables us to determine one of the coefficients c_i . Indeed, in each one, put $x = a$ on the right and on the left. Then all terms containing the factors $(x - a)$ will vanish and we get the equations for determining the coefficients:

$$\begin{array}{ll} y(a) = c_0, & \text{whence} & c_0 = y(a), \\ y'(a) = c_1, & & c_1 = y'(a), \\ y''(a) = 2c_2, & & c_2 = \frac{1}{2} y''(a), \\ y'''(a) = 3 \cdot 2c_3, & & c_3 = \frac{1}{2 \cdot 3} y'''(a), \\ \dots & & \dots \\ y^{(n)}(a) = n(n-1) \dots 3 \cdot 2 \cdot c_n, & & c_n = \frac{y^{(n)}(a)}{2 \cdot 3 \dots (n-1)n}, \\ \dots & & \dots \end{array}$$

We thus have

$$y(x) = y(a) + y'(a)(x-a) + \frac{y''(a)}{2!}(x-a)^2 + \frac{y'''(a)}{2 \cdot 3}(x-a)^3 + \frac{y^{(4)}(a)}{2 \cdot 3 \cdot 4}(x-a)^4 + \dots + \frac{y^{(n)}(a)}{2 \cdot 3 \dots (n-1)n}(x-a)^n + \dots \quad (3.17-12)$$

* Except for the case of a polynomial; see the end of Sec. 3.18.

The first n terms of this formula and (3.17-10) coincide. Note also the special case of the formula when $a = 0$,

$$y(x) = y(0) + y'(0)x + \frac{y''(0)}{2}x^2 + \frac{y'''(0)}{2 \cdot 3}x^3 + \dots \quad (3.17-13)$$

We have a convenient designation for the product of a succession of natural numbers $n(n-1) \dots 3 \cdot 2$. It is $n!$ and is read n -factorial. For example $3! = 3 \cdot 2 = 6$, $4! = 4 \cdot 3 \cdot 2 = 24$, $5! = 120$. It is customary, in defining the factorial, to include the number 1 as a factor as well:

$$n! = n(n-1) \dots 3 \cdot 2 \cdot 1$$

The product naturally remains unchanged but it is easier to remember that $n!$ is a product of n successive natural numbers from n to 1. For example, $3!$ is the product of three factors, $3 \cdot 2 \cdot 1$, from 3 to 1. With this definition, we naturally get $1! = 1$. Using the factorial notation, we can write formulas (3.17-12) and (3.17-13) very compactly:

$$y(x) = y(a) + \sum_{n=1}^{n=\infty} \frac{y^{(n)}(a)}{n!} (x-a)^n \quad (3.17-14)$$

$$y(x) = y(0) + \sum_{n=1}^{n=\infty} \frac{y^{(n)}(0)}{n!} x^n \quad (3.17-15)$$

These two formulas yield the expansion of a function $y(x)$ in a series of integral powers of $x-a$ (or of x). The formula (3.17-14) is called Taylor's series: (3.17-15) is called Maclaurin's series. Suppose, for example, $y(x) = e^x$. Then

$$y' = e^x, y'' = e^x, \dots, y^{(n)} = e^x, \dots$$

Taking advantage of the formula (3.17-15) of Maclaurin's series, we have

$$y(0) = y'(0) = y''(0) = \dots = 1$$

Substituting into (3.17-15), we get the expansion of the function $y = e^x$ in a series of powers of x :

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots + \frac{x^n}{n!} + \dots$$

Let us examine the formula obtained from the Taylor series if we confine ourselves to, say, three terms:

$$y(x) = y(a) + (x-a)y'(a) + \frac{(x-a)^2}{2}y''(a)$$

Removing brackets on the right side and arranging the result in powers of x , we have

$$y(x) = \left[y(a) - ay'(a) + \frac{1}{2} a^2 y''(a) \right] + [y'(a) - ay''(a)]x + \frac{1}{2} y''(a) x^2 \quad (3.17-16)$$

On the right is a polynomial of degree two. Note that this expression does not coincide with what we would have if we took three terms in the Maclaurin series:

$$y(x) = y(0) + y'(0)x + \frac{y''(0)}{2} x^2 \quad (3.17-17)$$

This will become clear if we recall that formula (3.17-16) yields a good result if x is close to a , while formula (3.17-17) is good when x is close to zero.

In Chapter 2 we gave a definition of a derivative as the limit of the ratio of the increment of the function to the increment of the independent variable.

Now that we have expressed the function as a series, we state generally the law according to which the ratio $\frac{\Delta y}{\Delta x}$ approaches $\frac{dy}{dx}$ as Δx tends to zero.

Let us take the Taylor series and denote $(x - a) = \Delta x$. Then $y(x) - y(a) = \Delta y$ and we get

$$\frac{\Delta y}{\Delta x} = y'(a) + \frac{1}{2} y''(a) \Delta x + \frac{1}{6} y'''(a) (\Delta x)^2 + \dots$$

For small Δx , the second term with Δx is greater than the third term with $(\Delta x)^2$. Disregarding the latter, we conclude that the difference of the ratio $\frac{\Delta y}{\Delta x}$ from the value of the derivative at the endpoint of the interval is proportional to the interval Δx and to the second derivative $y''(a)$. Here, we compare the ratio of the increment over the interval from $x = a$ to $x = a + \Delta x$ with the derivative $y'(a)$ at the endpoint of the interval.

The derivative may be evaluated differently. Take the increment as x varies from $a - \frac{\Delta x}{2}$ to $a + \frac{\Delta x}{2}$ and, dividing by Δx , compare this ratio with the derivative $y'(a)$, which is to say, with the derivative at the midpoint of the interval. We get

$$\begin{aligned} \Delta y &= f\left(a + \frac{\Delta x}{2}\right) - f\left(a - \frac{\Delta x}{2}\right), \\ f\left(a + \frac{\Delta x}{2}\right) &= f(a) + \frac{\Delta x}{2} f'(a) + \frac{1}{2} \left(\frac{\Delta x}{2}\right)^2 f''(a) + \frac{1}{6} \left(\frac{\Delta x}{2}\right)^3 f'''(a), \\ f\left(a - \frac{\Delta x}{2}\right) &= f(a) - \frac{\Delta x}{2} f'(a) + \frac{1}{2} \left(\frac{\Delta x}{2}\right)^2 f''(a) - \frac{1}{6} \left(\frac{\Delta x}{2}\right)^3 f'''(a), \\ \frac{\Delta y}{\Delta x} &= f'(a) + \frac{2}{6 \cdot 2} \left(\frac{\Delta x}{2}\right)^2 f''(a) = f'(a) + \frac{(\Delta x)^2}{24} f''(a) \end{aligned}$$

This method is much more exact: the difference between the ratio of the increments and the derivative is proportional to $(\Delta x)^2$ and not to (Δx) and, what is more, contains the coefficient $\frac{1}{24}$.

Exercises

1. Expand the third-degree polynomial $y = ax^3 + bx^2 + cx + d$ in a series in powers of $x - x_0$. Compare the first two, three, and four terms with the polynomial.

2. Expand the function $y = xe^x$ in a Maclaurin series. Verify that the expansion can be obtained from the expansion of e^x .

3. Expand the function e^x in a Taylor series in powers of $(x - 1)$.

4. Find numerically the derivative of the function e^x when $x = 0$, given the interval $\Delta x = 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$.

5. Determine the accuracy of the formula $(1 + r)^m = e^{mr}$. To do this, write the left-hand member as $(1 + r)^m = e^{m \ln(1+r)}$ and expand $\ln(1 + r)$ in a series.

3.18 COMPUTING THE VALUES OF FUNCTIONS BY MEANS OF SERIES

Let us dwell briefly on the principles underlying the formulas of Sec. 3.17. When we began the study of higher mathematics, we assumed as known the concept of a function and we proceeded from the fact that we could compute the value of the function for any value of the argument. That is why, when we considered derivatives, we found them directly, empirically so to say, by computing the values of the function for close-lying values of the argument. Later on we learned how to find derivatives by formulas and it turned out that setting up formulas for derivatives is a rather simple matter. And so finding the values of a function by means of a formula involving derivatives turns out to be even simpler than a direct computation of the function.

Since only in the case of a polynomial does the Taylor series terminate, contain a finite number of terms, it follows that any function different from a polynomial will be represented by an infinite series. The practical value of such a series for computational purposes is due to the possibility of confining oneself to two or three terms of the series in order to obtain a sufficiently accurate result. This requires that the discarded terms of the series be small.

Let us consider a few very simple examples. Let $y = e^x$. In the preceding section we obtained the formula

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots + \frac{x^n}{n!} + \dots \quad (3.18-1)$$

In particular, substituting $x = 1$, we get an expression of the number e in the form of a series:

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \dots + \frac{1}{n!} + \dots \quad (3.18-2)$$

This formula enables us to compute e^x rapidly and to a high degree of accuracy, as witness Table 2.

Table 2

x	e^x	$1 + x$	$1 + x + \frac{x^2}{2}$	$1 + x + \frac{x^2}{2} + \frac{x^3}{6}$	$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$
0.10	1.1052	1.10	1.1050	1.1052	1.1052
0.25	1.2840	1.25	1.2812	1.2838	1.2840
0.50	1.6487	1.50	1.6250	1.6458	1.6484
0.75	2.1170	1.75	2.0312	2.1015	2.1147
1.00	2.7183	2.00	2.5000	2.6667	2.7083
1.25	3.4903	2.25	3.0312	3.3568	3.4585
1.50	4.4817	2.50	3.6250	4.1876	4.3986
2.00	7.3891	3.00	5.0000	6.3333	7.0000

The first two terms of the formula yield an accuracy of 0.5% when $x = 0.1$.

The first three terms of the formula yield an accuracy of 1.4% for $x = 0.5$.

The first four terms of the formula yield an accuracy of 1.8% for $x = 1.0$.

Such a high accuracy is plainly due to the fact that the terms of the series fall off rapidly. Each subsequent term of the series is less than the preceding one primarily because the denominator of the $(n + 1)$ th term is n times the denominator of the preceding n th term. If $x < 1$, then in addition we have that x^n is the smaller, the greater n is.

But even when $x > 1$, the increase of the denominator in the distant terms of the series will inevitably overcome the increase in the numerator. As can be seen from Table 2, when $x = 2$ the sum of five terms of the series yields an error of 5%. But if we add a sixth term $\left(\frac{x^5}{120}\right)$, then we get 7.3500, which is in error by 0.5%.

Let us construct formulas of the same type for trigonometric functions:

$$\begin{aligned} y(x) &= \sin x, \quad y'(x) = \cos x, \quad y''(x) = -\sin x, \\ y'''(x) &= -\cos x, \quad y^{(4)}(x) = \sin x \end{aligned}$$

The law for subsequent derivatives is obvious.

Substituting $x = 0$, we get

$$y(0) = 0, y'(0) = 1, y''(0) = 0, y'''(0) = -1, \dots$$

Consequently

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots \quad (3.18-3)$$

In similar fashion we get the formula

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots \quad (3.18-4)$$

Figs. 82 and 83 show the graphs of the sine, cosine, and also the graphs of polynomials obtained if we take one, two, and three terms

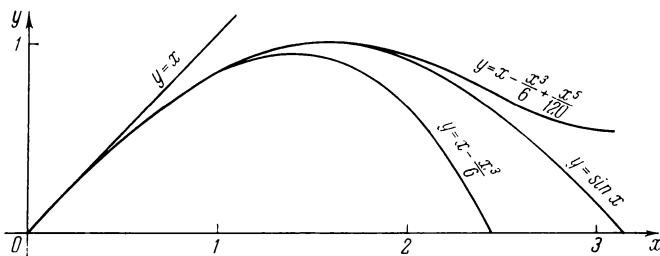


Fig. 82

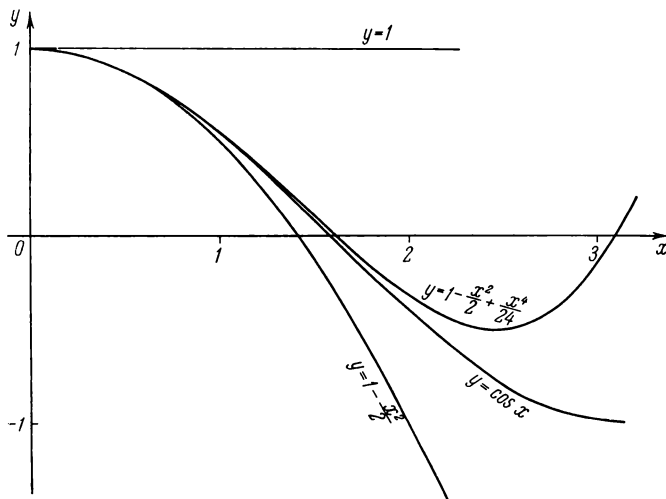


Fig. 83

of the corresponding series. Accuracy improves visibly when we take more and more terms of the series.

Tables 3 and 3a list the values of the sine and cosine functions respectively. It is evident from the tables that two or three terms of

Table 3

x	φ^*	$\sin x$	x	$x - \frac{x^3}{6}$	$x - \frac{x^3}{6} + \frac{x^5}{120}$
0	0°	0.0000	0.0000	0.0000	0.0000
$\frac{\pi}{20}$	9°	0.1564	0.1571	0.1564	0.1564
$\frac{\pi}{10}$	18°	0.3090	0.3142	0.3090	0.3090
$\frac{3\pi}{20}$	27°	0.4540	0.4712	0.4538	0.4540
$\frac{4\pi}{20}$	36°	0.5878	0.6283	0.5869	0.5878
$\frac{5\pi}{20}$	45°	0.7071	0.7854	0.7046	0.7071
$\frac{6\pi}{20}$	54°	0.8090	0.9425	0.8029	0.8091
$\frac{7\pi}{20}$	63°	0.8910	1.0996	0.8780	0.8914
$\frac{8\pi}{20}$	72°	0.9510	1.2566	0.9258	0.9519
$\frac{9\pi}{20}$	81°	0.9877	1.4137	0.9427	0.9898
$\frac{\pi}{2}$	90°	1.0000	1.5708	0.9248	1.0045

* φ is an angle corresponding to x but expressed in degrees.

the series suffice to obtain excellent accuracy in the interval from 0 to $\frac{\pi}{4}$. Thus a power series offers a very convenient practical method for computing the values of trigonometric functions. Note that in absolute value the nonzero terms of the series for the sine and cosine are exactly equal to the corresponding terms of the series for the function e^x . For this reason, everything that has been said pertaining to the falling off of terms with high powers of x in formula (3.18-1) for e^x refers also to the series (3.18-3) and (3.18-4) for the sine and cosine.

Note that if we substitute $x = \varphi\sqrt{-1}$ into expression (3.18-1) and if we replace x by φ in (3.18-3) and (3.18-4), we get the relation $e^{\varphi\sqrt{-1}} = \cos \varphi + \sqrt{-1} \sin \varphi$ which is mentioned on page 124.

Table 3a

x	φ	$\cos x$	$1 - \frac{x^2}{2}$	$1 - \frac{x^2}{2} + \frac{x^4}{24}$	$1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}$
0	0°	1.0000	1.0000	1.0000	1.0000
$\frac{\pi}{20}$	9°	0.9877	0.9877	0.9877	0.9877
$\frac{\pi}{10}$	18°	0.9510	0.9506	0.9510	0.9510
$\frac{3\pi}{20}$	27°	0.8910	0.8890	0.8911	0.8910
$\frac{4\pi}{20}$	36°	0.8090	0.8026	0.8091	0.8090
$\frac{5\pi}{20}$	45°	0.7071	0.6916	0.7075	0.7071
$\frac{6\pi}{20}$	54°	0.5878	0.5558	0.5887	0.5877
$\frac{7\pi}{20}$	63°	0.4540	0.3954	0.4563	0.4539
$\frac{8\pi}{20}$	72°	0.3090	0.2105	0.3144	0.3089
$\frac{9\pi}{20}$	81°	0.1564	0.0007	0.1672	0.1561
$\frac{\pi}{2}$	90°	0.0000	-0.2337	0.0200	-0.0009

If the function $y(x)$ is a polynomial of degree n , then $y'(x)$ is a polynomial of degree $(n - 1)$, $y''(x)$ is a polynomial of degree $(n - 2)$, . . . , $y^{(n)}(x)$ is a constant, and $y^{(n+1)}(x)$ and all higher derivatives are zeroes. That is why for a polynomial the Taylor series (3.17-14) terminates. It consists of a finite number of terms. We obtain a polynomial arranged in powers of $(x - a)$. For polynomials of degree n the sum of the first $n + 1$ terms of the Taylor series yields an *exact* equation which is true for all x and not only for the x near a .

3.19 CONDITION FOR APPLICABILITY OF SERIES. THE GEOMETRIC PROGRESSION

In the preceding section we set up formulas for the three functions e^x , $\sin x$ and $\cos x$. In these formulas the functions are represented as the sums of a series of powers of x with constant coefficients. In these three cases it turned out that for arbitrary x , each subsequent

term of the series, with the possible exception of the first few terms, is less than the preceding one, and the greater the number-label of the term, the closer this term is to zero. In these examples, we can compute the value of the function for any x by means of a series if a large enough number of terms of the series is taken so that the discarded terms have practically no effect on the result.

To summarize, then, we began with the problem of approximating a function in a small range of the variable and constructed more and more exact formulas by taking into account the first, second, third, and higher derivatives. The accuracy of each formula,

$$y(x) = y(a) \quad (0)$$

$$y(x) = y(a) + (x-a)y'(a) \quad (I)$$

$$y(x) = y(a) + (x-a)y'(a) + \frac{(x-a)^2}{2}y''(a) \quad (II)$$

is the greater, the smaller the quantity $(x-a)$. On the other hand, for a given $(x-a)$, formula (I) is more accurate than formula (0), the accuracy of formula (II) exceeds that of (I), etc. Hence if we increase the number of terms of a series, this permits increasing the quantity $(x-a)$ while preserving a given accuracy.

The question now arises as to whether it is always possible to attain a given accuracy for any value of $(x-a)$ merely by increasing the number of terms of the series. We will use a very important example to illustrate this point and will show that this is not so. A power series constructed so as to yield a good approximation in a small range of x , for arbitrary $(x-a)$, can have a natural limit of applicability, the limit of admissible increase in $(x-a)$ (the limit not depending on the number of terms taken) although this was not evident in the examples of the preceding section. Consider the function

$$y = \frac{1}{1-x} = (1-x)^{-1}$$

Taking the derivatives in succession, we have

$$y' = \frac{1}{(1-x)^2}, \quad y'' = \frac{1 \cdot 2}{(1-x)^3}, \quad \dots, \quad y^{(n)} = \frac{n!}{(1-x)^{n+1}}$$

Substituting $x = 0$, we get

$$y(0) = 1, \quad y'(0) = 1, \quad y''(0) = 2, \quad \dots, \quad y^{(n)}(0) = n!$$

We thus get the series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots + x^n + \dots \quad (3.19-1)$$

The example of the function $\frac{1}{1-x}$ is remarkable not only due to the unusually simple form of the resulting power series (all coefficients are equal to 1). Here it is easy to give an exact formula for

the sum of the first n terms of the series (3.19-1):

$$1 + x + x^2 + \dots + x^{n-1} = \frac{1 - x^n}{1 - x} \quad (3.19-2)$$

The truth of this formula is apparent if we multiply both sides of (3.19-2) by $(1 - x)$. Formula (3.19-2) can then be rewritten as

$$1 + x + x^2 + \dots + x^{n-1} = \frac{1}{1-x} - \frac{x^n}{1-x} \quad (3.19-3)$$

Comparing this formula with (3.19-1) we see that $\frac{x^n}{1-x}$ is the quantity we neglect if we confine ourselves to the first n terms of the series:

$$1 + x + x^2 + x^3 + \dots + x^n + \dots \quad (3.19-4)$$

If $-1 < x < 1$, then the greater n , the closer x^n is to zero and, consequently, if we take a sufficiently large number of terms of the series, we discard a small quantity. Note that the closer x is to 1, the more terms we have to take in order to obtain a given accuracy.

The whole picture changes if we take $x > 1$. In this case, each subsequent term of (3.19-4) is greater than the preceding one. Formula (3.19-3) remains valid, but for $x > 1$, x^n increases without bound together with the growth of n and for this reason we cannot disregard the fraction $\frac{x^n}{1-x}$. Here, (3.19-1) does not hold true.

There is not even any qualitative similarity between the sum of the positive terms of (3.19-4) and the negative (since $x > 1$) quantity $\frac{1}{1-x}$. From (3.19-3) we see that when $x > 1$ the sum of the series (3.19-4) increases without bound as n increases. Such series are termed convergent series.

The terms of the series (3.19-4) form a geometric progression. We have established that the sum of the terms of an infinite geometric progression is equal to $\frac{1}{1-x}$ if $|x| < 1$. But if $x \geq 1$, then the infinite geometric progression does not have a finite sum.

Also note that any periodic fraction is a sum of terms of a geometric progression. For instance

$$\begin{aligned} 1.(1) &= 1.111 \dots = 1 + 0.1 + 0.01 + 0.001 + \dots \\ &= 1 + \frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \dots = \frac{1}{1 - \frac{1}{10}} = \frac{1}{0.9} = \frac{10}{9} = 1 \frac{1}{9} \end{aligned}$$

Thus we have already encountered an elementary series (geometric progression) in arithmetic and algebra.

The function $y = \frac{1}{1-x}$ (Fig. 84) has a discontinuity at $x = 1$; if x is close to 1 but greater than 1, then $\frac{1}{1-x}$ is a large (in absolute

value) negative number, if x is close to 1 but less than 1, then $\frac{1}{1-x}$ is a large positive number. Thus, when x passes through the value $x = 1$, the value of the function $\frac{1}{1-x}$ moves from large positive numbers to large (in absolute value) negative numbers. The series

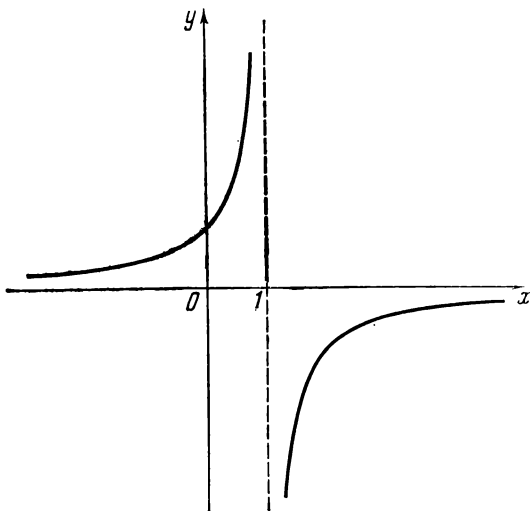


Fig. 84

cannot describe this peculiarity in the behaviour of the function.

We note yet another circumstance. When $x = 1$, the function $y = \frac{1}{1-x}$ becomes infinite (the closer x is to 1, the greater y is in absolute value), and again for $x = 1$ the terms of the series (3.19-4) cease to decrease. A series is suitable for computational purposes only if its terms diminish in absolute value.* For $x = 1$, the series is unsuitable for computational purposes because its terms do not decrease. This means that the series is unsuitable for computing the values of the function for $x = -1$ as well (since for $x = -1$ the terms do not diminish in absolute value either) although the function itself does not have a discontinuity when $x = -1$ and is equal to $\frac{1}{1-(-1)} = \frac{1}{2}$.

No matter how we choose the coefficients of a polynomial, the graph will always be a solid continuous line. A polynomial does not have discontinuities. Therefore if some function $f(x)$ has a discon-

* Of course if one or two or several of the first terms of the series increase, there is no harm done if the subsequent terms of the series fall off rapidly; see the example involving e^x for $x = 2$, Table 2.

tinuity at $x = x_0$ ($x_0 = 1$ in the example involving $\frac{1}{1-x}$), then for the value $x = x_0$ the series constructed for $f(x)$ is definitely unsuitable for computations. Since the greater the absolute value of x , the greater (in absolute value) each term of the series $c_n x^n$ is, it follows that for arbitrary x which is greater in absolute value than x_0 , the series is likewise unsuitable for computations.

Thus, in the case of a discontinuity of $f(x)$ we can indicate beforehand an x_0 such that for all x exceeding x_0 in absolute value, the series will prove to be unsuitable for computational purposes.

Note that the presence of a discontinuity of a function is a sufficient condition for the series to cease to converge, but it is not a necessary condition. By way of an illustration let us consider the function $y = \frac{1}{1+x}$. Applying formula (3.17-13), we get

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots \quad (3.19-5)$$

Take $x = 2$ for instance. Then

$$\left. \frac{1}{1+x} \right|_{x=2} = \frac{1}{1+2} = \frac{1}{3}$$

The sum of the terms of the series

$$1 - x + x^2 - x^3 + \dots \quad (3.19-6)$$

however changes sharply depending on the number of terms n :

n	1	2	3	4	5	6	7
sum of terms:	1	-1	3	-5	11	-21	43

The series is clearly unsuitable for computations for $x = 2$. Why does this occur, particularly since the function itself, $y = \frac{1}{1+x}$ has no discontinuity either for $x = 2$ or anywhere between $x = 0$ and $x = 2$ (Fig. 85)?

However, the function $y = \frac{1}{1+x}$ has a discontinuity when $x = -1$. Therefore, for $x = -1$ the terms of (3.19-6) do not diminish. Also note that the absolute values of the terms of (3.19-6) do not depend on the sign of x . Consequently, for $x = 1$ (and all the more so for $x > 1$) the series is not suitable for computation.

Therefore even if we are interested in the behaviour of a series only for $x > 0$, we still have to take into account all the values of x , including negative values as well, for which the function undergoing expansion has a discontinuity.

Indeed, the convergence of a series is affected even by the behaviour of a function for complex values of the argument. Here is an example. Replacing x by x^2 in (3.19-5) we get

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots \quad (3.19-7)$$

The graph of the function $y = \frac{1}{1+x^2}$ (Fig. 86) does not have any discontinuities and does not go to infinity anywhere either for positive or for negative x , but the series (3.19-7) is suitable for computations only if $x^2 < 1$, that is, for $-1 < x < 1$. The reason for this

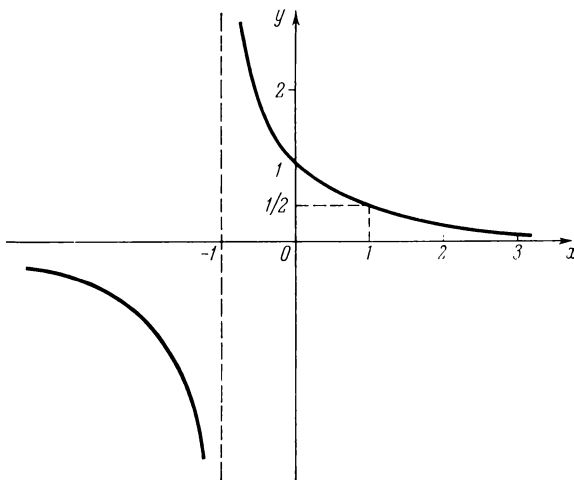


Fig. 85

is that when $x = \pm\sqrt{-1} = \pm i$, i.e., for $x^2 = -1$, the function $y = \frac{1}{1+x^2}$ becomes infinite and therefore the terms of the series do not decrease in absolute value for $x^2 = -1$. Hence, neither do

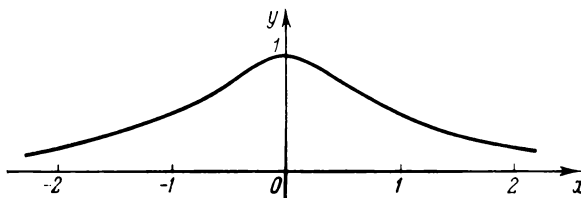


Fig. 86

they decrease in absolute value for $x^2 = 1$. However, a detailed and comprehensible discussion of the problem of the behaviour of a function for complex values of x is beyond the scope of this book. The interested reader is referred to "Elements of Applied Mathematics" by Ya. B. Zeldovich and A. D. Myškis.

Let us consider one more example. We will find the Maclaurin series for the function $y = \tan x$. By general rules we find

$$\begin{aligned}y &= \tan x = \frac{\sin x}{\cos x}, & y'(x) &= \frac{1}{\cos^2 x}, \\y''(x) &= \frac{2 \sin x}{\cos^3 x}, & y'''(x) &= \frac{2 + 4 \sin^2 x}{\cos^4 x}, \\y^{(4)}(x) &= \frac{16 \sin x + 8 \sin^3 x}{\cos^5 x}, \\y^{(5)}(x) &= \frac{16 + 88 \sin^2 x + 16 \sin^4 x}{\cos^6 x}\end{aligned}$$

whence

$$\begin{aligned}y(0) &= 0, & y'(0) &= 1, & y''(0) &= 0, & y'''(0) &= 2, \\y^{(4)}(0) &= 0, & y^{(5)}(0) &= 16\end{aligned}$$

Therefore

$$\tan x = 0 + 1 \cdot x + 0 \cdot x^2 + \frac{2}{3 \cdot 2 \cdot 1} x^3 + 0 \cdot x^4 + \frac{16}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} x^5 + \dots$$

Thus

$$\tan x = x + \frac{1}{3} x^3 + \frac{2}{15} x^5 + \frac{7}{315} x^7 + \frac{62}{2835} x^9 + \dots \quad (3.19-8)$$

The coefficients of x^7 and x^9 in this last expression can be obtained in the same way as the coefficients of x , x^3 , x^5 were in the text.

What can be said about the range of applicability of the series (3.19-8)? The graph of the tangent (see Fig. 78) shows at once that the series (3.19-8) is suitable for computations only when $|x| < \frac{\pi}{2}$ since for $x = \frac{\pi}{2}$ the function $\tan x$ behaves just as badly as the function $\frac{1}{1-x}$ for $x = 1$.

Just looking at the series itself, $x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots$, it would be difficult to say for what value of x the series cannot be applied because the law obeyed by the coefficients of the series is not a simple one, in contrast to the earlier considered series $1 + x + x^2 + \dots$.

Exercises

1. Write the Maclaurin series for the function $y = \frac{x+1}{1-x}$.
2. Write the Maclaurin series for the function $y = \ln(1+x)$.
3. Write the Taylor series for the function $y = \ln x$ in powers of $(x-1)$. What range of applicability do the series obtained in Problems 1 to 3 have?
4. Obtain the first three terms of the series expansion in powers of x of the function $f(x)g(x)$. Construct the same series by multiplying together the series for $f(x)$ and the series for $g(x)$.

3.20 THE BINOMIAL THEOREM FOR INTEGRAL AND FRACTIONAL EXPONENTS

Let us form the Maclaurin series expansion of a binomial $(a + x)$ to an arbitrary power m : $y = (a + x)^m$.

Using the general rule, let us first find the derivatives

$$\left. \begin{aligned} y' &= m(a + x)^{m-1}, & y'' &= m(m-1)(a + x)^{m-2}, \\ &\dots\dots\dots \\ y^{(n)} &= m(m-1)\dots(m-n+1)(a + x)^{m-n}, \\ &\dots\dots\dots \end{aligned} \right\} \quad (3.20-1)$$

and the values of the function and of the derivatives for $x = 0$:

$$\left. \begin{aligned} y(0) &= a^m, & y'(0) &= ma^{m-1}, \\ y''(0) &= m(m-1)a^{m-2}, & \dots, \\ y^{(n)}(0) &= m(m-1)\dots(m-n+1)a^{m-n}, & \dots \end{aligned} \right\} \quad (3.20-2)$$

From this we get Maclaurin's series:

$$\begin{aligned} (a + x)^m &= a^m + \frac{m}{1} a^{m-1}x + \frac{m(m-1)}{1 \cdot 2} a^{m-2}x^2 + \dots \\ &\quad + \frac{m(m-1)(m-2)\dots(m-n+1)}{n!} a^{m-n}x^n + \dots \end{aligned} \quad (3.20-3)$$

If the exponent m is a positive integer, then $(a + x)^m$ is a polynomial of degree m so that in this case the series (3.20-3) is finite. The $(m + 1)$ th derivative of the function $(a + x)^m$ and, hence, all higher derivatives are zero. The formulas (3.20-1), (3.20-2) and (3.20-3) reflect this circumstance. Indeed, for $n = m + 1$ the factor $(m - n + 1)$ vanishes; for $n > m + 1$ there will be, some place in the sequence of factors $m(m-1)\dots$, a factor equal to zero and, consequently, the product will be equal to zero too.

For a positive integer m , the product in the numerator can be written in a more convenient form:

$$\begin{aligned} m(m-1)\dots(m-n+1) \\ = \frac{m(m-1)\dots(m-n+1)(m-n)(m-n-1)\dots 3 \cdot 2 \cdot 1}{(m-n)(m-n-1)\dots 3 \cdot 2 \cdot 1} = \frac{m!}{(m-n)!} \end{aligned}$$

Thus, for positive integral m we finally have

$$\begin{aligned} (a + x)^m &= a^m + \frac{m!}{1!(m-1)!} a^{m-1}x + \frac{m!}{2!(m-2)!} a^{m-2}x^2 + \dots \\ &\quad + \frac{m!}{n!(m-n)!} a^{m-n}x^n + \dots + \frac{m!}{(m-2)!2!} a^2x^{m-2} \\ &\quad + \frac{m!}{(m-1)!1!} ax^{m-1} + x^m \end{aligned} \quad (3.20-4)$$

In formula (3.20-4) we have polynomials of degree m on the right and on the left. Thus, for the case of a positive integer m , we obtain an exact equation that is valid for arbitrary values of x . Formula (3.20-4) is symmetric with respect to x and a : the coefficients of the terms $a^{m-n}x^n$ and $a^n x^{m-n}$ are the same. This is clear since $(x+a)^m$ does not depend on the order of the summands in the parentheses:

$$(x+a)^m = (a+x)^m.$$

Formula (3.20-4) is called the binomial theorem (Newton's binomial theorem) or the binomial expansion. It can be obtained without resorting to higher mathematics and derivatives. We have to take the product $\underbrace{(a+x)(a+x)\dots(a+x)}_{m \text{ times}}$, perform the multiplica-

tion and collect like terms. But when m is specified in the general form by a literal symbol and not a number, the collection of like terms is rather complicated. On the whole the derivation of the binomial expansion using Maclaurin's series is simpler.

Newton obtained the general formula (3.20-3), that is, the expansion of $(x+a)^m$, for the case of an arbitrary exponent m . It would therefore be more appropriate to call formula (3.20-3) Newton's binomial theorem, instead of (3.20-4), which is a simple particular case of the formula (3.20-3).

Let us return to the general formula (3.20-3). Suppose m is not a positive integer. In Maclaurin's series (3.20-3) the powers of the variable x , i.e., the numbers n , are positive integers. This means that the numerator in (3.20-3), if m is not a positive integer, does not vanish for any n , and (3.20-3) yields an infinite series. In particular, for $m = -1$ this series is of the form

$$\frac{1}{a+x} = \frac{1}{a} - \frac{x}{a^2} + \frac{x^2}{a^3} - \frac{x^3}{a^4} + \dots \quad (3.20-5)$$

Note that for $a = 1$ (3.20-5) passes into the familiar formula

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$$

From (3.20-5) we also find

$$\frac{1}{a-x} = \frac{1}{a} + \frac{x}{a^2} + \frac{x^2}{a^3} + \frac{x^3}{a^4} + \dots$$

For $m = \frac{1}{2}$ we have

$$\begin{aligned} \sqrt{a+x} &= \sqrt{a} + \frac{1}{2} \frac{x}{\sqrt{a}} - \frac{1}{8} \frac{x^2}{a\sqrt{a}} + \frac{1}{16} \frac{x^3}{a^2\sqrt{a}} \\ &\quad - \frac{5}{128} \frac{x^4}{a^3\sqrt{a}} + \frac{7}{256} \frac{x^5}{a^4\sqrt{a}} - \frac{21}{1024} \frac{x^6}{a^5\sqrt{a}} + \dots \end{aligned} \quad (3.20-6)$$

In the expansion of $(a+x)^m$, for any m , all the terms have the same sum of powers of a and x , each subsequent term differing from

the preceding one by the factor $\left(\frac{x}{a}\right)$ and the coefficient. A physicist would say that a and x in formula (3.20-3) must have the same dimensions and so $\frac{x}{a}$ is dimensionless. From the very beginning we could take a outside the brackets:

$$(a+x)^m = a^m \left(1 + \frac{x}{a}\right)^m$$

and expand $\left(1 + \frac{x}{a}\right)^m$ in powers of $\frac{x}{a}$.

It turns out that for all m (negative and positive fractional) the series (3.20-3) is suitable only for $\left|\frac{x}{a}\right| < 1$, i.e., for $|x| < |a|$. For $\left|\frac{x}{a}\right| \geq 1$ the series (3.20-3) is divergent. The positive integers m are an exception because in that case formula (3.20-3) contains a finite number of terms.

Formula (3.20-6) offers a good method for taking roots. Here, the smaller $\left|\frac{x}{a}\right|$, the fewer terms one has to take in (3.20-6) to attain a specified accuracy.

Exercises

1. Using a series expansion, find $\sqrt[3]{1.1}$ and $\sqrt[3]{1.5}$ as $\sqrt[3]{1+x}$ for $x = 0.1$, and for $x = 0.5$ retaining two, three and four terms in the expansion. Compare the results with the tabular values.

2. Show that for $|x| < 1$ the approximate formula $\sqrt[n]{1+x} = 1 + \frac{x}{n}$ is valid and that the smaller x is, the more accurate the formula.

3. Using the formula of the preceding exercise, find $\sqrt[3]{1.2}$, $\sqrt[3]{1.1}$, $\sqrt[3]{1.05}$. Compare with tabulated values.

4. Find $\sqrt[3]{6}$ to three decimal places.

Hint. Take advantage of the fact that $6 = 4 + 2$, $\sqrt[3]{4} = 2$ and apply formula (3.20-6).

5. Why is it impossible to expand $y = \sqrt{x}$ by Maclaurin's formula?

3.21 THE ORDER OF INCREASE AND DECREASE OF FUNCTIONS

The series expansion of functions yields a general method for reducing different functions to the same form and enables one to compare the functions. This method of comparison is needed, for example, when we consider the ratio of two functions $\frac{f(x)}{g(x)}$ for a value of the argument x for which the values of the two functions are close to zero.

In the computation of derivatives it was demonstrated that the ratio of two almost-zero quantities can be a quite definite number. In certain cases, this ratio may be equal to zero or infinity. A few examples will suffice. For the sake of simplicity of notation, we take examples in which the value of x that interests us is equal to zero.

For small x , the functions $\sin x$ and $\tan x$ are also small. The functions e^x and $\cos x$ are close to 1 and hence $e^x - 1$ and $1 - \cos x$ are small. Here the smaller the $|x|$, the closer are the values of the functions $\sin x$, $\tan x$, $e^x - 1$, $1 - \cos x$ to zero.

Let us compare these functions with the value of x . To do this, write out their Maclaurin-series expansions:

$$\left. \begin{aligned} \sin x &= x - \frac{x^3}{6} + \dots, \\ \tan x &= x + \frac{x^3}{3} + \dots, \\ 1 - \cos x &= \frac{x^2}{2} - \frac{x^4}{24} + \dots, \\ e^x - 1 &= x + \frac{x^2}{2} + \dots \end{aligned} \right\} \quad (3.21-1)$$

whence we find

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \dots$$

Consequently

$$\frac{\sin x}{x} \xrightarrow{x \rightarrow 0} 1 \quad \text{or} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Similarly, from (3.21-1) we find

$$\begin{aligned} \frac{\tan x}{x} &= 1 + \frac{x^2}{3} + \dots \xrightarrow{x \rightarrow 0} 1, \\ \frac{1 - \cos x}{x} &= \frac{x}{2} - \frac{x^3}{24} + \dots \xrightarrow{x \rightarrow 0} 0, \\ \frac{1 - \cos x}{x^2} &= \frac{1}{2} - \frac{x^2}{24} + \dots \xrightarrow{x \rightarrow 0} \frac{1}{2}, \\ \frac{e^x - 1}{x} &= 1 + \frac{x}{2} + \dots \xrightarrow{x \rightarrow 0} 1 \end{aligned}$$

More complicated relations can also be found. For instance, from

$$\begin{aligned} \sin x &= x - \frac{x^3}{6} + \frac{x^5}{120} - \dots, \\ \tan x &= x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \end{aligned}$$

there follows

$$\tan x - \sin x = \frac{1}{2}x^3 + \frac{1}{8}x^5 + \dots$$

and

$$\frac{\tan x - \sin x}{x^3} \xrightarrow{x \rightarrow 0} \frac{1}{2}$$

A scale can be constructed of the order of decrease of various functions as x tends to zero. Let us use the term order of decrease for the power of x which decreases just as rapidly as the quantity in question. If a function $f(x)$ has k th order of decrease for small x , this means that it decreases as x^k , that is, the ratio $\frac{f(x)}{x^k}$ has for its limit, as $x \rightarrow 0$, a finite nonzero number.

Thus, $\sin x$, $\tan x$, $e^x - 1$ decrease by order one, $1 - \cos x$ decreases by order two, $\tan x - \sin x$ decreases by order three for small x .

In certain concrete cases it is possible to determine the order of decrease without a series expansion. For instance, by drawing the lines of the sine and the cosine, we see from the figure that $\sin x \approx x$, $\tan x \approx x$ for small x , that is, $\sin x$ and $\tan x$ have the first order of decrease. The formula $1 - \cos x = 2 \sin^2 \frac{x}{2}$ is valid and since $\sin \frac{x}{2}$ is of the first order, it is therefore evident that $1 - \cos x$ is of the second order of decrease. The function $\tan x - \sin x$ may be written as $\frac{\sin x}{\cos x} - \sin x = \frac{\sin x}{\cos x} (1 - \cos x)$. Since for small x , $\cos x$ is close to 1, $\sin x$ is of the first order, and $1 - \cos x$ is of the second order, it is clear that $\tan x - \sin x$ is of the third order. However, these concrete devices require a great deal of ingenuity and so precisely for this reason a general method that operates without failure is particularly useful.

Such relationships between ingenious solutions of individual problems and general methods are in evidence everywhere: the properties of tangent lines to a parabola, the area of a circle, the volume of a pyramid, and the volume of a sphere were all familiar to the ancient Greeks, but only differential and integral calculus provided us with general and simple methods for solving all problems of that type.

Using series, it is possible not only to find the ratio of a function to a power of x , but also the ratio of one function to another.

Here are some examples.

$$\begin{aligned}\frac{e^x - 1}{\sin x} &= \frac{x + \frac{x^2}{2} + \frac{x^3}{6} + \dots}{x - \frac{x^3}{6} + \dots} = \frac{1 + \frac{x}{2} + \dots}{1 - \frac{x^2}{6} + \dots} \xrightarrow{x \rightarrow 0} 1 \\ \frac{e^x - 1}{1 - \cos x} &= \frac{x + \frac{x^2}{2} + \dots}{\frac{x^2}{2} - \frac{x^4}{24} + \dots} = \frac{1 + \frac{x}{2} + \dots}{\frac{x}{2} - \frac{x^3}{24} + \dots} \xrightarrow{x \rightarrow 0} \infty \\ \frac{e^x - 1}{\sqrt{x}} &= \frac{x + \frac{x^2}{2} + \dots}{\sqrt{x}} = \sqrt{x} + \frac{1}{2}x\sqrt{x} + \dots \xrightarrow{x \rightarrow 0} 0\end{aligned}$$

The coefficients of the Maclaurin series are expressed in terms of derivatives. It is therefore possible to state the results obtained by means of series in the form of rules referring to derivatives. If $f(0) = g(0) = 0$, then from the formulas

$$f(x) = f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \dots$$

and

$$g(x) = g(0) + g'(0)x + \frac{1}{2}g''(0)x^2 + \dots$$

we get

$$f(x) = f'(0)x + \frac{1}{2}f''(0)x^2 + \dots$$

and

$$g(x) = g'(0)x + \frac{1}{2}g''(0)x^2 + \dots$$

From this we have

$$\frac{f(x)}{g(x)} = \frac{f'(0)x + \frac{1}{2}f''(0)x^2 + \dots}{g'(0)x + \frac{1}{2}g''(0)x^2 + \dots} = \frac{f'(0) + \frac{1}{2}f''(0)x + \dots}{g'(0) + \frac{1}{2}g''(0)x + \dots} \xrightarrow{x \rightarrow 0} \frac{f'(0)}{g'(0)}$$

that is,

$$\frac{f(x)}{g(x)} \xrightarrow{x \rightarrow 0} \frac{f'(0)}{g'(0)}$$

for $f(0) = g(0) = 0$.

Thus, instead of considering the ratio of two functions whose values are almost zero (since both functions vanish for the same value of the argument near which the ratio is being considered), we can consider the ratio of their derivatives. This result is called *l'Hospital's rule*.

After studying series, it is more convenient not to bother remembering some special rule but, for small x , to use series in which the function is expanded in powers of x . Wherever there is a sum of different powers of x , we leave only the lowest-degree term when passing to small x .

Just as we considered, for small x , the order of decrease of functions equal to zero when $x = 0$, we can examine the behaviour of functions when x increases without bound, that is, as $x \rightarrow \infty$. If we are dealing with a polynomial, then it is obvious that for large x only the highest-degree term in x is of importance. We can speak of an order of increase of a function as x , as x^2 , etc.

A fact of prime importance is that the function e^x increases faster than any power x^n for x increasing without bound. To prove this, use the series expansion of e^x which, as was pointed out in Sec. 3.19, is valid for arbitrary x . We have

$$\frac{e^x}{x^n} = \frac{1}{x^n} + \frac{1}{x^{n-1}} + \dots + \frac{1}{n!} + \frac{x}{(n+1)!} + \frac{x^2}{(n+2)!} + \dots \quad (3.21-2)$$

For a given n and a sufficiently large x , the fraction $\frac{e^x}{x^n}$ will become as large as we wish due to terms with positive powers of x in the formula (3.21-2). Clearly, the same goes for the function e^{kx} with positive k : setting $kx = y$, we find that

$$\frac{e^{kx}}{x^n} = k^n \frac{e^{ky}}{(ky)^n} = k^n \frac{e^y}{y^n} \xrightarrow{y \rightarrow \infty} \infty \quad (3.21-3)$$

It remains to note that if $y \rightarrow \infty$ then $x \rightarrow \infty$ as well. Considering a fractional n does not change anything in the result. We conclude that the exponential function grows faster than any power function as the argument tends to ∞ .

As x tends to ∞ , the exponential function with negative exponent decreases, in the limit, faster than any negative power function. This assertion, for arbitrary n , is written

$$f = \frac{e^{-x}}{x^n} = x^n e^{-x} \rightarrow 0 \text{ as } x \rightarrow \infty$$

We cannot use the series expansion of e^{-x} for large x to prove this because the expansion is an alternating expansion. We therefore consider the reciprocal: $\frac{1}{f} = \frac{1}{x^n e^{-x}} = \frac{e^x}{x^n}$

According to (3.21-3), for arbitrary n the quantity $f^{-1} = e^x/x^n \rightarrow \infty$ as $x \rightarrow \infty$. From the fact that $f^{-1} \rightarrow \infty$ it follows that $f \rightarrow 0$, which completes the proof.

To summarize: in the limit, for large absolute values of the argument in the exponent, the exponential function e^x depends more strongly on x than any constant power of x ; e^x increases faster than x^n and e^{-x} decreases faster than x^{-n} . This is vividly illustrated in the table given below for x^5 and e^x :

x	1	3	5	10	20	50	100
x^5	1	32	3125	10^5	$3 \cdot 10^6$	$3 \cdot 10^8$	10^{10}
e^x	2.72	20	150	$2 \cdot 10^4$	$4 \cdot 10^8$	$5 \cdot 10^{21}$	10^{43}
$\frac{x^5}{e^x} = \frac{e^{-x}}{x^{-5}}$	0.37	1.6	24	5	0.01	10^{-13}	10^{-33}

Exercises

Find the following limits.

- $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$.
- $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2}$.
- $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$.
- $\lim_{x \rightarrow 0} \frac{e^x - 1 - \tan x}{x^3}$.
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$.
- $\lim_{x \rightarrow 0} \frac{\sin x - x}{x - \tan x}$.

Chapter 4

The Application of Differential and Integral Calculus to Geometry and the Investigation of Functions

4.1 INVESTIGATING MAXIMA AND MINIMA OF FUNCTIONS WITH THE AID OF THE SECOND DERIVATIVE

The problem of finding the value of x for which a given function $y = f(x)$ attains a maximum or minimum is not solvable in general form by the tools of elementary algebra.

In Chapter 2 we established that at points where a function has a maximum or minimum, the derivative is equal to zero. It was also shown there how, using the derivative y' , to establish exactly what the function has at the given point x_0 , a maximum, minimum or inflection. To do that we had to compute the values of y' for values of x close to x_0 on the right and on the left.

In this section we give another method of investigation that invokes the second derivative y'' , but here the only value of it we need is at $x = x_0$.

We will show that if at the point $x = x_0$

$$f'(x_0) = 0, \quad f''(x_0) < 0$$

then the function $f(x)$ at that point has a maximum. Indeed, from the condition $f'(x_0) = 0$ it follows that the tangent at the point $x = x_0$ is horizontal. From the inequality $f''(x_0) < 0$ it follows (also see Sec. 4.5, page 195) that the point $x = x_0$ is a point of convexity, that is to say that the graph near $x = x_0$ is located under the tangent line, and these two facts together mean that the function $f(x)$ has a maximum at $x = x_0$. Using the same reasoning, it is easy to see that if at the point $x = x_1$

$$f'(x_1) = 0, \quad f''(x_1) > 0$$

then at that point the function $f(x)$ has a minimum. These conclusions are also obtained when considering the Taylor series

$$f(x) = f(x_0) + f'(x_0) \cdot (x - x_0) + \frac{1}{2!} f''(x_0) \cdot (x - x_0)^2 + \dots \quad (4.1-1)$$

Let $f'(x_0) \neq 0$. For example, let $f'(x_0) > 0$. For x close to x_0 , the quantities $(x - x_0)^2$, $(x - x_0)^3$, ... may be neglected when

compared with $(x - x_0)$. We obtain

$$f(x) = f(x_0) + f'(x_0) \cdot (x - x_0)$$

or

$$f(x) - f(x_0) = f'(x_0) \cdot (x - x_0) \quad (4.1-2)$$

From this equation we see that for $x > x_0$, $f(x) - f(x_0) > 0$, that is, $f(x) > f(x_0)$. But if $x < x_0$, then $f(x) < f(x_0)$. Therefore when $x = x_0$ there is neither a maximum nor a minimum. Similarly when $f'(x_0) < 0$. But if $f'(x_0) = 0$, then we cannot neglect the term containing $(x - x_0)^2$. Ignoring terms in $(x - x_0)^3$, $(x - x_0)^4$, etc., as compared with $(x - x_0)^2$, we get, from (4.1-1)

$$f(x) = f(x_0) + \frac{1}{2!} f''(x_0) (x - x_0)^2$$

From this we see that for $f''(x_0) > 0$, $f(x) > f(x_0)$ irrespective of whether $x < x_0$ or $x > x_0$. Hence, $f(x_0)$ is less than any adjacent value of $f(x)$ and therefore $f(x_0)$ is a minimal value of the function. If $f''(x_0) < 0$, then $f(x) < f(x_0)$ and $f(x_0)$ is a maximal value of the function.

It may however happen that $f''(x_0) = 0$. How do we investigate the values of the function near $x = x_0$ in that case? We then have to take the next derivatives of the function $f(x)$. If $f'''(x_0) \neq 0$, then, neglecting $(x - x_0)^4$, etc., as compared with $(x - x_0)^3$, we get from (4.1-1)

$$f(x) = f(x_0) + \frac{1}{6} f'''(x_0) (x - x_0)^3$$

The difference $f(x) - f(x_0)$ changes sign depending on whether $x > x_0$ or $x < x_0$. For $x = x_0$ we have neither a maximum nor a minimum.

But if $f'''(x_0) = 0$ and $f^{(4)}(x_0) \neq 0$, then

$$f(x) = f(x_0) + \frac{1}{24} f^{(4)}(x_0) (x - x_0)^4$$

The sign of the expression $f(x) - f(x_0)$ is the same for $x < x_0$ and for $x > x_0$; it is determined by the sign of $f^{(4)}(x_0)$. If $f^{(4)}(x_0) > 0$, then we have a minimum, if $f^{(4)}(x_0) < 0$, then a maximum.

The attentive reader has probably already guessed that if for $x = x_0$ the first nonzero derivative is of odd order (first, third, fifth, etc.), then there is neither a maximum nor a minimum. But if the first nonzero derivative is of even order (second, fourth, etc.), then we either have a maximum or a minimum depending on the sign of the derivative.

Let us consider some examples.

1. It is required to build an open-at-the-top box of maximum volume, using a square sheet of tin with side $2a$ by cutting out equal squares at the corners of the sheet and then bending the tin to form

the sides of the box (Fig. 87). What is the length of the side of the squares to be cut out?

Let the sides of the cut-out squares be x . The volume of the box will depend on what kind of square we cut out and therefore it is natural to write $V(x)$. Let us compute this volume:

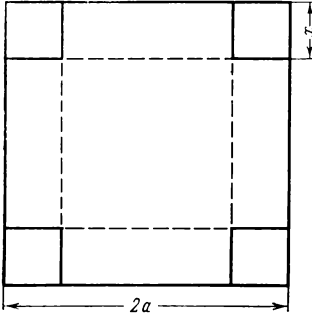


Fig. 87

$$V(x) = (2a - 2x)^2 x = 4(a - x)^2 x$$

Now find the derivative of this function:

$$V'(x) = -8(a - x)x + 4(a - x)^2$$

Solve the equation $V'(x) = 0$:

$$-8(a - x)x + 4(a - x)^2 = 0$$

$$\text{or } (a - x)(a - 3x) = 0$$

$$\text{whence } x_1 = a, x_2 = \frac{a}{3}.$$

We note at once that the value $x_1 = a$ does not interest us because then we wouldn't have a box by cutting the sheet in that fashion.

There remains $x = \frac{a}{3}$. Then

$$V\left(\frac{a}{3}\right) = 4 \cdot \frac{4a^2}{9} \cdot \frac{a}{3} = \frac{16a^3}{27}, \quad V'\left(\frac{a}{3}\right) = 0,$$

$$V''(x) = 8x - 8(a - x) - 8(a - x) = 24x - 16a,$$

$$V''\left(\frac{a}{3}\right) = -8a < 0$$

Consequently, the function $V(x)$ has a maximum at $x = \frac{a}{3}$.

To summarize, the maximum value is obtained for $x = \frac{a}{3}$, that is, we have to cut out squares whose sides are $1/3$ the side of the original square.

Let us compute $V(x)$ for several x close to $\frac{a}{3}$ and tabulate the results.

x	$V(x)$	x	$V(x)$
0.25a	0.562a ³	0.40a	0.576a ³
0.30a	0.588a ³	0.45a	0.540a ³
0.33a	0.592a ³		

From the table it is clear that small variations of x near $x = \frac{a}{3}$, i.e., near the value of x to which corresponds the maximum of the function, bring about very small changes in V , which means the function near the maximum varies very slowly.

This is also evident from the Taylor formula (4.1-1). Since $f'(x_0) = 0$ at the point of maximum, (4.1-1) takes the form

$$f(x) = f(x_0) + \frac{1}{2}f''(x_0) \cdot (x - x_0)^2 + \frac{1}{6}f'''(x_0) \cdot (x - x_0)^3 + \dots$$

The series does not contain $(x - x_0)$. The smallest power is $(x - x_0)^2$ and it is extremely small for x close to x_0 . In our example, a change in x by 9% (from $0.33a$ to $0.30a$) causes a change in V by less than 1%, while a change in x by 24% causes a change in V by 5%.

Therefore, if we are interested in the maximal value of a function and if we make a small error in finding x_0 from the equation $f'(x) = 0$ (for example, if we solved this equation in an approximate fashion), then this has but slight effect on the maximal value of the function. The values of the function for x close to x_0 will be very close to the value for $x = x_0$.

2. $y = A + B(x - a)^3$. Find the maxima and minima of the function.

$$y' = 3B(x - a)^2, \quad y'(a) = 0,$$

$$y'' = 6B(x - a), \quad y''(a) = 0,$$

$$y''' = 6B \neq 0$$

The first nonzero derivative is of order three. At the point $x = a$ there is neither a maximum nor a minimum but there is an inflection. This is evident from a glance at Fig. 88 (here, $A = 2$, $B = 1$, $a = 1$).

3. $y = A + B(x - a)^4$. Investigate the function for a maximum and minimum.

$$y' = 4B(x - a)^3, \quad y'(a) = 0,$$

$$y'' = 12B(x - a)^2, \quad y''(a) = 0,$$

$$y''' = 24B(x - a), \quad y'''(a) = 0,$$

$$y^{(4)} = 24B \neq 0$$

The first nonzero derivative is of order four. If $B > 0$, then it is positive, the function has a minimum. If $B < 0$, it has a maximum.

This conclusion could easily have been drawn directly. Indeed, for $B < 0$, $B(x - a)^4$ is negative for all $x \neq a$; for $x = a$ it is zero. Therefore, a positive quantity is always subtracted from A but for $x = a$ nothing is subtracted. This means that there is a maximum at $x = a$.

Similarly, if $B > 0$, then at $x = a$ we have a minimum.

4. Using available boards we can build a fence of length l metres. How can we fence out a rectangular yard of maximum area using for one side the wall of an adjacent building (Fig. 89)?

Let two sides have length x metres. Then the third side is $l - 2x$ metres long. The area of the yard is $S(x) = (l - 2x)x = -2x^2 + lx$, $S'(x) = -4x + l$. Solving the equation $S'(x) = 0$, we get $x = \frac{l}{4}$, $S''(x) = -4 < 0$. For $x = \frac{l}{4}$ $S(x)$ has a maximum.

We write down $S(x)$ using formula (4.1-1) and setting $x_0 = \frac{l}{4}$

$$S(x) = \frac{l^2}{8} - 2 \left(x - \frac{l}{4} \right)^2 \quad (4.1-3)$$

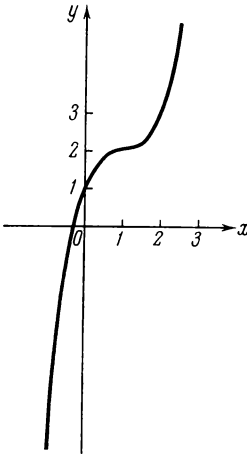


Fig. 88

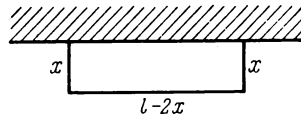


Fig. 89

Since $S(x)$ is a polynomial of second degree, (4.1-3) is an exact equation (see Sec. 3.17). It is immediately evident that $S(x)$ has a maximum for $x = \frac{l}{4}$.

The equation (4.1-3) can also be obtained without resorting to higher mathematics. Indeed, suppose we have to seek the maximum (or minimum) of a second-degree polynomial

$$y = ax^2 + bx + c \quad (4.1-4)$$

Transform the polynomial as follows:

$$\begin{aligned} y &= a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) = a \left[x^2 + 2 \frac{b}{2a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a} \right] \\ &= a \left[\left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right] = a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a} \end{aligned}$$

Thus

$$y = a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a} \quad (4.1-5)$$

Noting that $\left(x + \frac{b}{2a} \right)^2 \geq 0$ for all x , equality occurring only for

$x = -\frac{b}{2a}$, we find from (4.1-5), that y has a maximum if $a < 0$ and this maximum is at $x = -\frac{b}{2a}$; y has a minimum if $a > 0$ and this minimum is at $x = -\frac{b}{2a}$.

We obtained the value $x = -\frac{b}{2a}$ by performing special artificial manipulations with the polynomial (4.1-4). Using the derivative, we find $x = -\frac{b}{2a}$ automatically, as witness: equating to zero the derivative of (4.1-4), we get $2ax + b = 0$, whence $x = -\frac{b}{2a}$. The second derivative of (4.1-4) is $y'' = 2a$. And so the question of whether there is a maximum or a minimum is settled depending on the sign of a .

5. A man walking from A wishes to reach a river (the straight line A_1B_1 in Fig. 90) and then go to B . How can he do this by traveling the shortest possible distance?

We have $AA_1 = a$, $BB_1 = b$, $A_1B_1 = c$; the numbers a , b , c are given. Let the broken line AMB be the path taken. Our aim is to find out for what position of the point M on the line A_1B_1 is this path the shortest. To determine the position of M it suffices to specify the distance from M to the point A_1 , the foot of the perpendicular dropped from A onto the straight line representing the river. Denote this distance A_1M by x . Then

$$AM = \sqrt{a^2 + x^2}, \quad MB = \sqrt{b^2 + (c - x)^2}$$

The path traversed is then denoted by $s(x)$,

$$s(x) = \sqrt{a^2 + x^2} + \sqrt{b^2 + (c - x)^2} \quad (4.1-6)$$

and we find

$$s'(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{c - x}{\sqrt{b^2 + (c - x)^2}}$$

Equating $s'(x)$ to zero we get

$$\frac{x}{\sqrt{a^2 + x^2}} = \frac{c - x}{\sqrt{b^2 + (c - x)^2}} \quad (4.1-7)$$

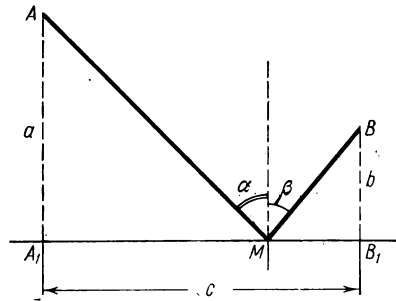


Fig. 90

It is easy to solve this equation. Squaring both sides, we have

$$\frac{x^2}{a^2 + x^2} = \frac{(c-x)^2}{b^2 + (c-x)^2}$$

or

$$x^2 b^2 + x^2 (c-x)^2 = a^2 (c-x)^2 + x^2 (c-x)^2,$$

$$x^2 b^2 = a^2 (c-x)^2, \quad \frac{x^2}{(c-x)^2} = \frac{a^2}{b^2}$$

Taking the root of both members, we find

$$\frac{x}{c-x} = \pm \frac{a}{b}$$

whence

$$x_1 = \frac{ac}{a+b}, \quad x_2 = \frac{ac}{a-b}$$

Substituting the values x_1 and x_2 into the original equation (4.1-7), we see that the second root does not satisfy the equation. This is an extraneous root generated by the squaring process. Thus, $x = \frac{ac}{a+b}$.

It is possible, however, to give a pictorial geometrical representation that will enable us to obtain the answer without solving the equation. Rewrite the condition (4.1-7) as

$$\frac{A_1 M}{AM} = \frac{MB_1}{MB} \quad (4.1-8)$$

But $\frac{A_1 M}{AM} = \cos \angle A_1 M A = \sin \alpha$. Similarly, $\frac{MB_1}{MB} = \cos \angle B_1 M B = \sin \beta$. The condition (4.1-8) yields

$$\sin \alpha = \sin \beta \quad (4.1-9)$$

But α and β are acute angles, therefore from (4.1-9) we get

$$\alpha = \beta$$

The man must take the path of a ray of light bounced off the river: the angle of incidence is equal to the angle of reflection.

For a complete solution to the problem it remains to demonstrate that for such a position of point M the distance is indeed minimal (and not maximal). This can be done by computing the second derivative of (4.1-6).

But it is also possible to reason differently. From the expression (4.1-6) for $s(x)$ we see that $s(x)$ is positive for any x . Then $s(x)$ increases without bound together with the growth in the absolute value of x , irrespective of whether $x > 0$ or $x < 0$. And since $s'(x)$ vanishes only for one value of x , it is clear that at this value of x the function $s(x)$ will then have a minimum. If in the interval at hand the first derivative has only one root, then obvious considera-

tions frequently permit dispensing with a formal investigation by means of the second derivative.

Problem (5) can be solved in a purely geometrical manner without resorting to methods of higher mathematics. Referring to Fig. 91, extend the segment AA_1 to A' ($A_1A' = AA_1$) and join A' to B . Then $AM = A'M$ since triangle $AA_1M =$ triangle $A_1A'M$. Therefore $AM + MB = A'M + MB = A'B$. For any other point D on the segment A_1B_1 we will have $AD + DB = A'D + DB$ and $A'D + DB > A'B$ since a polygonal line is longer than any segment of a straight line. Consequently, the desired point M is the point of intersection of the straight lines $A'B$ and A_1B_1 , whence follows $\alpha = \beta$.

The last two examples show that certain problems involving the finding of maxima and minima may be solved by the tools of elementary mathematics. But, first, not all problems can be tackled without appealing to higher mathematics and, second, the solution by elementary means requires a good deal of ingenuity, whereas higher mathematics offers a standard method of solution of such problems.

Do not get the idea that higher mathematics does not require ingenuity! It will now simply be used for still harder problems.

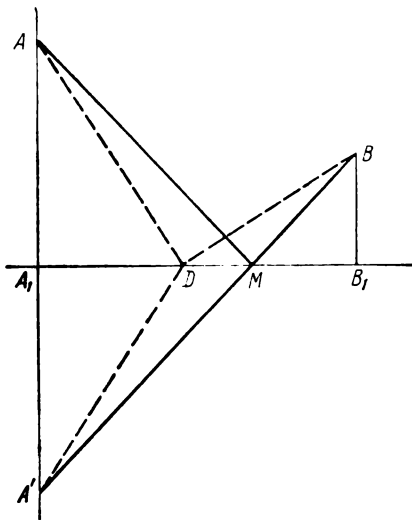


Fig. 91

Exercises

1. We want to build a box out of a rectangular sheet of tin of sides a and b cutting out equal squares at the corners. What must the side of a square be so that the box is of maximum volume?
2. Inscribe in an acute-angled triangle with base a and altitude H a rectangle of the largest area, two vertices of which lie on the base of a triangle.
3. Determine the greatest area of a rectangle which can be inscribed in a circle of radius R .
4. For what radius of the base and for what altitude will a closed cylindrical can of a given volume V have a minimum total surface area?
5. Two bodies are moving along the sides of a right angle with constant speeds v_1 and v_2 (metres per second) in the direction of the vertex, from which, at the beginning, the first was distant a metres and the second, b metres. How many seconds after they started will the distance between the bodies be a minimum?

6. Prove that the product of two positive numbers whose sum is a constant is greatest when the factors are equal.

7. A straight line l divides a plane into two parts (medium I and medium II). A body moves in medium I at a rate of v_1 , in medium II at a rate of v_2 . What path must a point take so as to get, in minimum time, from a given point A of medium I to a given point B of medium II?

4.2 OTHER TYPES OF MAXIMA AND MINIMA. SALIENT POINTS AND DISCONTINUITIES

Up to now we have said that maxima and minima of a function occur at values of x for which the first derivative vanishes. However, maxima and minima can also arise for values of the argument that

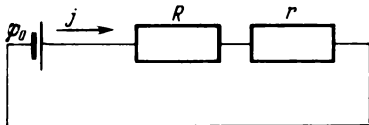


Fig. 92

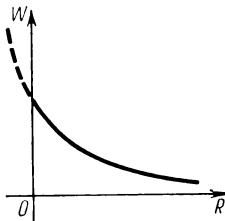


Fig. 93

do not make the first derivative vanish. Let us consider the following problem.

Determine for what value of resistance R in series with resistance r that interests us, the maximum power is released on r (Fig. 92). The resistance r and the battery voltage φ_0 are taken to be constant. We get the current j in the circuit using Ohm's law:

$$j = \frac{\varphi_0}{R+r}$$

The power $W(R) = j\varphi_r$ where φ_r is the voltage drop across the resistance r . By Ohm's law, $\varphi_r = jr$ and so

$$W(R) = \frac{\varphi_0^2 r}{(R+r)^2}$$

To determine the maximum of $W(R)$, solve the equation $\frac{dW}{dR} = 0$ to get

$$-2\varphi_0^2 \frac{r}{(R+r)^3} = 0$$

This equation does not have a solution. But does this mean that the power can increase without bound and that the problem of maximum power does not have a solution? From the physical essence of the problem it is clear that the power will be a maximum when

$R = 0$ (in which case $W = \frac{\varphi_0^2}{r}$). Why did we not get the value $R = 0$ from the equation $\frac{dW}{dR} = 0$? .

To see why, consider the graph of $W(R)$ (Fig. 93).

It is evident from the graph that if R could assume negative values, then for $R = 0$ there would be no maximum. But negative R have no meaning. Every physical problem presupposes that $R \geq 0$. Thus, the quantity W has a maximum at $R = 0$ because the range of the argument is bounded. This means that if the range of the argument is bounded, then we must take into consideration the boundary values of the argument when testing for maximum and minimum.

When the maximum (minimum) is attained at the end point of the range of the argument, the series

$$f(x) - f(x_0) = f'(x_0)(x - x_0) + \frac{1}{2!} f''(x_0)(x - x_0)^2 + \dots$$

may begin with $(x - x_0)$ instead of $(x - x_0)^2$. Therefore, if the maximum of a function is obtained when $x = x_0$ and we have departed somewhat from x_0 , then we may err considerably in determining the maximum. This error will be proportional to $(x - x_0)$ and not to $(x - x_0)^2$ as was the case in Sec. 4.1. Hence, even a slight departure from the value of the argument that yields a maximal value is undesirable in this case.

It is assumed in the case at hand that the function $f(x)$ is defined by a formula for $x < x_0$ as well, but the values of the function for $x < x_0$ do not interest us in this concrete problem (they are devoid of any *physical* meaning).

It may happen that $f(x)$ is simply meaningless for certain values of the argument. For example, if a function contains an even-degree root, say a square root, then the range of the argument is as a rule bounded (the radicand cannot be negative). Hence, boundary values are values of the argument that make the radicand vanish. They merit special consideration when investigating for maxima.

Consider the following example. Let

$$y = a - \sqrt{b - x}, \quad y' = \frac{1}{2\sqrt{b - x}} \quad (4.2-1)$$

Although y' does not vanish, the investigation is not over. The value $x = b$ makes the radicand vanish. From (4.2-1) we see that $y = a$ for $x = b$; but if $x < b$ then $y < a$ since a positive number is subtracted from a ($\sqrt{b - x}$ is understood to be the positive root). Therefore, y has a maximum at $x = b$.

A maximum (or minimum) may also occur at interior points where the derivative does not vanish. This is the case when the curve has a salient point (corner). Such points occur, in particular, when

the curve consists of two parts described by different formulas for $x < x_0$ and for $x > x_0$. Here is an instance of a physical problem of this nature. Suppose a teakettle is being heated on an electric hot plate. Our problem is to determine the instant of time when the teakettle has the greatest amount of heat. For the sake of simplicity, we assume that the coefficient of efficiency of the hot plate is 100%, which means that all the heat is delivered to the teakettle. We put

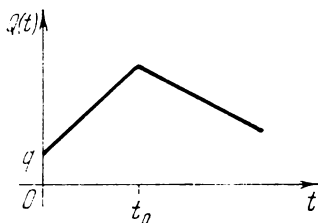


Fig. 94

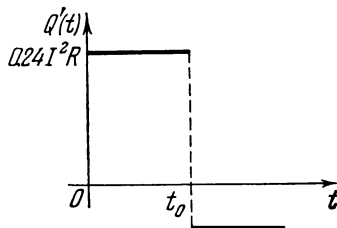


Fig. 95

the teakettle on to heat at time $t = 0$, at which time it had q calories of heat (for zero we take the thermal energy of water at 0°C). The quantity of heat released by the hot plate is given by

$$Q = 0.24I^2Rt$$

where I is the current in amperes, R the resistance in ohms, t the time in seconds and Q is in calories.

Thus, at time t the quantity of heat in the teakettle is

$$Q = q + 0.24I^2Rt$$

At time $t = t_0$ the water in the teakettle begins to boil. At this time a quantity $q + 0.24I^2Rt_0$ of heat has accumulated.

When the water boils it begins to turn into steam (the formation of steam starts at less than 100°C but we ignore this fact). The formation of one gram of steam requires 539 calories. Therefore, denoting by dm the quantity of water that has boiled off in time dt , we get

$$dm = \frac{0.24I^2R dt}{539}$$

And so in 1 second a total of $\frac{dm}{dt} = \frac{0.24I^2R}{539}$ grams of water boil away.

The amount of water that boils away in 1 second carries out of the teakettle $\frac{dQ_1}{dt} = 100 \cdot \frac{dm}{dt} = \frac{24}{539} I^2R = 0.041I^2R$ calories. Therefore, by time t ($t > t_0$) the boiled-off water has carried out of the teakettle a total of $Q_1 = 0.041I^2R (t - t_0)$ calories.

Thus, the quantity of heat in the teakettle is

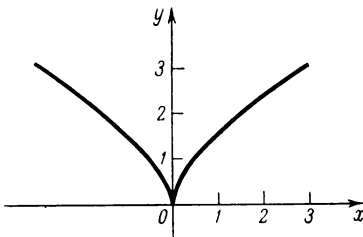
$$Q = q + 0.24I^2Rt$$

if $t \leq t_0$ (prior to onset of boiling);

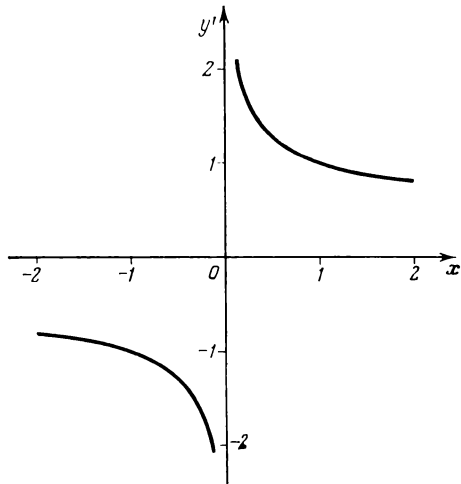
$$\begin{aligned} Q &= q + 0.24I^2Rt_0 - 0.041I^2R(t - t_0) \\ &= q + I^2R(0.281t_0 - 0.041t) \end{aligned}$$

if $t \geq t_0$ (after the water has begun to boil). The graph of $Q(t)$ is shown in Fig. 94. It is clear from the drawing that $Q(t)$ has a maximum when $t = t_0$ although the derivative does not vanish for this value of t .

The derivative is discontinuous at $t = t_0$. Indeed, $Q'(t) = 0.24I^2R$ if $t < t_0$;



(a)



(b)

Fig. 96

$Q'(t) = -0.041I^2R$ if $t > t_0$. The graph of the derivative is shown in Fig. 95.

This example shows that a maximum may occur if the derivative is discontinuous, that is, if the curve has a salient point (corner).

Finally, from Fig. 96 a (the curve here is that given by the function

$y = x^{\frac{2}{3}} = \sqrt[3]{x^2}$) it is clear that a minimum (or maximum) can occur for those values of the argument x_0 at which the derivative has an infinite discontinuity. This point of the curve is called a cusp. The graph of the derivative for this case is depicted in Fig. 96 b. Here, as in the case of an ordinary minimum, $y' < 0$ for $x < x_0$; the function falls off as x approaches x_0 from the left. For $x > x_0$, $y' > 0$. The function increases as x increases after the value $x = x_0$ has been passed. But at $x = x_0$ the derivative becomes meaningless. It becomes arbitrarily large for x close to x_0 and $x > x_0$; it becomes arbitra-

rily large in absolute value but negative for x close to x_0 and $x < x_0$. The maxima and minima attained for those values of the argument when the derivative is discontinuous are called cuspidal.

In connection with this consideration of singular points on curves, primarily salient points (see Fig. 94), we can make precise our reasoning that led us to the concept of the derivative. In Chapter 2 we considered smooth curves without specially stipulating this fact.

The derivative $y'(t)$ taken at the point t is equal to the limit of the ratio

$$\frac{y(t_2) - y(t_1)}{t_2 - t_1} \quad (4.2-2)$$

as t_2 and t_1 tend to t (it is clear then that the difference $t_2 - t_1$ tends to zero). We have specially emphasized that this limit does not depend on how t_2 and t_1 are chosen; they can both be greater than t or both smaller than t , or one greater and the other smaller than t , or one equal to t and the other greater or smaller than t . Indeed, when we take

$$\frac{y(t + \Delta t) - y(t)}{\Delta t} \quad \text{and} \quad \Delta t > 0$$

then this expression corresponds to the case where $t_1 = t$, $t_2 = t + \Delta t > t$ in (4.2-2). When we take

$$\frac{y(t) - y(t - \Delta t)}{\Delta t}, \quad \Delta t > 0$$

this corresponds to $t_1 = t - \Delta t < t$, $t_2 = t$ in (4.2-2). Finally, we also computed the derivative as the limit of the ratio

$$\frac{y\left(t + \frac{\Delta t}{2}\right) - y\left(t - \frac{\Delta t}{2}\right)}{\Delta t}$$

which corresponds to $t_1 = t - \frac{\Delta t}{2} < t$, $t_2 = t + \frac{\Delta t}{2} > t$.

In the case of a smooth curve, all three expressions yield the same limit, which is equal to the derivative at the given point. The situation changes when we deal with a curve with a salient point. If by t_0 we denote the value of t at which the salient point occurs, then, taking

$$\frac{y(t_0 + \Delta t) - y(t_0)}{\Delta t}$$

we get, for Δt positive and tending to zero, a definite quantity—in the example on page 185 this quantity is equal to $-0.041I^2R$ —which is called the “derivative on the right”. Taking

$$\frac{y(t_0) - y(t_0 - \Delta t)}{\Delta t}$$

we get, for Δt positive and tending to zero, another limit, equal in the aforementioned example to $+0.24I^2R$. It is called the “derivative on the left”.

Taking t_2 and t_1 on different sides of t_0 we can obtain different values of the ratio (4.2-2) as $t_2 \rightarrow t_0$, $t_1 \rightarrow t_0$. To summarize, then,

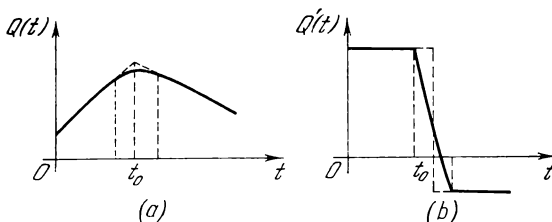


Fig. 97

the derivative does not have a definite value at the salient point, but we can determine the derivative on the left and the derivative on the right.

In Chapter 2, when we first began studying derivatives, we simplified matters by not assuming all the time that a definite value of the derivative that is independent of the mode of approach of Δt to zero (from the left or from the right) exists only for points at which the curve is smooth. As is evident from Fig. 95, the curve of the derivative $y'(t)$ has a *discontinuity* at the point where the curve $y(t)$ has a *salient point*. Now if we replace the salient point on the curve $y(t)$ by an arc of small radius that is tangent to the curve on the left and on the right (what draftsmen call *conjugation*), then on the range of t where the curve $y(t)$ is replaced by the arc, the curve $y'(t)$ changes direction sharply (Fig. 97).

If the curve $y(t)$ has a *discontinuity* at the point t_0 (see Fig. 98), then we can say that at t_0 the derivative $y'(t)$ is infinite. Indeed, if the discontinuity is replaced by a variation of y from y_1 to y_2 on a small interval from $t_0 - \varepsilon$ to $t_0 + \varepsilon$, then on this interval the derivative is equal to $\frac{y_2 - y_1}{2\varepsilon}$, which is to say, it is very large, increasing as ε decreases (Fig. 98).

Now how does the integral $\int_a^b y(t) dt$ behave if the function $y(x)$ is not smooth? If the function has a salient point, then no new problems arise when we compute the area bounded by the curve $y(t)$. In Sec. 2.10, we split up the definite integral—the area under the curve—into a sum of areas of rectangular strips of the form

$$y(t_n)(t_{n+1} - t_n) \text{ or } y(t_{n+1})(t_{n+1} - t_n)$$

In the limit, as the intervals, that is the differences $(t_{n+1} - t_n)$, are decreased, it makes no difference whether one takes $y(t_n)$ or $y(t_{n+1})$ either in the case of a smooth curve $y(t)$ or in the case of a curve $y(t)$ with a salient point.

If the curve $y(t)$ is discontinuous at the point $t = t_0$, but remains bounded, then for the interval that contains the discontinuity $(t_n < t_0 < t_{n+1})$ the quantities $y(t_n)$ and $y(t_{n+1})$ remain distinct no matter how t_n and t_{n+1} approach one another. To summarize, then, in the expression of the integral as a sum, the value of one of

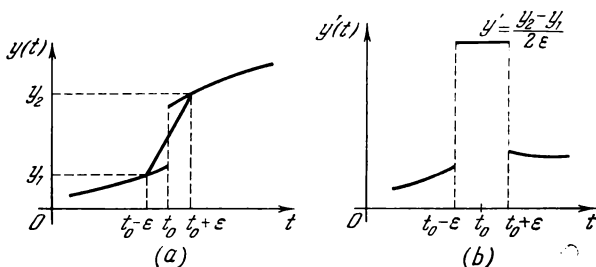


Fig. 98

the summands in this case depends on how the sum is taken: by formula (2.8-1) or (2.8-2). However, as the interval $t_{n+1} - t_n$ tends to zero, the summand itself tends to zero, and so the limit of the sum, that is the integral, has a definite value (independent of the mode of computing the sum) also in the case where the integrand has a discontinuity in the region of integration.

The relationship between the integral and the derivative is likewise preserved.

Referring to Figs. 94 and 95, let us take the function $Q'(t)$, the graph of which is given in Fig. 95, $Q'(t) = f(t)$. Then the function $Q(t)$, the graph of which is given in Fig. 94, is an indefinite integral: $Q(t) = \int f(t) dt$. This example illustrates that a discontinuity in the integrand function $f(t)$ leads to a salient point in the integral $Q(t)$ of this function.

The definite integral of a function with a finite discontinuity can be found with the aid of the indefinite integral by the general rule

$$\int_a^b f(t) dt = Q(b) - Q(a)$$

We may continue: consider Fig. 98. We can say that for a function tending to infinity on an interval tending to zero (Fig. 98b) the integral is a discontinuous function (Fig. 98a). However, in this case,

we must make precise the law by which the function tends to infinity and the interval to zero. We will not dwell on that here. Examples of this kind lead to the concept of the delta function (δ -function) (see Chapter 9).

Exercises

1. Find the smallest value of the function $y = x^2 - 2x + 3$ as x varies from 2 to 10.

Find the cuspidal maxima of the following functions.

2. $y = (x-5)\sqrt[3]{x^2}$.

3. $y = 1 - \sqrt[3]{x^2}$.

4.3 COMPUTING AREAS

In Chapter 2 we showed that the value of a definite integral $\int_a^b f(x) dx$ yields the area of a figure bounded from above by the curve $y = f(x)$, from below by the x -axis and on the sides by the vertical lines $x = a$ and $x = b$ (Fig. 99). Thus, being able to find definite integrals enables us to use standard techniques in computing various areas, whereas elementary mathematics only allows for calculating the areas of rectilinear figures and the circle.

Let us find the area of a figure bounded from above by the curve $y = cx^n$ ($n > 0$), from below by the x -axis, and on the right by the straight line $x = x_0$ (Fig. 100, $n=2$, $c = 0.25$):

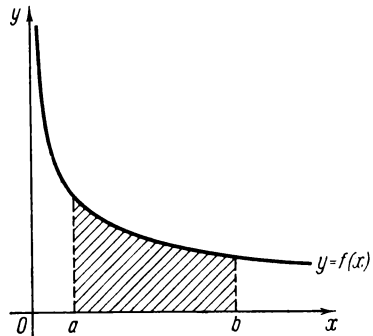


Fig. 99

$$S = \int_0^{x_0} cx^n dx = \left[\frac{cx^{n+1}}{n+1} \right]_0^{x_0} = \frac{cx_0^{n+1}}{n+1} \quad (4.3-1)$$

Let us rewrite formula (4.3-1) as

$$S = \frac{1}{n+1} cx_0^n x_0$$

or, since $cx_0^n = y(x_0)$, we have

$$S = \frac{1}{n+1} y(x_0) x_0 \quad (4.3-2)$$

The quantities y and x have the dimensions of length. From (4.3-2) we see that S is indeed measured in units of area. We see that the area is, as to order of magnitude, $y(x_0) \cdot x_0$. The area differs from this product solely in the factor $\frac{1}{n+1}$, which, as to order of magnitude, is close to unity for n not too large.

In the next example, we find the area bounded from above by the curve

$$y = ce^{-\frac{x}{a}} \quad (a > 0) \quad (4.3-3)$$

from below by the x -axis, on the left by the straight line $x = x_0$ and on the right by the straight line $x = A$ ($A > x_0$) (Fig. 101). This area is

$$S_A = \int_{x_0}^A ce^{-\frac{x}{a}} dx = -cae^{-\frac{x}{a}} \Big|_{x_0}^A = ca \left[e^{-\frac{x_0}{a}} - e^{-\frac{A}{a}} \right] \quad (4.3-4)$$

If A is great compared with x_0 , then $e^{-\frac{x_0}{a}} \gg e^{-\frac{A}{a}}$. It will be seen from (4.3-4) that increasing A hardly at all changes S_A . As A increases

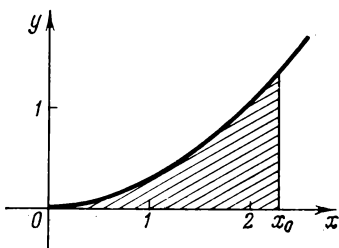


Fig. 100

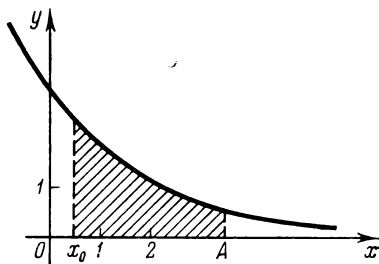


Fig. 101

without bound, the value of $e^{-\frac{A}{a}}$ approaches zero without bound. And so we can speak of the area of the figure in Fig. 101 as being unbounded on the right. The area here is

$$S_\infty = \int_{x_0}^{\infty} ce^{-\frac{x}{a}} dx = cae^{-\frac{x_0}{a}} = y(x_0)a \quad (4.3-5)$$

In formula (4.3-3), the exponent must be a dimensionless number. For this reason the dimensions of a are the same as those of x ; they are the dimensions of length; y also has the dimensions of length. The dimensions of S are those of area.

It turns out that the area under one arch of a sine curve (Fig. 102) is expressed very simply. It is

$$S = \int_0^{\pi} \sin x \, dx = -\cos x \Big|_0^{\pi} = 2$$

Let us determine the area S of an ellipse. Note that by virtue of symmetry it suffices to find the area S_1 of that portion which lies in the first quadrant and then multiply by 4. Thus, $S = 4S_1$. To



Fig. 102

compute S_1 we find y from the equation of the ellipse. The equation of an ellipse with semiaxes a and b and with centre at the origin is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(also see Sec. 1.7). Since $y > 0$ in the first quadrant,

$$y = +\frac{b}{a} \sqrt{a^2 - x^2}$$

and so

$$S_1 = \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx \quad (4.3-6)$$

The integral (4.3-6) can easily be found by making the change of variable $x = a \sin t$ to get

$$\begin{aligned} \int_0^a \sqrt{a^2 - x^2} \, dx &= \int_0^{\frac{\pi}{2}} a \sqrt{1 - \sin^2 t} \, a \cos t \, dt = \int_0^{\frac{\pi}{2}} a^2 \cos^2 t \, dt \\ &= a^2 \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2t}{2} \, dt = a^2 \left[\frac{t}{2} + \frac{\sin 2t}{4} \right]_0^{\frac{\pi}{2}} = \frac{\pi a^2}{4} \quad (4.3-7) \end{aligned}$$

Using this equation, from (4.3-6) we get $S_1 = \frac{b}{a} \frac{\pi a^2}{4} = \frac{\pi ab}{4}$. The area of the entire ellipse is $S = \pi ab$. If $a = b = r$, then we have $S = \pi r^2$ (the area of a circle) in complete accord with the fact that for $a = b = r$ an ellipse becomes a circle.

Note an important circumstance. In Chapter 2 we already pointed out that the area (the integral) can be either positive or negative.

This calls for a certain amount of care when finding areas. Suppose we want to know the amount of paint needed to paint an area bounded by two arches of a sine curve and the x -axis (see Fig. 58) if unit area requires a grams of paint. As was shown on page 91, one integral cannot be used to compute the entire area. We have to take separate integrals over the intervals from 0 to π and from π to 2π .

Generally, if the integrand $y = f(x)$ changes sign, then to solve the problem in paint consumption one has to split the interval of integration into parts in which $f(x)$ preserves sign, then evaluate the integral over the separate parts, and finally sum the absolute values of the resulting integrals.

Let us find the area of a figure bounded from above by the curve $y = x^n e^{-x}$ (n a positive integer), from below by the x -axis for $x \geq 0$ (the figure is not bounded on the right).

This area is expressed by the integral

$$S = \int_0^{\infty} x^n e^{-x} dx$$

To compute this integral we use integration by parts setting

$$e^{-x} dx = dg, \quad x^n = f$$

Then

$$g = -e^{-x}, \quad df = nx^{n-1} dx,$$

$$\int x^n e^{-x} dx = -x^n e^{-x} + \int nx^{n-1} e^{-x} dx$$

and so

$$\int_0^{\infty} x^n e^{-x} dx = [-x^n e^{-x}]_0^{\infty} + \int_0^{\infty} nx^{n-1} e^{-x} dx$$

In Sec. 3.21 it was established that $x^n e^{-x} = \frac{x^n}{e^x} \xrightarrow{x \rightarrow \infty} 0$. Since $x^n e^{-x} = 0$ for $x = 0$, it follows that $[-x^n e^{-x}]_0^{\infty} = 0$, hence

$$\int_0^{\infty} x^n e^{-x} dx = \int_0^{\infty} nx^{n-1} e^{-x} dx$$

Denote $\int_0^{\infty} x^n e^{-x} dx = I_n$. Then $I_n = nI_{n-1}$.

Using integration by parts with respect to I_{n-1} , we likewise get $I_{n-1} = (n-1)I_{n-2}$ and so forth. For this reason

$$I_n = n(n-1)(n-2) \dots 3 \cdot 2 I_0$$

But $I_0 = \int_0^{\infty} e^{-x} dx$. We get the value of this integral by putting $c = 1$, $a = 1$, $x_0 = 0$ in (4.3-5). Then $I_0 = e^0 = 1$. Thus

$$I_n = \int_0^{\infty} x^n e^{-x} dx = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 = n! \quad (4.3-8)$$

Exercises

1. Find the area of a figure bounded by a single arch of the curve $y = \sin^2 x$ and by the x -axis (the graph of the function $y = \sin^2 x$ is given in Fig. 103).

Hint. Take advantage of the formula $\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$.

2. Do the same for the curve $y = \cos^2 x$.

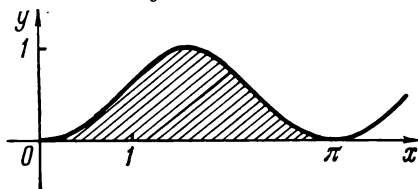


Fig. 103

3. Find the area of a figure bounded from above by the curve $y = x(1-x)$ and from below by the x -axis.

4. Find the areas into which the parabola $y = \frac{1}{2}x^2$ divides the circle $x^2 + y^2 = 8$.

5. Find the amount of paint needed to cover an area bounded by the curve $y = \frac{x}{1+x^2}$, the x -axis and the vertical lines $x = 1$ and $x = -1$.

6. The same for an area bounded by the curve $y = x^3 + 2x^2 - x - 2$ and the x -axis.

Hint. First construct the graph of the function

$$y = x^3 + 2x^2 - x - 2$$

7. Find the area of the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$.

4.4 MEAN VALUES

The reader will recall that the mean value of a function $f(x)$ on an interval from $x = a$ to $x = b$ is

$$\bar{f}(a, b) = \frac{\int_a^b f(x) dx}{b-a} \quad (4.4-1)$$

We note two simple facts relating to mean values.

1. The mean value of a constant on any interval is that constant. This is clear physically: if the instantaneous velocity does not change, then the average velocity over an interval is equal to the constant value of the instantaneous velocity.

This is also very simply obtained from formula (4.1-1):

$$\bar{C}(a, b) = \frac{\int_a^b C dx}{b-a} = \frac{C(b-a)}{b-a} = C$$

2. The mean value of a sum of two functions is equal to the sum of the mean values of the summands:

$$\overline{y_1 + y_2} = \bar{y}_1 + \bar{y}_2$$

Indeed,

$$\begin{aligned} \overline{y_1 + y_2} &= \frac{1}{b-a} \int_a^b [y_1(x) + y_2(x)] dx \\ &= \frac{1}{b-a} \int_a^b y_1(x) dx + \frac{1}{b-a} \int_a^b y_2(x) dx = \bar{y}_1 + \bar{y}_2 \end{aligned}$$

Let us find the mean value of the function $y = \sin x$ on the interval from $x = 0$ to $x = \pi$:

$$\bar{y}(0, \pi) = \frac{\int_0^\pi \sin x dx}{\pi - 0} = \frac{2}{\pi} = 0.637$$

The mean value of $y = \sin x$ on the interval between $x = 0$ and $x = b$ is

$$\bar{y}(0, b) = \frac{\int_0^b \sin x dx}{b-0} = \frac{1 - \cos b}{b} \quad (4.4-2)$$

What will happen if we increase the number b without bound, that is, if we increase without bound the interval?

In (4.4-2) the numerator does not exceed 2 for arbitrary b (it is equal to 2 if $\cos b = -1$, that is, for $b = \pi, 3\pi, 5\pi, 7\pi, \dots$). The denominator in (4.4-2) will increase without bound, and so the fraction as a whole will approach zero without bound. Therefore, the larger the interval, the closer to zero is the mean value of $\sin x$.

We will show that the mean value of the function $y = \cos x$ on an infinite interval is also equal to zero. Indeed,

$$\bar{y}(0, b) = \frac{\int_0^b \cos x dx}{b-0} = \frac{\sin x|_0^b}{b} = \frac{\sin b}{b} \quad (4.4-3)$$

Now if we increase the number b without bound, then the denominator of (4.4-3) will increase without bound and the numerator will remain not more than unity. Hence the whole fraction tends to zero: $\overline{y}(0, \infty) = 0$.

Literally in the same way we find that the mean value of the function $y = \cos kx$ is also equal to zero on an infinite interval.

Let us find the mean value of the function $y = \sin^2 x$ on the infinite interval from $x = 0$ to $x = \infty$.

Using a familiar formula of trigonometry, we have

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

whence

$$\overline{\sin^2 x} = \overline{\frac{1}{2}} - \overline{\frac{1}{2} \cos 2x} = \frac{1}{2} - 0 = \frac{1}{2}$$

Taking advantage of the formula $\sin^2 x + \cos^2 x = 1$, we get the mean value of $\cos^2 x$ on the same interval:

$$\overline{\cos^2 x} = 1 - \overline{\sin^2 x} = 1 - \frac{1}{2} = \frac{1}{2}$$

Exercises

1. Find the mean value of the function $y = x^n$ on the interval from $x = 0$ to $x = x_0$.

2. Find the mean value of the function $y = Ce^{kx}$ on the interval in which y varies from n to m ; express this mean value in terms of n and m , eliminating C and k from the answer. Investigate the resulting expression when m is close to n : $m = n + v$, $v \ll n$.

3. Find the mean values of the functions $y = \sin^2 x$ and $y = \cos^2 x$ on the intervals: (a) $x = 0$ to $x = \pi$, (b) $x = 0$ to $x = \frac{\pi}{4}$.

4. Determine the period of the function $y = \sin(\omega t + \alpha)$ where ω , α are constants. Find the mean value of the function y^2 over its period.

4.5 ARC LENGTH AND CURVATURE

Let us pose the problem of finding the arc length s of a curve $y = f(x)$ between points $x = a$ and $x = b$ (Fig. 104).

We replace the length of a small portion of AC by straight-line segment connecting A and C . We will only consider curves without discontinuities and salient points. By the Pythagorean theorem,

$$\Delta s = \sqrt{(\Delta x)^2 + (\Delta y)^2} = (\Delta x) \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

whence

$$\frac{\Delta s}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \quad (4.5-1)$$

Pass to the limit in (4.5-1) as $\Delta x \rightarrow 0$; $\frac{\Delta y}{\Delta x}$ becomes the derivative $y' = f'(x)$, where $y = f(x)$ is the equation of the curve. We get*

$$ds = \sqrt{1 + f'^2(x)} dx$$

The desired arc length is

$$s = \int_a^b \sqrt{1 + f'^2(x)} dx \quad (4.5-2)$$

Because of the radical under the integral sign it is rarely possible to take the integral in (4.5-2) with ease.

Some cases follow in which the computations can be completed with relative ease.

1. The circumference of a circle. We seek the circumference of the circle $x^2 + y^2 = R^2$. We will find the length s of one-fourth of the circumference in the first quadrant and then multiply by 4.

From the equation of the circle, we have

$$y = \sqrt{R^2 - x^2}, \quad y' = \frac{-x}{\sqrt{R^2 - x^2}}$$

By formula (4.5-2),

$$s = \int_0^R \sqrt{1 + \frac{x^2}{R^2 - x^2}} dx = \int_0^R \frac{R dx}{\sqrt{R^2 - x^2}} \quad (4.5-3)$$

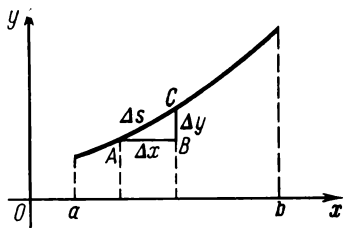


Fig. 104

We introduce a new variable t : $x = R \sin t$; then $dx = R \cos t dt$ and from (4.5-3) we get (see Sec. 3.16)

$$s = \int_0^{\frac{\pi}{2}} R dt = \frac{\pi R}{2}$$

whence we obtain the circumference $C = \frac{\pi R}{2} \cdot 4 = 2\pi R$.

2. Catenary curve. This is a curve whose equation is

$$y = \frac{a}{2} \left(e^{+\frac{x}{a}} + e^{-\frac{x}{a}} \right) \quad (4.5-4)$$

where a is a constant. The word “catenary” comes from the Latin *catena*, meaning “chain”; the curve has the form of a freely hanging

* The difference between the arc length and the length of the line segment is of the order of $(\Delta x)^2$ and so can be neglected when passing to the limit (to differentials).

flexible, inextensible cable (or chain) suspended from two fixed points. The graph of the catenary curve is given in Fig. 105 (for $a = 2$).

Let us find the length of an arc of the catenary from $x = 0$ to $x = x_0$.

From (4.5-4), $y' = \frac{e^{\frac{x}{a}} - e^{-\frac{x}{a}}}{2}$ and so

$$\sqrt{1+y'^2} = \sqrt{1 + \frac{e^{\frac{2x}{a}} - 2 + e^{-\frac{2x}{a}}}{4}} = \sqrt{\frac{\left(e^{\frac{x}{a}} + e^{-\frac{x}{a}}\right)^2}{4}} = \frac{e^{\frac{x}{a}} + e^{-\frac{x}{a}}}{2},$$

$$s = \int_0^{x_0} \frac{e^{\frac{x}{a}} + e^{-\frac{x}{a}}}{2} dx = \frac{1}{2} a \left[e^{\frac{x}{a}} - e^{-\frac{x}{a}} \right]_0^{x_0} = \frac{a}{2} \left(e^{\frac{x_0}{a}} - e^{-\frac{x_0}{a}} \right)$$

Related to the arc length is the problem of determining the radius R of curvature of a curve at a point; $1/R$ is simply referred to as the curvature (the smaller the radius, the more bent is the curve).

Let us take a small section of a curve (Fig. 106) of length ds and find the angle between the tangents to the curve at the endpoints of the section. This angle may be regarded as the increment $d\alpha$ in the angle α of inclination of the tangent to the x -axis. Draw normals (perpendicular lines) to the tangents at the two points. The angle between the normals is equal to the angle $d\alpha$ between the tangent lines, in accordance with a familiar theorem of geometry. From this we can find the distance R from the curve to the point of intersection of the normals.

We regard the small portion of the curve as an arc of a circle. The normal to the circle is clearly a radius. The point of intersection of the normals is the centre of the circle. If the curve were a circle, then $ds = R d\alpha$ or $\frac{d\alpha}{ds} = \frac{1}{R}$. This quantity is constant for any portion of an arc of the circle. For an arbitrary curve, this quantity, $\frac{d\alpha}{ds}$, for an infinitesimal portion of the curve, can serve as a definition of the curvature at a given point. Using the formula for ds and the fact that $\alpha = \arctan y'$, we can find an expression for the curvature:

$$d\alpha = d \arctan y' = \frac{dy'}{1+y'^2}, \quad ds = \sqrt{1+y'^2} dx,$$

$$\frac{d\alpha}{ds} = \frac{1}{(1+y'^2)^{3/2}} \cdot \frac{dy'}{dx} = \frac{y''}{(1+y'^2)^{3/2}}$$

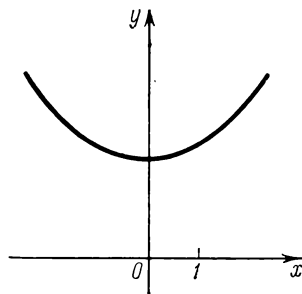


Fig. 105

The sign of the curvature $\frac{d\alpha}{ds}$ coincides with the sign of the second derivative y'' and characterizes the direction of convexity of the curve. If at the point x_0 the quantity $y'' > 0$ (Fig. 107), then the curve near this point lies above the tangent at this point and is convex

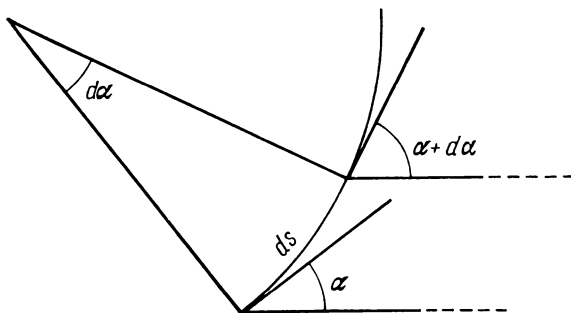


Fig. 106

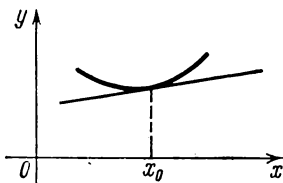


Fig. 107

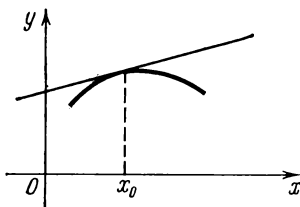


Fig. 108

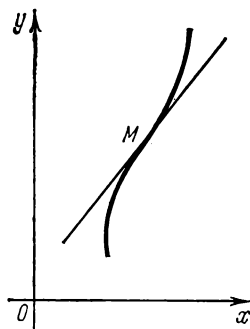


Fig. 109

down. If $y''(x_0) < 0$ (Fig. 108), then the curve lies below the tangent and is convex up.

It may happen that $y''(x_0) = 0$, and to the right of x_0 (i.e., when $x > x_0$) $y''(x) > 0$, whereas for $x < x_0$ we will have $y''(x) < 0$. This means that the curve to the right of x_0 is convex up and to the left is convex down (Fig. 109). At such a point (point M in Fig. 109), the curve moves over from one side of the tangent line to the other. The curve at this point changes the sense of convexity (the direction of bending). Such points are called points of inflection.

Exercises

1. Write as an integral the arc length of the parabola $y = x^2$ between point $(0, 0)$ and point $(1, 1)$.
2. Write as an integral the arc length of the curve $y = e^x$ between $x = 0$ and $x = 1$.

3. Write the arc length of an ellipse in the form of an integral.
4. Complete Problem 2 by making the change of variable $1 + e^{2x} = z^2$ in the integral.

•

4.6 APPROXIMATION OF ARC LENGTH

In Sec. 4.5 we obtained a formula for computing the arc length of a curve:

$$s = \int_a^b \sqrt{1 + y'^2(x)} dx \quad (4.6-1)$$

It was also pointed out that in most cases it is difficult (or even impossible) to integrate the function $\sqrt{1 + y'^2(x)}$ in terms of elementary functions due to the radical. For this reason, approximate formulas for computing arc length are of particular interest.

Suppose that $y'^2(x)$ is small compared with unity: $|y'(x)| \ll 1$. Then, neglecting $y'^2(x)$ in (4.6-1), we get

$$s \approx \int_a^b \sqrt{1} dx = b - a \quad (4.6-2)$$

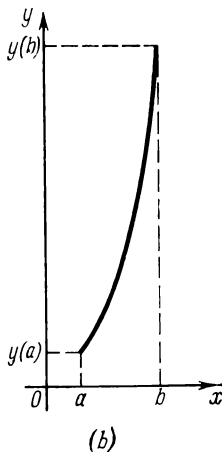
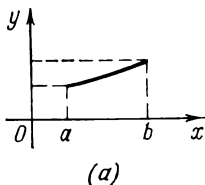


Fig. 110

The difference $b - a$ is the length of the horizontal portion with endpoints $x = a$ and $x = b$. Formula (4.6-2) shows that if y' is small in absolute value (the curve differs but slightly from the horizontal), then also the length of the arc of this curve is close to the length of the horizontal line segment (Fig. 110a).

If $y'^2(x) \gg 1$, then in (4.6-1) we ignore unity in comparison with $y'^2(x)$ to get

$$s \approx \int_a^b \sqrt{y'^2(x)} dx = \int_a^b y'(x) dx = y(b) - y(a) \quad (4.6-3)$$

This formula shows us that in the given case the arc length of the curve is close to that of a vertical line segment with endpoints $y(a)$ and $y(b)$ (Fig. 110b). Indeed, if the derivative y' is great, then the curve bends steeply upwards and so is similar to a vertical straight line (for a vertical line the derivative is infinite).

The formulas (4.6-2) and (4.6-3) yield simple approximate formulas for arc length. But these are very rough approximations which can be obtained without appealing to (4.6-1).

We wish to obtain more exact formulas.

Let $|y'(x)| < 1$. Retaining two terms in the binomial expansion (see Sec. 3.20), we get

$$\sqrt{1 + y'^2(x)} = 1 + \frac{1}{2} y'^2(x)$$

Formula (4.6-1) yields

$$s \approx \int_a^b \left[1 + \frac{1}{2} y'^2(x) \right] dx = (b-a) + \frac{1}{2} \int_a^b y'^2(x) dx$$

If $|y'(x)| > 1$, then

$$\sqrt{1 + y'^2(x)} = y'(x) \sqrt{1 + \frac{1}{y'^2(x)}}$$

We apply the binomial theorem to the last radical since $\frac{1}{y'^2} < 1$:

$$\begin{aligned} y'(x) \sqrt{1 + \frac{1}{y'^2(x)}} &= y'(x), \\ \left[1 + \frac{1}{2y'^2(x)} \right] &= y'(x) + \frac{1}{2y'(x)} \end{aligned}$$

Substituting this into (4.6-1) we obtain

$$\begin{aligned} s &= \int_a^b \left[y'(x) + \frac{1}{2y'(x)} \right] dx = \int_a^b y'(x) dx + \frac{1}{2} \int_a^b \frac{dx}{y'(x)} \\ &= y(b) - y(a) + \frac{1}{2} \int_a^b \frac{dx}{y'(x)} \end{aligned}$$

We now have approximate formulas:

$$\left. \begin{aligned} s &= (b-a) + \frac{1}{2} \int_a^b y'^2(x) dx & \text{if } |y'(x)| < 1, \\ s &= y(b) - y(a) + \frac{1}{2} \int_a^b \frac{dx}{y'(x)} & \text{if } |y'(x)| > 1 \end{aligned} \right\} \quad (4.6-4)$$

The integrals here are simpler than the integral in (4.6-1) and it is easier to perform the computations by these formulas than by (4.6-1). But these formulas are approximate.

What errors result from their use? The first formula is best for small $|y'|$, the second for large $|y'|$. Both formulas yield the worst results for $|y'| = 1$. To estimate the error here we will examine the worst case, which is $y'(x) = 1$.*

By the exact formula (4.6-1)

$$s = \int_a^b \sqrt{1+1} dx = \sqrt{2} (b-a) \quad (4.6-5)$$

By the first of the formulas (4.6-4),

$$s = (b-a) + \frac{1}{2} \int_a^b dx = \frac{3}{2} (b-a) \quad (4.6-6)$$

The second formula of (4.6-4) yields the same: $s = \frac{3}{2} (b-a)$. Comparing (4.6-5) and (4.6-6), we see that the maximum error in the approximate formula is 6%.

When computing arc length, the curve should be divided into portions over which either $|y'| \leq 1$ or $|y'| \geq 1$. Then the error will at least not exceed 6%. And since $y'^2(x)$ assumes a value equal to 1 only at certain points of the curve, a proper partition of the curve into portions will reduce the error below 6%. Of course there is no sense in finding the length of rectilinear portions by means of an approximate formula.

Let us take a look at some examples.

1. Find the arc length of the parabola $y = x^2$ between points with abscissas $x = 0$ and $x = 2$.**

We find the derivative $y' = 2x$. It is equal to 1 when $x = 0.5$ and is greater than 1 for $x > 0.5$. Therefore the arc length (s_1) corresponding to a variation of x from 0 to 0.5 can be found from the first formula in (4.6-4), and the arc length (s_2) corresponding to a variation of x from 0.5 to 2, by the second formula:

$$s_1 = (0.5-0) + 0.5 \int_0^{0.5} 4x^2 dx = 0.5 + 2 \cdot \frac{0.5^3}{3} = 0.58,$$

$$s_2 = 4 - 0.25 + 0.5 \int_{0.5}^2 \frac{dx}{2x} = 3.75 + 0.25 (\ln 2 - \ln 0.5) = 4.10$$

* If $y'(x) = 1$, then $y(x) = x + c$. The graph of the function is a straight line.

** In these examples, the computations are carried to two decimal places.

The sought-for arc length is

$$s = s_1 + s_2 = 0.58 + 4.10 = 4.68$$

Now let us compute the exact value of the arc length using (4.6-4):

$$s = \int_0^2 \sqrt{1+4x^2} dx$$

Making the change of variable $2x = z$, we get, by formula No. 33 (see Appendix, Table II),

$$\int \sqrt{1+4x^2} dx = \frac{1}{2} \left[x \sqrt{4x^2+1} + \frac{1}{2} \ln(2x + \sqrt{4x^2+1}) \right] \quad (4.6-7)$$

Any doubting reader can convince himself of the truth of the formula by taking the derivative of the right member of (4.6-7).

Using (4.6-7) we get

$$s = \frac{1}{2} \left[2 \sqrt{17} + \frac{1}{2} \ln(4 + \sqrt{17}) \right] = 4.65$$

The error committed by using formulas (4.6-4) came out to about 0.7%.

2. Find the arc length of the curve $y = e^x$ between points with abscissas $x = 0$ and $x = 1$.

In this case, $y' = e^x$, and the derivative grows from 1 to e as x ranges from 0 to 1. So we use the second formula of (4.6-4):

$$s = e^1 - e^0 + 0.5 \int_0^1 \frac{dx}{e^x} = 2.72 - 1 - 0.5e^{-x} \Big|_0^1 = 2.04$$

The exact formula for the arc length yields the value (see exercises 2 and 4 in Sec. 4.5)

$$s = \sqrt{1+e^2} - \sqrt{2} + \frac{1}{2} \ln \frac{\sqrt{1+e^2}-1}{\sqrt{1+e^2}+1} - \frac{1}{2} \ln \frac{\sqrt{2}-1}{\sqrt{2}+1} = 2.00$$

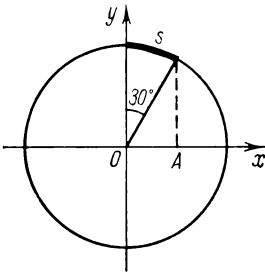


Fig. 111

The approximate formula has an error of 2%.

The arc length may also be approximated occasionally by a series expansion of the integrand in x in (4.6-4). By retaining an appropriate number of terms of the expansion, we can obtain the arc length to any degree of accuracy.

Let us consider an example. We will determine the circumference of a circle by seeking the arc length s of the circle cor-

responding to a central angle of 30° (Fig. 111). The circumference is $C = 12s$. Quite obviously we will obtain the same kind of integral as in (4.6-3) but with a different upper limit:

$$s = \int_0^{OA} \frac{R \, dx}{\sqrt{R^2 - x^2}}$$

Note that $OA = R \sin 30^\circ = \frac{1}{2}R$, and so

$$s = \int_0^{R/2} \frac{R \, dx}{\sqrt{R^2 - x^2}} \quad (4.6-8)$$

The integrand is transformed as follows:

$$\frac{R}{\sqrt{R^2 - x^2}} = \frac{R}{R \sqrt{1 - \left(\frac{x}{R}\right)^2}} = \frac{1}{\sqrt{1 - \left(\frac{x}{R}\right)^2}}$$

But

$$\frac{1}{\sqrt{1 - \left(\frac{x}{R}\right)^2}} = \left[1 - \left(\frac{x}{R}\right)^2\right]^{-\frac{1}{2}} \quad (4.6-9)$$

We put $\left(\frac{x}{R}\right)^2 = t$ and expand this in a binomial series [see formula (3.20-3)] to get

$$\begin{aligned} \left[1 - \left(\frac{x}{R}\right)^2\right]^{-\frac{1}{2}} &= (1-t)^{-\frac{1}{2}} = 1 + \frac{1}{2}t + \frac{3}{8}t^2 + \frac{5}{16}t^3 + \frac{35}{128}t^4 + \dots \\ &= 1 + \frac{1}{2}\left(\frac{x}{R}\right)^2 + \frac{3}{8}\left(\frac{x}{R}\right)^4 + \frac{5}{16}\left(\frac{x}{R}\right)^6 + \frac{35}{128}\left(\frac{x}{R}\right)^8 + \dots \end{aligned} \quad (4.6-10)$$

Substituting (4.6-10) into (4.6-8) and integrating, we obtain

$$\begin{aligned} s &= \frac{1}{2}R + \frac{1}{6 \cdot 8}R + \frac{3}{40 \cdot 32}R + \frac{5}{16 \cdot 7 \cdot 2^7}R + \frac{35}{128 \cdot 9 \cdot 2^9}R + \dots \\ &= R \left[\frac{1}{2} + \frac{1}{48} + \frac{3}{40 \cdot 32} + \frac{5}{16 \cdot 7 \cdot 2^7} + \frac{35}{128 \cdot 9 \cdot 2^9} + \dots \right] \end{aligned} \quad (4.6-11)$$

It is clear that the terms of the series (4.6-11) decrease rather rapidly and so we need only a few terms of the series to get s . Taking one term, we have $s = \frac{1}{2}R$, whence the circumference is $C = 6R$. Taking two terms, we have $s = 0.521R$, $C = 6.252R$. Three terms yield $s = 0.523R$, $C = 6.276R$, and so on.

We know that the circumference of a circle is $C = 2\pi R$. Comparing this with the results we have obtained, we can approximate the value of the number π :

$$3, 3.126, 3.138, \dots$$

The more terms of the series (4.6-11) we take, the more exact the value of π we obtain. The value of π to seven decimal places is 3.1415926.

Exercises

1. Use approximate formulas to find the arc length of a catenary curve between the points $x = 0$ and $x = 2$ ($a = 1$). Compare this with the exact value of the arc length.

2. Find the arc length of the hyperbola $xy = -1$ between the points $x = 0.5$, and $x = 1$.

Note. In this case we cannot obtain an exact value because the integral in (4.6-1) is not expressible in terms of elementary functions.

3. Obtain approximate values of the number π by computing the arc length of a circle with central angle 45° (retain three, four and five terms of the series).

4.7 COMPUTING VOLUMES. THE VOLUME AND SURFACE AREA OF A SOLID OF REVOLUTION

In Sec. 2.14 we obtained the formula

$$V = \int_{x_0}^{x_h} S(x) dx \quad (4.7-1)$$

where $S(x)$ is the area of a section of a solid by a plane perpendicular to the x -axis and passing through the point x (we advise the reader to repeat the derivation of this formula).

This formula was used to obtain an expression for the volume of a pyramid. The volume of a cone was also obtained in exactly this fashion. Put the origin of coordinates at the centre of the circle of the cone base and send the x -axis along the altitude of the cone (Fig. 112).

Let $S(x)$ be the area of a section of the cone by a plane perpendicular to the altitude and distant x from the base. This section is a circle of radius r_x . From the similarity of triangles we have

$$\frac{r_x}{r} = \frac{H-x}{H}$$

where r is the radius of the base and H is the altitude of the cone, whence $r_x = \frac{r}{H}(H - x)$ and, consequently,

$$\begin{aligned} V &= \int_0^H \pi \frac{r^2}{H^2} (H - x)^2 dx = \frac{\pi r^2}{H^2} \int_0^H (H - x)^2 dx \\ &= -\frac{\pi r^2}{H^2} \frac{(H - x)^3}{3} \Big|_0^H = \frac{\pi r^2 H^3}{3H^2} = \frac{\pi r^2 H}{3} \end{aligned}$$

To obtain the volume of the sphere we place the origin at the centre of the great circle and direct the x -axis along a diameter of the sphere, which diameter is perpendicular to the plane of the great circle.

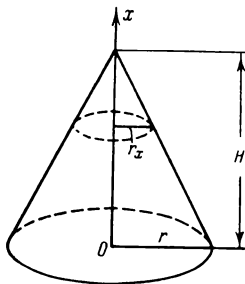


Fig. 112

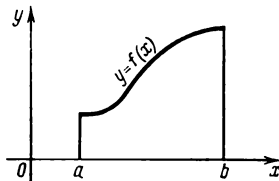


Fig. 113

The section cut by a plane perpendicular to the x -axis and distant x from the origin is a circle of radius R_x . By the Pythagorean theorem, $R_x = \sqrt{R^2 - x^2}$ and so

$$\begin{aligned} S(x) &= \pi R_x^2 = \pi (R^2 - x^2), \\ V &= \int_0^R \pi (R^2 - x^2) dx = \pi \left[R^2 x - \frac{x^3}{3} \right]_0^R = \frac{4}{3} \pi R^3 \end{aligned}$$

From formula (4.7-4) follows Cavalieri's principle:* given two parallel planes P and Q between which are located two solids; if, in the section of these solids by an arbitrary plane parallel to P and Q , we obtain equivalent figures [equal integrals $S(x)$], then the volumes of these solids are equal.

Let a solid be generated by revolving the figure depicted in Fig. 113 about the x -axis (such a figure is called a curvilinear trapezoid). In this case, the section is a circle of radius $y = f(x)$ and $S(x) =$

* Cavalieri was an Italian mathematician of the 17th century. His principle was formulated (essentially without proof) in his book *Geometria indivisibilibus* (The geometry of indivisibles) (1635).

$= \pi y^2$. Using (4.7-1) we find

$$V = \pi \int_a^b y^2 dx \quad (4.7-2)$$

Let us find, say, the volume of a solid generated by the revolution of the upper half of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about the x -axis. This solid is called an ellipsoid of revolution.

From the equation of an ellipse, $y = \frac{b}{a} \sqrt{a^2 - x^2}$, and from (4.7-2) we have

$$V = \pi \int_{-a}^a \frac{b^2}{a^2} (a^2 - x^2) dx = \frac{\pi b^2}{a^2} \left[a^2 x - \frac{x^3}{3} \right]_{-a}^a = \frac{4}{3} \pi a b^2$$

For $a = b = R$ we obtain the volume of a sphere of radius R .

Now let us derive the formula for the surface area of a solid of revolution (Fig. 114). We consider a solid bounded by sections

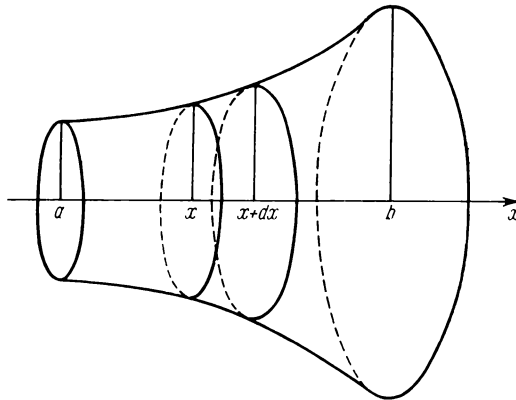


Fig. 114

passing through the points x and $x + dx$. Denote by dF the lateral surface area of this solid. Regarding it as the frustum of a cone, we get

$$dF = \pi [y(x) + y(x + dx)] ds$$

where ds is the length of a small portion of the curve, $ds = \sqrt{1 + y'^2(x)} dx$ (see Sec. 4.5). The sum $y(x) + y(x + dx)$ may be replaced by $2y(x)$, disregarding the quantity $y'(x) dx$ as compared with $y(x)$.*

* Note that in the expression dF the sum $y(x) + y(x + dx)$ is multiplied by ds so that the quantity we ignore is of the order of $dx \cdot ds \approx dx^2$.

Therefore

$$dF = 2\pi y(x) \sqrt{1 + y'^2(x)} dx$$

The entire surface area of the solid of revolution is

$$F = 2\pi \int_a^b y(x) \sqrt{1 + y'^2(x)} dx \quad (4.7-3)$$

The surface area of a sphere is readily found by means of this formula. A sphere is generated by the revolution of the upper semi-circle about the x -axis. The equation of a circle is $x^2 + y^2 = a^2$, whence

$$y = \sqrt{a^2 - x^2}, \quad y' = \frac{-x}{\sqrt{a^2 - x^2}}$$

Substituting into (4.7-3) we get

$$F = 2\pi \int_{-a}^a \sqrt{a^2 - x^2} \frac{a}{\sqrt{a^2 - x^2}} dx = 2\pi ax \Big|_{-a}^a = 4\pi a^2$$

Exercises

1. Find the volume of a cylindrical piece, or wedge, that is, a solid cut from a right circular cylinder of radius R by a plane drawn through a diameter of the base of the cylinder at an angle of α (Fig. 115).

2. Find the volume of a cone using the fact that the cone is a solid generated by rotating a right triangle about one of the legs.

3. Find the volume of a solid generated by revolving a figure bounded from above by the curve $y = \sqrt{x}$, from below by the x -axis, on the right by the vertical straight line $x = 2$.

4.8 CURVE SKETCHING

The most primitive method of constructing the graph of a function $f(x)$ is to compute the value of $f(x_n)$ for a large number of points x_n . The usual procedure is to choose points x_n in the form $x_n = x_0 + na$, $n = 0, \pm 1, \pm 2, \dots$. This is quite an extravagant method. In order to see the variation of the function on an interval Δx , we have to choose a step a much less than Δx : $a \ll \Delta x$. And if the step (subinterval) is small, a very large number of points are required to embrace the whole range of interest.

The techniques considered in Secs. 4.1 and 4.2 make it possible to construct graphs much faster and more reliably and to gain a general picture of the shape of the curve. This requires, first of all, that

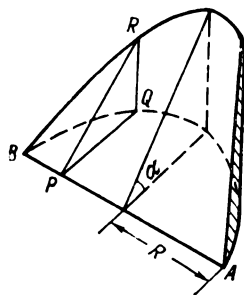


Fig. 115

we find the characteristic points of the graph—maxima, minima, salient points, points of inflection, etc.

Let us illustrate this fact by using the example of the graph of a third-degree polynomial, i.e., the graph of the function

$$y = ax^3 + bx^2 + cx + d \quad (4.8-1)$$

A knowledge of the graph permits obtaining some important information about the function, say the number of real roots, the intervals in which the roots lie, and more.

By way of an illustration, let us construct the graph of the function

$$y = 0.5x^3 - 0.75x^2 - 3x + 2.5 \quad (4.8-2)$$

First we find the maxima and minima. Equating to zero the derivative of (4.8-2), we have

$$y' = 1.5x^2 - 1.5x - 3 = 0 \quad (4.8-3)$$

whence we find $x_1 = -1$, $x_2 = 2$.

Let us investigate each of these values separately. To do so, we find y'' : $y'' = 3x - 1.5$, $y''(-1) = -4.5 < 0$. Hence, for $x_1 = -1$ the function has a maximum:

$$y_{max} = -0.5 - 0.75 + 3 + 2.5 = 4.25$$

$y''(2) = 6 - 1.5 = 4.5 > 0$ and so for $x_2 = 2$ the function has a minimum:

$$y_{min} = -2.5$$

Now let us see how the polynomial behaves for very large absolute values of x . Note that for very large x , the term containing x^3 will appreciably exceed the other terms in absolute value. Therefore the sign of the polynomial (4.8-1) is determined by the sign of the expression ax^3 .

If $a > 0$, then $ax^3 > 0$ for $x > 0$, the right-hand branch of the graph goes upwards; $ax^3 < 0$ for $x < 0$, the left branch of the curve goes down. It is clear that for $a < 0$ the left branch goes up and the right branch goes down.

Let us find points of inflection. From what was said in Sec 4.5 it is clear that to find the points of inflection we have to solve the equation $f''(x) = 0$. Using (4.8-3), we find $y'' = 3x - 1.5$. The equation $y'' = 0$ yields $x = 0.5$. Since $y''' = 3$, it follows that $y - \tilde{y} = \frac{1}{2}(x - 0.5)^3$, where $\tilde{y}$ denotes the ordinate of the tangent line.

Therefore, if $x > 0.5$, then $y - \tilde{y} > 0$ but if $x < 0.5$, then $y - \tilde{y} < 0$. Hence, for $x = 0.5$ we have an inflection.

It will be noted that the graph of a third-degree polynomial always has a point of inflection, and it is unique. Indeed when y is a polynomial of degree 3, the equation $y'' = 0$ is a first-degree equation.

It always has a unique root, x_0 . Since $y''' = \text{const}$, it follows that $y - \tilde{y} = A(x - x_0)^3$. It is clear that $y - \tilde{y}$ changes sign when x goes through the value x_0 .

We return to the construction of the graph and compute the ordinate y of the point of inflection to get $y \approx 0.88$. Let us also determine the direction of the tangent to the curve at the point of inflection.

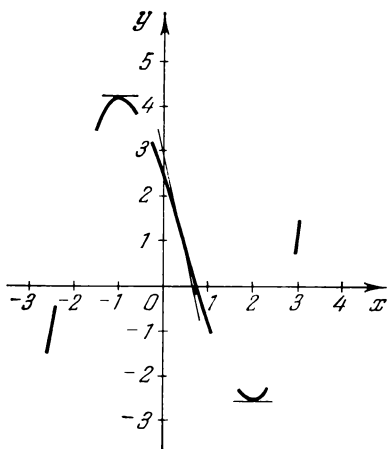


Fig. 116

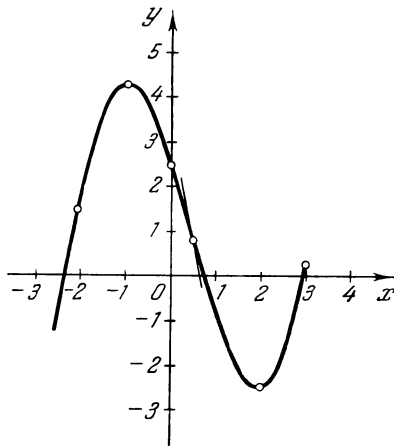


Fig. 117

Using (4.8-3), we get $\tan \alpha = y'(0.5) = 3.38$. Using all the foregoing arguments, we get Fig. 116 for (4.8-2).

Of course if we do not compute any other values of the function, then the resulting graph will give only a very rough qualitative picture of the behaviour of the function. But even such a graph enables us to count the number of roots (that is, the number of points of intersection of the graph with the x -axis) and to draw certain conclusions about their values. In our example, Fig. 116, we see that there are three roots, that one of the roots lies somewhere between 0.5 and +2, that the second root must definitely be positive (it exceeds 2) while the third root is negative (it is less than -4).

The graph may be improved by computing a few more values of the function for certain values of x .

Let us compute three more values of the function in this example. For $x = 0$, $y = 2.5$. This permits us to get a better picture of the variation of the curve between maximum and minimum. For $x = 3$, $y = 0.25$. We computed this value so as to get an idea of the rate of climb of the right branch of the curve. Similarly, to get an idea of the rate of fall of the left-hand branch of the curve we take $x = -2$ and get $y = 1.5$. Using these values we obtain the curve shown in Fig. 117.

With this graph we can draw more accurate conclusions concerning the roots: one root lies between $x = 0.5$ and $x = 1$; the second between $x = 2$ and $x = 3$, closer to $x = 3$; the third is less than $x = -2$ (its value is most likely close to $x = -2.5$).

It may happen that after equating the derivative to zero we will not obtain any real roots. This will mean that the polynomial does not have a maximum or a minimum. Since all that has been said about the behaviour of the polynomial for very large absolute values of x remains valid, the graph will intersect the x -axis only at one point (the polynomial has one real root).

Finally, the derivative may have only one (double) root x_0 . Then it will be of the form

$$y' = A(x - x_0)^2 \quad (4.8-4)$$

Integrating (4.8-4), we get

$$y = \frac{A}{3}(x - x_0)^3 + C \quad (4.8-5)$$

From (4.8-5) we see that in this case the polynomial differs from a perfect cube only in the constant summand. From (4.8-5) we conclude that y has neither maximum nor minimum (see Example 2 of Sec. 4.1). The graph intersects the x -axis in one point. We find this point by equating y to zero in (4.8-5):

$$\frac{A}{3}(x - x_0)^3 + C = 0,$$

$$(x - x_0)^3 = -\frac{3C}{A},$$

$$x = x_0 - \sqrt[3]{\frac{3C}{A}}$$

Finding the maximum and minimum of a third-degree polynomial and, hence, the investigation of its graph can always be completed because by equating the derivative to zero we get a quadratic equation whose roots are not hard to find.

Exercises

Find the maxima and minima of the following functions and sketch their graphs.

1. $y = x^3 - 3x^2 + 2$. 2. $y = x^3 - 3x^2 + 3x - 15$. 3. $y = x^3 - 3x^2 + 6x + 3$.

Determine the number of real roots of the following equations.

4. $2x^3 - 3x^2 - 12x + 15 = 0$. 5. $4x^3 + 15x^2 - 18x - 2 = 0$. 6. $2x^3 - x^2 - 4x + 3 = 0$. 7. $x^3 - x^2 + 2 = 0$.

Chapter 5

Water Flow. Radioactive Decay and Nuclear Fission. Absorption of Light

5.1 WATER FLOW FROM A VESSEL. STATEMENT OF THE PROBLEM

We now consider the flow of water from a vessel with an opening near the bottom. The vessel can also receive an inflow of water from some outside source. The statement of this problem is very simple and pictorial. At the same time, the mathematical methods required to describe the flow of water are also employed in more complicated and interesting problems.

So let us imagine a vessel with water flowing in and out. We denote the volume of water in the vessel by V (cm³). This volume varies with time, which means V is a function of time t (sec). What meaning has the quantity $\frac{dV}{dt}$?

It is quite clear that $dV = V(t + dt) - V(t)$ is the amount of water that has entered the vessel during time dt . Therefore $\frac{dV}{dt}$ is the amount of water entering the vessel in unit time, or the rate of change of the amount of water in the vessel. This quantity has a special name, "water flow". We will denote the flow by $q(t)$. If $q > 0$, then the water is entering the vessel, if $q < 0$, the water is being discharged from the vessel, and the amount in the vessel is diminishing.

If we know the dependence of water flow on time, i.e., if we know the function $q(t)$, then

$$\frac{dV}{dt} = q(t) \quad (5.1-1)$$

In this case the problem of finding V is similar to that of determining distance traveled from a given velocity. In Chapter 2 we learned that this problem is solved by integration.

For this problem to have a definite solution, we must know the amount of water V_0 in the vessel at a definite initial time t_0 . The condition that $V = V_0$ at time $t = t_0$ is called the initial condition.

The amount of water admitted during time t_0 to t_1 is $\int_{t_0}^{t_1} q(t) dt$.

Whence the amount of water in the vessel at time t_1 is

$$V(t_1) = V_0 + \int_{t_0}^{t_1} q(t) dt \quad (5.1-2)$$

This expression holds true for any time t_1 and, consequently, fully determines the desired dependence of V upon t_1 . Observe, in particular, that for $t_1 = t_0$ the integral in (5.1-2) is equal to zero and $V(t_0) = V_0$. Thus, the condition (5.1-2) does indeed satisfy the stated condition relative to the amount of water at time t_0 (the initial condition).

Note also that formula (5.1-2) can be used for $t_1 < t_0$. But then the meaning of this formula for $t_1 < t_0$ and $t_1 > t_0$ is different. For $t_1 > t_0$ the quantity $V(t_1)$ is the amount of water that will be in the vessel at time t_1 if at time t_0 the amount of water was V_0 and the water flow is given by the function $q(t)$. For $t_1 < t_0$, $V(t_1)$ is the amount of water that must be in the vessel at time t_1 so that at a later time, by time t_0 , there should be V_0 water, given the flow specified by the function $q(t)$.

Instead of writing t_1 we can simply write t . Then formula (5.1-2) takes the form

$$V(t) = V_0 + \int_{t_0}^t q(t) dt \quad (5.1-3)$$

Strictly speaking, the letter t then denotes the upper limit of integration and is thus "engaged". It would therefore be advisable to denote the variable of integration by some other letter, say, τ , and write (5.1-3) as

$$V(t) = V_0 + \int_{t_0}^t q(\tau) d\tau$$

which coincides with (5.1-2) if we replace t_1 by t and t by τ . This is not usually done however and the formula is written in the form (5.1-3). This is not really confusing; remember only that $q(t)$ in (5.1-3) is not the value of q at the upper limit but the function of the variable of integration that runs through all values from t_0 to t .

Formula (5.1-3), which yields the solution of the water discharge problem if the flow $q(t)$ is given and also the amount of water at the initial time $t = t_0$, can also be obtained by somewhat different

reasoning. From (5.1-1), by virtue of the definition of the indefinite integral, it follows that

$$V(t) = \int q(t) dt$$

Suppose that the indefinite integral of the function $q(t)$ has been found in some way. Denote it by $I(t)$. Then

$$\int q(t) dt = I(t) + C$$

where C is the constant of integration. From this,

$$V(t) = I(t) + C \quad (5.1-4)$$

To determine the constant of integration let us take advantage of the initial condition, that is, let us require that for $t = t_0$ we have $V = V_0$. Substituting $t = t_0$ into (5.1-4), we get

$$V_0 = I(t_0) + C$$

whence

$$C = V_0 - I(t_0)$$

Substituting the value of C into (5.1-4) we find

$$V(t) = V_0 + I(t) - I(t_0)$$

This coincides with (5.1-3) since

$$\int_{t_0}^t q(t) dt = I(t) \Big|_{t_0}^t = I(t) - I(t_0)$$

The formula (5.1-4) may be termed the general solution of equation (5.1-1).

By choosing one or another value of C , we can, from formula (5.1-4), obtain a variety of particular solutions that correspond to different initial conditions.

Ordinarily however the flow is not known as a function of time. More often we know a physical law describing the discharge of water. This law gives the flow as a function of the water head, that is, of the height of water level z (Fig. 118).

Thus, for example, when water is flowing out through a long thin pipe,

$$q = -kz$$

where the coefficient k is a positive constant, the minus sign meaning that the water is being discharged. When the water flows out through

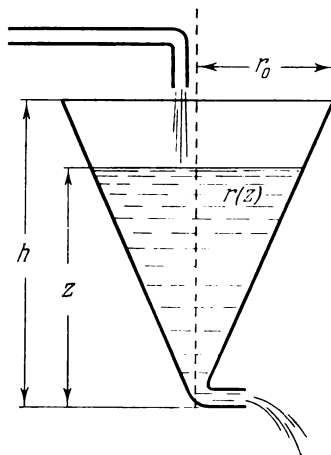


Fig. 118

an opening in a thin wall, then

$$q = -a\sqrt{z}$$

In each of these cases, until the problem has been solved, we do not know the dependence of water level in the vessel on time, $z(t)$, and hence we do not know the flow. For this reason, the problem of determining V from the equation

$$\frac{dV}{dt} = q(z) \quad (5.1-5)$$

cannot be reduced to the preceding problem. Here we stated the problem in the general case for an arbitrary relation of the flow q as a function of the level z .

Equation (5.1-5) involves two unknown quantities: the amount (volume) of water V and the water level z . It is clear that these quantities are not independent. A definite water level is associated with a very definite amount of water so that V is a known function* of z , $V(z)$.

Substituting $V(z)$ into equation (5.1-5) we get

$$\frac{dV(z)}{dt} = \frac{dV(z)}{dz} \cdot \frac{dz}{dt} = q(z)$$

The derivative of the volume with respect to height is equal to the area of the cross-section at height z [see formula (2.14-8)]. Denote this derivative by $S(z)$:

$$\frac{dV(z)}{dz} = S(z)$$

We finally get the equation

$$S(z) \frac{dz}{dt} = q(z) \quad (5.1-6)$$

A method for solving this equation is considered in the following section.

* The form of this function is determined by the shape of the vessel. For a cylindrical vessel, for example, $V = \pi r_0^2 z$. For a conical vessel (see Fig. 118), $V = \frac{1}{3} S(z) z$, where S is the area of a cross-section of the vessel at height z , $S = \pi r^2(z)$, where $r(z)$ is the radius of the cross-section at level z . From the similarity of triangles, we find $r(z) = r_0 \frac{z}{h}$, where r_0 is the radius of the upper

base of the cone and h is the total height so that $V = \frac{1}{3} \frac{\pi r_0^2}{h^2} z^3$.

5.2 THE SOLUTION OF AN EQUATION WHEN THE DERIVATIVE DEPENDS ON THE DESIRED FUNCTION

The problem of water flow (discharge) reduced to determining the function $z(t)$ from the equation in which the derivative $\frac{dz}{dt}$ is given as a function of z :

$$S(z) \frac{dz}{dt} = q(z) \quad (5.1-6)$$

Rewrite this equation as $\frac{dz}{dt} = \frac{q(z)}{S(z)}$. Put $\frac{q(z)}{S(z)} = f(z)$; then we finally have

$$\frac{dz}{dt} = f(z) \quad (5.2-1)$$

Equations that contain the desired function and its derivatives are called differential equations. If the equation involves only the first derivative, then it is called an equation of the first order. We have already dealt with differential equations of the form

$$\frac{dz}{dt} = f(t)$$

To solve such an equation means to find the function from its derivative. This problem is solved by integration.

Let us consider equation (5.2-1). We rewrite it

$$\frac{dt}{dz} = \frac{1}{f(z)} \quad (5.2-2)$$

This notation fits the fact that we temporarily regard t as a function of z ; we will seek the inverse function $f(z)$ [see Sec. 3.2, in particular formula (3.2-3)] and after finding it will express z in terms of t .

Integrate the left and right members of (5.2-2):

$$\int_{z_0}^z \frac{1}{f(z)} dz = \int_{t_0}^t dt$$

whence

$$t = t_0 + \int_{z_0}^z \frac{dz}{f(z)} \quad (5.2-3)$$

We have the solution to the problem: on the right is a function of z , on the left is the time t . This equation enables us, for each value of t , to find the corresponding value of z . The solution (5.2-3) satisfies the initial condition: $z = z_0$ for $t = t_0$ (at the initial time $t = t_0$ the level of water in the vessel, z_0 , is given).

We conclude this section with two examples.

1. Water is flowing out of a conical vessel through a thin pipe:

$$q = -kz, \quad V = \frac{1}{3} \pi \frac{r_0^2}{h^2} z^3,$$

$$\frac{dV}{dz} = S(z) = \frac{\pi r_0^2}{h^2} z^2, \quad \frac{dz}{dt} = \frac{q(z)}{S(z)} = -\frac{kzh^2}{\pi r_0^2 z^2} = -\frac{kh^2}{\pi r_0^2} \frac{1}{z},$$

$$-\frac{\pi r_0^2}{kh^2} z \, dz = dt, \quad -\frac{\pi r_0^2}{kh^2} \int_{z_0}^z z \, dz = -\frac{\pi r_0^2}{kh^2} \left(\frac{z^2}{2} - \frac{z_0^2}{2} \right) = t - t_0$$

In this example it is easy to express z as a function of t :

$$\frac{z^2}{2} = \frac{z_0^2}{2} - \frac{kh^2}{\pi r_0^2} (t - t_0),$$

$$z = \sqrt{z_0^2 - \frac{2kh^2}{\pi r_0^2} (t - t_0)} \quad (5.2-4)$$

This formula completely solves the problem. It is readily verified that

$$\frac{dz}{dt} = -\frac{kh^2}{\pi r_0^2} \frac{1}{\sqrt{z_0^2 - \frac{2kh^2}{\pi r_0^2} (t - t_0)}} = -\frac{kh^2}{\pi r_0^2 z}$$

so that z does indeed satisfy the equation. It is also clear that $z = z_0$ for $t = t_0$. The expression (5.2-4) permits finding the time when the vessel is completely emptied: $z = 0$ when $t = t_0 + \frac{\pi r_0^2}{kh^2} \frac{z_0^2}{2}$.

2. A cylindrical vessel with a tube. In this case

$$V(z) = \pi r_0^2 z, \quad \frac{dV}{dt} = \pi r_0^2 \frac{dz}{dt} = -kz$$

whence

$$-\pi r_0^2 \frac{dz}{z} = k \, dt$$

$$-\pi r_0^2 (\ln z - \ln z_0) = -\pi r_0^2 \ln \frac{z}{z_0} \quad \pi r_0^2 \ln \frac{z_0}{z} = k (t - t_0) \quad (5.2-5)$$

From (5.2-5) it is quite easy to express z in terms of t . Indeed, $\ln z = \ln z_0 - \frac{k}{\pi r_0^2} (t - t_0)$ and from this

$$z = z_0 e^{-\frac{k}{\pi r_0^2} (t - t_0)}$$

We consider two instants of time, t and $t + \Delta t$, and find the ratio $\frac{z(t + \Delta t)}{z(t)}$:

$$\frac{z(t + \Delta t)}{z(t)} = e^{-\frac{k}{\pi r_0^2} (t + \Delta t - t_0 - t + t_0)} = e^{-\frac{k \Delta t}{\pi r_0^2}}$$

We see that this ratio depends only on Δt and is independent of t . Therefore, the water level z falls in equal ratio during equal intervals of time.

There is an interesting qualitative difference between these two examples. In the second solution there is no instant of time when z becomes exactly zero, z decreases with time but tends to zero only when $t \rightarrow \infty$.

5.3 RADIOACTIVE DECAY

The basic law of radioactive decay states that the ratio of the number of atoms disintegrated in unit time to the total number of atoms is a constant which is only dependent on the species of atom. It is understood here that the total number of atoms is extremely large.

This ratio is called the probability of disintegration. Denote by $N(t)$ the quantity of atoms that have not disintegrated by time t . At time $t + dt$ there will be $N(t + dt)$ untransformed atoms. For this reason, during time dt (from t to $t + dt$) there will be $N(t) - N(t + dt) = -dN$ disintegrations of atoms. The probability of disintegration $\omega = \frac{dN}{N dt}$, whence

$$\frac{dN}{dt} = -\omega N \quad (5.3-1)$$

From this relation, recalling that the dimensions of $\frac{dN}{dt}$ are the same as those of the ratio $\frac{N}{t}$, we see that the probability of disintegration ω has the dimensions of 1/sec.*

The initial condition consists in specifying the number of atoms at the initial time: $N = N_0$ for $t = t_0$.

Solving equation (5.3-1) by the method given in the preceding section and using the initial condition, we get

$$N(t) = N_0 e^{-\omega t} \quad (5.3-2)$$

* Consequently, probability here is not to be understood in the sense of the assertion that, as in coin tossing, the probability is one half that the coin will fall heads. The definition of the probability of disintegration as the ratio of the number of disintegrations per unit time to the initial number of atoms holds true only for the case when the number of disintegrations per unit time (say, per second) constitutes a small fraction of the total number of atoms. The exact definition of probability of disintegration is given by the formula $\omega = -\frac{1}{N} \frac{dN}{dt}$, that is, the probability of disintegration is equal to the ratio of the number of disintegrations during a small time interval to the total number of atoms and to the magnitude of the time interval.

(we advise the reader to perform all the computations). However, when the derivative is proportional to the desired function, a simpler solution to the equation can be offered.

In Chapter 3 we discovered that the derivative of an exponential function is proportional to the function itself:

$$\frac{d(a^x)}{dx} = \text{const} \cdot a^x$$

in particular,

$$\frac{d(Ce^{kx})}{dx} = Cke^{kx}$$

if C and k are constants. Recalling this property of the exponential function, let us suppose that the solution of equation (5.3-1) is of the form

$$N = Ce^{kt} \quad (5.3-3)$$

and let us try to choose C and k so that the equation and the initial condition are satisfied. Differentiating (5.3-3), we get

$$\frac{dN}{dt} = Cke^{kt} = kN$$

Substitute this into (5.3-1) to get $kN = -\omega N$, whence $k = -\omega$. Assuming $t = 0$ in (5.3-3) and using the initial condition, we get $C = N_0$. And so $N = N_0 e^{-\omega t}$.

The quantity $-\omega t$ in the exponent is nondimensional as it should be.

Radioactive atoms are characterized by their half-life T , which is the time during which the number of atoms N diminishes via disintegration by one half the original amount.

Let us determine the half-life T . From (5.3-2), $N(T) = N_0 e^{-\omega T}$. On the other hand, $N(T) = \frac{1}{2} N_0$, by definition. Therefore $N_0 e^{-\omega T} = \frac{1}{2} N_0$, $e^{-\omega T} = \frac{1}{2}$, whence

$$-\omega T = -\ln 2, \quad T = \frac{\ln 2}{\omega} \approx \frac{0.69}{\omega} \quad (5.3-4)$$

The half-life is inversely proportional to the probability of disintegration.

Prior to disintegration, every atom exists a certain period of time, which is called the lifetime of the atom.

We find the mean lifetime $\bar{t}$ of an atom of a given radioactive element. Suppose, at the initial time $t = 0$, when the atoms were created, there were N_0 atoms. During time t to $t + dt$, the quantity of atoms that disintegrated was

$$-dN = \omega N dt$$

All the atoms of this group lived roughly the same lifetime, t . Among the atoms taken at the initial time there are groups of atoms that will have various lifetimes: from the common-to-all-atoms time of creation to the distinct-for-various-atoms time of disintegration. To find the mean lifetime, we have to multiply the lifetime of each group by the number of atoms in the group, then add these quantities for all groups and divide by the total number of atoms in all groups.

Since we have to add a very large number of terms, the integral takes the place of the sum, and we have

$$\bar{t} = \frac{\int_0^{\infty} t \cdot \omega N dt}{\int_0^{\infty} \omega N dt} \quad (5.3-5)$$

We substitute the expression for N from (5.3-2). The denominator is

$$\int_0^{\infty} \omega N dt = \int_0^{\infty} \omega N_0 e^{-\omega t} dt = \omega N_0 \int_0^{\infty} e^{-\omega t} dt = -\omega N_0 \frac{e^{-\omega t}}{\omega} \Big|_0^{\infty} = N_0$$

as was to be expected, since the integral in the denominator yields the total number of all disintegrated atoms, which, clearly, is equal to the number of atoms existing at the initial time.

We integrate the integral in the numerator of (5.3-5) by parts, setting $t = f$, $e^{-\omega t} dt = dg$, to get

$$\begin{aligned} \omega N_0 \int_0^{\infty} t e^{-\omega t} dt &= \omega N_0 \left[-\frac{1}{\omega} t e^{-\omega t} + \int \frac{1}{\omega} e^{-\omega t} dt \right]_0^{\infty} \\ &= \omega N_0 \left[-\frac{1}{\omega} t e^{-\omega t} - \frac{1}{\omega^2} e^{-\omega t} \right]_0^{\infty} = \frac{N_0}{\omega} \end{aligned}$$

From formula (5.3-5) we now obtain

$$\bar{t} = \frac{N_0}{\omega N_0} = \frac{1}{\omega} \quad (5.3-6)$$

Using this fact we can write the basic equation (5.3-1) and its solution (5.3-2) as

$$\frac{dN}{dt} = -\frac{N}{\bar{t}} \quad (5.3-7)$$

$$N = N_0 e^{-\frac{t}{\bar{t}}} \quad (5.3-8)$$

We must not forget that the time t is the independent variable: the number of atoms depends on t . Now, the quantity $\bar{t}$ is a constant that describes the given type of radioactive atom.

From (5.3-8) it is evident that during time $t = \bar{t}$ the number of atoms diminishes from N_0 to $N_0 e^{-1} = \frac{N_0}{e}$, by a factor of e , which is roughly equal to 2.72.

By formula (5.3-7), the initial rate of disintegration is such that if the number of atoms decaying per unit time did not fall off, all the atoms would disintegrate in time t . Indeed, when $t = 0$ there were N_0 atoms the rate of disintegration was $\left. \frac{dN}{dt} \right|_{t=0} = -\frac{N_0}{\bar{t}}$.

At that rate, complete disintegration requires a time equal to $\bar{t}$. From (5.3-4), $\omega = \frac{\ln 2}{T}$ and so

$$\bar{t} = \frac{T}{\ln 2} \approx 1.45T$$

Computationally, the quantity $\bar{t}$ is more convenient than the half-life T .

Exercises

1. The mean lifetime of radium is 2400 years. Determine the half-life of radium.
2. We start with 200 grams of radium. How much radium will be left in 300 years?
3. Ten grams of radium disintegrated in 500 years. How much was there at the beginning?
4. Determine how much time will elapse for 1%, 10%, 90%, and 99% of an original supply of radium to disintegrate.
5. The amount of radium in the earth in various rocks comes out to about $1/10^{12}$ (by atoms). What was the content of radium in the rocks 10 000 years ago, 10^6 years ago, 5×10^9 years ago (5×10^9 years is the age of the earth)?

5.4 MEASURING THE MEAN LIFETIME OF RADIOACTIVE ATOMS

The mean lifetime $\bar{t}$ of various radioactive atoms is extremely diversified. To illustrate, let us take the several known isotopes of uranium. One, with atomic weight 238 (U^{238}), has a mean lifetime of $\bar{t} = 7 \times 10^9$ years. Another (U^{235}) has a mean lifetime $\bar{t} = 10^9$ years (the fission of uranium-235 in nuclear power plants is the main source of atomic energy). The mean lifetime of radium is 2400 years.*

However, it would be wrong to think that the mean lifetimes of all radioactive atoms are measured in thousands of years. Among radioactive substances that occur in nature and were studied by Marie and Pierre Curie and Ernest Rutherford, we find polonium with a mean lifetime of about 200 days, radium A with a mean lifetime of 4 minutes and radium C' with a mean lifetime of 2×10^{-4} second.

During the past 30 odd years encompassing the recent period of development of nuclear physics and the use of atomic energy, over

* Reference books frequently give the half-life as $T = 0.69 \bar{t}$, see Sec. 5.3.

400 different radioactive substances with a vast range of mean lifetimes have been discovered.

If at time t there are $N(t)$ untransformed atoms, then there will be $n(t) = \omega N(t)$ disintegrations (of atoms) in unit time. The quantity $n(t)$ is the rate of disintegration of the atoms.

Multiply both sides of (5.3-2) by ω to get

$$\omega N = \omega N_0 e^{-\omega t}$$

or

$$n(t) = n_0(t) e^{-\omega t} \quad (5.4-1)$$

where $n_0(t)$ is the rate of disintegration at the initial time.

If an element has a long mean lifetime, then formula (5.3-2) cannot be verified experimentally. Suppose we take uranium-238. Present-day measuring techniques enable us to detect every instance of radioactive decay. It turns out that 1.2×10^4 disintegrations take place in one gram of U^{238} every second. One gram of U^{238} contains 2.5×10^{21} atoms and so

$$\omega = \frac{1.2 \times 10^4}{2.5 \times 10^{21}} = 5 \times 10^{-18} \frac{1}{\text{sec}}$$

Using (5.3-6) we find $\bar{t} = \frac{2.5 \times 10^{21}}{1.2 \times 10^4} = 2 \times 10^{17} \text{ sec} = 7 \times 10^9 \text{ years}$.

Suppose we observe the disintegration of uranium for 10 years. During this time, 4×10^{12} atoms will have disintegrated in one gram. It would be very hard to detect that the original amount of 2.5×10^{21} atoms was diminished by $(2.5 \times 10^{21} - 4 \times 10^{12})$.

However, by experiments involving radioactive substances with relatively short mean lifetimes (from a few minutes to a few days) it has been possible to verify formula (5.4-1) with great accuracy and, thus, to corroborate formulas (5.3-1) and (5.3-2). To do this, let us compute the number of disintegrations over small periods of time. Dividing the number of disintegrations by the time interval, we get the rate of disintegration at various instants of time.

We construct the graph of the disintegration rate as a function of time. How can we be sure that this curve is the graph of the exponential function? We can compute the logarithms of the resulting values of the rate of disintegration and, on this basis, construct a graph of $\ln n$ as a function of the time t . The result should be a straight line, which can be checked roughly by eye. Numerous experiments do indeed yield a straight line.

Thus, $\ln n(t)$ is a linear function of time, or

$$\ln n(t) = a + bt \quad (5.4-2)$$

Now this means that $n(t) = e^{a+bt} = e^a e^{bt} = ce^{bt}$. The quantity b turns out to be negative on the graphs: $b = -\omega$ where ω is the pro-

bability of disintegration. Thus, experiment confirms the basic result of the preceding section and enables us to determine ω by computing the tangent of the angle of inclination of the straight line (5.4-2) to the t -axis.

Actually, this is a very remarkable result.* Imagine N_0 radioactive atoms "manufactured" simultaneously at time $t = 0$. They are all prepared in the same fashion and at the same time. We know that radioactive atoms are unstable and are capable of disintegrating. We can suppose that the disintegration of the atoms requires a definite time. Imagine that after the atoms are ready, a certain time must elapse before they are mature enough to disintegrate. But then we should expect all the atoms to mature in the same period of time, and at the expiration of that time, disintegrate simultaneously. Imagine, further, that we have models of guns with stretched springs and gears (or clockwork time devices) that release their shells when the gears (or clocks) reach a particular position (time). Firing will be regarded as disintegration of the model. If all models are the same, manufactured at the same time, the shells should be fired after the lapse of an identical time interval.

But this picture has nothing whatsoever in common with the actual behaviour of radioactive atoms. Though created at the same time, they disintegrate at all imaginable times. Let us try to find out what percentage disintegrates during a time less than the mean lifetime. From (5.3-2) we find that the rate of disintegration (the number of atoms that decay in unit time) is $\frac{dN}{dt} = -\omega N_0 e^{-\omega t}$. During time dt there will be

$$\frac{dN}{dt} \cdot dt = dN = -\omega N_0 e^{-\omega t} dt$$

atomic disintegrations, and during time from $t = 0$ to $t = \bar{t}$ the following number of atoms will disintegrate:

$$M = - \int_0^{\bar{t}} \omega N_0 e^{-\omega t} dt = -N_0 e^{-\omega t} \Big|_0^{\bar{t}} = N_0 (1 - e^{-\omega \bar{t}})$$

Since $\omega = \frac{1}{\bar{t}}$, it follows that

$$M = N_0 \left(1 - \frac{1}{e}\right) \approx 0.63 N_0$$

* Niels Bohr, speaking in 1905 on radioactive transformations, said in this connection: "The meaning of the discussions on the mean lifetimes of atoms without any indication of a definite instant of time lies in the fact that they, so to say, do not grow old until they begin to decay; this means that the probability of disintegration is the same for them at any time."

Thus, 63% of the atoms will disintegrate during a time less than $\bar{t}$. Similarly, we compute that during time from $\bar{t}$ to $2\bar{t}$, 23% of the atoms disintegrate and during time exceeding $2\bar{t}$, 14% of the atoms.

In Fig. 119 we see two curves: one for the number of disintegrations in unit time for radioactive atoms (1) and for the gun models (2). The gun-model curve has a certain width here. We can figure perhaps that the models were not quite exact and therefore did not fire off quite at the same time. The more precise the models, the narrower the curve in Fig. 119.

The area under the curve represents the total number of disintegrated atoms for one curve and the total number of all models for the other curve. We can take the number of models to be equal to the number of atoms. Then both curves will have the same area.

The abscissa of the centre of gravity of both curves is also the same.* This means that we consider models in which the mean lifetime (prior to firing) is the same as the mean lifetime of the radioactive atoms.

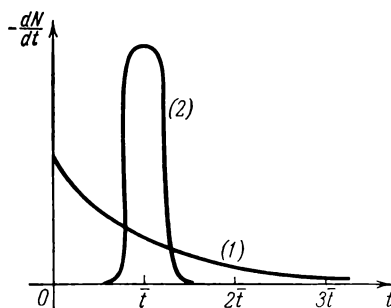


Fig. 119

We have thus done everything in our power to make the curves similar: we took as many models and with mechanisms such that the total number of models and atoms and the mean lifetimes of the models and atoms are the same. And yet the curves are so strikingly different! Experimentation with radioactive nuclei irrefutably rejects the type of curve obtained for the models. The more accurate the experiment, the more precisely is the law (5.3-2) confirmed.

We examined this system with models so that the reader would not accept as ordinary and natural the relationship (5.3-2) for radioactive decay and would have cause for surprise and curiosity: "Indeed, why does radioactive disintegration proceed in this fashion?"

What is the physical meaning of the probability of disintegration? A long time ago, at the beginning of the century, it was suggested that radioactive decay requires some kind of external action, say the entry of a particle from outside. Then we would image that one atom disintegrated earlier since it was hit by an incoming particle, while some other atom remained untouched. But this hypothesis did not fit the facts which stated that radioactive disintegration proceeds at the same rate under all manner of conditions, irrespective of temperature, collisions of atoms among themselves, the action

* As will be demonstrated in Sec. 6.15, this follows from formula (5.3 5).

of cosmic radiation. Also, energy is strictly conserved in radioactive disintegration, which likewise rejects the idea of some kind of outside influence.

A second possible hypothesis is that at the initial time of creation of the radioactive atoms they were actually not quite alike and for this reason decayed at different times. This is in keeping with the clock-driven models with clocks set for different times. This hypothesis presumes that an exact knowledge of the state of every atom completely determines the whole subsequent history of the atom and, in particular, determines with exactitude when the given atom will disintegrate. If atoms disintegrate at different times after their creation, this means that the whole business was foreordained: when created, the different atoms of one and the same radioactive substance were not exactly the same and the diverse decay times were predetermined in the creation stage.

This view does not hold water either. For each specific mode of generation of atoms of a radioactive element we should have a definite relationship between the rate of disintegration and time. Experiment refutes this supposition.

One and the same type of radioactive atom can often be obtained in a variety of ways: say, atoms of Mo^{99} (molybdenum-99, molybdenum with atomic weight 99) are produced in nuclear reactors in the process of the fission of uranium atoms. These same atoms were earlier obtained under the action of the nuclei of heavy hydrogen (deuterium) on the atoms of ordinary, naturally occurring nonradioactive molybdenum. At the present time we know of numerous instances where one and the same type of radioactive atom is produced by a variety of methods. Experiment has shown that, irrespective of the mode of production of the atoms, the disintegration rate is given by formula (5.3-2) with a constant value of ω , which characterizes the given type of atom. Consequently, it is precisely the basic equation

$$\frac{dN}{dt} = -\omega N$$

that all experiments confirm.

This equation is pregnant with meaning: all radioactive atoms are identical. The probability of disintegration does not depend on how and when the atoms were obtained. One hundred freshly produced atoms disintegrate in exactly the same fashion as in the case where 10^6 are generated, and a time interval elapses such that 100 atoms are left and we consider the fate of these 100 remaining atoms.*

* Characteristic of the exponential function is that any portion of the curve is similar to the whole curve. Indeed, let $N = N_0 e^{-\omega t}$. At time $t = t_1$, $N = N_1 = N_0 e^{-\omega t_1}$. Let us begin reckoning time anew from time t_1 . Denote by τ the time reckoned from this instant: $\tau = t - t_1$, $t = t_1 + \tau$. Then $N =$

What is so remarkable in the fact that 100 atoms with a given atomic weight and a given number of electrons are the same? If these were nonradioactive atoms there would indeed be no cause for surprise. But for radioactive atoms there certainly is cause enough when we recall that out of 100 atoms, 63 disintegrate in time $\bar{t}$ and the other 37 disintegrate after $\bar{t}$. What is strange here is that the disintegration time is different although the atoms are the same.

It is not fruitless to wonder in this fashion. In the phenomenon of radioactive decay we already perceive certain peculiarities in the laws of motion of atomic and nuclear particles that differ from the laws of motion of the bodies we are accustomed to in ordinary life. These peculiarities are studied in quantum mechanics. All this is of course outside the scope of our book. Our aim is modest enough. It is to show that the necessity for elaborating radically new conceptions which differ drastically from those of ordinary mechanics stems from very simple facts about radioactivity that can be comprehended by any school child. To realize that the old conceptions were insufficient, it was necessary to doubt, to wonder, to be surprised.

In his autobiography, Albert Einstein—the greatest physicist of the twentieth century—notes the surprise and wonder that he experienced when he first saw a compass and perceived the mysterious action of a magnetic force that passes through paper, wood, the earth, and acts on the compass needle without any direct contact. He wrote that this wonderment served as a tremendous impetus for a further search. He wrote of curiosity, which, he claims “the modern methods of teaching have all but stifled.” Einstein himself evinced an extraordinary capability for wonderment and he was able to derive inspiration and impetus for the creation of theories out of the most mundane facts of everyday life. Thus, underlying the brilliant general theory of relativity is the simple fact that caused Einstein to wonder why different bodies fell with the same acceleration.

Quite naturally, to wonder is not enough, and to merely pose a problem does not suffice. Einstein combined the ability to pose a problem and to solve it, which means mastering the requisite mathematical techniques. And yet, among a galaxy of outstanding scientists, Einstein is the celebrated physicist of the twentieth century because of his capacity to wonder and pose a problem where others were not able to see anything out of the ordinary.

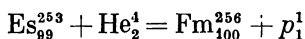
$= N_0 e^{-\omega t} = N_0 e^{-\omega(t_1 + \tau)} = N_0 e^{-\omega t_1} e^{-\omega \tau} = N_1 e^{-\omega \tau}$. We consider the disintegration of N_0 particles prepared at time $t = 0$ and we are interested in the portion of the process that develops after $t = t_1$. For this portion we get $N = N_0 e^{-\omega t_1} e^{-\omega \tau} = N_1 e^{-\omega \tau}$, where τ is the time reckoned from t_1 . Thus, the law of disintegration for the number of particles $N_1 = N_0 e^{-\omega t_1}$ remaining after the previous decay is exactly the same as the law of disintegration of N_1 freshly obtained particles. That is precisely what is asserted in the text.

Perhaps this analysis of radioactive decay will help the reader to see what depths of content are hidden behind simple facts and formulas.

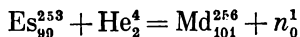
To conclude, we will illustrate with the curves of radioactive decay obtained experimentally in 1955 by Glenn Seaborg and his associates in the United States. They were first to observe Element No. 101 of the periodic table, to which they gave the name *mendelevium* (element symbol: Md) in honour of the great Russian chemist Dmitri Mendeleyev.

In this study, the 98th element, *californium*, with atomic weight 252 was irradiated with neutrons in a nuclear reactor to yield californium-253, which ejects an electron and turns into Element No. 99, which is called einsteinium (element symbol: Es) and has the same atomic weight of 253.

About 10^9 atoms of einsteinium (this is equivalent to 4×10^{-13} gram) were deposited on a gold plate and subjected to alpha-particle (α -particles are nuclei of helium) bombardment in a cyclotron. This generates Element No. 100, fermium, Fm, in accordance with the reaction



and mendelevium via the reaction



In this notation, each chemical symbol has a subscript indicating the number in the periodic table (that is, the number of protons in the nucleus). The superscript indicates the atomic weight (rounded to a whole number) which is the sum of the number of neutrons and protons in the nucleus. He_2^4 is a helium nucleus, or α -particle, p_1^1 is the nucleus of an atom of hydrogen, or simply, the proton, n_0^1 is a neutron. In a nuclear reaction, the sum of the subscripts on the right equals that on the left, also the sum of superscripts on the right is equal to that on the left, since in nuclear reactions all we have is an exchange of neutrons and protons between nuclei. After the α -particle bombardment, the gold plate together with the newly formed fermium and mendelevium was dissolved in acid, and then the fermium and mendelevium were extracted chemically. As Seaborg writes, it was the periodic table of Mendeleyev that permitted foreseeing the chemical properties of an element that had never before existed in nature and had never been studied. After chemical separation, measurements were made of the radioactive decay characteristics. Fermium-256 (with atomic weight of 256, that is) disintegrates radioactively with a half-life of about 3.5 hours. It breaks up (that is, fissions spontaneously) into two nuclear fragments of roughly the same mass (the details of the fission process are given in Sec. 5.5).

The upper curve in Fig. 120 shows the number of nuclei of fermium as a function of time in this experiment. Plotted on the axis of abscissas is the time in minutes. Plotted on the axis of ordinates is the number of atoms available at a given time.* The scale on this axis is not uniform: an ordinate is proportional to the logarithm of the number of atoms. In particular, the axis of abscissas ($y = 0$) corresponds to one remaining atom ($\ln 1 = 0$), while for zero atoms we have $-\infty$ on the axis of ordinates. The disintegration of each separate atom changes the number of atoms by 1; in between disintegrations, the number of atoms is constant. In this experimental set-up in which each separate disintegration is recorded, we have a polygonal (step-like) curve instead of a smooth curve. On the step-like curve, each disintegration is associated with a vertical line connecting two steps. The straight line drawn in Fig. 120 corresponds to the decay law

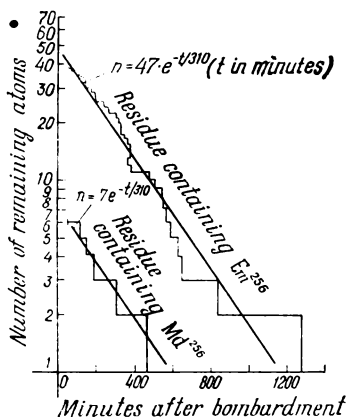


Fig. 120

$$n = n_0 e^{-\frac{t}{\tau}}$$

where $\tau = \frac{T}{\ln 2} \approx 5$ hours, $T \approx 3.5$ hours. It will be seen from Fig. 120 that, in all, there were recorded 40 disintegrations of fermium. The more atoms there are, the closer is the polygonal line to a straight line. When there are fewer than five atoms left, then quite naturally the probabilistic nature of radioactive decay leads to appreciable deviations from the exponential law which holds true for large numbers of atoms.

After chemical separation, the nucleus of mendelevium rapidly (in half an hour) captures an atomic electron and turns into a nucleus of fermium. And so the precipitate containing mendelevium also yields (when the radioactivity is measured) a disintegration of atoms into two fragments with a half-life of 3.5 hours. The decay curve of fermium obtained from mendelevium lies in the lower left-hand corner of Fig. 120. Six disintegrations were experimentally observed. Special experiments demonstrated that these six atoms could not

* It is not possible to count the number of atoms available at a given instant. What is recorded experimentally are the disintegration events of the atoms. The number of atoms N at time t is computed after the experiment when all N atoms have disintegrated.

have appeared as a fermium impurity in the precipitate being measured but most definitely had formed from the mendelevium.

Seaborg and his associates observed a total of 17 atoms of mendelevium in that series of experiments.

The foregoing example is not very good as an illustration of the how exactly the exponential law holds true in radioactive decay. Experiments demonstrating the validity of the exponential law were successfully carried out with more common radioactive substances. On the other hand, the example of mendelevium and fermium shows to what peaks of experimental technique modern physicists have attained in synthesizing new elements and recording the disintegration of each separate atom.

In Seaborg's experiments, the counter recording mendelevium disintegrations was hooked up to an amplifier in the loudspeaker system of the institute and every disintegration event was heard by workers in the various laboratories on different floors so they could celebrate the birth (actually the recorded death) of every atom of the new element created by man (true, before the work was over, the local fire department got interested in these goings-on and the disintegration news broadcast was stopped).

5.5 SERIES DISINTEGRATION (RADIOACTIVE FAMILY)

In a number of cases, radioactive decay produces atoms which again decay radioactively, so what we have is a chain of disintegrations: an atom of element A turns into an atom of element B , which in turn disintegrates into an atom of element C , and so on. Let us consider the mathematical problem of determining the dependence on time of quantities of elements A , B , C and ways of solving the problem. We denote the quantities of atoms of substances (elements) A , B , C that have not yet decayed at time t by the letters A , B , C .

Let the probabilities of disintegration of A , B , C be respectively equal to ω , ν , u . Then

$$\frac{dA}{dt} = -\omega A \quad (5.5-1)$$

(here, A is termed the parent element). We write the equation for element B . In unit time a total of νB atoms of element B disintegrate. On the other hand, during the same time we have ωA disintegrations of element A , and since each disintegration of an atom of A gives rise to one atom of B , then ωA atoms of B are formed in unit time. Therefore

$$\frac{dB}{dt} = -\nu B + \omega A \quad (5.5-2)$$

Similar reasoning gives

$$\frac{dC}{dt} = -uC + \nu B \quad (5.5-3)$$

The equations (5.5-1) to (5.5-3) form a system of differential equations. In the given instance, we can solve these equations one by one, having to deal each time only with one equation in one unknown. True enough, B and C do not enter equation (5.5-1). From it we therefore determine $A(t) = A_0 e^{-\omega t}$, where A_0 is the number of atoms of element A at the initial time $t = 0$.

Substituting the expression for $A(t)$ into (5.5-2), we get an equation involving only one unknown function, B :

$$\frac{dB}{dt} = -vB + \omega A(t) \quad (5.5-4)$$

How does one go about solving this equation? We can find a solution if we first consider the fate of the group of atoms of B that have formed during the same interval of time, from τ to $\tau + \Delta\tau$. We will consider the number of atoms of this group that are still "alive", ΔB (that is to say, atoms that have not disintegrated by time t), as a function of time t . So as to avoid confusion about the time t when we measure the number of atoms, and the time of formation of the group, let us denote these times by different letters: t and τ , respectively. At time τ , the rate of formation of atoms of element B was $\omega A(\tau)$. During the small time interval $\Delta\tau$, a total of $\Delta B_0 = \omega A(\tau) \Delta\tau$ atoms of element B were formed.

How does the number of atoms in the group at hand depend on time t ? For $t < \tau$ it is equal to zero: the atoms of interest have not yet formed since the group itself is still nonexistent, $\Delta B = 0$. Let $t > \tau$. Observe that a time $t - \tau$ has already passed since the group began to form. The decay probability of element B is v . Therefore, after the lapse of time $t - \tau$ from the time of formation of the group the number of untransformed atoms will be

$$\Delta B(\tau) = \Delta B_0 e^{-v(t-\tau)} = \omega A(\tau) e^{-v(t-\tau)} \Delta\tau$$

To find the total number of atoms of element B at time t , we have to add the number of atoms in all groups that formed prior to t . If we take $\Delta\tau$ (and hence ΔB as well) very small, then the sum turns into the integral

$$B(t) = \int_0^t \frac{\Delta B(\tau)}{\Delta\tau} d\tau = \int_0^t \omega A(\tau) e^{-v(t-\tau)} d\tau$$

Observe that here the variable of integration is denoted by τ . The argument t , upon which B depends, enters into the integral twice: as the upper limit and in the integrand. When integrating with respect to τ the quantity t is to be regarded as a constant. We can therefore write

$$e^{-v(t-\tau)} = e^{-v t} e^{v \tau}$$

and take ωe^{-vt} out from under the integral sign as a factor that is independent of τ . We then get

$$B(t) = \omega e^{-vt} \int_0^t A(\tau) e^{v\tau} d\tau \quad (5.5-5)$$

It is easy to verify, without evaluating the integral, that the solution (5.5-5) satisfies the original equation (5.5-4) for any function $A(\tau)$. Indeed, we find the derivative $\frac{dB(t)}{dt}$. By the rule for differentiating a product we get

$$\frac{dB(t)}{dt} = -\omega v e^{-vt} \int_0^t A(\tau) e^{v\tau} d\tau + \omega e^{-vt} \frac{d}{dt} \left(\int_0^t A(\tau) e^{v\tau} d\tau \right)$$

Since, by the property of the derivative of an integral (see Sec. 2.9),

$$\frac{d}{dt} \left(\int_0^t A(\tau) e^{v\tau} d\tau \right) = A(t) e^{vt}$$

it follows that

$$\frac{dB}{dt} = -\omega v e^{-vt} \int_0^t A(\tau) e^{v\tau} d\tau + \omega A(t) = -vB + \omega A$$

Now if we set $A(\tau) = A_0 e^{-\omega\tau}$ then we get the concrete solution

$$B(t) = \frac{A_0 \omega}{v - \omega} (e^{-\omega t} - e^{-vt})$$

The solution could also have been found without resorting to a consideration of the separate groups of atoms. Now that the solution has been found, it is already a simple matter to guess the mathematical technique that will lead us to our goal. The solution (5.5-5) is of the form

$$B(t) = e^{-vt} I(t) \quad (5.5-6)$$

where $I(t)$ denotes the integral that depends on t . We will seek the solution in the form of a product of e^{-vt} by the unknown function I and we will set up an equation for I :

$$\frac{dB}{dt} = \frac{d}{dt} (e^{-vt} I) = -v e^{-vt} I + e^{-vt} \frac{dI}{dt} \quad (5.5-7)$$

Substituting expressions (5.5-7) and (5.5-6) into (5.5-4) we get

$$e^{-vt} \cdot \frac{dI}{dt} = \omega A(t)$$

or

$$\frac{dI}{dt} = \omega e^{vt} A(t). \quad (5.5-8)$$

By hypothesis, at the initial time $t = 0$, $B = 0$, and hence $I = 0$.

With this initial condition, the solution of equation (5.5-8) has the form

$$I(t) = \int_0^t \omega e^{v\tau} A(\tau) d\tau$$

and, finally,

$$B(t) = e^{-vt} I(t) = e^{-vt} \int_0^t \omega A(\tau) e^{v\tau} d\tau$$

In this formula it is essential, so as to avoid confusion, to retain strict designations and not to denote the variable of integration τ by the same letter we use for the upper limit of integration, t .

5.6 INVESTIGATING THE SOLUTION FOR A RADIOACTIVE FAMILY (SERIES)

In the preceding section we brought to completion the solution of the problem in the case of two radioactive substances (elements). Let us now investigate this solution for two particular cases:

(1) a short-lived parent element A and long-lived daughter element B ;

(2) a long-lived parent element A and short-lived daughter element B .

Below, in addition to the decay probabilities ω and ν , we will make use of the mean lifetimes $\bar{t}_A = \frac{1}{\omega}$, $\bar{t}_B = \frac{1}{\nu}$. In the first case, the nature of the solution can be readily grasped without calculations and formulas. The entire process breaks down into two stages.

First, when t is of the order of $\bar{t}_A$ (here, by hypothesis, $\bar{t}_A \ll \bar{t}_B$ and so also $t \ll \bar{t}_B$ in the first stage), A is converted into B . During this time there is hardly any disintegration of element B . During this period the amount of B is equal to the difference between the original amount A_0 and the amount A remaining at time t :

$$B(t) = A_0 - A(t) = A_0 - A_0 e^{-\omega t} = A_0 (1 - e^{-\omega t}), \quad (t \ll \bar{t}_B)$$

By the end of this period, practically the whole of element A has been converted into B , the quantity of B becomes equal to the original amount of the parent element A_0 . Then follows a slow and protracted disintegration of B .

$$B(t) = A_0 e^{-\nu t}, \quad t \gg \bar{t}_A$$

We will show how these expressions are obtained from the exact formula. For the case of two radioactive elements A and B , we obtained in the preceding section the formula

$$B(t) = A_0 \frac{\omega}{v - \omega} (e^{-\omega t} - e^{-vt})$$

In our case, $\bar{t}_A \ll \bar{t}_B$, $\omega \gg v$ and so it is more convenient to interchange the signs so as to be dealing with positive quantities in the parentheses and in the denominator of the fraction. Then

$$B(t) = A_0 \frac{\omega}{\omega - v} (e^{-vt} - e^{-\omega t}) \quad (5.6-1)$$

Since $v \ll \omega$, it follows that $\frac{\omega}{\omega - v} \approx \frac{\omega}{\omega} = 1$.

We consider the expression $e^{-vt} - e^{-\omega t}$ for two successive stages. First, when $t \ll \bar{t}_B = \frac{1}{v}$, it will be true that $vt \ll 1$. Then $e^{-vt} \approx 1$. Since t can be a quantity of the order of $\bar{t}_A$ and ωt , consequently, of the order of unity, it follows that the quantity $e^{-\omega t}$ can be computed exactly. From formula (5.6-1) we get

$$B(t) = A_0 (1 - e^{-\omega t})$$

In the second stage, when $t \gg \bar{t}_A = \frac{1}{\omega}$, it will be true that $\omega t \gg 1$. We can disregard $e^{-\omega t}$ in this stage. Then $e^{-\omega t}$ is small not only with respect to unity but also in comparison with e^{-vt} , since $v \ll \omega$. We get

$$B(t) = A_0 e^{-vt}$$

Thus the exact formula does indeed yield the same results as those obtained from simple qualitative reasoning.

Now let us take up the second case, that of the long-lived parent element A and the short-lived daughter element B :

$$\bar{t}_A \gg \bar{t}_B, \quad \omega \ll v$$

We consider the period when a time t , considerably exceeding $\bar{t}_B$, has passed since the onset of the process. In that case, the element B that was formed at the beginning of the process has already fully disintegrated by time t . Since B disintegrates rapidly, in a short time, there is always some of the recently formed element B available at every given instant of time. What we have here is a *steady state* (also known as a *stationary state*): element B is formed from A and straightway disintegrates, element B does not accumulate because it decays rapidly, but does not disappear completely because A is producing new quantities of B all the time. In a steady-state system, there are just as many atoms of B disintegrating in unit time as there

are atoms of B being formed from A . Mathematically, this condition is written thus:

$$\nu B = \omega A \quad \bullet$$

whence

$$B(t) = \frac{\omega}{\nu} A(t) = \frac{\bar{t}_B}{\bar{t}_A} A(t)$$

In a steady-state system, the instantaneous quantity of B is proportional to the quantity of A and always represents some small portion of A . This portion is small because in the case at hand (the second case), $\bar{t}_B \ll \bar{t}_A$ and hence $\frac{\bar{t}_B}{\bar{t}_A} \ll 1$, for otherwise there would be no steady state.

How do we obtain the steady-state equation from the exact differential equation $\frac{dB}{dt} = -\nu B + \omega A$? Evidently, if we take it that $\frac{dB}{dt}$ is small in comparison with each of the two terms on the right, then, replacing $\frac{dB}{dt}$ by 0, we approximately get

$$0 = -\nu B + \omega A, \quad \nu B = \omega A$$

Let us now examine the beginning of the process. For $t = 0$, $A = A_0$, $B = 0$. This means that at the beginning we do not have a steady state, since by the steady-state formulas we should first have

$$B_s = \frac{\omega}{\nu} A_0$$

(the subscript s on B denotes a steady state). At time $t = 0$, element B is forming at the rate of $\frac{dB}{dt} = \omega A_0$, while there is no disintegration of B at all at the initial time, since $B = 0$.

It is possible to determine the time t_1 during which, for the initial rate of buildup of B , the quantity B_s will be attained. Indeed, if the rate of formation of the substance remains constant, equal to $\left(\frac{dB}{dt}\right)_{t=0}$, then

$$B = t \left(\frac{dB}{dt}\right)_{t=0}$$

Here, putting $B_s = \frac{\omega}{\nu} A_0$, $\left(\frac{dB}{dt}\right)_{t=0} = \omega A_0$ we get the desired time

$$t_1 = \frac{\omega A_0}{\nu \omega A_0} = \frac{1}{\nu} = \bar{t}_B$$

Thus, the steady state is attained in a time roughly equal to the mean decay time of element B . From the condition $\bar{t}_A \ll \bar{t}_B$ it is

evident that the quantity of element A changes but slightly during this time.

On the whole, the approximate examination, in the second case, of a short-lived daughter element yields the following:

$$\text{for } t < \bar{t}_B \quad B(t) = \left(\frac{dB}{dt} \right)_{t=0} t = \omega A_0 t,$$

$$\text{for } t > \bar{t}_B \quad B(t) = B_s = \frac{\omega}{v} A_0 e^{-\omega t} = \omega \bar{t}_B A_0 e^{-\omega t}$$

We get the function $B = B(t)$ in the form of two lines: first a straight line, then as an exponential curve (Fig. 121, $\bar{t}_A = 10\bar{t}_B$). It

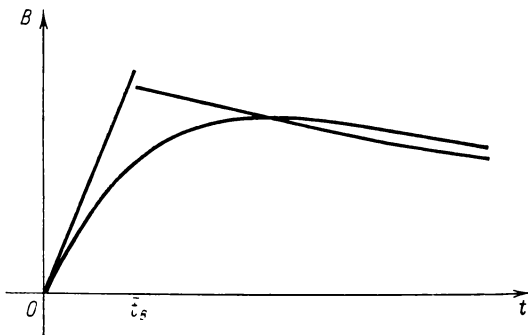


Fig. 121

is easy to verify that for $t = \bar{t}_B$ the two formulas yield almost the same values.

Let us see what the exact solution to equation (5.6-1) gives us,

$$B = A_0 \frac{\omega}{v - \omega} (e^{-\omega t} - e^{-vt})$$

in the case at hand when $v \gg \omega$, $\bar{t}_B \ll \bar{t}_A$. In the denominator, we neglect ω compared with v . For $vt \gg 1$, we disregard e^{-vt} in the brackets to get

$$B = A_0 \frac{\omega}{v} e^{-\omega t}$$

which is precisely the steady-state solution.

We determine the approach of the solution to a steady state by how fast e^{-vt} falls off. For very small t , when $vt < 1$, so that ωt is all the more so small, we get the following by expanding e^{-vt} and $e^{-\omega t}$ in a series and confining ourselves to the first two terms:

$$B = A_0 \frac{\omega}{v - \omega} (1 - \omega t - 1 + vt) = A_0 \omega t$$

which coincides with the approximate result. Actually, however, the exact formula yields a single smooth curve without discontinuities and salient points (it is shown in Fig. 121). The approach of this curve to a steady-state solution depends on how fast e^{-vt} diminishes. Thus, for e^{-vt} to make a correction of the order of 10%, we must have $vt = 2.3$, $t = 2.3 \cdot \frac{1}{v} = 2.3 \bar{t}_B$. Here, due to the smallness of ωt , we assume that $e^{-\omega t} \approx 1$. Thus, true enough, the transition from the stage of initial buildup to the stage where the solution is equal, with sufficient exactitude, to the steady-state solution takes place in a time of the order of the time of disintegration, $\bar{t}_B$.

The example of a radioactive family (series) is very instructive in the sense that obtaining a general exact solution does not in the least signify that the work is at an end. The construction of approximate theories for various limiting cases is an absolutely necessary part of the work and the existence of an exact formula does not at all take the place of an approximate theory. Approximate, yet clear-cut and pictorial conceptions serve as a check on an exact formula.

Also, approximate theories give us important new qualitative concepts such as that of the steady state. These can more easily be remembered and they possess a broader range of application than do the exact formulas. For instance, in the case of a radioactive family consisting of several generations $A \rightarrow B \rightarrow C \rightarrow D$, the exact formula is extremely unwieldy. But if $\bar{t}_A$ is greater than all other times, then all the results referring to the steady state are obtained just as easily as in the case of two elements A and B .

Frequently, the easiest way consists in obtaining an exact solution valid for arbitrary v and ω (in our case), from which we then (for $v \ll \omega$, or $v \gg \omega$) obtain, via mathematical manipulations, some simpler approximate formulas for the two extreme cases. But this is not yet all! If a simple approximate formula has been obtained in a simple yet long-winded manner via the general solution, then alongside this there should be another, simple, way of obtaining an approximate formula. One should always attempt to find simple pathways because there will invariably appear problems in which the approach to an exact solution is insuperably complicated and only a simple approximate approach makes it possible to advance in the solution.

In practical situations, exact formulas come up just as rarely as equations with solutions in whole numbers, although most of the textbook problems lead to exact formulas, just as problem books for junior classes abound in equations that can always be solved in whole numbers.

Observe that the conceptions of radioactive families account for the strange result of the exercise in Sec. 5.3 about the amount of

radium in the past: radium is a descendent (true, not direct, but via a number of intermediate substances) of uranium-238. It is therefore not correct to regard the present-day supply of radium as the result of the decay of primordial radium. Actually, radium is in a steady state with uranium. From the equation

$$B = \frac{\omega}{\nu} A$$

we find that the quantity of radium $B = 10^{-12}$ corresponds to the uranium content:

$$A = \frac{\bar{t}_A}{\bar{t}_B} B = 3 \times 10^6 B = 3 \times 10^{-6}$$

We have approximately found the present-day amount of uranium-238 in rocks. The original abundance, 5×10^9 years ago, was twice as much, of the order of 6×10^{-6} . These magnitudes are quite reasonable, unlike the results of the exercise in Sec. 5.3.

5.7 THE CHAIN REACTION IN THE FISSION OF URANIUM

In 1938, Otto Hahn and Fritz Strassmann in Germany and Curie and Joliot in France demonstrated that when a neutron enters a nucleus of uranium, fission occurs in which the nucleus breaks up into two large fragments with the simultaneous emission of two or three new neutrons. Uranium with an atomic weight of 235 (uranium-235 for short) is very active in this respect. Naturally-occurring uranium contains about 0.7% of uranium-235 atoms and 99.3% of uranium-238 atoms.* The fission fragments of uranium-235 are medium atomic-weight nuclei from 75 to 160. The charge on these nuclei lies within the range from 35 to 57, the sum of the charges of two fragments always being equal to the charge on the nucleus of uranium, that is 92 elementary charges. The sum of the atomic weights of the two fragments is equal to $235 + 1 - \nu$, where 235 is the atomic weight of uranium, 1 is the atomic weight of the neutron that caused the fission, and ν is the number of neutrons generated in the act of

* This was followed in 1939, in the laboratory of I. V. Kurchatov in Leningrad, by the demonstration (carried out by the Soviet scientists G. N. Flyorov and K. A. Petrzhak) that uranium-238 is capable of undergoing spontaneous fission without the entry of any neutron, although the probability of this event is extremely small. The probability of radioactive decay (with the emission of an α -particle) of uranium-238 corresponding to a half-life of 4.5×10^9 years is equal to $\omega = 5 \times 10^{-18}$ 1/sec while the probability of the spontaneous fission of uranium-238 is less by a factor of 10^8 , i.e., it is equal to 5×10^{-26} 1/sec. Thus, in one second in one kilogram of uranium (which is about 2.5×10^{21} atoms) there occur roughly 10^7 radioactive disintegrations and only 10 events of spontaneous fission.

On the other hand, in the very heaviest elements, spontaneous fission becomes predominant, the most probable decay process (see end of Sec. 5.4 where the decay curve of mendelevium is given). The problem of the chain reaction that we consider below does not involve spontaneous fission at all.

fission. An enormous energy of 6×10^{17} erg/g (per gram of fissioned uranium) is released in the fission process. Thanks to this great energy, the fragments rush apart at speeds of about 10^9 cm/sec.

The source of this energy is the electric repulsion of two like-charged fragments. Before the nucleus is separated into two parts, the nuclear forces between particles that make up the nucleus balance the electric repulsive forces. But as soon as the nucleus has broken up into two separate fragments, the repulsion of these two fragments is not countered in any way and so they fly apart at high speed. The fragments are very quickly brought to rest in a dense substance. Their time of flight is between 10^{-13} and 10^{-12} sec. In this time they traverse distances from 10^{-4} to 10^{-3} cm. The kinetic energy of the fragments is converted into heat. The neutrons produced in fission have velocities of about the same order as the fragments (about 2×10^9 cm/sec).

Of crucial importance for the practical utilization of the energy of nuclear fission is the fact that a fission event caused by one neutron gives rise to more than one neutron. It is quite clear that if the neutrons do not leave the system, their number will increase in geometric progression with time, i.e., in accordance with the law of the exponential function. The rate of energy release will build up by the same law, in proportion to the number of neutrons. And even if at the onset of the process there were few neutrons, their number builds up so fast that the energy will be released at a rate convenient for practical use (for instance as a source of energy for a nuclear power plant), and in just a short additional space of time the energy release will build up to such an extent that an atomic explosion will take place. In reality, part of the neutrons leave the system, some are captured by other nuclei without causing fission. We can utilize this to control the number of neutrons and, in a particular case, attain a steady-state system in which the number of newly formed neutrons in unit time is equal to the number of used up neutrons so that the number of neutrons in the system remains the same in the course of time, and the energy can be released at a constant rate. That precisely is the regime we need if atomic energy is to be used for peaceful purposes.

Our immediate task is to set up and investigate the equation describing the number of neutrons in a system as a function of time.

5.8 MULTIPLICATION OF NEUTRONS IN A LARGE SYSTEM

Let us first derive an equation for the variation in the number of neutrons with time in a very large system (say in a large chunk of uranium-235) when loss of neutrons to the outside can be neglected.*

* We will consider the simplest case of metallic uranium-235 without graphite moderator.

The neutrons can all be regarded as having the same speed, denote it by v .

Fission of a nucleus occurs in roughly half of all cases when a neutron enters a nucleus of uranium-235. In the other half, the neutron emerges leaving the nucleus in the same state, the number of neutrons has not changed. The uranium nucleus is a sphere of radius R of the order of 10^{-12} cm.

How often will a neutron in flight inside the metal hit a nucleus of uranium?

In the small time interval dt a neutron traverses a distance of vdt . Let us picture a cylinder whose axis is the route covered by the neutron; the radius of the cylinder is equal to the radius R of the uranium nucleus. The neutron collides with the nuclei whose centres lie inside the cylinder. If the centre of the nucleus lies inside the cylinder, the path of the neutron will pass at a distance less than R from the centre of the nucleus and so the neutron will hit the nucleus and enter it. The volume of the cylinder is equal to

$$\pi R^2 v dt$$

There are N atoms in a unit volume of metallic uranium and hence N nuclei (the dimensions of N are $1/\text{cm}^3$, or cm^{-3}). Therefore, in the volume $\pi R^2 v dt$ that interests us there are $N\pi R^2 v dt$ nuclei. There will be just as many events of a neutron hitting a nucleus during the small time interval dt . Not every neutron hit makes the nucleus fission. Let α be the portion of cases a neutron hitting a nucleus causes fission (in the case of uranium-235, $\alpha \approx \frac{1}{2}$). Then the number of fissions during time dt is equal to

$$N\alpha\pi R^2 v dt$$

The quantity $\alpha\pi R^2$, which has the dimensions of area since α and π are nondimensional, is called the cross section for fission and is denoted by σ_f (the subscript f on the Greek letter sigma stands for "fission").

If there are n neutrons in the bulk of metallic uranium, then the number of fissions in time dt is equal to

$$nN\sigma_f v dt$$

Each act of fission produces ν neutrons, but this involves the absorption of one neutron, and so the variation in number of neutrons in every fission event is equal to $(\nu - 1)$. Associated with the above number of fissions is the variation in the number of neutrons

$$dn = nN(\nu - 1)\sigma_f v dt \quad (5.8-1)$$

Thus, from this equation we get

$$\frac{dn}{dt} = nN(\nu - 1)\sigma_f v$$

Set

$$N (\nu - 1) \sigma_f v = a \quad (5.8-2)$$

Then

$$\frac{dn}{dt} = an$$

We already know that the solution of this equation is

$$n(t) = n_0 e^{at} \quad (5.8-3)$$

where n_0 is the number of neutrons in the system at $t = 0$.

To summarize, then, if the number of neutrons in a system varies solely because of fission, then the number of neutrons increases in geometric progression if the time increases in arithmetic progression.

Indeed, if we take a number of equally spaced intervals of time

$$t_1, t_1 + \Delta t, t_1 + 2\Delta t, t_1 + 3\Delta t, \dots$$

then the corresponding number of neutrons is

$$n_1 = n_0 e^{at_1}, f n_1, f^2 n_1, f^3 n_1, \dots$$

where $f = e^{a\Delta t}$. This way of describing the process—growth in geometric progression—is common in the popular literature. Physicists and engineers speak rather of an exponential growth (in accord with the law of exponential increase). The exponential law is characterized by the growth rate a [formula (5.8-3)].

Let us find the dimensions of a . In (5.8-3) at is in the exponent. This means that at is a nondimensional quantity and, consequently, the dimensions of a are 1/sec. This same result can be obtained if we recall that

$$a = N \frac{1}{\text{cm}^3} (\nu - 1) \sigma_f \text{cm}^2 v \frac{\text{cm}}{\text{sec}}$$

Let us find the approximate value of the constant a . The density of uranium is roughly equal to 18 g/cm³. The number of nuclei per cubic centimetre, N , can be computed by recalling the Avogadro number, which is equal to 6×10^{23} atoms in 1 gram-atom of any substance. Hence, 235 grams of uranium-235 contain 6×10^{23} atoms, or 6×10^{23} nuclei. In one cubic centimetre there are $\frac{18}{235} \times 6 \times 10^{23} \approx 4 \times 10^{22}$ nuclei, $N = 4 \times 10^{22}$ 1/cm³. Substituting the mean value $\nu \approx 2.5$, $v = 2 \times 10^9$ cm/sec, $\sigma_f = \frac{1}{2}\pi (10^{-12})^2 = 1.6 \times 10^{-24}$ cm², we get $a = 4 \times 10^{22} \times 1.5 \times 1.6 \times 10^{-24} \times 2 \times 10^9 = 2 \times 10^8$ 1/sec; $\frac{1}{a} = 5 \times 10^{-9}$ sec.

To summarize, if the neutrons do not leave the system, their number increases by a factor of e in 5×10^{-9} second.

At this rate of buildup, in one microsecond (10^{-6} sec), the number of neutrons has increased by a factor of

$$e^{2 \times 10^8 \times 10^{-6}} = e^{200}, \text{ or } 10^{0.43 \times 200} = 10^{86}$$

One metric ton of uranium-235 contains 2.5×10^{27} nuclei. If the neutrons do not leave the system, then this quantity of uranium will fission in less than one microsecond. This process is an explosion.

Such a rate of buildup is not permissible if we want to use the fission process for generating electric power. It is necessary that neutrons leave the system and thus reduce the rate of neutron buildup.

5.9 ESCAPE OF NEUTRONS

Picture a mass of uranium-235 in the form of sphere of radius r . We have to set up an equation for the number n of neutrons inside the sphere. Assume for the sake of simplicity that the sphere is fixed to a thin support so that it is surrounded by a complete void and a neutron that has left the sphere will never enter it again.

How can we determine the neutron flux (the number of neutrons leaving the sphere in unit time)? We make a rough calculation. Consider a small time interval dt . During this time, each neutron covers a distance of $v dt$. Where are the neutrons that leave the sphere in time dt ? Evidently they will have to be inside the sphere in a thin layer adjacent to the surface of the sphere but at a distance not exceeding $v dt$ from the surface, otherwise during time dt they will not reach the surface, cross it and leave for good. But neither will those neutrons that are inside the layer of thickness $v dt$ be able to leave in time dt since not all neutrons inside the layer have velocity directed outwards along a radius. In our very rough calculation we will ignore this latter circumstance.

How is it possible to find the number of neutrons in the layer? There are a total of n neutrons in the whole sphere. The volume of the sphere is $V = \frac{4}{3} \pi r^3$, the volume of the thin layer that interests us near the surface is approximately equal to $Sv dt$ if $v dt$ is small. Here, $S = 4\pi r^2$ (the surface area of a sphere).

The mean density of neutrons (the number per unit of volume) is $C = \frac{n}{V}$. Suppose that the density near the surface in the thin layer does not differ from the mean density. Then the number of neutrons in this layer is

$$CSv dt = \frac{nS}{V} v dt$$

Therefore, the flux (number of neutrons leaving in unit time) is

$$q = \frac{nS}{V} v = \frac{n \cdot 4\pi r^2}{\frac{4}{3} \pi r^3} v = \frac{3v}{r} n$$

Actually, the neutron density near the surface is less than the mean density and, what is more (this was noted above) the neutron

velocities have different directions. The neutron flux is thus less than we obtained,

$$q = \frac{3kv}{r} n \quad (5.9-1)$$

where k is a numerical coefficient less than 1. Later on, in Sec. 5.12, we will compare our results with experiment and find that k is close to 0.3. If nuclear fission does not occur inside the sphere and no new neutrons are generated, then for the number of neutrons inside the sphere we get the equation $\frac{dn}{dt} = -q$ or, using (5.9-1),

$$\frac{dn}{dt} = -\frac{3kv}{r} n$$

Setting

$$\frac{3kv}{r} = b \quad (5.9-2)$$

we get

$$\frac{dn}{dt} = -bn$$

The solution of this equation is familiar:

$$n = n_0 e^{-bt} \quad (5.9-3)$$

The mean residence time of neutrons inside the sphere is, by (5.9-3),

$$\bar{t} = \frac{1}{b} = \frac{r}{3kv}$$

Observe that $\bar{t} \approx \frac{r}{v}$. Therefore the mean time is roughly equal to the time during which a neutron moving at a speed of v travels a distance equal to the radius r of the sphere. An exact consideration of the escape of neutrons requires extraordinarily laborious computations. It is very important from the very start of one's studies to get used to approximate calculations of all quantities of interest. Exact calculations are frequently very involved and require quite a different range of knowledge, at times even the collective efforts of many workers and the use of electronic computers, and so on. But does this mean that a student engaged in self-instruction and living far away from higher educational institutions should give up the desire to consider a problem? There always exist simple, even though rough, methods (similar to the one just considered) for an approximate approach to a problem. To stop short of an approximate solution because the exact computations are complicated is merely to hide one's lack of courage. Very often, just such hesitancy is destructive to the first steps of a scientist or inventor!

5.10 CRITICAL MASS

Up to now we have considered separately two processes: the multiplication of neutrons without regard for their escape, and the escape of neutrons without regard for their multiplication.

Let us now consider a system in which neutrons multiply and can escape. As we know, in unit time there are formed an neutrons and bn neutrons escape from the system. Since the variation of the number of neutrons in unit time is $\frac{dn}{dt}$, it follows that

$$\frac{dn}{dt} = an - bn$$

or

$$\frac{dn}{dt} = cn \quad (5.10-1)$$

where $c = a - b$. For a given initial number of neutrons n_0 , equation (5.10-1) has the solution

$$n = n_0 e^{ct} \quad (5.10-2)$$

This solution gives quite different results depending on whether c is positive or negative. Indeed, from (5.10-2) it is evident that when $c < 0$ the number of neutrons n falls off with increasing t , which means n tends to zero with time. But if $c > 0$, then n increases with t , that is, n grows without bound in the course of time. Only the effect of new physical factors not accounted for in the equation can halt the growth of n .

Thus, the value $c = 0$ is a "critical value", for it separates the distinct types of solution with increasing and decreasing number of neutrons. Since $c = a - b$, for a given a we can speak of the critical value of b : $b_{cr} = a$ since for $b < b_{cr} = a$, $c = a - b > 0$, and for $b > b_{cr} = a$, $c = a - b < 0$. The quantity a is determined by the properties of the fissionable substance: according to (5.8-2), $a = N\nu\sigma_f(\nu - 1)$. The quantity b depends on the amount of fissionable substance taken:

$$b = \frac{3k\nu}{r} \quad (5.9-2)$$

The concept is therefore introduced of the critical value of the radius, r_{cr} , for which $b = b_{cr} = a$. From formulas (5.8-2) and (5.9-2) it follows that $\frac{3k\nu}{r_{cr}} = N\nu\sigma_f(\nu - 1)$, whence

$$r_{cr} = \frac{3k}{N\sigma_f(\nu - 1)}$$

The mass of a sphere of radius r_{cr} is called the critical mass, m_{cr} . It is clear that

$$m_{cr} = \frac{4}{3} \pi r_{cr}^3 \rho \quad (5.10-3)$$

where ρ is the density of the fissionable substance (as before, we consider a mass of fissionable material, say, uranium, in the form of a sphere).

For $r > r_{cr}$ (this is the same as $m > m_{cr}$), $c > 0$ and we have a multiplication of neutrons. For $r < r_{cr}$ ($m < m_{cr}$), $c < 0$ and the original quantity of neutrons diminishes. Suppose we have a sphere of radius r . Its surface area is

$$S = 4\pi r^2$$

the volume

$$V = \frac{4}{3} \pi r^3$$

$$\frac{S}{V} = \frac{4\pi r^2}{\frac{4}{3} \pi r^3} = \frac{3}{r}$$

If r is small, this ratio is great, but if r is great, the ratio is small. No wonder then that when the radius is small, when the ratio of surface area to volume is great, the neutron escape increases and the

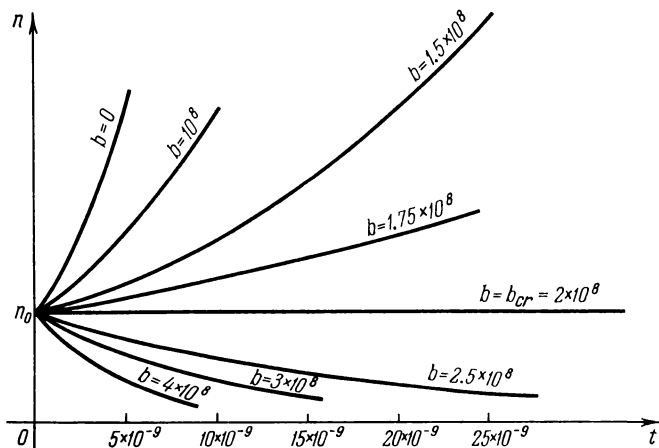


Fig. 122

conditions for neutron multiplication deteriorate. It is surprising how sharply the number of neutrons varies with b : if $b > b_{cr}$, then in a short time the number of neutrons becomes practically zero, irrespective of whether $b = 1.01 b_{cr}$ or $b = 2b_{cr}$. If $b < b_{cr}$, then

the number of neutrons increases without bound both for $b = 0.99b_{cr}$ and for $b = 0.5b_{cr}$, although the rate differs. That is precisely why

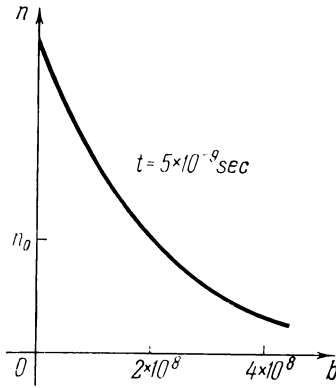


Fig. 123

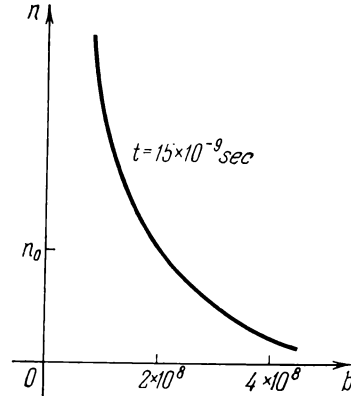


Fig. 124

one speaks of the critical value of b , the critical value of r or the critical value of mass. When above critical, the mass is called a supercritical mass, when less than critical, it is called a subcritical mass.

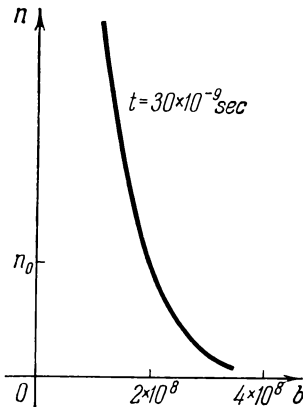


Fig. 125

In Fig. 122 are given the curves $n = n_0 e^{(a-b)t}$ for a number of values of b . Let us construct the curves of n as a function of b for a few definite values of time t . In the computations, a is taken equal to $2 \times 10^8 \text{ sec}^{-1}$. Figure 123 shows the curve of $n(b)$ for $t = 5 \times 10^{-9} \text{ sec}$. Figure 124 shows the curve of $n(b)$ for $t = 15 \times 10^{-9} \text{ sec}$, Fig. 125 shows the curve of $n(b)$ for $t = 30 \times 10^{-9} \text{ sec}$.

The intersection of the curves with the axis of ordinates ($b = 0$) in Figs. 124 and 125 lies outside the drawing: in Fig. 124 $n = 20n_0$ for $b = 0$, in Fig. 125 $n = 400n_0$ for $b = 0$.

As is seen in Fig. 122, and also from a comparison with Figs. 123 to 125, the greater the time interval t , the more divergent are the curves of $n(t)$ (Fig. 122), the steeper are the curves of $n(b)$ (Figs. 123 to 125), and the more sharply is the criticality of the value $b = 2 \times 10^8$ (in this example) manifested.

If we take $t > 10^{-8} \text{ sec}$, then the curve of $n(b)$ cannot be distinguished from the vertical line $b = b_{cr} = 2 \times 10^8$; $n = 0$, for $b > b_{cr}$, $n = \infty$ for $b < b_{cr}$.

5.11 SUBCRITICAL AND SUPERCRITICAL MASS FOR A CONSTANT SOURCE OF NEUTRONS

In the preceding section we considered the problem of the variation with time of the number of neutrons for a given initial number n_0 of neutrons. We now pose a somewhat different problem. Suppose at an initial time $t = 0$ the number of neutrons is zero and a neutron source is switched on at this instant of time emitting q_0 neutrons per unit time. This problem leads to the equation

$$\frac{dn}{dt} = cn + q_0 \quad (5.11-1)$$

where $c = a - b$. We seek a solution to this equation with the initial condition $n = 0$ for $t = 0$.

A method of solution was given in Sec. 5.5 for a similar problem. We give a brief review of the reasoning there. We seek the number of neutrons at time t . The entire time interval from 0 to t is partitioned into subintervals $\Delta\tau$. We consider one such subinterval from τ to $\tau + \Delta\tau$. During this time the source emitted $q_0\Delta\tau$ neutrons. If the source operated only during one subinterval of time $\Delta\tau$, then we would be dealing with the problem of the preceding section with the initial number of neutrons $n_0 = q_0\Delta\tau$, the sole difference being that these neutrons are emitted at time $t = \tau$ and not at time $t = 0$. Therefore, instead of the solution $n = n_0e^{ct}$ we would have the solution $n = n_0e^{c(t-\tau)} = q_0\Delta\tau e^{c(t-\tau)}$ (this solution refers to $t > \tau$; for $t < \tau$, $n = 0$), since clearly it is precisely on the time that elapsed after the initial number of neutrons was given that the number of neutrons depends. That is, in the given case, on the quantity $t - \tau$.

Actually, the neutron source is in constant operation during the whole time from 0 to t , and so we have to add the contributions of all neutrons emitted by the source in the various subintervals of time $\Delta\tau$ and their sum covers the entire time interval from 0 to t . Such a sum, given small subintervals $\Delta\tau$, is an integral, and so

$$n(t) = \int_0^t q_0 e^{c(t-\tau)} d\tau$$

It is easy to evaluate this integral:

$$\begin{aligned} n(t) &= q_0 e^{ct} \int_0^t e^{-c\tau} d\tau = q_0 e^{ct} \cdot \frac{-1}{c} e^{-c\tau} \Big|_0^t = q_0 e^{ct} \cdot \frac{-1}{c} (e^{-ct} - 1) \\ &= \frac{q_0}{c} (e^{ct} - 1) \end{aligned} \quad (5.11-2)$$

It is readily seen that this solution satisfies the equation

$$\frac{dn}{dt} = \frac{d}{dt} \left[\frac{q_0}{c} (e^{ct} - 1) \right] = q_0 e^{ct} = cn + q_0$$

and the condition $n = 0$ at $t = 0$.

The same formula (5.11-2) yields a solution for a positive or negative value of c , though the shape of the curve $n = n(t)$ is essentially different. For $c > 0$, (i.e., for $a > b$), the exponent ct is positive so that e^{ct} quickly exceeds unity with increasing t . For large t and $c > 0$,

$$n \approx \frac{q_0}{c} e^{ct}$$

For $c < 0$, $ct < 0$ and so e^{ct} becomes much less than unity with increasing t , and the values of n approach the number $-\frac{q_0}{c}$ (this number is positive since $c < 0$):

$$n \approx -\frac{q_0}{c}$$

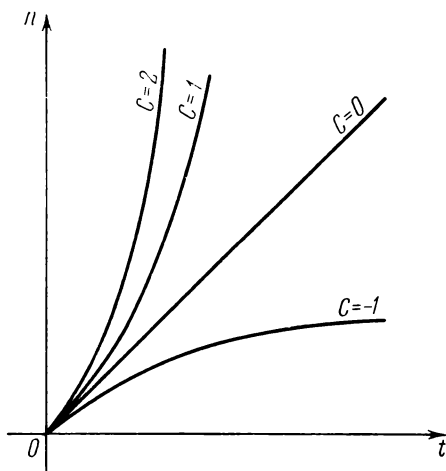


Fig. 126

The curves are shown in Fig. 126.

Note the curious particular case of $c = 0$. If $c = 0$, the formula (5.11-2) cannot be used directly.

Expand e^{ct} in a series:

$$e^{ct} = 1 + ct + \frac{(ct)^2}{2} + \dots$$

Substituting this into (5.11-2), we get

$$n(t) = \frac{q_0}{c} \left[1 + ct + \frac{(ct)^2}{2} + \dots - 1 \right] = q_0 \left[t + \frac{1}{2} ct^2 + \dots \right]$$

This formula may be used when $c = 0$. We then have

$$n(t) = q_0 t \quad (5.11-3)$$

This result is also readily obtainable from (5.11-1). Indeed, for $c = 0$, (5.11-1) has the form $\frac{dn}{dt} = q_0$, whence $n(t) = q_0 t + A$, where A is the constant of integration. For $t = 0$ it must be true that $n = 0$, and so $A = 0$ and we get (5.11-3).

As was shown above, when $c < 0$ the concentration of neutrons in the course of time attains a constant value $-\frac{q_0}{c}$ or, what is the same

thing, $\frac{q_0}{|c|}$. The smaller $|c|$ (the closer we are to the critical state), the greater this constant value. Thus, even for a very weak source (small q_0), a mass close to critical can yield an arbitrarily large number of neutrons, a large number of fissions, and a great quantity of energy. Such in principle is the mode of operation of nuclear reactors.

The maintenance of such a regime is no easy task, since small variations of b and c drastically alter the magnitude of $\frac{q_0}{c}$, when c is close to zero, and operation at c close to zero is necessary when we want to obtain a big power output for small q_0 . However, this engineering problem can be solved by means of automatic control: when n gets out of bounds, the control system changes a or b . Besides, there are also natural factors that facilitate control: thus, for instance, when n increases, the temperature of the active material rises and then, it turns out, c diminishes, so that to a certain extent the system is self-regulating.

5.12 THE CRITICAL MASS

We now know how sensitive the properties of a system are depending on whether we have a supercritical or subcritical mass. Let us examine in more detail the critical size, this is the size of a reactor having critical mass:

$$r_{cr} = \frac{3k}{N\sigma_f(\nu-1)}$$

Substituting the numbers for uranium-235, $\sigma_f = 1.6 \times 10^{-24}$, $\nu = 2.5$, $N = 4 \times 10^{22}$, we get

$$r_{cr} = k \frac{3}{4 \times 10^{22} \times 1.6 \times 10^{-24} \times 1.5} \approx k \cdot 30 \text{ cm}$$

We do not know how to determine the coefficient k , all we know is that it is less than unity. Let us find this coefficient by comparing the formula with experiment. Experiments show that the critical mass of uranium-235 is about 50 kg. A uranium sphere weighing 50 kilograms has a radius of about 8.5 cm, so, in the given case,

$$k \approx \frac{8.5}{30} = 0.3$$

Let us examine the physical significance of the formula for the critical radius. The neutron velocities cancelled out in the expression of r_{cr} , which means that the formula for r_{cr} can be obtained without regarding the course of the process in time and without examining the rate of neutron multiplication and the rate of neutron escape from the system.

If we disregard the nondimensional factor $3k$ (it is of the order of unity), the formula for the critical radius becomes

$$r_{cr}N\sigma_f \approx \frac{1}{v-1} \quad (5.12-1)$$

What is this quantity on the left? The volume of a cylinder of height equal to the radius and with area of the base equal to σ_f is $r_{cr}\sigma_f$. Recall that if a neutron is in motion along the axis of such a cylinder, then it causes fission of those nuclei of uranium-235 whose centres lie inside the cylinder. N is the number of nuclei in unit volume. Consequently, $Nr_{cr}\sigma_f$ is the mean number of nuclei in the volume of the cylinder.

We can now give a different statement of the criticality condition. Earlier, we learned that the mean path, inside a fissionable material, of a neutron born inside the material (via fission) is of the order of the radius r . After a neutron has traversed a distance of about r , it leaves the fissionable material and is lost to the process. The critical size of a reactor means that, on the average, prior to leaving the system, a neutron should produce one neutron over this distance. In fission, $v - 1$ new neutrons are generated. Hence, it is necessary that the neutron, prior to escape, produce approximately $\frac{1}{v-1}$ fissions, i.e., that there be roughly $\frac{1}{v-1}$ nuclei in the volume of the cylinder $r\sigma_f$. This is the condition that leads to formula (5.12-1).

Quite naturally, these arguments are not rigorous but they are necessary for an understanding of the physical essence of the matter and cannot be replaced by any kind of calculations, even the most precise ones performed on modern electronic computers. Computer executed computations do not replace but merely *supplement* a clear-cut grasp of the qualitative physical aspect of the matter. In particular, the reader should pay special attention to the principle expressed at the beginning of the section: if some quantity (v) enters into the derivation of a formula, but is cancelled out in the final result, then this means that there is a derivation of the formula that dispenses altogether with that quantity. And one should always find that simpler derivation because a different derivation of a formula is tantamount to a fresh view of the process being investigated.

5.13 ABSORPTION OF LIGHT. STATEMENT OF THE PROBLEM AND A ROUGH ESTIMATE

Let us consider the absorption of light in air containing black particles of soot. Suppose a unit volume contains N particles. The area of a section of one particle by a plane perpendicular to the ray of light is denoted by σ . For short, we call σ the cross section. For

example, for a particle in the shape of a sphere of radius r , σ is the area of a cross section passing through the centre of the sphere, i.e., $\sigma = \pi r^2$.*

We will assume that the light incident on the surface of the particle of soot is completely absorbed. The problem consists in determining the portion of absorbed light and the portion of transmitted light as a function of the quantities N , σ and the path length x that a light ray traverses through air containing the soot.

We begin with the roughest kind of estimate of the distance over which an appreciable portion of light is absorbed. We denote this distance by L . Just what the pregnant expression "appreciable portion of light" means will be examined later on in the sections that follow. Let us not be upset by the clumsy statement of the problem.

Consider a cylinder with base area S and height L . We require that the sum of the cross sections of all particles in this cylinder be equal to S .

In the volume SL of the cylinder there are NSL particles, the sum of the cross sections of which is σNSL , and so we require that

$$\sigma NSL = S$$

whence

$$L = \frac{1}{\sigma N} \quad (5.13-1)$$

Let us verify the dimensions of (5.13-1): σ is the area, so its dimensions are cm^2 , N is the number of particles per unit volume and has the dimensions cm^{-3} . Consequently, $[L] = \frac{1}{\text{cm}^2 \cdot \frac{1}{\text{cm}^3}} = \text{cm}$, as re-

quired.

What is the physical meaning of the condition thus posed? If it were possible to arrange the particles so that the areas covered by various particles do not overlap, then using the particles in the cylinder of height L and base area S it would be possible to cover the whole base of the cylinder and achieve a complete absorption of all the light. For $x < L$, total absorption of the light is clearly impossible: no matter how the particles of soot are placed, the total area of their cross sections does not suffice to cover the whole base of the cylinder.

It is clear that for $x = L$ and even for $x > L$ there will not really be complete absorption. For a random arrangement of soot particles and for arbitrary x , there will remain certain directions along which there will not be a single particle in the path of the light, which will then pass through.

* An exact definition of the cross section σ for particles of intricate shape is this: σ is the mean area of the shadow cast by a particle on a surface perpendicular to a ray of light.

The energy transmitted through an area in 1 second is called the flux of light energy. Let I be the flux of light energy through an area of 1 square centimetre. This is called the energy-flux density, which has the dimensions $\text{erg/sec} \cdot \text{cm}^2$. Below we consider the energy-flux density of light $I(x)$ as a function of the thickness x of a layer. Clearly,

$$I(x) = I_0 f(x) \quad (5.13-2)$$

where I_0 is the energy of incident light and $f(x)$ the desired function which characterizes the attenuation of the light.

What can we say about the properties of the function $f(x)$? If $x = 0$, there is no attenuation of light, $I(0) = I_0$ and so $f(0) = 1$. If $x > 0$, then the light is attenuated, $I(x) < I_0$ and therefore $f(x) < 1$.

Clearly, $f(x)$ decreases with increasing x and approaches zero, that is, $f(x)$ is a decreasing function. Thus, its derivative is negative,

$$\frac{df}{dx} < 0$$

We have already said that there will not be complete absorption either for $x = L$ or for $x > L$, and so we do not expect $f(x)$ to vanish when $x = L$. However, we may assume that the value $x = L$ is a characteristic length. This means that when light is transmitted over a path $x \ll L$, the fraction of absorbed light is extremely small when compared with the fraction of transmitted light. Over a path $x \approx L$ a perceptible portion of the light is absorbed, and over a path $x \gg L$, most of the light is absorbed and only a very small portion is transmitted.

As may be seen from formula (5.13-2), the function $f(x)$ is non-dimensional. We can assume that if a dimensionless variable $\frac{x}{L}$ is introduced, then the function $f\left(\frac{x}{L}\right)$ will always be the same for any kind of particles and for arbitrary N and σ . These suppositions will be corroborated and made precise in the sections that follow.

5.14 THE ABSORPTION EQUATION AND ITS SOLUTION

We consider a thin layer of air between x and $x + dx$. We conduct all calculations for a column of air in the form of a cylinder with base area 1 cm^2 (in the preceding section, when we considered a cylinder with base area $S \text{ cm}^2$, the quantity S cancelled out anyway).

A beam of light consists of parallel rays and is characterized by the energy-flux density I . If no light were absorbed by the soot particles, then I would be constant.

The layer under consideration contains Ndx particles covering an area of σNdx of the total area (1 cm^2) of the base of the layer. Hence, the layer absorbs a portion σNdx of the energy incident from the left

on the layer. Therefore $dQ = IN\sigma dx$. When light passes through the layer dx , the flux of light energy is attenuated by an amount equal to the quantity of absorbed energy dQ . Prior to entry into the layer, the energy-flux density was $I(x)$; after emergency from the layer, it became $I(x + dx)$ and so

$$I(x) - I(x + dx) = I\sigma N dx \quad (5.14-1)$$

Noting that $I(x + dx) - I(x) = dI$, we get, from (5.14-1),

$$\frac{dI}{dx} = -IN\sigma$$

The solution of this equation is

$$I = I_0 e^{-\sigma N x} \quad (5.14-2)$$

This solution is obtained in the very same way as the solution $N = N_0 e^{-\omega t}$ in radioactive decay (see Sec. 5.3). Here, I_0 is the value of I when $x = 0$.

If the layer thickness is increased in arithmetic progression, $x_1 = a$, $x_2 = 2a$, $x_3 = 3a$, etc., the light intensity diminishes in geometric progression. Indeed, denoting $e^{-\sigma N a} = \alpha$ (then $\alpha < 1$), we find, using (5.14-2), $I(x_1) = I_0 \alpha$, $I(x_2) = I_0 \alpha^2$, $I(x_3) = I_0 \alpha^3$ and so on.

5.15 RELATIONSHIP BETWEEN EXACT AND APPROXIMATE CALCULATIONS

It will be very instructive to compare the exact solution (Sec. 5.14) and the rough estimate (Sec. 5.13). Such a comparison will help us to make use of rough estimates in complicated problems where an exact solution is hard to find. Also such a comparison helps one to understand the range of applicability of a rough solution.

In the rough solution we found the distance over which appreciable absorption takes place,

$$L = \frac{1}{N\sigma}$$

With the aid of the quantity L , the exact solution (5.14-2) can be expressed as follows:

$$I = I_0 e^{-\frac{x}{L}} \quad (5.15-1)$$

Thus the supposition that the quantity L , found in a crude chain of reasoning, enters into the exact solution is fully corroborated.

The exact solution is indeed of the form

$$I = I_0 f\left(\frac{x}{L}\right)$$

From the exact solution (5.15-1) we find the concrete form of the function $f\left(\frac{x}{L}\right)$. True enough, $f\left(\frac{x}{L}\right) = e^{-\frac{x}{L}}$.

We consider the distance $x = L$. Approximate reasoning gave complete absorption of light over this distance. Actually, from the exact solution (5.15-1), putting $x = L$, we find $I = I_0 e^{-1} = 0.37 I_0$,

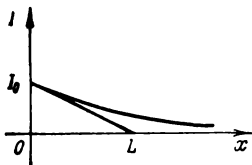


Fig. 127

which means that 37% of the light is transmitted and, hence, the absorption is 63%. For small $\frac{x}{L}$ we express $e^{-\frac{x}{L}}$ by means of Maclaurin's formula, confining ourselves to two terms, to get

$$e^{-\frac{x}{L}} \approx 1 - \frac{x}{L} \quad (5.15-2)$$

Geometrically, this is tantamount to replacing the curve by the tangent to the curve at $x = 0$ (Fig. 127). As can be seen from (5.15-2), the tangent line intersects the x -axis at $x = L$. Therefore, if absorption occurred at the same rate, that is, so that the same amount of light is absorbed on every unit of length, all the light would be absorbed over the distance $x = L$.

To summarize, then, the quantity L which was obtained via rough considerations is indeed of extreme importance in the exact solution as well.

The question of rough solutions is very important in practical work and one should take every opportunity to develop skill in finding and understanding approximate solutions. This is far more important and fruitful than malicious snickering over the drawbacks of rough solutions. We will be pleased that a rough solution yields 100% of absorption where the exact solution gives 63%—the error is only by a factor of $1\frac{1}{2}$. The rough solution, for $x = L$, yields 0%

of light transmission in place of the exact value of 37%. But that isn't so bad either because from the very start it was evident that we couldn't expect good accuracy from a rough solution.

If it has been established that a problem does not have an exact solution in the form of an explicit formula, one should not be deterred

in the least. Seek even for a very very rough solution of the problem. But when using it, be sure to remember that the solution is a rough one, an approximate one, and by no means an exact one.

Let us again dwell for a moment on the question of dimensions. We have verified the dimensions of $L = 1/N\sigma$ and have established that this is length. It is often possible to find an approximate expression of the quantity that interests us when all we know are its dimensions and the dimensions of the initial quantities given in the statement of the problem. In the case at hand however this is not possible. Indeed, a quantity having the dimensions of length can be constructed by proceeding solely from the concentration $N \left(\frac{1}{\text{cm}^3} \right) : l_1 = \frac{1}{\sqrt[3]{N}}$.

The quantity l_1 is the mean distance between particles. A quantity having the dimensions of length can also be constructed out of the cross section $\sigma \text{ (cm}^2\text{)} : l_2 = \sqrt{\sigma}$. The quantity l_2 characterizes the particle size. It is obvious that the quantity $l_\alpha = l_1^\alpha l_2^{1-\alpha}$, for any value of the exponent α , also has the dimensions of length. In particular, the L that interests us is obtained for $\alpha = 3$. Thus, in the problem at hand, dimensional analysis does not give a definite answer. To find L , i.e., the quantity with dimensions of length entering into the exact solution, it is precisely the rough solution to the problem that one, it turns out, has to find. A formal application of dimensional analysis does not in this case yield an unambiguous answer. But even when dimensional reasoning yields a unique answer, it is also desirable to get a rough solution to the problem so as to obtain a clearer picture of the phenomenon.

5.16 EFFECTIVE CROSS SECTION

In the problem of attenuation of light passing through dusty air, the quantity σ has a simple geometric meaning of the area of the shadow cast by a single dust particle. The law of attenuation of light (5.14-2) is the same for light of different wavelengths (i.e., different colours) since σ is independent of the wavelength.

In the absorption of light by separate molecules and atoms there is observed a strong dependence of the law of attenuation of light upon the wavelength of the light. For example, in clean air, visible light is hardly at all attenuated (attenuation is less than 1% per kilometre of path length; accordingly, the attenuation is by a factor of e over a distance of about 100 km). Ultraviolet rays of wavelength $1800 \times 10^{-8} \text{ cm} = 1.8 \times 10^{-5} \text{ cm} = 1800 \text{ \AA}$ ($\text{\AA}$ stands for Angström, $1 \text{ \AA} = 10^{-8} \text{ cm}$) are attenuated by a factor of e over a distance of $L = 0.1 \text{ cm}$. Still shorter ultraviolet rays of wavelength $1.1 \times 10^{-5} \text{ cm} = 1100 \text{ \AA}$ are attenuated e times over a path length of $L = 0.01 \text{ cm}$.

Consequently, the absorption of light by air is not like the absorption of light by a black dust particle, which absorbs light of any wavelength in the same degree.

The light energy q absorbed by a single atom in unit time is proportional to the energy-flux density of the light I at the point where the atom is located:

$$q = \sigma I \quad (5.16-1)$$

Here, σ is the constant of proportionality. Let us determine the dimensions of σ . The dimensions of q are erg/sec. The dimensions of energy flux I are erg/cm² sec. Hence, the dimensions of σ are cm². The quantity σ is called the effective cross section. For a black dust particle, the constant of proportionality coincides with the geometric area of the shadow. For molecules and atoms, σ is strongly dependent on the wavelength of the light.

In rough fashion, we can picture the cause of this dependence as follows. The amount of energy absorbed by an atom when acted upon by light proves to be particularly great when the frequency of the light oscillations coincides with the frequency of motion of the electrons in the atom. This is resonance: the electron oscillates intensively and absorbs a particularly large amount of light energy.

Such a resonance is attained for instance in the absorption of sodium atoms (in the vapour state) of yellow light of wavelength $5890 \text{ \AA} = 5.89 \times 10^{-5} \text{ cm}$. The very same yellow light is emitted by sodium atoms at higher temperatures when electron oscillations are caused by energetic collisions of atoms among themselves.

At resonance, σ reaches 10^{-10} cm^2 . Atoms and molecules are of size 10^{-8} - 10^{-7} cm , which corresponds to a cross section of the order of 10^{-16} - 10^{-14} cm^2 .

Thus, the maximum effective cross sections are many times greater than the true cross-sectional areas of atoms and molecules. On the other hand, for light whose frequency does not correspond to the natural frequency of the atom, the effective cross section is small, much less than the cross-sectional area of the atom.

5.17 ATTENUATION OF A CHARGED-PARTICLE FLUX OF ALPHA AND BETA RAYS

The exponential law of diminution of particle flux as a function of distance

$$I = I_0 e^{-\frac{x}{L}} \quad (5.17-1)$$

is based on a very general supposition that the attenuation of the flux over a small distance dx is proportional to the intensity itself of the flux:

$$\frac{dI}{dx} = -\frac{1}{L} I \quad (5.17-2)$$

where the constant of proportionality $\frac{1}{L}$ is dependent solely on the kind of particle.

The solution of equation (5.17-2) is

$$I = I_0 e^{-\frac{x}{L}}$$

The formulas (5.17-1) and (5.17-2) are therefore equivalent, one being a consequence of the other.

Experiments show that in certain cases the exponential law (5.17-1) is quite exact; but sometimes deviations from the law are observed. Let us consider carefully the reasons that can give rise to deviations from formula (5.17-1) or (what is the same thing) from (5.17-2).

It is easy to answer the question about the meaning of deviations from (5.17-2). Formula (5.17-2) presumes that when x and I vary, the light (or other radiation) under consideration does not vary qualitatively, otherwise the number L would change. Rewrite (5.17-2) as

$$\frac{1}{I} \frac{dI}{dx} = -\frac{1}{L}$$

From this we see that the quantity $\frac{1}{I} \frac{dI}{dx}$ is constant. If it turns out that at different points in space the quantity $\frac{1}{I} \frac{dI}{dx}$ is different, this means that at such points not only is the intensity of radiation different but also its physical characteristics (say a different colour light, that is, having a different mean wavelength).

When considering problems of protection against radioactive radiations and questions of the passage of α -, β - and γ -rays and neutrons through various substances, there is a different reason for departures from the simple law (5.17-2).

As applied to the process of light absorption, the law (5.17-2) signifies the following: if the light encounters a dust particle, some passes by the particle without any change while the other portion of light is completely absorbed by the dust particle. The situation is more complicated in the case of radioactive radiations: an α -particle is a nucleus of the helium atom flying out of the radioactive parent nucleus at high speed (of the order of $0.07 c$, where c is the speed of light, that is, at a speed of about 2×10^9 cm/sec). In passing through an atom, the α -particle gives up part of its energy to the electrons. After roughly 50,000 collisions with atoms the α -particle will have lost half of its energy. It will not cease to exist, but its energy and speed will have changed. After 100,000 collisions the α -particle comes to a halt, ceases to collide with atoms and to knock out electrons. This is the number of collisions the α -particle experiences in air over a distance (flight path) of a few centimetres.

Actually, different α -particles (having the same initial energy) experience different numbers of collisions; they are not necessarily equal exactly to 100,000. However, since 100,000 is a big number, over a given path length the departures of the number of collisions of separate α -particles from the mean (100,000) are but slight (of the order of 300, which is about 0.3% of the mean number of collisions). For this reason, α -particles of the same energy always lose all their energy over roughly the same distance. This path length depends on the initial energy of the α -particle. If a flux of α -particles of the same energy flies along the x -axis, the relationship between the intensity of the flux and the path length x is shown by the

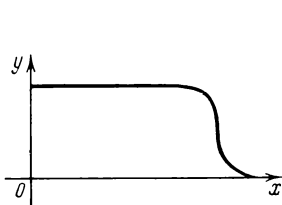


Fig. 128

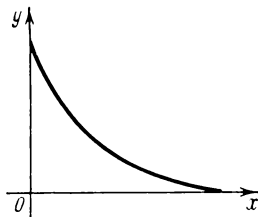


Fig. 129

curve in Fig. 128. This curve is quite different from the graph of the exponential function. Over a considerable portion of the path length, the intensity of the flux of particles does not change: the same number of α -particles fly through an area of 1 cm^2 in the same intervals of time. Then the intensity falls off sharply. This drastic fall was prepared over the section where the intensity remained constant, because over this portion the energy of the α -particles diminished with increasing path length x . The sharp drop in the flux occurs where the energy of the α -particles becomes extremely small.

The picture is similar in the case of fast electrons (β -particles emitted when a neutron is converted into a proton in the nucleus of an atom). Here the situation is complicated by the fact that in radioactive decay there is an emission of electrons with different energies; what is more, the electrons give up part of their energy to the atom near which they fly and also experience a considerable lateral deviation.

The curve for β -particles is of the shape shown in Fig. 129. Already for small x , some of the electrons fall out of the beam. These are mostly electrons which had low initial velocities. Therefore, near $x = 0$ the behaviour of the curve is similar to that of the exponential function. Later on however the curve reaches the x -axis, the intensity I becomes zero for a very definite value of x corresponding to the maximal energy of the electrons generated in the given type of radioactive decay.

The most important practical problems (they are also the most difficult ones) are those connected with protection against γ -rays (gamma rays) emitted by radioactive substances and against neutrons produced in the fission of nuclei in atomic reactors and nuclear explosions. The situation here is confused and complicated in the extreme by the fact that γ -rays and neutrons give up energy in large portions and are strongly deflected from their original directions in the process. Even in a thick layer of air (100 to 200 metres) there is a considerable probability (of the order of 37%) of the passage of unaltered γ -rays and neutrons. That is why they require thick shielding. A flux of γ -rays and neutrons does not become zero for a definite thickness of the protective layer, as was the case for α - and β -rays. As experiments and complicated calculations show, for a given large thickness of the protective layer, a flux of gamma rays and neutrons falls off in rough accord with the exponential law.

Chapter 6

Mechanics

6.1 FORCE, WORK AND POWER

The relations existing between the most important quantities of mechanics admit of exact formulations only by means of integrals and derivatives. In Chapter 2 we examined the relationship between the distance covered (or the position) of a body and its velocity, and also between the velocity and the acceleration of the body. Without repeating this material, we now go on to examine the relationships between quantities such as force, work, energy, power. Let us consider the rectilinear motion of a body along the x -axis. Suppose a force F acting on a body is also directed along the x -axis. In elementary physics, the work A performed by this force is defined as the product of the force F by the distance traversed by the body: $l = x_k - x_n$, where x_n is the initial position of the body and x_k is the terminal position:

$$A = Fl = F (x_k - x_n)$$

Obviously, the situation is the same here as in the case of the relationship between velocity and distance: the simple formula—work is equal to the product of force by distance—is valid only for the case where the force is constant. Now if the force varies during the process of translation, then the whole process has to be partitioned into separate small intervals (subintervals) so that over every subinterval the force may be taken to be constant. Then for the subinterval

$$\Delta A_i = F_i \Delta x_i = F_i (x_{i+1} - x_i)$$

Thus, in the general case of a variable force, the work is expressed not as a product but as an integral:

$$A = \int_{x_n}^{x_k} F dx$$

We assume as known the motion of a body given by a known function $x = x(t)$. The translation of the body during a small time dt

is equal to the product of the instantaneous velocity v by the time dt :

$$dx = v dt = \frac{dx}{dt} dt$$

Therefore, the expression of work can be written thus:

$$A = \int_{t_n}^{t_k} F \frac{dx}{dt} dt = \int_{t_n}^{t_k} Fv dt \quad (6.1-1)$$

The product Fv in this formula is the work performed in unit time; it is called the power. Indeed, in the case of constant velocity and force, the distance is equal to $x = vt$, the work is $A = Fx = Fvt$ and the ratio of work to the time elapsed is equal to $\frac{A}{t} = Fv$. Denoting power by W , $Fv = W$, we can write

$$A = \int_{t_n}^{t_k} W dt$$

Recall that in the mks (mkg force s or meter-kilogram force-second system of units, the force is measured in kilograms kgf: the unit of force is the force with which a mass of 1 kilogram is attracted to the earth), the length is measured in metres, the unit of work is measured in kilogram-force-metres (kgf-m), the unit of power is kilogram-metres per second (kgf-m/s). A frequently used unit of power is the so-called metric horsepower, which is the power required to raise 75 kg against the force of gravity through a distance of one metre in one second (75 kgf-m/s).

In the cgs (centimetre-gram-second) system of units, force is measured in dynes (the force that will accelerate a mass of 1 gram 1 centimetre per second per second, or 1 g per 1 cm/sec²), the work is in ergs (1 erg = 1 dyne · 1 cm), power is in erg/sec. There is also a unit of work called the joule, equal to 10⁷ ergs, and the unit of power called the watt, equal to 1 joule per second = 10⁷ erg/sec.

A body can be acted upon by several forces, say, F_1 and F_2 . Then we can speak of the work performed by the first force (A_1) and the second force (A_2) during the time that the body was translated from the initial position x_n to the terminal position x_k . Regarding F_1 and F_2 as constant, we get

$$A_1 = (x_k - x_n) F_1, \quad A_2 = (x_k - x_n) F_2$$

Note the signs of the quantities in these expressions. A force is taken to be positive when it acts in the direction of increasing x , that is, in the direction indicated by the arrowhead on the x -axis (Fig. 130), to the right. A force acting in the opposite direction, to the left, is regarded as negative.

If a body is translated in the direction of the acting force, the work of that force is positive. Imagine a body moving in the direction opposite that of the force so that F_1 and $(x_k - x_n)$ have different signs. Then the work A_1 of the force is negative. Now picture two forces acting on a body (Fig. 131a): the force F_1 of a stretched spring and the force F_2 of the tension of the rope which you (the reader)

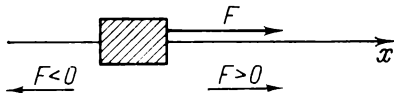


Fig. 130

hold in your hand. F_1 acts leftwards, $F_1 < 0$; you are pulling rightwards, $F_2 > 0$. If you pull with more strength than the spring,* then the body will move from left to right. In Fig. 131a is shown the initial position of the body, and in Fig. 131b the terminal position:

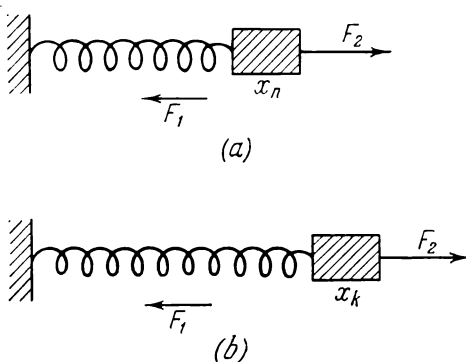


Fig. 131

$(x_k - x_n) > 0$, $F_1 < 0$. The work A_1 performed upon the body by the tension force of the spring, or, more briefly, by the work of the spring, in this translation is negative. Here, the work which you have performed is positive, $A_2 > 0$. The total work, $A = A_2 + A_1$ is also positive. But $A < A_2$ since $A_1 < 0$. This means that only part of the work performed by you (A) was received by the body, the other part ($|A_1|$) having gone into stretching the spring. Observe that in all cases, the force of friction against a stationary surface is directed against the velocity of motion of the body, and so the

* Mathematically, this means that the absolute value of the force with which you pull is greater than the absolute value of the force with which the spring pulls the body leftwards: $|F_2| > |F_1|$.

work of the friction force against a fixed surface is always negative, irrespective of the direction of the motion of the body.

A force F_1 with which a spring, one end of which is fixed, acts on a body differs in one very important respect: this force depends exclusively on the position of the body. Not all forces, by any means, have this property. For example, the force of friction between a moving body and a fixed surface always retards the motion of a body. It is directed leftwards if the body is in motion rightwards, and it is directed rightwards if the body is in motion leftwards. Thus, the direction of the force of friction depends on the direction of motion

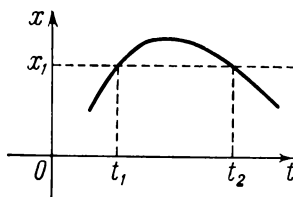


Fig. 132

of the body. Besides, the force of friction can also depend on the magnitude of the velocity of the body. Thus, the force of friction depends on the magnitude and direction of the velocity of a body.

The force F_2 with which you pull the rope in the example of Fig. 131, *a* and *b*, can vary in any fashion, at your pleasure. The body can, say, move to the right and then to the left. In so doing, it will twice pass through the same position: the first time in the rightward movement at t_1 , the second time on the return route at time t_2 .

A possible graph of the motion of the body (the dependence of the x -coordinate on the time t) is shown for this case in Fig. 132. We can, at time t_1 pull to the right, $F_2(t_1) > 0$, and at time t_2 let go of the rope so that $F_2(t_2) = 0$ or even push the body leftwards so that $F_2(t_2) < 0$. But $x(t_1) = x(t_2) = x_1$ and so, speaking generally, an arbitrary force F_2 cannot be regarded as a function of the x -coordinate.

The foregoing examples of the force of friction and the force applied by a person acting of his own free will serve to demonstrate that the dependence of force solely on the position of a body, $F_1 = F_1(x)$, which is characteristic of the force F_1 with which a spring acts on a body, is not a general property of all forces, but is a particular property associated with the elasticity of a spring.

To find the work performed by the given force F_i using one of the formulas $A_i = \int_{x_n}^{x_k} F_i dx$, or $A_i = \int_{t_n}^{t_k} F_i v dt$ we have to know two

things: (1) what the motion of the body was, that is, the dependence of the coordinate of the body on time, $x(t)$, and (2) the expression of the force F_i , which in the general case depends on x , t , and v .

Knowing the functions $x(t)$ and $v(t)$ and substituting them into the expression $F_i(x, t, v)$, we get an expression for F_i as a function of time and we can describe the work as an integral with respect to time.

Example. Let a force $F(x) = -kx$ and let the motion of the body be defined by the equation $x = b \sin \omega t$; then

$$\begin{aligned} v &= \frac{dx}{dt} = b\omega \cos \omega t, \quad F(x(t)) = -kx(t) = -kb \sin \omega t, \\ A_i &= -b^2 k \omega \int_{t_n}^{t_h} \sin \omega t \cos \omega t dt = \frac{-b^2 k \omega}{2} \int_{t_n}^{t_h} \sin 2\omega t dt \\ &= + \frac{b^2 k}{4} \cos 2\omega t \Big|_{t_n}^{t_h} = \frac{b^2 k}{4} (\cos 2\omega t_h - \cos 2\omega t_n) \quad (6.1-2) \end{aligned}$$

In this case, when the force depends solely on the coordinate, it is much easier and convenient to take advantage of the expression of work as an integral with respect to x :

$$A = \int_{x_n}^{x_h} F(x) dx = -k \int_{x_n}^{x_h} x dx = \frac{-kx^2}{2} \Big|_{x_n}^{x_h} = \frac{kx_n^2}{2} - \frac{kx_h^2}{2}$$

Substituting $x = b \sin \omega t$, it is also easy to get the expression for work over a specified interval of time from t_n to t_h :

$$A = \frac{kb^2 \sin^2 \omega t_n}{2} - \frac{kb^2 \sin^2 \omega t_h}{2} \quad (6.1-3)$$

It is easy to see that this expression coincides exactly with the preceding one, since

$$\begin{aligned} \cos 2\omega t &= \cos^2 \omega t - \sin^2 \omega t = 1 - 2 \sin^2 \omega t, \\ \cos 2\omega t_h - \cos 2\omega t_n &= 1 - 2 \sin^2 \omega t_h - (1 - 2 \sin^2 \omega t_n) \\ &= 2 (\sin^2 \omega t_n - \sin^2 \omega t_h) \end{aligned}$$

Substituting this identity into (6.1-2) we get (6.1-3).

A good deal of caution is required when using the expression for work as an integral with respect to the x -coordinate in the general case of a force $F(x, v, t)$ depending on x , v , t . Indeed, in principle, if the motion $x = x(t)$ is given, then this equation can be solved for t and we can determine $t(x)$. But one must bear in mind that t may not be a single-valued function of x ; x may have one and the same position for two distinct instants of time, which means that one and the same value of x is associated with two distinct values of t (see

Fig. 132). Then the overall motion has to be divided into separate periods during which the velocity does not change sign and t is a single-valued function of x . But for different periods, t is expressed by unlike functions of x . For example, let a body be moving via the law $x = b \sin \omega t$, as in the preceding example, but the force is given as a function of time, $F = f \cos \omega t$, the force not being a single-valued function of the position x . Indeed, let $t = 0$, then $x = 0$, $F = f$. If we put $t = \frac{\pi}{\omega}$, again $x = 0$ but then $F = -f$ so that the body will be in the same position $x = 0$ at different times ($t = 0$ and $t = \frac{\pi}{\omega}$), though the force will not be the same. This difficulty is absent when integrating with respect to time: to every instant of time t there corresponds one definite value of the x -coordinate, the force F and of all other quantities.

It is easy to find the work by integrating with respect to time,

$$A = \int_{t_n}^{t_k} Fv \, dt = \int_{t_n}^{t_k} f \cos \omega t b \omega \cos \omega t \, dt = fb\omega \int_{t_n}^{t_k} \cos^2 \omega t \, dt$$

Let us take advantage of the above trigonometric formula

$$\cos 2\varphi = 2 \cos^2 \varphi - 1$$

Whence

$$\cos^2 \varphi = \frac{1}{2} + \frac{\cos 2\varphi}{2}$$

Therefore

$$\begin{aligned} A &= fb\omega \int_{t_n}^{t_k} \left(\frac{1}{2} + \frac{\cos 2\omega t}{2} \right) dt \\ &= \frac{1}{2} fb\omega (t_k - t_n) + \frac{fb}{4} (\sin 2\omega t_k - \sin 2\omega t_n) \quad (6.1-4) \end{aligned}$$

As is evident from (6.1-4), the work increases without bound with the passage of time. This is due to the fact that the force and the oscillations are in resonance (resonance will be examined in detail in Sec. 6.11).

Motion in accordance with the law $x = b \sin \omega t$ represents oscillation of the body. Consider the work of a force during one half-period, choosing for the initial time $t_n = 0$, $x_n = 0$ and the terminal time $t_k = \frac{\pi}{\omega}$, $\sin \omega t_k = \sin \pi = 0$, $x_k = 0$. Then, in (6.1-4), $\sin 2\omega t_k = \sin 2\omega t_n = 0$ and the work is

$$A = \frac{1}{2} fb\omega \frac{\pi}{\omega} = \frac{\pi}{2} fb \quad (6.1-5)$$

The body has returned to its initial state, while the work performed by the force is not equal to zero but has a definite magnitude. How is this result to be understood from the viewpoint of the first formula

$A = \int_{x_n}^{x_k} F dx$? At first glance, if we substitute $x_n = x_k = 0$, we get

$$A = \int_0^0 F dx = 0$$

Actually, however, we have to consider separately the process of buildup of x from 0 to $x_{max} = b$ and the process of decline of x from $x_{max} = b$ to 0. During buildup, each value of x is associated with a definite value of the force F , which we denote by F_1

$$F_1 = f \cos \omega t = f \sqrt{1 - \sin^2 \omega t} = f \sqrt{1 - \left(\frac{x}{b}\right)^2} > 0$$

During decline of x , the same positive values of x are associated with a negative value of force,* which we denote by F_2 :

$$F_2(x) = -f \sqrt{1 - \left(\frac{x}{b}\right)^2}$$

Thus the integral with path x for variable of integration breaks up into two:

$$A = \int_0^b F_1(x) dx + \int_b^0 F_2(x) dx \quad (6.1-6)$$

These two integrals cannot be combined by the formula

$$\int_a^b \varphi dx + \int_b^c \varphi dx = \int_a^c \varphi dx$$

since the integrands in the two integrals of (6.1-6) are defined by *different* formulas, although their meaning is the same (force). This is due to the fact that F is given as a function of t , while t is expressed in terms of x by different formulas for increase of x from 0 to b and for decrease of x from b to 0. In this case, $F_2(x) = -F_1(x)$. Substituting the expressions for $F_1(x)$ and $F_2(x)$ into

* The equation $\cos^2 \omega t + \sin^2 \omega t = 1$ is true for all values of ωt . From this it follows that $\cos \omega t = \pm \sqrt{1 - \sin^2 \omega t}$ and the sign in front of the radical depends on the value of ωt . It is easy to see that for $-\frac{\pi}{2} < \omega t < \frac{\pi}{2}$ one has to take the plus sign, and for $\frac{\pi}{2} < \omega t < \frac{3\pi}{2}$ the minus sign, which was done above.

(6.1-6), we get

$$A = f \int_0^b \sqrt{1 - \left(\frac{x}{b}\right)^2} dx - f \int_b^0 \sqrt{1 - \left(\frac{x}{b}\right)^2} dx$$

In the second integral we can interchange the limits of integration (this changes the sign of the integral) to get

$$A = 2f \int_0^b \sqrt{1 - \left(\frac{x}{b}\right)^2} dx \quad (6.1-7)$$

Putting $z = \frac{x}{b}$, $dx = b dz$, we find

$$A = 2bf \int_0^1 \sqrt{1 - z^2} dz$$

The integral $I = \int_0^1 \sqrt{1 - z^2} dz = \frac{\pi}{4}$ (the area of a quadrant of a circle of radius 1), and so from (6.1-7) we have

$$A = 2bf \frac{\pi}{4} = \frac{\pi}{2} bf$$

which coincides with formula (6.1-5) obtained by integrating with respect to time.

Thus, in the case of a force that is dependent on time and can assume different values for the same value of x , the work A is also not a single-valued function of x . In the foregoing case of oscillatory motion, $F = f \cos \omega t$, $x = b \sin \omega t$, the quantity x again and again passes through the same values in the course of time, and the work performed by the force continues to increase all the time (for positive f).

If the force is a function of velocity (as is the case of friction), the situation will be similar: the body can return to its original position, but the work of the force will not be zero. In the case of the force of friction, the work is negative (see exercises).

Exercises

1. Find an expression in the form of an integral for the work of friction, the force being proportional to the velocity of motion of the body and in the opposite direction, $F = -h v$, $h > 0$. Demonstrate that the work is negative.
2. The force of friction is constant in magnitude and opposite in direction to the velocity, that is

$$F = \begin{cases} -h & \text{if } v > 0 \\ +h & \text{if } v < 0 \end{cases}$$

The body moves in accordance with the law $x = b \sin \omega t$. Find the work of the force of friction during a time interval from $t = 0$ to $t = \frac{\pi}{\omega}$.

3. The force acting on a body is given by the formula $F = f_0 \sin \omega_0 t$, f_0 a constant. Since the body is also acted upon by other forces, it moves according to the law $x = b \sin \omega_1 t$. Determine the work performed by the force F during the time interval from $t = 0$ to $t = t_k$. Consider the case $\omega_0 = \omega_1$.

4. A body is falling according to the law $x = \frac{gt^2}{2}$ (the x -axis is directed downwards). Find the formula for the work resulting from the air-resistance force $F = -aS\rho \frac{v^2}{2}$, where a is the constant of proportionality dependent on the shape of the body (it can vary between 0.1 and 1), S is the cross-sectional area of the body in cm^2 , ρ is the air density ($1.3 \times 10^{-3} \text{ gm/cm}^3$), v the rate of fall in cm/sec . Also find the formula for the work done by the force of gravity $F = mg$, where m is the mass of the body.

Perform the computations and compare the results for a wooden ball of diameter 1 cm, $a = 0.8$ and for a steel bullet of length 3 cm, diameter 0.7 cm, $a = 0.2$, for $t = 1 \text{ sec}$, 10 sec , 100 sec .

Remark. The idea behind the calculation is that we assume that the force of air resistance is to be small compared with the force of gravity and does not noticeably affect the law of free fall. Computing the work done by the air resistance and comparing it with the work performed by gravity, we verify the correctness of our starting assumption concerning the small role of the force of air resistance. In Sec. 6.7 we give the exact solution of the problem of free fall with air resistance taken into account.

5. A wind blowing with a speed of v_0 acts on the sail of a boat with a force equal to

$$F = \begin{cases} +aS\rho \frac{(v_0 - v)^2}{2} & \text{for } v < v_0, \\ -aS\rho \frac{(v_0 - v)^2}{2} & \text{for } v > v_0 \end{cases}$$

where v is the rate of motion of the boat, S is the area of the sail, ρ is the air density, a is a dimensionless coefficient (for the sail perpendicular to the wind direction, $a \approx 1$). Find the work done by the force of the wind in moving the boat b metres. Find the power of the wind force. Assume that the boat is in uniform motion at a constant speed v . Determine work and power as functions of v . Find the maximum power for $v_0 = 30 \text{ m/sec}$, $a = 1$, $S = 10 \times 10 \text{ m}^2$ and express it in metric horsepower.

6. A body is in motion in accord with the law $x = c \cos(\omega t + \alpha)$ under the action of a force $F = f \cos \omega t$. Find the work done by the force during the time interval from $t = t_n$ to $t = t_k$. Find the work done by the force during one period of operation (from $t = 0$ to $t = \frac{2\pi}{\omega}$). Determine the mean power.

6.2 ENERGY

We consider the case of a force that depends solely on the position (coordinate) of the body, $F = F(x)$. As we have already mentioned, an instance of this kind of force is the force with which a spring acts

on a body, the other end of the spring being fixed.* In that case, the expression $A = \int_{x_n}^{x_h} F dx$ may be applied without any complications (compare this with the preceding section). In particular, in this case if the body first moves in one direction from x_n to x_m , and then in the opposite direction returning to the initial position, then $x_h = x_n$, and the total work done by the force is actually equal to zero,

$$A = \int_{x_n}^{x_h=x_n} F(x) dx = 0$$

Dividing the path length into sections only corroborates this conclusion:

$$A = \int_{x_n}^{x_m} F dx + \int_{x_m}^{x_h} F dx = \int_{x_n}^{x_m} F dx - \int_{x_h}^{x_m} F dx$$

and $A = 0$ for $x_h = x_n$.

In mechanics, *potential energy* is defined as the capacity to do work. A spring possesses a definite reserve of potential energy depending on how compressed or stretched it is. If one end is fixed in position, the potential energy of the spring depends on the position of the body to which the free end of the spring is attached. Thus, the potential energy $u = u(x)$ is a function of the coordinate x . If in the initial position, the potential energy is $u(x_n)$, then after the body has been displaced to x_h , when the spring has performed work A equal to

$$A = \int_{x_n}^{x_h} F(x) dx$$

the remaining potential energy is equal to $u(x_n) - A$. Thus

$$u(x_h) = u(x_n) - A = u(x_n) - \int_{x_n}^{x_h} F(x) dx \quad (6.2-1)$$

Get a good feeling of the sign affixed to A in this expression: if the spring does work, then the reserve capacity of the spring to do work will diminish! The work performed by the spring is taken from the reserve of potential energy. For this reason, the work done (that given up by the spring) is equal to the difference between the initial and final energy of the spring:

$$A = u(x_n) - u(x_h)$$

* If the second end of the spring is allowed to move at random, the force acting on the body will depend not only on the position of the body but also on the position of the second end of the spring and this does not satisfy the stated condition.

All formulas involve the difference of potential energy in two positions of a body. Therefore, if we replace $u(x)$ by $u(x) + C$, where C is an arbitrary constant, this will in no way affect the physical results. Indeed,

$$[u(x_n) + C] - [u(x_k) + C] = u(x_n) - u(x_k)$$

The value of $u(x)$ at some given point, call it x_0 , can be chosen quite arbitrarily. Denote it by u_0 . Then at some other point x , the value of the function $u(x)$ is determined from the formula (6.2-1) if in it we put $x_n = x_0$, $x_k = x$,

$$u(x) = u_0 - \int_{x_0}^x F(x) dx \quad (6.2-2)$$

That is how the problem of determining the potential energy from a given force is solved.

We can pose the converse problem: knowing the potential energy as a function of x , $u(x)$, find the force $F(x)$. To solve this problem,

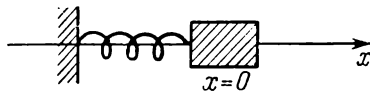


Fig. 133

take the derivative of both sides of (6.2-2). The derivative of the integral is equal to the integrand, so that

$$\frac{du(x)}{dx} = -F(x)$$

The minus sign here is very essential. The force is positive (in the direction of increasing x) if $\frac{du}{dx}$ is negative; that is, as x increases, the potential energy u decreases. The force is negative (in the direction of decreasing x) if $\frac{du}{dx} > 0$; that is, when x increases, the energy u increases. In this case, obviously, as x decreases, the energy u also decreases. This means that the force is always in the direction of diminishing potential energy.

Let us examine in more detail the example of the spring. Let the body be at the origin when the spring is not under tension (Fig. 133). When the body is pulled to the right, the force is proportional to the displacement of the body and is directed leftwards:

$$F = -kx, \quad k > 0$$

Assume $u_0 = 0$ for $x_0 = 0$, that is, regard the potential energy for the nontense spring as zero. This yields

$$u(x) = - \int_0^x F dx = k \int_0^x x dx = k \frac{x^2}{2}$$

It is easy to see that this $u(x)$ is associated by the formula $F = - \frac{du}{dx}$ with a force $F = - \frac{d}{dx} \left(k \frac{x^2}{2} \right) = - kx$.

We consider a second example, the force of gravity. Send the z -axis upwards. The force of gravity acts downwards and is equal to $-mg$, where g is the acceleration of gravity. It is independent of the height z , but a constant quantity is merely a special case of a function. The important thing is that the force of gravity does not depend on time and velocity. We can therefore make use of the formulas derived above. We take as zero the potential energy of the body at the earth's surface when $z = 0$. Then

$$u(z) = - \int_0^z F dz = - \int_0^z (-mg) dz = mgz \quad (6.2-3)$$

The potential energy grows linearly with increasing height of the body above the earth's surface.

In the preceding example we assumed that the distance z is small compared with the radius of the earth. Let us now examine the attractive force on the assumption that the distances can be arbitrarily large. By Newton's law of gravitation, the attractive force is inversely proportional to the square of the distance between the bodies. We know that for a body above the earth's surface, the force of gravitation towards the entire globe is equal to the force of attraction to a mass equal to the earth's mass and concentrated at the centre of the earth.* It is therefore convenient to reckon the distance from the centre of the earth. Denote it by r . Then the force acting on a body is

$$F = - \frac{C}{r^2}$$

The constant C is taken to be positive. The force is negative since it is directed towards the centre of the earth, and the coordinate r increases as the body recedes from the earth; the force acts in the direction of diminishing r .

The constant C can readily be determined from the condition that the force acting at the earth's surface ($r = r_0 = 6400$ km =

* This does not hold true for a body inside the earth, in which case we have regard only for that portion of the earth's mass between the earth's centre and the body.

$= 6.4 \times 10^8$ cm) is known:

$$F(r_0) = -mg = -\frac{C}{r_0^2}, \quad C = mgr_0^2 \quad (6.2-4)$$

where g is the acceleration of gravity at the earth's surface, $g = 981$ cm/sec². We finally have

$$F = -\frac{mgr_0^2}{r^2}$$

For zero we again take the potential energy of the body at the surface of the earth. Then

$$\begin{aligned} u &= -\int_{r_0}^r F dr = mgr_0^2 \int_{r_0}^r \frac{dr}{r^2} = mgr_0^2 \left(-\frac{1}{r} \right) \Big|_{r_0}^r \\ &= mgr_0^2 \left(-\frac{1}{r} + \frac{1}{r_0} \right) = mg \left(r_0 - \frac{r_0^2}{r} \right) = mg \frac{r_0}{r} (r - r_0) \end{aligned} \quad (6.2-5)$$

At a small height $z = r - r_0 \ll r_0$, $\frac{r_0}{r}$ differs but slightly from unity and, approximately, we have

$$u(r) = mg(r - r_0) = mgz$$

which coincides with formula (6.2-3) obtained earlier. But, as can be seen from (6.2-5), the potential energy does not increase without bound with increasing r , as would have been the case in accordance with the approximate formula (6.2-3), but tends to a definite limit

$$u(\infty) = mgr_0$$

Thus, making allowance for the decrease of gravity with distance, the energy of a body at an infinite distance is the same as, by the approximate formula, at a distance of r_0 from the earth's surface, or at a distance of $2r_0$ from the centre of the earth.

In this problem we encountered a physical situation involving infinite distance. In this respect we must note that in any physical problem we are always interested in finite quantities, finite distances. For instance, if we consider the motion of a body and the energy of the body as dependent on the earth's gravity, then we can be interested in attaining the moon, Mars or other planets, even stars. All these objects involve distances that are very very great relative to that of the earth's radius, but they are finite!

Suppose we consider the problem of launching a rocket to a great height, to a considerable distance from the earth. We are interested in the energy required and the time of flight. Here are two cases:

(1) a space vehicle is to traverse a distance of $R = 10 r_0$ where r_0 is the earth's radius,

(2) a space vehicle is to traverse $R = 100 r_0$.

The time of flight is roughly proportional to the distance. Accordingly, in the second case the vehicle will be in flight 10 times longer than in the first case. A change in R produces an essential change in the time of flight. For this reason, one cannot replace R by infinity when considering time of flight.

The work needed to tear away from the earth and go a distance R from the centre of the earth is

$$A = mgr_0^2 \left(\frac{1}{r_0} - \frac{1}{R} \right)$$

Recalling that $r_0 = 6.4 \times 10^8$ cm, we get $A_1 = mg \times 5.76 \times 10^8$ in the first case and $A_2 = mg \times 6.34 \times 10^8$ in the second.

A 10-fold change in distance caused a relatively small change in the requisite energy. If we replaced R by infinity, we would get

$$A_\infty = mg \times 6.4 \times 10^8$$

A_1 differs from A_∞ by 10%, A_2 differs by 1%. That is why, when computing work, R may be replaced by infinity.

To summarize, then, one and the same quantity R in one and the same problem can either be replaced by infinity or not, depending on the aspect considered. The possibility of such a substitution depends not only on the quantity R itself (and its comparison with other quantities of the same dimensions entering into the formulas, r_0 in the given case). The possibility of replacement depends on the structure of the formula in which it occurs.

Returning to the question of potential energy of a body attracted to the earth, let us find the numerical value of $u(\infty)$ per unit mass: in the cgs system of units it is equal to $gr_0 = 981 \times 6.4 \times 10^8 \approx \approx 6.28 \times 10^{11}$ erg/gm, in the mks system, $gr_0 = 6.28 \times 10^8 \frac{\text{kgf-m}}{\text{kg}}$.

It will be interesting to see what this quantity looks like in thermal units: 1 kilocalorie is equal to 427 kgf-m so that $u(\infty) = 15 \times \times 10^3$ kilocalories per kilogram. This is 30 times the evaporation heat of water and 10 times the chemical energy of explosives.

In problems of celestial mechanics and in physics it is advisable to choose for zero the potential energy of a body located at an infinite distance from the mass attracting it. Then for the potential energy of a body at distance r we get

$$u(r) = u(\infty) - \int_{\infty}^r F(r) dr = -\frac{C}{r}$$

where C is the constant in the expression of the force ($F = -\frac{C}{r^2}$); it is determined from formula (6.2-4) if we know the acceleration of gravity, g , at the earth's surface and the radius of the earth, r_0 .

We can obtain a different expression for C . By Newton's law of gravitation, $F = -G \frac{mM}{r^2}$, where m is the mass of a body attracted

to the earth, M is the mass of the earth, r the distance to the centre of the earth and G the gravitation constant equal to 6.7×10^{-8} dyne $\text{cm}^2/\text{g}^2 = 6.7 \times 10^{-8} \text{ cm}^3/\text{g} \cdot \text{sec}^2$. Therefore $C = G m M$. Using this formula, we can easily determine C if we know G and M .

The problem of potential energy of two electric charges e_1 and e_2 is completely analogous to the preceding one. The interaction force between them is equal to

$$F = k \frac{e_1 e_2}{r^2} \quad (6.2-6)$$

Here if the charges are expressed in the electrostatic system of units (the unit of charge $= \frac{1}{3 \times 10^9}$ coulomb), the force in dynes, then $k = 1$ in formula (6.2-6). There is no minus sign here that we see in the expression for the gravitational force. Indeed, if e_1 and e_2 are like charges (both positive or both negative), the product $e_1 e_2$ is positive. But then the charges repulse one another, i.e., the force F is positive.

Again defining $u(r)$ so that $u(\infty) = 0$, we get

$$u(r) = \frac{e_1 e_2}{r}$$

The potential energy of two like charges separated by a finite distance is positive: they repulse and, moving from r to ∞ , can perform work equal to

$$u(r) - u(\infty) = u(r)$$

The potential energy of two unlike charges is negative. Indeed, $e_1 e_2 < 0$ if $e_1 > 0$, $e_2 < 0$; this is clear physically: since unlike charges attract, energy must be expended to pull them apart to infinity.

Observe that thanks to the law of conservation of energy, the potential energy may be defined not only as the capacity to do work but also as the work required to bring a system to a given state. A stretched spring can do a definite amount of work in returning to the unstretched state. That, clearly, was the work that had to be done in stretching the spring. Similar assertions may be made in the case of a body raised a definite height above the earth, or for a system of two charges.

6.3 EQUILIBRIUM AND STABILITY

We consider a body that can move without friction along a straight line, which we take for the x -axis. Let the body be acted upon by a force directed along this axis and dependent on the x -coordinate. We can again picture the spring. Below we will examine other examples as well.

The equilibrium position of a body is defined as that position for which the force is zero and the body is at rest. Denote by x_0 the point of equilibrium, then $F(x_0) = 0$. Expanding the function $F(x)$ in a Taylor series and ignoring all powers of $(x - x_0)$ except the first, we see that two versions of the function $F(x)$ are possible in the neighbourhood of point x_0 [provided $F(x_0) = 0$]: $F(x) = k_1(x - x_0)$, $F(x) = -k_2(x - x_0)$. In both formulas, it is assumed that k_1 and k_2 are positive quantities. The first case is shown in Fig. 134a, the second in Fig. 134b.

These two cases are associated with an entirely different character of equilibrium. In the case of Fig. 134a, if the body is somewhat to the right of point x_0 , then it is acted upon by a positive force, i.e., a force which pulls it farther rightwards. Thus, the equilibrium at

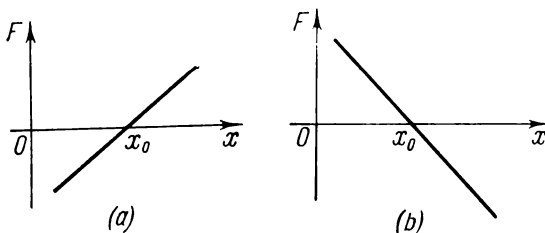


Fig. 134

point $x = x_0$ in Fig. 134a, is unstable. A slight deviation of the body (whether to the right or left makes no difference) suffices for a force to begin to act on the body that will increase the deviation. On the contrary, in the case of Fig. 134b, the force is negative (pulls leftwards) when the body deviates to the right. Deviation of the body from the equilibrium position gives rise to a force which tends to return the body to the position of equilibrium. Here we have to do with stable equilibrium. It is easy to see that the second case occurs for a body attached to a spring.

In accordance with the above expressions for force, we find the expressions of potential energy via (6.2-2). In the case of unstable equilibrium,

$$u(x) = u(x_0) - \frac{1}{2} k_1 (x - x_0)^2$$

In the case of stable equilibrium

$$u(x) = u(x_0) + \frac{1}{2} k_2 (x - x_0)^2$$

The appropriate curves are shown in Figs. 135a, 135b.

Thus, in the case of unstable equilibrium, the potential energy has a maximum, in the case of stable equilibrium, it has a minimum. In both cases, the force is zero at the point of maximum or minimum,

$$F = -\left. \frac{du}{dx} \right|_{x=x_0} = 0.$$

This result is quite natural. If a body is in the state of maximum potential energy, then energy is released during displacements in both directions. This energy can be used to overcome inertia and is converted into kinetic energy. But if the body is in the state of minimum energy, then energy from an outside source is required to move it to any other position. This energy will go to increase the

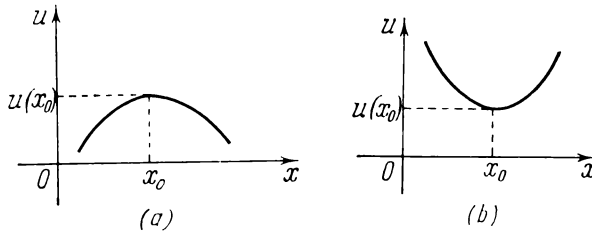


Fig. 135

potential energy. A small expenditure of energy will displace the body only a small distance. These properties of a body in a position of minimum potential energy fully accord with the concept of stable equilibrium.

When gravity acts near the earth's surface, the potential energy is mgz , where z is the height above the surface. The curves depicting

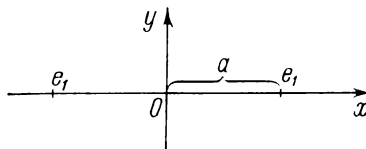


Fig. 136

the function $u(x)$ can be visualized as curves indicating the height of a body as a function of the horizontal x -coordinate. We have to imagine a body in motion along a curve like a bead on a stiff wire. The curve $u(x)$ corresponds to the shape of the wire if the plane of the drawing is vertical. Then it is clear that the maximum of $u(x)$ is (see Fig. 135a) a point on the wire from which the bead slides downwards at the slightest touch, and the minimum of $u(x)$ (Fig. 135b), the lowest point at which the bead is in a stable position, and any other beads on the wire would strive to take up that position.

The graph of $u(x)$ thus gives a pictorial visualization of the direction of forces and character of equilibrium.

Let us examine a few examples.

1. Referring to Fig. 136, let a charged body be in motion along a straight line (which we take for the x -axis) on which are fixed two identical charges symmetric about the origin at a separation of $2a$.

It is quite clear that at the origin the body is in a state of equilibrium. Indeed, in this case the forces acting on the body from the fixed charges are equal in magnitude and opposite in direction so that they balance, which means their resultant is zero.

The potential energy of the body is made up of two terms:

$$u(x) = \frac{e_1 e}{r'} + \frac{e_1 e}{r''}$$

where e is the charge on the body, e_1 is the fixed charge, r' is the distance to the left-hand charge, r'' to the right-hand charge,

$$r' = x + a, r'' = a - x$$

and so

$$u(x) = e_1 e \left(\frac{1}{a+x} + \frac{1}{a-x} \right) \quad (6.3-1)$$

The appropriate curves are shown in Fig. 137. The upper curve corresponds to $e_1 e > 0$, which is the case of a like charge of body and the fixed charges, the lower curve corresponds to $e_1 e < 0$, which means that the body has a charge opposite to the fixed charges.

In the case of $e_1 e < 0$, equilibrium at the origin is unstable. Indeed, the body is attracted both by the left and the right charge and at the origin the forces of attraction balance. But if the body is displaced the slightest bit in any direction, say to the right, then attraction on the right will exert a stronger effect and will continue to pull it rightwards.

We find $\left. \frac{d^2 u}{dx^2} \right|_{x=0}$. Using (6.3-1), we get

$$\frac{d^2 u}{dx^2} = 2e_1 e \left[\frac{1}{(a+x)^3} + \frac{1}{(a-x)^3} \right] \quad (6.3-2)$$

Putting $x = 0$ in (6.3-2) we have

$$\left. \frac{d^2 u}{dx^2} \right|_{x=0} = \frac{4e_1 e}{a^3}$$

Hence, for $x = 0$, $\frac{d^2 u}{dx^2} > 0$ if $e_1 e > 0$. In this case, $u(x)$ has a minimum for $x = 0$ and the equilibrium is stable. But if $e_1 e < 0$, then $\left. \frac{d^2 u}{dx^2} \right|_{x=0} < 0$ and $u(x)$ has a maximum for $x = 0$ and the equilibrium is unstable.

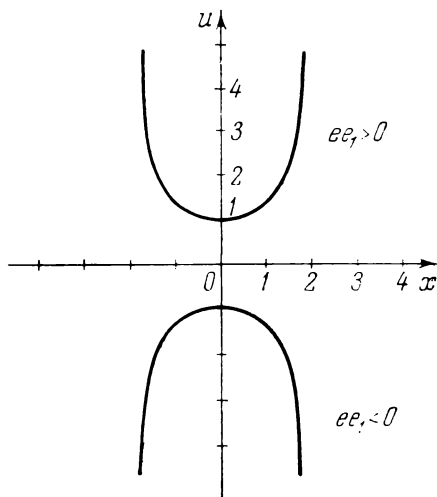


Fig. 137

2. Similarly, we consider a situation in which the charges are spaced in the same way from the origin, but along a straight line

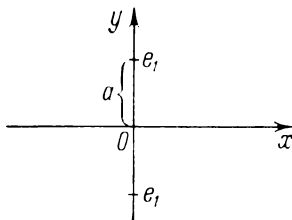


Fig. 138

perpendicular to the line (x -axis) along which the charged body is in motion (Fig. 138). In this case the potential energy is

$$u(x) = 2 \frac{e_1 e}{\sqrt{a^2 + x^2}}$$

[the graph of potential energy for $a = 1$, $|e_1 e| = 1$ is shown in Fig. 139]. In Example 2, equilibrium at the origin is unstable for

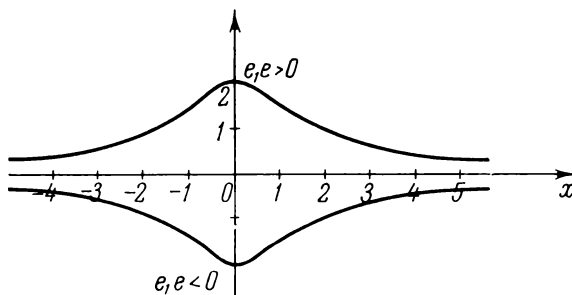


Fig. 139

$e_1 e > 0$. If the charge on the body is opposite to the fixed charges ($e_1 e < 0$), the equilibrium is stable.

This is easy to establish if we examine the force acting on a moving charge (Fig. 140). Let $e_1 e > 0$. Displace the body rightwards from the position of equilibrium. Then the resultant force of repulsion is also directed to the right, further increasing the deviation. The equilibrium is unstable. In the case of $e_1 e < 0$, the resultant force is in the direction of decreasing deviation. The equilibrium is stable.

These results are also readily arrived at by considering $\left. \frac{d^2 u}{dx^2} \right|_{x=0}$.

Observe that for $e_1 e > 0$, when stability occurred in Example 1 (Fig. 136), in Example 2 (Fig. 138) we had unstable equilibrium. For $e_1 e < 0$ (unlike charges), the situation was reversed: the equilibrium is unstable for the arrangement of charges as given in Fig. 136 and is stable for their arrangement shown in Fig. 138.

Turning Fig. 138 through 90° , we note that actually it refers to the same initial distribution of charges in the equilibrium position

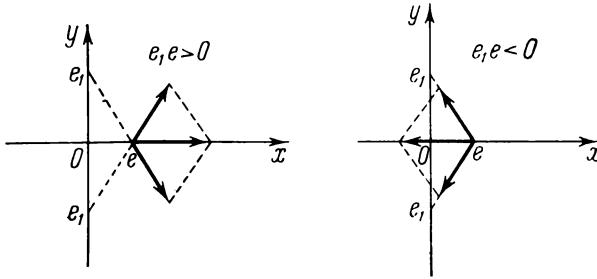


Fig. 140

as in Fig. 136. We can say that Figs. 136 and 138 refer to the same initial distribution of charges but the directions of motion under consideration differ (Fig. 141). Then the equilibrium will always (for any signs of the charges) be unstable in one direction or in the other.

Proof is given in electrostatics that this result is quite general: there is no point of equilibrium in the space between external fixed charges such that equilibrium is stable relative to displacements in any direction.

The general proof of this fact given below may appear too hard for the reader and he can skip it without any loss of continuity in the book.

For proof in the general form, note that the potential energy of a charge e at point (x, y, z) , depending on its distance r from a fixed charge e_1 located at point (x_1, y_1, z_1) is given by the formula

$$u = \frac{e_1 e}{r} = \frac{e_1 e}{\sqrt{(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2}}$$

Consider the motion along the x -axis and find $\frac{d^2 u}{dx^2}$ for y and z constant. This quantity is denoted by $\frac{\partial^2 u}{\partial x^2}$. Then in analogous fashion we find $\frac{\partial^2 u}{\partial y^2}$ and $\frac{\partial^2 u}{\partial z^2}$, which refer to motion along the y - and z -axes,

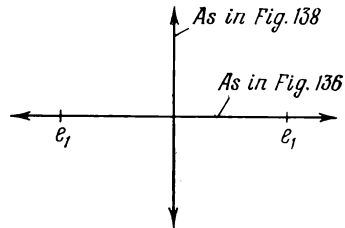


Fig. 141

respectively. It turns out (and the reader should convince himself of this) that for arbitrary x, y, z, x_1, y_1, z_1 the sum of the second derivatives along the three perpendicular axes is equal to zero!

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Obviously, this property will be preserved for the sum of any number of terms of the form $\frac{e_k e}{r_k}$, where e_k is a fixed charge at the point (x_k, y_k, z_k) and r_k is the distance of charge e from this point. Consequently, for any distribution of fixed charges in space, the following formula is valid:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad (6.3-3)$$

in particular, also at the point at which the charge e is in equilibrium. For equilibrium, it is necessary that the forces along any axis be equal to zero. For this we must have $\frac{\partial u}{\partial x} = 0$, $\frac{\partial u}{\partial y} = 0$, $\frac{\partial u}{\partial z} = 0$.

If the forces acting along three perpendicular axes are zero, then so also is the force in any direction.*

For equilibrium to be stable relative to motion along all three perpendicular axes, it must be true that

$$\frac{\partial^2 u}{\partial x^2} > 0, \quad \frac{\partial^2 u}{\partial y^2} > 0, \quad \frac{\partial^2 u}{\partial z^2} > 0$$

But this contradicts equation (6.3-3) since the sum of three positive quantities cannot be equal to zero.

During recent years (beginning with 1956) the question of whether it is possible to hold charged particles stable by acting on them with fixed charges has attracted considerable attention. To obtain nuclear energy via a deuterium reaction it is necessary to contain charged particles in space and not allow them to collide against the walls of the containing vessel. By what has just been proved, this cannot be achieved by arranging charges on the walls of the vessel, no matter how the charges are chosen. Inventions in this direction are therefore clearly hopeless. We now know that, fundamentally, the problem may be solved by applying a magnetic field.

Exercises

1. A charge e is moving along a straight line on which are fixed two positive charges e_1 and $e_2 = 4e_1$ at a separation of $2a$. Find the point on the straight line at which equilibrium of the charge is possible and determine the type of equilibrium. Consider two cases: $e > 0$ and $e < 0$.

2. Solve Problem 1 when the sign of charge e_2 is changed.

* If we have a nonzero force F acting in some direction, then there will be a force acting along each axis equal to the projection of the force F on the axis.

6.4 NEWTON'S SECOND LAW

Newton's second law of motion states that the product of the mass by the acceleration is equal to the force applied.* The acceleration a is the derivative of velocity v with respect to time; in turn, velocity is the derivative of the coordinate of the body with respect to time. Thus

$$ma = m \frac{dv}{dt} = F \quad (6.4-1)$$

or

$$m \frac{d^2x}{dt^2} = F \quad (6.4-2)$$

We begin with the case where the force is given as a function of time, $F = F(t)$. This means that the derivative $\frac{dv}{dt}$ is given as a function of time. Using Newton's law (6.4-1) it is easy to find the velocity at any given instant of time. Besides the applied force we also have to specify the velocity at some time t_0 . Then

$$v(t) = v(t_0) + \frac{1}{m} \int_{t_0}^t F(t) dt \quad (6.4-3)$$

Knowing the velocity as a function of time, $v(t)$, and the initial position $x(t_0)$ of a body, we can find the position of the body at any time:

$$x(t) = x(t_0) + \int_{t_0}^t v(t) dt \quad (6.4-4)$$

where $v(t)$ is given by the preceding formula.

The relationship between velocity and distance is considered in detail together with examples in Chapter 2.

On the whole, formulas (6.4-3) and (6.4-4) solve the problem of finding $x(t)$ from equation (6.4-2), which is a differential equation of the second order involving the second derivative of the unknown function $x(t)$. The answer includes not only the given function $F(t)$, but also two constants defined from the initial conditions: the position and the velocity of the body at a given time t_0 .

If the law of motion of a body is given or has been experimentally determined, that is, we know the function $x(t)$, then it is easy to find the force applied to the body: to do this we have to find the second derivative of the function $x(t)$ and multiply it by m [formula (6.4-2)].

* Newton's first law, the law of inertia, states that any body at rest or in uniform translational motion will maintain this condition unless acted upon by an external force. In this case, the acceleration is equal to zero for a force equal to zero. Thus, the first law is contained within the second law as a particular case.

Exercises

1. Find the law of motion of a body acted upon by a constant force F if at time $t = 0$ the body is at rest at the origin ($x = 0$).
2. The same provided that $x = 0$, $v = v_0$ at $t = 0$.
3. The same provided that $x = x_0$, $v = v_0$ at $t = 0$.
4. A body of mass 20 kg begins to move under the action of a force of 1 kgf from the origin without an initial velocity. What distance will it cover in 10 seconds?
5. A ball falls from a height of 100 metres (initial velocity, zero). How long will it take the ball to reach the ground? (Disregard air resistance.)
6. Under the conditions of the preceding problem, the ball starts falling at velocity $v_0 = 10$ metres per second. Examine two cases: (a) the initial velocity of the ball, v_0 , is directed downwards, (b) the initial velocity, v_0 , is directed upwards. Determine the time required to reach the ground; and the velocity it will have at the time of impact. Verify that in cases (a) and (b) the velocity at impact is the same.
7. A body is acted upon by a force which is proportional to the time that elapses from the beginning of motion (the constant of proportionality is equal to k). Find the law of motion of the body if it is known that the body begins moving from point $x = 0$ with initial velocity v_0 .
8. A body is acted upon by a force periodically varying in time, $F = f \cos \omega t$ (f , ω are constants).
 - (a) Find the law of motion of the body provided that $x = 0$, $v = 0$ at $t = 0$. Establish that this is oscillatory motion. Determine the period of oscillation, the maximum value of $x(t)$ and the greatest value of the velocity.
 - (b) The same for a force $F = f \sin \omega t$ and $x = 0$, $v = 0$ at $t = 0$.
9. A body is in motion under an applied constant force F . At time $t = t_0$ the body is at point $x = x_0$. Find the velocity the body must have at $t = t_0$ so that at $t = t_1$ it will reach point $x = x_1$.

6.5 IMPULSE

The problem of finding the law of motion of a body for a given dependence of the force on the time was, in principle, solved in the preceding section. Here we will examine the properties of this solution and certain new concepts associated with the solution.

The product $P = mv$ of mass by velocity is called the quantity of motion (momentum). The quantity

$$\int_{t_0}^t F(t) dt \quad (6.5-1)$$

is known as the impulse of the force during the time interval between t_0 and t .

Formula (6.4-3) may be written

$$\int_{t_0}^t F dt = P(t) - P(t_0) \quad (6.5-2)$$

In words: impulse equals change in momentum.

There are forces that act during very brief time intervals (an instance is the blow of a hammer and the rebound after striking a body). Both prior to and following the blow, the force is equal to zero. It is clear that in the absence of other forces (other than the

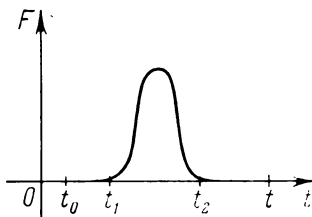


Fig. 142

brief blow) the body prior to the blow moves with a constant velocity and after the blow with another, also constant, velocity.

Let $F(t)$ be different from zero only during the interval between t_1 and t_2 (Fig. 142). We consider the integral

$$I = \int_{t_1}^{t_2} F(t) dt \quad (6.5-3)$$

It may be called the *total impulse* in the sense that the integral is taken over the entire interval of time during which the force acts.

The expression (6.5-1) involves an integral from t_0 to t . If $t_0 < t_2$ and $t > t_2$, then

$$\int_{t_0}^t F dt = I$$

Indeed, we write

$$\int_{t_0}^t F dt = \int_{t_0}^{t_1} F dt + \int_{t_1}^{t_2} F dt + \int_{t_2}^t F dt$$

The first and third integrals on the right are equal to zero since $F = 0$ over the respective intervals and the second (middle) integral is I . Thus, from (6.5-2) and (6.5-3) we get $P(t) = P(t_0) + I$ if $t_0 < t_1$, $t > t_2$.

From formula (6.4-3) we see that the velocity following the blow depends solely on the *impulse* of the force, that is, on the integral of the force but not on the particular type of the function $F(t)$. For example, several different curves of $F(t)$ shown in Fig. 143 all yield the same impulse, which is to say, they all change the velocity of a body by the same amount. It is not difficult to draw the appro-

priate graph of velocity, $v(t)$, for each of these curves of $F(t)$. Fig. 144 depicts these graphs under the general assumption that the initial velocity is equal to zero. The common element of all the

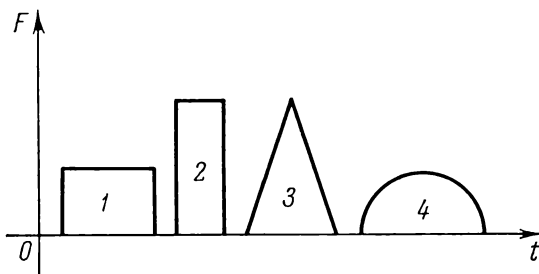


Fig. 143

curves in Fig. 144 is the finite value of velocity. All the curves go into a horizontal straight line on the right at a height of $v = \frac{I}{m}$.

Each of the curves of $F(t)$ in Fig. 143 may be compressed along the axis of time and proportionately stretched along the axis of

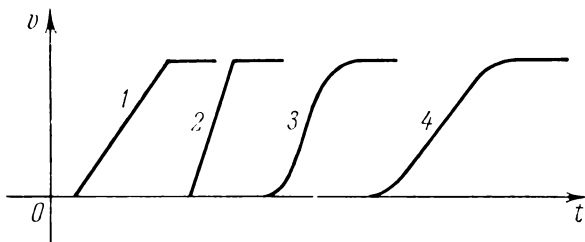


Fig. 144

force. The area under the curve of F , that is, $\int F dt$, the total impulse, does not change in the process. That is precisely how, say, Curve 2 in Fig. 143 was obtained from Curve 1.

The shorter the time of action of a force, the shorter the time interval during which the velocity of a body varies from the initial value $v_0 = 0$ to the final value, $v_k = \frac{I}{m}$ (Fig. 144). Thus, in the limit, for a very great force acting during a very small time interval, the graph of velocity takes the shape of a step (Fig. 145). It is not essential here which of the curves of Fig. 143 we compressed: the step is characterized by only one quantity, $v_k = \frac{I}{m}$, and this quantity is the same for all the curves.

If prior to the application of the force the body was at rest at point x_0 , then after a brief application of a large force the body begins to move with a constant velocity equal to $\frac{I}{m}$. If the force acted at time t_1 (we consider the interval from t_1 to t_2 to be small and we do not distinguish between t_2 and t_1), then the position of the body as a function of the time is given by the formulas

$$\begin{aligned} x &= x_0, & t < t_1 \\ x &= x_0 + \frac{I}{m} (t - t_1), & t > t_1 \end{aligned} \quad (6.5-4)$$

The appropriate graph is shown in Fig. 146. Observe that $x(t)$ satisfies the equation

$$m \frac{d^2x}{dt^2} = F(t)$$

Recall that on the graph of $x(t)$, the first derivative $\frac{dx}{dt}$ is connected with the slope of the tangent to the curve. The second derivative

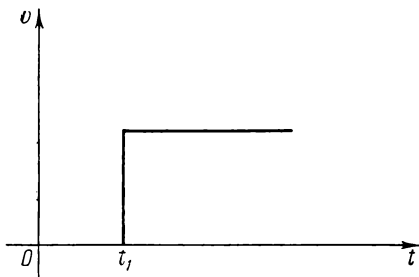


Fig. 145

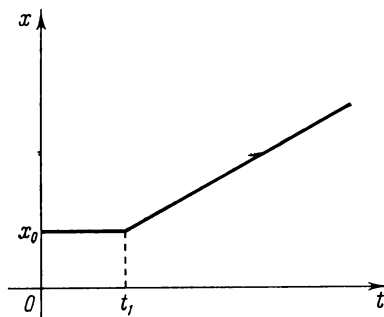


Fig. 146

describes the rate of change of the first derivative, that is, the second derivative is connected with the curvature of the curve $x(t)$.

In Fig. 146 the curve $x(t)$ has a salient point at $t = t_1$, $x = x_0$. The salient point can be regarded as a point at which the curvature is infinite so that the existence of a salient point corresponds to the consideration of a very large (infinite in the limit) force. However, both before and after the salient point the derivative $\frac{dx}{dt}$ is finite; hence, the very large force acted over a very small interval of time so that the impulse is finite. The impulse can readily be found from the graph (Fig. 146) by computing the velocity after the application of the force and utilizing formula (6.5-2).

The law that we have found concerning the motion of a body which up to time $t = \tau$ was at rest, and at that moment received an impulse I will help us to refine the formulas (6.4-3) and (6.4-4) of

the preceding section. For this we need a special case of the formula (6.5-4) when, at $t = \tau$, the body is at the origin ($x_0 = 0$). Let us introduce some special notation:

$$x_1, (t, \tau) = \begin{cases} 0, & t < \tau, \\ \frac{I}{m} (t - \tau), & t > \tau \end{cases} \quad (6.5-5)$$

If we substitute $v(t)$ from (6.4-3) into (6.4-4) and use more accurate designations (so that the upper limit and the variable of integration have different letters), we get an expression which at first glance is rather unwieldy:

$$x(t) = x(t_0) + (t - t_0) \cdot v(t_0) + \frac{1}{m} \int_{t_0}^t dt_1 \int_{t_0}^{t_1} F(t_2) dt_2 \quad (6.5-6)$$

The integral here can be transformed by the formal rules for handling repeated integrals. But we have not mentioned such rules anywhere and so we get the transformed expression (in the form of a single integral) by using the law (6.5-5) of the motion of a body under the action of a single impulse of force.

The action of a force $F(t)$ during the time interval from τ to $\tau + \Delta\tau$ may be approximately replaced by the action of the impulse $\Delta I = F(\tau) \cdot \Delta\tau$. We already know the motion of a body under the action of such an impulse: see formula (6.5-5) in which I is to be replaced by $\Delta I(\tau)$.

It then only remains to combine the contributions of all intervals Δt_i to the coordinate $x(t)$ to get

$$\begin{aligned} x(t) &= \sum x_1'(t, \tau) \Delta I(\tau) = \sum \frac{1}{m} (t - \tau) \cdot F(\tau) \cdot \Delta\tau \\ &= \frac{1}{m} \int_{t_0}^t F(\tau) \cdot (t - \tau) d\tau \end{aligned} \quad (6.5-7)$$

Here, as usual, we replaced the sum by the integral for sufficiently small subintervals $\Delta\tau$. Formula (6.5-7) does not take into account the initial coordinate $x(t_0)$ and the motion with the initial velocity: $(t - t_0) \cdot v(t_0)$; we simply add these terms to expression (6.5-7) and we finally have

$$x(t) = x(t_0) + (t - t_0) v(t_0) + \frac{1}{m} \int_{t_0}^t F(t_1) (t - t_1) dt \quad (6.5-8)$$

Formula (6.5-8) has the advantage over (6.5-6) that in (6.5-8) we have to integrate only once. We did not state why we can merely add the summands involving the individual impulses, the initial velocity, and the final coordinate. This is examined in more detail

in Sec. 9.5. Here it suffices for us that we can directly verify the values of $x(t_0)$, $\frac{dx}{dt}|_{t_0}$ and $\frac{d^2x}{dt^2}$ using formula (6.5-8). We have to differentiate $x(t)$ in (6.5-8) in a manner similar to what was done in the verification of formula (5.5-5) in Sec. 5.5.

The reader will recall that by Newton's third law, in the interaction of two bodies, the force with which the second body acts on the first (F_1)* is equal in magnitude and opposite in direction to the force with which the first body acts on the second one (F_2) (for every force there is an equal and opposite force):

$$F_2(t) = -F_1(t)$$

As applied to the first body and force F_1 the formula (6.5-2) yields

$$P_1(t) - P_1(t_0) = \int_{t_0}^t F_1 dt \quad (6.5-9)$$

This same formula applied to the second body and the force F_2 yields

$$P_2(t) - P_2(t_0) = \int_{t_0}^t F_2 dt \quad (6.5-10)$$

Since $F_2 = -F_1$ by Newton's third law, it follows that

$$\int_{t_0}^t F_2 dt = - \int_{t_0}^t F_1 dt$$

That is why (6.5-10) takes the form

$$P_2(t) - P_2(t_0) = - \int_{t_0}^t F_1 dt \quad (6.5-11)$$

Comparing (6.5-9) and (6.5-11) we find

$$P_1(t) - P_1(t_0) = P_2(t_0) - P_2(t)$$

whence

$$P_1(t) + P_2(t) = P_1(t_0) + P_2(t_0)$$

The latter formula shows that the action of one body on the other does not change the sum of the momenta of the bodies.

6.6 KINETIC ENERGY

Let us consider a body moving under the action of a known force $F(t)$ and find the relationship between the work done by the force and the rate of motion of the body.

* The subscript on F indicates the *body* acted upon by the force F ; the subscript on P also denotes the number of the body to which the momentum refers.

Multiplying both sides of the basic equation $m \frac{dv}{dt} = F(t)$ by velocity v , we get

$$mv \frac{dv}{dt} = F(t) v \quad (6.6-1)$$

The identity

$$v \frac{dv}{dt} = \frac{d}{dt} \left(\frac{v^2}{2} \right)$$

is valid no matter what the function $v(t)$. Using this fact, rewrite (6.6-1) as

$$m \frac{d}{dt} \left(\frac{v^2}{2} \right) = F(t) v$$

and since m is a constant, it follows that

$$\frac{d}{dt} \left(\frac{mv^2}{2} \right) = F(t) v$$

Introducing the notation

$$\frac{mv^2}{2} = K \quad (6.6-2)$$

we finally obtain

$$\frac{dK}{dt} = F(t) v \quad (6.6-3)$$

Using the expression for work (6.1-1) we have

$$A = \int_{t_0}^{t_1} F(t) v dt = \int_{t_0}^{t_1} \frac{dK}{dt} dt$$

whence

$$A = K(t_1) - K(t_0) \quad (6.6-4)$$

The quantity K is the kinetic energy of the body. Formula (6.6-4) expresses the law of conservation of energy: the change in the kinetic energy of a body is equal to the work done by a force. Formula (6.6-3) expresses the law: the rate of change of kinetic energy is equal to the power developed by a force.

When the force is given by a definite function of time, then the impulse and, hence, the change in momentum caused by the given force are dependent neither on the mass of the body nor on its initial velocity, since the impulse, the change in momentum, is $\int_{t_0}^{t_1} F dt$.

On the contrary, the work done by a force and the change in kinetic energy of a body under the action of a force are essentially dependent, as may be seen from (6.6-2), (6.6-3), (6.6-4), not only on the force itself but also on the mass of the body and its initial velocity. Indeed, by acting with a given force over a specified time inter-

val on a heavy body at rest at the start of motion, we impart only a small velocity that will result in only a small displacement, and the work done by the force will likewise be small. A light body will take up appreciable work and will acquire a large energy. If prior to the action of the force the body was in motion in the opposite direction to the force, then the force can reduce the energy of the body.

Picture a body participating in two motions at once. Say, a man walking in the cabin of a ship in motion, or a ball dropped in the cabin. Suppose one of these motions (that of the ship, in our case) is uniform. The question then arises whether it is possible, by observing the ball falling in the cabin or the motion of some other body under the action of an applied force, to establish whether the ship is moving or not. To put it differently, does the uniform motion of the ship influence the character of motion of objects on the ship? No, it does not affect such motion in any way. Experiments have demonstrated that the absence of any influence of uniform motion on physical phenomena refers not only to mechanics but also to the process of propagation of light, to electric and magnetic phenomena. From this fact, Einstein drew conclusions of tremendous importance in developing his theory of relativity (we do not explain the theory of relativity in this book).

In mechanics, it is not hard to establish the absence of the influence of uniform motion. Indeed, let a body be in motion in a train with velocity v , the train itself moving at a constant speed of v_0 . Then, relative to an observer standing on the tracks, the body has a velocity of $v_1 = v + v_0$. The acceleration of the body is the same both for the observer standing on the railway tracks and for another observer riding in the train:

$$a_1 = \frac{dv_1}{dt} = \frac{d}{dt} (v + v_0) = \frac{dv}{dt} + \frac{dv_0}{dt} = \frac{dv}{dt} = a$$

Thus, a constant summand in the expression for velocity does not change the acceleration. And therefore the force acting on the body does not change: $F = ma_1 = ma$. The difference in velocities of the body prior to and after the action of an applied force is likewise the same for an observer on the tracks and for another observer in the train. Indeed, suppose the velocity with respect to the observer in the train prior to the action of the force is v' , after the action of the force, v'' ; for the observer standing on the tracks, the respective velocities are v'_1 and v''_1 . Then $v'_1 = v' + v_0$, $v''_1 = v'' + v_0$ and so

$$v''_1 - v'_1 = v'' + v_0 - v' - v_0 = v'' - v'$$

The situation is more complicated in respect of kinetic energy. (What now follows can be skipped in a first reading.) Not only the kinetic energy itself but even the differences of kinetic energies are

distinct for different observers. For the observer standing on the railway tracks,

$$\begin{aligned} K_1'' - K_1' &= \frac{m(v_1'')^2}{2} - \frac{m(v_1')^2}{2} = \frac{m(v'' + v_0)^2}{2} - \frac{m(v' + v_0)^2}{2} \\ &= \frac{m(v'')^2}{2} - \frac{m(v')^2}{2} + mv_0v'' - mv_0v' = K'' - K' + mv_0(v'' - v') \end{aligned}$$

In this formula, K_1'' and K_1' are the final and initial kinetic energies computed by an observer on the tracks, and K'' and K' are, respectively, the kinetic energies computed by an observer in the train.

The work done by a force and the power are also different for different observers, since although the force is the same, the distance and the velocities are different for an observer on the tracks and for another one in the train.

However, the law of equality of change in kinetic energy and work is valid for any observer, although each of these quantities taken separately differs for different observers (see the exercises for examples corroborating this fact).

Note the following remarkable formula which is valid for a body moving under the action of only one given force $F(t)$,

$$\begin{aligned} A &= \int_{t_0}^{t_1} F(t) v(t) dt = \frac{mv_1^2}{2} - \frac{mv_0^2}{2} = \frac{m}{2} (v_1 + v_0) (v_1 - v_0) \\ &= \frac{v_1 + v_0}{2} (mv_1 - mv_0) = \frac{v_1 + v_0}{2} \int_{t_0}^{t_1} F(t) dt \end{aligned}$$

In this case, it turns out, the velocity $v(t)$ may be taken out from under the integral sign and replaced by the arithmetic mean of the initial and terminal velocities.

This conclusion holds true only for the case where $v(t)$ is a velocity acquired by a body under the action of only one force $F(t)$. If the body is acted upon by a number of forces, F_1, F_2, F_3 , then the work performed by all these forces is equal to the product of the mean velocity by the sum of the impulses of all forces:

$$\begin{aligned} A &= \frac{v_1 + v_0}{2} \int_{t_0}^{t_1} (F_1 + F_2 + F_3) dt \\ &= \frac{v_1 + v_0}{2} \int_{t_0}^{t_1} F_1 dt + \frac{v_1 + v_0}{2} \int_{t_0}^{t_1} F_2 dt + \frac{v_1 + v_0}{2} \int_{t_0}^{t_1} F_3 dt \quad (6.6-5) \end{aligned}$$

However the work done by each of these forces (say, F_2) separately is not equal to the corresponding summand, $\frac{v_0 + v_1}{2} \int_{t_0}^{t_1} F_2 dt$, in (6.6-5), since the force F_2 , acting separately, would impart a velocity to the body that differs from $v(t)$ (see Problem 6 below).

Exercises

1. Find the formula for the kinetic energy of a body moving under the action of a constant force F (the velocity was zero at the initial time) as a function of time and also as a function of the distance traveled.

2. A body is in motion under the action of a force $F = f \cos \omega t$, and $v = 0$ at $t = 0$. Find the expression for the kinetic energy of the body. Determine the maximum of kinetic energy.

3. A body is moving in accordance with the law $x(t) = A \cos(\omega t + \alpha)$, (A , ω , α are constants). Determine the mean kinetic energy provided t increases without bound from $t = 0$.

4. A ball falls from a height H from a state of rest. Demonstrate that the kinetic energy of the ball $K = mg(H - h)$, where h is the height of the ball above the ground at a given instant of time.

5. A train weighing 500 metric tons started out from a station, and in 3 minutes developed a speed of 45 km/hr traveling 1.5 km. Determine (a) the work and the mean power of the locomotive, on the assumption that friction on the rails is absent; (b) the same but having regard for friction. The coefficient of friction is $k = 0.004$. (The force of friction is equal to the force of attraction of the train to the earth, i.e., its weight, multiplied by k .)

6. A body is acted upon by two forces: $F_1 = at$ and $F_2 = a(\theta - t)$. The impulses of these forces during time 0 to θ are the same. At time $t = 0$, the body had a velocity of $v_0 = 0$. Find the work of each force during the time interval from 0 to θ and compare it with the product of the impulse by the average velocity.

7. A man standing still on the ground acts on a given mass m with a force F during a time t . As a result, the mass, which was originally at rest, acquires a velocity $v_1 = \frac{Ft}{m}$ and a kinetic energy $\frac{mv_1^2}{2}$ equal to the work performed by the man.

Consider the same experiment done in a train traveling at v_0 . The mass m had a velocity v_0 prior to the experiment, and $v_0 + v_1$ after the experiment. Find the change in the kinetic energy of the mass m . What work was done by the man? Assuming that the man rests against the wall of the railway car and its velocity is v_0 and does not change, find the work of the force done by the train (locomotive) during the experiment.

8. A man of mass M standing in skates on the ice (friction neglected) acts with a force F on a mass m during time t .

What kinetic energy will be imparted to the mass m ? What kinetic energy will the man acquire? What is the total work done by the force acting on mass m and on the man? Why is it greater than in Problem 7?

9. The same experiment as in Problem 8, but the man has an initial velocity of v_0 together with mass m . The velocity of mass m is, after the action of the force, $v_0 + \frac{Ft}{m}$, the velocity of the man is $v_0 - \frac{Ft}{M}$. Find the change in kinetic energy of the mass m and the man as a result of the action of the force. Find the work done by the force, which work is equal to the change in the overall kinetic energy, and compare it with the result of the preceding problem.

6.7 MOTION UNDER THE ACTION OF A FORCE DEPENDENT SOLELY ON THE VELOCITY

When in motion, every body experiences a counteraction from the medium in which the motion is taking place. If the resistance is slight, it can often be neglected. But in some cases this approach is not satisfactory and the resistance has to be taken into account. It

has been established that if a body is moving in a liquid or a gas, and the speed is small and the size of the body is small, then the force of resistance is proportional to the speed:

$$F(t) = -kv(t) \quad (6.7-1)$$

Here, the coefficient of proportionality $k > 0$, and the minus sign in (6.7-1) shows that the force of resistance is opposite that of the velocity of the body. The number k depends on the properties of the medium and is proportional to the viscosity of the medium. Besides, k is dependent on the shape and dimensions of the body. For example, in the case of a sphere of radius R , formula (6.7-1) takes the form

$$F = -6\pi R\eta v(t)^* \quad (6.7-2)$$

where η is the viscosity of the medium. For air, $\eta = 1.8 \times 10^{-4}$, for water $\eta = 0.01$ (at 20°C), $[\eta] = \frac{\text{g}}{\text{cm} \cdot \text{sec}}^{**}$.

We consider the problem of deceleration of a body. Suppose some force has imparted a velocity to a body and at time $t = t_0$ has ceased to act. The body continues in motion and is acted upon by the force of resistance alone.

By Newton's second law,

$$m \frac{dv}{dt} = -kv$$

Dividing both sides by m and setting $\frac{k}{m} = \alpha$ ($\alpha > 0$), we get

$$\frac{dv}{dt} = -\alpha v$$

The solution of this equation is (see Chapter 4)

$$v(t) = v_0 e^{-\alpha(t-t_0)} \quad (6.7-3)$$

* Formula (6.7-2) is valid for $\frac{vR\rho}{\eta} < 5$, where ρ is the density of the medium.

The reader can easily see that the quantity $\frac{vR\rho}{\eta}$ is dimensionless. It is called the Reynolds number.

** Viscosity η can be defined as follows. Let a liquid (or a gas) be in motion along the x -axis, but the velocities of the various particles are distinct and depend on the y -coordinate. It is clear that a rigid body could not move in that fashion and would be destroyed. In a liquid or gas, in this case, there arises, between adjacent layers, a force of friction which is proportional to the difference of the velocities of the adjacent layers, that is, to the derivative $\frac{dv}{dy}$. The proportionality constant in the expression for force per square centimetre of horizontal surface area is called the viscosity:

$$f \frac{\text{dyne}}{\text{cm}^2} = \eta \frac{dv}{dy}$$

Here, v_0 is the value of velocity at time $t = t_0$. Since $\alpha > 0$, it follows that for $t > t_0$ the exponent in (6.7-3) is negative, $e^{-\alpha(t-t_0)} < 1$, and hence $v(t) < v_0$, which means the velocity falls off with time. The medium retards the motion of the body.

Let us find an expression for the distance traveled by the body. From (6.7-3) we have

$$\frac{dx}{dt} = v_0 e^{-\alpha(t-t_0)}$$

or

$$dx = v_0 e^{-\alpha(t-t_0)} dt \quad (6.7-4)$$

Suppose at $t = t_0$ (initial time) the body is at the origin: $x(t_0) = 0$. Integrating (6.7-4) we obtain

$$x(t) = v_0 \int_{t_0}^t e^{-\alpha(t-t_0)} dt$$

whence

$$x(t) = \frac{v_0}{\alpha} [1 - e^{-\alpha(t-t_0)}] \quad (6.7-5)$$

Using formula (6.7-5), we can obtain the entire distance covered by the body after time t_0 , which is after the force ceased to act. In this connection, note that for very large values of t , the quantity $e^{-\alpha(t-t_0)}$ is extremely small and can be neglected in comparison with unity. Therefore, the body travels a total distance of $\frac{v_0}{\alpha}$.

Let us examine the fall of a body in air. The x -axis will point downwards towards the ground, we place the origin at a height of H above the ground (on the ground $x = H$). Let the motion begin at $t = 0$ with a velocity of v_0 . Then $x(0) = 0$, $v(0) = v_0$. The body is acted upon by two forces: the force of gravity (causes the motion) and the force of air resistance (inhibits the motion).

Newton's second law gives

$$m \frac{dv}{dt} = mg - kv \quad (6.7-6)$$

Dividing all terms of this equation by m , we get (since $\frac{k}{m} = \alpha$)

$$\frac{dv}{dt} = g - \alpha v \quad (6.7-7)$$

On the right, factor out α :

$$\frac{dv}{dt} = \alpha \left(\frac{g}{\alpha} - v \right) \quad (6.7-8)$$

We establish the dimensions of $\frac{g}{\alpha}$. Since $\alpha = \frac{k}{m}$ and $k = -\frac{F}{v}$, it follows that α has the dimensions of sec^{-1} . The dimensions of $\frac{g}{\alpha}$

are $\text{cm} \cdot \text{sec}/\text{sec}^2 = \text{cm}/\text{sec}$, which means $\frac{g}{\alpha}$ has the dimensions of velocity.*

Set $\frac{g}{\alpha} = v_1$. Equation (6.7-8) takes the form

$$\frac{dv}{dt} = \alpha (v_1 - v) \quad (6.7-9)$$

Suppose that $v_0 < v_1$. Then the right member of (6.7-9) is positive at the start of motion and, hence, the left member is positive too,

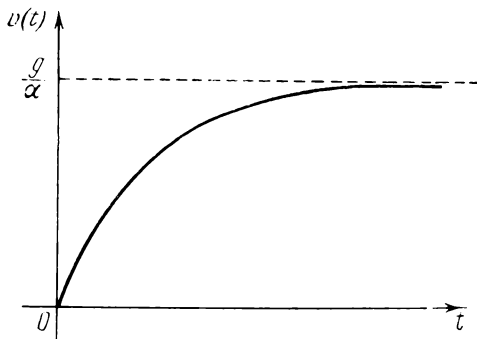


Fig. 147

$\frac{dv}{dt} > 0$; therefore the velocity $v(t)$ increases. And the closer v is to v_1 , the closer $\frac{dv}{dt}$ is to zero and, consequently, the slower v grows. If at some time t_1 it were true that $v(t_1) = v_1$, then v would remain constant since $v \equiv v_1$ is the solution of equation (6.7-9) with initial condition $v(t_1) = v_1$. Similarly if at the start of motion $v > v_1$, then v approaches v_1 but in this case v decreases. For this reason, a certain time after the start of motion the body falls at practically a constant speed of $v_1 = \frac{g}{\alpha}$, irrespective of the speed it had at the start of fall. The graph of velocity for $v_0 = 0$ is shown in Fig. 147.

The foregoing examination shows that a number of properties of $v(t)$ can be detected even without solving the equation (6.7-9). Now

* The calculation performed here of the dimensions of $\frac{g}{\alpha}$ is a check. The dimensions of $\frac{g}{\alpha}$ are evident from formula (6.7-8). Since only quantities having the same dimensions can be subtracted, it follows that $\frac{g}{\alpha}$ must have the dimensions of velocity.

let us solve this equation. Put $v_1 - v = z$. Then $\frac{dz}{dt} = -\frac{dv}{dt}$ and (6.7-9) can be rewritten as

$$\frac{dz}{dt} = -\alpha z$$

Here, at $t = 0$ it must be true that $z = v_1 - v_0$. The desired solution is $z(t) = (v_1 - v_0)e^{-\alpha t}$. Passing to the function $v(t)$, we get

$$v_1 - v(t) = (v_1 - v_0)e^{-\alpha t}$$

whence

$$v(t) = v_1 + (v_0 - v_1)e^{-\alpha t} \quad (6.7-10)$$

Considering this equation, it is easy to see that we can draw the same conclusions as were evident from an inspection of equation (6.7-9). Firstly, if $v_0 > v_1$, then $v(t) > v_1$, since $(v_0 - v_1)e^{-\alpha t} > 0$. But if $v_0 < v_1$, then $(v_0 - v_1)e^{-\alpha t} < 0$ and therefore $v(t) < v_1$. Secondly, no matter what v_0 is, the quantity $e^{-\alpha t}$ is small for sufficiently large t , and, practically speaking, $v(t) = v_1$.

Using (6.7-10), let us find the expression for the distance as a function of time:

$$\frac{dx}{dt} = v_1 + (v_0 - v_1)e^{-\alpha t}$$

whence, recalling that $x(0) = 0$,

$$x(t) = v_1 t + \frac{v_0 - v_1}{\alpha} (1 - e^{-\alpha t})$$

If a body has a large velocity or considerable dimensions, then the resistance is proportional to the square of the velocity. It has been experimentally established in this case that*

$$F = -kS\rho \frac{v^2}{2}$$

where S is the cross-sectional area of the body, ρ is the density of the medium. In this case, the force of resistance is practically independent of the viscosity of the medium. The coefficient k in this formula is a dimensionless number; its magnitude depends on the shape of the body (for streamlined bodies, k can drop to 0.03-0.05, for bodies with poor streamline characteristics, k can reach 1.0 to

* This formula is valid for the Reynolds number $\frac{Rv\rho}{\eta} > 100$. The meaning of the formula given in the text is that for the motion of a large body, the energy expended on overcoming the resistance of the medium is not spent on the friction of layers of the liquid but on the kinetic energy of the liquid, which is forced to move in order to let the body pass through it. The reader is advised to derive the formula for the force.

1.5). Denoting $\frac{kS\rho}{2} = G$ we obtain

$$F(t) = -Gv^2(t) \quad (6.7-11)$$

It is clear that G has the dimensions of g/cm.

Let us solve the problem of deceleration for the force of resistance (6.7-11). The appropriate equation is of the form

$$m \frac{dv}{dt} = -Gv^2$$

Dividing both sides by m and setting $\frac{G}{m} = \beta$ ($\beta > 0$), we get

$$\frac{dv}{dt} = -\beta v^2$$

whence $\frac{dv}{v^2} = -\beta dt$. Integrating, we get $-\frac{1}{v} \Big|_{v_0}^v = -\beta t \Big|_{t_0}^t$, where v_0 is the velocity of the body at time $t = t_0$.

Therefore $-\frac{1}{v} + \frac{1}{v_0} = -\beta(t - t_0)$ and from this we have

$$v = \frac{v_0}{1 + \beta v_0(t - t_0)} \quad (6.7-12)$$

From the formula $\beta = \frac{G}{m}$ we find that β has the dimensions 1/cm.

Find the formula for the distance. Using (6.7-12) we get

$$dx = \frac{v_0}{1 + \beta v_0(t - t_0)} dt$$

whence

$$x(t) = x(t_0) + \int_{t_0}^t \frac{v_0}{1 + \beta v_0(t - t_0)} dt \quad (6.7-13)$$

Assuming that the body begins to move from the origin, $x(t_0) = 0$, we get, from (6.7-13),

$$x(t) = \frac{1}{\beta} \ln [1 + \beta v_0(t - t_0)] \quad (6.7-14)$$

It is easy to see that this formula corresponds to velocity being an exponential function of the distance traveled: $v = v_0 e^{-\beta x}$. If we now want to find the entire distance traversed by the body after the force that imparted the velocity has ceased to act, we will see that this distance [formula (6.7-14)] is the greater, the more time, t , that elapses. [By formula (6.7-14) $x \rightarrow \infty$ as $t \rightarrow \infty$.] Actually, this is not so. The point is that when the velocity of the body is small, the relation (6.7-11) no longer holds true. We have to resort to (6.7-1) and, for the distance, to formula (6.7-5).

Let us consider the problem of a body falling in air for the case of air resistance being proportional to the square of the velocity.

At start, let the body fall from the origin having an initial velocity v_0 . In analogous fashion to the case where the resistance is proportional to the velocity, we get the equation.

$$\frac{dv}{dt} = g - \beta v^2 \quad (6.7-15)$$

We rewrite (6.7-15) in the form

$$\frac{dv}{dt} = \beta \left(\frac{g}{\beta} - v^2 \right)$$

It is easy to establish that $\sqrt{g/\beta}$ has the dimensions of velocity; set $\sqrt{g/\beta} = v_1$, $g/\beta = v_1^2$. Then

$$\frac{dv}{dt} = \beta (v_1^2 - v^2) \quad (6.7-16)$$

An exact solution is given in the answers to Exercise 1. Let us consider the general properties of the solution. Reasoning in similar fashion to the way we did for equation (6.7-9), we can show that in this case a velocity of $v_1 = \sqrt{g/\beta}$ should set in. We will show that after a long enough time lapse following the start of fall, the formula

$$v - v_1 = C e^{-2\beta v_1 t}, \quad (6.7-17)$$

where C is a constant, will hold true. Equation (6.7-16) can be rewritten thus:

$$\frac{dv}{dt} = \beta (v_1 + v) (v_1 - v) \quad (6.7-18)$$

For t large, $v \approx v_1$ and so in this equation we replace $v_1 + v$ by $2v_1$. If we replace v by v_1 in the difference $v_1 - v$, we get $\frac{dv}{dt} = 0$, whence $v = \text{const} = v_1$. Since we are interested precisely in the small difference between v and v_1 (the law of approach of v to v_1), it is not permissible to ignore the difference $v - v_1$. Thus, from (6.7-18) we get

$$\frac{dv}{dt} = 2\beta v_1 (v_1 - v) \quad (6.7-19)$$

Put $v_1 - v = z$, $\frac{dz}{dt} = -\frac{dv}{dt}$. Equation (6.7-19) then takes the form

$$\frac{dz}{dt} = -2\beta v_1 z$$

Its solution is

$$z = C e^{-2\beta v_1 t} \quad (6.7-20)$$

which coincides with (6.7-17).

The value of C in (6.7-20) cannot be determined from the initial condition $v(0) = v_0$ [that is $z(0) = v_1 - v_0$] because equation (6.7-19) holds true only for sufficiently large t (near $t = 0$ we cannot replace $v + v_1$ by $2v_1$).

Also observe that the formula $F(t) = -Gv^2(t)$ is valid only when $v > 0$. Indeed, if $v < 0$, it must be true that $F(t) = Gv^2(t)$ since the force of resistance is in the direction opposite to the velocity and, hence, is positive if the velocity is negative. Both cases ($v > 0$ and $v < 0$) are embraced by the formula

$$F(t) = -Gv(t) |v(t)|$$

Exercises

1. Find the expression for the velocity as a function of time from the equation $\frac{dv}{dt} = \beta(v_1^2 - v^2)$ for the initial condition $v(0) = v_0$. Using the formula

for v show that a velocity is established equal to $v_1 = \sqrt{\frac{g}{\beta}}$. Show that for the

formula (6.7-20), $C = \frac{2v_1(v_0 - v_1)}{v_1 + v_0}$.

2. In the problem on the falling of bodies (the resistance is proportional to velocity) have regard for the fact that the body is acted upon by an expulsive force in accord with the Archimedean law.

3. Applying the result of the preceding problem to a sphere and noting that for a sphere $k = 6\pi R\eta$, where R is the radius of the sphere and η is the viscosity of the medium, demonstrate that for large t the speed of fall is established for the sphere at $v = \frac{2R^2g(\rho - \rho')}{9\eta}$ (here, ρ is the density of the material of the sphere, ρ' is the density of the medium).

6.8 MOTION UNDER THE ACTION OF AN ELASTIC FORCE

Let us examine the case where the force acting on a body depends solely on the position of the body: $F = F(x)$. Above we considered in detail the work done by such a force and found out that in this case the system has a definite potential energy $u(x)$ with which the force is connected by the relation

$$F(x) = -\frac{du(x)}{dx}$$

Return to the problem on the motion of a body under the action of such a force. The basic equation is of the form

$$m \frac{dv}{dt} = F(x) \quad (6.8-1)$$

This equation cannot be solved directly since it involves the derivative with respect to time, while the force is given as a function of the x -coordinate. It is therefore natural to seek the quantities of interest as functions of the x -coordinate. For one thing, we will seek the velocity as a function of the coordinate, i.e., $v(x)$. We will then represent the derivative with respect to time, $\frac{dv}{dt}$, as the derivative

of a composite function, since the x -coordinate itself depends on the time:

$$\frac{dv[x(t)]}{dt} = \frac{dv[x(t)]}{dx} \cdot \frac{dx(t)}{dt}$$

But $\frac{dx}{dt}$ is nothing other than the velocity $v(x)$. Thus, $\frac{dv}{dt} = v \frac{dv}{dx}$, whence

$$m \frac{dv}{dt} = mv \frac{dv}{dx} = \frac{d}{dx} \left(\frac{mv^2}{2} \right)$$

Substituting this into the equation of motion (6.8-1), we get

$$\frac{d}{dx} \left(\frac{mv^2}{2} \right) = F(x) \quad (6.8-2)$$

Integrating we find

$$\int_{v_0}^{v_1} \frac{d}{dx} \left(\frac{mv^2}{2} \right) dx = \int_{x_0}^{x_1} F(x) dx$$

or

$$\frac{mv_1^2}{2} - \frac{mv_0^2}{2} = \int_{x_0}^{x_1} F(x) dx$$

The physical meaning of this expression is quite clear: the change in kinetic energy is equal to the work done by the force.

Using potential energy, we rewrite (6.8-2) as

$$\frac{d}{dx} \left(\frac{mv^2}{2} \right) = - \frac{du}{dx}$$

or

$$\frac{d}{dx} \left(\frac{mv^2}{2} + u(x) \right) = 0$$

If the derivative of an expression is identically zero, then the expression itself is a constant, therefore

$$\frac{mv^2}{2} + u(x) = \text{const} \quad (6.8-3)$$

In this form, formula (6.8-3) expresses the law of conservation of energy: when a body is in motion due to the action of a force dependent exclusively on the coordinate, the sum of the kinetic energy of the body, $\frac{mv^2}{2}$, and its potential energy $u(x)$ remains constant.

This sum is called the total energy of the body. The purpose of these lengthy transformations here and in the preceding section was to show that the law of conservation of energy (as applied to mechanics) is a consequence of the Newtonian law. Also a corollary to Newton's law is the fact that the kinetic energy of a body is precisely $\frac{mv^2}{2}$ and not some other function of the velocity of the body.

How do we continue the solution of the problem of the motion of the body? Using the values of the velocity v_0 and the coordinate x_0

of the body at the initial time, we find the total energy E of the body, which is a quantity that remains constant throughout the motion: $\frac{mv_0^2}{2} + u(x_0) = E$. With the aid of formula (6.8-3) and knowing E , we can find the velocity of the body as a function of x :

$$v(x) = \sqrt{\frac{2}{m} [E - u(x)]} \quad (6.8-4)$$

It remains to find the relationship between x and t . From (6.8-4) we get

$$\frac{dx}{dt} = \sqrt{\frac{2}{m} [E - u(x)]}$$

whence

$$\frac{dx}{\sqrt{\frac{2}{m} [E - u(x)]}} = dt, \quad t = t_0 + \int_{x_0}^x \frac{dx}{\sqrt{\frac{2}{m} [E - u(x)]}}$$

Thus, the time t is expressed as a function of the x -coordinate

$$t = t(x) \quad (6.8-5)$$

This function is given by an integral. We can find $x(t)$ by solving (6.8-5) for x . Since v is expressed as a function of x by means of

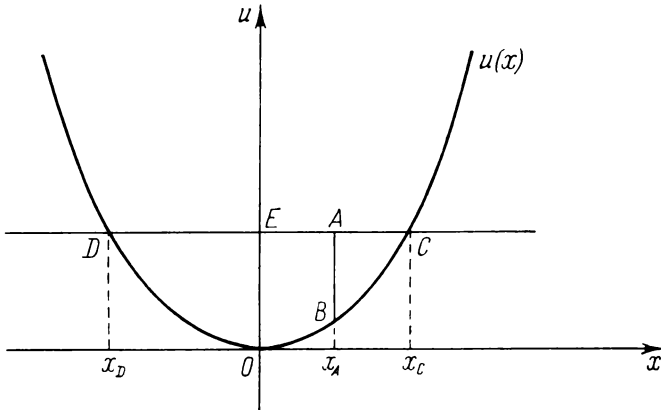


Fig. 148

a square root, even the simple expression $u(x)$ frequently leads to extremely awkward expressions for $t(x)$.

To get a general idea of the nature of the motion, it will be useful to draw the curve of $u(x)$. If on the same graph we plot the horizontal line at height E (Fig. 148), the picture is vivid in the extreme. The velocity is proportional to the square root of the difference

$E - u(x)$. For instance, when $x = x_A$ the velocity is proportional to the square root of the length of the line segment AB (Fig. 148). To the right of point C and to the left of point D is a region where $E < u(x)$, that is, a region into which a body with total given energy E cannot penetrate for lack of energy (formally speaking, the root of a negative quantity yields an imaginary value for the velocity). In the region where $E > u(x)$, the square root for $v(x)$ yields two possible values in accord with the two signs of the radical:

$$v = \pm \sqrt{\frac{2}{m} [E - u(x)]}$$

At the initial time, both the quantity and the sign of v_0 are determined by the initial conditions. From then on, the motion occurs in the direction specified by the sign of the initial velocity, v_0 . Evidently, when $v \neq 0$ the sign of the velocity cannot change suddenly. Thus, if at the initial time the body was located at the point $x = x_1^*$ and $v_0 > 0$, then the body will reach the extreme admissible point x_C . At this point the velocity of the body becomes zero and there occurs a transition from the formula $v = \sqrt{\frac{2}{m} [E - u(x)]}$ to

the formula $v = -\sqrt{\frac{2}{m} [E - u(x)]}$. Since $v = 0$ at this point, the change of sign takes place without a jump (discontinuity) in the velocity. The situation is similar at the point $x = x_D$. Thus, in the case depicted in Fig. 148, the motion of the body will be in the form of oscillations between two extreme positions x_C and x_D .

Let us consider another example. The potential energy of a body is given by the function $u = ax$ ($a > 0$). Find the law of motion of the body.

Suppose that at the initial instant of time, t_0 , $x = x_0$, $v = v_0$. Then the total energy $E = \frac{mv_0^2}{2} + ax_0$. Using (6.8-4) we get

$$v(x) = \sqrt{\frac{2}{m} \left[\frac{mv_0^2}{2} + ax_0 - ax \right]} = \sqrt{v_0^2 + \gamma(x_0 - x)}$$

where $\gamma = \frac{2a}{m}$. Knowing $v(x)$ we find the time

$$t = t_0 + \int_{x_0}^x \frac{dx}{\sqrt{v_0^2 + \gamma(x_0 - x)}}$$

In the integral make the change of variable $v_0^2 + \gamma(x_0 - x) = z^2$, $2z dz = -\gamma dx$ to get

$$t = t_0 - \int_{v_0}^{\sqrt{v_0^2 + \gamma(x_0 - x)}} \frac{2 dz}{\gamma} = t_0 - \frac{2}{\gamma} [\sqrt{v_0^2 + \gamma(x_0 - x)} - v_0]$$

* The point x_1 is not shown in Fig. 148, it can be located at any place between x_D and x_C .

From this we find x :

$$\frac{\gamma}{2} (t - t_0) = -\sqrt{v_0^2 + \gamma(x_0 - x)} + v_0$$

transposing v_0 to the left, squaring and cancelling γ , we obtain

$$x = -\frac{\gamma}{4} (t - t_0)^2 + v_0 (t - t_0) + x_0$$

Finding $\frac{d^2x}{dt^2}$ we are convinced that what we have is a case of uniformly decelerated motion. This was to be expected since $F = -\frac{du}{dx} = -a$; the force is constant and negative, which means that the motion is uniformly decelerated. In this elementary case where

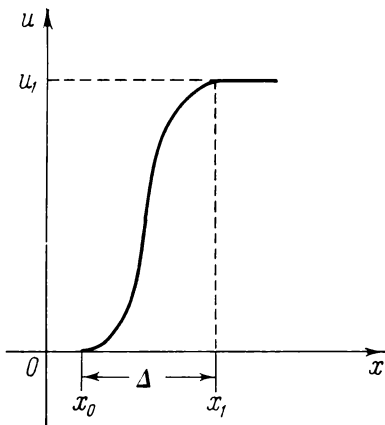


Fig. 149

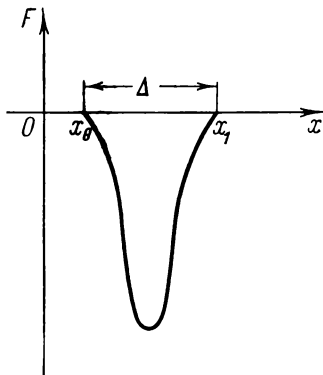


Fig. 150

the force was actually independent of x , there was of course no reason for employing such a complicated computational procedure.

In the next example we will consider the potential energy whose graph is in the form of a step (Fig. 149). Such a function $u(x)$ is associated with the graph of force given in Fig. 150 (to convince oneself that this is so, the reader should recall that $F = -\frac{du}{dx}$), the force is extremely great and negative, that is, in the direction of decreasing x . The steeper $u(x)$ is (the curve in Fig. 149), i.e., the shorter $\Delta = x_1 - x_0$ over which the rise in $u(x)$ occurs, the greater is the force in absolute value. The force is zero where the function $u(x)$ is constant (to the left of x_0 and to the right of x_1).

Let a body start moving from x_0 (Fig. 149) with a velocity v_0 . Suppose the total energy of the body is equal to E . For what values of E can the body reach point x_1 ? Since $u(x_0) = 0$, it follows that

$E = \frac{mv_0^2}{2}$. On the other hand, $E = \frac{mv_1^2}{2} + u_1$, where v_1 is the velocity of the body at the point x_1 and u_1 is the potential energy for $x = x_1$. Therefore

$$\frac{mv_1^2}{2} = E - u_1 \quad (6.8-6)$$

From this formula it is evident that if $E < u_1$, then the body cannot reach x_1 because then $v_1^2 < 0$, which is impossible. For this reason, the body can reach $x = x_1$ only if $E \geq u_1$.

For this case we determine the work done by the force F in displacing the body from x_0 to x_1 :

$$A = \frac{mv_1^2}{2} - \frac{mv_0^2}{2} = \frac{mv_1^2}{2} - E$$

Using (6.8-6) we find

$$A = E - u_1 - E = -u_1$$

The force F does not perform any work in further motion of the body to the right of the point x_1 , since $F=0$ for $x > x_1$.

Exercises

1. The potential energy is given by the formula $u = \frac{kx^2}{2}$ ($k > 0$). Construct a graph and show that the motion is oscillatory.
2. The potential energy is given by the formula

$$u(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 2x & \text{if } 0 \leq x \leq 1 \\ 2 & \text{if } x \geq 1 \end{cases}$$

At the initial time t_0 a body of mass 1 gram leaves the origin and moves rightwards with a velocity v_0 (cm/sec): (a) $v_0 = 1$, (b) $v_0 = 1.9$, (c) $v_0 = 2.1$. Indicate for each case whether the body can continue in motion indefinitely to the right. If it cannot, find the point at which it will stop.

3. $u(x) = -x^3 + 4x^2$. At the initial instant of time a body of mass 2 grams starts out from the point x_0 at a velocity of v_0 (cm/sec): (a) $x_0 = 1$, $v_0 = 1$, (b) $x_0 = -2$, $v_0 = 1$, (c) $x_0 = -2$, $v_0 = -1$. In each case investigate the nature of the motion (points at which the body stops, regions which the body cannot reach). In the case of stopping points, give at least a rough idea of their coordinates.

4. The same requirements relative to $u(x) = \frac{x^2}{1+x^2}$, $m = 2$: (a) $x_0 = 0$, $v_0 = 2$, (b) $x_0 = \frac{1}{2}$, $v_0 = \frac{1}{2}$. Express the time t as a function of x in terms of an integral.

6.9 OSCILLATIONS

Consider a body acted upon by the force

$$F = -kx$$

As we know, to this force corresponds a potential energy

$$u = \frac{kx^2}{2}$$

The origin is the position of stable equilibrium. The curve of potential energy (parabola) has the shape shown in Fig. 148.

The motion of a body under the action of such a force constitutes oscillations to the left and to the right of the position of equilibrium. We can imagine a ball rolling from one branch of the parabola building up speed and, by inertia, rolling up the other branch, rolling down again, and so forth. By Newton's second law, the equation of these oscillations is of the form

$$m \frac{d^2x}{dt^2} = -kx \quad (6.9-1)$$

We will not solve it by the general and rather involved procedure given in the preceding section. Instead we will guess the type of solution and concentrate our attention on investigating the properties of the solution.

Thus, we assume that

$$x = a \cos \omega t \quad (6.9-2)$$

This form of the solution is chosen because the cosine is one of the simplest periodic functions.

Put expression (6.9-2) into the basic equation (6.9-1). Since

$$v = \frac{dx}{dt} = -a\omega \sin \omega t, \quad \frac{d^2x}{dt^2} = -a\omega^2 \cos \omega t$$

it follows that

$$-ma\omega^2 \cos \omega t = -ka \cos \omega t \quad (6.9-3)$$

This relation will hold for any t if $m\omega^2 = k$. Therefore the function (6.9-2) does indeed satisfy the equation if $m\omega^2 = k$, whence

$\omega = \sqrt{\frac{k}{m}}$. Then

$$x = a \cos \left(t \sqrt{\frac{k}{m}} \right) \quad (6.9-4)$$

Observe that the square root in the expression for ω does not lead to two solutions since $\cos \omega t = \cos (-\omega t)$.

Let us find the period of oscillation, that is, the time during which the body returns to the initial position with the initial velocity. The function $\cos \varphi$ returns to its original position when the angle φ makes a complete revolution (changes by 2π). Thus, in the expression $a \cos \omega t$ the quantity ωt should vary through 2π in one period T . Therefore T is found from the condition

$$\omega(t + T) = \omega t + 2\pi$$

whence

$$\omega T = 2\pi, \quad T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}} \quad (6.9-5)$$

The quantity $\nu = \frac{1}{T}$ yields the number of oscillations per unit time and is called the frequency. It has dimensions 1/sec, or sec^{-1}

(reciprocal second). The unit of frequency—one cycle, or oscillation, per second—goes by the special name *hertz*, in honour of the German physicist Heinrich Hertz. It is evident, from formula (6.9-5), that $\nu = \frac{\omega}{2\pi}$. But it is more convenient in all formulas to deal with ω and not with ν , otherwise the coefficients 2π and $4\pi^2$ will appear throughout. The quantity $\omega = \frac{2\pi}{T}$ is called the circular frequency.*

The constant a cannot be determined from (6.9-1) because the equation is satisfied for arbitrary a [a can be cancelled from both members of (6.9-3)].

The velocity of the body is $v = \frac{dx}{dt} = -a\omega \sin \omega t$. From the relation $\cos^2 \omega t + \sin^2 \omega t = 1$ it follows that for $\cos \omega t = \pm 1$ it will be true that $\sin \omega t = 0$. Consequently, the velocity v is equal to zero at times of maximum deviation of the body in one direction or the other ($x = a$ or $x = -a$). Imagine that at $t < 0$ the body is placed at the point $x = a$ and is held at rest in this point with the aid of some external force (say a hook) until time $t = 0$, and then the hook is disengaged. At this instant the body is at rest, and oscillations are initiated under the action of a force $F = -kx$. In this case, the dependence of the coordinate x of the body upon time t is given by the formula $x = a \cos \omega t$. Since the absolute value of $\cos \omega t$ does not exceed 1, it follows that a is the largest value of x , or the maximum departure of the body from the position of equilibrium. The number a is called the amplitude of the oscillations. Thus the amplitude is equal to the original deviation of the body if at the onset of oscillations the body was at rest.**

Let us note in passing that, generally speaking, if $A(t) = L \cos \omega t$ [or $A(t) = L \sin \omega t$], then L is the greatest value of the quantity $A(t)$ and is termed the amplitude of that quantity $A(t)$.

* To grasp the origin of this term, consider a line segment of length a revolving counterclockwise. The similarity between rotation and oscillation is quite apparent, since a revolving hand of a clock returns to its original position after each revolution just as an oscillating body returns to its original position after one period. Here, the x -coordinate of, say, a second hand varies via the law $x = a \cos \omega t$, if the second hand revolves with angular velocity ω . In the case of rotation, if T is the period of one revolution, then $\nu = \frac{1}{T}$ is the number of revolutions per unit time, $\omega = 2\pi\nu$ is the angular velocity of rotation expressed in radians per second. Since the radian is a dimensionless quantity, ω has the dimensions of 1/sec. In view of the simple meaning of ω in the case of circular motion, this quantity in problems involving oscillations is termed the circular frequency.

** We defined the amplitude as *one half* of one oscillation period. Between the extreme left-hand point $x = -a$ and the extreme right-hand point $x = +a$, the body travels a distance of $2a$, which is *twice* the amplitude.

Also observe that the frequency ω is independent of the amplitude a .

Let $x = x_1(t)$ be a solution of the equation (6.9-1), that is, let it be true that $m \frac{d^2 x_1}{dt^2} = -kx_1$. We consider the function $x_2(t) = Cx_1(t)$, where C is a constant. Substituting into (6.9-1) the values x_2 and $\frac{d^2 x_2}{dt^2}$ we get $mC \frac{d^2 x_1}{dt^2} = -kCx_1(t)$ or, cancelling out C ,

$$m \frac{d^2 x_1}{dt^2} = -kx_1$$

Thus, if $x = x_1(t)$ satisfies (6.9-1), then also $x_2(t) = Cx_1(t)$ satisfies this equation.

It is easy to see that (6.9-1) also has another solution, $x(t) = b \sin \omega t$. Indeed, $\frac{d^2 (\sin \omega t)}{dt^2} = -\omega^2 \sin \omega t$. Substituting x and its second derivative into (6.9-1) and cancelling out $\sin \omega t$, we get $\omega = \sqrt{\frac{k}{m}}$, which is the same value as before.* For this reason

$$x(t) = b \sin \left(t \sqrt{\frac{k}{m}} \right) \quad (6.9-6)$$

As in the preceding case, we are not able to find b from (6.9-4). It is determined from the initial conditions. Suppose at the initial time $t = 0$ the body is at the point $x = 0$ and has acquired a definite initial velocity v_0 due to a brief external force (a blow, say). Then, since $v(t) = b\omega \cos \omega t$, it follows that for $t = 0$

$$v_0 = b\omega \quad (6.9-7)$$

whence $b = \frac{v_0}{\omega}$. Hence, the amplitude in this case is determined by the initial velocity.

The relation (6.9-7) offers a practical and convenient way of measuring impulse and velocity that is widely used in mechanics and goes by the name ballistic pendulum. If a body is suspended in the form of a pendulum or is held in a position of equilibrium by springs and its frequency is known, then the initial velocity following a blow may be determined from the amplitude of oscillations caused by the blow.

We will show how (6.9-7) can be approximately obtained via some general elementary reasoning. The dimensions of amplitude are cm, those of velocity, cm/sec, and the dimensions of time are sec.

* Here again, $\omega = -\sqrt{\frac{k}{m}}$ does not yield a new solution since $b \sin \left(-\sqrt{\frac{k}{m}} t \right) = -b \sin \left(\sqrt{\frac{k}{m}} t \right)$, the minus sign before the root expressing ω corresponds to the same form of function as before but for a different value of the constant b .

Therefore, for reasons of dimensionality, the amplitude must be a quantity of the same order as the product of the initial velocity into some portion of a period. Since motion caused by a blow lasts a quarter period up to maximum deviation and $v < v_0$ (since the motion is retarded), it follows that $b < v_0 T/4$. If the motion has a constant deceleration, the mean velocity would be equal to half the initial value and, consequently, $b \approx v_0 T/8$. Actually, however, as follows from formulas (6.9-7) and (6.9-5),

$$b = \frac{v_0}{\omega} = \frac{v_0 T}{2\pi} = \frac{v_0 T}{6.28}$$

The important thing here is that due to the fact that the period is independent of the amplitude, the latter is directly proportional to the initial velocity.

We have verified that two different functions (6.9-4) and (6.9-6) satisfy the equation $m \frac{d^2x}{dt^2} = -kx$. Suppose we want to solve the problem of the motion of a body having a given initial position and a given initial velocity: at $t = 0$, $x = x_0$, $v = v_0$ (and the values x_0 and v_0 are arbitrary, each may not be equal to zero). We will call this the general problem. Up to now, in contrast to the general problem, we have only considered particular problems in one of which $x = x_0$ at $t = 0$, $v = 0$ and in another $x = 0$ for $t = 0$, $v = v_0$.

Suppose we have taken the solution $x = a \cos \omega t$. Putting $t = 0$, we get $x_0 = a$, hence $x = x_0 \cos \omega t$. But then $v = \frac{dx}{dt} = -x_0 \omega \times \sin \omega t$ so that for $t = 0$, v equals 0 and not v_0 . Therefore, using the solution $x = a \cos \omega t$, we cannot solve the general problem. All we can solve is the problem with zero velocity.

Let's try the solution $x = b \sin \omega t$. Here, $v = \frac{dx}{dt} = b\omega \cos \omega t$. When $t = 0$ we get $v_0 = b\omega$, $b = \frac{v_0}{\omega}$ whence $x = \frac{v_0}{\omega} \sin \omega t$. But for $t = 0$, x equals 0 and not x_0 . Again, we are unable to solve the general problem with the aid of this solution.

It is easy to see that the sum

$$x = a \cos \omega t + b \sin \omega t \quad (6.9-8)$$

is also a solution of equation (6.9-1) for arbitrary a and b [the reader can verify this himself by finding the second derivative of the sum (6.9-8) and putting it into (6.9-1)]. We thus have a solution with two arbitrary constants: $x = a \cos \omega t + b \sin \omega t$. The corresponding velocity is $v = -a\omega \sin \omega t + b\omega \cos \omega t$.

With the help of (6.9-8) we can solve the general problem of the motion of a body with an arbitrary position and an arbitrary velocity at the initial time: $v = v_0$ for $t = 0$, $x = x_0$. Using the initial

data, we find from (6.9-8) $a = x_0$, $b = \frac{v_0}{\omega}$ and so

$$x = x_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t$$

From the foregoing it follows that the solutions $x = a \cos \omega t$, $x = b \sin \omega t$ do not make it possible to solve the general problem of motion, but only the particular problems involving special initial conditions. That is why these solutions are called particular solutions. Now the solution $x = a \cos \omega t + b \sin \omega t$ permits solving the general problem of motion, a problem with arbitrary initial conditions. That is why this solution is termed the general solution.

The general solution with two arbitrary constants may be obtained by other reasoning. In place of the independent variable t in equation (6.9-1) we introduce a new independent variable τ by the formula

$$\tau = t + t' \quad (6.9-9)$$

where t' is a constant. Then $\frac{dx}{d\tau} = \frac{dx}{dt} \frac{dt}{d\tau} = \frac{dx}{dt}$ since from (6.9-9) it is evident that $\frac{dt}{d\tau} = 1$. Similarly, $\frac{d^2x}{d\tau^2} = \frac{d^2x}{dt^2}$. Equation (6.9-1) takes the form

$$m \frac{d^2x}{d\tau^2} = -kx$$

We know its solution, $x = C \cos \omega \tau$, where C is a constant and $\omega = \sqrt{\frac{k}{m}}$. Returning to the variable t , we get $x = C \cos \omega (t + t') = C \cos (\omega t + \omega t')$. Putting $\omega t' = \alpha$, we have

$$x = C \cos (\omega t + \alpha) \quad (6.9-10)$$

Using the formula $\cos (\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$, let us compare the solutions (6.9-10) and (6.9-8):

$$C \cos (\omega t + \alpha) = C \cos \alpha \cos \omega t - C \sin \alpha \sin \omega t = a \cos \omega t + b \sin \omega t$$

Thus, for both solutions to be able to describe one and the same motion, they must satisfy the conditions

$$a = C \cos \alpha, \quad b = -C \sin \alpha$$

Since a and b are readily expressed in terms of initial position and initial velocity, it is useful to be able to solve the converse problem: to find C and α , knowing a and b . To do this, we set up the expressions

$$a^2 + b^2 = C^2 \cos^2 \alpha + C^2 \sin^2 \alpha = C^2$$

whence

$$C = \sqrt{a^2 + b^2} \quad (6.9-11)$$

$$-\frac{b}{a} = \frac{\sin \alpha}{\cos \alpha} = \tan \alpha, \quad \alpha = \arctan \left(-\frac{b}{a} \right)$$

If the solution is written in the form (6.9-10), then it is clear that the amplitude is equal to C . Hence, if the solution is of the form (6.9-8), then the amplitude is equal to $\sqrt{a^2 + b^2}$. Suppose $v = v_0$ at $t = 0$, $x = x_0$ then $a = x_0$, $b = \frac{v_0}{\omega}$ and so the amplitude is $\sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$.

Exercise

1. A body is in oscillation according to the law $\frac{d^2x}{dt^2} = -x$. Find the function $x(t)$ and determine the period for the following cases: (a) $v = 2$ cm/sec at $t = 0$, $x = 0$. (b) $v = 0$ at $t = 0$, $x = 1$. (c) $v = 2$ cm/sec at $t = 0$, $x = 1$. For Case (c), write the solution both in the form of (6.9-8) and as (6.9-10).

6.10 OSCILLATION ENERGY. DAMPED OSCILLATIONS

Let us write down the general solution for the equation (6.9-1) in the form

$$x = C \cos(\omega t + \alpha) \quad (6.9-10)$$

The potential energy of a body is equal, at every instant, to

$$u(x(t)) = \frac{kx^2(t)}{2} = \frac{kC^2}{2} \cos^2(\omega t + \alpha)$$

and the kinetic energy is

$$K(t) = \frac{mv^2}{2} = \frac{m}{2} [-C\omega \sin(\omega t + \alpha)]^2 = \frac{mC^2\omega^2}{2} \sin^2(\omega t + \alpha)$$

The frequency of the oscillation, as we already know, is determined by the formula $\omega^2 = \frac{k}{m}$. Substituting ω^2 into the expression for kinetic energy, we get

$$K(t) = \frac{kC^2}{2} \sin^2(\omega t + \alpha)$$

Thus, the factor in front of the trigonometric function in the expression for potential energy and in the expression for kinetic energy is the same. The functions themselves, $\cos^2(\omega t + \alpha)$ and $\sin^2(\omega t + \alpha)$, are very much alike, one being derivable from the other by a displacement along the time axis amounting to $\Delta t = \frac{\pi}{2\omega}$ (Fig. 151). Each of the quantities u and K oscillates between maximum and zero; when one is at maximum, the other is at zero. Observe that the functions $\cos^2(\omega t + \alpha)$ and $\sin^2(\omega t + \alpha)$ describe oscil-

lations about the mean value equal to half the maximum. This is clearly evident from Fig. 151 and also from the familiar formulas

$$\cos^2 \beta = \frac{1}{2} (1 + \cos 2\beta) = \frac{1}{2} + \frac{1}{2} \cos 2\beta,$$

$$\sin^2 \beta = \frac{1}{2} (1 - \cos 2\beta) = \frac{1}{2} - \frac{1}{2} \cos 2\beta$$

It is clear here that the quantity $\frac{1}{2} \cos 2\beta$ oscillates, taking on positive and negative values alternately, and $\frac{1}{2}$ is the mean value.

The sum of the potential and kinetic energies (that is, the total energy of the system),

$$E = K + u = \frac{kC^2}{2} [\cos^2 (\omega t + \alpha) + \sin^2 (\omega t + \alpha)] = \frac{kC^2}{2}$$

is constant, as was to be expected.

Note that if we specified the motion with a frequency that did not satisfy the formula $\omega = \sqrt{\frac{k}{m}}$, then the sum of the potential and kinetic energies would not be a constant, and the maximum

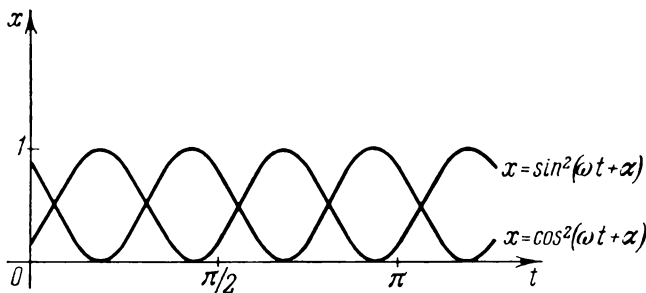


Fig. 151

kinetic energy would not be equal to the maximum potential energy. This is not surprising since oscillation with a frequency different from $\omega = \sqrt{\frac{k}{m}}$ does not satisfy the basic equation of motion. Hence, for such oscillation to occur, it is necessary that the body be acted upon by some kind of other, external, forces, besides the force $F = -kx$.* Because of the work of external forces, the total energy $\left(\frac{mv^2}{2} + \frac{kx^2}{2}\right)$ will no longer be preserved.

Now let us investigate the problem of damping of oscillations. Let a body be acted upon by the force of friction in addition to the

* This force is associated with the potential $u(x) = \frac{kx^2}{2}$.

force $F = -kx$ of a spring. Suppose the friction is small so that over one period the work of the force of friction is small compared with the oscillation energy. We can then assume, approximately, that the oscillations occur as in the case of no friction:

$$x(t) = C \cos(\omega t + \alpha)$$

The oscillation energy is $\frac{kC^2}{2}$. In the case of friction, the energy of the oscillations diminishes with time. Thus, friction will cause the coefficient C of $\cos(\omega t + \alpha)$ to decrease slowly; it will no longer be a constant. The law of decrease of C is defined by the condition that the decrease in energy is equal to the work of the force of friction.

With respect to unit time, these quantities are

$$\frac{dE}{dt} = \frac{d\left(\frac{kC^2}{2}\right)}{dt} = kC \frac{dC}{dt} = F_1 v = W_1 \quad (6.10-1)$$

where F_1 is the force of friction, v is the velocity of the body, and W_1 is the power of the force of friction. In the oscillation process, the velocity v and the force F_1 vary periodically. The product $F_1 v$ always remains negative. In the case at hand of small friction which is a slow damping of the oscillations, we can take it that the variation of amplitude $C(t)$ is small over several oscillation periods. The product $F_1 v$ may be understood as the mean value of this product during one period. Formula (6.10-1) holds true only for time intervals exceeding the oscillation period.

By way of an illustration, let us examine the friction proportional to the velocity of the body:

$$F_1 = -\hbar v, \quad F_1 v = -\hbar v^2$$

Substituting $v = -C\omega \sin(\omega t + \alpha)$ we get

$$F_1 v = -\hbar C^2 \omega^2 \sin^2(\omega t + \alpha)$$

Note that the mean value of $\sin^2(\omega t + \alpha)$ during one period is equal to $1/2$ (see Exercises in Sec. 4.4 and also the formulas on page 308). Using (6.10-1) we finally get

$$kC \frac{dC}{dt} = -\hbar C^2 \omega^2 \cdot \frac{1}{2}$$

whence

$$\frac{dC}{dt} = -\frac{\hbar \omega^2}{2k} C$$

Recalling that $\omega^2 = \frac{k}{m}$, we obtain a simpler expression:

$$\frac{dC}{dt} = -\frac{\hbar}{2m} C$$

The solution of this equation is

$$C = C_0 e^{-\frac{\hbar}{2m} t} \quad (6.10-2)$$

Here, C_0 is defined from the initial conditions. Multiplying both sides of (6.10-2) by $\cos(\omega t + \alpha)$ and using (6.9-10), we get

$$x(t) = C_0 e^{-\frac{h}{2m}t} \cos(\omega t + \alpha) \quad (6.10-3)$$

where $\omega = \sqrt{\frac{k}{m}}$. This is an approximate formula obtained on the assumption that the friction is slight and is proportional to the velocity.

When the force of friction is proportional to the velocity, the problem has an exact solution. In this case, the body is acted on by two forces: $-kx$ and $h \frac{dx}{dt}$. By Newton's second law,

$$m \frac{d^2x}{dt^2} = -kx - h \frac{dx}{dt} \quad (6.10-4)$$

We will seek the solution $x(t)$ in the same form as was obtained for a small friction:

$$x(t) = C_0 e^{-\gamma t} \cos(\omega_1 t + \alpha) \quad (6.10-5)$$

Then

$$\begin{aligned} \frac{dx}{dt} &= -\gamma C_0 e^{-\gamma t} \cos(\omega_1 t + \alpha) - C_0 \omega_1 e^{-\gamma t} \sin(\omega_1 t + \alpha), \\ \frac{d^2x}{dt^2} &= \gamma^2 C_0 e^{-\gamma t} \cos(\omega_1 t + \alpha) + \gamma C_0 \omega_1 e^{-\gamma t} \sin(\omega_1 t + \alpha) \\ &\quad + C_0 \omega_1 \gamma e^{-\gamma t} \sin(\omega_1 t + \alpha) - C_0 \omega_1^2 e^{-\gamma t} \cos(\omega_1 t + \alpha) \end{aligned}$$

Substituting into (6.10-4) the expression for x , $\frac{dx}{dt}$, $\frac{d^2x}{dt^2}$ and cancelling out $C_0 e^{-\gamma t}$, we get

$$\begin{aligned} m\gamma^2 \cos(\omega_1 t + \alpha) + m\gamma \omega_1 \sin(\omega_1 t + \alpha) + m\gamma \omega_1 \sin(\omega_1 t + \alpha) \\ - m\omega_1^2 \cos(\omega_1 t + \alpha) = -k \cos(\omega_1 t + \alpha) + h\gamma \cos(\omega_1 t + \alpha) \\ + h\omega_1 \sin(\omega_1 t + \alpha) \end{aligned}$$

or

$$\begin{aligned} [m\gamma^2 - m\omega_1^2] \cos(\omega_1 t + \alpha) + 2m\gamma \omega_1 \sin(\omega_1 t + \alpha) \\ = -[k - h\gamma] \cos(\omega_1 t + \alpha) + h\omega_1 \sin(\omega_1 t + \alpha) \end{aligned} \quad (6.10-6)$$

This equation holds true for arbitrary t if

$$\left. \begin{aligned} m\gamma^2 - m\omega_1^2 &= -k + h\gamma \\ 2m\gamma \omega_1 &= h\omega_1 \end{aligned} \right\} \quad (6.10-7)$$

Cancelling ω_1 out of the second equation, we obtain

$$\gamma = +\frac{h}{2m} \quad (6.10-8)$$

Then from the first equation

$$\omega_1^2 = \frac{k}{m} - \frac{h^2}{4m^2} \quad (6.10-9)$$

Consequently

$$x(t) = C_0 e^{-\frac{h}{2m}t} \cos \left(\sqrt{\frac{k}{m} - \frac{h^2}{4m^2}} t + \alpha \right) \quad (6.10-10)$$

When friction is low, that is, the number h is small in comparison with k , we can ignore the term $\frac{h^2}{4m^2}$ under the radical sign as compared with $\frac{k}{m}$. Then (6.10-10) becomes (6.10-3). Thus, in the approximate

consideration we correctly obtained the law of decrease of amplitude but did not notice the small variation of frequency due to friction.

If the friction is considerable, the radicand may become negative and the formulas become meaningless. This means that motion involving appreciable friction is no longer oscillatory. In that case the solution is to be sought in the form $x = Ce^{-\gamma t}$. Substituting this into the equations we get two values of γ . The sum of the two solutions corresponding to these γ will yield the general solution and will enable us to solve the problem for arbitrary initial data. This case is considered in detail in Sec. 8.10 in connection with electrical oscillations.

Exercises

1. Find the law of damping of oscillations for friction proportional to the square of the velocity (this friction is characteristic of rapid motion of a body in a low-viscosity liquid). Show that after the lapse of a large time interval, the amplitude $C(t) = 1/bt$, where b is a constant independent of C_0 (C_0 is the amplitude at the initial time).

2. Find the law of damping of oscillations for friction that is independent of the velocity (this is characteristic of the friction of hard dry surfaces). Determine the time during which the oscillations cease.

3. Obtain the equation for small vibrations of a pendulum, that is, a material point suspended by a thread of length l . Small vibrations are deviations through a small angle, in other words, vibrations such that the horizontal displacement x is small compared with l . Find the period.

Hint. Take advantage of the fact that the sum of the kinetic and potential energies is a constant.

6.11 FORCED OSCILLATIONS AND RESONANCE

Consider a body acted upon by an elastic force $F = -kx$. We have established that this force causes the body to oscillate with a definite frequency $\omega = \sqrt{k/m}$, which is the so-called natural (or free) frequency. From now on we will denote the natural frequency by ω_0 so that $\omega_0 = \sqrt{k/m}$.

Now let the body be acted upon by an elastic force and also a periodic external force with frequency ω . It then turns out that the amplitude of the oscillations brought about by the external force is very strongly dependent on how close the frequency ω of the external force is to the natural frequency. This phenomenon is called resonan-

ce and has a wide range of applications. It refers to any systems that admit oscillations and vibrations. In mechanical systems (machine tools, motors) such vibrations can result in deformation and destruction of the equipment.

At times, resonance is purposely used to produce, via a small force, vibrations of the operating tool with great amplitude.

In electric systems, resonance enables us, using several periodic forces with different frequencies (say, a number of radio transmitters), to achieve a situation in which the oscillations in our system depend solely on one of the periodic forces (the one whose frequency is close to the natural frequency of the system). This allows for tuning a radio set to a definite station.

Let us set up the oscillation equation:

$$m \frac{d^2x}{dt^2} = -kx - h \frac{dx}{dt} + f \cos \omega t \quad (6.11-1)$$

In this equation, $f \cos \omega t$ is the external force.

Divide through by m and set $\frac{k}{m} = \omega_0^2$ in accordance with the fact that (in the absence of friction) such is the natural frequency of the body. Denote the ratio $\frac{h}{m}$ by 2γ [see formulas (6.10-8) and (6.10-9)]. We obtain

$$\frac{d^2x}{dt^2} = -\omega_0^2 x - 2\gamma \frac{dx}{dt} + \frac{f}{m} \cos \omega t$$

It is natural to expect that under the action of a force having a frequency ω the body will oscillate with that frequency. We therefore seek the solution in the form

$$x = a \cos \omega t + b \sin \omega t \quad (6.11-2)$$

Substituting the expressions for x and its derivatives into (6.11-1), we get

$$\begin{aligned} -a\omega^2 \cos \omega t - b\omega^2 \sin \omega t = & -a\omega_0^2 \cos \omega t - b\omega_0^2 \sin \omega t \\ & + 2\gamma a\omega \sin \omega t - 2\gamma b\omega \cos \omega t + \frac{f}{m} \cos \omega t \end{aligned}$$

For this equation to be true for arbitrary t , the separate terms involving $\cos \omega t$ and $\sin \omega t$ must be equal. Equating these terms, we get

$$\left. \begin{aligned} -a\omega^2 &= -a\omega_0^2 - 2\gamma b\omega + \frac{f}{m}, \\ -b\omega^2 &= -b\omega_0^2 + 2\gamma a\omega \end{aligned} \right\} \quad (6.11-3)$$

From the latter equation of (6.11-3) we get

$$b = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} a$$

Substituting this into the first equation of (6.11-3), we find

$$a = \frac{f}{m} \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2} \quad (6.11-4)$$

Then

$$b = \frac{f}{m} \frac{2\gamma\omega}{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2} \quad (6.11-5)$$

Going over to the form $x = C \cos (\omega t + \alpha)$ and recalling that

$$C = \sqrt{a^2 + b^2} \quad (6.9-11)$$

we obtain the amplitude C of oscillations produced by an external force:

$$C = \frac{f}{m} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}} \quad (6.11-6)$$

It is then clear that C is the greater, the closer ω is to ω_0 . The curve of C as a function of ω for a given ω_0 is shown in Fig. 152 for two values of γ ($\frac{f}{m}=1$, $\omega_0=1$). The less the friction, the sharper the rise in amplitude for the frequency of the external force equal to the natural frequency.

It is not hard to see that the sum of the solution (6.10-5) of equation (6.10-4) and the general solution (6.11-2) of equation (6.11-1).

$$x = a \cos \omega t + b \sin \omega t + C_0 e^{-\gamma t} \cos (\omega_1 t + \alpha) \quad (6.11-7)$$

where a and b are given by the formulas (6.11-4) and (6.11-5), is also a solution of equation (6.11-1).

Using this solution, we can solve a problem with arbitrary initial data by choosing C_0 and α . Indeed, suppose that $v = v_0$ at $t = 0$, $x = x_0$. Then, using (6.11-7), we find

$$\left. \begin{aligned} x_0 &= a + C_0 \cos \alpha, \\ v_0 &= b \omega - C_0 (\gamma \cos \alpha + \omega_1 \sin \alpha) \end{aligned} \right\} \quad (6.11-8)$$

From this system of equations we can determine C_0 and α (see Exercises). Thus, (6.11-7) is the general solution to a problem involving oscillations of a body under the action of an elastic force and a periodic external force. This general solution confirms the assumption, made at the beginning of this section, that under the protracted action of an external force with frequency ω a body will oscillate

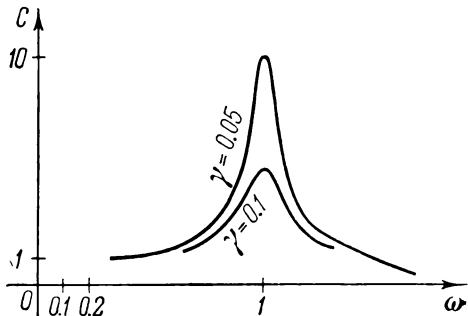


Fig. 152

with the same frequency ω . This is true because no matter what the initial conditions, they only affect the values C_0 and α , which is to say, only the last summand in the solution (6.11-7). However, in the course of time, this term, which has frequency ω_1 , becomes arbitrarily close to zero due to the factor $e^{-\gamma t}$ and we can neglect it for large t . The remaining terms describe oscillations with frequency ω , which do not decay with time since they are maintained by the action of an external force.

Exercises

1. Determine C_0 and α from the system (6.11-8).
2. Because of friction, the maximum amplitude C is for ω_{max}^* somewhat different from ω_0 .

Find the deviation of ω_{max}^*/ω_0 from unity as a function of γ .

Hint. Test the radicand in (6.11-6) for a minimum, denoting $\omega^2 = z$.

6.12 ON EXACT AND APPROXIMATE SOLUTIONS OF PHYSICAL PROBLEMS

In the preceding section we had the luck to find with comparative ease the exact solution of a problem involving oscillations of a body under the action of a periodic external force in the case of a restoring force ($-kx$) and friction ($-h \frac{dx}{dt}$). With this exact solution at our disposal it is easy to find a number of important limiting cases.

(1) The frequency ω of the external force is extremely small compared with ω_0 , where $\omega_0^2 = \frac{k}{m}$. Neglecting ω in (6.11-6) in comparison with ω_0 , we get $C = \frac{f}{m\omega_0^2} = \frac{f}{k}$.

(2) The frequency ω of the external force is extremely large and much larger than ω_0 . Then $C = \frac{f}{m} \frac{1}{\sqrt{\omega^4 + 4\gamma^2\omega^2}}$. But $\gamma^2\omega^2 \ll \omega^4$ (the friction is not very great); neglecting the term $4\gamma^2\omega^2$, we obtain $C = \frac{f}{m\omega^2}$.

(3) The force of friction is small. Disregarding the term containing γ , we get

$$C = \frac{f}{m} \frac{1}{|\omega_0^2 - \omega^2|} \quad (6.12-1)$$

We take the absolute value since we consider the positive value of the radical in formula (6.11-6).

(4) The phenomenon of exact resonance: the frequency of the external force is exactly equal to the natural frequency, that is, $\omega = \omega_0$. Then

$$C = \frac{f}{m} \frac{1}{2\gamma\omega} = \frac{f}{h\omega} = \frac{f}{h\omega_0} \quad (6.12-2)$$

These limiting cases actually make up over 90% of the content of all the results obtained. When one obtains a general result, it is always necessary to simplify it by considering various limiting cases, as we have just done. The simple formulas relating to the limiting cases are more easily remembered and more frequently used in practical situations. Only once in a while does one have to resort to general formulas. If we know the limiting cases, we do not know everything but we know *almost* everything that is contained in the more complicated exact formula.

The question arises as to the possibility of obtaining these limiting formulas directly via simplifications in the equation itself. To solve an involved equation in exact fashion and then simplify the solution is just as senseless as to use intricate machinery to package goods elegantly and then immediately tear open the package to get them.

To obtain limiting (approximate) expressions directly is particularly important for the added reason that an exact solution is very sensitive to the slightest variations in the statement of the problem. A slight complication in the problem and one finds it impossible to get an exact solution. An approximate solution is rougher and more stable with respect to variations in the problem.

Of particular importance to students are cases where it is possible to obtain and compare both solutions, exact and approximate. It is precisely in such cases that one can acquire some experience in the proper choice of approximations and be sure of the results.

Let us now return to the first case: the frequency of the external force is low. This is clearly a slow motion. Therefore, in the original equation

$$m \frac{d^2x}{dt^2} = -kx - h \frac{dx}{dt} + f \cos \omega t \quad (6.11-1)$$

we drop terms involving motion, $m \frac{d^2x}{dt^2}$ and $h \frac{dx}{dt}$, to get

$$0 = -kx + f \cos \omega t$$

whence

$$x = \frac{f \cos \omega t}{k} = C \cos \omega t, \quad C = \frac{f}{k}$$

Thus, for low frequencies, at each moment the applied external force is balanced by the force of elasticity. This result is clearly very general, for it refers to any motion with a low frequency. This limiting case is called a static case. In particular, the force of elasticity may be any function of the coordinate [$F(x)$], the external force may be any function of the time [$F_1(t)$]. The oscillation equation takes the form

$$m \frac{d^2x}{dt^2} = F(x) - h \frac{dx}{dt} + F_1(t) \quad (6.12-3)$$

It is not always possible to obtain the exact solution of this equation, but the approximate approach is preserved. Indeed, neglecting in the case of slow motion the terms involving velocity and acceleration, we get from (6.12-3)

$$F(x) + F_1(t) = 0$$

From this we find $x(t)$, or an approximate relationship between x and t . Substituting $x(t)$ into the exact equation (6.12-3), we can find the order of the error by neglecting the terms $m \frac{d^2x}{dt^2}$ and $h \frac{dx}{dt}$.

Let us take a look at the second limiting case, very high frequency ω . For a high frequency, the time of action of the external force and, hence, the impulse during each half-cycle (while the force is acting in one direction) are small because of the short duration of the half-cycle. Thus, for a given amplitude of the force f , the greater ω , the smaller the velocity that the body can acquire and the smaller the displacement of the body. Neglecting the terms kx and $h \frac{dx}{dt}$, we get an equation of motion of a free body with no forces acting except the external force,

$$m \frac{d^2x}{dt^2} = f \cos \omega t \quad (6.12-4)$$

We will seek the solution of (6.12-4) in the form

$$x = C \cos \omega t$$

Then

$$\frac{d^2x}{dt^2} = -C\omega^2 \cos \omega t$$

Substituting into (6.12-4) we get

$$-C m \omega^2 \cos \omega t = f \cos \omega t$$

whence

$$C = -\frac{f}{m\omega^2}$$

and so

$$x = -\frac{f}{m\omega^2} \cos \omega t \quad (6.12-5)$$

In the standard form $x = C \cos(\omega t + \alpha)$ the solution (6.12-5) may be written so that C is positive:

$$x = \frac{f}{m\omega^2} \cos(\omega t + \pi)$$

Here, the elastic force is

$$-kx = \frac{kf}{m\omega^2} \cos \omega t = \frac{\omega_0^2}{\omega^2} f \cos \omega t$$

The force of friction is

$$h \frac{dx}{dt} = \frac{hf}{m\omega} \sin \omega t$$

When comparing forces that depend periodically on time, one should not compare their instantaneous values but their amplitudes. The ratio of an external force to the elastic force (the ratio of the amplitudes) is

$$\frac{\omega^2 f}{\omega_0^2 f} = \frac{\omega^2}{\omega_0^2}$$

This ratio is the greater, the greater ω . Similarly, the ratio of the external force to that of friction grows without bound with the growth of ω . For this reason, given large ω , the external force appreciably exceeds both the elastic force and the force of friction. This supports the possibility of an approximate consideration of motion under the action of the external force alone.*

The third limiting case—neglect of friction—is easily obtained directly:

$$m \frac{d^2 x}{dt^2} = -kx + f \cos \omega t = -m\omega_0^2 x + f \cos \omega t \quad (6.12-6)$$

We seek the solution of (6.12-6) in the form $x = C \cos(\omega t + \alpha)$. Substituting into this equation the expressions for x and $\frac{d^2 x}{dt^2}$, we get

$$\alpha = 0, \quad m(\omega_0^2 - \omega^2) C \cos \omega t = f \cos \omega t$$

whence

$$C = \frac{f}{m(\omega_0^2 - \omega^2)}$$

We consider the fourth limiting case: the frequency of an external force is exactly equal to the natural frequency of the oscillations, $\omega = \omega_0$.

We will seek the solution to (6.11-1) in the form

$$x = C \cos(\omega_0 t + \alpha)$$

Then

$$m \frac{d^2 x}{dt^2} = -mC\omega_0^2 \cos(\omega_0 t + \alpha)$$

Recalling that $\omega_0^2 = k/m$, we get

$$m \frac{d^2 x}{dt^2} = -kC \cos(\omega_0 t + \alpha) = -kx$$

Substituting into (6.11-1) the expressions for x and its derivatives, we obtain

$$hC \omega_0 \sin(\omega_0 t + \alpha) + f \cos \omega_0 t = 0$$

* It is essential that the friction considered above is the closer to zero, the closer the velocity is to zero. In dry friction (force of friction independent of velocity) an external force less than that of friction will not cause oscillations at any frequency.

This equation will hold true for any t if

$$\alpha = -\frac{\pi}{2}, \quad C = \frac{f}{h\omega_0}$$

Hence the solution is

$$x = \frac{f}{h\omega_0} \cos \left(\omega_0 t - \frac{\pi}{2} \right) \quad (6.12-7)$$

The amplitude of oscillations at resonance is $C = f/h\omega_0$.

Referring to Fig. 153, let us look at the relationship between C and ω provided by the approximate formulas (6.12-1) and (6.12-2). Formula (6.12-1) yields two branches going off to infinity at $\omega = \omega_0$; formula (6.12-2) yields a finite value $C = A$ for $\omega = \omega_0$.

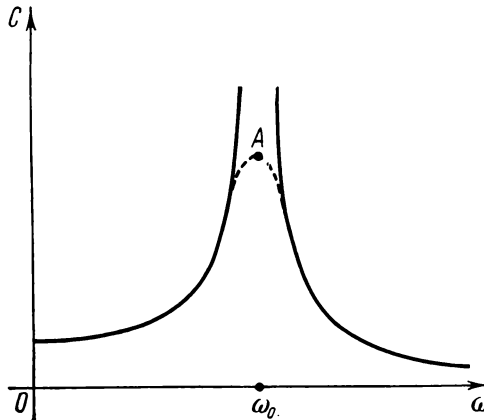


Fig. 153

If we construct the curves of (6.12-1) and place the point A that corresponds to formula (6.12-2), it is then easy to draw free-hand a smooth curve (dashed in Fig. 153) which, far from resonance, coincides with the curves of (6.12-1) and has a maximum A at the point ω_0 .

In the case of resonance (fourth case) the amplitude C and the initial phase α can be determined by means of energy considerations, the value of which consists in the fact that they also enable one to solve approximately certain problems that do not have exact solutions.

The power developed by an external force $f \cos \omega t$ in the case of motion specified by the expression $x = C \cos (\omega t + \alpha)$ is

$$W_{ex} = f \cos \omega t \frac{dx}{dt} = -fC\omega \cos \omega t \sin (\omega t + \alpha)$$

Let us determine the mean power of the external force during a large (more precisely, infinite) interval of time:

$$\overline{W}_{ex} = -fC\omega \overline{\cos \omega t \sin(\omega t + \alpha)}$$

Observe that

$$\cos \omega t \sin(\omega t + \alpha) = \frac{1}{2} \sin 2\omega t \cos \alpha + \cos^2 \omega t \sin \alpha$$

and so

$$\overline{\cos \omega t \sin(\omega t + \alpha)} = \frac{1}{2} \overline{\sin 2\omega t} \cos \alpha + \overline{\cos^2 \omega t} \sin \alpha = \frac{1}{2} \sin \alpha$$

Consequently

$$\overline{W}_{ex} = -\frac{fC\omega}{2} \sin \alpha$$

or

$$\overline{W}_{ex} = \frac{fC\omega}{2} \cos \left(\alpha + \frac{\pi}{2} \right) \quad (6.12-8)$$

Now let us determine the mean power of the force of friction. Since $F_{fr} = -h\dot{v}$, it follows that

$$\overline{W}_{fr} = -\overline{h\dot{v}^2} \quad (6.12-9)$$

But

$$\overline{\dot{v}^2} = \overline{\left(\frac{dx}{dt} \right)^2} = C^2 \omega^2 \overline{\cos^2(\omega t + \varphi)} = \frac{C^2 \omega^2}{2}$$

Therefore (6.12-9) yields

$$\overline{W}_{fr} = -h \frac{C^2 \omega^2}{2}$$

Since the work of the external force goes to overcome friction, the mean powers of the external force and the force of friction must be equal in absolute value:

$$|\overline{W}_{fr}| = |\overline{W}_{ex}| \quad (6.12-10)$$

that is,

$$\left| \frac{fC\omega}{2} \right| \cos \left(\alpha + \frac{\pi}{2} \right) = h \frac{C^2 \omega^2}{2}$$

or

$$f \left| \cos \left(\alpha + \frac{\pi}{2} \right) \right| = hC\omega$$

whence

$$C = \frac{f}{h\omega} \left| \cos \left(\alpha + \frac{\pi}{2} \right) \right| \quad (6.12-11)$$

The maximum possible amplitude (resonance) results, as may be seen from (6.12-11), when $\cos \left(\alpha + \frac{\pi}{2} \right) = 1$, that is, for $\alpha = -\frac{\pi}{2}$.

Here, $\omega = \omega_0$, and $C = \frac{f}{h\omega_0}$. Hence, the solution in the case of

resonance is

$$x = \frac{f}{h\omega_0} \cos \left(\omega_0 t - \frac{\pi}{2} \right)$$

We again have the formula (6.12-7).

Let us return to formula (6.12-8). From this formula we see that at resonance $\overline{W}_{ex}$ has the largest value since at resonance $\cos \left(\alpha + \frac{\pi}{2} \right) = 1$. For this reason, in the case of resonance an external force develops the maximum mean power and, consequently, performs maximum work.

These arguments of an energy nature make it possible to determine the amplitude at resonance also in the case of a more complicated dependence of the force of friction on velocity. Let the force of friction be given by the formula

$$F_{fr} = -h\nu|v|^{n-1} \quad (6.12-12)$$

For $v > 0$, (6.12-12) gives $F_{fr} = -h\nu^n$, for $v < 0$ we get $F_{fr} = h|v|^n$. Therefore, (6.12-12) yields the force of friction in opposition to velocity for any sign of the velocity v . The mean power of the external force is, as before, given by the formula (6.12-8). We determine $\overline{W}_{fr}$. The instantaneous value $W_{fr} = F_{fr}v = -h\nu^2|v|^{n-1}$; since $v^2 = |v|^2$, it follows that $W_{fr} = -h|v|^{n+1}$. Substituting here the value of v , we find

$$W_{fr} = -hC^{n+1}\omega_0^{n+1} |\sin(\omega_0 t + \alpha)|^{n+1} \quad (6.12-13)$$

Using this equation we get

$$\overline{W}_{fr} = -hC^{n+1}\omega_0^{n+1}A$$

where $A = |\overline{\sin(\omega_0 t + \alpha)}|^{n+1}$.* The condition (6.12-10) yields

$$hC^{n+1}\omega_0^{n+1}A = \frac{1}{2}fC\omega_0 \left| \cos \left(\alpha + \frac{\pi}{2} \right) \right|$$

From this we have

$$C = \sqrt[n]{\frac{f}{2hA\omega_0^n} \left| \cos \left(\alpha + \frac{\pi}{2} \right) \right|}$$

The maximum amplitude attained at resonance is equal to

$$C = \frac{1}{\omega_0} \sqrt[n]{\frac{f}{2hA}} \quad (6.12-14)$$

A particular case of formula (6.12-14) for $n = 1$ (friction proportional to velocity) is the earlier found formula

$$C = \frac{f}{\omega_0 h}$$

* For reference we give values of A for a few n : $n \rightarrow 0$, $A \rightarrow \frac{2}{\pi} = 0.64$; $n = 1$, $A = 0.5$; $n = 2$, $A = \frac{4}{3\pi} = 0.42$; $n = 3$, $A = \frac{3}{8} = 0.375$.

6.13 JET PROPULSION AND TSIOLKOVSKY'S FORMULA

In the case of motion in airless space, the only method of flight control (changing speed and direction) consists in ejecting a portion of the mass of the flying body itself, which means applying the reaction (jet) principle.

The Russian scientist K. E. Tsiolkovsky was the first to fully realize the significance of the jet principle and to investigate the fundamental regularities of reaction motion (jet propulsion). From him, via his pupils and followers—Soviet scientists and engineers—stems the scientific tradition that saw final embodiment in the artificial earth satellites and space vehicles of recent years.

Let us derive the basic equation of the rectilinear motion of a rocket. The propellant, whether gunpowder or a mixture of fuel (alcohol, gasoline) and oxidizer (oxygen, nitric acid), possesses a definite supply Q of chemical energy per unit mass (Q is of the order of 1000 kilocalories per kilogram for smokeless powder and 2500 kcal/kg for a gasoline (or petrol) and oxygen mixture.* In burning, this chemical energy is converted into the thermal energy of the products of combustion, which stream out of a nozzle, the thermal energy turning into the kinetic energy of motion.

When a reaction (rocket) engine is fixed on a test bed, the products of combustion are exhausted at a definite velocity u_0 . The kinetic energy they have per unit mass constitutes a definite portion of the chemical energy of the propellant:

$$\frac{u_0^2}{2} = \alpha Q \quad (6.13-1)$$

where α is a dimensionless number, the coefficient of efficiency of the processes of combustion and exit of gases.** From now on we will consider the exhaust velocity u_0 to be a given known quantity. It is roughly 2 km/sec for powder and about 3 km/sec for liquid propellant. It is easy to see that these quantities are associated with the values of $\alpha \approx 0.5$ (efficiency of the order of 50%).

Prior to combustion, the propellant was at rest. Suppose a mass dm of propellant is burnt and exits from the nozzle. In so doing, it acquires a momentum of $u_0 dm$. Clearly, the impulse dI of the force with which the rocket acts on this mass is equal to the momentum

* The heating value of gasoline is about 10 000 kcal/kg. However, burning 1 kg of gasoline (or petrol) (CH_2) requires 3.4 kg of oxygen. In a rocket launched into airless space, the oxygen has to be carried along and the energy must be referred to the sum of the weights of the fuel and oxidizer.

** In formula (6.13-1), Q has to be expressed in mechanical units (erg/g) and then u_0 is obtained in cm/sec. We then have $1 \text{ kcal/kg} = 1 \text{ cal/g} = 4.18 \times 10^7 \text{ erg/g}$.

acquired by the mass dm ,*

$$dI = Fdt = u_0 dm$$

By the law of every action having an equal and opposite reaction, the impulse of the force with which the mass dm of the products of combustion acts on the rocket vehicle is equal to that same quantity with sign reversed. Suppose, for instance, the exhaust velocity u_0 is in the direction of decreasing x . Then u_0 is negative, $u_0 = -|u_0|$. For the impulse of the force acting on the rocket we have

$$dI_r = F_r dt = -u_0 dm = |u_0| dm \quad (6.13-2)$$

The quantity

$$I' = \frac{dI}{dm} = |u_0| \quad (6.13-3)$$

is the impulse per unit mass, the so-called unit impulse. This quantity is equal to the exhaust velocity of gases from a rocket at rest.

Let us check the dimensions in formula (6.13-3). The force F has the dimensions of $\text{g} \cdot \text{cm}/\text{sec}^2$ (dyne), the impulse I is the product of force by time, and so its dimensions are $\text{g} \cdot \text{cm}/\text{sec}$. The dimensions of $\frac{dI}{dm}$ are $\text{g} \cdot \text{cm}/\text{sec} \cdot \text{g} = \text{cm}/\text{sec}$, which are the dimensions of velocity.

For powder gases, $u_0 = 2 \times 10^5 = 2 \text{ km}/\text{sec}$, for liquid fuel $u_0 = 3 \text{ km}/\text{sec}$. In the mks system of units, the unit impulse is expressed in $\text{kgf} \cdot \text{sec}/\text{kg}$, where kgf denotes the kilogram force and kg denotes the kilogram mass. A force of 1 kgf is equal to the force expressed in dynes divided by 1000 g , where g is the acceleration of gravity. A mass of 1 kg is equal to the mass expressed in grams divided by 1000.

Therefore I' expressed in $\text{kgf} \cdot \text{sec}/\text{kg}$ is numerically equal to $\frac{u_0 \cdot 1000}{1000g} = \frac{u_0}{g}$. Assuming, roughly, that $g \approx 1000 \text{ cm}/\text{sec}^2$, we get

200 $\text{kgf} \cdot \text{sec}/\text{kg}$ for powder gases and 300 $\text{kgf} \cdot \text{sec}/\text{kg}$ for liquid fuel.

The force acting on a rocket is, by formula (6.13-2),

$$F_r = |u_0| \frac{dm}{dt}$$

It is proportional to the quantity of gases exhausted in unit time.

Now let us examine the derivation of the formula for the velocity of the rocket vehicle. If the rocket is itself in motion with a velocity u , then the exhaust velocity of the gases differs from u_0 and is equal to $u + u_0 = u - |u_0|$ (recall that when the vehicle is at rest, the exhaust velocity of gases is equal to $-|u_0|$). It is obvious that such quantities as the *difference* between the velocity of powder prior to combustion and the velocity of the exhausted powder gases and as the *force* with which the powder gases act on the rocket are independent of whether the rocket vehicle is in motion or at rest.

* The designation dI is due to the fact that we consider a small mass dm .

Let us denote the initial mass of the rocket together with the powder by M_0 , the mass of exhausted powder gases by m . The quantity m is a function of the time, $m = m(t)$. The designation m is in accord with the fact that the small exhausted mass was denoted by dm , and the quantity of powder gases exhausted in unit time was denoted by $\frac{dm}{dt}$. The mass of a rocket with powder is at time t equal to

$$M = M(t) = M_0 - m(t) \quad (6.13-4)$$

The equation of motion (Newton's second law) is

$$M \frac{du}{dt} = F = |u_0| \frac{dm}{dt}$$

This equation can be written thus:

$$M du = |u_0| dm$$

or, using (6.13-4),

$$(M_0 - m) \frac{du}{dm} = |u_0| \quad (6.13-5)$$

The possibility of cancelling out dt has the physical meaning that (in the absence of other forces acting on the rocket) the velocity of the rocket depends only on the amount of exhausted powder gases (for a fixed value of u_0). By the time a given amount m of powder gases is exhausted through the nozzle, the rocket has acquired a definite velocity u , irrespective of the time during which the given amount of powder gases was released.

It is easy to solve equation (6.13-5). At start, $u = 0$ for $m = 0$. We thus have

$$\begin{aligned} u &= |u_0| \int_0^m \frac{dm}{M_0 - m} = -|u_0| \ln(M_0 - m) \Big|_0^m \\ &= |u_0| [-\ln(M_0 - m) + \ln M_0] = |u_0| \ln \frac{M_0}{M_0 - m} = |u_0| \ln \frac{M_0}{M} \end{aligned}$$

Thus

$$u = |u_0| \ln \frac{M_0}{M} \quad (6.13-6)$$

This formula was first derived by K. Tsiolkovsky and bears his name.

If we are interested in the terminal velocity u_{ter} at burnout, then in formula (6.13-6) we put M_{ter} (terminal mass of the rocket after all the fuel has burnt out) in place of M : $M_{ter} = M_0 - m_{tot}$, where m_{tot} is the total mass of the propellant. We get

$$u_{ter} = |u_0| \ln \frac{M_0}{M_{ter}} \quad (6.13-7)$$

This formula can also be used to solve the converse problem: what initial mass of the rocket vehicle must be taken so that to a given

terminal mass M_{ter} is imparted a definite velocity u_{ter} :

$$\ln \frac{M_0}{M_{ter}} = \frac{u_{ter}}{|u_0|}$$

whence

$$M_0 = M_{ter} e^{\frac{u_{ter}}{|u_0|}} \quad (6.13-8)$$

For a body to be able to orbit the earth as a satellite, it is necessary that its centrifugal force balance the gravitational force of the earth. The corresponding velocity, u_1 , is called the orbital velocity (or first cosmic velocity). To determine it we obtain

$$M_{ter} \frac{u_1^2}{R} = M_{ter} g \quad (6.13-9)$$

where R is the radius of the orbit. It is approximately equal to the radius of the earth and so for the gravitational force in the right-hand member of (6.13-9) we take the force of gravity at the earth's surface. From (6.13-9) we find

$$u_1 = \sqrt{gR} \approx \sqrt{gr_0} \approx 8 \text{ km/sec}$$

For a satellite at a distance r from the earth's centre that is appreciably different from r_0 , one must bear in mind that the acceleration of gravity (equal to g at the surface of the earth) varies with altitude. Indeed, by Newton's law of gravitation, a body distant r from the centre of the earth is attracted to the earth with a force $F = G \frac{mM}{r^2}$, where m is the mass of the body and M is the mass of the earth. On the other hand, by Newton's second law, $F = ma$, where a is the acceleration of gravity at a distance r from the centre of the earth. Comparing these two expressions for F , we find $a = \frac{GM}{r^2}$. If $r = r_0$, then $a = g$, and so $g = \frac{GM}{r_0^2}$ whence $G = \frac{gr_0^2}{M}$. We finally get $a = g \frac{r_0^2}{r^2}$. In this case the equality of the centrifugal force and the force of gravity yields

$$M_{ter} \frac{u^2}{r} = M_{ter} g \frac{r_0^2}{r^2}$$

whence the orbital velocity of the satellite is

$$u = \sqrt{\frac{gr_0^2}{r}}$$

The greater the distance r , the smaller the velocity u required for a satellite to stay in orbit. However this does not at all mean that it is easier to launch a vehicle into orbit with a very great r than into one with r close to r_0 : the point is that in launching a rocket

vehicle into an orbit with large r more energy is required to overcome the force of gravity over the trajectory from the earth's surface to the orbit.

Now let us consider the next (in order of difficulty of execution) problem. For a body to be able to leave the sphere of terrestrial gravitation, it is necessary that its initial kinetic energy be greater than the difference between the potential energy of the launched body and the body at the earth's surface. We found this quantity in Sec. 6.2 [formula (6.2-5)]. Here it is assumed that the propellant is used up and the requisite velocity is acquired rapidly over a portion of the flight path that is small compared with the earth's radius, so that we can ignore the change in potential energy on this portion of the flight path. This means that the reactive force is very great during the time of combustion of the fuel, and we can also disregard the action of gravity during this time. It has been demonstrated that it is more advantageous to burn the fuel quickly (less fuel will be required) than to stretch out the process of combustion over a time necessary to cover a distance of the order of the earth's radius.*

The initial velocity needed for a body to be able to leave the sphere of the earth's gravitational pull is called the earth escape velocity (also, second cosmic velocity), u_2 . Let us find it. From formula (6.2-5) the initial energy required to reach a distance r from the centre of the earth by a body originally on the ground (r_0) is $K_0 = mg \frac{r_0}{r} (r - r_0)$. In our case, r is much greater than r_0 and so $r - r_0 \approx r$, which yields $K_0 = mgr_0$. Equating to this quantity the kinetic energy of the rocket,

$$M_{ter} \frac{u_2^2}{2} = M_{ter} gr_0$$

we get

$$u_2 = \sqrt{2gr_0} \approx 11.2 \text{ km/sec}$$

Finally, the initial velocity that a body must have in order to be able to leave the gravitational field of the sun, that is, to be able to leave the solar system, is called the solar escape velocity (or third cosmic velocity), u_3 . Let us find it using the fact that the velocity v_1 of the earth's revolution about the sun is known to be $v_1 = 30 \text{ km/sec}$.

By Newton's law of gravitation, the force of attraction of a mass m to the sun is $F = -k \frac{M_{\odot} m}{r^2}$, where $M_{\odot}$ is the mass of the sun,** r is

* Only over sections of the flight path where the air density is extremely high (and hence air resistance too) is it disadvantageous to develop high velocities. The thickness of the atmosphere is small in comparison to the radius of the earth (see Chapter 7) and we will disregard it.

** Astronomers use the symbol $\odot$ to denote the sun. Numerically, $M_{\odot} = 2 \times 10^{33} \text{ g}$.

the distance from the centre of the sun to the body, and k is a constant coefficient. The potential energy of a body carried to a distance r from the centre of the sun is equal to

$$u(r) = -\frac{kM_{\odot}}{r}m \quad (6.13-10)$$

Here, the value of potential energy at infinity is taken to be zero (see Sec. 6.2).

The magnitude of potential energy of a body at the radius of the earth's orbit can easily be expressed in terms of the velocity of the earth in its orbit. In orbit, the earth's attraction to the sun is balanced by the centrifugal force,

$$M \frac{v_1^2}{r_1} = k \frac{M_{\odot}M}{r_1^2}$$

where v_1 is the earth's speed in revolution about the sun, and r_1 is the radius of the earth's orbit, $150\,000\,000\text{ km} = 1.5 \times 10^{13}\text{ cm}$. From this we get

$$kM_{\odot} = v_1^2 r_1$$

and the formula (6.13-10) takes the form

$$u(r_1) = -v_1^2 m$$

For a body at a distance r_1 from the sun to leave the solar field of gravitation it is necessary that, at this distance, the sum of the kinetic and potential energies of the body be nonnegative. This leads to the condition

$$M_{ter} \frac{v_2^2}{2} + u(r_1) = M_{ter} \frac{v_2^2}{2} - M_{ter} v_1^2 \geq 0$$

Here, v_2 is the sought-for velocity of escape from the solar system, v_1 is the known orbital velocity of the earth.

We have already encountered a similar situation when we considered the motion of a body in the gravitational field of the earth: the velocity of escape from the terrestrial field of gravitation is associated with a kinetic energy twice that corresponding to the velocity needed for maintaining a vehicle in earth orbit.

From the last relation we find the minimum required velocity to be

$$v_2 = v_1 \sqrt{2} \approx 42\text{ km/sec}$$

To summarize, then: in order to escape from the solar system, a body at earth orbit must have an initial velocity (relative to the sun) of 42 km/sec. It then turns out that a body with a velocity exceeding 42 km/sec will leave the solar system, *irrespective* of the direction of the velocity: along a radius from the sun [1] or along a tangent to the orbit of the earth [2], [3] or even towards the sun [4] (but at a certain angle so as not to fall into the sun). The only thing the

direction of the initial velocity charges is the shape of the flight path (Fig. 154). The numbers in brackets correspond to the numbers of the flight paths in Fig. 154.

It is clear that the most advantageous trajectory for launching a rocket vehicle from the earth is [2]. The earth itself is moving at 30 km/sec and so to obtain a velocity of 42 km/sec in the same direction the rocket needs to attain a velocity $v'_2 = 12$ km/sec. The rocket has to have a velocity of 12 km/sec when it leaves the earth's field of gravitation, which is to say when it has receded from the earth to a distance large compared to the radius of the earth but small compared to the radius of the earth's orbit.

For this purpose, what must the initial velocity be at the earth's surface? It is precisely u_3 , the solar escape velocity (or the third cosmic velocity). We determine it from the relation

$$M_{ter} \frac{u_3^2}{2} = M_{ter} g r_0 + M_{ter} \frac{v_2'^2}{2} \quad (6.13-11)$$

Here, the first term on the right is the energy needed to overcome the earth's gravity, the second term is the energy that must remain afterwards so that the rocket has (after adding in the orbital velocity of the earth) velocity v_2 , which is necessary for escape from the solar system. Formula (6.13-11) yields

$$u_3^2 = 2gr_0 + v_2'^2 = u_2^2 + v_2'^2$$

whence

$$u_3 = \sqrt{u_2'^2 + v_2'^2} = \sqrt{11.2^2 + 12^2} \approx 16.4 \text{ km/sec}$$

Note that the earth escape velocity (second cosmic velocity) does not suffice for approaching the sun or landing on Mercury or Venus. Indeed, at this velocity the rocket tears away from the earth and will continue in that orbit with a velocity that of the earth, which is 30 km/sec.

Although the potential energy diminishes as the rocket approaches the sun, it cannot come close to the sun because of the centrifugal force of its orbital motion. In order to penetrate into the depths of the solar system, we have to reduce the speed of the rocket relative to the sun, but this is just as hard to do as to increase the speed. For instance, to hit the sun, the rocket must be brought to a stop, which means it must have a speed of 30 km/sec relative to the earth (after

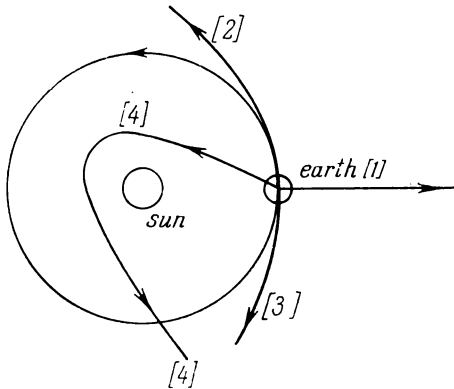


Fig. 154

leaving the field of gravitation). What this means is an initial velocity at the earth's surface of

$$u_4 = \sqrt{30^2 + 11.2^2} = 32 \text{ km/sec}$$

It is harder to fall into the sun than to get away from it! Better variants can be obtained by utilizing the change in rocket speed caused by the influence of other planets, but we will not go into that problem here.

Let us find the magnitudes of the ratio $\frac{M_0}{M_{ter}}$ for which the velocities u_1, u_2, u_3 are attained. For gunpowder $|u_0| = 2 \text{ km/sec}$. Using formula (6.13-8) we get for $u_1 = 8 \text{ km/sec}$, $\frac{M_0}{M_{ter1}} = e^4 = 54$; for $u_2 = 11.2 \text{ km/sec}$, $\frac{M_0}{M_{ter2}} = e^{5.6} = 270$; for $u_3 = 16.4 \text{ km/sec}$, $\frac{M_0}{M_{ter3}} = e^{8.2} = 3641$. In the case of liquid fuel, $|u_0| = 3 \text{ km/sec}$. Analogous computations yield: $\frac{M_0}{M_{ter1}} = 14.5$, $\frac{M_0}{M_{ter2}} = 42$, $\frac{M_0}{M_{ter3}} = 245$. From the foregoing we see that the magnitude of $\frac{M_0}{M_{ter}}$ is strongly dependent on the exhaust velocity (u_0) of the gases. To get an idea of the difficulty of the problem of launching a rocket, one should bear in mind that M_{ter} includes the weight of the fuel tanks, etc.

Let us find the efficiency of a rocket as a whole. We define this quantity as the ratio of the kinetic energy of the rocket at burnout, $M_{ter} \frac{u_{ter}^2}{2}$, to the chemical energy of the burnt fuel $mQ = (M_0 - M_{ter})Q$. The efficiency is

$$\eta = \frac{M_{ter} u_{ter}^2}{2Q(M_0 - M_{ter})} \quad (6.13-12)$$

Substituting into (6.13-12) the expression for u_{ter} from (6.13-7) and expressing u_0^2 from (6.13-1), we finally obtain

$$\eta = \alpha \frac{M_{ter}}{M_0 - M_{ter}} \left(\ln \frac{M_0}{M_{ter}} \right)^2$$

The efficiency turns out to be the product of the "internal efficiency" α (which characterizes the completeness of burning of the fuel and the conversion of thermal energy into the kinetic energy of gases) and the second factor, which depends solely on the choice of the ratio between the mass m of fuel and the mass M_{ter} of the payload. Denote $\frac{m}{M_{ter}} = z$. Then $M_0 = M_{ter} + m = M_{ter}(1 + z)$ and the efficiency is

$$\eta = \alpha \frac{M_{ter}}{m} \left(\ln \frac{M_{ter} + m}{M_{ter}} \right)^2 = \alpha \frac{1}{z} [\ln(1 + z)]^2$$

At first glance it might appear that due to the fraction $\frac{1}{z}$ the efficiency is very great for small z . In reality, for small z we have $\ln(1+z) \approx z$ and so

$$\eta \approx \alpha \frac{1}{z} z^2 = \alpha z$$

The efficiency is proportional to z and hence is small for small z . For small z the rocket moves slowly and almost all the energy is carried away by the gases. For very great z , the efficiency again falls because of diminished payload mass.* Since the terminal velocity of the rocket is also dependent solely on z , we can say that the efficiency of the vehicle is determined by the requisite velocity. At small velocities, the efficiency of the rocket vehicle is small and so it is disadvantageous to employ jet propulsion in automobiles and other cases of relatively slow motion. At high velocities, the energy efficiency of the rocket again diminishes, but the use of the rocket is nevertheless justified since we do not possess any other means of accelerating bodies to high velocities.

Exercises

1. Find the value of z which yields maximal efficiency η . Find the magnitude of this maximum.
2. Find the radius of the orbit of an artificial satellite with an orbital period of 24 hours. A vehicle launched into such an orbit in the plane of the equator will remain hanging over one spot on the earth's surface.

6.14| THE PATH OF A PROJECTILE

Let us consider the problem of the flight path of a projectile (shell) fired from a gun with initial velocity v_0 . We take the point of ejection of the shell from the barrel of the gun for the origin and sent the y -axis vertically upwards. For the sake of simplicity, we disregard air resistance because this would introduce considerable complications.

By Newton's second law,

$$m \frac{dv}{dt} = F$$

We applied this law earlier only for rectilinear motion. But in the flight-path problem the direction of v varies with time (the velocity is always directed along a tangent to the path of the shell). For this reason we will do as follows. We will decompose the force F into components along the x -axis and along the y -axis, F_x and F_y , respectively. We do the same with respect to the velocity v .

* For large z , the quantity $[\ln(1+z)]^2$ grows more slowly than z . Indeed, denoting $y = \ln(1+z)$, we get $z = e^y - 1$ and the function e^y grows faster than any power of y (see Chapter 3).

Any motion in the xy -plane may be regarded as the result of combining two motions: one occurring along the x -axis under the action of the force F_x with velocity v_x , and the other along the y -axis under the action of the force F_y with velocity v_y . Applying Newton's second law to each of these motions separately, we get

$$\left. \begin{aligned} m \frac{dv_x}{dt} &= F_x, \\ m \frac{dv_y}{dt} &= F_y \end{aligned} \right\} \quad (6.14-1)$$

We have obtained two equations, (6.14-1), but in each of them the force and the velocity are directed along a single straight line (the x -axis in the first equation and the y -axis in the second).

Denote by φ the angle which the barrel of the gun makes with the horizontal; call φ the angle of departure. Since we are considering the most elementary case in which the shell in flight is acted upon solely by the force of gravity (directed earthwards), it follows that $F_x = 0$, $F_y = -mg$. And so the equations of (6.14-1) have the form

$$\left. \begin{aligned} m \frac{dv_x}{dt} &= 0, \\ m \frac{dv_y}{dt} &= -mg \end{aligned} \right\} \quad (6.14-2)$$

Substitute the initial conditions for the functions $v_x(t)$ and $v_y(t)$. At the time of emergence of the shell from the barrel, $t = 0$,

$$\begin{aligned} v_x(0) &= v_0 \cos \varphi, \\ v_y(0) &= v_0 \sin \varphi \end{aligned}$$

The first of the equations of (6.14-2) yields $\frac{dv_x}{dt} = 0$, whence it follows that v_x is constant and so

$$v_x(t) = v_x(0) = v_0 \cos \varphi \quad (6.14-3)$$

The second equation in (6.14-2) yields $\frac{dv_y}{dt} = -g$, whence, integrating from 0 to t , we get

$$v_y(t) - v_y(0) = -gt$$

or

$$v_y(t) = -gt + v_0 \sin \varphi \quad (6.14-4)$$

To determine the displacements of x and y along the coordinate axes, we take advantage of the obvious relations

$$\left. \begin{aligned} \frac{dx}{dt} &= v_x, \\ \frac{dy}{dt} &= v_y \end{aligned} \right\} \quad (6.14-5)$$

Using the formulas (6.14-3) and (6.14-4), we get from (6.14-5)

$$\left. \begin{aligned} \frac{dx}{dt} &= v_0 \cos \varphi, \\ \frac{dy}{dt} &= -gt + v_0 \sin \varphi \end{aligned} \right\} \quad (6.14-6)$$

At the initial instant of time, the shell was at the origin and so for $t = 0$ $x = 0$, $y = 0$ (6.14-7)

Integrating (6.14-6) from 0 to t and using the initial conditions of (6.14-7), we find

$$\left. \begin{aligned} x &= v_0 t \cos \varphi, \\ y &= v_0 t \sin \varphi - \frac{gt^2}{2} \end{aligned} \right\} \quad (6.14-8)$$

These formulas enable us to determine the position of the shell at any instant of time t .

Taking various values of t , we can find the position of the shell, from formulas (6.14-8), at different times and we can plot a graph of the path of the shell. Thus, equations (6.14-8) yield a curve in the xy -plane. Representation of a curve by means of two equations like

$$\begin{aligned} x &= f_1(t) \\ y &= f_2(t) \end{aligned}$$

is called parametric representation of the curve. The number t is called the parameter.

From equations (6.14-8) we can easily get rid of t and obtain an equation of the path in the ordinary form, as a function y of x . Indeed, the first equation in (6.14-8) yields $t = \frac{x}{v_0 \cos \varphi}$; then from the second equation of (6.14-8) we get

$$y = x \tan \varphi - x^2 \frac{g}{2v_0^2 \cos^2 \varphi} \quad (6.14-9)$$

From (6.14-9) we see that y is a second-degree polynomial in x , the graph of which is a parabola. Consequently, disregarding air resistance, the path of a shell is in the shape of a parabola. Fig. 155 depicts the path of (6.14-9) for the case $v_0 = 80$ m/sec,* $\varphi = 45^\circ$.

From (6.14-9) it is evident that for one and the same v_0 , the shape of the trajectory depends on the angle of departure φ . We will find the maximum altitude of ascent of the shell and the range of fire for given φ and v_0 . To determine the maximum altitude, we set up

* For a small initial velocity, the air resistance is indeed small. But if we take an initial velocity $v_0 = 800$ m/sec, then a 305-mm calibre (diameter) shell at an angle of departure $\varphi = 55^\circ$ will meet air resistance that cuts the range of fire from 61 to 22.2 km.

the equation $\frac{dy}{dx} = 0$ to get

$$\tan \varphi - x \frac{g}{v_0^2 \cos^2 \varphi} = 0$$

whence

$$x = \frac{v_0^2 \sin \varphi \cos \varphi}{g} = \frac{v_0^2 \sin 2\varphi}{2g}$$

For this value of x , the height y is a maximum (it is physically clear

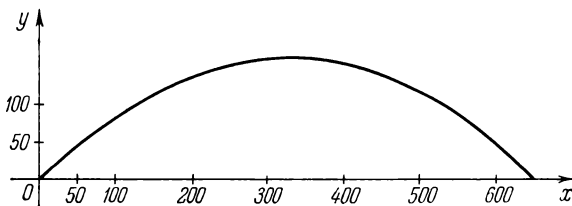


Fig. 155

that this is precisely the maximum; but, incidentally, this can be verified from $\frac{d^2y}{dx^2}$). Substituting the x found into (6.14-9), we get

$$y_{\max} = \frac{v_0^2 \sin^2 \varphi}{2g}$$

To determine the range of the shell, it suffices to determine the value of x for which $y = 0$ (see Fig. 155):

$$x \tan \varphi - x^2 \frac{g}{2v_0^2 \cos^2 \varphi} = 0$$

Discarding the solution $x = 0$ which does not interest us, we find

$$x_{\max} = \frac{v_0^2 \sin 2\varphi}{g} \quad (6.14-10)$$

The range of fire depends on the initial velocity and on the angle of departure.

For what angle of departure (initial velocity v_0 unchanged) is the range of fire the greatest? Evidently when $\sin 2\varphi = 1$, that is for $\varphi = 45^\circ$.

Let us determine the time during which the shell rises upwards. All we need to do is solve the equation $\frac{dy}{dt} = 0$ because at time t when y attains its maximum value, the shell will cease to rise and will begin to fall. The condition $\frac{dy}{dt} = 0$ yields $v_0 \sin \varphi - gt = 0$,

whence

$$t = \frac{v_0 \sin \varphi}{g} \quad (6.14-11)$$

The total flight time t_{tot} can be found from the fact that the flight stopped when $x = x_{max}$. Using (6.14-8) and (6.14-10), we find

$$v_0 t_{tot} \cos \varphi = \frac{v_0^2 \sin 2\varphi}{g}$$

whence

$$t_{tot} = \frac{2v_0 \sin \varphi}{g} \quad (6.14-12)$$

Comparing (6.14-12) and (6.14-11), we see that the total flight time t_{tot} is twice the time of ascent. The ascent time of a shell is equal to the descent time.

Note, in conclusion, that the actual flight paths of shells are not exact parabolas and the distortion is due to air resistance. The range of fire, altitude of ascent of the shell, flight time and the like depend on the weight of the shell, its shape and the air density.

Exercises

1. A shell leaves the gun with a velocity of 80 m/sec. Determine the range of fire and the maximal height reached by the shell if the angle of departure $\varphi = 30^\circ, 45^\circ, 60^\circ$.

2. Determine the maximum altitude at which a shell with initial velocity $v_0 = 80$ m/sec can hit a target located 500 metres from the gun.

6. 15 THE MASS, CENTRE OF GRAVITY AND MOMENT OF INERTIA OF A ROD

We consider a narrow rod. The x -axis will lie along the rod. Denote by ρ the mass per unit length of rod. Thus, on a portion dx between x and $x + dx$ there will be a mass

$$dm = \rho dx$$

The rod may be made of a material whose density depends on x , or it can have a cross section that is variable in length (that is, dependent on x). Therefore ρ is a function of the x -coordinate. The quantity ρ is a product of the volume density d (g/cm^3) and the cross section of the rod S (cm^2):

$$\rho (\text{g/cm}) = Sd$$

The quantity ρ should be called the density per unit length. However, since the real density d (volume density) does not enter into the subsequent computations, we will call ρ , for short, the density. We consider the thickness of the rod to be small and depict it merely

as a straight line, a line segment of the x -axis. The mass of the rod is clearly

$$m = \int_a^b \rho(x) dx \quad (6.15-1)$$

where a and b are the coordinates of the ends of the rod.

Let the rod be fixed on the x -axis, which is horizontal, the y -axis being directed vertically upwards. The force of gravity acts on the rod (as indicated in Fig. 156 by the arrow) tending to pull the rod downwards.

Imagine that the x -axis is a weight lever. Indicated schematically in the drawing is a prism supporting the x -axis at the origin. The

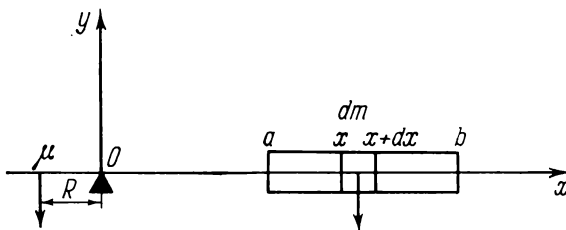


Fig. 156

x -axis can thus rotate about this axis perpendicular to the plane of the drawing. Let us find the weight μ on the left at a distance R that is needed to balance the rod on the right.

By the laws of a lever, the element of mass dm distant x to the right of the axis is balanced by element of mass $d\mu$ to the left if the masses are inversely proportional to the distances, that is, if

$$\frac{d\mu}{dm} = \frac{x}{R}$$

or

$$R d\mu = x dm \quad (6.15-2)$$

The element of mass dm is equal (as we learned above) to ρdx . To balance the entire rod we need a mass μ that satisfies the equation

$$R\mu = \int_a^b x\rho(x) dx \quad (6.15-3)$$

This equation is the result of integrating the left and right members of (6.15-2). To the right of the axis, different elements of mass dm are located at different distances x from the support. That is why the quantity x appears under the integral sign. To the left of the axis, all elements of mass $d\mu$ (which balance the distinct elements

dm of the rod) are collected together at the same distance R from the support. R is a constant and so

$$\int R d\mu = R \int d\mu = R\mu$$

It will be seen, from a comparison of (6.15-3) and (6.15-1), that the integral upon which depends the mass μ balancing the rod differs from the integral that expresses the mass of the rod.

The question now is: if we concentrate the total mass of the rod at one point, then at what distance x_C must this point be from the

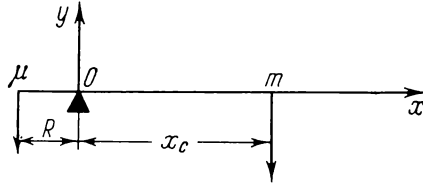


Fig. 157

support (from the origin, that is) in order to balance the same mass μ at distance R that is balanced by the rod (Fig. 157). We find

$$R\mu = x_C m = \int_a^b x\rho dx \quad (6.15-4)$$

whence

$$x_C = \frac{1}{m} \int_a^b x\rho dx = \frac{\int_a^b x\rho dx}{\int_a^b \rho dx} \quad (6.15-5)$$

The quantity x_C is the coordinate of the centre of gravity or, as it is also called, the centre of mass of the rod. It is very important that the point x_C is indeed a definite point of the rod: if we displace the rod as a whole along the x -axis, say, to the right a distance l (Fig. 158), then x_C too will increase by this same quantity l , so that the point with coordinate $x = x_C$ is always (for a given rod) at a very definite distance from the endpoints of the rod. We will prove this.

Let us consider a rod displaced a distance l to the right from the original position (Fig. 158, bottom). The quantities referring to the new (displaced) position will be denoted by the same letters as those referring to the original position, but labelled with the subscript 1. Then

$$a_1 = a + l, \quad b_1 = b + l, \quad \rho_1(x) = \rho(x - l)$$

Note particularly the minus sign in the last formula. Indeed, from Fig. 158 it is evident that to each value of x in the new position of the rod (bottom figure) there corresponds the same value of density

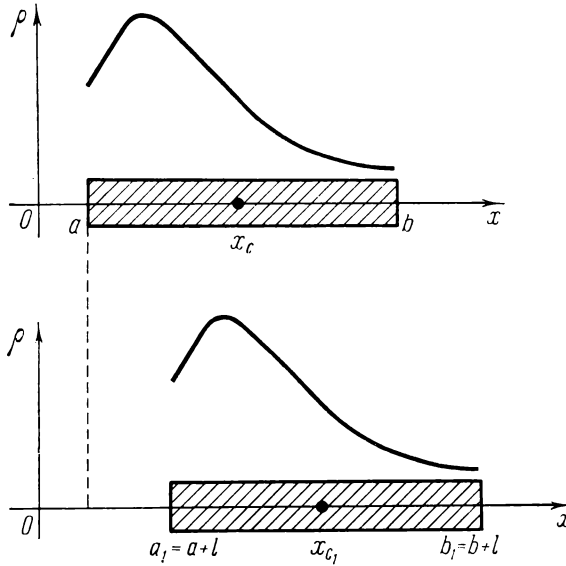


Fig. 158

as to the value $x - l$ in the original position of the rod (top figure). By formula (6.15-5)

$$x_C = \frac{\int_a^b x \rho(x) dx}{\int_a^b \rho(x) dx}, \quad x_{C_1} = \frac{\int_{a_1}^{b_1} x \rho_1(x) dx}{\int_{a_1}^{b_1} \rho_1(x) dx} \quad (6.15-6)$$

Make the change of variables $z = x - l$ or $x = z + l$ in the integrals in the formula for x_C . Whence $dx = dz$, then $\rho_1(x) = \rho(x - l) = \rho(z)$. For $x = a_1 = a + l$ we get $z = a$; for $x = b_1 = b + l$ we get $z = b$. And so

$$\int_{a_1}^{b_1} \rho_1(x) dx = \int_a^b \rho(z) dz = m$$

This means that the mass of the rod is independent of the position of the rod.

Let us examine the integral in the numerator:

$$\int_{a_1}^{b_1} x \rho_1(x) dx = \int_a^b (z+l) \rho(z) dz = \int_a^b z \rho(z) dz + l \int_a^b \rho(z) dz$$

Observe that from the formula (6.15-4)

$$\int_a^b z \rho(z) dz = \int_a^b x \rho(x) dx = x_C m$$

and so

$$\int_{a_1}^{b_1} x \rho_1(x) dx = x_C m + l m$$

Now we find, using (6.15-6), that

$$x_{C_1} = \frac{x_C m + l m}{m} = x_C + l$$

This result was obvious from the start, but the foregoing formal transformations are good exercises in changing variables in a definite integral.

The most convenient thing is to choose the system of coordinates with origin at the centre of gravity of the rod (Fig. 159). The quanti-

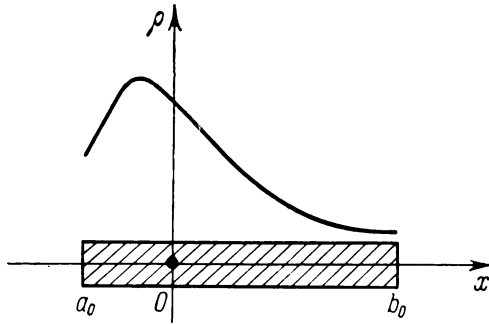


Fig. 159

ties in this coordinate system will be denoted by a zero subscript. It is clear that

$$\int_{a_0}^{b_0} \rho_0(x) dx = m$$

The coordinate of the centre of gravity x_{C_0} is zero in this system and so

$$\int_{a_0}^{b_0} x \rho_0(x) dx = 0 \quad (6.15-7)$$

We will show that for any position of the rod its potential energy in the field of gravity is equal to the potential energy of its entire mass concentrated at the centre of gravity of the rod. We consider the position of the rod as indicated in Fig. 160. The potential energy of an element of rod with mass dm is equal to $gz \, dm$, where z is the altitude and g is the acceleration of gravity. The potential energy u

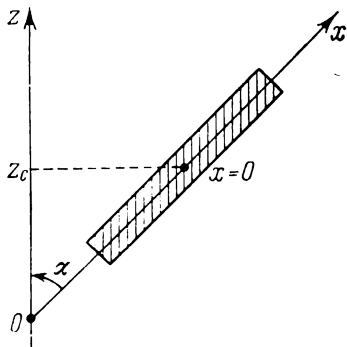


Fig. 160

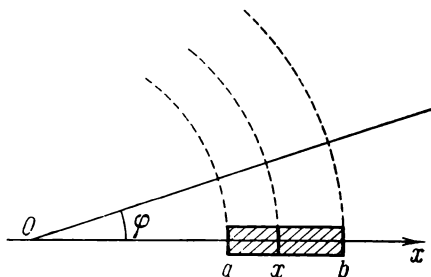


Fig. 161

of the whole rod is found by integrating. For the variable of integration we choose a length reckoned along the rod from its centre of gravity. Then the density at the point x is denoted by $\rho_0(x)$. We express the height z in terms of x . As is evident from Fig. 160, $z(x) = z_C + x \cos \alpha$, where z_C is the height of the centre of gravity of the rod. We get

$$\begin{aligned} u &= \int_{a_0}^{b_0} gz \rho_0(x) \, dx = g \int_{a_0}^{b_0} (z_C + x \cos \alpha) \rho_0(x) \, dx \\ &= gz_C \int_{a_0}^{b_0} \rho_0(x) \, dx + g \cos \alpha \int_{a_0}^{b_0} x \rho_0(x) \, dx = gz_C m \end{aligned}$$

since the second integral is equal to zero by formula (6.15-7). Thus, the potential energy depends only on the mass of the rod and the height of its centre of gravity.

Now let us investigate the so-called moment of inertia. This concept comes up in a consideration of the rotational motion of a rod. Let a rod be in rotation about an axis perpendicular to the plane of the drawing and passing through the origin. Then each point of the rod will describe a circle of radius equal to the abscissa of the given point x in the initial (horizontal) position of the rod (Fig. 161). Denote by ω the angular velocity of rotation expressed in radians per second. This means that during time dt the x -axis rotates through the angle $d\varphi = \omega \, dt$. The arc length traversed by an arbitrarily

chosen point with abscissa x is equal to

$$dl = x d\varphi = x\omega dt$$

hence the linear velocity of every point of the circle is

$$v(x) = \frac{dl}{dt} = \omega x$$

Let us find the kinetic energy of rotation of the whole rod. An element of mass dm distant x from the origin (in the interval dx from x to $x + dx$) has kinetic energy

$$\frac{v^2}{2} dm = \frac{\omega^2 x^2}{2} dm = \frac{\omega^2 x^2}{2} \rho(x) dx$$

Hence the kinetic energy of the whole rod is

$$E = \frac{\omega^2}{2} \int_a^b x^2 \rho(x) dx$$

The integral in this formula is called the moment of inertia of the rod about the axis passing through the origin and is symbolized by I ,

$$I = \int_a^b x^2 \rho(x) dx$$

Thus

$$E = \frac{I\omega^2}{2}$$

The kinetic energy of rotation is expressed in terms of the moment of inertia and the angular velocity in exactly the same way that the kinetic energy of simple translation is expressed in terms of mass and linear velocity,

$$E = \frac{mv^2}{2}$$

We now take up the evaluation of I . For a rod whose centre of gravity lies at the origin, the moment of inertia assumes the value I_0 :

$$I_0 = \int_{a_0}^{b_0} x^2 \rho_0(x) dx \quad (6.15-8)$$

Observe that I_0 is positive since the integrand in (6.15-8) is positive.

Let us determine the moment of inertia of a rod for the case where the centre of gravity is distant l rightwards from the origin, so that $x_{C_1} = l$. In this case

$$a = a_0 + l, \quad b = b_0 + l, \quad \rho(x) = \rho_0(x - l), \quad I = \int_a^b x^2 \rho(x) dx$$

Set $z = x - l$, then $x = z + l$, $dx = dz$. When x varies from a to b , z varies from a_0 to b_0 . Therefore

$$I = \int_{a_0}^{b_0} (z+l)^2 \rho_0(z) dz = l^2 \int_{a_0}^{b_0} \rho_0(z) dz + 2l \int_{a_0}^{b_0} z \rho_0(z) dz + \int_{a_0}^{b_0} z^2 \rho_0(z) dz \quad (6.15-9)$$

Note that $\int_{a_0}^{b_0} \rho_0(z) dz = m$ and the second integral on the right of (6.15-9) is zero by formula (6.15-7), finally, the third integral is I_0 by (6.15-8). Thus, formula (6.15-9) becomes

$$I = ml^2 + I_0 \quad (6.15-10)$$

The quantity ml^2 is clearly the moment of inertia of a point mass distant l from the axis of rotation (from the origin). Thus, the moment of inertia of a rod with respect to rotation about an arbitrary axis

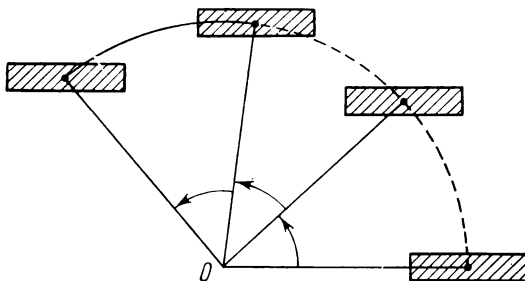


Fig. 162

perpendicular to the rod is equal to the sum of the moment of inertia of the rod with respect to rotation about the centre of gravity and the moment of inertia of a mass equal to the rod mass at a distance from the axis equal to the distance of the centre of gravity of the rod from the axis.

We can picture a rod hinged at the centre of gravity. Then rotation of the axis need not be accompanied by rotation of the rod and we can visualize a motion with successive stages as indicated in Fig. 162. The kinetic energy of such motion is equal to $E' = 1/2 m v_{C_1}^2$, where v_{C_1} is the velocity of the centre of gravity of the rod. But $v_{C_1} = \omega l$ so that $E' = \frac{\omega^2}{2} ml^2$.

The motion we considered earlier (Fig. 161) differs from that of Fig. 162 in that in the former case the rod itself was in rotation with

angular velocity ω about its centre of gravity. For this reason, the kinetic energy of rotation in Fig. 161 turns out to be equal to the sum of the energy of rotation of the type in Fig. 162 and of the energy of rotation about the centre of gravity that is equal to $I_0 \frac{\omega^2}{2}$.

It is evident from the derivation of the formula that such a simple addition of energies in the combination of two motions only results when we consider the motion of the centre of gravity; only then do we find the integral (6.15-7) to be equal to zero.

Exercises

1. Find the moment of inertia about the centre of gravity of a rod of length l with a uniform distribution of mass.

2. A rod is made up of two pieces: one piece of length l_1 has constant density ρ_1 , the other one of length l_2 has a constant but different density ρ_2 . Find the position of the centre of gravity of the rod.

3. Find the position of the centre of gravity and the magnitude of the moment of inertia relative to the centre of gravity for a rod in the form of a thin triangle of length L . Express them in terms of the length L and the mass m of the rod.

Hint. If the x -axis is along the median and the origin is chosen at the vertex of the triangle, then $\rho(x) = ax$, where a is a constant.

6.16 THE OSCILLATIONS OF A SUSPENDED ROD

Consider a rod suspended at the point A (Fig. 163). Let the centre of gravity be below the point of suspension, the distance between the point of suspension and the centre of gravity being l . Such a rod is called a pendulum. Let us determine the period of oscillation.

If the pendulum is deflected from its position of equilibrium through a small angle φ , then its potential energy is

$$u = -mgl \cos \varphi$$

We expand $\cos \varphi$ in a series and, due to the smallness of φ , confine ourselves to the first two terms: $\cos \varphi = 1 - \frac{\varphi^2}{2}$. Therefore

$$u = -mgl \left(1 - \frac{\varphi^2}{2}\right) = -mgl + mgl \frac{\varphi^2}{2}$$

Thus, the increase in potential energy upon deflection of a pendulum through an angle φ from the equilibrium position ($\varphi = 0$) is $\Delta u = mgl \frac{\varphi^2}{2}$.

The kinetic energy of rotation of the rod about the axis is

$$E = I \frac{\omega^2}{2} = I \cdot \frac{1}{2} \left(\frac{d\varphi}{dt} \right)^2$$

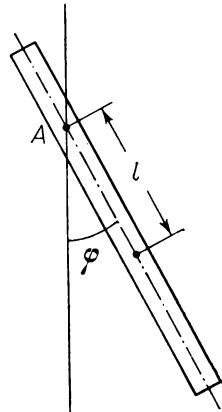


Fig. 163

By formula (6.15-10), $I = ml^2 + I_0$ and so

$$E = \frac{1}{2} (ml^2 + I_0) \left(\frac{d\varphi}{dt} \right)^2$$

where I_0 is the moment of inertia of the pendulum about the centre of gravity.

Suppose that the rod performs harmonic oscillations, that is, $\varphi = a \cos \omega t$. By the law of conservation of energy, $\Delta u_{max} = E_{max}$. Since $\frac{d\varphi}{dt} = -a\omega \sin \omega t$, it follows that

$$E_{max} = \frac{1}{2} (ml^2 + I_0) a^2 \omega^2, \quad \Delta u_{max} = mgl \frac{a^2}{2}$$

whence

$$mgl \frac{a^2}{2} = \frac{1}{2} (ml^2 + I_0) a^2 \omega^2$$

which yields

$$\omega = \sqrt{\frac{mgl}{ml^2 + I_0}}$$

and the period of oscillation is

$$T = \frac{2\pi}{\omega}$$

In particular if the entire mass of the rod is concentrated at its centre of gravity, then $I_0 = 0$. In this case we obtain the ordinary formulas for frequency and period of oscillation of a so-called simple pendulum:

$$\omega = \sqrt{\frac{g}{l}}, \quad T = 2\pi \sqrt{\frac{l}{g}}$$

From the formulas obtained it follows that the larger I_0 is, the lower the frequency and hence the larger the period of oscillation.

If $I_0 \neq 0$, then there is a definite position of the point of suspension for which the frequency is a maximum. Since the position of the point of suspension is characterized by l , then to find the position we desire let us solve the equation $\frac{d\omega}{dl} = 0$. This yields

$$mg(ml^2 + I_0) - mgl \cdot 2ml = 0$$

whence we find

$$l_{max} = \sqrt{\frac{I_0}{m}}$$

For a rod of length L with uniform distribution of mass $I_0 = \frac{mL^2}{12}$ (see Exercises of Sec. 6.15); therefore $l_{max} = \frac{L}{\sqrt{12}} \approx 0.3 L$.

Exercise

1. A pendulum is in the form of a triangular lamina (tin or cardboard) (see Fig. 164). Determine the period of oscillation if the pendulum is suspended

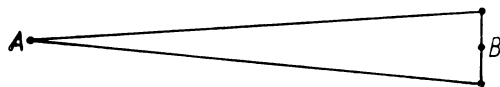


Fig. 164

(a) from the acute end A , (b) from the middle of the base B . In both cases, indicate how the point of suspension is to be displaced so as to obtain a minimum period of oscillation.

Chapter 7

The Thermal Motion of Molecules and the Distribution of Air Density in the Atmosphere

7.1 THE CONDITION FOR EQUILIBRIUM IN THE ATMOSPHERE

Let us consider the question of the law of distribution of air density in the atmosphere in altitude. It is common knowledge that at high altitudes the air is less dense and the air pressure is lower than at sea level. The reason for the dependence of pressure upon altitude is obvious: let us mentally select a cylindrical volume (altitude Δh , base area S , volume $S\Delta h$). The air in this volume (mean density $\bar{\rho}$, mass $m = \bar{\rho}\Delta h S$) is attracted to the earth, i.e., experiences the force of gravity directed downwards and equal to $mg = \bar{\rho}\Delta h Sg$. However, this volume does not fall and is in a state of rest for the reason that at altitude h it is acted upon from below by a pressure $p(h)$ which exceeds the pressure from above at altitude $h + \Delta h$, which pressure is equal to $p(h + \Delta h)$ (Fig. 165). The pressure on the lower base of the cylinder is $S p(h)$; this pressure balances the sum of the pressure on the upper base and the force of gravity:

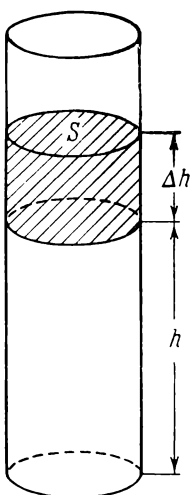


Fig. 165

$$S p(h) = S p(h + \Delta h) + \bar{\rho} \Delta h S g \quad (7.1-1)$$

Formula (7.1-1) can be written as

$$p(h) - p(h + \Delta h) = \bar{\rho} \Delta h g \quad (7.1-2)$$

We will assume that Δh is very small. Then there is no need to speak of the mean density $\bar{\rho}$ since the altitudes h and $h + \Delta h$ are almost the same and $\bar{\rho}$ is hardly at all different from $\rho(h)$. Therefore, (7.1-2) assumes the form

$$\frac{dp}{dh} = -g\rho \quad (7.1-3)$$

We have obtained a differential equation for the function $p(h)$. This equation involves the air density ρ .

The quantities p and ρ are connected by the law of Boyle-Mariotte. We will assume that the temperature of the atmosphere is the same at all altitudes. Actually, the air temperature depends on the heat flux from the sun and the removal of heat due mainly to heat radiation by the air into outer space or, to be more exact, by the water vapour and carbon dioxide in the air. A small portion of the solar radiation is absorbed by the upper rarified layers of the air. A large portion of the energy of solar radiation reaches the earth's surface and is absorbed by the ground, which in turn heats the air. This explains the actually rather complicated distribution of temperature in the atmosphere: at ground level the temperature is known to fluctuate roughly between -40° and $+40^\circ \text{C}$ depending on the geographical location and time of year; at an altitude of about 15 km the temperature is minimal (about -80°C) and is approximately the same both summer and winter round the globe. At considerable altitudes the air temperature increases reaching $+60^\circ \text{C}$ to $+75^\circ \text{C}$ at altitudes between 50 and 60 km.

Recent measurements by means of artificial earth satellites show that at altitudes of 300 to 1000 km the air density is slight but still is greater than was earlier thought. As we will see later on, high air density indicates a very high temperature of the air in these upper strata. What is more, a substantial portion of the molecules of oxygen and nitrogen break up at these altitudes into atoms, ions and electrons.

If there were no influx of heat from without or any removal of heat, that is, if we were to consider a heat-insulated column of air, then the temperature throughout the column would eventually even out. Below we will consider precisely this kind of idealized case of total equilibrium, both thermal and mechanical. Thermal equilibrium means that the temperature is everywhere the same and so there are no fluxes of heat (if the temperature differed at distinct points in the air column, the hotter points would move to the cooler ones, with a resultant flow of heat). Mechanical equilibrium consists in the resultant of all forces acting on any volume of air chosen in the atmosphere being equal to zero. Here we have to consider the force of gravity of the air in the volume and the pressure on the entire surface bounding the given volume.

For the pressure distribution that satisfies equation (7.1-3), the atmosphere can be in a state of rest.

Since we consider altitudes h small by comparison with the earth's radius, g (the acceleration of gravity) can be regarded as constant.

7.2 THE RELATIONSHIP BETWEEN DENSITY AND PRESSURE

By the law of Boyle-Mariotte the product of the pressure and volume of a gas is constant for a given mass m_0 of the gas and for a given temperature:

$$pv = a$$

where a is a constant. Denoting the gas density by ρ , we get $m_0 = v\rho$. Hence, $v = m_0/\rho$ and since $p = a/v$, it follows that

$$p = b\rho \quad (7.2-1)$$

where we put $b = a/m_0$. Thus the gas pressure is directly proportional to the density.

It is easy to find the constant of proportionality for air at room temperature. We know that the air pressure at sea level, p_0 , is roughly equal to $1 \text{ kgf/cm}^2 = 10^6 \text{ dynes/cm}^2$. The air density ρ_0 at pressure p_0 is equal* approximately to $1.3 \times 10^{-3} \text{ g/cm}^3$. Substituting p_0 and ρ_0 into formula (7.2-1), we get $p_0 = b\rho_0$ whence

$$b = \frac{10^6}{1.3 \times 10^{-3}} = 7.7 \times 10^8$$

Observe that b has the dimensions of the square of the velocity. Actually, this quantity is closely connected with the velocity of molecules and of sound: the square of the speed of sound is equal to $1.4 b$ (we will not derive this relation).

Later on we will need not only the numerical value of b for air at room temperature but also the general expression of the constant b for any gas and any temperature. To this end, we take advantage of the Clapeyron law,

$$pV = RT \quad (7.2-2)$$

where V is the volume occupied by one gram-molecule** of gas, T is the absolute temperature (reckoned from absolute zero -273°C^{***}), R is the so-called universal gas constant. We know that at 0°C (equal to 273° on the absolute scale) and at atmospheric pressure, i.e., when $p_0 = 10^6 \text{ dynes/cm}^2$, one gram-molecule of gas occupies a volume equal to 22.4 litres, or $2.24 \times 10^4 \text{ cm}^3$ (Avogadro's law),**** whence $10^6 \times 2.24 \times 10^4 = R \cdot 273$. We then have

$$R = 8.3 \times 10^7 \frac{\text{dyne} \cdot \text{cm}^3}{\text{mole} \cdot \text{cm}^2 \cdot \text{deg}} = 8.3 \times 10^7 \frac{\text{erg}}{\text{deg} \cdot \text{mole}} \text{ or erg mole}^{-1} \text{ deg}^{-1}$$

We denote the molecular weight of the gas by M . For hydrogen H_2 , $M = 2$, for helium He , $M = 4$, for nitrogen N_2 , $M = 28$, for air the mean value is $M = 29.4$. By definition, V contains M grams

* This quantity can easily be found experimentally by weighing. A hermetically sealed vessel of known volume is weighed with and without air (a vacuum pump is used to evacuate the vessel).

** One gram-molecule of a gas is called, for short, one mole.

*** Ordinarily, temperatures on the absolute scale are measured in degrees Kelvin, $^\circ\text{K}$, after the English scientist [Lord Kelvin: $20^\circ\text{C} = 293^\circ\text{K}$ (read: "20 degrees Celsius is equal to 293 degrees Kelvin").

**** Here, we ignore small differences between 1 atmosphere and 1 kgf/cm^2 and between 1 kgf and 10^6 dynes .

of gas. Hence, the density ρ is connected with V by the relation

$$\rho = \frac{M}{V} \text{ or } V = \frac{M}{\rho}$$

Substituting this expression for V into (7.2-2), we get

$$p = \rho \frac{RT}{M} \quad (7.2-3)$$

Comparing (7.2-3) and (7.2-1), we find

$$b = \frac{RT}{M} \quad (7.2-4)$$

Finally, let us express the pressure in terms of the number of molecules n contained in a unit volume of gas. We know that one gram-molecule of any substance contains 6×10^{23} molecules. This quantity is called the Avogadro number and is denoted by A . Thus, a mass of one molecule

$$m = \frac{M}{A} = \frac{1}{6 \times 10^{23}} M \quad (7.2-5)$$

If one gram-molecule occupies a volume V , then the number n of molecules in unit volume is $n = \frac{A}{V}$. The gas density $\rho = nm$. Clapeyron's law (7.2-2) yields

$$p = n \frac{RT}{A} = nkT$$

where k is the Boltzmann constant:

$$k = \frac{R}{A} = \frac{8.3 \cdot 10^7}{6 \cdot 10^{23}} = 1.38 \cdot 10^{-16} \text{ erg/deg}$$

The quantity R refers to a conventionally chosen amount of substance, one gram-molecule, and so the dimensions of R involve the mole. The quantity k refers to one molecule, and so k has the dimensions of erg/deg. The quantity kT has the dimensions of energy (erg). In Sec. 7.4 it will be shown that in the atmosphere the quantity kT is equal to the mean potential energy of one molecule in the field of gravity at temperature T . The mean kinetic energy of translational motion of one molecule is $\frac{3}{2} kT$.

7.3 DENSITY DISTRIBUTION

From formula (7.2-4), we find $\rho = \frac{p}{b}$. Putting this into the differential equation for air density (7.1-3), we get

$$\frac{dp}{dh} = -\frac{g}{b} p$$

The solution of this equation is $p = Ce^{-\frac{g}{b}h}$, where C is determined from the initial condition. Let $p = p_0$ for $h = 0$, then

$$p = p_0 e^{-\frac{g}{b}h} \quad (7.3-1)$$

Dividing (7.3-1) by b we get

$$\rho = \rho_0 e^{-\frac{g}{b}h} \quad (7.3-2)$$

where ρ_0 is the air density at $h = 0$ (sea level). From formula (7.3-1) it is evident that at altitude $H = \frac{b}{g}$ above sea level the air pressure diminishes by a factor of e .

We obtain a formula relating H and kT : $H = \frac{b}{g}$. Using (7.2-4) and (7.2-5), we find $H = \frac{RT}{Amg}$, whence

$$H = \frac{kT}{mg} \quad (7.3-3)$$

Let us compute H using the formula $H = \frac{b}{g}$:

$$H = \frac{7.7 \cdot 10^8 \text{ cm}^2/\text{sec}^2}{10^3 \text{ cm}/\text{sec}^2} = 7.7 \cdot 10^5 \text{ cm} = 7.7 \text{ km}$$

Using H , we can write formulas (7.3-1) and (7.3-2) as

$$p = p_0 e^{-\frac{h}{H}}, \quad \rho = \rho_0 e^{-\frac{h}{H}}$$

If the altitude increases in arithmetic progression, the pressure and density fall in geometric progression:

if $h = 0$, then $p = p_0$, $\rho = \rho_0$;

if $h = H$, then $p = \frac{p_0}{e} = 0.368p_0$, $\rho = 0.368\rho_0$;

if $h = 2H$, then $p = \frac{p_0}{e^2} = 0.135 p_0$, $\rho = 0.135\rho_0$;

if $h = 3H$, then $p = \frac{p_0}{e^3} = 0.05 p_0$, $\rho = 0.05\rho_0$.

Knowing the dependence of density on altitude, we can express the total mass of air m_a in a column with base area 1 cm^2 in terms of ρ_0 and H . Indeed,

$$m_a = \int_0^\infty \rho \, dh = \int_0^\infty \rho_0 e^{-\frac{h}{H}} \, dh$$

Make the change of variable $z = \frac{h}{H}$; then $dz = \frac{1}{H} \, dh$,

$$m_a = \rho_0 H \int_0^\infty e^{-z} \, dz = -\rho_0 H e^{-z} \Big|_0^\infty = \rho_0 H$$

Using the relation $m_a = \rho_0 H$ we can compute H once again (by way of a check). Since the atmospheric pressure is 1 kgf/cm^2 , the mass of air in a column with base area 1 cm^2 is precisely equal to 1 kg . Thus, $m_a = 1 \text{ kg/cm}^2 = 1000 \text{ g/cm}^2$. Knowing $\rho_0 = 1.3 \times 10^{-3} \text{ g/cm}^3$ we then get

$$H = \frac{m_a}{\rho_0} = \frac{10^3}{1.3 \cdot 10^{-3}} = 7.7 \cdot 10^5 \text{ cm}$$

in accord with the earlier computation.

We will find the mean altitude at which the air is located, that is, the altitude of the centre of gravity of a vertical cylindrical column of air. So as not to introduce extra quantities, we consider a column of air with base 1 cm^2 , although it is clear that the altitude of the centre of gravity is independent of the base of the cylinder. At a height h between h and $h + dh$ is a mass $dm = \rho dh$. The mean altitude is equal to

$$\bar{h} = \frac{\int_0^\infty h \, dm}{m_a} = \frac{\int_0^\infty h \rho(h) \, dh}{\int_0^\infty \rho(h) \, dh}$$

Let us find the integral in the numerator. Using formula (7.3-3), we get

$$\int_0^\infty h \rho(h) \, dh = \int_0^\infty h \rho_0 e^{-\frac{h}{H}} \, dh = \rho_0 H^2 \int_0^\infty z e^{-z} \, dz = \rho_0 H^2$$

$\left[\int_0^\infty z e^{-z} \, dz = 1, \text{ see formula (4.3-8), Chapter 4} \right]$. Finally we have

$$\bar{h} = \frac{\rho_0 H^2}{\rho_0 H} = H \quad (7.3-4)$$

Thus, the altitude H at which the density and the pressure of the air diminish e -fold is at the same time the *mean* altitude at which the air is located.

A similar result was obtained earlier when we considered radioactive decay (Sec. 5.3). If the probability of decay is equal to $\omega \frac{1}{\text{sec}}$, $\frac{dn}{dt} = -\omega n$, $n = n_0 e^{-\omega t}$, then during time $\tau = \frac{1}{\omega}$ the amount of radioactive substance decreases by a factor of e , and the mean life-time of a radioactive atom is equal to the same quantity: $\bar{t} = \tau = \frac{1}{\omega}$.

Remember that the simple dependence of density and pressure on altitude, (7.3-3), refers to the case of a constant temperature. Actually, the distribution of density and pressure departs somewhat from formula (7.3-3) and depends on the time of year and other factors.

Exercises

1. Find the air pressure in a mine at the following depths: 1 km, 3 km, 10 km.
 2. Find the dependences of air pressure on altitude for air temperature equal to -40°C and $+40^\circ\text{C}$.

3. Suppose the air temperature varies with altitude by the law $\frac{dT}{dh} = -\alpha T_0$, where T_0 is the air temperature at ground level and α is a constant coefficient. Find the air pressure as a function of altitude.

4. We know that under the conditions of Problem 3 the quantity $\alpha \approx 0.07 \times 10^{-5} \text{ cm}^{-1}$. Using the result of Problem 3, determine the air pressure in a mine at depths of 1 km, 3 km, 10 km. The temperature at ground level is taken to be zero. Compare the results with those of Problem 1.

7.4 THE MOLECULAR KINETIC THEORY OF DENSITY DISTRIBUTION

In the preceding sections we found the distribution of air density in altitude under the action of gravity in a state of equilibrium. We regarded the air as a continuous medium with a given dependence of pressure on density.

Now let us take that result and approach it from a different angle, namely, the viewpoint of molecular theory. We will consider the separate molecules and their motion. The idea that matter consists of individual atoms was first expressed in ancient Greece. However, the *motion* of the molecules and its connection with heat was first examined by the great Russian scholar M. V. Lomonosov, who is thus the founder of the molecular kinetic theory.

The gaseous state differs from the liquid and solid states in that in a gas the molecules may be regarded as independent and noninteracting. The motion of molecules in a gas is that of free flight by inertia. From time to time the molecules collide. Under ordinary conditions, such collisions occur with extreme frequency and the path lengths which molecules traverse between collisions are extremely small.

At atmospheric pressure and a temperature of 0°C , 22.4 litres of gas comprise 1 gram-molecule or 6×10^{23} molecules; 1 cm^3 of gas contains $n = \frac{6 \times 10^{23}}{2.24 \times 10^4} = 2.7 \times 10^{19}$ molecules.

For our crude purposes we will regard molecules as spheres of radius about $2 \times 10^{-8} \text{ cm}$.^{*} Then for a collision between two molecules to take place it is necessary that the trajectory of the centre of one molecule hit a target of radius $4 \times 10^{-8} \text{ cm}$ about the centre of the other molecule. The area of such a target is $\sigma = \pi r^2 \approx 5 \times 10^{-15} \text{ cm}^2$. This means that over a path length of 1 cm a given molecule will collide with all molecules whose centres lie in a cylinder with base area $5 \times 10^{-15} \text{ cm}^2$ and altitude 1 cm. The volume of

^{*} In reality, diatomic molecules, say of oxygen or nitrogen, are more like pairs of merged spheres, something reminiscent of peanuts (two nuts to a shell).

such a cylinder is equal to $\sigma \text{ cm}^3$, and the number of molecules in it is $n\sigma$, where n is the number of molecules in 1 cm^3 .

Thus, a molecule experiences $n\sigma$ collisions over a path length of 1 cm. Therefore, the mean distance of free flight between collisions is

$$l = \frac{1}{n\sigma} = 0.7 \cdot 10^{-5} \text{ cm}$$

This quantity is known as the free path length.

Because of collisions, a molecule moves in a polygonal line, but the volume of a cylinder formed from polygonal lines does not differ from the volume of a right cylinder and so our computations remain valid.

Actually, one has also to consider the motion of those molecules that are hit in collisions. It can be proved that this circumstance changes but slightly the free path length of a molecule reducing it by a factor of only 1.5.

Molecules have velocities of the order of 300 to 500 m/sec. Hence the free path time (which is the mean time between collisions) is of the order of

$$\tau = \frac{0.5 \cdot 10^{-5}}{4 \cdot 10^4} \approx 10^{-10} \text{ sec}$$

At first glance, the quantities $l \approx 10^{-5} \text{ cm}$ and $\tau \approx 10^{-10} \text{ sec}$ are extremely small. But they have to be compared with the size of a molecule whose radius is $r \approx 2 \times 10^{-8} \text{ cm}$ and with the duration of the collision itself, which is less than $\frac{r}{v} \approx 10^{-13} \text{ sec}$. If we do that it will be apparent that the molecules of a gas collide *very rarely*. At atmospheric pressure, the molecules of the air spend 99.9% of the time in free flight and only 0.1% of the time in a state of collision.

Molecular collisions in a gas do not affect the pressure of the gas and do not influence the law of distribution of density of the gas in the atmosphere. Confirmation of this fact lies in the laws of Boyle-Mariotte and Clapeyron. In Sec. 7.2 these laws are written as $p = nkT$.

The gas pressure depends on the number of molecules in unit volume, but the radius r of the molecules and their cross section σ do not enter into the formula. This means that the quantities r and σ cannot enter into the formula for the density distribution in altitude.

Let us rewrite the formula for density distribution (7.3-2) and express b in terms of molecular quantities. Since $b = \frac{RT}{M} = \frac{AkT}{Am} = \frac{kT}{m}$, it follows that

$$\rho = \rho_0 e^{-\frac{gh}{b}} = \rho_0 e^{-\frac{mgh}{kT}} \quad (7.4-1)$$

Divide both sides of (7.4-1) by m , where m denotes the mass of one molecule. Note that $\frac{\rho}{m} = n$ is the number of molecules in unit volume at height h and $\frac{\rho_0}{m} = n_0$ is the number of molecules in unit volume at sea level. The formula (7.4-1) assumes the form

$$n = n_0 e^{-\frac{mgh}{kT}} \quad (7.4-2)$$

The quantity mgh is the potential energy of a molecule of mass m located at altitude h if for zero we take the potential energy of a molecule at sea level. The potential energy $u(0)$ of a molecule at sea level can be chosen arbitrarily (see Sec. 6.2). Then

$$(h) = u(0) + mgh$$

whence

$$mgh = u(h) - u(0)$$

Formula (7.4-2) may be written as

$$n(h) = n(0) e^{-\frac{u(h) - u(0)}{kT}}$$

This is the law of distribution in altitude of the number of molecules. We can write it like this:

$$n(h) = Be^{-\frac{u(h)}{kT}}$$

where B is a constant defined from the value of density at sea level ($h = 0$),

$$n(0) = Be^{-\frac{u(0)}{kT}}$$

A remarkable fact is that the density of molecules at a certain altitude is only dependent on the potential energy of the molecules at the given site: the mass m of a molecule, the acceleration g of gravity and the altitude h entered into the formula (7.4-2) in exactly the same combination (mgh) as they entered into the expression for the potential energy u .

Let us find the mean value of the potential energy of a molecule: $\bar{u} = \overline{mgh} = m\bar{gh} = mgH$ [see formula (7.3-4)]. Using formula (7.3-3), we get

$$\bar{u} = mgH = mg \frac{kT}{mg} = kT$$

Thus, the mean potential energy of a single molecule is kT .

We have established that the distribution of air molecules in the atmosphere depends on the temperature and on the potential energy of the molecules. But for a given mean potential energy equal to kT , we get a *definite* distribution of molecules in potential energy. Part

of the molecules—those below altitude H —have a potential energy less than kT . Let us find the ratio of the number of such molecules to the total number of molecules. This ratio is

$$\frac{\int_0^H n dh}{\int_0^\infty n dh} = \frac{n_0 \int_0^H e^{-\frac{mgh}{kT}} dh}{n_0 \int_0^\infty e^{-\frac{mgh}{kT}} dh}$$

Let us evaluate these integrals:

$$\int_0^H e^{-\frac{mgh}{kT}} dh = -\frac{kT}{mg} e^{-\frac{mgh}{kT}} \Big|_0^H = \frac{kT}{mg} \left(1 - e^{-\frac{mgH}{kT}} \right) = \frac{kT}{mg} (1 - e^{-1})$$

$$\int_0^\infty e^{-\frac{mgh}{kT}} dh = -\frac{kT}{mg} e^{-\frac{mgh}{kT}} \Big|_0^\infty = \frac{kT}{mg}$$

Therefore

$$\frac{\int_0^H n dh}{\int_0^\infty n dh} = \frac{\frac{kT}{mg} (1 - e^{-1})}{\frac{kT}{mg}} = 1 - e^{-1} \approx 0.63$$

To summarize, then, 63 % of all the molecules have a potential energy less than the mean energy, and 37 % have a potential energy exceeding the mean value. It is now easy to calculate that 14 % of all molecules have a potential energy exceeding $2kT$, 5 % of all molecules, exceeding $3kT$, and so on. Generally speaking, the portion of molecules whose potential energy is greater than a given value u is equal to $e^{-\frac{u}{kT}}$.

7.5 THE BROWNIAN MOVEMENT AND KINETIC-ENERGY DISTRIBUTION OF MOLECULES

Over a hundred years ago, an English botanist Robert Brown observed the random movement of microscopic particles suspended in a fluid medium. Einstein advanced the idea that this movement of particles is due to their thermal agitation. From this the conclusion was drawn that, for one thing, the particles would not all lie on the bottom of a vessel but would be distributed in height by the same law as the distribution of molecules.

If a suspended particle has the shape of a sphere of diameter $d = 5 \times 10^{-5}$ cm, then its volume is $\frac{\pi d^3}{6} \approx 6.5 \times 10^{-14}$ cm³, and with density $\rho = 1$ g/cm³ the mass of a particle is equal to $6.5 \times$

$\times 10^{-14}$ g. At room temperature, $T = 17^\circ\text{C} = 290^\circ\text{K}$, such particles are distributed in height [by formula (7.4-2)] in accordance with the law

$$n = n_0 e^{-\frac{6.5 \cdot 10^{-14} \cdot 981}{290 \cdot 1.38 \cdot 10^{-16}} h}$$

or $n = n_0 e^{-1.6 \times 10^{13} h}$. Thus, the number of particles per unit volume decreases by a factor e when the altitude is increased by $\frac{1}{1.6 \times 10^3}$ cm $= 0.62 \times 10^{-3}$ cm.

By observing the distribution, in altitude, of particles of known size and density, it is possible to obtain the Boltzmann constant k . On the other hand, the Clapeyron law yields a magnitude of $R = kA$, after which we can find the Avogadro number. This work was carried out by Einstein and Perrin in 1903-1907 and served as a crucial experimental corroboration of the entire atomic-molecular theory and played a tremendous part in the development of physics.

There is a constant transformation of energy taking place when molecules move under the force of gravity: if a molecule is moving



Fig. 166



Fig. 167



Fig. 168

downwards at a given time, then potential energy is being converted into kinetic energy; but if a molecule is in motion upwards, then kinetic energy is being converted into potential energy. When a gas is in a state of equilibrium, that is, the pressure of the gas is balanced by gravity, then the gas molecules are actually moving at random with high speeds. However, if we picture to ourselves a horizontal plane in the gas, then the number of molecules passing through it in unit time upwards is equal to the number of molecules passing through the plane in the downward direction, so that, on the average, the gas is at rest. In the equilibrium state, the transition of kinetic energy into potential energy and the transition of potential energy into kinetic energy balance since the number of molecules moving up equal the number moving down.

It is to be noted that in random motion the individual (identical) molecules have different velocities, i.e., different kinetic energies. This is true because if two balls having identical speeds collide at an angle, the velocities of the balls after the collision may differ. Let us examine Figs. 166 to 168. Illustrated is a collision after which one of the balls (on the left) is brought to a halt, while the other

one, moving upwards, shoots off with double energy (Fig. 166, prior to collision, Fig. 167, collision, Fig. 168, after collision). Note the positions of the balls at the instant of collision; if the second one were, at collision, located below the first one, then it would stop and give up all its energy to the first ball.

Since in molecular motion there is a mutual conversion of kinetic and potential energy, it is natural to suppose that the distribution of the molecules as to kinetic energy is similar to that with respect to potential energy.

We give without proof the results of computations carried out at the end of last century by Maxwell and Boltzmann. The number of molecules having velocity components

along the x -axis between v_x and $v_x + dv_x$,

along the y -axis between v_y and $v_y + dv_y$,

along the z -axis between v_z and $v_z + dv_z$

is equal to

$$dn = \frac{n_0}{\left(\frac{2\pi kT}{m}\right)^{\frac{3}{2}}} e^{-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2kT}} dv_x dv_y dv_z \quad (7.5-1)$$

where n_0 is the total number of molecules and m is the mass of one molecule. Observe that $v_x^2 + v_y^2 + v_z^2 = v^2$, where v is the velocity of a molecule. Therefore, (7.5-1) has, in the exponent, the quantity $\frac{mv^2}{2} : kT$, which is the ratio of kinetic to potential energy. The mean kinetic energy calculated on the basis of (7.5-1) turned out equal to $\frac{3}{2} kT$. For the number of molecules n whose kinetic energy exceeds the given value E we have a rather unwieldy relationship. True, this complicated relationship can approximately be described by the simple formula

$$n = n_0 e^{-\frac{E}{kT}} \quad (7.5-2)$$

The law (7.5-2) yields an incorrect value for the mean kinetic energy of the molecules:

$$\bar{E}_{kin} = \frac{\int_0^\infty n dE}{n_0} = \int_0^\infty e^{-\frac{E}{kT}} dE = kT$$

instead of $\frac{3}{2} kT$. This law gives perceptible departures from the true value if E is of the order of kT . However, when $E \gg kT$, the divergence between the exact and the approximate law is not essential.

It will be noted that for the same temperature, molecules with different masses have the same mean kinetic energies and have the same distribution as to magnitude of kinetic energy, since the mean velocity of a molecule is proportional to $\frac{1}{\sqrt{m}}$, where m is the mass of a molecule.

Considering the collisions of molecules against the walls of the containing vessel, we can find the gas pressure to be

$$p = \frac{2}{3} n_0 \bar{E}_{kin}$$

Putting $\bar{E}_{kin} = \frac{3}{2} kT$, we get Clapeyron's law

$$p = n_0 kT$$

Mutual collisions of molecules give rise not only to an exchange of kinetic energy between the molecules, but also to a conversion of the kinetic energy of motion of the molecules into the energy of rotation of a molecule and into the energy of vibrations of the atoms of the molecule, which is to say, into the internal energy of the molecule. The converse process is also possible: in a collision, part of the internal energy of molecules is transformed into kinetic energy. It is therefore natural that the distribution of molecules as to their internal energy W also obeys the law of proportionality to the quantity $e^{-\frac{W}{kT}}$. The fact that the number of particles having a given energy is an exponential function of the magnitude of the energy is an all-embracing universal law of nature.

7.6 RATES OF CHEMICAL REACTIONS

Of what use is the law of distribution of molecules as to kinetic energy? Such important characteristics of a gas as the pressure it exerts on the walls of the containing vessel, its heat capacity, the total reserve of energy in the volume of the gas are defined by mean quantities, which is to say they are defined by the bulk of the molecules whose energy is close to the mean value. For example, why do we have to know that a minute portion (of the order of 0.00001 %) of the molecules have kinetic energy exceeding $17 kT$? We don't for the simple reason that these separate molecules with very large energies have practically no perceptible effect on the pressure and the general supply of energy of the gas.

However, the picture changes drastically if we consider chemical reactions. It turns out that precisely these rare molecules with high energy completely determine the course of chemical reactions. The mystery of chemical reactions stems from the fact that molecules entering into a reaction collide every 10^{-10} sec whereas a reaction

frequently requires several minutes (sometimes hours). Which means that only an extremely small portion of all collisions result in a chemical reaction.

The idea was advanced that molecules have a certain very small "sensitive spot" that must be touched in order for a reaction to occur. This is reminiscent of the Greek hero Achilles who was vulnerable only in the heel.

A proper explanation was finally given at the end of the 19th century by the Swedish scientist Svanté Arrhenius. It is this: reactions are initiated only by collisions of molecules whose energy exceeds a definite value, the so-called activation energy, E_A .

For instance, when molecules of hydrogen and iodine collide, they form two molecules of hydrogen iodide HI , the energy of the colliding molecules must exceed 3×10^{-12} erg. Compare this with the fact that at 0°C the magnitude of $kT = 1.38 \times 10^{-16} \times 273 \approx 3.8 \times 10^{-14}$ erg. This means that at room temperature only a minute fraction of the molecules possess the needed energy: $\alpha = e^{-\nu}$, where $\nu = 3 \times 10^{-12} / 3.8 \times 10^{-14} \approx 80$, whence we get $\alpha = \frac{1}{10^{35}}$.

We get the reaction time by multiplying the time between collisions (it is of the order of 10^{-10} sec) by the mean number of collisions, among which there will be one collision involving the required energy. This mean number of collisions is of the order of $\frac{1}{\alpha} = 10^{35}$. We obtain the reaction time at 0°C of the order of 10^{25} sec $\approx 3 \times 10^{17}$ years. This result accords with the fact that at 0°C the reaction $\text{H}_2 + \text{I}_2 = 2\text{HI}$ is practically unobservable.

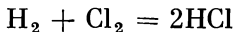
From the reasoning given above it follows that, depending on the temperature, the reaction time is expressed by the formula

$$t = \tau e^{\frac{E_A}{kT}}$$

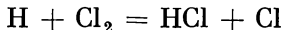
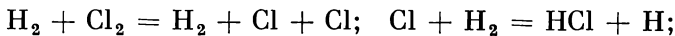
where τ is the time between two collisions, E_A is the activation energy. This formula gives a true picture of the dependence of the rate of chemical reactions on the temperature. A characteristic feature of this formula is the extremely sharp decrease in reaction time and increase in reaction rate for slight variations in temperature.

However, it frequently happens that chemical reactions are much more involved because they may proceed via diverse intermediate stages. The Soviet scientist, Academician N. N. Semenov has made a thorough investigation of complex (chain) chemical reactions and has elucidated the laws governing the course of such reactions and the general causes that lead to such complicated reaction schemes.

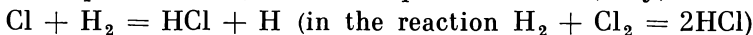
By way of an illustration, let us examine the reaction



This reaction does not proceed via collisions of molecules of hydrogen and a molecule of chlorine, but by the scheme



As a result the actually observed reaction rate involves complicated relationships. However, for each separate reaction, say, for



the Arrhenius law holds true, and the reaction rate is proportional to $e^{-\frac{E_A}{kT}}$, the activation energy E_A having different values for each reaction.

7.7 EVAPORATION. THE EMISSION CURRENT OF A CATHODE

The idea of Svanté Arrhenius concerning the role of a small number of molecules whose energy greatly exceeds the mean value of energy is helpful in analyzing not only chemical reactions but also a series of other phenomena including the evaporation of a liquid.

Evaporation requires the expenditure of a considerable amount of energy. For example, the evaporation of 1 gram of water at 100 °C requires the consumption of about 540 calories.* Per molecule, this comes out to**

$$Q = \frac{18 \times 540 \times 4.18 \times 10^7}{6 \times 10^{23}} \approx 7 \times 10^{-13} \text{ erg at } T = 0^\circ \text{C} = 273^\circ \text{K}, \quad kT = 3.8 \times 10^{-14}; \text{ therefore } \frac{Q}{kT} \approx 20. \text{ Only those}$$

molecules can tear away from the surface of the liquid and evaporate whose energy exceeds the evaporation heat Q . The portion of such

molecules is equal to $e^{-\frac{Q}{kT}}$. Therefore the rate of evaporation is also

proportional to $e^{-\frac{Q}{kT}}$. For computational convenience it is common practice to multiply the numerator and the denominator of the expression $\frac{Q}{kT}$ by the Avogadro number A :

$$\frac{Q}{kT} = \frac{QA}{kAT} = \frac{QA}{RT}$$

The quantity QA is the evaporation heat of 6×10^{23} molecules, which is to say the evaporation heat of one gram-molecule. The

* The evaporation heat is but slightly dependent on temperature; for water, $Q = 540 \text{ cal/g}$ at 100 °C and 600 cal/g at 0 °C. We henceforth disregard this relation.

** Water has a molecular weight of 18, the Avogadro number is 6×10^{23} , 1 cal = 4.18×10^7 ergs.

quantity $kA = R$ is the (universal) gas constant. In thermal units (small calories per gram-mole)

$$R = \frac{8.3 \cdot 10^7}{4.18 \cdot 10^7} \approx 2 \frac{\text{cal}}{\text{deg} \cdot \text{mole}}$$

The evaporation heat of one mole of water is equal to

$$Q_m = 18.540 \approx 10,000 \text{ cal/mole}$$

Thus, the rate of the evaporation of water is proportional to

$$e^{-\frac{10,000}{2T}} = e^{-\frac{5000}{T}}$$

Let us consider the saturated vapour above a water surface. If the vapour is saturated, the number of molecules of water evaporating per unit time is equal to the number of molecules in the vapour and adhering to the surface of the water (condensing) in unit time. The rate of evaporation is

$$Ce^{-\frac{Q_m}{RT}}$$

where C is a constant proportional to the area of the water surface. The rate of condensation is proportional to the pressure of water vapour and is also proportional to the area of the surface. Hence, in the case of saturated vapour, when the rates of evaporation and condensation are equal,

$$Dp = Ce^{-\frac{Q_m}{RT}}$$

where D and C are quantities proportional to the area of the surface and only slightly dependent on the temperature and totally independent of the pressure, whence

$$p = Fe^{-\frac{Q_m}{RT}}$$

where the constant F does not depend on the surface area of the water. Thus a relationship is established between the pressure of saturated vapour and evaporation heat.

Let us consider yet another process similar to evaporation, that of the emission of electrons from a heated surface. This process occurs on the cathode of an electron tube. A cold cathode in vacuo does not emit electrons.* But at high temperature the cathode does emit electrons. Then if the anode (also called plate) has a sufficiently high positive potential, it will attract the electrons and each electron torn out of the surface of the cathode will fall onto the anode. The electric current flowing in a circuit through an electron tube is equal

* Here we do not consider the case of a very strong electric field (10^6 V/cm and more) capable of tearing electrons even out of a cold cathode. Neither do we discuss the ejection of electrons from a cathode by the action of light or bombardment of a cathode by electrons, ions or other particles.

to the product of the number of electrons released by the cathode in unit time into the magnitude of the charge of a single electron.

Experiment shows that in these conditions the following relationship exists between current, j , and temperature:

$$j = ge^{-\frac{Q}{kT}}$$

Q differs for different cathodes. For instance, for a cathode made of pure tungsten, $\frac{Q}{k} = 55\,000\text{ }^{\circ}\text{C}$, for a barium oxide cathode, $\frac{Q}{k} = 30\,000\text{ }^{\circ}\text{C}$ and, hence, such a cathode can operate at lower temperatures. Using the dependence of j on T , we can determine $\frac{Q}{k}$. Here, the quantity Q which enters the latter formula coincides with the energy necessary for tearing an electron out of the cathode (this electron-ejection energy can also be determined in other ways).

An electron tube offers a marvelous method for measuring the distribution of electrons leaving the surface of a cathode in accordance with their speeds at a given temperature. When the cathode is heated, we will impress a small negative potential φ on the anode. With this potential, the anode will repulse the electrons ejected by the cathode. For this reason, a large portion of the electrons will not reach the anode and will fall back onto the cathode. However, there will be some electrons which will reach the anode over the repulsive force. For this to occur, the kinetic energy of the electron ejected from the cathode must exceed the difference in potential energy of the anode and cathode, i.e., the quantity $e\varphi$. The portion of such electrons is equal to $e^{\frac{-e\varphi}{kT}}$. Thus for a negative potential φ

of the anode, the current is equal to $j = j_0 e^{\frac{-e\varphi}{kT}}$, where j_0 is the current for a positive potential. In this experiment, it is necessary that the distance between the cathode and anode be small so that the number of electrons between them should not be great and the mutual repulsion of electrons should not affect the result of the experiment.

The Soviet scientist, Academician A. F. Ioffe proposed using this phenomenon for the direct conversion of thermal energy into electric energy. If electrons go from the cathode to a negatively charged anode, such a system is a source of voltage: the current in an external circuit between the negatively charged anode and the positive cathode is in a direction such that it performs work. This method of obtaining electric current is remarkable in that there are no moving parts and the circuit is fundamentally simple. In this respect, it resembles the generation of electric power by means of thermoelectric cells, which was also proposed by Academician Ioffe. This new method is presently under careful study in many countries with the aim of practical utilization.

Chapter 8

Electric Circuits and Oscillatory Phenomena in Them

8.1 BASIC CONCEPTS AND UNITS OF MEASUREMENT

In this chapter we consider phenomena that occur in electric circuits. The principal elements of an electric circuit are resistance, capacitance, inductance, and sources of current (voltage).

As in the other parts of this book devoted to the application of mathematics to physical problems, our exposition is not designed to take the place of a standard physics textbook but rather to supplement, develop and refine some of the knowledge contained in any such school text. We will therefore confine ourselves to a brief review of the definitions of resistance, capacitance and so forth and to their units, on the understanding that the reader is sufficiently acquainted with the basic notions.

The *quantity of electricity* is determined as the difference between the positive charge and the negative charge. We denote it by q . We use the mksa electromagnetic system of units (also called the Giorgi system of units). Here the unit for the quantity of electricity is the *coulomb* (abbreviated: C). The elementary charge—the charge on the proton—is equal to $e_p = 1.6 \times 10^{-19}$ C, the charge on the electron is $e_e = -1.6 \times 10^{-19}$ C.

The *current* is defined as the quantity of electricity flowing in unit time through a cross section of a conductor. We will denote current by j . The unit of current in the mksa system is that current in which 1 coulomb passes through a cross section of a conductor in one second. This unit bears the name *ampere* (A):

$$\text{ampere} = \text{coulomb/second}$$

For the direction of current we take the direction in which positive charges would have to move in order to produce a given current. Actually, in metallic conductors positive charges are stationary, and the current flows due to the motion of electrons. As a rule, a positively charged body is one which has lost part of its electrons (only in rare cases is a positive charge the result of a body acquiring positive charges). A negatively charged body is one which has acquir-

ed a surplus of electrons. The direction of current is *opposite* to that in which the electrons move in a conductor.

The *electric potential* of a given point is the potential energy which a positive charge of 1 C possesses when placed at the given point. The electric potential of the ground (earth) is taken to be zero. Hence, the point of a circuit connected to the ground by a metal conductor (we say it is grounded, or earthed) has potential zero. In the practical system of units, the unit of potential is the *volt*. The potential of a point is equal to 1 volt (1 V) if a charge of 1 coulomb placed at

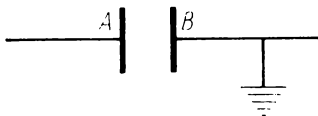


Fig. 169

this point has a potential energy of 1 joule. A joule is equal to 10^7 ergs. The potential energy u of a charge q placed at a point where the potential is equal to ϕ is

$$u \text{ (joule)} = q \text{ (coulomb)} \cdot \phi \text{ (volt)} \quad (8.1-1)$$

We have to imagine here that q is small, because if a large charge (say 1 C) is placed at the given point, then the potential ϕ itself will change. For this reason, it is better to say that the potential is the coefficient of q in (8.1-1).

The work A performed by a field in transferring a charge from a point where the potential is equal to ϕ_1 to a point where the potential is ϕ_2 is

$$A = u_1 - u_2 = q(\phi_1 - \phi_2)$$

Just as in mechanics only the difference of potential energies enters into all physical results, so in electricity, the formulas always involve a difference of potential. There will be no change in the potential difference if to all potentials at all points we add an identical summand. It is therefore possible to choose the potential of any point in a circuit or a piece of equipment in arbitrary fashion, say, set it equal to zero. However, after this has been done, the potentials of all other points become quite definite quantities. It is precisely for this reason that we can take the ground potential as zero.

Let us consider a capacitor (Fig. 169) consisting of two parallel plates. One of the plates (the left one) can be connected to some source of voltage. The quantity of electricity on the left-hand plate is directly proportional to the potential difference of the plates of the capacitor, ϕ_C :

$$q = C\phi_C$$

the difference of potential being defined as the potential of the left plate minus the potential of the right plate. Since in Fig. 169 the right plate is grounded, it follows that φ_C in this case is equal to the potential of the left plate.

The coefficient of proportionality C is called the *capacitance* of the capacitor. The unit of capacitance is the *farad* (F). It is the capacitance of a capacitor in which the potential difference of the plates is 1 volt for a charge of 1 coulomb. 10^{-6} F is called a *microfarad*, 10^{-9} F, a *nanofarad*, 10^{-12} F, a *picofarad*.

An equal quantity of electricity (but opposite in sign) accumulates (builds up) on the right-hand plate of the capacitor. Denoting one plate of the capacitor by A and the other by B , we get

$$q_A = C\varphi_C, \quad q_B = -q_A = -C\varphi_C$$

An electric charge is a conserved quantity. Electric charges of the same sign never appear or disappear in any processes whatsoever.* A change in the charge on plate A of the capacitor is due to the fact

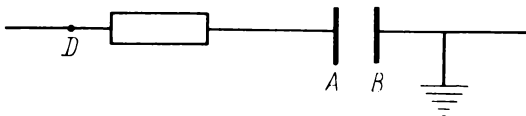


Fig. 170

that a portion of the charge left the plate and moved to some other site, say point D along the conductor AD (Fig. 170).

If a current j is flowing from D to A (from left to right), then in time dt a quantity of electricity $j dt$ will flow through the cross section of the conductor; therefore

$$dq_A = j dt \quad \text{or} \quad \frac{dq_A}{dt} = j$$

Let us now find out what a current flowing in a conductor depends on. By Ohm's law the current is proportional to the difference of potential across the terminals of the conductor, the current flowing from higher to lower potential. Thus

$$j = k (\varphi_D - \varphi_A) \quad (8.1-2)$$

The quantity k is positive and is called the *conductance*.

We denote the quantity $\varphi_D - \varphi_A$ by φ_R . This is the potential difference across the resistance R . The value of φ_R is defined (like that of φ_C) as the left-hand potential minus the right-hand potential.

* The total electric charge of a system remains unchanged when two particles of equal and opposite charge appear or disappear.

The reciprocal, $\frac{1}{k}$, is called the *resistance* of a conductor and is denoted by R . The unit of resistance is the *ohm*, which is the resistance of a conductor through which a current of 1 ampere is flowing when a potential difference of 1 volt is impressed across its terminals. Ohm's law (7.1-2) may be written thus:

$$j = \frac{\varphi_R}{R} \quad \text{or} \quad \varphi_R = Rj \quad (8.1-3)$$

As a *source of voltage* in a circuit we can take a voltaic cell. There is a definite potential difference across the terminals of the cell. We can assume, roughly, that the potential difference is independent of the current flowing through the cell. In particular, in a cell the current can flow from a low potential to a higher potential. Through a resistance, the current always flows from a high potential to a lower potential, like water in a tube connecting two vessels flows from high level to low level.

The cell is like a pump that can take in water in a low-level vessel and pump it up to a high-level vessel, that is to say, make the water

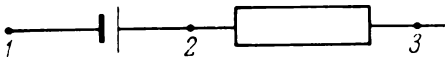


Fig. 171

move uphill. To operate the pump we need some kind of external source of energy. The same applies to the cell. When the current flows from low to high potential, chemical reactions take place in the cell. The energy of these chemical reactions in the cell is transformed into electric energy.

The potential difference which the cell yields is termed the *electromotive force* (also called the *electromotance*) which we abbreviate to *emf*.

The potential difference across the cell taken as left-hand potential minus right-hand potential (Fig. 171) is equal to minus electromotive force of the cell:

$$\varphi_1 - \varphi_2 = -E$$

In reality, the emf is slightly dependent on the current flowing through the cell. When the current is flowing (from left to right in Fig. 171) in the direction from low potential to high potential (which is the normal operating conditions of the cell when it is generating electric energy), the emf E diminishes with increasing current flow. Approximately, we can take it that the emf is constant, but more exactly

$$E = a - bj \quad (8.1-4)$$

We will call a cell whose emf is independent of the current j an ideal cell.

Let us consider a series connection of an ideal cell with an emf equal to a and a resistance b (Fig. 171). Then

$$\varphi_E = \varphi_1 - \varphi_2 = -a, \quad \varphi_b = \varphi_2 - \varphi_3 = bj$$

and therefore

$$\varphi_1 - \varphi_3 = (\varphi_1 - \varphi_2) + (\varphi_2 - \varphi_3) = -a + bj = -(a - bj) = -E$$

The quantity b in formula (8.1-4) is for this reason called the internal resistance of the cell: the real cell that we are dealing with that has an emf which satisfies formula (8.1-4) yields the same dependence of E on j as the series connection of an ideal cell and a resistance b . The name for a remains the same: the emf of a real cell, bearing in mind that $E = a$ when $j = 0$, and the emf drop for $j \neq 0$ is characterized by b .*

In the sequel, when considering electric circuits involving current sources, for example a cell and various resistances, we can imagine that we are dealing with an ideal cell with constant emf independent of the current, and the internal resistance b may be combined with the external resistance R . Thus, a real cell with internal resistance b connected in series with a resistance R is equivalent to an ideal cell connected in series with a resistance $R_1 = (R + b)$.

It is worth once again paying special attention to the difference between resistance and source of voltage. If in a circuit there is a potential difference across a resistance, such that $\varphi_2 > \varphi_1$ (Fig. 172), then, by our definition $\varphi_R = \varphi_1 - \varphi_2 < 0$, which is to say that φ_R is negative. Hence, by formula (8.1-3), the current is negative as well. Which means that the current flows from right to left, from 2 to 1. Now suppose that there is a potential difference of the same sign across the terminals of the voltage source, and the dependence of E on j is given by the formula (8.1-4) (Fig. 171). Here, let $\varphi_3 > \varphi_1$ but $\varphi_3 - \varphi_1 < a$. Then $bj = \varphi_1 - \varphi_3 + a = a - (\varphi_3 - \varphi_1) > 0$, that is, $j > 0$. Therefore the current flows from left to right despite the fact that the potential φ_1 on the left is less than the potential φ_3 on the right. Thus, the voltage source is capable of overcoming the potential difference and yielding a positive current (from left to right) for a negative difference of potential [$(\varphi_1 - \varphi_3) < 0$], provided that this negative potential difference does not exceed in absolute value the emf of the source. Yet for a negative potential difference, the resistance always yields a negative current.

In the particular case (Fig. 173) of a cell having internal resistance b and, in series, external resistance R , the current is determined by the formula $j = \frac{E}{R + b}$.

* Since the current is zero for an open circuit, the emf may be defined as the potential difference of a disconnected cell.

Now let us consider inductance. The phenomenon of inductance is connected with the magnetic field that is produced in the space surrounding a conductor carrying a current. This magnetic field is particularly great if the conductor has the form of a coil with a large number of turns. The field is further increased if the coil is wound on an iron core.

The magnetic field in turn gives rise to electric phenomena. As we know, given a varying magnetic field, each turn (even every portion of a turn) of the coil becomes a source of voltage, something like

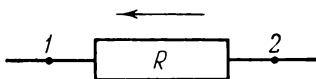


Fig. 172

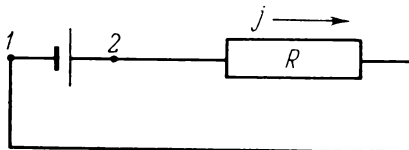


Fig. 173

a voltaic cell. In a coil in which the turns are wound so that the current traverses the core of the coil in the same direction throughout the length of the coil, all these voltage sources are connected in series so that the overall voltage builds up (the voltages are additive in a series connection).

On the whole, a coil is equivalent to a voltage source with a potential difference proportional to the rate of change of the magnetic



Fig. 174

field. But the magnetic field in a coil is proportional to the current flowing in the coil.* For this reason, the rate of change of the magnetic field is proportional to the rate of change of current flow, that is, to the derivative $\frac{dj}{dt}$. Referring to Fig. 174, we finally get

$$\varphi_L = \varphi_1 - \varphi_2 = L \frac{dj}{dt} \quad (8.1-5)$$

Here the positive direction of the current is taken to be from 1 to 2 inside the coil, and the quantity φ_L is the potential difference across

* We will not discuss the case of two coils wound on one core, which is a transformer connecting two electric circuits carrying different currents.

** Neither do we consider cases of a more complicated dependence of the magnetic field on the current when an iron core is inserted in a coil and the current is so great that the iron is saturated.

the coil. It is defined as the potential φ_1 on the left minus the potential φ_2 on the right. When considering in detail the direction of the magnetic field and the emf induced by its variation, it may be proved that the coefficient L in this formula (the so-called inductance) is always positive.

From formula (8.1-5) it follows that if $\frac{dj}{dt} < 0$, then $\varphi_1 - \varphi_2 < 0$, that is, $\varphi_2 > \varphi_1$. Thus, if the current is positive (flows from 1 to 2) and decreases in magnitude, then the coil plays the role of a cell sustaining a positive current in the circuit, despite the fact that $\varphi_L < 0$. But if the current is positive and increasing, then $\frac{dj}{dt} > 0$, and so $\varphi_L > 0$. In this case, the coil plays the part of an additional resistance, since the potential difference across the coil is positive for a positive current [compare with (8.1-3)].

A coil differs essentially from a voltage source and from a resistance in that the quantity φ_L depends not on the current intensity j but on the rate of change of the current, $\frac{dj}{dt}$.

The coefficient L in the equation bears the name "coil inductance" (also self-inductance).

The unit of inductance is the *henry* (H). If the inductance of a coil is equal to 1 henry, this means that when the current is changing at the rate of 1 A/sec, a potential difference of 1 volt is induced in the coil. We obtain the dimensions of inductance from formula (8.1-5):

$$\text{henry} = \frac{\text{volt} \cdot \text{second}}{\text{ampere}}$$

For short, one often says "inductance L " instead of "a coil with inductance equal to L ". We also say "capacitance C " instead of "a capacitor whose capacitance is equal to C ". In the same way, we speak of an emf E_0 instead of spelling it all out to "voltaic cell" or "voltage source".

From the foregoing it is clear that inductance influences the current in a circuit just like an inert mass (flywheel) affects velocity: inductance impedes any change in the current, and a mass (by Newton's second law) tends to impede any change in velocity. This similarity will be discussed in more detail in Sec. 8.4.

From the standpoint of subsequent computations, capacitance, resistance, emf (electromotance) and inductance have one thing in common, they all require two terminals for connection in a circuit (unlike, say, a transformer, which requires four leads, or an electron tube, which has three leads: anode, cathode, and control grid). Devices with circuit connections involving two leads are called two-terminal networks. If there are four leads, they are two-terminal pair networks (or four-pole networks). Each circuit element—capaci-

tance, resistance, electromotance and inductance—is characterized at any given time by a specific current passing through it and a definite potential difference at input and output.

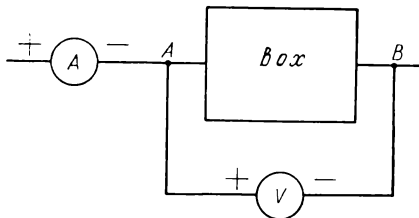


Fig. 175

We can imagine a closed box (labelled "Box" in Fig. 175) with two leads A and B . The interior of the box can contain anything: R, E, L, C . Connect an ammeter A and a voltmeter V . With the circuit connections as shown in Fig. 175 (the "+" and "-" signs correspond to the labels at the terminals of the ammeter and voltmeter), the ammeter indicates the current j passing in the direction from A to B . The voltmeter indicates the potential difference:

$$\varphi_{Box} = \varphi_A - \varphi_B$$

The relationship between φ_{Box} and j depends on what is inside the box:

$$\text{in the case of a resistance, } R, \varphi_{Box} = Rj, \quad (8.1-6)$$

$$\text{in the case of an electromotance (emf), } E_0^*, \varphi_{Box} = -E_0, \quad (8.1-7)$$

$$\text{in the case of an inductance, } L, \varphi_{Box} = L \frac{dj}{dt}, \quad (8.1-8)$$

$$\text{in the case of a capacitance, } C^{**}, \varphi_{Box} = (\varphi_{Box})_0 + \frac{1}{C} \int_{t_0}^t j dt \quad (8.1-9)$$

$$\left(\text{or } \frac{d\varphi_{Box}}{dt} = \frac{1}{C} j \right) \quad (8.1-9a)$$

There are of course cases in which more complicated relationships are involved. For example, a rectifier (a vacuum-tube diode or semiconductor diode) does not fit any of the foregoing formulas. However, in a large number of important problems we can confine ourselves to considering the circuit elements for which formulas (8.1-6) to (8.1-9) are valid to a high degree of accuracy. These are the circuits that we will investigate (with the exception of Sec. 8.16, where we give special consideration to the properties of a circuit with a device

* The internal resistance of the emf is disregarded.

** $q_A = C\varphi_{Box}$, $\frac{dq_A}{dt} = j$, whence $\frac{d\varphi_{Box}}{dt} = \frac{1}{C} j$. If at the initial time $t = t_0$, $\varphi_{Box} = (\varphi_{Box})_0$, then $\varphi_{Box} = (\varphi_{Box})_0 + \frac{1}{C} \int_{t_0}^t j dt$.

which exhibits a complicated relationship between current and potential difference).

Let us study the circuit shown in Fig. 176. We will first write down the voltage drops on the separate elements of the circuit:

$$\left. \begin{aligned} \varphi_C &= \varphi_{A_C} - \varphi_{B_C}, & \varphi_L &= \varphi_{A_L} - \varphi_{B_L}, \\ \varphi_R &= \varphi_{A_R} - \varphi_{B_R}, & \varphi_E &= \varphi_{A_E} - \varphi_{B_E} \end{aligned} \right\} \quad (8.1-10)$$

Also observe that $\varphi_{B_C} = \varphi_{A_R}$, $\varphi_{B_R} = \varphi_{A_L}$, $\varphi_{B_L} = \varphi_{A_E}$. For this

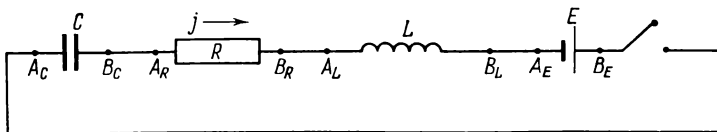


Fig. 176

reason, adding all the equations (8.1-10) termwise, we get

$$\varphi_C + \varphi_R + \varphi_L + \varphi_E = \varphi_{A_C} - \varphi_{B_E}$$

If the circuit in Fig. 176 is closed, then $\varphi_{A_C} = \varphi_{B_E}$. In this case, consequently,

$$\varphi_C + \varphi_R + \varphi_L + \varphi_E = 0 \quad (8.1-11)$$

This general equation, together with the expressions (8.1-6) to (8.1-9), fully describes all the processes that occur in the circuit. Below we will use this equation to examine a variety of circuits beginning with the very simplest which consists of only two elements.

8.2 DISCHARGE OF A CAPACITOR THROUGH A RESISTOR

Let us examine the process occurring in a circuit with capacitance C and resistance R (Fig. 177). We denote by φ the potential of point A (the opposite plate of the capacitor will be grounded). To begin with, let $\varphi = \varphi_0$. The corresponding quantity of electricity on plate A is $q_A = C\varphi_0$.

Can we speak of a current flowing through a capacitor? A capacitor consists of two plates separated by an insulator (say, air) so that in reality an electron cannot go through the capacitor, say, from A to B .

However, if a positive charge is impressed on plate A , then plate B will have a negative charge, and a positive charge will flow out of plate B along the wire (the current also goes from left to right). Two

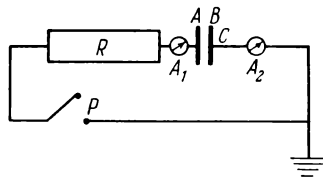


Fig. 177

ammeters A_1 and A_2 , one of which measures current in the wire connected to plate A , the other in the wire connected to plate B , will have identical readings. What precisely it is that flows (positive charges or electrons) through different portions of the electric circuit does not interest us, just as we are not interested in whether the same electrons pass through A_2 that have passed through A_1 or not. For this reason we will henceforth only speak of the current passing through the capacitor and will have in mind the current flowing in the conductors connected to the plates of the capacitor. We can speak of current flowing in an electric circuit through a capacitance in the same way that we speak of current flowing through a resistance or an inductance. The difference lies in the different type of relationship between current and potential difference as expressed by the formulas (8.1-9) and (8.1-9a).

When we close the switch P (Fig. 177), a current

$$j = \frac{1}{R} \varphi_R$$

will flow through the resistance R . By formula (8.1-14), $\varphi + \varphi_R = 0$, whence $\varphi_R = -\varphi$ and so

$$j = -\frac{1}{R} \varphi \quad (8.2-1)$$

Since current flowing from left to right is taken to be positive, it follows, as may be seen from formula (8.2-1), that for $\varphi > 0$ the current is negative, it flows from right to left and the capacitor becomes discharged.* Recalling that $j = \frac{dq}{dt}$ (current flowing through a capacitor) and $q = C\varphi$, we find

$$j = C \frac{d\varphi}{dt} \quad (8.2-2)$$

Comparing (8.2-1) and (8.2-2), we find

$$\frac{d\varphi}{dt} = -\frac{1}{RC} \varphi \quad (8.2-3)$$

We solved an equation like this in connection with the problem of radioactive decay. If $\varphi = \varphi_0$ when $t = 0$, then

$$\varphi(t) = \varphi_0 e^{-\frac{t}{RC}} \quad (8.2-4)$$

whence

$$j(t) = -\frac{\varphi_0}{R} e^{-\frac{t}{RC}}$$

* Observe that in all circuits having the form of a rectangle (Fig. 176 and subsequent figures), we speak of the direction of current in the *upper* side of the rectangle; the current flow in the bottom side that closes the circuit will clearly be in the opposite direction.

It can be seen from formula (8.2-3) that the quantity RC has the dimensions of time. Let us verify this:

$$[R] = \text{ohm} = \frac{\text{volt}}{\text{ampere}} = \frac{\text{volt} \cdot \text{second}}{\text{coulomb}}; \quad [C] = \frac{\text{coulomb}}{\text{volt}}$$

whence

$$[RC] = \frac{\text{volt} \cdot \text{second}}{\text{coulomb}} \cdot \frac{\text{coulomb}}{\text{volt}} = \text{second}$$

During time $t = RC$ the charge q on the capacitor and also the current j diminish by a factor of e .

The discharging process in a capacitor can easily be observed experimentally. Buy a capacitor with capacitance $C = 20$ micro-

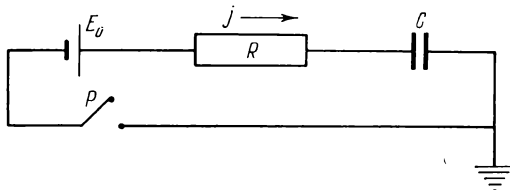


Fig. 178

farads $= 20 \times 10^{-6}$ F and a resistor $R = 20$ megohms $= 20 \times 10^6$ ohms. For an RC circuit of this type we get $RC = 400$ sec, which is a very convenient time for observational purposes.

The quantity RC is called the *time constant* of a circuit consisting of a capacitance and a resistance (recall that in the case of radioactive decay the mean lifetime was an analogous quantity).

We will consider the problem of charging a capacitor through a resistance. The circuit diagram is shown in Fig. 178. If switch P is closed, then, by (8.1-11), $\varphi_E + \varphi_R + \varphi = 0$, where φ is the potential of the nongrounded plate of the capacitor. Since $\varphi_E = -E_0$, $\varphi_R = Rj$, it follows that $-E_0 + Rj + \varphi = 0$. The current flowing through the capacitor is $j = \frac{dq}{dt} = C \frac{d\varphi}{dt}$ and so

$$-E_0 + RC \frac{d\varphi}{dt} + \varphi = 0$$

or

$$\frac{d\varphi}{dt} = -\frac{1}{RC} (\varphi - E_0) \quad (8.2-5)$$

To find out how φ varies with time, it will be convenient to make the change of variable $z = \varphi - E_0$; then $dz = d\varphi$. Equation (8.2-5) can be rewritten

$$\frac{dz}{dt} = -\frac{z}{RC}$$

Its solution is

$$z = z_0 e^{-\frac{t}{RC}} \quad (8.2-6)$$

where z_0 is the value of z at the initial time.

Let us find the solution for the case where at the initial time the capacitor is not charged: $\varphi = 0$ for $t = 0$. Then $z_0 = -E_0$. From (8.2-6) we get $z = -E_0 e^{-\frac{t}{RC}}$,

$$\varphi = z + E_0 = -E_0 e^{-\frac{t}{RC}} + E_0 = E_0 \left(1 - e^{-\frac{t}{RC}}\right) \quad (8.2-7)$$

The graph of φ as a function of t is given in Fig. 179. The curve corresponds to the formula (8.2-7), the dashed horizontal line represents the value $\varphi = E_0$ to which the solution approaches in the

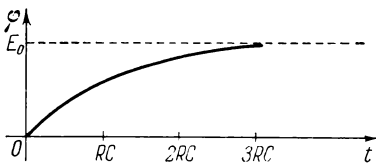


Fig. 179

course of time. The quantity z has the geometric meaning of vertical distance from the curve to the dashed line. This distance diminishes exponentially with time.

During a time equal to RC , the charge on the capacitor reaches 63% of its final value; during time $2RC$, 86%, and during time $3RC$, 95% of the final value.

From formulas (8.2-4) and (8.2-7) it is evident that charging and discharging the capacitor is the faster, the smaller the resistance R .

Exercises

1. Referring to Fig. 177, $C = 10^{-6}$ F, $R = 10^7$ ohms, $R = 10^8$ ohms, $R = 10^9$ ohms. For each of these cases, determine the time lapse during which a current flowing through the capacitor at the initial time t_0 falls off by 10%; decreases by a factor of 2.

2. Consider the process of equalizing the potential across a resistance R in series with two capacitors C_1 and C_2 , one of which at time $t = 0$ is charged to a potential difference $\varphi_{C_1}(0) = a$, while the other is not charged at all, that is, $\varphi_{C_2}(0) = 0$ (Fig. 180).

3. Determine the variation in the time constant of the circuit depicted in Fig. 177 if all linear dimensions of the circuit diagram are increased n -fold (for the case of a plane capacitor).

Remark 1. The condition of the problem is to be understood in this way: the dimensions of the capacitor and the resistance are increased but the materials of which they are made are not changed.

Remark 2. The formula for the capacitance of a plane capacitor is $C = \epsilon S/4\pi d$, where S is the area of a plate of the capacitor, d is the distance between the plates, and ϵ is a constant dependent on the material between the

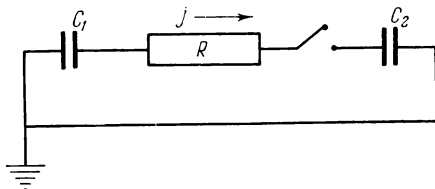


Fig. 180

plates (the dielectric constant). The magnitude of a wire resistor is found from the formula $R = \rho \frac{l}{\sigma}$, where l is the length, σ is the cross-sectional area, and ρ is a constant that depends on the kind of wire used.

8.3 OSCILLATIONS IN A CAPACITANCE CIRCUIT WITH SPARK GAP

A typical circuit diagram involving a capacitor is shown in Fig. 181. The circuit includes a voltage source with emf E and resistance R (the role of R may be played by the internal resistance of the voltage source). Underneath is a spark gap. For a potential

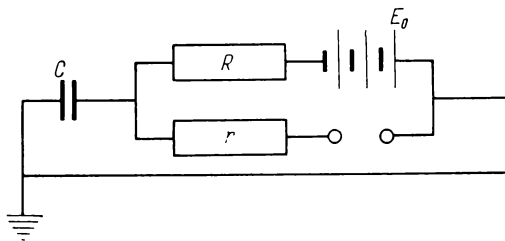


Fig. 181

difference less than a definite value φ_1 , the spark gap is an insulator. At $\varphi = \varphi_1$ a spark jumps the gap, the air between the wires heats up and becomes a good conductor. We denote the total resistance of the leads and the incandescent air by r . The quantity r is small and remains small as long as current flows maintaining a high temperature of the air. For a definite small value of current j_2 the air cools and the spark gap again becomes an insulator. This current value is associated with the potential difference $\varphi_2 = j_2 r$. Here $\varphi_1 > \varphi_2$: a higher voltage is needed to initiate a spark than to keep it burning.

Fig. 182 shows the dependence of φ on t for such a circuit. The capacitor is charged over the section OA , there is no current flowing

through the spark gap. In this case the formula (8.2-6) is valid:

$$\varphi = E \left(1 - e^{-\frac{t}{RC}} \right) \quad (8.3-1)$$

The potential difference at point A at time $t = t_A$ reaches the value φ_1 , the spark gap begins to conduct current and the capacitor discharges. Since for this case $R \gg r$, the current from the voltage source can be ignored as compared to the current passing through the spark gap. Therefore, for φ we get the equation

$$\frac{d\varphi}{dt} = -\frac{\varphi}{rC}$$

and $\varphi = \varphi_1$ for $t = t_A$, whence we get

$$\varphi = \varphi_1 e^{-\frac{(t-t_A)}{rC}} \quad (8.3-2)$$

At time $t = t_B$ (at point B), $\varphi = \varphi_2$ and the spark gap again becomes an insulator. The charging process is initiated (section BC).

Let us isolate the time $t_B - t_A$ during which the capacitor is discharging. To do this, we take advantage of the fact that $\varphi = \varphi_2$ at $t = t_B$. Putting $\varphi = \varphi_2$, $t = t_B$ in (8.3-2), we get

$$\varphi_2 = \varphi_1 e^{-\frac{t_B - t_A}{rC}}$$

whence

$$t_B - t_A = rC \ln \frac{\varphi_1}{\varphi_2}$$

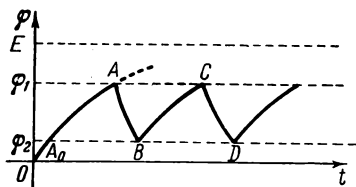


Fig. 182

Over BC (charging) the relation (8.3-1) displaced in time by the amount τ holds true (in Fig. 182, τ is depicted by the line segment A_0B). For this reason

$$\varphi = E \left(1 - e^{-\frac{t-\tau}{RC}} \right)$$

Putting $t = t_B$ here, we get

$$\varphi_2 = E \left(1 - e^{-\frac{t_B - \tau}{RC}} \right)$$

Similarly, setting $t = t_C$, we find

$$\varphi_1 = E \left(1 - e^{-\frac{t_C - \tau}{RC}} \right)$$

From the last two formulas we have

$$\frac{E - \varphi_2}{E - \varphi_1} = e^{\frac{t_C - t_B}{RC}} \quad \text{or} \quad t_C - t_B = RC \ln \frac{E - \varphi_2}{E - \varphi_1}$$

The complete period (a charge-discharge cycle) is

$$T = t_C - t_A = (t_C - t_B) + (t_B - t_A) = R\mathcal{G} \ln \frac{E - \varphi_2}{E - \varphi_1} + rC \ln \frac{\varphi_1}{\varphi_2}$$

Ordinarily, the resistance R in the circuit of the voltage source is many times greater than that of the spark gap, and for this reason the charging time is much longer than the time of discharge. On the other hand, the discharge current is many times greater than the charging current, greater than the maximum current obtainable from the voltage source (with an internal resistance of R_1 the voltage source does not produce a current exceeding $\frac{E}{R_1}$). The circuit shown in Fig. 181 transforms a long-term small current generated by the voltage source into a strong current, which however is of short duration (it is customary to speak of "short pulses" of current).

This circuit operates like a system in which a tiny flow of water fills a vessel (Fig. 183). The vessel is fixed so that when a sufficient amount of water has accumulated, it turns over pouring out the water. The vessel then rights itself and the process begins anew. In the figure, the vessel is fixed on a horizontal axis OO' below the midpoint. At the bottom of the vessel is attached a weight so that the centre of gravity of the empty vessel lies below the axis. But when the vessel fills up with water, the centre of gravity of the full vessel lies above the axis and the vessel tips over.

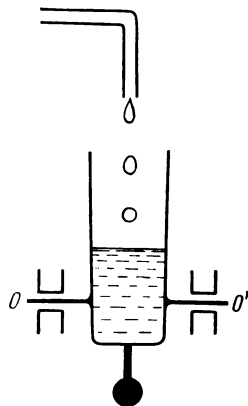


Fig. 183

Let us return to the circuit diagrams in Figs. 177 and 178. In these circuits, which consist of capacitances, resistances and electromotances, the potentials even out in the course of time. Indeed, in Fig. 177, $\varphi = 0$ and in Fig. 178, $\varphi = E_0$ [see formulas (8.2-4) and (8.2-7)]. The situation is quite different in the case of spark-gap circuits. Here we have undamped oscillations of magnitude φ (true, they are very different from those that we studied earlier). These oscillations are connected with certain specific properties of the spark gap, in particular with the fact that until a definite potential is reached (the so-called breakdown potential φ_1), no current flows through the spark gap.

Thick books have been written about the properties of discharge through air in a spark gap. All we have given here is a smattering of information—only enough to understand the operation of the circuit shown in Fig. 181. This information does not even suffice

to answer the simple question: What will happen if we connect the spark gap to a voltage source without a capacitor?

Indeed, if no current flows, the voltage on the spark gap will be E_0 . Since $E_0 > \varphi_1$ breakdown should occur. But if this occurred, the resistance of the spark gap would become small, equal to r . Then a potential difference equal to $E_0 \cdot \frac{r}{r+R}$ would appear across the spark gap and the current would be $j = \frac{E_0}{r+R}$. If R is great, the current j is small, less than j_2 , the potential difference across the spark gap is small, less than φ_2 . But then the air will not heat up and the resistance of the spark gap will not become the small quantity r , which means the potential difference will be great and equal to E . We have a contradiction.

Actually, under these conditions we have an electric discharge of a different type, the so-called glow discharge (small current without heating of the air), instead of the spark with incandescent air.

8.4 THE ENERGY OF A CAPACITOR

A charged capacitor has a definite supply of energy, which can be given up very quickly if the capacitor is discharged through a small resistance.

Let us find the supply of energy of a capacitor of capacitance C , one plate of which is grounded and the other has potential φ_0 . Then the quantity of electricity $q_0 = C\varphi_0$.

It would appear at first glance that the energy is equal to the product $q_0\varphi_0$. In reality, this expression is not exact, though it is correct as to order of magnitude; it differs from the true value by a factor of 2. Let us consider the charging process of the capacitor. When its potential is φ and the charge is q , the addition of a small quantity of electricity dq increases the energy by

$$dW = \varphi dq \quad (8.4-1)$$

The essential thing is that during charging the potential φ itself changes since $\varphi = \frac{1}{C} q$. Substituting this value of φ into (8.4-1), we get

$$dW = \frac{1}{C} q dq \quad (8.4-2)$$

Integrating (8.4-2) from $q = 0$ (uncharged capacitor) to $q = q_0$, we get

$$W(q_0) = \frac{1}{C} \int_0^{q_0} q dq = \frac{1}{2} \frac{q_0^2}{C} = \frac{1}{2} \varphi_0 q_0 = \frac{1}{2} C \varphi_0^2 \quad (8.4-3)$$

Thus an exact evaluation yields the coefficient $\frac{1}{2}$.

Now let us examine the charging process of a capacitor from a voltage source through a resistance (see Sec. 8.2, the circuit in Fig. 178). The voltage source has a constant emf, E_0 . Therefore, when a quantity dq of electricity flows, the voltage source does work $E_0 dq$ (this work is performed at the expense of the chemical energy of the voltage source, which diminishes). Hence, the total work done by the voltage source is equal to $E_0 q_0$, where q_0 is the total amount of electricity that has flowed. When the capacitor is being charged, the process stops when $\varphi = E_0$. In this process, the voltage source will have performed work

$$E_0 q_0 = E_0 C E_0 = C E_0^2$$

What supply of energy will the capacitor possess? This can readily be computed from formula (8.4-3):

$$W = \frac{1}{2} C E_0^2$$

Where has half the work performed by the source gone? We will show that it went to heat up the resistance R . Recall that if a quantity dq of electricity flows through a resistance, the energy released will be

$$dA = \varphi_R dq \quad (8.4-4)$$

where φ_R is the potential difference across the resistance. Using the fact that $dq = j dt$, $j = \frac{\varphi_R}{R}$, we can transform (8.4-4) to the familiar form

$$dA = \frac{(\varphi_R)^2}{R} dt = j^2 R dt$$

The quantity $j^2 R = \frac{(\varphi_R)^2}{R}$ is the quantity of energy released on the resistance in unit time, which is to say, it is the thermal capacity.

The function $j(t)$ in the case of the charging of a capacitor through a resistance was found in Sec. 8.2 to be

$$j(t) = \frac{E_0}{R} e^{-\frac{t}{RC}}$$

Therefore

$$dA = \frac{E_0^2}{R} e^{-\frac{2t}{RC}} dt$$

The energy released in time t is

$$A(t) = \frac{E_0^2}{R} \int_0^t e^{-\frac{2t}{RC}} dt$$

whence

$$A(t) = -\frac{C E_0^2}{2} e^{-\frac{2t}{RC}} \Big|_0^t = \frac{1}{2} C E_0^2 \left(1 - e^{-\frac{2t}{RC}} \right) \quad (8.4-5)$$

We know that as the time interval t increases without bound the potential φ approaches the value E_0 without bound. Then, as may be seen from (8.4-5), A approaches $\frac{1}{2}CE_0^2$ without bound. Therefore the total energy released on the resistance is

$$A = \frac{1}{2}CE_0^2 \quad (8.4-6)$$

Thus, calculations confirm the fact that in the charging process of a capacitor, half the energy is lost on the resistance. The efficiency of charging is only 50%. Note that if we make a direct connection from the voltage source to the capacitor, nothing will change, the efficiency remains at 50%, the role of the resistance R being taken by the internal resistance of the voltage source, which will then heat up. From formula (8.4-6) it is evident that the energy lost uselessly on the resistance in charging the capacitor is independent of the magnitude of the resistance R and hence does not depend on how fast the charging takes place.

Since R did not appear in (8.4-6), this formula may be obtained without introducing R into the intermediate transformations. Indeed, for the circuit of Fig. 178, $\varphi_E + \varphi_R + \varphi_C = 0$, whence $\varphi_R = -\varphi_E - \varphi_C = E_0 - \frac{q}{C}$. Therefore $dA = \left(E_0 - \frac{q}{C}\right) dq$. Integrating this expression from $q = 0$ to $q = q_0 = E_0C$, we get

$$A = \frac{1}{2}CE_0^2$$

This last derivation holds true also for the case where the resistance R varies with time. The previous derivation held true only for $R = \text{const}$ since only in this case could the formulas of Sec. 8.2 be applied.

In order to reduce losses in charging a capacitor we should have done as follows: first take a voltage source with small emf E_1 and charge the capacitor to the potential E_1 , then disconnect the first voltage source and connect a second source with greater emf E_2 . Having charged the capacitor to potential E_2 , disconnect the second source and connect a third with emf E_3 , and so on. The gain is clearly seen in the graph: lay off the charge q of the capacitor on the axis of abscissas, the potential φ on the axis of ordinates. They are connected by the relation $\varphi = \frac{1}{C}q$, which represents a straight line (Fig. 184).

The energy of a capacitor is equal to the area of the triangle OAB . The work done by the voltage source is equal to the area of the rectangle $OABD$. The energy lost on the resistance is equal to the area of the triangle ODB . If the capacitor is charged in stages, the sum of the works of all voltage sources is equal to the shaded area in Fig. 185. We leave it to the reader to find the efficiency for the case

where the charging process is divided into n stages:

$$E_1 = \frac{\varphi}{n}, \quad E_2 = \frac{2\varphi}{n}, \quad E_3 = \frac{3\varphi}{n}, \quad \dots, \quad E_n = \varphi$$

In the foregoing case, one plate of the capacitor was grounded, i.e., its potential was $\varphi_1 = 0$. Here, the energy of the capacitor

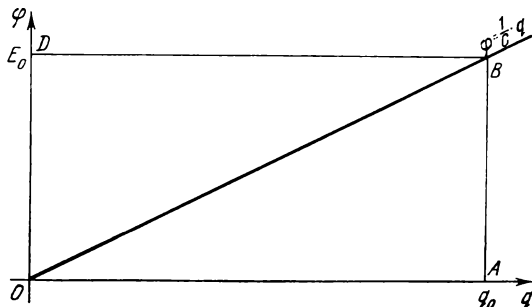


Fig. 184

depends on the potential of the other plate, φ_2 , $W = \frac{1}{2} C \varphi_2^2$. If neither plate is grounded, the energy of the capacitor depends on the

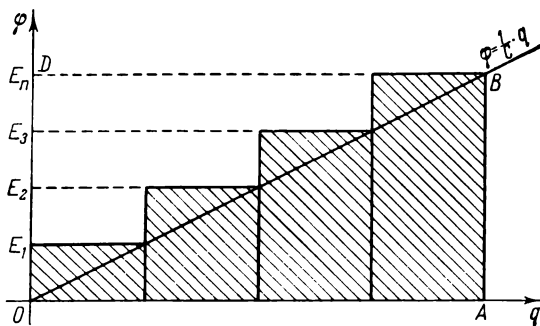


Fig. 185

difference of potential across the plates, φ_C :

$$W = \frac{1}{2} C \varphi_C^2$$

Indeed, we know that the charge q on each plate of the capacitor depends on the potential difference, the charges on the plates being equal in magnitude but opposite in sign:

$$q_A = C \varphi_C, \quad q_B = -C \varphi_C = -q_A, \quad dq_A = -dq_B$$

When computing the variation of energy in the charging process, one has to take into account the variation of charge on both plates. Let the potential of plate A be φ_1 , the potential of plate B , φ_2 , $\varphi_1 - \varphi_2 = \varphi_C$. Then

$$\begin{aligned} dW &= \varphi_1 dq_A + \varphi_2 dq_B = \varphi_1 dq_A - \varphi_2 dq_A = \\ &= (\varphi_1 - \varphi_2) dq_A = \varphi_C dq_A \end{aligned}$$

Since $\varphi_C = \frac{q_A}{C}$, it follows that

$$dW = \frac{1}{C} q_A dq_A \quad (8.4-7)$$

Integrating (8.4-7) from 0 to q_A , we get

$$W = \frac{q_A^2}{2C} = \frac{1}{2} C \varphi_C^2$$

Knowing the expression for the energy of a charged capacitor as a function of the capacitance, we can find the mechanical forces acting between the plates of the capacitor. Imagine the plates to be connected mechanically with some kind of lever and suppose the capacitance C depends on the position of the lever. If the position of the lever is characterized by the value of the x -coordinate, then the capacitance is $C(x)$. At a definite position x_0 of the lever the capacitance of the capacitor is $C(x_0) = C_0$. If in this position the capacitor is charged to the potential φ_0 , then the charge on the plates $q_0 = C_0 \varphi_0$ and the energy of the capacitor is

$$W = \frac{C_0 \varphi_0^2}{2} = \frac{q_0^2}{2C_0}$$

Let us disconnect the capacitor from the voltage source and move the lever. The charge will then remain constant (the potential varies inversely as the capacitance), the energy will vary:

$$W(x) = \frac{q_0^2}{2C(x)}$$

The electric energy of a charged capacitor is similar to the elastic energy of a spring. $W(x)$ increases if an external force applied to the level does work. Then the external force overcomes the forces with which the plates of the capacitor act on the lever. Contrariwise, if $W(x)$ decreases, the lever is displaced and does work in opposition to the external applied forces. We may conclude that the force with which the plates act on the lever is equal to

$$F = -\frac{dW}{dx} = -\frac{d}{dx} \left(\frac{q_0^2}{2C(x)} \right) = \frac{q_0^2}{2[C(x)]^2} \cdot \frac{dC(x)}{dx} = \frac{\varphi^2(x)}{2} \cdot \frac{dC(x)}{dx} \quad (8.4-8)$$

The force is directed towards increasing capacitance. Thus, if the capacitor consists, say, of two equal parallel plates, the capacitance

is inversely proportional to the distance between the plates. This means the capacitance increases when the plates are brought closer together. Quite true, because when a capacitor is charged the charges on the plates are of opposite sign and so the plates attract with more force if they are close together.

Formula (8.4-8) enables us to find the force in more complicated situations, as, for example, in the case of a variable capacitor in which one plate can move in and out between two fixed plates.

It is important to note that we took the derivative $\frac{dW}{dx}$ for a given constant charge q . However, it is *not permissible*, when seeking the force by the formula $F = -\frac{dW}{dx}$, to take the derivative of $W = \frac{C(x)\varphi^2}{2}$, assuming φ constant and having regard solely for the fact that C depends on x . We would then obtain an incorrect sign for the force. Indeed, if the capacitor is disconnected from the voltage source, then φ is not constant, $\varphi = \frac{q}{C}$, $C = C(x)$. If the capacitor is connected to a voltage source, then φ remains constant as the capacitance varies. But then the charge q varies, which means that a current is flowing through the voltage source, that is, that the voltage source is doing work equal to φdq (as C increases). Hence, when applying the law of conservation of energy for constant φ and variable capacitance, one has to take into consideration not only the variation in energy of the capacitor and the work of the force but also the work done by the voltage source.

8.5 INDUCTANCE CIRCUIT

Let us consider a circuit consisting of resistance R and inductance L (Fig. 186). By formula (8.1-14)

$$\varphi_R + \varphi_L = 0 \quad (8.5-1)$$

Since $\varphi_R = Rj$ and $\varphi_L = L \frac{dj}{dt}$, using (8.5-1), we find

$$Rj + L \frac{dj}{dt} = 0$$

Thus, the current in the circuit of Fig. 186 satisfies the equation

$$\frac{dj}{dt} = -\frac{R}{L}j \quad (8.5-2)$$

The solution of this equation is

$$j(t) = j_0 e^{-\frac{R}{L}t} \quad (8.5-3)$$

Thus, the current in the circuit of Fig. 186 falls off exponentially. The current diminishes e times during a time

$$T = \frac{L}{R}$$

Let us verify the dimensions of $\frac{L}{R}$. L is measured in henrys, that is, in $\frac{\text{volt} \cdot \text{second}}{\text{ampere}}$, R in ohms, or in $\frac{\text{volt}}{\text{ampere}}$. Therefore the dimensions of $\frac{L}{R}$ are $\frac{\text{volt} \cdot \text{second}}{\text{ampere}} : \frac{\text{volt}}{\text{ampere}} = \text{second}$, so that actually $\frac{L}{R}$ has the dimensions of time. We will call $\frac{L}{R}$ the buildup time. In the

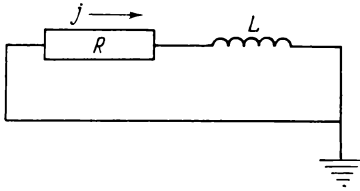


Fig. 186

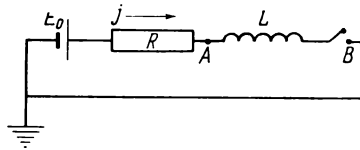


Fig. 187

circuit diagram shown in Fig. 186, in which there is no voltage source, the current tends to zero with time. How to set up an initial current of j_0 will be discussed later on.

For the present, let us consider a circuit consisting of a source having emf equal to E_0 , a resistance R and an inductance L (Fig. 187). From the condition

$$\varphi_E + \varphi_R + \varphi_L = 0$$

recalling that $\varphi_E = -E_0$, we find

$$-E_0 + Rj + L \frac{dj}{dt} = 0 \quad (8.5-4)$$

We rewrite this equation as

$$\frac{dj}{dt} = \frac{R}{L} \left(\frac{E_0}{R} - j \right)$$

This equation is similar to the equation (8.2-5) (see Sec. 8.2) and is solved by the very same procedure. We get

$$j(t) = \frac{E_0}{R} + Ae^{-\frac{Rt}{L}} \quad (8.5-5)$$

where the value A is determined from the initial condition. Suppose the switch is closed at $t = 0$. Then $j(0) = 0$ because there was no current flowing in the circuit when the switch was open. Given this

condition, we find $A = -\frac{E_0}{R}$ and (8.5-5) assumes the form

$$j = \frac{E_0}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \quad (8.5-6)$$

In the course of time the current approaches the value

$$j(\infty) = \frac{E_0}{R} \quad (8.5-7)$$

This current value is independent of the inductance L and is simply obtained from Ohm's law in a circuit with emf E_0 and resistance R . However, this current value is not established at once, but gradually,

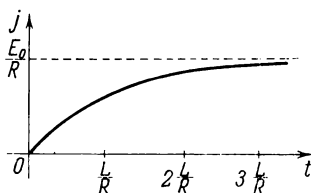


Fig. 188

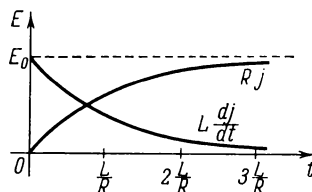


Fig. 189

and the *time* required to establish the current depends on the inductance L . In time $\frac{L}{R}$ the current is equal to $0.63 j(\infty)$, in time $2 \frac{L}{R}$ the current is equal to $0.86 j(\infty)$, in time $3 \frac{L}{R}$ the current is $0.95 j(\infty)$, and so forth (Fig. 188).

According to the basic equation (8.5-4), the sum of the differences of potential on the resistance Rj and on the inductance $L \frac{dj}{dt}$ is equal to the emf E_0 . It is interesting to follow each term separately. They are shown in Fig. 189. At the initial time, $j = 0$, $Rj = 0$, $E_0 = L \frac{dj}{dt}$. We say the voltage is absorbed entirely by the inductance.

As time passes the current approaches a constant value, $\frac{dj}{dt}$ tends to zero, and the voltage is absorbed by the resistance.

It is interesting to compare the solutions which formula (8.5-6) yields for the same E_0 and for different R and L . Let R_1 be small, R_2 great, L_1 small, L_2 great. For different combinations of R and L we get four curves of current as functions of time (see Fig. 190). The final current $j(\infty)$ depends on R alone, it is the same for R_1, L_1 and for R_1, L_2 ; $j(\infty)$ is also the same for the pairs of curves R_2, L_1 and R_2, L_2 . The initial rate of buildup of current depends only on the inductance L and does not depend on the resistance.

Arguing from dimensions, it is clear that the established current is proportional to the initial rate of current buildup and the time of buildup. For our definition of buildup time, the formula is correct

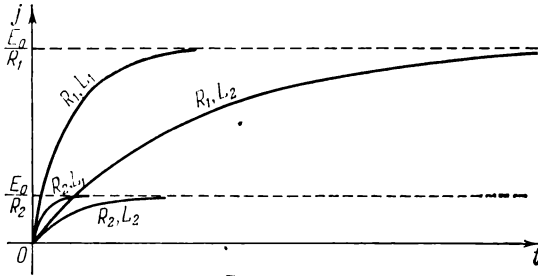


Fig. 190

without any additional coefficients. Indeed, the initial rate of current buildup $\left. \frac{dj}{dt} \right|_{t=0}$ is equal to $\frac{E_0}{L}$, the buildup time $T = \frac{L}{R}$, whence the established current is

$$j(\infty) = T \left. \frac{dj}{dt} \right|_0 = \frac{L}{R} \frac{E_0}{R} = \frac{E_0}{R}$$

Now comes the question we posed at the beginning of our discussion of setting up the initial current j_0 in the circuit in Fig. 186. We

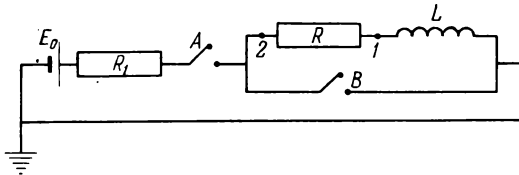


Fig. 191

can take the circuit diagram of Fig. 191. We start by closing switch A and leaving switch B open. Then a current will flow and soon attain the value $\frac{E_0}{R+R_1}$ in accordance with formula (8.5-7). We choose E_0 so that $\frac{E_0}{R+R_1} = j_0$. We wait for the steady state, when the current is equal to j_0 with A closed and B open, to be established. Then in this state we close B and open A . The result is the circuit shown in Fig. 186. At the initial instant of time (when closing B), a current j_0 flows. The potential at point 1 , prior to closing the switch, is $\varphi_1 = 0$, since in the steady state, when j_0 is constant, the voltage drop on L is zero. Prior to closing the switch, the potential at point 2 is equal to $\varphi_2 = Rj_0$. When switch B is closed, the point 2 becomes grounded

and so the potential of point 2 is $\varphi_2 = 0$. There is then a corresponding readjustment of potentials at all other points of the circuit. In particular, the potential at point 1 is now $\varphi_1 = -Rj$.

8.6 BREAKING AN INDUCTANCE CIRCUIT

Above we considered the process of establishing a current in the circuit of Fig. 187, which consisted of a voltage source, a resistance R , an inductance L and a switch. Fig. 188 shows the curve of current buildup when the switch is closed at time $t = 0$. In time, the current reaches the value $j_0 = \frac{E_0}{R}$. What will happen now if we suddenly open switch B ? If the current ceases to flow in a very short time τ , then the derivative of the current with respect to time $\frac{dj}{dt} \approx \frac{j(t+\tau) - j(t)}{\tau} = -\frac{j_0}{\tau}$, which is to say, that the derivative will be very great in absolute value if τ is very small. And at point A there will appear a very large (in absolute value) negative potential: $\varphi_A = L \frac{dj}{dt} \approx -L \frac{j_0}{\tau}$. The potential difference across the resistance R (it is equal to Rj) and the emf of the source change but slightly when the switch is opened. For this reason, the great potential difference that appears on the inductance L when the switch is opened falls entirely on the switch; by this we mean that the potential difference across the open contacts of the switch becomes very great, of the order of $L \frac{j_0}{\tau}$ and can exceed many times over the emf of the current source, E_0 . If the potential difference is great, the air gap between the open contacts will break down and a spark will jump across.

The problem of current change in a circuit when a switch is opened proves to be very complicated; this is due to the involved nature of the laws of electric discharge in air between plates. Indeed, prior to breakdown, when $\varphi < \varphi_b$, there was no current; but when breakdown occurs, the resistance of the spark falls drastically, a big current flows at a potential difference considerably less than φ_b . Here we only note the basic fact: large potential differences arise in inductance circuits in circuit breaking. When such a circuit is closed the potential difference does not exceed E_0 (the emf of the source) anywhere.

The two circuits shown in Figs. 192 and 193 give a quantitative idea of the phenomenon of sudden brief increases in potential difference. These circuits differ from that of Fig. 187 in that current can also flow in the inductance L when the switch B is open, so that circuit breaking occurs without any spark. However, if the resistance R

is much greater than the resistance r , a large potential difference appears across the inductance at break.

As an example, let us consider the circuit shown in Fig. 192. We assume $R \gg r$. If the switch is closed, then at an arbitrary time the

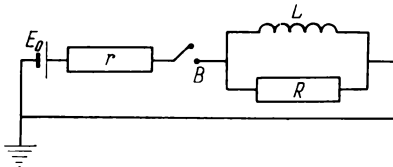


Fig. 192

current in the left portion of the circuit (r, E) is equal to the sum of the currents in the parallel connection RL :

$$j_r = j_E = j_R + j_L$$

It is then always true that $\varphi_L = \varphi_R$. Let the switch be closed at time $t = 0$. At this instant all the current will go through the resistance

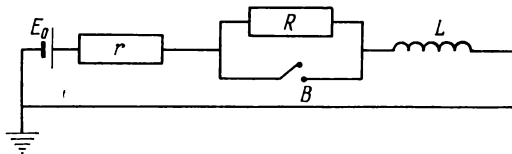


Fig. 193

R so that $j_{r0} = j_{R0} = \frac{E_0}{R+r}$ by Ohm's law. Then $\varphi_{r0} = E_0 \frac{r}{r+R}$, $\varphi_{R0} = E_0 \frac{R}{r+R}$ and consequently $\left. \frac{dj_L}{dt} \right|_{t=0} = \frac{\varphi_{R0}}{L} = \frac{E_0}{L} \frac{R}{r+R}$. After a sufficient time lapse after making the circuit, a constant current will flow. In the steady state, the entire current will go through the inductance. Indeed, if the current j does not vary with time, then $\frac{dj}{dt} = 0$, therefore $\varphi_L = 0$, and so $\varphi_R = 0$, whence $j_R = 0$.

In the steady state, $\varphi_{r\infty} = E_0$, $j_{r\infty} = j_{L\infty} = \frac{E_0}{r}$ whence it is easy to obtain the order of the time τ_1 during which the current is established:

$$j_{L\infty} - j_{L0} \approx \left. \frac{dj_L}{dt} \right|_{t=0} \tau_1$$

or

$$\frac{E_0}{r} \approx \frac{E_0}{L} \frac{R}{r+R} \tau_1$$

whence

$$\tau_1 = \frac{L(R+r)}{rR} \approx \frac{L}{r}$$

Now let us examine breaking the circuit after a time lapse of $t \gg \tau_1$ after the circuit is closed, that is to say, after a constant current $j_\infty = \frac{E_0}{r}$ has been set up in the circuit. When the circuit is broken, $j_r = j_E = 0$ and $j_R + j_L = 0$, whence $j_R = -j_L$. This means that the entire current passing through L must pass through R in the reverse direction. As before, of course, $\varphi_L = \varphi_R$. Therefore, $\varphi_R = Rj_R = -Rj_L$ or $\varphi_L = -Rj_L$. Since $\varphi_L = L \frac{dj_L}{dt}$, it follows that

$$L \frac{dj_L}{dt} = -Rj_L$$

We have equation (8.5-2), which is quite natural since the right-hand part of the diagram in Fig. 192 (after the circuit is broken) does not differ from the circuit diagram in Fig. 186.

The current diminishes by a factor of e during time $\tau_2 = \frac{L}{R}$. Here, $\tau_2 \ll \tau_1$ since $R \gg r$. At the time of break, the current has the value $j_{L\infty} = \frac{E_0}{r}$. After the break has taken place, but prior to the current falling off perceptibly, i.e., for break time $t < \tau_2$, we get

$$\varphi_R = \varphi_L = -Rj_{L\infty} = -E_0 \frac{R}{r}$$

Thus, at break we can obtain a potential difference that is many times greater than the emf of the voltage source. This principle is extensively employed in engineering, in particular in the ignition systems of internal combustion engines. Observe that this large potential difference occurs over an extremely small time interval.

The foregoing is a rough consideration of the problem without the use of derivatives and higher mathematics. An exact consideration of the problem of closing a switch in the circuit of Fig. 192 yields the following. Proceeding from the relations

$$\varphi_E + \varphi_r + \varphi_L = 0, \quad j = j_R + j_L, \quad \varphi_R = \varphi_L$$

we get the differential equation

$$\frac{dj_L}{dt} + \frac{rR}{(r+R)L} j_L = \frac{E_0 R}{(r+R)L}$$

At the initial time, $t = 0$, the current flowing through the inductance is zero: $j_L = 0$ at $t = 0$. Therefore

$$j_L = \frac{E_0}{r} [1 - e^{-\frac{rR}{(r+R)L} t}] = \frac{E_0}{r} (1 - e^{-\frac{t}{\tau_1}})$$

The current in the circuit is

$$j = j_L + j_R = \frac{E_0}{r} (1 - e^{-\frac{t}{\tau_1}}) + \frac{E_0}{R+r} e^{-\frac{t}{\tau_1}}$$

In Fig. 194, the approximate solution is shown by the broken line, the exact solution, by the smooth curve.

We advise the reader to examine the process of variation of current and potential difference at make and break in the circuit of Fig. 193. It is useful to solve the problem twice: once by setting up the differential equation and seeking its solution in the form of the exponential function, and the second time, in approximate fashion, like we did for the circuit depicted in Fig. 192.

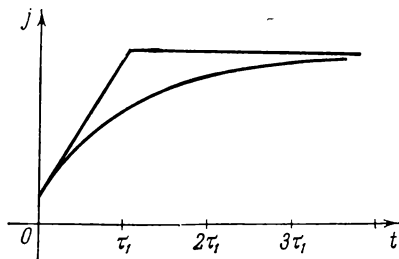


Fig. 194

8.7 THE ENERGY OF INDUCTANCE

We have seen that in a circuit consisting only of an inductance L and a resistance R , current continues to flow after the voltage source has been disconnected. The current gradually falls off in time. In the process, a heat Rj^2 is released on the resistance in unit time.

What is the source of the electric energy that is converted into heat in the resistance? It is given up by the inductance, which has a certain supply of energy.

Let us find that supply of energy by considering the elementary circuit diagram shown in Fig. 186 and let us compute the entire heat energy released on R . Suppose at the initial time, $t = 0$, this circuit has a current j_0 . The current will then decay in time in accordance with the law

$$j(t) = j_0 e^{-\frac{Rt}{L}} \quad (8.5-3)$$

The quantity of energy being released on the resistance R in unit time, that is, the rate of energy release, is the instantaneous thermal capacity h . Using (8.5-3), we find

$$h = Rj^2 = Rj_0^2 e^{-\frac{2Rt}{L}} \quad (8.7-1)$$

Knowing h , it is easy to find the total amount of heat released from time $t = 0$ to infinity (to complete decay of the current). To do this, it suffices to integrate (8.7-1) from $t = 0$ to $t = \infty$. This yields

$$Q = \int_0^{\infty} Rj_0^2 e^{-\frac{2Rt}{L}} dt = Rj_0^2 \int_0^{\infty} e^{-\frac{2Rt}{L}} dt = Rj_0^2 \frac{L}{2R} = \frac{Lj_0^2}{2} \quad (8.7-2)$$

This heat is equal to the supply of energy of the inductance through which the current j_0 flows. This supply of energy does not depend on the magnitude of the resistance R . An inductance L with current

j_0 has a definite supply of energy which, ultimately, is transformed completely into heat, irrespective of the magnitude of the resistance R . R only affects the rate of transformation of energy into heat but not the total amount of energy.

Formula (8.7-2) can also be obtained by considering the process of current buildup in an inductance. Indeed, the power of the current (that is, work per unit time) is equal to φj . This work is done by external sources of current and goes to increase the inductance energy W :

$$h = \frac{dW}{dt} = \varphi j \quad (8.7-3)$$

Using the fact that $\varphi = L \frac{dj}{dt}$ we get, from (8.7-3),

$$\frac{dW}{dt} = Lj \frac{dj}{dt} = \frac{1}{2} L \frac{d(j^2)}{dt} \quad (8.7-4)$$

We will assume that $W = 0$, at $t = 0$, $j = 0$, and $W = W_0$, at $t = t_0$, $j = j_0$. Then, integrating (8.7-4) between the limits $t = 0$ and $t = t_0$, we get

$$W_0 = \frac{1}{2} L j_0^2$$

To be specific, imagine the circuit in Fig. 187 ($\varphi = \varphi_A$) and carry through the detailed computations for buildup of the energy of inductance. In a steady-state regime, when the current has reached a constant value j_0 , $\varphi_A = 0$, the energy of inductance does not vary, but the source of emf needed to maintain the constant current j_0 must continue to generate energy, which is released as heat on the resistance R .

The energy W of inductance is proportional to the square of the current, which is to say, it is proportional to the square of the rate of motion of the electrons. Therefore, externally, W resembles kinetic energy. But is W the kinetic energy of the electrons? Let us compare the order of magnitude of W and electron energy. Using a copper wire of length $100 \text{ m} = 10^4 \text{ cm}$ and diameter 0.35 mm (cross section, 10^{-3} cm^2), we can wind a coil having inductance 0.02 henry . A current of 1 A flowing in this coil will release $W = 0.02 \times 1^2 \times 0.5 = 10^{-2} \text{ joule} = 10^5 \text{ ergs}$. We will now find the kinetic energy of the electrons. We will assume that for each atom of copper there is one electron carrying current ("conduction electron"). The atomic weight of copper is about 63, so 63 g of copper contain 6×10^{23} conduction electrons, or roughly 10^{22} electrons per gram. Copper has a density of about 8 g/cm^3 and so 1 cm^3 contains about $n = 8 \times 10^{22}$ conduction electrons. Imagine a piece of copper wire of length $v dt$ and cross-sectional area S to the left of cross section O (Fig. 195). If the electron velocity* is $v \text{ cm/sec}$, then through $S \text{ cm}^2$ there will flow $Snv dt$

* We have in view the mean velocity of their motion in the direction of current flow and not the velocity of random thermal motion.

electrons in time dt . In time dt the electrons at cross section A will move to cross section O , which means that during this time all the electrons in the volume between O and A will pass through O . That is, they will pass through the volume of a cylinder of altitude $v dt$ and base S .

Denote by e the charge in coulombs on one electron, $e = -1.6 \times 10^{-19}$ C. The amount of electricity which these $Snv dt$ electrons

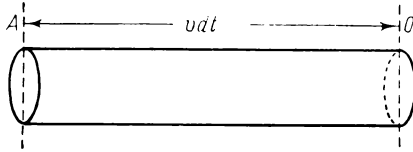


Fig. 195

transfer in time dt is equal to the current in amperes multiplied by the time dt . Therefore, $Snve dt = j dt$, whence $j = Snve$ or $v = \frac{j}{Sne}$. Substituting $j = 1$ A, $S = 10^{-3}$ cm², $n = 8 \times 10^{22}$, $e = -1.6 \times 10^{-19}$ C, we find

$$v = \frac{1}{10^{-3} \cdot 8 \cdot 10^{22} \cdot 1.6 \cdot 10^{-19}} = 0.08 \text{ cm/sec}$$

Now we have to find the kinetic energy of the electrons. The electron mass $m = 9 \times 10^{-28}$ g. The total number of electrons moving in the wire is

$$10^4 \text{ cm} \cdot 10^{-3} \text{ cm}^2 \cdot 8 \cdot 10^{22} \frac{1}{\text{cm}^3} \approx 10^{24}$$

The kinetic energy is

$$T = \frac{mv^2}{2} = 9 \cdot 10^{-28} \cdot 10^{24} \cdot \frac{0.08^2}{2} \approx 3 \cdot 10^{-6} \text{ erg}$$

Thus, the kinetic energy of the electrons constitutes a minute fraction of the inductance energy, although it depends on the current via the same law (it is proportional to j^2) as the inductance energy. The physical energy of inductance is the energy of the *magnetic field* which appears in the coil when current flows through it.

Let us point out some similarities and differences between capacitance and inductance. Both capacitance and inductance can serve as reservoirs of energy. Inductance and capacitance can both be used to accumulate electric energy from a weak primary source of current and then release it quickly at the required place and time.

A capacitor can be charged with a small current j_1 during a long time t_1 ; then rapidly discharging it through a small resistance during a short time t_2 , we can obtain a large current $j_2 \approx \frac{j_1 t_1}{t_2}$, and the potential difference across the capacitor does not exceed the emf

of the primary source. A capacitor enables us to increase the current but not the voltage.

We can send a large current through an inductance under a small voltage (small emf) E_0 of the primary source. The only requirement here is that the 'resistance' of the inductance (called inductive reactance) and of the primary current source be sufficiently small. Then it takes a comparatively long time t_3 for a large current to build up in the inductance. When an inductance is shunted by a high resistance, we can obtain a large potential difference ϕ for a short time t_4 , here $\phi \approx E_0 \frac{t_3}{t_4}$. Inductance enables us to increase the voltage but not the current.

The essential practical difference between a capacitance and an inductance is that a capacitor disconnected from a current source can retain its supply of energy for a very long time—hours and even days. The discharge time of a capacitor is equal to RC , where C is the capacitance and R is the so-called leakage resistance. Using good insulators, we can obtain enormous values of R , i.e., long discharge times. An inductance in the form of a coil and short-circuited (minimal resistance, more precisely, reactance) retains its energy (if current is flowing) for only a fraction of a second.

The decay time of the current is of the order of $\frac{L}{R}$, but even with the best conductors (copper, silver), it is impossible to make $\frac{L}{R}$ greater than a few seconds for an ordinary laboratory-type coil. It will be noted that if we increase the number of turns in the coil for a given volume by using thinner wire, L will increase, but so will R , their ratio, however, to within order of magnitude, does not change. Therefore, under laboratory conditions, inductance is conveniently used for increasing voltage but not for long-term storage of energy.

Circuits involving capacitance and inductance can be used to accumulate energy from a flashlight battery, which, with an internal resistance of several ohms, yields a few volts so that the maximum power output is of the order of 1 to 2 watts. Using circuits of this kind we are able to obtain powers up to hundreds of kilowatts. But a power output of this kind lasts for a time interval of the order of 10^{-6} sec.

It has been noted that the electric energy in an inductance is quickly converted into heat due to its resistance. This assertion holds true for coils of the ordinary laboratory kind and at ordinary (normal) temperatures. In two extreme cases however this does not hold true.

1. At very low temperatures of the order of -260°C on down to absolute zero (-273°C), many metals (for instance, lead, mercury,

but not copper) pass into the so-called superconducting state. Their specific resistance (resistivity) becomes exactly equal to zero.

The Dutch scientist Kamerlingh Onnes, who discovered this phenomenon in 1911, observed a constant current in a ring circuit of superconducting material that lasted many days without any decrease in intensity. The presence of current in such a ring circuit is detected via the magnetic field of the current.

The practical application of superconductors is limited not only by the difficulty of generating low temperatures. A strong magnetic field converts a superconductor to the normal state (with finite resistance). That is why large currents cannot be transmitted through a superconductor.*

2. The relationship between inductance and resistance and the conditions of current decay vary drastically when all the dimensions of the coil are increased, particularly when passing to astronomical phenomena.**

Picture two geometrically similar coils, one of which is n times the other in size, the number of turns in both coils being the same. In the large coil, the diameter of the coil is n times greater, but so also is the height of the coil and the diameter of the wire. Suppose the coils are made of the same material. Quantities relating to the small coil will be labelled with the subscript 1, those referring to the large coil will have the subscript 2. Let us compute the relation between the resistances of the coils:

$$R_1 = \rho \frac{l_1}{S_1}, \quad R_2 = \rho \frac{l_2}{S_2}$$

where ρ is the resistivity of the coil material, l is the length of the wire, and S the cross-sectional area.

Geometrically, it is clear that

$$l_2 = nl_1, \quad S_2 = n^2 S_1$$

and, hence, that

$$R_2 = \frac{1}{n} R_1$$

The resistance is inversely proportional to n , that is, to the dimensions.

It can be proved that the inductance of the large coil is exactly n times the inductance of the small coil,

$$L_2 = nL_1$$

* In 1961 an alloy was discovered of the rare element niobium and tin in which a current up to 100 000 A/cm² and a magnetic field up to 250 000 gauss are not yet able to destroy superconductivity.

** Compare with Problem 3 of Sec. 8.2: for a capacitance with a resistance discharge time does not change when all dimensions are altered.

which means that increasing the linear dimensions of the coil n times increases the inductance n times too. The decay time of the current, τ , is of the order of $\frac{L}{R}$; consequently

$$\tau_1 = \frac{L_1}{R_1}, \quad \tau_2 = \frac{L_2}{R_2} = n^2 \frac{L_1}{R_1} = n^2 \tau_1$$

Thus the decay time of the current is proportional to the square of the dimensions. If the earth consisted of copper, the decay time of a current flowing in it would be of the order of 10^{15} to 10^{16} sec, or 10^8 years.

The conductivity of ionized gases is of the same order as the conductivity of copper. For this reason, the decay time of a current in astronomical phenomena is enormous. This means that resistance and Ohm's law do not play any role whatsoever in these phenomena. Recall Fig. 190: the current on the initial portion of the curve depends on L alone but not on R ; in astronomy we are always on the "initial portion".

Terrestrial magnetism is a magnetic field of currents flowing in the viscous molten mass of the central core of the earth. The slow motions of this molten mass in the magnetic field sustain the currents, just as the motion of the armature in a dynamo in a magnetic field sustains the current in the armature winding and in the winding of the electro-magnet.

8.8 THE OSCILLATORY CIRCUIT

Let us consider a circuit consisting of a capacitance C and an inductance L (Fig. 196). Let point B of the circuit be grounded. By formula (8.1-11), $\varphi_C + \varphi_L = 0$. Here $\varphi_L = L \frac{dj}{dt}$, $\varphi_C = \frac{q}{C}$. The voltage drop on the capacitance, φ_C , will be denoted simply φ . Then

$$\varphi + L \frac{dj}{dt} = 0 \quad (8.8-1)$$

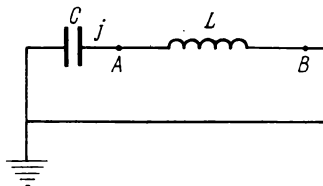


Fig. 196

Note that $j = \frac{dq}{dt}$ and so $\frac{dj}{dt} = \frac{d^2q}{dt^2}$.

Since $\frac{d^2q}{dt^2} = C \frac{d^2\varphi}{dt^2}$, it follows that, using the relation (8.8-1), we find

$$\varphi + LC \frac{d^2\varphi}{dt^2} = 0$$

or

$$\frac{d^2\varphi}{dt^2} = -\frac{1}{LC} \varphi \quad (8.8-2)$$

We considered a similar equation in Chapter 6 in the study of mechanical vibrations. It was established there that the functions

$\varphi = A \sin \omega t$ and $\varphi = B \cos \omega t$ are solutions of (8.8-2) for arbitrary A and B and a suitably chosen ω . Let us verify this, say, for $\varphi = A \sin \omega t$ and in passing we will define ω . Substitute φ and $\frac{d^2\varphi}{dt^2}$ into (8.8-2) to get

$$-ALC\omega^2 \sin \omega t = -A \sin \omega t$$

or, cancelling out $-A \sin \omega t$,

$$LC\omega^2 = 1$$

whence

$$\omega = \frac{1}{\sqrt{LC}} \quad (8.8-3)$$

Consequently, for a solution of equation (8.8-2) we have functions that describe oscillations with a circular frequency $\frac{1}{\sqrt{LC}}$. The oscillation period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC} \quad (8.8-4)$$

Let us check the dimensions of (8.8-4): the dimensions of capacitance are $[C] = \text{farad} = \frac{\text{coulomb}}{\text{volt}} = \frac{\text{ampere} \cdot \text{second}}{\text{volt}}$, the dimensions of inductance are $[L] = \text{henry} = \frac{\text{volt}}{\text{ampere/second}} = \frac{\text{volt} \cdot \text{second}}{\text{ampere}}$ so that $\sqrt{LC}$ does indeed have the dimensions of time (*second*).

Let us examine in detail the solution of equation (8.8-2). The solutions $\varphi = A \sin \omega t$ and $\varphi = B \cos \omega t$ are actually indistinguishable since the sine curve is obtained from the cosine curve by a shift along the t -axis, and so we consider one of the solutions, say,

$$\varphi = B \cos \omega t$$

The amplitude B may be arbitrary. For a given $\varphi(t)$ we find the dependence of current on time:

$$j = C \frac{d\varphi}{dt} = -CB\omega \sin \omega t$$

Let us find the energy of capacitance and the energy of inductance:

$$W_C = \frac{C\varphi^2}{2} = \frac{CB^2}{2} \cos^2 \omega t, \quad W_L = \frac{Lj^2}{2} = \frac{LC^2B^2\omega^2}{2} \sin^2 \omega t$$

Substituting the expression (8.8-3) for ω we get

$$W_L = \frac{CB^2}{2} \sin^2 \omega t$$

The total energy is independent of time, as was to be expected. Indeed,

$$P = W_C + W_L = \frac{CB^2}{2} (\cos^2 \omega t + \sin^2 \omega t) = \frac{CB^2}{2}$$

To summarize, the motion of charges in a capacitance-inductance circuit is similar to the motion of a mass attached to a spring. The energy of a charged capacitor may be likened to the elastic energy of a spring, which is a maximum when the mass is in the extreme position of maximum separation from the equilibrium position. The energy of inductance may be likened to the kinetic energy of a moving mass. When the charge on a capacitance is equal to zero, the current reaches its maximum (in absolute value); at this instant the capacitance energy is zero and the inductance energy is equal to the total energy ($\cos^2 \omega t = 0$, $\sin^2 \omega t = 1$). This is exactly what happens in the oscillations of a mass on a spring: when the mass passes through the position of equilibrium, the potential energy is zero and the kinetic energy is equal to the total energy of the oscillations.

Let us use the term general problem for the problem of finding the potential in a circuit provided that at the initial time $t = 0$, $j = j_0$, $\varphi = \varphi_0$. Neither the solution $\varphi = A \sin \omega t$ nor the solution $\varphi = B \cos \omega t$ enables us to solve the general problem. To solve the general problem we form the sum

$$\varphi = A \sin \omega t + B \cos \omega t \quad (8.8-5)$$

It is easy to verify that this sum is the solution of equation (8.8-2).

Here

$$j = C \frac{d\varphi}{dt} = CA\omega \cos \omega t - CB\omega \sin \omega t \quad (8.8-6)$$

Setting $t = 0$ in (8.8-5) and (8.8-6), we get

$$\varphi(0) = B = \varphi_0, \quad j(0) = CA\omega = j_0$$

whence we find the solution with given φ_0 and j_0 :

$$\begin{aligned} \varphi &= \frac{j_0}{C\omega} \sin \omega t + \varphi_0 \cos \omega t, \\ j &= j_0 \cos \omega t - C\varphi_0\omega \sin \omega t \end{aligned} \quad (8.8-7)$$

We leave it to the reader to verify that for such oscillations the total energy is constant and is equal to the initial energy

$$\frac{C\varphi_0^2}{2} + \frac{Lj_0^2}{2}$$

The expression (8.8-7) for φ may be written as

$$\varphi = \varphi_m \cos(\omega t + \alpha) \quad (8.8-8)$$

with amplitude φ_m . When $\cos(\omega t + \alpha) = \pm 1$, the entire energy is the capacitance energy, that is, $\frac{C\varphi_m^2}{2}$. Since the total energy is

conserved, we find $\frac{C\varphi_m^2}{2} = \frac{C\varphi_0^2}{2} + \frac{Lj_0^2}{2}$, whence $\varphi_m = \sqrt{\varphi_0^2 + \frac{L}{C} j_0^2}$.

From the expression (8.8-8), we find $j = -C\varphi_m\omega \sin(\omega t + \alpha) + C\varphi_m\omega \sin(\omega t + \alpha_1) = j_m \sin(\omega t + \alpha_1)$, where $\alpha_1 = \pi + \alpha$, $j_m = C\varphi_m\omega$. The law of conservation of energy yields

$$\frac{C\varphi_0^2}{2} + \frac{Lj_0^2}{2} = \frac{Lj_m^2}{2}$$

whence

$$j_m = \sqrt{j_0^2 + \frac{C}{L} \varphi_0^2}$$

The values of φ_m and j_m can of course be obtained without resorting to energy reasoning, simply by using the formulas of trigonometry.

The circuit diagram for generating oscillations is shown in Fig. 197. Here we have the voltage source E_0 . If we close A and leave B open, then after a lapse of time $\tau \gg RC$ after closing the circuit the capacitance will be charged to potential E_0 . Open A and at time $t = 0$

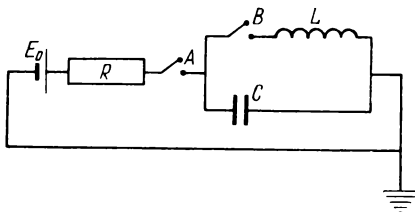


Fig. 197

close B . Then oscillations in an LC circuit will set in with $\varphi = \varphi_0 = E_0$, $j = 0$ at time $t = 0$. Note that with these oscillations, the potential difference across the electrodes of the open switch A will vary periodically from 0 to $2E_0$.

There is another way of setting up oscillations in the circuit of Fig. 197. First close both switches A and B . Then the current flowing in the circuit will be $j_0 = \frac{E_0}{R}$. At $t = 0$ open A . Then in an LC circuit, oscillations will set in, and at the initial time $\varphi_0 = 0$, $j_0 = \frac{E_0}{R}$. For these oscillations, the maximum amplitude of the potential will reach

$$\varphi_m = j_0 \sqrt{\frac{L}{C}} = E_0 \frac{1}{R} \sqrt{\frac{L}{C}}$$

It will be recalled that in a circuit without capacitance, when we break the circuit containing inductance L , the potential difference developed on the switch is the larger, the greater the resistance of the air gap between the electrodes of the switch. When such a circuit (with no capacitance) is broken, there is always a discharge in the air gap between the contacts of the switch. In the case of

capacitance, the maximum potential difference between the contacts of switch A does not exceed a definite value ($E_0 + \varphi_m$). If this value is less than what is required to ignite the discharge in the air gap of the switch, there will be no discharge. We say that the capacitance C extinguishes the discharge when an inductance circuit is broken.

Note that the quantity $\frac{1}{R} \sqrt{\frac{L}{C}}$ may be greater than unity. Then by opening the switch B a quarter-period after opening A , we obtain a potential on capacitance C that is higher than the potential E_0 of the current source.

Exercises

1. Determine α in formula (8.8-8).
2. Consider the variation of potential with time in the circuit shown in Fig. 198. Determine the greatest value of φ and the time required to attain it.

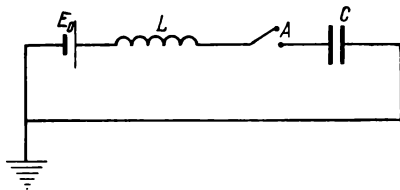


Fig. 198

Assume that switch A is closed at time $t = 0$.

3. In the preceding problem, find the energy of capacitance and the energy released by the current source when φ is a maximum.

8.9 DAMPED OSCILLATIONS

Let us consider a circuit with a resistance R in series with an inductance (Fig. 199). We assume R to be small. If R is not taken into

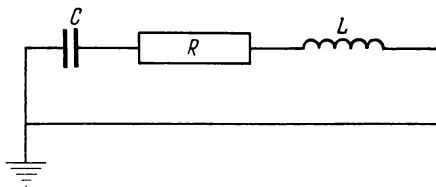


Fig. 199

consideration at all, then we get the circuit diagram of Fig. 196, which was studied in Sec. 8.8. If at $t = 0$, $\varphi = \varphi_0$ and $j = 0$, then by (8.8-7), we have

$$\varphi = \varphi_0 \cos \omega t, \quad j = j_m \sin (\omega t + \pi) \quad (8.9-1)$$

where we put

$$j_m = C\varphi_0\omega, \quad \omega = \frac{1}{\sqrt{LC}} \quad (8.9-2)$$

The total energy is then $P = \frac{C\varphi_0^2}{2}$ or, using (8.9-2), we can also write

$$P = \frac{Lj_m^2}{2} \quad (8.9-3)$$

When a resistance is present, electric energy is converted into thermal energy. The thermal capacity h is equal to

$$h = Rj^2 = Rj_m^2 \sin^2(\omega t + \pi) = Rj_m^2 \sin^2 \omega t = \frac{Rj_m^2}{2} (1 - \cos 2\omega t) \quad (8.9-4)$$

In the case of electric oscillations, the thermal capacity does not remain constant; over each period (cycle), h twice reaches a maximum and twice becomes zero (the sign of course does not change). Let us find the mean value of h over one period. From formula (8.9-4) we find $\bar{h} = \frac{Rj_m^2}{2} (1 - \overline{\cos 2\omega t})$. Recalling that the mean value of the cosine over one period is zero, we get

$$\bar{h} = \frac{Rj_m^2}{2}$$

Heat release on resistance R can only occur as the result of a reduction in the electric energy P . Therefore

$$\frac{dP}{dt} = -h \quad (8.9-5)$$

We assumed that R was small and so h is small. The oscillation energy falls off slowly and an appreciable change in energy becomes noticeable only after several cycles. Considering time intervals that are large compared with the oscillation period T , we replace h by $\bar{h}$ in the right member of (8.9-5):

$$\frac{dP}{dt} = -\bar{h} = -\frac{Rj_m^2}{2} \quad (8.9-6)$$

Since the energy P varies slowly, from (8.9-3) we see that j_m too is a slowly varying quantity. Expressing j_m from (8.9-3), we get

$$j_m = \sqrt{\frac{2P}{L}} \quad (8.9-7)$$

Using (8.9-7), from (8.9-6) we get

$$\frac{dP}{dt} = -\frac{R}{L} P$$

The solution of this equation is

$$P = P_0 e^{-\frac{R}{L}t}$$

where P_0 is the value of P at $t = 0$. Therefore, according to (8.9-7),

$$j_m = \sqrt{\frac{2P_0}{L}} e^{-\frac{Rt}{2L}}$$

Then

$$j = \sqrt{\frac{2P_0}{L}} e^{-\frac{Rt}{2L}} \sin(\omega t + \pi) \quad (8.9-8)$$

Recalling that $\varphi = \varphi_0 \cos \omega t$ and $\varphi_0 = \frac{j_m}{C\omega}$, we get

$$\varphi = \frac{j_m}{C\omega} \cos \omega t = \frac{1}{C\omega} \sqrt{\frac{2P_0}{L}} e^{-\frac{Rt}{2L}} \cos \omega t \quad (8.9-9)$$

Formulas (8.9-8) and (8.9-9) show that if a small resistance is present the electric oscillations damp out via the exponential law.

The solution given above was obtained by means of an approximate calculation. Note that in this approximate solution the relation $j = C \frac{d\varphi}{dt}$ is not satisfied, although it holds the more exactly the smaller R is. Now let us try to solve the problem exactly. For the circuit shown in Fig. 199 we have the relation $\varphi + \varphi_R + \varphi_L = 0$, whence

$$\varphi + Rj + L \frac{dj}{dt} = 0 \quad (8.9-10)$$

and $j = C \frac{d\varphi}{dt}$. Substituting the expression for j and $\frac{dj}{dt}$ into (8.9-10), we find

$$LC \frac{d^2\varphi}{dt^2} = -\varphi - RC \frac{d\varphi}{dt} \quad (8.9-11)$$

We will seek the solution of equation (8.9-11) in the form obtained in the approximate consideration, i.e.,

$$\varphi = Ae^{-\lambda t} \cos \omega t \quad (8.9-12)$$

where λ , ω , A are constants that must be determined. We put the expressions for φ and its derivatives into equation (8.9-11) and cancel the common factor $Ae^{-\lambda t}$ out of all terms to get

$$\begin{aligned} LC\lambda^2 \cos \omega t + 2LC\lambda\omega \sin \omega t - LC\omega^2 \cos \omega t \\ = -\cos \omega t + RC\lambda \cos \omega t + RC\omega \sin \omega t \end{aligned}$$

For this equation to be valid for arbitrary t , it is necessary that the coefficients of $\cos \omega t$ and $\sin \omega t$ be equal separately on the right and on the left:

$$LC\lambda^2 - LC\omega^2 = RC\lambda - 1, \quad (8.9-13)$$

$$2LC\lambda\omega = RC\omega \quad (8.9-14)$$

The condition (8.9-14) yields $\lambda = \frac{R}{2L}$. Then, from (8.9-13) we get

$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad (8.9-15)$$

The constant A was not determined from equation (8.9-11). The magnitude of this constant is determined from the initial condition: at $t = 0$, $\varphi = \varphi_0$. Finally, knowing $\varphi(t)$, we can easily find $j = C \frac{d\varphi}{dt}$. We then have

$$j = -CAe^{-\lambda t} (\omega \sin \omega t + \lambda \cos \omega t) \quad (8.9-16)$$

Comparing the exact solution with the approximate solution, we note the following: (1) in the approximate consideration of the problem we correctly determined the number λ , which describes the rate of decay of the oscillations. However, the approximate solution does not yield the dependence of frequency ω on the magnitude of the resistance R ; (2) the formula for current is somewhat different from the one that was obtained in approximate fashion.

In exactly the same way we can show that the equation (8.9-11) has yet another solution:

$$\varphi = Be^{-\lambda t} \sin \omega t \quad (8.9-17)$$

where ω and λ are the same. The corresponding current is

$$j = CBe^{-\lambda t} (\omega \cos \omega t - \lambda \sin \omega t) \quad (8.9-18)$$

The sum of the solutions (8.9-12) and (8.9-17) is also a solution of equation (8.9-11). It is only with the aid of this sum that we can solve the general problem: to find the solution of equation (8.9-11) with the initial condition $\varphi = \varphi_0$, $j = j_0$ at $t = 0$. Indeed, for the coefficients A and B we then get the equations $\varphi_0 = A$, $j_0 = CA\lambda - CB\omega$, whence

$$A = \varphi_0, \quad B = \frac{C\lambda\varphi_0 - j_0}{C\omega}$$

Exercises

1. Find $j(t)$ in the circuit of Fig. 199 if $C = 1$, $L = 1$, $R = 0.1, 0.5, 1$. At $t = 0$, $\varphi = 1$, $j = 0$.

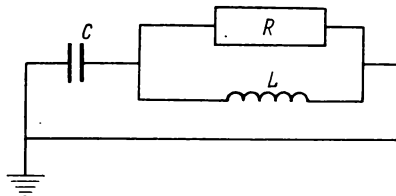


Fig. 200

2. The same question if at $t = 0$, $\varphi = 0$, $j = 1$.
3. Using the approximate method, find the rate of decay λ of oscillations in the circuit shown in Fig. 200 on the assumption that R is very great.

8.10 THE CASE OF A LARGE RESISTANCE

The case considered here of a large resistance is mainly of mathematical interest and is not connected with the sequel. It may therefore be skipped in a first reading.

The solution of equation (8.9-11) obtained in the preceding section is valid only for R that are not too large. Indeed, from (8.9-15) it is seen that if $R > 2\sqrt{\frac{L}{C}}$, then ω is meaningless since the radicand is negative. In that case, (8.9-11) has a different kind of solution. We will seek the solution in the form $\varphi = Ae^{-\beta t}$ (and, accordingly, $j = -AC\beta e^{-\beta t}$). Substituting into (8.9-11) the expressions for φ and its derivatives and cancelling $Ae^{-\beta t}$ out of all terms, we get

$$LC\beta^2 = -1 + RC\beta$$

This is a quadratic equation in β . Solving it, we find

$$\beta = \frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} \quad (8.10-1)$$

The radicand in (8.10-1) differs in sign from the radicand in (8.9-15) for ω . Hence, in those cases where it is impossible to find ω we can find β . Formula (8.10-1) yields two distinct values of β , and so we can set up two solutions to equation (8.9-11):

$$\varphi = Ae^{-\beta_1 t} \quad \text{and} \quad \varphi = Be^{-\beta_2 t}$$

Their sum is also a solution:

$$\varphi = Ae^{-\beta_1 t} + Be^{-\beta_2 t} \quad (8.10-2)$$

Accordingly

$$j = -AC\beta_1 e^{-\beta_1 t} - BC\beta_2 e^{-\beta_2 t} \quad (8.10-3)$$

If $j = j_0$, at $t = 0$, $\varphi = \varphi_0$, then, assuming $t = 0$ in (8.10-2) and (8.10-3), we get

$$A + B = \varphi_0, \quad -AC\beta_1 - BC\beta_2 = j_0$$

We can find A and B from this system of equations.

Let us consider in more detail the expression for β . Let $R \gg 2\sqrt{\frac{L}{C}}$. Then $\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} = \frac{R}{2L} \sqrt{1 - \frac{4L}{R^2C}}$ can be expanded by the binomial theorem. We confine ourselves to two terms: $\frac{R}{2L} \times \sqrt{1 - \frac{4L}{R^2C}} = \frac{R}{2L} \left(1 - \frac{1}{2} \cdot \frac{4L}{R^2C}\right) = \frac{R}{2L} - \frac{1}{RC}$. Therefore, $\beta_1 = \frac{R}{2L} + \frac{R}{2L} - \frac{1}{RC} = \frac{R}{L} - \frac{1}{RC} \approx \frac{R}{L}$; since R is great, $\beta_2 = \frac{R}{2L} - \frac{R}{2L} + \frac{1}{RC} = \frac{1}{RC}$. These values β_1 and β_2 are familiar from

Secs. 8.4 to 8.5. Indeed, β_1 corresponds to current decay by the law $e^{-\frac{R}{L}t}$, which means this is an RL circuit (see Sec. 8.5). The second root β_2 corresponds to current decay by the law $e^{-\frac{t}{RC}}$, which means this is an RC circuit (see Sec. 8.2)

Of mathematical interest is the particular case where the radicand in (8.10-1) is exactly zero:

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

so that both roots β_1 and β_2 coincide. We obtain only one solution to equation (8.9-11). But in order to solve the problem with initial conditions $\varphi = \varphi_0$ and $j = j_0$ at $t = 0$, we need two solutions.

How can we find the second solution? Suppose that $\beta_1 \neq \beta_2$ but $\beta_1 - \beta_2$ is a small quantity. Then we have two solutions: $e^{-\beta_1 t}$ and $e^{-\beta_2 t}$. Their difference is also a solution. We write this solution as

$$e^{-\beta_1 t} - e^{-\beta_2 t} = e^{-\beta_2 t} [e^{(\beta_2 - \beta_1)t} - 1]$$

Since $\beta_2 - \beta_1$ is small, it follows that (in the Taylor series only two terms can be retained) $e^{(\beta_2 - \beta_1)t} \approx 1 + (\beta_2 - \beta_1)t$, whence

$$e^{-\beta_1 t} - e^{-\beta_2 t} = e^{-\beta_2 t} t (\beta_2 - \beta_1)$$

This last expression suggests that in case $\beta_2 = \beta_1 = \beta$ the second solution should be taken in the form $\varphi = Bte^{-\beta t}$. Substituting this φ into equation (8.9-11) and noting that $\beta = \frac{R}{2L}$, we see that the equation is indeed satisfied. Thus, when $\beta_1 = \beta_2 = \beta$ we must take φ in the form

$$\varphi = Ae^{-\beta t} + Bte^{-\beta t}$$

This φ (and the corresponding j) permits solving the problem with arbitrary initial φ_0 and j_0 .

Exercises

1. Find $\varphi(t)$ when $t = 0$, $\varphi_0 = 1$, $j_0 = 0$ for $L = 1$, $C = 1$, $R = 2, 6, 10$.
2. Find $\varphi(t)$ for $L = 1$, $C = 1$, $R = 2, 4$ provided that at $t = 0$, $\varphi_0 = 1$, $j_0 = 1$.

8.11 ALTERNATING CURRENT

In contrast to the circuit diagrams considered up to now, we will examine circuits in which the voltage source has an emf that varies periodically with time with a definite given frequency ω . These problems are very important in radio-circuit work. The frequency of alternating current exerts quite a different effect on the passage of current through an inductance and a capacitance. The higher the frequency, the faster the current varies and the "harder" it is for the current to pass through an inductance and the greater the potential

difference that a current of a given intensity can set up. Contrariwise, the potential difference on the plates of a capacitor is the smaller, the greater the frequency. When the frequency is increased, the period diminishes and consequently the time interval decreases during which the current flowing in one direction can charge up the capacitor. Therefore, as the frequency increases, the charge on the capacitor decreases and so also the potential difference on the plates of the capacitor.

We have already pointed out (Sec. 8.8) that the movement of charges in an LC circuit (inductance and capacitance) may be likened to the oscillations of a body suspended from a spring: whereas in the case of a vibrating body the distance of the body from the origin and also its velocity vary periodically with time, in a circuit the potential and current vary periodically.

The frequency with which a body vibrates under the action of the elastic force of the spring (in the absence of any other forces) is called the natural frequency. Similarly, the frequency of oscillation of potential in an LC circuit is termed the natural frequency of the circuit.

Developing this analogy further, we can assume that if the circuit is connected to an alternating current network (which means the potential impressed on the circuit will vary periodically), then we will have what is called resonance. What resonance means is that the amplitude of oscillation is a maximum when the frequency ω of the current is equal to the natural frequency ω_0 of the circuit. The amplitude increases sharply when $(\omega - \omega_0)$ approaches zero. Resonance actually does take place and we will consider it in Sec. 8.13.

For every two-terminal network (see page 368, Fig. 175) connected to an alternating-current circuit, there is a definite relationship between the potential difference and the current. Let us find this relation first for the simplest case of separate elements R , L , C and then (in Secs. 8.13 and 8.14) for more complicated circuits.

We will consider alternating current of a definite frequency ω ; as before, the frequency ω is connected with the period by the relation

$$\omega = \frac{2\pi}{T}$$

Thus, for example, in the USSR the standard current is 50 cycles per second: $T = \frac{1}{50}$ sec, $\omega = 2\pi \cdot 50 = 314 \frac{1}{\text{sec}}$.

We refer to the circuit diagram in Fig. 201, which contains an ammeter A that indicates current flow j , a voltmeter V measuring the voltage (difference of potential). Suppose that the ammeter and voltmeter are so inertialess (high-speed) that they permit measuring the instantaneous value of current at each instant and, hence, their readings are measured at periods equal to the period of the current.

This experiment is usually accomplished with the aid of an oscillograph (a so-called loop oscillograph with two loops) or with the help of a cathode-ray oscillograph with two beams. The positive direction of current is shown by the arrow. The voltmeter V measures

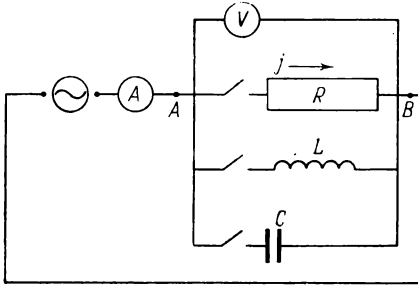


Fig. 201

$\varphi = \varphi_A - \varphi_B$. By closing one or another of the switches, we can investigate the current flowing through the resistance, inductance or capacitance.

Suppose the current is varying with time in accordance with the law

$$j = j_0 \cos(\omega t + \alpha) \quad (8.11-1)$$

If this current flows through resistance R , then, by Ohm's law,

$$\varphi_R = Rj = Rj_0 \cos(\omega t + \alpha) \quad (8.11-2)$$

For the sake of generality, let us write this equation as

$$\varphi_R(t) = \varphi_1 \cos(\omega t + \alpha_1), \quad \text{where } \varphi_1 = Rj_0, \alpha_1 = \alpha$$

Let the current (8.11-1) flow through inductance L . Then

$$\varphi_L = L \frac{dj}{dt} = -L\omega j_0 \sin(\omega t + \alpha)$$

Set

$$\varphi_L = \varphi_2 \cos(\omega t + \alpha_2) \quad (8.11-3)$$

Then $\varphi_2 = L\omega j_0$, $\alpha_2 = \alpha + \frac{\pi}{2}$. Indeed, we know that $\cos\left(\beta + \frac{\pi}{2}\right) = -\sin \beta$ for arbitrary β and so

$$\cos\left(\omega t + \alpha + \frac{\pi}{2}\right) = -\sin(\omega t + \alpha)$$

Thus, in the case of alternating current the relationship between the amplitude of the current j and the amplitude of the voltage φ_2 in the inductance is the same as in a resistance equal to $R_2 = L\omega$. If L is expressed in henrys and ω in reciprocal seconds, then R_2 will be expressed in ohms.

Inductance differs from resistance in that the curve of the voltage is displaced a quarter-period from the current curve (Fig. 202). This is quite evident from the formula

$$-\sin(\omega t + \alpha) = \cos\left(\omega t + \alpha + \frac{\pi}{2}\right) = \cos\left[\omega\left(t + \frac{\pi}{2\omega}\right) + \alpha\right]$$

Let the function $\cos(\omega t + \alpha)$, to which the current is proportional, reach a definite value at time t_1 :

$$\cos(\omega t_1 + \alpha) = a$$

The function $\cos\left(\omega t + \alpha + \frac{\pi}{2}\right)$, to which the voltage on the inductance is proportional, reaches the same value a at a different time t_2 so that

$$\cos\left(\omega t_2 + \alpha + \frac{\pi}{2}\right) = a = \cos(\omega t_1 + \alpha)$$

Therefore, $\omega t_2 + \frac{\pi}{2} = \omega t_1$, whence $t_2 = t_1 - \frac{\pi}{2\omega} = t_1 - \frac{T}{4}$, which means the voltage leads the current by one quarter of a period.

Quite naturally, we can add any integral number of periods to t_1 and write $t_2 = t_1 - \frac{T}{4} + T = t_1 + \frac{3}{4}T$ or $t_2 = t_1 + \frac{7}{4}T$. The

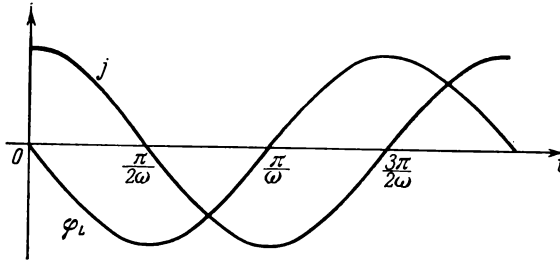


Fig. 202

formula indicates the smallest (in absolute value) time shift that carries the current curve into the voltage curve.

Let us consider the case of capacitance. Here, $j = C \frac{d\varphi_C}{dt}$ and so*

$$\varphi_C = \frac{1}{C} \int j dt = \frac{1}{C} \int j_0 \cos(\omega t + \alpha) dt = \frac{j_0}{C\omega} \sin(\omega t + \alpha) \quad (8.11-4)$$

Writing φ_C as $\varphi_C = \varphi_3 \cos(\omega t + \alpha_3)$, we get

$$\varphi_3 = \frac{1}{C\omega} j_0, \quad \alpha_3 = \alpha - \frac{\pi}{2}$$

Thus, in an alternating-current circuit the relationship between amplitude of current and amplitude of voltage is the same on a capacitance as on a resistance equal to $R_3 = \frac{1}{C\omega}$. Expressing capacitance in farads and frequency in reciprocal seconds, we obtain R_3 in ohms.

In a capacitance, the voltage curve is shifted *forwards* with respect to the current curve by a quarter-period (Fig. 203). Thus the curve of voltage φ_C in the capacitance is shifted in the opposite direction to the curve of voltage φ_L in the inductance.

* The constant of integration is equal to $\overline{\varphi_C}$. For alternating current it is always true that $\overline{\varphi_C} = 0$.

For a given identical current, φ_L and φ_C are of opposite sign. If the curves of φ_L and φ_C are brought to coincidence, we will see that the current flowing through the capacitance and the current flowing through the inductance have opposite signs. Indeed, all formulas

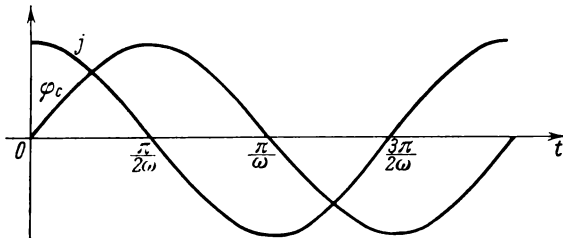


Fig. 203

expressing φ as a function of j can readily be inverted to express j as a function of φ . We write them side by side:

$j = j_0 \cos(\omega t + \alpha)$ $\varphi_R = R j_0 \cos(\omega t + \alpha)$ $\varphi_L = L \omega j_0 \cos\left(\omega t + \alpha + \frac{\pi}{2}\right)$ $= -L \omega j_0 \sin(\omega t + \alpha)$ $\varphi_C = \frac{1}{C \omega} j_0 \cos\left(\omega t + \alpha - \frac{\pi}{2}\right)$ $= \frac{1}{C \omega} j_0 \sin(\omega t + \alpha)$	$\varphi = \varphi_0 \cos(\omega t + \alpha)$ $j_R = \frac{\varphi_0}{R} \cos(\omega t + \alpha)$ $j_L = \frac{\varphi_0}{L \omega} \cos\left(\omega t + \alpha - \frac{\pi}{2}\right)$ $= \frac{\varphi_0}{L \omega} \sin(\omega t + \alpha)$ $j_C = \varphi_0 C \omega \cos\left(\omega t + \alpha + \frac{\pi}{2}\right)$ $= -\varphi_0 C \omega \sin(\omega t + \alpha)$
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The opposite phase shift and the opposite signs in the formulas referring to inductance and capacitance are of crucial importance when considering L and C connected in one circuit (LC circuits).

In alternating-current experiments, one frequently makes use of a single-beam cathode-ray oscillograph. A voltage proportional to the current is impressed on the pair of deflection plates (deflection along the x -axis) and a voltage proportional to φ is applied to the other pair of deflection plates (deflection along the y -axis). The beam moves along a line whose equation has the form $x = aj$, $y = b\varphi$; the coefficients a and b depend on the sensitivity of the oscillograph. Since j and φ are periodic functions of time, the beam sweeps out the same curve on the screen all the time. At 50 cycles per second, the human eye cannot detect any motion of the ray and sees a solid luminous curve.

If the potential difference from the resistance, φ_R , is impressed on the vertical-deflection plates of the oscillograph, then the ray des-

cribes a straight line. True enough, for

$$x = aj = aj_0 \cos(\omega t + \alpha), \quad y = b\varphi_R = bRj_0 \cos(\omega t + \alpha)$$

Eliminating t , we find $y = \frac{a}{bR} x$. If to these plates we apply the potential difference from the capacitance, φ_C , then the result is an ellipse:

$$x = aj_0 \cos(\omega t + \alpha), \quad y = b \frac{1}{C\omega} j_0 \sin(\omega t + \alpha),$$

$$\left(\frac{x}{aj_0}\right)^2 + \left(\frac{y}{bj_0/C\omega}\right)^2 = \cos^2(\omega t + \alpha) + \sin^2(\omega t + \alpha) = 1$$

Also, an ellipse results from φ_L (inductance). If we connect $\varphi_R + \varphi_L$ or $\varphi_R + \varphi_C$, the axes of symmetry of the ellipse no longer coincide with the x - and y -axes. Thus, the shape of an oscillogram tells us what the circuit is made up of (C , L or R), what the "innards of the box" consist of (see page 368, Fig. 175).

8.12 MEAN QUANTITIES, POWER AND PHASE SHIFT

In the preceding section, the current and voltage were regarded as functions of time. However, in many cases it suffices to know the mean values of these quantities.

As an elementary case, let us consider a heating device with resistance R . We know that in a direct-current circuit the power (that is, the quantity of energy released per unit of time) is equal to $h = \varphi j = Rj^2 = \varphi^2/R$. In an alternating-current circuit the instantaneous power is

$$h(t) = \varphi(t) j(t) = R[j(t)]^2 = [\varphi(t)]^2/R$$

Therefore

$$h(t) = Rj_0^2 \cos^2(\omega t + \alpha) \quad (8.12-1)$$

Over one period, $h(t)$ becomes zero twice and reaches a maximum (equal to Rj_0^2) twice. When considering electric heaters we are usually interested in the amount of heat generated during time t which is many times greater than the period T of alternating current. We therefore find the mean value of the power for a large time interval t . By virtue of (8.12-1),

$$\bar{h} = \overline{Rj_0^2 \cos^2(\omega t + \alpha)} = Rj_0^2 \overline{\cos^2(\omega t + \alpha)} = Rj_0^2/2 \quad (8.12-2)$$

This equation is approximate, being the more exact the greater t is.

The mean value of an alternating current $\bar{j}$ is ordinarily defined as the intensity of a direct current that generates an equivalent power on a resistance R :

$$R\bar{j}^2 = \frac{Rj_0^2}{2} \quad (8.12-3)$$

From (8.12-3) we find

$$\bar{j} = \frac{1}{\sqrt{2}} j_0 \approx 0.71 j_0 \quad (8.12-4)$$

In the same way, the mean value of the voltage, $\bar{\varphi}$, is determined from the condition

$$h = \frac{\bar{\varphi}^2}{R} = \frac{\varphi_0^2}{2R} \quad \text{whence} \quad \bar{\varphi} = \frac{1}{\sqrt{2}} \varphi_0 \quad (8.12-5)$$

Instruments that measure alternating current (ammeters and voltmeters) are calibrated so that they give the mean value of current and voltage: $\bar{j}$, $\bar{\varphi}$.

From formulas (8.12-4) and (8.12-5) it follows that the maximal values of current and voltage attained in an alternating-current circuit exceed the mean values by a factor of $\sqrt{2}$. For example, in a circuit with mean voltage 220 volts the maximum instantaneous voltage reaches ± 310 volts.

From the relations (8.12-2), (8.12-4), (8.12-5) and the formula $\varphi_0 = Rj_0$ it follows that $\bar{\varphi} = R\bar{j}$ and $\bar{h} = \bar{\varphi} \cdot \bar{j}$ so that Ohm's law and the relationship between power, current and voltage on a resistance hold true for mean values.

When we considered alternating current flowing through a capacitance and an inductance, we saw that the current and voltage vary along curves that are shifted with respect to one another, although the frequency is the same. Let us consider the power in the general case of an arbitrary phase shift.

Let $j = j_0 \cos(\omega t + \beta)$, $\varphi = \varphi_0 \cos(\omega t + \beta + \alpha)$. Then

$$h(t) = j_0 \varphi_0 \cos(\omega t + \beta) \cos(\omega t + \beta + \alpha)$$

Taking advantage of a familiar trigonometric formula, we have

$$\cos(\omega t + \beta + \alpha) = \cos(\omega t + \beta) \cos \alpha - \sin(\omega t + \beta) \sin \alpha$$

Substituting, we get

$$\begin{aligned} \cos(\omega t + \beta) \cos(\omega t + \beta + \alpha) &= \cos \alpha \cos^2(\omega t + \beta) \\ &\quad - \sin \alpha \cos(\omega t + \beta) \sin(\omega t + \beta) \end{aligned}$$

Since $\overline{\cos^2(\omega t + \beta)} = 1/2$ and

$$\overline{\cos(\omega t + \beta) \sin(\omega t + \beta)} = \frac{1}{2} \overline{\sin(2\omega t + 2\beta)} = 0$$

it follows that

$$\bar{h} = j_0 \varphi_0 \cos \alpha \cdot \frac{1}{2} = \bar{j} \bar{\varphi} \cos \alpha$$

Thus the mean value of power in the general case, when there is a phase shift α , is proportional to $\cos \alpha$. In the particular case of a resistance, $\alpha = 0$, $\cos \alpha = 1$, and we return to formula (8.12-2).

In the case of a capacitance, $\alpha = -\pi/2$, $\cos \alpha = 0$, in the case of an inductance, $\alpha = +\pi/2$, $\cos \alpha = 0$. Hence, in both cases the mean power is equal to zero. This result is quite understandable,

physically speaking. In a capacitance and an inductance, electric energy is not transformed into heat, it is merely stored up. In an alternating-current circuit, a capacitance during one portion of a period takes electric energy from the circuit and stores it, only to release it back into the circuit during the other portion of the period. The same goes for inductance in an alternating-current circuit.

An ordinary transformer without any load is actually a pure inductance (if we ignore the slight losses in the wires). A current flows through the transformer with amplitude $j = \varphi/L\omega$. However, as already stated, no power is taken from the circuit due to the fact that the phase $\alpha = \pi/2$ and $\cos \alpha = 0$. It is an interesting fact that electric meters are designed to measure just the quantity $j\varphi \cos \alpha$. For this reason, a nonloaded transformer will hardly add anything to your electric bill, it will only increase the total current flowing in the wires.

If a large number of inductances (transformers, unloaded electric motors, and the like) are connected in parallel, the total current can become rather large, and then the losses in the electric wiring will be rather noticeable. This effect is an important factor relative to the electric networks of a whole city.

8.13 AN ALTERNATING-CURRENT OSCILLATORY CIRCUIT. SERIES RESONANCE

Let us now consider an alternating-current circuit comprising resistance, inductance, and capacitance connected in series (Fig. 204).

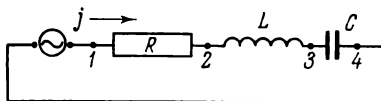


Fig. 204

It is obvious that in this system the current flowing through R , L and C is the same. We write it in the form

$$j = j_0 \cos(\omega t + \alpha) \quad (8.13-1)$$

The potential difference in the series is $\varphi = \varphi_1 - \varphi_4 = \varphi_R + \varphi_L + \varphi_C$.

Recalling the formulas (8.11-2) to (8.11-4), we get

$$\begin{aligned} \varphi &= Rj_0 \cos(\omega t + \alpha) - L\omega j_0 \sin(\omega t + \alpha) \\ &+ \frac{j_0}{C\omega} \sin(\omega t + \alpha) = Rj_0 \cos(\omega t + \alpha) + j_0 \left[\frac{1}{C\omega} - L\omega \right] \sin(\omega t + \alpha) \end{aligned} \quad (8.13-2)$$

We see from this formula that the potential difference on the inductance and on the capacitance have different signs, and so the coefficient of $\sin(\omega t + \alpha)$ is the difference of two terms. We write φ as

$$\varphi = b \cos(\omega t + \alpha + \beta) \quad (8.13-3)$$

Then b is the amplitude of the potential difference, which is to say, the maximum value of potential difference (voltage). To find b , we rewrite (8.13-3) as follows:

$$\varphi = b \cos \beta \cos(\omega t + \alpha) - b \sin \beta \sin(\omega t + \alpha)$$

Comparing this expression with (8.13-2), we find

$$b \cos \beta = R j_0, \quad b \sin \beta = j_0 \left(L\omega - \frac{1}{C\omega} \right) \quad (8.13-4)$$

Squaring (8.13-4) and adding, we obtain

$$b^2 = R^2 j_0^2 + j_0^2 \left(L\omega - \frac{1}{C\omega} \right)^2$$

whence

$$b = j_0 \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega} \right)^2} \quad (8.13-5)$$

From formula (8.13-5) it is seen that for a given amplitude of the current j_0 the amplitude of voltage b is minimal when

$$L\omega = \frac{1}{C\omega} \quad (8.13-6)$$

Writing (8.13-5) as $j_0 = \frac{b}{\sqrt{R^2 + \left(L\omega - \frac{1}{C\omega} \right)^2}}$, we see that for a gi-

ven amplitude of voltage the amplitude of current is a maximum if condition (8.13-6) is fulfilled. This condition may be written thus:

$\omega = \frac{1}{\sqrt{LC}}$. But this is nothing other than the natural frequency of an LC circuit. Therefore, the condition (8.13-6) is the condition of resonance, the condition of coincidence of the natural frequency of the circuit and the impressed frequency of the alternating current. Observe that at resonance the circuit voltage is equal to

$$\varphi = R j_0 \cos(\omega t + \alpha) \quad (8.13-7)$$

Using (8.13-1) we find that at resonance

$$\varphi = R j \quad (8.13-8)$$

Let us now pass to mean values. We determine the mean values of current and potential difference in accord with formulas (8.12-4) and (8.12-5). From the formula $\varphi_L = L\omega j_0 \sin(\omega t + \alpha)$ we obtain

$$\bar{\varphi}_L = L\omega \bar{j} = \frac{L\omega}{R} \bar{\varphi} \quad (8.13-9)$$

Similarly, from the formula $\varphi_C = \frac{1}{C\omega} j_0 \sin(\omega t + \alpha)$ we get

$$\bar{\varphi}_C = \frac{1}{C\omega} \bar{j} = \frac{\bar{\varphi}}{C\omega R} \quad (8.13-10)$$

Formulas (8.13-7) to (8.13-10) are valid only in the case of resonance, that is, when $\omega = \frac{1}{\sqrt{LC}}$. Putting this value of ω into (8.13-9) and (8.13-10), we get

$$\bar{\varphi}_L = \bar{\varphi}_C = \frac{1}{R} \sqrt{\frac{L}{C}} \bar{\varphi}$$

For this reason, when we have resonance, the voltage on an inductance and a capacitance is the greater, the smaller the resistance R ; and the quantities $\bar{\varphi}_L$ and $\bar{\varphi}_C$ can exceed the a-c source voltage $\bar{\varphi}$ many times over.

In a series circuit the resistances are additive. But the "resistances" (reactances) of a capacitance and inductance are of opposite sign and are different functions of the frequency. At resonance frequency they are equal in absolute value and hence cancel each other.

Thus, at resonance, $\omega = \omega_0$, an RLC series circuit has a *minimal* "resistance" (impedance) and carries a maximum current for a given amplitude of alternating voltage as compared with that at any non-resonance frequency, $\omega \neq \omega_0$.

It is of interest to investigate in detail how the amplitude of voltage and the amplitude of current vary in the case of departure from exact resonance, that is, when we consider $\omega \neq \frac{1}{\sqrt{LC}}$. To do this, let us take advantage of formula (8.13-3). We find that $\bar{\varphi} = b/\sqrt{2}$, whence $b = \sqrt{2} \bar{\varphi}$. Substituting this value into (8.13-5), we get

$$\bar{\varphi} = \frac{j_0}{\sqrt{2}} \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}$$

but $j_0/\sqrt{2} = \bar{j}$, and so

$$\bar{\varphi} = \bar{j} \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}$$

Finding $\bar{j}$ from this formula and putting it into (8.13-9) and (8.13-10), we obtain

$$\bar{\varphi}_L = \frac{L\omega}{\sqrt{R^2 + \left(\frac{1}{C\omega} - L\omega\right)^2}} \bar{\varphi}, \quad \bar{\varphi}_C = \frac{1}{C\omega} \frac{1}{\sqrt{R^2 + \left(\frac{1}{C\omega} - L\omega\right)^2}} \bar{\varphi}$$

Denote by ω_0 the natural frequency of the circuit. Then $\omega_0^2 = \frac{1}{LC}$.

These formulas can now be written in the form

$$\bar{\varphi}_L = \frac{\omega^2}{\sqrt{\frac{R^2\omega^2}{L^2} + (\omega_0^2 - \omega^2)^2}} \bar{\varphi}, \quad \bar{\varphi}_C = \frac{\omega_0^2}{\sqrt{\frac{R^2\omega^2}{L^2} + (\omega_0^2 - \omega^2)^2}} \bar{\varphi} \quad (8.13-14)$$

In this form it is quite evident how the amplitude depends on the closeness of the natural frequency of the circuit, ω_0 , to the alternating-current frequency ω .

The ratio $\frac{\bar{\Phi}_L}{\bar{\Phi}}$ as a function of ω near $\omega = \omega_0$ is shown in Fig. 205.

The graph is constructed for the case of $\frac{R}{L\omega_0} = 0.05$. It is illustrative of the typical resonance curve.

If $\frac{R}{L\omega} \ll 1$, then the dependence of $\frac{\bar{\Phi}_L}{\bar{\Phi}}$ and $\frac{\bar{\Phi}_C}{\bar{\Phi}}$ on ω is mainly determined by the second term of the radicand, $(\omega_0^2 - \omega^2)^2$. When $\omega = \omega_0$, this term vanishes, and for the assumption that $\frac{R}{L\omega} \ll 1$ the denominator has a minimum and the amplitude is a maximum.

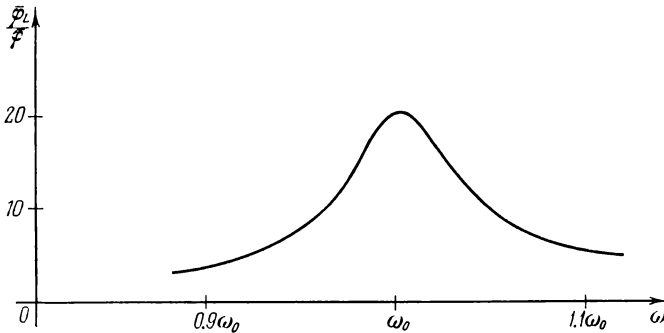


Fig. 205

The amplitude constitutes 70% of the maximum value when $(\omega_0^2 - \omega^2)^2 = \frac{R^2\omega^2}{L^2}$, that is, for $\omega_0^2 - \omega^2 = \pm \frac{R\omega}{L}$, whence

$$\omega_0 - \omega = \pm \frac{R}{L} \frac{\omega}{\omega_0 + \omega} \approx \pm \frac{R}{2L}$$

The variation of frequency for which the square of the amplitude falls to one half of the maximum value is called the bandwidth of resonance. If the amplitude is 70% (0.7 of maximum), then the square of the amplitude comes to $0.7^2 \approx 0.5$ of maximum. Therefore the bandwidth of the resonance curve $\omega - \omega_0$ is $R/2L$, which means the width is equal to a quantity that characterizes the rate of decay of oscillations in that circuit (see Sec. 8.9).

Consequently, the smaller the resistance R , the smaller the bandwidth of resonance and the steeper the curve near $\omega = \omega_0$. From the

formulas (8.13-14) it is evident that the smaller R , the greater the maximum $\frac{\bar{\varphi}_L}{\bar{\varphi}}$. That is why the phenomenon of resonance is particularly strong if R is small.

8.14 INDUCTANCE AND CAPACITANCE IN PARALLEL. PARALLEL RESONANCE

Consider the circuit in Fig. 206, which differs from that of Fig. 204 in that L and C are in parallel. We take the resistance of the circuit to be extremely small and we neglect it. Then φ_C and φ_L are the same and equal to the voltage φ in the circuit (that is, in the a-c

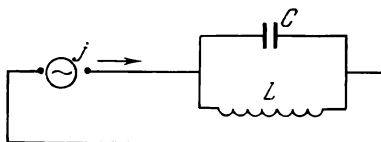


Fig. 206

source), while the current j is made up of the current j_C flowing through C combined with the current j_L flowing through L . Let $\varphi_L = \varphi_C = \varphi = \varphi_0 \cos(\omega t + \alpha)$. Using the formulas of Sec. 8.11, we find

$$j_C = -C\omega\varphi_0 \sin(\omega t + \alpha), \quad j_L = \frac{\varphi_0}{L\omega} \sin(\omega t + \alpha)$$

Therefore

$$j = j_C + j_L = \varphi_0 \left(\frac{1}{L\omega} - C\omega \right) \sin(\omega t + \alpha)$$

whence, assuming $\omega_0 = \frac{1}{\sqrt{LC}}$,

$$\bar{j} = \bar{\varphi} \left(\frac{1}{L\omega} - C\omega \right) \quad \text{or} \quad \bar{\varphi} = \frac{\bar{j}}{\frac{1}{L\omega} - C\omega} = \frac{\omega \bar{j}}{C(\omega_0^2 - \omega^2)}$$

In this case too we see a typical resonance relationship: for a given current $\bar{j}$, the voltage $\bar{\varphi}$ is the greater, the closer ω is to ω_0 . It is easy to see that when ω is close to ω_0 , $\bar{j}_L$ and $\bar{j}_C$ are much greater than the current j in the circuit. Actually, a circuit containing L and C experiences strong oscillations. A small external current suffices to sustain much stronger currents in the circuit.

It will be recalled that in a parallel circuit, conductances (which are the reciprocals of resistances) are additive:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

The “conductances” (i.e. ratios of current to potential difference) of a capacitance and an inductance have opposite sign and depend differently on the frequency. At resonance ($\omega = \omega_0$) they cancel each other and the total “conductance” is a minimum, which is to say, the current is the smallest for a given potential difference and, hence, the potential difference φ_{AB} is a maximum for the given current in the external circuit.

In a simplified circuit without resistance, the amplitude of oscillation grows without bound as ω approaches ω_0 . In reality, the resistance in the circuit makes the amplitude finite when $\omega = \omega_0$.

If R is connected in parallel with L and C , then all calculations become very similar to those of the preceding section. But this case

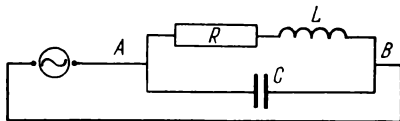


Fig. 207

is rarely encountered in practical situations. Ordinarily, the inductance has a perceptible “resistance” (inductive reactance) and therefore the typical circuit diagram is that shown in Fig. 207. In this case the calculations are somewhat longer than in the preceding section and we will not carry them through in detail. The result

for ω close to ω_0 and for small R/L ω is $\frac{\bar{I}_L}{\bar{I}} \approx \frac{\bar{I}_C}{\bar{I}} = \frac{\omega_0^2}{\sqrt{(R\omega/L)^2 + (\omega_0^2 - \omega^2)^2}}$.

It thus turns out that current amplification at resonance in a parallel circuit obeys the same law as voltage amplification in a series circuit that we discussed in Sec. 8.13.

There we obtained a formula for the bandwidth of resonance, $\omega - \omega_0 = R/2L$, which shows that the slower the decay of oscillations, the smaller the bandwidth. This does not occur only with respect to electric oscillations. We can consider any system capable of oscillation. Let an external force give rise to oscillations in such a system and then cease to operate. The system is now on its own. The oscillations begin to decay. If the amplitude of oscillation is proportional to $e^{-\gamma t}$, then the rate of decay may be characterized by the quantity γ , which has the dimensions of 1/sec. During time $\tau = 1/\gamma$ the amplitude diminishes by a factor of e , or by 63 %.

Let us now consider the resonant step-up of such a system by a periodic external force. The amplitude of oscillation at a given time is the sum of the amplitudes acquired during the time of step-up. In the case of damping, an amplitude acquired a long time before will have time to decay and will not play any part or make any contribution to the amplitude of oscillation at the given time.

The decay time is clearly $1/\gamma$. During this time the amplitude of free oscillations will have diminished e times, or by 63%. Hence, even when the step-up force is constantly operating from $t = -\infty$, the amplitude of oscillation will still be determined solely by the time interval from $t - 1/\gamma$ to t , where t is the time of observation. The action of the force at earlier times will have already decayed.

For the difference between two periodic forces with somewhat distinct periods, $F_0 \sin \omega_0 t$ and $F_0 \sin \omega t$, to manifest itself conspicuously, a time T of observation is needed during which their phases will have separated by approximately π units: $\omega T \pm \pi$ so that $|\omega - \omega_0| = \pi/T$. Consequently, if the oscillations of a system "remember" only the action of the force during time $T = 1/\gamma$, then in such a system a difference less than $|\omega - \omega_0| < \pi/T = \pi\gamma$ in the frequencies of the exciting force hardly affects the amplitude. From this we see that the bandwidth of the resonance curve is proportional to the decay γ .

On the other hand, since the system "remembers" and accumulates the action of a force during time $1/\gamma$, the amplitude itself at resonance (hence also the height of the resonance curve) is inversely proportional to γ . Calculations confirm this reasoning.

8.15 DISPLACEMENT CURRENT AND THE ELECTROMAGNETIC THEORY OF LIGHT

Up to now we have almost everywhere considered current flow through a capacitor without any reservations. Indeed, if we connect a capacitor in an alternating-current circuit in series with an ammeter, the ammeter will indicate a definite current $\bar{j} = C\omega\bar{\varphi}$. On the other hand, no current flows through a capacitor because the plates of the capacitor are separated by an insulator (air or a material) and so the individual current carriers (electrons) in the left-hand conductor and plate will never get over to the right-hand plate and conductor. Consequently, no charged particles are in motion in the space between the plates, that is, there is no electric current in the sense that we have spoken of current up to now. All there is in this space is an electric field which varies when the charge on the plates varies, which is to say, when a current flows in the right- and left-hand conductors. We can now do one of two things:

(1) either beg the reader's pardon and explain that wherever we have spoken of current flowing through a capacitor (capacitance) this was not so; actually there was no current, the only current flow being in the conductors to the right and left;

(2) or make the following assumptions. When current flows in the conductors on the right and left, the electric field in the space between the plates must vary. What this means is that a varying electric field must be regarded on a par with ordinary current (the moti-

on of charged particles). Maxwell, who suggested this view, was able to draw conclusions of tremendous significance.

It had long since been known that electric current (the motion of charged particles) gives rise to a *magnetic field*. But if a varying electric field is similar to an electric current, then an electric field varying in a vacuum should also set up a magnetic field. This hypothesis of Maxwell led to a remarkable symmetry between electric and magnetic fields. Faraday experimentally discovered induction, that is, the fact that any variation of a magnetic field gives rise to an electric field. Maxwell, in strictly theoretical fashion, hypothesized the existence of a similar phenomenon in which any variation of an electric field gives rise to a magnetic field. Only then did the theory of electric and magnetic fields acquire its modern form.

The mathematical theory of Maxwell is written in the form of differential equations that are too complicated for this book and so we do not give them. The solutions of these equations describe the propagation of electric and magnetic fields in empty space. Both fields must be present at all times: a variation in the electric field gives rise to a magnetic field, any change in the magnetic field generates an electric field.

At the time when Maxwell worked, Faraday's experiments had already been completed, the relationship between a varying magnetic field and the emf induced by it was known. Also known was the magnetic field of a current. Finally, the relationship between the charge on a capacitor and the electric field between its plates was likewise known. These findings sufficed for writing down the equations for the fields in empty space.

Maxwell found the rate of propagation of the fields in vacuo. This velocity proved to be equal to the velocity of light! From this it was natural to conclude that light is nothing other than electromagnetic oscillations. Furthermore, the theory predicted the possibility of the existence of electromagnetic oscillations of any wavelength including the X-rays (whose wavelength is thousands of times shorter than that of visible light) and radio waves with very large wavelengths. It was thus that the investigations of Faraday and Maxwell began the work which culminated in the discovery of radio waves by Hertz and the invention of radio as a means of communication by A. S. Popov.

8.16 NONLINEAR RESISTANCE AND THE TUNNEL DIODE

Let us consider a two-terminal network (a "box") which is similar to a resistance in the sense that current flowing through the box depends solely on the instantaneous value of potential difference. In this respect, the "box" is not like an inductance, where φ depends

on $\frac{dj}{dt}$, neither is it like a capacitance, where φ depends on $\int j dt$.

However the "box" differs from an ordinary resistance in that the function $j(\varphi)$ differs from Ohm's law, $j = \varphi/R$. The "box" has a more involved function $j(\varphi)$. This function is called the *characteristic curve* of the box.

The only general assertion that can be made with respect to $j(\varphi)$ is that φ and j cannot have different signs if batteries or some other sources of energy are not hidden in the box. If φ and j are of the same sign, then energy inside the box is *absorbed* during current flow, and the box takes up electric energy from the circuit in which it is connected. In the box this electric energy is converted into thermal energy and is dissipated. Since φ is constant and $\varphi < 0$ when $j < 0$ and $\varphi > 0$ when $j > 0$, it follows that $\varphi = 0$ for $j = 0$. Otherwise, the functional relation $j(\varphi)$

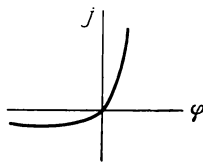


Fig. 208

can be of any kind. For example, for current rectifiers we have "boxes" whose characteristic curve is shown in Fig. 208: the current flows easily in one direction for a small potential difference and hardly at all in the other direction. As can be seen from the graph,

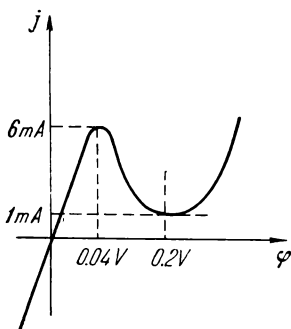


Fig. 209

the current is small even for a large negative potential difference. Such are the properties possessed by so-called diodes made of two semiconductors. In 1958 the Japanese devised a "box" made up of specially chosen semiconductors the so-called tunnel diode (in reality, this "box" is in the form of a minute cylinder just a few millimetres in diameter and altitude), which has an unusual curve of $j(\varphi)$ with a minimum (see Fig. 209 in which typical values of φ and j are indicated). This curve does not contradict the principle expressed above: the sign of φ is the same as that of j

everywhere, which means the "box" only absorbs energy. We will not go into the physical reasons for such a strange curve but we will examine the consequences for a circuit involving a tunnel diode. For the sake of brevity we will continue to call it a box.

We start with the simplest type of circuit consisting of three parts: a battery with emf E , a resistance R (the ordinary kind that obeys Ohm's law), and the box (Fig. 210). We include the internal resistance of the battery in R .

The equation defining the current and distribution of potential in the circuit is of the form $-E + Rj(\varphi) + \varphi = 0$, where φ is the

potential difference across the box and $j(\varphi)$ is the function defined by the properties of the box (see Fig. 209). The current through R is equal to the current through the box and so $\varphi_R = Rj = Rj(\varphi)$. This equation is conveniently solved by graph. We write it down as $\varphi = E - Rj(\varphi)$ and in the φj -plane we construct the straight line

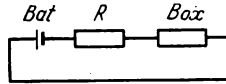


Fig. 210

$\varphi = E - Rj$. This line may be called the load curve of the battery-resistance system. The solution to the problem is given by the intersection of the straight line $\varphi = E - Rj$ (for $j = \frac{E - \varphi}{R}$) with the curve of $j(\varphi)$, which is the characteristic curve of the box. In Fig. 211 we have a graphic solution of the problem involving one battery and three different resistances: small, R_1 , medium, R_2 , and large, R_3 .

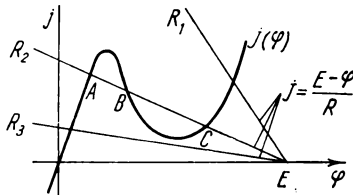


Fig. 211

From the graph we can see that for a sufficiently large E we can choose an R such that it will not be too small or too large and there will be three points of intersection, A , B , C and thus three solutions!

For three solutions to exist, the curve $j(\varphi)$ must have a descending portion. It is clear that the line on which $\frac{dj}{d\varphi} > 0$ everywhere can only once intersect the load curve, no matter what E and $R > 0$.

Now let us consider a somewhat more complicated circuit diagram involving capacitance in parallel with the box (Fig. 212). For this

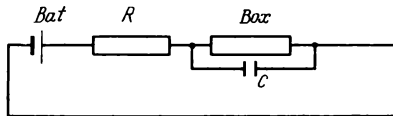


Fig. 212

circuit we find that the current through the box, $j(\varphi)$, and the current flowing through the capacitance, $C \frac{d\varphi}{dt}$, together equal the current flowing through the resistance $j(\varphi) + C \frac{d\varphi}{dt} = \frac{E - \varphi}{R}$, whence $\frac{d\varphi}{dt} = \frac{1}{C} \left[\frac{E - \varphi}{R} - j(\varphi) \right]$. The intersection points of the characteristic

curve $j(\varphi)$ of the box with the load curve $\frac{E-\varphi}{R}$ correspond to the solutions $\varphi = \text{const}$, $\frac{d\varphi}{dt} = 0$. Let us examine the sign of $\frac{d\varphi}{dt}$ near these points. A glance at Fig. 211 shows us that

$$\begin{aligned} \frac{d\varphi}{dt} > 0 \quad \text{for} \quad \varphi < \varphi_A, \quad \frac{d\varphi}{dt} > 0 \quad \text{for} \quad \varphi_B < \varphi < \varphi_C \\ \frac{d\varphi}{dt} < 0 \quad \text{for} \quad \varphi_A < \varphi < \varphi_B, \quad \frac{d\varphi}{dt} < 0 \quad \text{for} \quad \varphi > \varphi_C \end{aligned}$$

The arrows in Fig. 213 indicate the direction of variation of φ with time. This means the mean solution B is unstable: all we need to do is depart slightly to the left or to the right, and $\frac{d\varphi}{dt}$ appears with a sign such that the deviation of φ from φ_B increases!

A and C are the two stable solutions that correspond to stable states of the system.

The existence of two stable states permits using these boxes in mathematical machines as memory cells. By making a lot of such

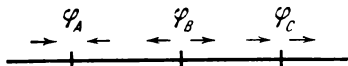


Fig. 213

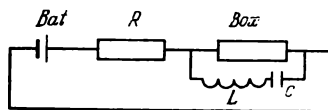


Fig. 214

circuits and transferring (by an external means) some into the A state and others into the C state, we can record ("remember") any desired number or other information. Using such systems, we record information in coded form as $AACACACCC\dots$, where each letter A or C indicates a state of the appropriate system (the first in A , the second in A , the third in C , the fourth in A , etc.).

We now consider a system (Fig. 214) consisting of an inductance and a capacitance connected in parallel with the box. We again denote by φ the potential difference across the box, $j(\varphi)$ its characteristic curve, and j_1 the current flowing through L and C :

$$j(\varphi) + j_1 = \frac{E-\varphi}{R}, \quad C \frac{d\varphi_C}{dt} = j_1, \quad \varphi - \varphi_C = \varphi_L = L \frac{dj_1}{dt}$$

We consider the process in the circuit when the current and the potential in the box are close to the middle point B of intersection. Let

$$\varphi = \varphi_B + f, \quad \varphi_C = \varphi_B + g, \quad j = j(\varphi_B) + f \frac{dj}{d\varphi} \Big|_{\varphi=\varphi_B} = j(\varphi_B) + kf$$

where k is shorthand for the derivative $\frac{dj}{d\varphi}$ when $\varphi = \varphi_B$. We used the first two terms of the Taylor series to represent current; the mean

potential of the capacitance is equal to φ_B , and φ_B and $j(\varphi_B)$ satisfy the condition $j(\varphi_B) = \frac{E - \varphi_B}{R}$. Substituting these expressions into the equations and simplifying, we get

$$kf + j_1 = -\frac{1}{R}f, \quad C \frac{dg}{dt} = j_1, \quad f - g = L \frac{dj_1}{dt} \quad (8.16-1)$$

From these equations we get $f = -\frac{1}{\frac{1}{R} + k}j_1 = -rj_1$, where

$\frac{1}{r} = \frac{1}{R} + k = \frac{1}{R} + \frac{dj}{d\varphi} \Big|_{\varphi=\varphi_B}$. From the equations (8.16-1) we find $L \frac{d^2j_1}{dt^2} = \frac{df}{dt} - \frac{dg}{dt} = -r \frac{dj_1}{dt} - \frac{1}{C}j_1$ and finally

$$\frac{d^2j_1}{dt^2} + \frac{r}{L} \frac{dj_1}{dt} + \frac{1}{LC}j_1 = 0$$

This last equation is the ordinary equation of an oscillatory circuit with capacitance C , inductance L and resistance r .

The resistance r corresponds to the fact that the capacitance and inductance are in parallel to two circuits: the circuit involving the battery and resistance R and the circuit with box and resistance $\frac{1}{k} = \frac{d\varphi}{dj}$. Since these two circuits are connected in parallel, the conductances (reciprocals of resistances) are additive, whence follows the expression for r .

The currents and potentials in the system break up into a sum of two terms: the constant term φ_B , $j(\varphi_B)$ and the oscillatory term $j_1(t)$, $f(t)$, $g(t)$. Here, for the oscillatory term the role of resistance of the box is played by the derivative $\frac{d\varphi}{dj}$ taken with respect to the characteristic curve of the box. If the box consisted of an ordinary (ohmic) resistance, $\varphi = Rj$ ($\frac{d\varphi}{dj} = R$), the derivative would be equal to the resistance.

What does this unusual characteristic curve of $j(\varphi)$ of the box (tunnel diode) of Fig. 211 lead to? At point B the derivative $\frac{dj}{d\varphi} < 0$, which means that with respect to oscillations the box has a negative resistance! What is more, from Fig. 211 it is evident that at the point of intersection $|k| = \left| \frac{dj}{d\varphi} \right| > \frac{1}{R}$ since $\frac{1}{R}$ is precisely the slope of the load curve $j = \frac{E - \varphi}{R}$ intersecting the characteristic curve at the point B . Consequently, the total impedance of the oscillatory circuit is

$$r = \frac{1}{R} + k < 0$$

The equation for an oscillatory circuit involving L , C , r for $r > 0$ yielded damped oscillations. For $r < 0$, this equation will yield step-up oscillations that reinforce with time.

Thus, a tunnel diode is capable of generating oscillations in a circuit.

The capability of generating oscillations is a consequence of the instability of point B . The oscillation energy is taken from the battery. The amplitude of oscillation increases with time by the exponential law only so long as it may be considered small and we can use the Taylor series expansion of the characteristic curve of $j(\varphi)$ about the point B . Roughly speaking, the maximum amplitude is bounded by the points A and C (Fig. 211). Already by 1961 oscillators with efficiencies up to 25% and power outputs of 0.5 milliwatt had been tested at a frequency of 7500 megahertz (at a 4-cm wavelength). For us, tunnel-diode circuits are interesting from the standpoint of a mathematical consideration of a nonlinear problem, questions of power, of stability of solutions and the representation of currents in a system as a superposition of a constant solution and oscillations.

Chapter 9

Dirac's Remarkable Delta Function

9.1 VARIOUS WAYS OF DEFINING A FUNCTION

The functions we have been studying up to now have ordinary been defined by formulas. This means that a procedure was always indicated for computing the values of the function for any given value of the independent variable. We could call this an algorithmic representation (algorithm meaning a procedure for computing something). To illustrate, take the function $y = f(x) = 2x + 3x^2$. It actually amounts to this: "take x , multiply by 2, square x , multiply by 3 and then add the two numbers to get the value of y for the given value of x ". Trigonometric functions were defined differently, by means of geometric concepts, measuring arcs and line segments in a circle.

Up to now we have studied the properties of functions specified in this fashion, the laws of their increase and decrease, maxima and minima, etc. The study of these properties leads to new modes of defining functions. For instance, the function $f = e^x$ may be defined as a function whose derivative is equal to the function itself:

$$\frac{df}{dx} = f$$

with the supplementary condition that $f(0) = 1$. The sine, the function $\varphi = \sin x$, and the cosine, or $\psi = \cos x$, may be defined as functions which satisfy one and the same equation:

$$\frac{d^2\varphi}{dx^2} = -\varphi, \quad \frac{d^2\psi}{dx^2} = -\psi$$

under different initial conditions:

$$\varphi(0) = 0, \quad \left. \frac{d\varphi}{dx} \right|_0 = 1; \quad \psi(0) = 1, \quad \left. \frac{d\psi}{dx} \right|_0 = 0$$

These definitions are in many respects more to the point, so to say, and more closely related to the applications of the exponential function and of trigonometric functions to many problems of physics, say, to problems involving oscillations, then are the definitions of $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ and of the sine and cosine in the circle.

It is curious to note that the definitions of e^x , $\sin x$, $\cos x$ by means of differential equations turn out to be convenient for electronic computers. If in a calculation the calculator has to substitute the values of e^x for different x , he copies them out of a table. When operating a computer it is more convenient and faster to have the machine compute e^x step by step by the approximate formula $e^{(x+\Delta x)} = e^x (1 + \Delta x)$ in accordance with the equation (or via more exact formulas based on the same equation) than it is to refer to a table of values. The same goes for the functions $\sin x$ and $\cos x$. It is easier to compute them every time:*

$$\sin(x + \Delta x) = \sin x + \Delta x \cos x, \quad \cos(x + \Delta x) = \cos x - \Delta x \sin x$$

Thus, one general approach to the concept of a function lies in specifying a procedure for computing it and in subsequently investigating it. There is another approach. We can seek a function with definite general properties, the aim being later on to attempt (on the basis of these properties) to find the formula that describes the function at hand. Such is the usual procedure when handling experimental data and finding empirical formulas by trial. Our object here is to construct in this way a remarkable function that is useful and important both in mathematics and in its applications.

9.2 DIRAC AND HIS FUNCTION

Paul Adrien Maurice Dirac, the celebrated English theoretical physicist, came to fame in 1929. He had elaborated a theory capable of describing the motions of electrons in electric and magnetic fields with arbitrary velocities almost up to that of light. This was the quantum theory which also accounts for the fact that the electrons in an atom move only in specific orbits with definite energy values. Dirac knew an electron possesses a definite rotational moment, or is similar to a spinning top, and he took this into account in building his theory. When the theory was constructed, it turned out that a conclusion could be drawn that Dirac had not foreseen: namely, the existence of particles with mass the same as the electron mass but with opposite (positive) charge. For two years it was thought that Dirac's theory was good for describing electron motion but the conclusion concerning particles with positive charge was erroneous and that as soon as he got rid of it the theory would be a very good one.

But in 1932, a positively charged particle—called the positron (also called the antiparticle of the electron)—was discovered! The big drawback of Dirac's theory became its triumph, its principal contribution: Dirac's discovery was the first instance of a new par-

* Where do you think these formulas come from? Check to see whether they accord with the equation for the second derivative of the sine and cosine.

ticle being discovered “at the tip of a pencil”. This is an instructive example from the standpoint of the relationships of theory and experiment. Theory rests on the findings of experiment, but a consistent, logical and mathematical development of a theory takes the investigator beyond the confines of the material used as its foundation and leads to fresh predictions.

Dirac is not only one of the best theoretical physicists in the world, he is a marvelous mathematician. In his classical “The Principles of Quantum Mechanics” Dirac introduced and made wide use of a new function which he denoted by $\delta(x)$. It is called Dirac’s delta function or, simply, the delta function.*

The delta function is defined as follows: $\delta(x) \equiv 0$ for any $x \neq 0$, that is, for $x < 0$ and for $x > 0$. When $x = 0$, $\delta(0) = \infty$. Besides, the following condition is stipulated:

$$\int_{-\infty}^{+\infty} \delta(x) dx = 1 \quad (9.2-1)$$

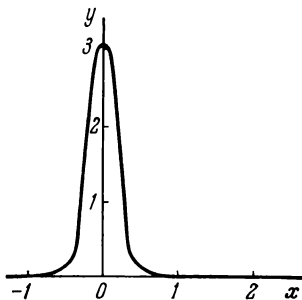


Fig. 215

Figure 215 gives a pictorial view of the graph of a function similar to the delta function, $\delta(x)$. The narrower we make the strip between the left and right

branches, the higher must it be for the strip (the integral, that is) to retain its given value of 1. As the strip becomes ever narrower, we approach the condition $\delta(x) = 0$ for $x \neq 0$, and the function approaches the delta function. In one of the following sections, these arguments will be utilized in the construction of formulas that yield the delta function. Here we continue the study of its general properties.

The most important formula of an integral involving $\delta(x)$ is of the form

$$\int_{-\infty}^{+\infty} f(x) \delta(x) dx = f(0) \quad (9.2-2)$$

Indeed, since $\delta(x) = 0$ for $x \neq 0$, it follows that the value of the integral does not depend on the values of $f(x)$ no matter what $x \neq 0$. The only essential value of $f(x)$ is that where $\delta \neq 0$, that is, for $x = 0$. This means that in the narrow domain where $\delta(x) \neq 0$ (Fig. 215), $\delta(x)$ is multiplied by $f(0)$. Hence, from the condition (9.2-1) follows the formula (9.2-2). We can also argue in reverse. We can say that $\delta(x)$ is a function such that no matter what form the

* Academician S. L. Sobolev has given a mathematical justification of functions of this kind which are called generalized functions.

auxiliary function $f(x)$, we always have the formula (9.2-2). This condition alone brings us to all the conclusions concerning the form of $\delta(x)$ that have been employed in its definition so far. From formula (9.2-2) it also follows that $\delta(x) = 0$ for $x \neq 0$ and also that* $\int \delta(x) dx = 1$ and also that $\delta(0) = \infty$.

Let us carry through a few more obvious consequences of the definition of $\delta(x)$. By the general rule of a change of variables (discussed in detail in Sec. 1.7), the function $\delta(x - a)$ is displaced a units to the right of $\delta(x)$, that is, $\delta(x - a) = \infty$ when $x = a$. Accordingly,

$$\int_{-\infty}^{+\infty} f(x) \delta(x - a) dx = f(a)$$

Now it is easy to see, if we consider a curve of the form of Fig. 215, that $b\delta(x)$ is b times higher than $\delta(x)$ and $\delta(cx)$ is $|c|$ times narrower than $\delta(x)$ so that the area under the curve $\delta(cx)$ is $|c|$ times smaller than the area under $\delta(x)$. Therefore

$$\int f(x) b\delta(x) dx = bf(0) \quad (9.2-3)$$

$$\int f(x) \delta(cx) dx = \frac{1}{|c|} f(0) \quad (9.2-4)$$

and we can say that $\delta(cx) = \frac{1}{|c|} \delta(x)$. Formula (9.2-3) is quite obvious but we can also hope that the reader who has gone through the trials and tribulations of the preceding chapters of this book will be able to grasp the formula (9.2-4) as well. Formally, it is readily obtained by the change of variable

$$y = |c|x, \quad dx = \frac{1}{|c|} dy$$

Here we also make use of the fact that the function $\delta(x)$ defined by formula (9.2-1) is an even function of its argument: $\delta(x) = \delta(-x)$.

Exercises

1. Show that for a function $\varphi(x)$ having a unique zero x_0 such that $\varphi(x_0) = 0$ we have the formula $\delta(\varphi(x)) = \frac{1}{|\varphi'(x_0)|} \delta(x - x_0)$. For the function ψ with several zeros, $\delta(\psi(x))$ is equal to the sum of such expressions over all zeros.

2. Evaluate $\int_{-\infty}^{+\infty} \psi(x) \delta(\sin x) dx$.

* Here, the integral sign without any limits of integration will always be understood as being over the range of the variable of integration from $-\infty$ to $+\infty$.

9.3 DISCONTINUOUS FUNCTIONS AND THEIR DERIVATIVES

Let us consider the integral of the function $\delta(x)$ as a function of its upper limit, that is, the function

$$\theta(x) = \int_{-\infty}^x \delta(z) dz \quad (9.3-1)$$

It is easy to see that the graph of this function has the form of a step (Fig. 216). As long as $x < 0$, the domain of integration in formula (9.3-1) is wholly located where $\delta(x) = 0$. Hence, $\theta(x) = 0$ ($x < 0$). But if $x > 0$, then the domain of integration involves the neighbourhood of the origin where $\delta(0) = \infty$. On the other hand, since

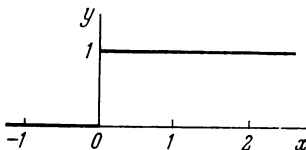


Fig. 216

for $x > 0$ $\delta(x) = 0$ as well, the value of the integral does not change when the upper limit varies from $+0.1$ to 1 or to 10 or to ∞ . Hence, for $x > 0$,

$$\theta(x) = \int_{-\infty}^x \delta(x) dx = \int_{-\infty}^{+\infty} \delta(x) dx \equiv 1$$

as is shown in Fig. 216.

Thus, with the aid of the delta function we have constructed an elementary discontinuous function $\theta(x)$ such that when $x < 0$, $\theta(x) = 0$ and in the domain $x > 0$, $\theta(x) = 1$. It is clear that when $x = 0$, θ is discontinuous between 0 and 1 . These simple considerations enable us to approach the problem of the derivative of a function having discontinuities in a more consistent fashion, without apparent exceptions and extensive reservations.

If we did not know about the delta function, we would have to say that derivatives cannot be found at points where the function is discontinuous. But we have just constructed a discontinuous function, $\theta(x)$. The general rule on the relationship between an integral and the derivative is:

$$\text{if } F(x) = \int_{x_0}^x g(z) dz, \text{ then } g(x) = \frac{dF}{dx}$$

Let us apply it to the expression (9.3-1) to get

$$\frac{d\theta(x)}{dx} = \delta(x) \quad (9.3-2)$$

Thus we do not need to make an exception for the derivative of a discontinuous function. We merely say that at the point of discontinuity the derivative is equal to a "singular" function, the delta function. We have learned to handle the derivative of an elementary discontinuous function and can now very simply find derivatives in more complicated situations. Here are some examples. Let

$$y = x, x < 1; y = x - 2, x > 1 \quad (9.3-3)$$

We refer to the graph of the function in Fig. 217. The jump occurs at $x = 1$. The magnitude of the jump is $y(1+0) - y(1-0) = -2$. Here we use the notation $y(1+0)$ to denote the limiting value of y as x approaches 1 from the right (from the direction of $x > 1$), $y(1-0)$ denotes the same on the left (see Fig. 217). From this we get

$$\frac{dy}{dx} = 1 - 2\delta(x-1) \quad (9.3-4)$$

This notation is better than the dreary statement that $\frac{dy}{dx} = 1$ everywhere except at the point $x = 1$, where the function has a discontinuity and does not have a derivative.

The delta function is a typical brainchild of the twentieth century. The nineteenth century had a passion for investing all its arguments—true, false and not quite true—in the form of "impossibilities". It is impossible to invent a perpetual motion machine, it is impossible to determine the composition of the stars, it is impossible to

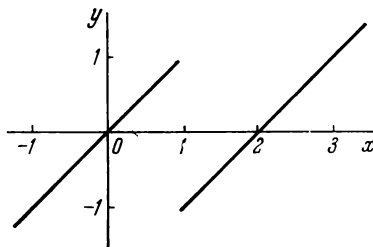


Fig. 217

find the derivative of a discontinuous function. Our present century has found numerous constructive solutions to what appeared to be impossible in the 19th century. To take an example, the delta function resolves the problem of the derivative at a point of discontinuity (at any rate for a discontinuity in the form of a finite jump). Indeed, the notation of (9.3-4) contains in one line the fact of discontinuity (since δ is involved), the site of the discontinuity ($x = 1$) and its value [the coefficient (-2) of δ].

Integrating (9.3-4) with the condition $x = 0, y = 0$, we can restore the graph of $y(x)$ in its entirety. True, in the case of the function (9.3-3) we had it easy in the sense that a simple case was specially chosen where the derivative on the left and on the right is expressed by a single formula. This of course is not obligatory. Why should the derivative be continuous if the function itself suffers a discontinuity?

Let us consider a more complicated example: $y = -x^2$, $x < 1$; $y = x^2$, $x > 1$. When $x < 1$, $y' = -2x$, for $x > 1$, $y' = +2x$. The discontinuity is associated with $y' = 2\delta(x - 1)$. We can now, at our pleasure, adjoin the point $x = 1$ to the left-hand region and then write $y' = -2x + 2\delta(x - 1)$, $x \leq 1$; $y' = +2x$, $x > 1$. Or, another version, we can adjoin $x = 1$ to the right-hand region and then, with the same full justification, we can write $y' = -2x$, $x < 1$; $y' = 2x + 2\delta(x - 1)$, $x \geq 1$. Note how the signs $<$ (less than) and $\leq$ (less than or equal to), $>$ (greater than) and $\geq$ (greater

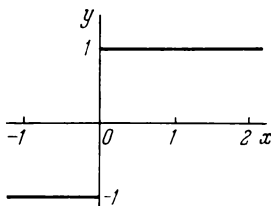


Fig. 218

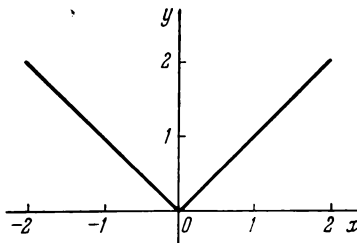


Fig. 219

than or equal to) are placed in the formulas. Be careful not to write the delta function twice.* We can also write

$$y' = \varphi(x) + 2\delta(x-1), \text{ where } \varphi(x) = \begin{cases} -2x & (x < 1), \\ 2x & (x > 1) \end{cases}$$

To verify the notation, integrate the expression of the derivative and again obtain the original discontinuous function.

Sometimes, use is made of the so-called *signum function*, $\text{sgn}(x)$, which is defined thus: $\text{sgn}(x) = -1$ for $x < 0$ and $\text{sgn}(x) = +1$ for $x > 0$. We can write $\text{sgn}(x) = \frac{x}{|x|}$, where $|x|$ is the modulus (absolute value) of x . The curve of the signum function, $y = \text{sgn } x$, is shown in Fig. 218. It is easy to see that

$$\text{sgn } x = -1 + 2\theta(x) = 1 + 2 \int_{-\infty}^x \delta(x) dx \quad (9.3-5)$$

Now let us examine the function $|x|$ itself. Its graph is shown in Fig. 219. We find the derivatives to be $\frac{d|x|}{dx} = -1$, $x < 1$; $\frac{d|x|}{dx} = +1$, $x > 1$, or, briefly, with the aid of the new function

$$\frac{d|x|}{dx} = \text{sgn } x$$

* As for the discontinuous function itself, there is no sense in asking for its value at the actual point of discontinuity. At any rate, this question is meaningless in nearly all applied problems.

From formula (9.3-5) it then follows that

$$\frac{d^2|x|}{dx^2} = 2\delta(x)$$

This is a very important formula that it would do well to get a good feeling of. We know that the second derivative is connected with the curvature of a curve on a graph and is equal to zero for the straight line. It would appear then that if the graph of $|x|$ consists of two straight lines, on each of which $\frac{d^2|x|}{dx^2} = 0$, then why not simply say that $\frac{d^2|x|}{dx^2}$ is equal to zero everywhere? Of course the crux of the matter lies in the salient point at $x = 0$ where the two straight lines meet. How are we to be sure that it is the salient point

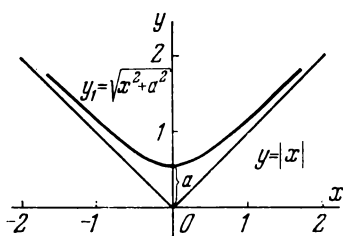


Fig. 220

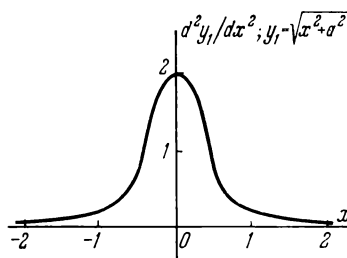


Fig. 221

that corresponds to the expression of the second derivative $2\delta(x)$? To assure ourselves of this, let us round off the salient point: take the function $y_1 = +\sqrt{x^2 + a^2}$. The smaller a (see Fig. 220), the closer this function is to the broken line $y = |x|$. It is easy to find $\frac{dy_1}{dx} = \frac{x}{\sqrt{x^2 + a^2}}$, $\frac{d^2 y_1}{dx^2} = \frac{a^2}{(x^2 + a^2)^{3/2}}$. Figure 221 depicts the graph of $\frac{d^2 y_1}{dx^2}$ (both Figs. 220 and 221 are constructed for $a = 0.5$). It is easy to see that this curve becomes higher and narrower as a decreases; the integral $\int \frac{d^2 y_1}{dx^2} dx = 2$, whence, in the limit when $a = 0$ we obtain the expression $\frac{d^2|x|}{dx^2} = 2\delta(x)$ given above.

Exercise

1. Write down the first derivatives of the following discontinuous functions:

$$(a) \ y = x, \ x < 1; \ y = x - 1, \ x > 1; \quad (b) \quad = \frac{e^{1/x}}{1 + e^{1/x}}.$$

9.4 REPRESENTING THE DELTA FUNCTION BY FORMULAS

At the end of Sec. 9.3 we inadvertently obtained a formula, the expression for the function

$$y = \frac{a^2}{2(x^2 + a^2)^{3/2}} \quad (9.4-1)$$

which approaches $\delta(x)^*$ in the limit as $a \rightarrow 0$. Let us examine this problem in detail. Let us take the function $\varphi(x)$ of which we only demand that it vanish for $x = \pm\infty$ and that the integral $I = \int \varphi(x) dx$ be nonzero. It is always possible to make this integral equal to one by multiplying φ by an appropriate constant. Suppose that this has already been done so that $\int \varphi(x) dx = 1$. It is clear that φ has a maximum somewhere between $-\infty$ and $+\infty$. The

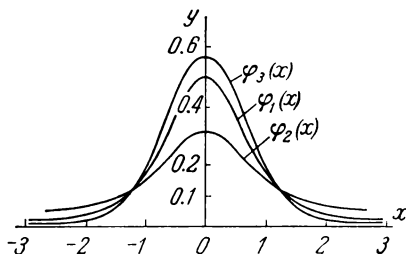


Fig. 222

simplest examples are functions that are everywhere positive and even, that is, symmetric about the y -axis [this means that $\varphi(x) = \varphi(-x)$]. Here are some concrete instances:

$$\varphi_1(x) = \frac{1}{2(1+x^2)^{3/2}}, \quad \varphi_2(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad \varphi_3(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}$$

The graph of each of these functions is bell-shaped (Fig. 222). If they are brought to the same height (see below), it is hard to distinguish them at a glance. Incidentally, the last (exponential) function is much closer than the others to the axis of abscissas for large $|x|$ far away from the maximum.** Now recall what needs to be done to increase the height of the bell n -fold and decrease the width m -fold: take $n\varphi(mx)$. If the area under the bell is to be preserved, choose $n = m$. To summarize, then, the function $n\varphi(nx) \rightarrow \delta(x)$ as $n \rightarrow \infty$ or, to put it otherwise, $\delta(x) = \lim_{n \rightarrow \infty} n\varphi(nx)$. It is also easy

* Note the "2" in the denominator of (9.4-1); without this two we would have $2\delta(x)$.

** The intersection of all three curves $\varphi_1, \varphi_2, \varphi_3$ at just about the same point is of course purely accidental.

formally to verify that $\int n\varphi(nx) dx = \int \varphi(z) dz = \int \varphi(x) dx = 1$ via the substitution $z = nx$.

Thus, to the three variants of $\varphi(x)$ correspond the following three representations of the delta function:

$$\lim_{n \rightarrow \infty} \frac{n}{2(1+n^2x^2)^{3/2}}, \quad \lim_{n \rightarrow \infty} \frac{n}{\pi(1+n^2x^2)}, \quad \lim_{n \rightarrow \infty} \frac{n}{\sqrt{\pi}} e^{-n^2x^2} \quad (9.4-2)$$

Let us verify that the procedure that was proposed earlier,

$$\lim_{a \rightarrow 0} \frac{a^2}{2(x^2 + a^2)^{3/2}},$$

fits this definition. To do this, we rewrite

$$\frac{a^2}{2(x^2 + a^2)^{3/2}} = \frac{a^2}{2a^3 \left(\frac{x^2}{a^2} + 1\right)^{3/2}} = \frac{1}{2a \left(\frac{x^2}{a^2} + 1\right)^{3/2}}$$

and set $n = 1/a$ to get the first representation of δ in accord with (9.4-2).

We conclude that there is no single definite simple formula that can yield $\delta(x)$. Clearly, the fact that $\delta(0) = \infty$ is not enough. To define δ , it still remains to demonstrate that this is precisely the infinity that is needed. However $\delta(x)$ can be obtained as the result of a passage to the limit ($n \rightarrow \infty$) from quite well-behaved (well defined) functions of x which involve the auxiliary quantity n as a parameter. We must stress particularly here that δ may be obtained by such a limit process from different functions φ . As long as n is finite, the functions $n\varphi(nx)$ differ from one another and, in particular,

$$I = \int f(x) n\varphi(nx) dx \neq f(0) \quad (9.4-3)$$

And only in the limit, as $n \rightarrow \infty$, do all the distinct functions $n\varphi(nx)$ tend to a single limit $\delta(x)$ and the corresponding integrals* (9.4-3) tend to $f(0)$:

$$\lim_{n \rightarrow \infty} I = f(0) \quad (9.4-4)$$

The arbitrariness that is evident in the choice of the original $\varphi(x)$ from which we obtain $\delta(x)$ is in full accord with the essence of the matter. In the section what follows we will consider examples of

* Strictly speaking, if $f(x) \rightarrow \infty$ for certain values of x or as $x \rightarrow \infty$ or as $x \rightarrow -\infty$, then not all $\varphi(x)$ can be used to obtain (9.4-4). Furthermore, the function $f(x)$ must not be discontinuous or at least its discontinuities must not fall on the point ($x = 0$), where $\delta(x) = \infty$, otherwise we will then have those meaningless questions about the value of the function at the point of discontinuity.

the application of $\delta(x)$ to physics. The description of some kind of action, that is to say, some kind of finite function $\psi(x)$, with the aid of the delta function is possible and desirable precisely when the detailed form of the action (which is to say, the true dependence of it on x) is inessential, the important thing being only the integral.

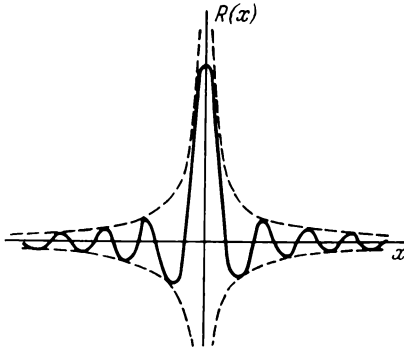


Fig. 223

The examples given above do not exhaust by any means the diverse $\varphi(x)$ from which we can “manufacture” $\delta(x)$. We can even give up the symmetry of $\varphi(x)$: as we pass to $n\varphi(nx)$ and increase n , the distance of the maximum from $x = 0$ diminishes as well; that is, even an asymmetric function approaches $\delta(x)$. Here is an example: $\frac{1}{\sqrt{\pi}} e^{-(x-1)^2}$ passes

into the function $\frac{n}{\sqrt{\pi}} e^{-(nx-1)^2}$

whose maximum lies at $x = 1/n$. We can give up the notation of $\varphi(x)$ by means of a simple unified formula that ensures the smoothness of $\varphi(x)$. Thus, we can take the function $\varphi(x)$ itself to be discontinuous:

$$\varphi = 1/2, -1 < x < 1; \varphi = 0, x < -1 \text{ and } x > 1 \quad (9.4-5)$$

The limit process consists in our taking

$$\left. \begin{aligned} \varphi_n &= n/2, -1 < nx < 1, \text{ that is, } -1/n < x < 1/n \\ \varphi_n &= 0, x < -1/n, x > 1/n \end{aligned} \right\} \quad (9.4-6)$$

and allowing $n \rightarrow \infty$. [Sketch the graph of $\varphi(x)$ according to (9.4-5) and also $\varphi_n(x)$ according to the formula (9.4-6) for $n = 3$ and $n = 10$.]

Finally, we can reject the condition that $\varphi(x)$ be positive. A curious and very important example is

$$R(x) = \frac{1}{2\pi} \int_{-\omega}^{+\omega} \cos \xi x \, d\xi = \frac{1}{\pi} \frac{\sin \omega x}{x}$$

The graph of $R(x)$ for given ω is shown in Fig. 223. The value $R(0)$ is equal to ω/π (the indeterminate form involved in the vanishing, simultaneously, of the numerator and the denominator at $x = 0$ is evaluated in elementary fashion). $R(x)$ passes through zero and changes sign for $x = \pm \pi/\omega, \pm 2\pi/\omega, \pm 3\pi/\omega, \dots$. The oscillations of

$R(x)$ damp out as they recede from $x = 0$ due to the denominator. The curve does not go beyond the lines $y = \pm 1/\pi |x|$ shown dashed in Fig. 223. It can be verified that $\int_{-\infty}^{+\infty} R^\bullet(x) dx = 1$ for arbitrary ω .

It turns out that as $\omega \rightarrow \infty$ we can regard $R(x)$ as a delta function! This is likely since as ω is increased, the altitude ω/π of the principal maximum on the R -axis grows, and there is a decrease in the width of this half-wave — $\pi/\omega < x < \pi/\omega$. But how are we to deal with the fact that as ω increases, the amplitude of oscillation

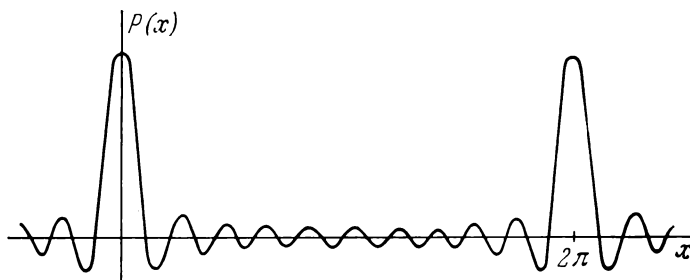


Fig. 224

does not decrease; as before, R attains $\pm 1/\pi |x|$, the dashed lines do not become narrower? Let us consider $\int f(x) R(x) dx$. The greater ω , the more frequent the oscillations, the more exactly the positive and negative half-waves compensate each other, yet the contribution of the first half-wave and the ones closest to it are all the time the same. It is for this reason (we do not give the proof) that $\lim_{\omega \rightarrow \infty} \int f(x) R(x) dx = f(0)$ but this means that $\lim_{\omega \rightarrow \infty} R(x)$ has the properties of the delta function.

Of interest is a similar function:

$$P(x) = \frac{1}{\pi} \left(\frac{1}{2} + \sum_{k=1}^{h=q} \cos kx \right)$$

Using the formulas of elementary trigonometry we can obtain the expression

$$P(x) = \frac{1}{2\pi} \frac{\sin \left(q + \frac{1}{2} \right) x}{\sin \frac{x}{2}}$$

The graph of $P(x)$ for $q = 10$ is shown in Fig. 224. As $q \rightarrow \infty$, $P(x)$ behaves near $x = 0$ just like $R(x)$ does as $\omega \rightarrow \infty$. $P(x)$ is different in that its high maxima repeat periodically at $x = 0, x = \pm 2\pi,$

$x = \pm 4\pi$, etc. In other words, $P(x)$ is a sum of delta functions: $P(x) = \delta(x) + \delta(x - 2\pi) + \delta(x + 2\pi) + \delta(x - 4\pi) + \delta(x + 4\pi) + \dots$

The functions R and P and their connection with the delta function are not mathematical oddities. Recall how R and P were constructed: R is an integral of cosines; P is a sum of cosines. If from cosines it is possible by addition (integration is a kind of addition!) to construct $\delta(x)$, then $\delta(x - a)$ may be constructed from $\cos \omega(x - a) = \cos \omega x \times \cos \omega a - \sin \omega x \sin \omega a$, that is, from cosines and sines with constant coefficients. But then any function $f(x)$ can be represented as a sum of cosines and sines. Any function can be replaced by a series of steps $f(x_i) \Delta x_i$ and each such step is actually $\delta(x - x_i) f(x_i) \Delta x_i$. Thus, with the aid of R and P , that is, essentially via delta functions, the possibility is proved of expanding functions in a Fourier series (if the function is periodic) and into the Fourier integral (if the function is nonperiodic).

Reread this section when you are in the second or third course of the university and are studying Fourier series. Ordinarily, the textbooks do not mention the delta function. Many mathematicians prefer to keep such physical heresy away from their students as long as possible, like books by Maupassant are kept out of the hands of schoolboys. The realization that actually the delta function is being used in the proofs will help you to grasp the meaning of these proofs.

9.5 APPLICATION OF THE DELTA FUNCTION

First of all, we will show you how the delta function permits abridging and making more convenient the writing of the conditions in many problems.

Let us consider a rod with a variable cross section* to which are attached a number of separate point loads (Fig. 225). Let the mass per unit length of the rod be expressed by the function $\rho(x)$. The

mass of the rod without loads is $\int_a^b \rho(x) dx$, with the loads it is

$$M = \int_a^b \rho(x) dx + \sum m_i$$

The position of the centre of gravity is

$$X = \frac{1}{M} \left(\int x \rho(x) dx + \sum x_i m_i \right)$$

The moment of inertia about the origin is

$$I = \int x^2 \rho(x) dx + \sum x_i^2 m_i$$

* Before tackling this example, go over Sec. 6.15 once again.

But with the aid of the delta function it is possible to include the separate masses in the generalized function of the density. We denote the new function by $\eta(x)$. It is defined by the formula

$$\eta(x) = \rho(x) + \sum m_i \delta(x - x_i)$$

Indeed, if we consider the general distribution of mass along the rod, we can say that at the points of the loads the density exhibits infinite jumps. With the aid of the new function, all the quantities can be written in uniform fashion and more succinctly:

$$M = \int_a^b \eta(x) dx,$$

$$X = \frac{1}{M} \int x \eta(x) dx,$$

$$I = \int x^2 \eta(x) dx$$

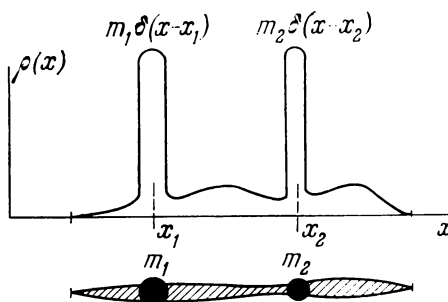


Fig. 225

The concept of the delta function permits combining continuously distributed masses and point masses in a single general expression.

Another example of the use of the delta function refers to the motion of a mass point. The basic equation, it will be recalled, is

$$m \frac{d^2x}{dt^2} = F(t)$$

Reread in Sec. 6.5 the material on impulse and the motion of a particle under the effect of a brief impulse, say, a blow. Recall the arguments to the effect that the action of the impulse is independent of the law of variation of the force, provided the force is sufficiently brief. These considerations are similar to the reasoning of Sec. 9.4 that the delta function can be constructed out of a variety of functions $\varphi(x)$ and concerning the conditions when it is possible to replace a finite function $\psi(x)$ by the generalized, singular function $\delta(x)$.

If the concrete form of the function of the force is not essential in the problem of a blow, this means that $F(t)$ may be replaced by the delta function, $F(t) \rightarrow J \delta(t - \tau)$, where τ is the instant of the blow, and $J = \int F(t) dt$ is the impulse of the force. We will carry out the integration of the equation of motion under the action of a unit delta force formally and according to all the rules. Let the particle, prior to the blow, be at rest at the origin: $t = -\infty$, $x = 0$,

$v = \frac{dx}{dt} = 0$. The equation is of the form

$$m \frac{d^2x}{dt^2} = m \frac{dv}{dt} = \delta(t - \tau)$$

Integrating we get

$$v(t) = \frac{1}{m} \int_{-\infty}^t \delta(t - \tau) dt = \frac{1}{m} \theta(t - \tau)$$

The velocity is expressed by the step-like function of the time (Fig. 226): $v = 0$, $t < \tau$, $v = \frac{1}{m}$, $t > \tau$. The next step consists in determining the path. From $v = \frac{dx}{dt}$ we get the answer $x = 0$, $t < \tau$, $x = \frac{1}{m}(t - \tau)$, $t > \tau$. The graph of the path is shown in Fig. 227.

Characteristic of the curve $x(t)$ of the path is the salient point at $t = \tau$. Here again we are convinced that the second derivative of

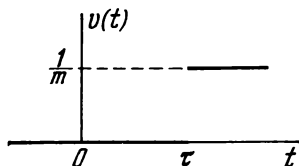


Fig. 226

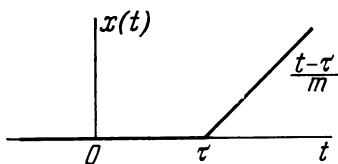


Fig. 227

a function having a salient point contains the delta function: the function $x(t)$ has a salient point; according to the equation of motion, the force is proportional to $\frac{d^2x}{dt^2}$; $x(t)$ with the salient point was obtained precisely for a force that was proportional to $\delta(t - \tau)$ so that in the case of a salient point, $\frac{d^2x}{dt^2}$ contains δ , which is what we set out to prove.

Now let us take the next step. The problem of the motion of a body under the action of a given force is a linear problem. This means that if there are two solutions $x_1(t)$ and $x_2(t)$ under the action of two distinct forces $F_1(t)$ and $F_2(t)$, then the sum of the solutions $x_3(t) = x_1(t) + x_2(t)$ is a solution that corresponds to the action of the sum of the forces $F_3(t) = F_1(t) + F_2(t)$. This property is a consequence of the simple fact that the second derivative of a sum of functions is the sum of the second derivatives of the functions:

$$\frac{d^2x_3}{dt^2} = \frac{d^2(x_1 + x_2)}{dt^2} = \frac{d^2x_1}{dt^2} + \frac{d^2x_2}{dt^2}$$

Taking into account that $\frac{d^2x_1}{dt^2} = \frac{F_1(t)}{m}$, $\frac{d^2x_2}{dt^2} = \frac{F_2(t)}{m}$, we get

$$\frac{d^2x_3}{dt^2} = \frac{F_1(t)}{m} + \frac{F_2(t)}{m} = \frac{F_3(t)}{m}$$

which is what we set out to prove—that the sum of the solutions, x_3 , describes the motion under the action of the sum of the forces.

Only one reservation is in order: the solutions of the equations of motion depend not only on the law of force but also on the initial conditions, that is, the initial position and the initial velocity of the mass at hand. If we choose these conditions thus: $t = -\infty$, $x_1 = 0$, $\frac{dx_1}{dt} = 0$; $t = -\infty$, $x_2 = 0$, $\frac{dx_2}{dt} = 0$, then the sum of the solutions, x_3 , will also satisfy the same condition: $t = -\infty$, $x_3 = 0$, $\frac{dx_3}{dt} = 0$.

Let us now combine the reasoning concerning linearity and the familiar solution of the delta function so as to obtain the general solution of the equation for a force that is arbitrarily dependent on the time. We partition the graph of the force $F(t)$ into strips of width $\Delta\tau$ (Fig. 228). What does a separate strip located between τ and $\tau + \Delta\tau$ represent? Let us change the designations, leaving t for "current" time varying from $-\infty$ to $+\infty$, whereas τ will refer to the given chosen strip. The height of the strip is $F(\tau)$, the width is $\Delta\tau$, the area (the impulse of the force, that is) is $F(\tau) \Delta\tau$. Since the strip is located at $t = \tau$, it is obvious that it can be replaced by the delta function with coefficient equal to the impulse $F(\tau) \Delta\tau \delta(t - \tau)$. We already know the solution of the equation of motion for the delta function. We denote it by $x_1(t, \tau)$. The solution as the function of time t depends on the instant τ of application of the force. Recall that

$$x_1(t, \tau) = 0, t < \tau; \quad x_1(t, \tau) = \frac{1}{m}(t - \tau), \quad t > \tau \quad (9.5-1)$$

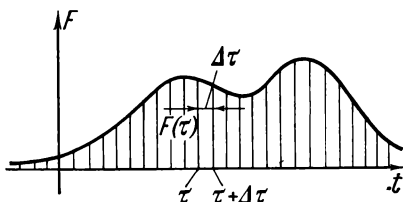


Fig. 228

One of the strips into which the force has been decomposed, from τ to $\tau + \Delta\tau$, is $\delta(t - \tau)$ with the coefficient $F(\tau) \Delta\tau$. Thanks to the linearity of the equation, the solution for the force in the form of such a strip is obtainable by multiplication of x_1 into that coefficient: $F(\tau) \Delta\tau x_1(t, \tau)$. This is the solution as referred to the action of a single strip. Now let us take advantage of linearity and write out the solution for the function $F(t)$, which we regard as the sum of the strips. It is clear here that the summation should actually be repla-

ced by the integral:

$$x(t) = \int_{-\infty}^{+\infty} x_1(t, \tau) F(\tau) d\tau$$

At first glance, this formula is rather strange: the x -coordinate at time t is expressed by an integral from $-\infty$ to $+\infty$ with respect to τ , i.e., the force enters into this expression at all instants of time. Yet it is clear that the law of force subsequent to time t does not affect the preceding motion. However, there is no error in the expression $x(t)$. The properties of the function $x_1(t, \tau)$ ensure reasonable properties of the solution. Indeed, $x_1(t, \tau) = 0$ when $t < \tau$. Hence, when integrating with respect to τ we actually do not need to take $\tau > t$, since the integrand is identically zero due to the factor $x_1(t, \tau)$ being equal to zero. Recalling the expression $x_1(t, \tau)$ (9.5-1), we get

$$x(t) = \frac{1}{m} \int_{-\infty}^t (t - \tau) F(\tau) d\tau$$

This method of obtaining a solution is very important generally. To summarize, then: if for a linear system we know a solution referring to the action of the delta function, then the solution referring to the action of an arbitrary function [$F(t)$ in the example] is obtained by simple summation or integration.

The ideas of linearity and addition (the real term is superposition) of solutions apply not only to such simple problems as the motion of a point, they hold true in vast areas of mathematics, physics and the natural sciences. It sometimes happens that a system is very complicated and it is impossible to solve the equations even for the most simple action by the delta function. A solution corresponding to the delta function can occasionally be obtained experimentally. In other cases, such a solution can be obtained from physical reasoning (see the problem below). Then linearity comes into play and we obtain the answer for any acting function. The solution that corresponds to the delta function [$x_1(t, \tau)$ in our example above] is so important that it has a special name, Green's function of the problem. Curiously enough, the English mathematician Green, for which the function was named, lived in the 19th century and quite naturally knew nothing about the delta function. But it was only the introduction of the delta function that clearly and succinctly explained the essence of Green's function.

Examples of this nature abound in mathematics, for we know of numerous results pertaining to tangent lines, areas and volumes that were obtained before the invention of derivatives and integrals. The advance of science lies not only in the attainment of new heights

and fresh results, but also in popularizing and simplifying the derivations of earlier times. The aim of this book which you are now finishing is precisely that: to simplify the understanding of our classical heritage—the fundamentals of higher mathematics.

Exercises

1. Consider a string held taut by a force k with ends fixed at points $x = 0$ and $x = l$. Regarding the deviation as small, determine by the parallelogram

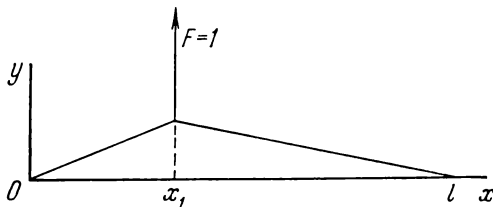


Fig. 229

of forces law the form of the string under the action of a unit load at point $x = x_1$ (Fig. 229). Obtain the formula for deviation of the string under the

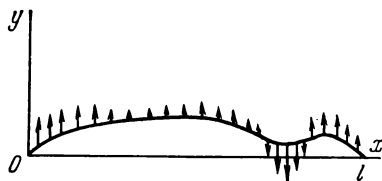


Fig. 230

action of a force distributed along its length via an arbitrary law $f(x)$ dynes/cm (Fig. 230).

2. Find the motion of a pendulum under the action of a force expressed by the delta function, that is, solve the equation $m \frac{d^2x}{dt^2} = -kx + \delta(t - \tau)$ provided $t = -\infty$, $x = 0$, $\frac{dx}{dt} = 0$. Using this solution, find the motion of the pendulum due to a force dependent on time via an arbitrary law.

Conclusion

What Next?

Higher mathematics, or, to be more exact, the differential and integral calculus makes it possible to solve a large class of problems that are not amenable to solution by the methods of arithmetic, algebra and geometry. Of tremendous importance is the very formulation of the new concepts such as instantaneous velocity, acceleration, impulse. These notions (and numerous others in diverse fields) are formulated exactly only in the language of derivatives and integrals.

The knowledge which you have gained in reading this book constitutes only a small portion of the whole of mathematical science and a small part of those divisions of mathematics that find applications in physics.

Here I wish to outline in brief the fields of physics and the associated divisions of mathematics that you will most likely study in the future.

Up to now the exposition has been that of a textbook and if you put your mind to the matter at hand, you will have mastered the material in all its details. What now follows is a very short outline of difficult problems that lie ahead, and the style is no longer that of the textbook. We do not expect to explain the *content* of mathematical physics, but merely to give the reader a general impression about the problems of this science and to show how exciting it can be.

For a better understanding of what follows, let us briefly state the general property of the problems we have dealt with up to now. These were problems involving the motion of a single particle in mechanics, problems on the variation of one or two quantities with time: the coordinate and velocity of a body or a charge on a capacitor and current in a circuit. We dealt with functions of one variable (time). The number of functions was one (current as a function of time) or two [the position of a body $x(t)$ and the velocity of the body $v(t)$].

Quite naturally follows the purely quantitative generalization: problems involving the motion of two bodies, three bodies, etc. Which must lead to the problems of the motion of a gas or a liquid,

the point being that one gram of hydrogen consists of 3×10^{23} molecules, hence 3×10^{23} separate bodies, no less.

It must be clear at this point that new methods are needed. Not only is it impossible to solve 3×10^{23} equations, there is neither paper nor time enough to write them down.

The new fields of hydrodynamics and gas dynamics appear with their new method (different from that of the mechanics of a point) of posing and solving problems. We ask how many molecules there are in some portion of the volume under consideration.

The solution of the problem consists in determining the distribution of density of the gas in space, $\rho(x, y, z)$, the pressure of the gas $p(x, y, z)$, the velocity of the gas at distinct points of space. Add to this that all these quantities are also functions of time, for example, $p(x, y, z, t)$. What is more, the velocity of the gas is a vector quantity, which means that at every point the magnitude and direction of the velocity are specified. In other words, we can say that three components of the vector are given. Thus, from problems of several functions of one variable we pass to functions of *several* independent variables.

Accordingly, in setting up these equations we have derivatives with respect to time and to the spatial coordinates, for instance, $\frac{\partial \rho}{\partial t}$ and $\frac{\partial \rho}{\partial x}$, $\frac{\partial \rho}{\partial y}$, $\frac{\partial \rho}{\partial z}$. It will be recalled that this is the notation of partial derivatives, that is, when we regard the variation of the function when one variable changes and the others remain fixed. Here is an example:

$$\frac{\partial \rho}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{\rho(x, y + \Delta y, z, t) - \rho(x, y, z, t)}{\Delta y}$$

An extremely important division of mathematical physics is the investigation of partial differential equations. These equations describe the motions of liquids, gases, solids, the propagation of heat in media, the phenomena of diffusion of atoms and molecules.

In all these cases, as we have already pointed out, it is possible in principle to continue regarding the separate particles and many functions of one variable (time). But there are other physical theories, primarily the theory of electromagnetism, where this is not possible.

Suppose we are considering two point charges at rest. The force acting between them depends on their position (their distance apart). This would seem to be a problem involving six functions (x_1, y_1, z_1) (x_2, y_2, z_2) of one variable (time). To a first approximation, the motion of charges introduces but little change: one need only take into account that a magnetic interaction appears between the charges dependent on their velocities.

An extremely important fact, which demands a fundamental new

approach, is the existence of a lag in interaction—propagation of the interaction with the velocity of light. The action of one charge on another depends on the position (and velocity) of the first charge at some earlier time. The theory in which everything that happens to charges at time $t + \Delta t$ is fully determined by the state at time t is the theory of the electromagnetic field. In this theory, besides separate charges we consider the electric field E and the magnetic field H that are given and fill all space. The quantities E and H are vectors and at the same time are functions of the coordinates and of time. Mathematically, the theory of the electromagnetic field is a theory of partial differential equations similar to the theory of elasticity, acoustics and gas dynamics. The only difference is that in the latter instance the equations are obtained by means of idealization (abstraction): when we speak of the density of a gas, we ignore the separate molecules. It is only in this approximate sense that a gas can be regarded as a continuous medium characterized by the continuous function $\rho(x, y, z, t)$. An electric field is indeed a continuous function.

Hydrodynamics, which developed in the 18th century, prepared the mathematical apparatus for the electromagnetic theory. No wonder then that at the beginning attempts were made to transfer the ideas of mechanics to the electromagnetic theory. A special substance called the ether was hypothesized as being responsible for electric and magnetic phenomena. We know that the mathematical analogy has remained, while the physical meaning of the electromagnetic theory has proved to be different and does not reduce to mechanics.

When speaking of a mathematical theory, one must not only speak about the statement of the problem and the initial equations but also about the nature of the results.

We can name two types of solutions for partial differential equations. One type is characteristic of a limited volume. This type includes natural oscillations with definite frequencies. A body of a given shape has a certain set of frequencies.

Recall the pendulum and its definite frequency of oscillation. If an external force acts on the pendulum, we get the characteristic phenomena of resonance when the frequency of the external force is almost the same as the frequency of the pendulum. All this is found

in the theory of ordinary differential equations: $m \frac{d^2x}{dt^2} = -kx + f(t)$. In the theory of partial differential equations, the body has many frequencies and behaves like a set or collection of many pendulums with distinct frequencies. There are many resonances. You can verify this at once if you have a piano at home. Depress one of the keys slowly and soundlessly, so as to release the string without striking it with the hammer. Now strike the other keys sharply and listen to the response of the free string....

The other type of solution of partial differential equations has to do with the matter that fills all space (propagation of waves). These are the waves of radio and light (in the electromagnetic theory), and sound waves in elastic media. Waves have the remarkable property of being able to carry information: pressure or an electric field in one point (near a receiver) as a function of time turns out to be similar to the curve of that same source quantity (transmitter) as a function of time.

It is possible to construct the solutions of equations describing the directed beam of a searchlight or a laser. A searchlight beam and the jet of water from a hose are strikingly similar. A knowledge of the properties of a solution of different kinds of problems has always been extremely important in the development of physics.

For a long time, atomic spectra were a mystery to physicists. It was not so much the specific laws and numerical values of the frequencies but the very fact that one and the same atom emits or absorbs via resonance the oscillations of several distinct but quite definite frequencies. The similarity to the oscillations of elastic bodies enabled scientists to approach the formulation of the equations of quantum mechanics. Likewise, the similarity between a stream of particles and the solutions for waves found its application in quantum mechanics.

Fundamentally, mathematics can be regarded as a variety of refined logic. The remarkable thing is that having set up the rules of this logic and learned them, man has at his disposal a more powerful tool than ordinary "common sense".

Using his hands, man makes simple tools with the aid of which he makes machine tools, with the aid of which he constructs more complicated devices, and using these he does things which he could not do with his hands alone. Mathematics is very much like that. It develops more and more complicated theories, introduces fresh notions and enables us to comprehend and master the most unusual phenomena of nature.

Above I cited some examples that pertain to the theory of equations of a definite type.

Geometry offers another marvelous case.

Human experience teaches us that in space it is convenient to introduce three coordinates: x , y , z . Any further complication would seem to be superfluous, "a trick of the devil". Yet, coordinates can be introduced in a different way so that the coordinate $\xi = \text{const}$ corresponds to some curved surface (whereas $x = \text{const}$ for arbitrary y and z is the equation of a plane perpendicular to the x -axis).

To summarize, then, we can introduce curvilinear coordinates ξ , η , ζ and with a lot of effort, agonizingly, learn to compute the distances between points and other quantities with the aid of these new coordinates.

At first glance this is a dull and totally useless effort. One must possess a peculiar bent to be able to see beauty in the mere fact of overcoming difficulties, in the development of a theory with arbitrary coordinates of a most general nature.

And then, like a bolt out of the blue, comes the general theory of relativity—perhaps the theories of Lobachevsky, Bolyai and Riemann were silent flashes of lightning that preceded this thunderbolt. Generalized coordinates are just as convenient (or just as inconvenient) to describe ordinary space (in which Euclid's geometry holds true) as they are to describe curved space. Rectangular coordinates x , y , z are convenient for ordinary space but are no good at all for describing curved space. The x , y , z coordinates do not even hint at the existence of any other kinds of space.

The study of curvilinear coordinates which had seemed to be such a needless complication actually prepared us for a vast range of spaces whose very existence was totally unknown to us. Then it turns out that the force of universal gravitation is linked up with the very fact that space is somewhat curved. True, this "somewhat" has to do with the conditions here on earth and in the solar system. In certain phenomena of a larger scale (catastrophic explosions of stars, evolutionary processes in the universe) space may turn out to be highly curved. In the study of nature, a diversity of approach is desired: the overcoming of mathematical difficulties, the mastering of the mathematical apparatus, physical intuition, boldness of conception, experimentation and observation. All these alloyed together make it possible to advance science.

Let us return to mathematics, more specifically, to mathematical physics.

One occasionally hears the small-minded remark that "mathematics is a mill that grinds up only what is put in." In this way, poor results are explained by the fact that the original premises were faulty. In reality, the mill quite often turns out much more than is put in and the results are somewhat totally unexpected!

At the end of the course of mathematical physics we can again start a new chapter entitled "WHAT NEXT", but do not lose heart, for not far off is the boundary line that marks the end of study and the beginning of creativity and the development of new theories.

I try to picture the reader who in a few moments will close this book with a sigh of relief. Most likely, you are finishing school or are in your first year at college, a sort of "unicellular protozoan", to quote a popular play of student life.

May mathematics always remain for you an exact and beautiful language, a means of expressing ideas, a way of thought. May mathematics be more than merely another subject that has to be "passed" at an examination and then left behind without a trace. Love mathematics and mathematics will love you back.

Answers and Solutions

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CHAPTER 1

Sec. 1.2

1. $A(2, 1)$, $B(1, 2)$, $C(0, 3)$, $D(-1, 2)$, $E(-2, 1)$, $F(-3, 0)$, $G(-2, -1)$, $H(-1, -2)$, $K(0, -3)$, $L(1, -2)$, $M(2, -1)$, $N(3, 0)$, $O(0, 0)$.

Sec. 1.3

1. See Fig. 231. 2. See Fig. 232. 3. See Fig. 233. 4. $r = \sqrt{2}$, $\alpha = 45^\circ$; $r = 2\sqrt{2}$, $\alpha = -45^\circ$; $r = 3\sqrt{2}$, $\alpha = -135^\circ$; $r = 4\sqrt{2}$, $\alpha = 135^\circ$. 5. 2, $2\sqrt{2}$,

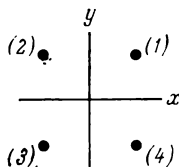


Fig. 231

$2\sqrt{2}$, $2\sqrt{2}$. 6. $A_1(0, 0)$, $A_2(2, 3)$, $A_3(4, 6)$ are collinear; $A_1(0, 0)$, $A_2(2, 3)$, $A_3(-2, -3)$ are collinear; $A_1(0, 0)$, $A_2(2, 3)$, $A_3(-2, 3)$ are not collinear.

7. $\left(0, \frac{a}{\sqrt{2}}\right)$, $\left(\frac{a}{\sqrt{2}}, 0\right)$, $\left(0, -\frac{a}{\sqrt{2}}\right)$, $\left(-\frac{a}{\sqrt{2}}, 0\right)$. See Fig. 234.

8. $(a, 0)$, $\left(\frac{a}{2}, \frac{a\sqrt{3}}{2}\right)$, $\left(-\frac{a}{2}, \frac{a\sqrt{3}}{2}\right)$, $(-a, 0)$, $\left(-\frac{a}{2}, -\frac{a\sqrt{3}}{2}\right)$, $\left(\frac{a}{2}, -\frac{a\sqrt{3}}{2}\right)$. See Fig. 235. 9. (a) Two cases: $\left(-\frac{a}{2}, 0\right)$, $\left(\frac{a}{2}, 0\right)$, $\left(0, \frac{a\sqrt{3}}{2}\right)$. See Fig. 236a; $\left(-\frac{a}{2}, 0\right)$, $\left(\frac{a}{2}, 0\right)$, $\left(0, -\frac{a\sqrt{3}}{2}\right)$.

(b) Four cases: $(0, 0)$, $(a, 0)$, $\left(\frac{a}{2}, \frac{a\sqrt{3}}{2}\right)$. See Fig. 236b; $(0, 0)$, $(a, 0)$, $\left(\frac{a}{2}, -\frac{a\sqrt{3}}{2}\right)$; $(0, 0)$, $(-a, 0)$, $\left(-\frac{a}{2}, \frac{a\sqrt{3}}{2}\right)$; $(0, 0)$, $(-a, 0)$, $\left(-\frac{a}{2}, -\frac{a\sqrt{3}}{2}\right)$. 10. $A_2(x_1, -y_1)$, $A_3(-x_1, y_1)$, $A_4(-x_1, -y_1)$. See Fig. 237.

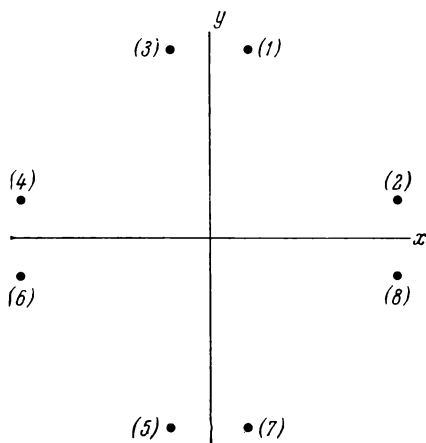


Fig. 232

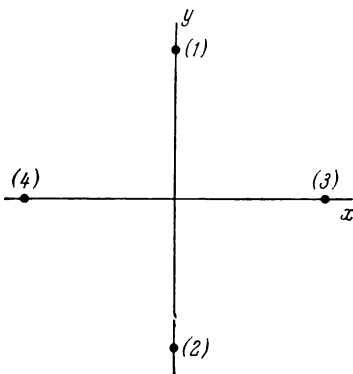


Fig. 233

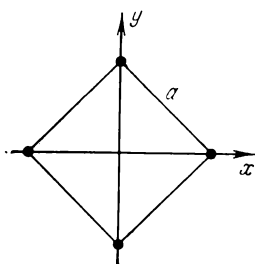


Fig. 234

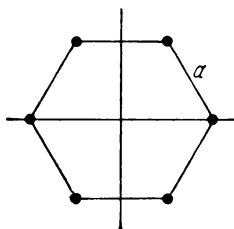


Fig. 235

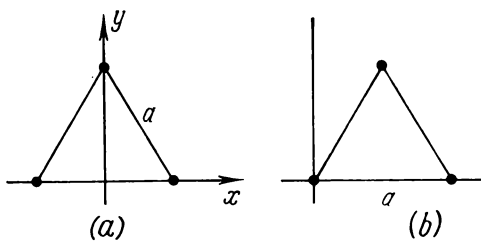


Fig. 236

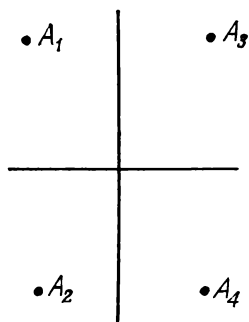


Fig. 237

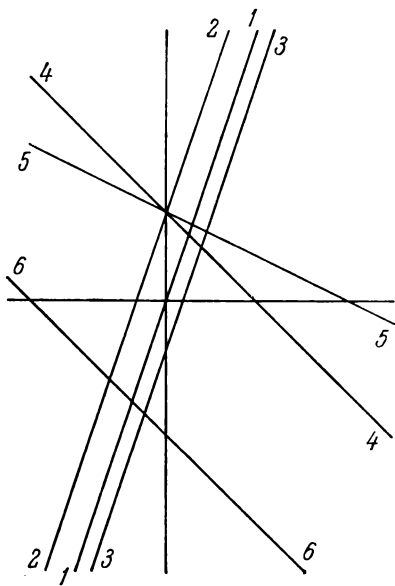


Fig. 238

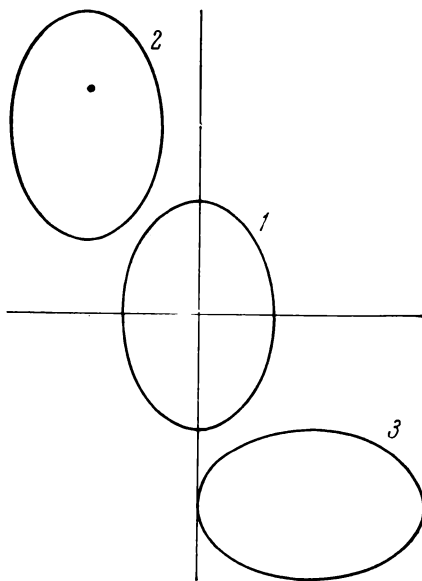
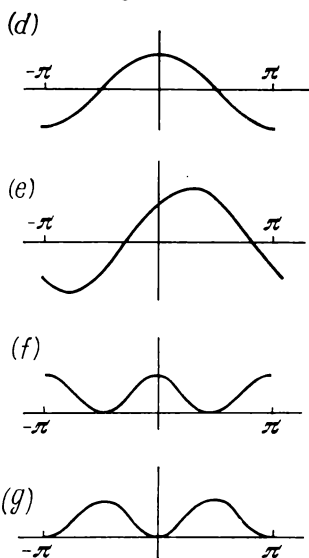
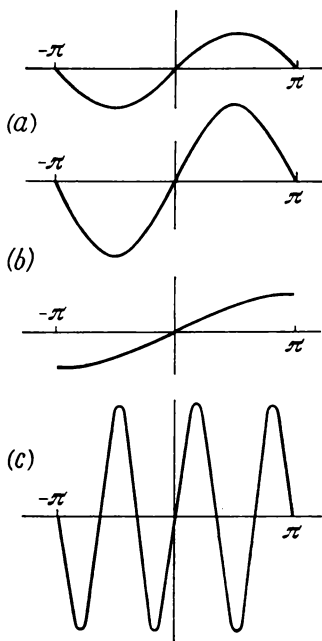


Fig. 239



(g)

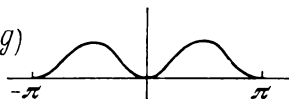


Fig. 240

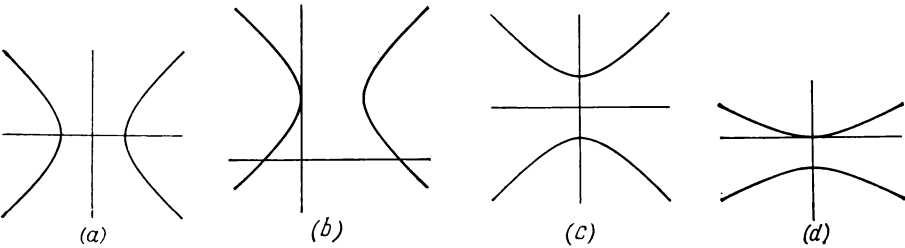


Fig. 241

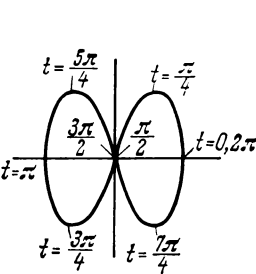


Fig. 242

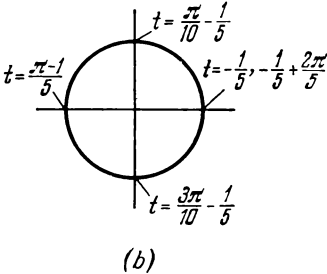
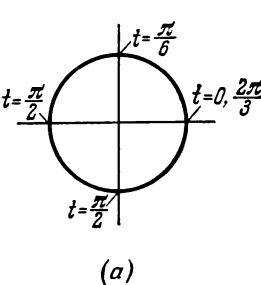


Fig. 243

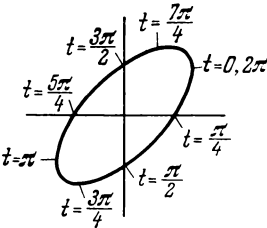


Fig. 244

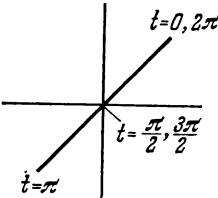


Fig. 245

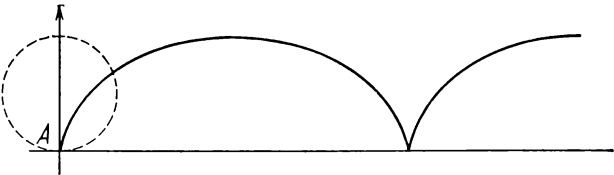


Fig. 246

Sec. 1.4

See Fig. 238.

Sec. 1.7

1. See Fig. 239. 2. See Fig. 240. 3. See Fig. 241.

Sec. 1.8

1. The first curve is shown in Fig. 242. 2. See Fig. 243. 3. See Fig. 244. 4. The desired graph is the straight line $y=x$, but not the whole line; only the portion between $x=-1$, $y=-1$ and $x=+1$, $y=+1$, see Fig. 245. 5. See Fig. 246. *Hint.* The parametric equations of the curve are $x=t-\sin t$, $y=1-\cos t$.

CHAPTER 2

Sec. 2.3

1. (a) $z = t^2$, $\Delta z = \left(t + \frac{\Delta t}{2}\right)^2 - \left(t - \frac{\Delta t}{2}\right)^2 = 2t \cdot \Delta t$, $\frac{\Delta z}{\Delta t} = 2t$, $\frac{dz}{dt} = 2t$;
 (b) $z = t^3$, $\Delta z = \left(t + \frac{\Delta t}{2}\right)^3 - \left(t - \frac{\Delta t}{2}\right)^3 = 3t^2 \cdot \Delta t + \frac{(\Delta t)^3}{4}$, $\frac{\Delta z}{\Delta t} = 3t^2 + \frac{(\Delta t)^2}{4}$,
 $\frac{dz}{dt} = 3t^2$. The results coincide with the computation in Sec. 2.3. Note that in the case $z = t^2$ the ratio $\frac{\Delta z}{\Delta t}$ does not depend on Δt , but when $z = t^3$ the ratio $\frac{\Delta z}{\Delta t}$ contains only $(\Delta t)^2$. 2. $y' = 4x^3$. 3. $y' = 2x + 2$. 4. $y' = -\frac{2}{x^3}$.
 5. $y' = -\frac{2b}{x^3}$. 6. $y' = \frac{1}{2\sqrt{x}}$.

Sec. 2.4

1. Find $(1.2)^2$. Consider the function $z = t^2$; let $t = 1$ and $\Delta t = 0.2$; $z' = 2t$, $z'(1) = 2$; therefore $\Delta z = 2 \cdot 0.2 = 0.4$; $(1.2)^2 = 1^2 + 0.4 = 1.4$. The exact value is $(1.2)^2 = 1.44$. The error is roughly 3%. $(1.1)^2 = 1.2$. The exact value is 1.21. The error is approximately 1%. $(1.05)^2 = 1.1$. The exact value is 1.1025. The error is about 0.2%. $(1.01)^2 = 1.02$. The exact value is 1.0201. The error is about 0.01%. 2. See Table 1.

Table 1

t	z	z_{true}	Error in % (rounded)
1.1	18.0	17.5	0.3
1.05	17.5	17.4875	0.07
0.98	16.8	16.798	0.01

Sec. 2.5

1. See Fig. 247. 2. See Fig. 248. 3. See Fig. 249. 4. See Fig. 250. 5. See Fig. 251. 6. $y = (3/4)x - 1/4$, $(1/3, 0)$, $(0, -1/4)$; $y = 3x - 2$, $(2/3, 0)$, $(0, -2)$.

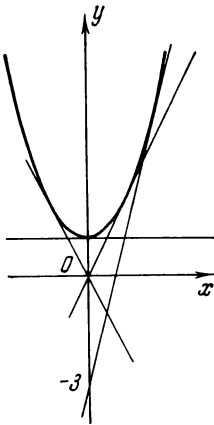


Fig. 247

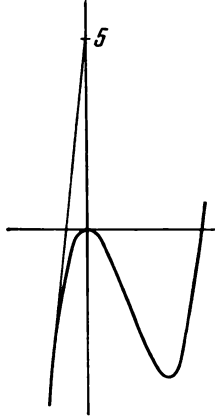


Fig. 248

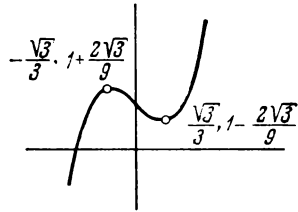


Fig. 249

7. For the curve $y = ax^2$, the tangent at the point (x_0, y_0) cuts the coordinate axes at the points $(x_0/2, 0)$ and $(0, -y_0)$ (see page 60). For the curve $y = ax^3$,

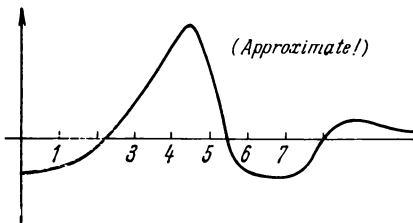


Fig. 250

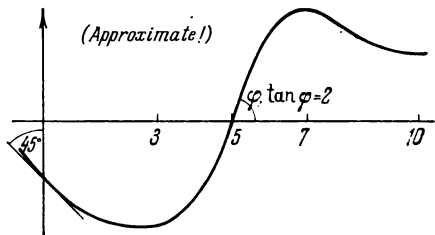


Fig. 251

the equation of the tangent at the point (x_0, y_0) is $y = 3ax_0^2x - 2y_0$. The points of intersection of the tangent and the axes are $(2/3 x_0, 0)$, $(0, -2y_0)$.

Sec. 2.6

1. $x=0$, minimum for $a > 0$, maximum for $a < 0$. 2. $x = -1$, maximum; $x = 1$, minimum. 3. $x = -\sqrt{a}$ ($a > 0$), maximum; $x = \sqrt{a}$ ($a > 0$), minimum. For $a < 0$ there is neither minimum nor maximum. 4. $x = -\frac{1}{\sqrt{3}}$, maximum; $x = \frac{1}{\sqrt{3}}$, minimum. 5. (a) $a > 0$, $x = 0$, minimum; (b) $a = 0$, $x = 0$, minimum; (c) $a < 0$, $x = 0$, maximum, $x = -\sqrt{-a/2}$, minimum, $x = \sqrt{-a/2}$, minimum.

Sec. 2.8

2. Table 2 gives the values of the sums for a partition of the interval into m parts, where $m=10, 20, 50, \infty$. It is evident that already for $m=50$ both sums differ but slightly from the limiting value as $m \rightarrow \infty$.

Table 2

m	10	20	50	∞
Δt	0.1	0.05	0.02	0
$\Sigma t_{i-1}^2 \Delta t$	2.18	2.26	2.30	2.33
$\Sigma t_i^2 \Delta t$	2.49	2.41	2.37	2.33

Sec. 2.11

1. $\frac{1}{3}$. 2. 0.11033 3. $\frac{1}{2}$. 4. $2(\sqrt{3}-1)$. 5. $S = \int_0^b y(x) dx$, where

$y=y(x)$ is the equation of the hypotenuse. Take an arbitrary point A on the

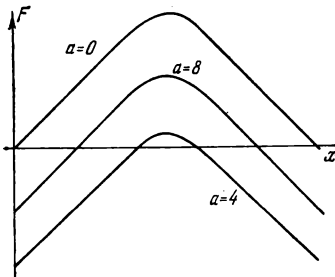


Fig. 252

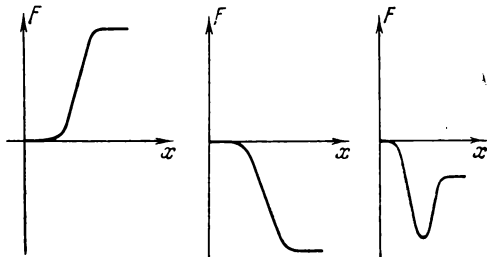


Fig. 253

hypotenuse with coordinates x and y . Draw a vertical line through this point (see Fig. 49). From the similarity of the triangles, $\frac{x}{b} = \frac{y}{h}$. Whence $y = \frac{h}{b} x$.

$$S = \int_0^b \frac{h}{b} x dx = \frac{h}{b} \int_0^b x dx = \frac{h}{b} \frac{x^2}{2} \Big|_0^b = \frac{1}{2} bh. \quad 7. S = \frac{1}{3} x_0 y_0. \quad 8. S = \frac{2}{3} x_0 y_0.$$

$$9. S = \int_{-r}^r \sqrt{r^2 - x^2} dx. \quad 10. 0.7837 \text{ for } m=5, 0.7850 \text{ for } m=10. \quad 11. \text{ See}$$

Fig. 252. 12. See Fig. 253.

Sec. 2.13

1. $\bar{y} = \bar{x^2} = \int_0^2 \frac{x^2 dx}{2} = \frac{x^3}{3} \Big|_0^2 \cdot \frac{1}{2} = \frac{4}{3}$. 2. $y(0) = 0, y(2) = 4, \frac{y(0) + y(2)}{2} = 2, \dot{y}(1) = 1, y(1) = 1 < \bar{y} = 1.33 < \frac{y(0) + y(2)}{2} = 2$. 3. $\bar{y} = \frac{1}{6} y(0) + \frac{2}{3} y(1) + \frac{1}{6} y(2) = \frac{2}{3} + \frac{1}{6} \cdot 4 = 1.33$. 4. $\int_a^b y dx = \left(\frac{1}{3} rx^3 + \frac{1}{2} px^2 + qx \right) \Big|_a^b = \frac{1}{3} r \times (b^3 - a^3) + \frac{1}{2} p (b^2 - a^2) + q (b - a) = (b - a) \left[\frac{1}{3} r (b^2 + ab + a^2) + \frac{1}{2} p (b + a) + q \right]$. By formula (2.13-2) it should be true that

$$\int_a^b y dx = (b - a) \bar{y} = (b - a) \left[\frac{1}{6} y(a) + \frac{2}{3} y\left(\frac{a+b}{2}\right) + \frac{1}{6} y(b) \right].$$

Substituting $y(a), y\left(\frac{a+b}{2}\right)$ and $y(b)$ and comparing it with the expression

obtained above, we see that they are identical. 5. $\bar{F} = \frac{1}{2R - R} \int_R^{2R} \frac{A}{r^2} dr = \frac{1}{R} \times \left(-\frac{A}{r} \right) \Big|_R^{2R} = \frac{1}{R} \left(-\frac{A}{2R} + \frac{A}{R} \right) = 0.5 \frac{A}{R^2}$. The mean value of the force on this portion is half the force on the ground, $\bar{F} = 0.5F_0$. 6. $F(R) = F_0, F(2R) = \frac{1}{4} F_0; \frac{F(R) + F(2R)}{2} = 0.625F_0 > 0.5F_0$. 7. $F\left(\frac{R+2R}{2}\right) = F\left(\frac{3}{2}R\right) = \frac{4}{9} F_0 \cdot \frac{1}{6} F_0 + \frac{2}{3} \cdot \frac{4}{9} F_0 + \frac{1}{6} \cdot \frac{1}{4} F_0 = \frac{109}{216} F_0 = 0.505F_0$. Error 1%!

CHAPTER 3

Sec. 3.3

1. Find the derivative $z = (ax + b)^2$ after removing the brackets: $z = a^2x^2 + 2abx + b^2, z' = 2a^2x + 2ab = 2a(ax + b)$. Now find the same derivative by the rule for finding the derivative of a composite function: $z = y^2, y = ax + b, \frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = 2ya = 2ay = 2a(ax + b)$. 2. $z' = -\frac{a}{(ax + b)^2}, z' = -\frac{2a}{(ax + b)^3}, z' = \frac{1}{(x + 1)^2}$.

Sec. 3.4

1. $y = x^4 = x^2x^2, y' = 2x \cdot x^2 + x^2 \cdot 2x = 4x^3$. 2. $y' = (4x + 1) \sqrt{x} + (2x^2 + x) \times \frac{1}{2\sqrt{x}}$. 3. $y' = \frac{(3x^2 + 10x)(x + 1) - (x^3 + 5x^2)1}{(x + 1)^2} = \frac{2x(x^2 + 4x + 5)}{(x + 1)^2}$. 4. $y' = \frac{-x^2 + 2x + 2}{(x^2 + 2)^2}$.

Sec. 3.5

1. $y' = 5x^4 - 12x^3 + 3x^2 + 14x - 2$. 2. $y' = 2(x^3 + x + 1)(3x^2 + 1)$. 3. $y' = 4(x^2 - x + 1)^3(2x - 1)$. 4. $y' = 10(3x^2 - 1)^9 6x$. 5. $y' = \frac{x}{\sqrt{x^2 - 1}}$. 6. It is convenient to write y via a fractional power, $y = x^{2/5}$. Now, by the general formula for the derivative of a power, find $y' = \frac{2}{5}x^{-3/5}$. 7. (a) If x changes by 1%, then $\Delta y = n \cdot 0.01y$; therefore, as x changes by $k\%$, $\Delta y = n \cdot 0.01 \cdot y \cdot k$. In the given case, x changes by 10%, $\Delta y = n \cdot 0.1y$. Since $n = \frac{1}{2}$, it follows that

$\Delta y = \frac{1}{2} \cdot 0.1y = 0.05y$. Therefore $y(11) = y(10) + 0.05 \cdot 5 = 5.25$, $y(9) = y(10) - 0.05 \cdot 5 = 4.75$. We obtain an exact solution. Denote the proportionality constant by k ; then $y = k\sqrt{x}$. Since for $x = 10$, y must be equal to 5, it follows that $5 = k\sqrt{10}$, whence $k = 1.58$ (use two decimal places in the computations). Therefore $y = 1.58\sqrt{x}$, $y(11) = 1.58\sqrt{11} = 5.24$, $y(9) = 4.74$. (b) The approximate values are $y(11) = 4.50$, $y(9) = 5.50$. The exact values are $y(11) = 4.54$, $y(9) = 5.56$. (c) The approximate values are $y(11) = 6.00$, $y(9) = 4.00$. The exact values are $y(11) = 6.05$, $y(9) = 4.05$.

Sec. 3.6

$$\begin{aligned}
 & 1. \quad y' = 3x^2(x^2 - 1)^2 + x^3 \cdot 2(x^2 - 1)2x = x^2(x^2 - 1)(7x^2 - 3). \quad 2. \quad y' = 3x^2 \\
 & \times \sqrt{x^2 + x} + \frac{(2x + 1)x^3}{2\sqrt{x^2 + x}}. \quad 3. \quad y' = 5x^4 \sqrt[3]{x^2 - 1}(x^3 - 2x)^{1/5} + \frac{1}{3} \frac{\sqrt[3]{x^2 - 1}}{x^2 - 1} 2x \\
 & \times x^5(x^3 - 2x)^{1/5} + \frac{1}{5} \frac{(x^3 - 2x)^{1/5}}{x^3 - 2x} (3x^2 - 2)x^5 \sqrt[3]{x^2 - 1}. \quad 4. \quad y' = \left(1 - \frac{1}{2\sqrt{x^3}}\right) \\
 & \times \sqrt{x^3 - 2} + \left(x + \frac{1}{\sqrt{x}}\right) \frac{3x^2}{2\sqrt{x^3 - 2}}. \quad 5. \quad y' = 2x\sqrt[3]{x + x} + x^2 \frac{\frac{1}{3\sqrt[3]{x^2}} + 1}{2\sqrt[3]{x + x}}. \\
 & 6. \quad y' = 5\left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}\right)^4 \left(\frac{1}{3\sqrt[3]{x^2}} - \frac{1}{3\sqrt[3]{x^4}}\right)x + \left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}\right)^5. \quad 7. \quad y' = \frac{x^2 + 1}{(x^2 - 1)^2}. \\
 & 8. \quad y' = 2 \frac{1 - x^2}{(x^2 - x + 1)^2}. \quad 9. \quad y' = \frac{x^2 + 2x + 5}{(x + 1)^2}. \quad 10. \quad y' = \frac{-12x + 5}{x^6} \cdot \sqrt{x^3 + 2} \\
 & + \frac{3(3x - 1)}{2x^3 \sqrt{x^3 + 2}}. \quad 11. \quad y' = -\frac{1}{(x^2 - 1)\sqrt{x^2 - 1}}. \quad 12. \quad y' = \frac{1}{\sqrt[3]{x + 1}} - \frac{x}{3\sqrt[3]{(x + 1)^4}} \\
 & = \frac{2x + 1}{3\sqrt[3]{(x + 1)^4}}. \quad 13. \quad y' = \frac{4x + 3\sqrt{x}}{4\sqrt{x^2 + x}\sqrt{x}}. \quad 14. \quad y' = \sqrt{x^2 + \sqrt[3]{x}} + \frac{6x^2 + \sqrt[3]{x}}{6\sqrt{x^2 + \sqrt[3]{x}}} \\
 & = \frac{7x^2 + 2\sqrt[3]{x}}{6\sqrt{x^2 + \sqrt[3]{x}}}. \quad 15. \quad y' = \frac{(8x - 33)\sqrt{1 + x^2}}{(1 + x^2)^3}. \quad 16. \quad y' = \sqrt[3]{(2x + 3)^2} + x \\
 & \times \frac{4}{3\sqrt[3]{(2x + 3)}}. \quad 17. \quad y' = 3x^2\sqrt{x - 1} + \frac{x^3 - 1}{2\sqrt{x - 1}} + \sqrt[3]{x^2 - 1} + \frac{2x^2}{3\sqrt[3]{(x^2 - 1)^2}}.
 \end{aligned}$$

$$\begin{aligned}
 18. \quad y' &= \frac{-2x^2 - x + 9}{3(x-1)^3 \sqrt[3]{2x-3}}. & 19. \quad y' &= \sqrt{\frac{x+1}{x-1}} \frac{1}{(x+1)^2}. & 20. \quad y' &= \\
 &= \frac{4x^3 - 10x^2 - 22x - 11}{3(x-2)^2 \sqrt[3]{(x+1)^2}}. & 21. \quad y' &= \frac{-x^5 + 2x^3 + 2x^2 - 1}{(x^3 + 1)^2 \sqrt{x^2 - 1}}. & 22. \quad y' &= \frac{1}{3} \\
 &\times \sqrt[3]{\left(\frac{x+1}{x^2+x+1}\right)^2 \frac{x^2+2x}{(x+1)^2}}. & 23. \quad y' &= \sqrt{x^2-1} \sqrt[3]{x+\sqrt{x}} + \frac{x^2}{\sqrt{x^2-1}} \\
 &\times \sqrt[3]{x+\sqrt{x}} + \frac{(1+2\sqrt{x})x\sqrt{x^2-1}}{6\sqrt{x}\sqrt[3]{(x+\sqrt{x})^2}}. & 24. \quad y' &= \frac{1}{7} \left(x + \frac{1}{\sqrt{x}}\right)^{-6/7} \\
 &\times \left(1 - \frac{1}{2\sqrt{x^3}}\right) x^2 + 2x \left(x + \frac{1}{\sqrt{x}}\right)^{1/7}. & 25. \quad y' &= \frac{\frac{1}{3\sqrt[3]{x^2}} - 2}{\sqrt[3]{x+1}} - \frac{1}{3} \\
 &\times \frac{\sqrt[3]{x}-2x}{(x+1)\sqrt[3]{x+1}} = \frac{1-6\sqrt[3]{x^2}-4x\sqrt[3]{x^2}}{3(x+1)\sqrt[3]{x^2}(x+1)}.
 \end{aligned}$$

Sec. 3.7

$$\begin{aligned}
 1. \quad y' &= 2.3 \cdot 10 \sqrt[3]{x} \frac{1}{2\sqrt{x}}. & 2. \quad y' &= 2.3 \log_{10} 2 \cdot 2^x. & 3. \quad y' &= 2.3 \log_{10} 5 \cdot 5^{x+1}. \\
 4. \quad y' &= -2.3 \log_{10} 2 \left(\frac{1}{2}\right)^x.
 \end{aligned}$$

Sec. 3.8

$$\begin{aligned}
 1. \quad y' &= -e^{-x}. & 2. \quad y' &= 2xe^{x^2}. & 3. \quad y' &= (3x^2-3)e^{x^3-3x+1}. & 4. \quad y' &= \frac{1}{2\sqrt{x}} e^{\sqrt{x}}. \\
 5. \quad y' &= 5e^x - 3e^{3x}.
 \end{aligned}$$

Sec. 3.9

$$\begin{aligned}
 1. \quad 2.3026, \quad 4.6052. & \quad 2. \quad \log_5 15 = \frac{\log_{10} 15}{\log_{10} 5} = 1.6825. & 3. \quad & \text{Differentiating both} \\
 \text{sides, we find } \frac{(uv)'}{uv} &= \frac{u'}{u} + \frac{v'}{v}, \text{ whence } (uv)' = u'v + uv'. & 5. \quad y' &= \frac{1}{2x} (2x)' \\
 &= \frac{1}{x}. \text{ The same result may also be obtained thus: } y = \ln 2x = \ln 2 + \ln x, \\
 y' &= (\ln 2)' + (\ln x)' = \frac{1}{x}. & 6. \quad y' &= \frac{1}{x+3}. & 7. \quad y' &= \frac{1}{x}. & 8. \quad y' &= \frac{2x}{x^2+1}. \\
 9. \quad y' &= \frac{6x-1}{3x^2-x+1}. & 10. \quad y' &= \frac{2}{(x+1)(x-1)} = \frac{2}{(x^2-1)}. & 11. \quad y' &= \frac{1-x}{2x(x+1)}. \\
 12. \quad y' &= \ln x + 1. & 13. \quad y' &= 3x^2 \ln(x+1) + \frac{x^3}{x+1}. & 14. \quad & \text{To find the derivative,} \\
 & \text{take the logs of both sides of the equation (any logarithmic base will do, we} \\
 & \text{choose natural logarithms): } \ln y = x \ln x. \text{ Now take the derivatives of both} \\
 & \text{sides of this equation, taking into account that } y \text{ is a function of } x \text{ and } \ln y, \\
 & \text{consequently, it is a composite function: } \frac{1}{y} y' = \ln x + 1, \text{ whence we find } y' = \\
 & = y(\ln x + 1) \text{ or, finally, } y' = x^x(\ln x + 1). \text{ The next example is done in simi-} \\
 & \text{lar fashion. } 15. \quad y' = x^{\sqrt{x^2-1}} \left(\frac{x \ln x}{\sqrt{x^2-1}} + \frac{\sqrt{x^2-1}}{x} \right).
 \end{aligned}$$

Sec. 3.10

1. $y' = 2 \cos(2x+3)$. 2. $y' = -\sin(x-1)$. 3. $y' = -(2x-1) \sin(x^2-x+1)$.
 4. $y' = 2 \sin x \cos x$. 5. $y' = 3 \cos 3x \cos^2 x - 2 \cos x \sin x \sin 3x$. 6. $y' = (\sin 2x)^x$
 $\times \left[\ln \sin 2x + \frac{2x \cos 2x}{\sin 2x} \right]$ (see the solution of Problem 14 of Sec. 3.9). 7. $y' =$
 $= \tan x + \frac{x}{\cos^2 x}$. 8. $y' = \frac{2}{\cos^2 2x} e^{\tan 2x}$. 9. $y' = -\frac{1}{2} \frac{1}{\sin^2 \frac{x}{2}}$.

Sec. 3.11

1. (a) $y' = -\frac{1}{\sqrt{1-x^2}}$, (b) $y' = -\frac{1}{1+x^2}$. 3. $y' = \frac{2}{\sqrt{1-4x^2}}$. 4. $y' =$
 $= \frac{3}{9x^2+6x+2}$. 5. $y' = \frac{2x-1}{x^4-2x^3+x^2+1}$. 6. $y' = \frac{1}{2\sqrt{x}(1+x)} e^{\arctan \sqrt{x}}$.

Sec. 3.12

1. -1 , 1. 2. -1 .

Sec. 3.15

1. To perform the integration, remove the brackets; we get an integral of a polynomial: $\int x(x-1)^2 dx = \int x(x^2-2x+1) dx = \int (x^3-2x^2+x) dx$
 $= \frac{x^4}{4} - \frac{2}{3}x^3 + \frac{1}{2}x^2 + C$. 2. Write the integrand as follows: $\frac{x^2+2x-3}{x}$
 $= x+2-\frac{3}{x}$. Now it is easy to perform the integration: $\int \frac{x^2+2x-3}{x} dx$
 $= \frac{x^2}{2} + 2x - 3 \ln x + C$. 3. Make the change of variable $3x-5=t$, $dt=3dx$;
 $\int \cos(3x-5) dx = \frac{1}{3} \int \cos t dt = \frac{1}{3} \sin t + C = \frac{1}{3} \sin(3x-5) + C$. 4. The
 solution is similar to the preceding one. We get $-\frac{1}{2} \cos(2x+1) + C$.
 5. $\frac{2}{9} \sqrt{(3x-2)^3} + C$. 6. $\cos x + x \sin x + C$. 7. $x(\ln x - 1) + C$. Problems 8 to
 11 can be solved by the integration-by-parts formula. However, it is more
 convenient to use the method of undetermined coefficients. 8. $\frac{1}{2} x \sin 2x$
 $-\left(\frac{1}{2}x^2 - \frac{1}{4}\right) \cos 2x + C$. 9. $(-x^3 - 3x^2 - 6x - 6)e^{-x} + C$. 10. $(2x+1) \cos x$
 $+ (x^2+x-1) \sin x + C$. 11. $\int (2x^2+1) \cos 3x dx = (a_1x^2 + b_1x + c_1) \cos 3x$
 $+ (a_2x^2 + b_2x + c_2) \sin 3x$. Take the derivatives of both sides of the equation:
 $(2x^2+1) \cos 3x = (2a_1x + b_1) \cos 3x - (3a_1x^2 + 3b_1x + 3c_1) \sin 3x + (2a_2x + b_2)$
 $\times \sin 3x + (3a_2x^2 + 3b_2x + 3c_2) \cos 3x$, or thus: $(2x^2+1) \cos 3x = (-3a_1x^2 - 3b_1x$
 $- 3c_1 + 2a_2x + b_2) \sin 3x + (3a_2x^2 + 3b_2x + 3c_2 + 2a_1x + b_1) \cos 3x$. We must there-
 fore have $2x^2+1 = 3a_2x^2 + 3b_2x + 3c_2 + 2a_1x + b_1$, $0 = -3a_1x^2 - 3b_1x - 3c_1$
 $+ 2a_2x + b_2$. For two polynomials to be equal, the coefficients of identical

powers of x must be equal. Equating the coefficients, we get $3a_2=2$, $3b_2+2a_1=0$, $3c_2+b_1=1$, $-3a_1=0$, $-3b_1+2a_2=0$, $-3c_1+b_2=0$. From this system we find $a_1=0$, $b_2=0$, $c_1=0$, $a_2=\frac{2}{3}$, $b_1=\frac{4}{9}$, $c_2=\frac{5}{27}$ and so

$$\int (2x^2+1) \cos 3x \, dx = \frac{4}{9} x \cos 3x + \left(\frac{2}{3} x^2 + \frac{5}{27} \right) \sin 3x + C. \quad 12. \frac{x}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}.$$

Reducing to a common denominator and then dropping it, we get $A(x-3)+B(x-2)=x$ or $x(A+B)-3A-2B=x$. Equating the coefficients of the same powers of x , we get $A+B=1$, $-3A-2B=0$, whence

$$A=-2, B=3; \int \frac{x \, dx}{(x-2)(x-3)} = \int \left\{ \frac{-2}{x-2} + \frac{3}{x-3} \right\} dx = -2 \ln(x-2) + 3 \ln(x-3) + C. \quad 13. \frac{x+1}{x^2-3x+2} = \frac{x+1}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2},$$

$$A(x-2)+B(x-1)=x+1. \text{ Put } x=2 \text{ in the last equation to get } B=3. \text{ Then put } x=1 \text{ to get } A=-2, \int \frac{x+1}{x^2-3x+2} dx = -2 \ln(x-1) + 3 \ln(x-2) + C. \quad 14. -\ln(x-1) + \ln(x-2) + C. \quad 15. \text{ Put } \sqrt{x}=z. \text{ Then } x=z^2 \text{ and } dx=2z \, dz;$$

$$\int \frac{x \, dx}{x+\sqrt{x}} = \int \frac{z^2 2z \, dz}{z^2+z} = 2 \int \frac{z^2 \, dz}{1+z} = 2 \int \frac{z^2-1+1}{1+z} \, dz = 2 \int \left[(z-1) + \frac{1}{1+z} \right] dz = z^2-2z + \ln(1+z) + C = x-2\sqrt{x} + \ln(1+\sqrt{x}) + C.$$

$$16. \sqrt{x^2-5} + C. \quad 17. -\frac{1}{2} \cos^2 x + \frac{1}{4} \cos^4 x + C. \quad 18. -\frac{1}{3 \sin^2 x} + \frac{1}{\sin x} + C.$$

$$19. -\ln \cos x + C. \quad 20. \frac{1}{a} \arctan \frac{x}{a} + C. \quad 21. \arcsin \frac{x}{a} + C. \quad 22. x \arcsin x + \sqrt{1-x^2} + C. \quad 23. x \arctan x - \frac{1}{2} \ln(x^2+1) + C. \quad 24. \text{ Perform integration by parts setting } f=\sin 3x, \, dg=e^{2x} \, dx \text{ to get } \int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \int e^{2x} \cos 3x \, dx. \text{ In the last integral, again perform integration by parts, setting } f=\cos 3x, \, dg=e^{2x} \, dx \text{ to get } \int e^{2x} \sin 3x \, dx = \frac{1}{2} e^{2x} \sin 3x - \frac{3}{2} \left(\frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \int e^{2x} \sin 3x \, dx \right). \text{ Regarding the last equation as an equation in } \int e^{2x} \sin 3x \, dx, \text{ we find } \int e^{2x} \sin 3x \, dx = \frac{e^{2x} (2 \sin 3x - 3 \cos 3x)}{13}.$$

$$25. \frac{e^x (\cos 2x + 2 \sin 2x)}{5}.$$

Sec. 3.17

1. $y = ax_0^3 + bx_0^2 + cx_0 + d + (3ax_0^2 + 2bx_0 + c)(x-x_0) + (3ax_0 + b)(x-x_0)^2 + a(x-x_0)^3$. The subsequent terms are all equal to zero. The sum of the four written terms is equal to the polynomial. 2. $y(0)=0$, $y'(0)=1$, $y''(0)=2$, ..., $y^{(n)}(0)=n$; $y = x + x^2 + \frac{x^3}{2!} + \dots = x \left(1 + x + \frac{x^2}{2!} + \dots \right) =$

$=xe^x$. 3. $y=e \left[1+(x-1)+\frac{1}{2!}(x-1)^2+\frac{1}{3!}(x-1)^3+\dots \right]$. 4. First method: $y(0)=1$, $\Delta y=y(\Delta x)-1$.

Δx	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\Delta x \rightarrow 0$
$y(\Delta x)$	2.7183	1.6487	1.2840	1.1331	$1+\Delta x$
$\frac{\Delta y}{\Delta x}$	1.718	1.297	1.136	1.065	1

Second method: $\Delta y=y\left(\frac{\Delta x}{2}\right)-y\left(-\frac{\Delta x}{2}\right)$.

Δx	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\Delta x \rightarrow 0$
$y\left(\frac{\Delta x}{2}\right)$	1.6487	1.2840	1.1331	1.064494	$1+\frac{\Delta x}{2}$
$y\left(-\frac{\Delta x}{2}\right)$	0.6065	0.7788	0.8825	0.939412	$1-\frac{\Delta x}{2}$
Δy	1.042	0.5052	0.2506	0.125082	Δx
$\frac{\Delta y}{\Delta x}$	1.042	1.010	1.002	1.0006	1

5. Find the derivatives: $\frac{d \ln(1+r)}{dr} = \frac{1}{1+r}$, $\frac{d^2 \ln(1+r)}{dr^2} = -\frac{1}{(1+r)^2}$, $\frac{d^3 \ln(1+r)}{dr^3} = \frac{2}{(1+r)^3}$, ... The values of these derivatives for $r=0$ are equal, respectively, to 1, -1, 2, ... Maclaurin's formula yields $\ln(1+r) = r - \frac{1}{2}r^2 + \frac{1}{6}r^3 + \dots = r - \frac{r^2}{2} + \frac{r^3}{3} + \dots$, $e^{m \ln(1+r)} = e^{mr} \cdot e^{-\frac{mr^2}{2} + \frac{mr^3}{3} + \dots}$;

for small r , e^{mr} differs from the true value by the factor $e^{-\frac{mr^2}{2}}$. In the problem of Sec. 3.8 on page 121, $m=50$, $r=0.02$; $e^{-\frac{mr^2}{2}} = e^{-0.01} = 0.99$, the error is 1%. m can be any large number so long as mr^2 is small. The smallness of mr^3 , mr^4 , ... is ensured.

Sec. 3.19.

1. $y = \frac{x+1}{1-x} = 1 + 2x + 2x^2 + 2x^3 + 2x^4 + \dots$. 2. $y = \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$. 3. $y = \ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \dots$.

In the first and second problems the series are suitable for computation if $|x| < 1$; in the third, if $0 < x < 2$. 4. $f(x)g(x) = f(0)g(0) + [f'(0)g(0) + g'(0)f(0)]x + \frac{1}{2}[f''(0)g(0) + 2f'(0)g'(0) + g''(0)f(0)]x^2 + \dots$

Sec. 3.21

$$1. 1. \quad 2. -\frac{1}{2}. \quad 3. \frac{1}{3}. \quad 4. \infty. \quad 5. 1. \quad 6. \frac{1}{2}.$$

CHAPTER 4

Sec. 4.1

1. $x = \frac{a+b-\sqrt{a^2+b^2-ab}}{6}$. 2. Let the base of the triangle be $AC=a$, the altitude $BH=h$, and let $DEFG$ be the desired rectangle. From the similarity (Fig. 254) $\frac{DE}{AC} = \frac{BH_1}{BH}$, denoting $DE=x$, we get $\frac{x}{a} = \frac{h-H_1H}{h}$ whence $H_1H = h\left(1 - \frac{x}{a}\right)$. The area of the rectangle is $S(x) = xh\left(1 - \frac{x}{a}\right) = hx -$

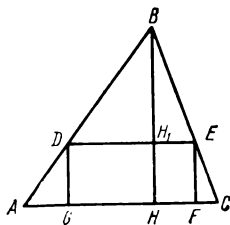


Fig. 254

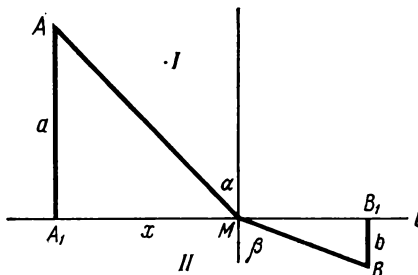


Fig. 255

$-\frac{h}{a}x^2$. Solving the equation $S'(x) = 0$, we get $x = \frac{a}{2}$, then $H_1H = \frac{h}{2}$.

3. The desired rectangle is the square $S = \frac{1}{2}R^2$. 4. The radius of the base

of the can is $r = \sqrt[3]{\frac{V}{2\pi}}$ and the altitude, $H = 2r$. 5. $t = \frac{av_1 + bv_2}{v_1^2 + v_2^2}$. 7. The

time of motion $T = \frac{1}{v_1}\sqrt{a^2 + x^2} + \frac{1}{v_2}\sqrt{b^2 + (c-x)^2}$ where $c = A_1B_1$ (Fig. 255).

The condition $\frac{dT}{dx} = 0$ yields $\frac{x}{v_1\sqrt{a^2 + x^2}} = \frac{c-x}{v_2\sqrt{b^2 + (c-x)^2}}$. Noting that

$\frac{x}{\sqrt{a^2 + x^2}} = \sin \alpha$, $\frac{c-x}{\sqrt{b^2 + (c-x)^2}} = \sin \beta$, we find $\frac{\sin \alpha}{\sin \beta} = \frac{v_1}{v_2}$. This is the Snell law; that is, the point must move like the beam of light passing from one medium to another. To prove that we indeed obtain the minimum of T , it suffices to write $\frac{d^2T}{dx^2}$. It is easy to see that for all x we have $\frac{d^2T}{dx^2} > 0$.

Sec. 4.2

1. $y_{\min}=3$. 2. For $x=0$, $y_{\max}=0$. 3. For $x=0$, $y_{\max}=1$.

Sec. 4.3

1. $\frac{\pi}{2}$. 2. $\frac{\pi}{2}$. 3. $\frac{1}{6}$. 4. $2\pi + \frac{4}{3}$ and $6\pi - \frac{4}{3}$. 5. $a \ln 2$, where a is the amount of paint needed for unit area. 6. $\frac{37}{12}a$. 7. 10π .

Sec. 4.4

1. $\frac{x_0}{n+1}$. 2. $\frac{m-n}{\ln m - \ln n}$. For $m = n + v$ $\ln m = \ln(n+v) = \ln n + \ln\left(1 + \frac{v}{n}\right) = \ln n + \frac{v}{n} - \frac{v^2}{2n^2} + \dots$; $\bar{y} = \frac{v}{\frac{v}{n} - \frac{v^2}{2n^2}} = n\left(1 + \frac{v}{2n}\right) = \frac{n+m}{2}$.

3. (a) Both mean values are equal to $\frac{1}{2}$; (b) $\frac{1}{2} - \frac{1}{\pi}$ and $\frac{1}{2} + \frac{1}{\pi}$. 4. If T is the period, then it should be true that $\sin[\omega(t+T) + \alpha] = \sin(\omega t + \alpha)$, whence $\omega(t+T) + \alpha = \omega t + \alpha + 2\pi$, $\omega T = 2\pi$, $T = \frac{2\pi}{\omega}$. But the period of the function y^2 is equal to $\frac{T}{2} = \frac{\pi}{\omega}$. Hence we have to find the mean value of the function $y = \sin^2(\omega t + \alpha)$ on the interval from $t=0$ to $t = \frac{\pi}{\omega}$.

$$\bar{y} = \frac{\int_0^{\frac{\pi}{\omega}} \sin^2(\omega t + \alpha) dt}{\frac{\pi}{\omega}} = \frac{\omega}{\pi} \int_0^{\frac{\pi}{\omega}} \left\{ \frac{1}{2} - \frac{1}{2} \cos 2(\omega t + \alpha) \right\} dt = \frac{1}{2}.$$

Sec. 4.5

1. $S = \int_0^1 \sqrt{1+4x^2} dx$. 2. $S = \int_0^1 \sqrt{1+e^{2x}} dx$. 3. $S = \frac{4}{a} \int_0^a \sqrt{a^2 + \frac{b^2 x^2}{a^2 - x^2}} dx$.

4. Making the indicated change of variable, we get $S = \int_{\sqrt{2}}^{\sqrt{1+e^2}} \frac{z^2 dz}{z^2 - 1}$.

We find $\int \frac{z^2 dz}{z^2 - 1} = \int \frac{z^2 - 1 + 1}{z^2 - 1} dz = \int \left(1 + \frac{1}{z^2 - 1}\right) dz = z + \int \frac{dz}{z^2 - 1}$.

In the last integral, we write the integrand thus:

$$\frac{1}{z^2 - 1} = \frac{1}{(z-1)(z+1)} = \frac{A}{z-1} + \frac{B}{z+1}$$

We find the numbers A and B by reducing to a common denominator and equating the numerators $A(z+1) + B(z-1) = 1$, whence $A = \frac{1}{2}$, $B = -\frac{1}{2}$.

Finally $\int \frac{z^2 dz}{z^2-1} = z + \frac{1}{2} \ln \frac{z-1}{z+1}$ and so $S = \left[z + \frac{1}{2} \ln \frac{z-1}{z+1} \right]_{\sqrt{2}}^{\sqrt{1+e^2}} =$
 $= \sqrt{1+e^2} - \sqrt{2} + \frac{1}{2} \ln \frac{\sqrt{1+e^2}-1}{\sqrt{1+e^2}+1} - \frac{1}{2} \ln \frac{\sqrt{2}-1}{\sqrt{2}+1}.$

Sec. 4.6

1. Partitioning into sections from $x=0$ to $x=0.9$ and from $x=0.9$ to $x=2$, we find $S_1=1.043$, $S_2 = \frac{e^2+e^{-2}}{2} - \frac{e^{0.9}+e^{-0.9}}{2} + \frac{1}{2} \int_{0.9}^2 \frac{2}{e^x-e^{-x}} dx$. In the last integral, we can put $e^x=t$. We finally get $S_2=2.624$, $S=S_1+S_2+3.667$. By the exact formula we find $S=3.627$. The error amounts to 1%. 2. $S=1.146$. 3. The arc length in question is

$$S = \int_0^{\frac{R}{\sqrt{2}}} \frac{R dx}{\sqrt{R^2-x^2}} = \int_0^{\frac{R}{\sqrt{2}}} \left\{ 1 + \frac{1}{2} \left(\frac{x}{R} \right)^2 + \frac{3}{8} \left(\frac{x}{R} \right)^4 + \frac{5}{16} \left(\frac{x}{R} \right)^6 + \frac{35}{128} \left(\frac{x}{R} \right)^8 + \dots \right\} dx = R \frac{1}{\sqrt{2}} \left\{ 1 + \frac{1}{12} + \frac{3}{160} + \frac{5}{896} + \frac{35}{18432} + \dots \right\}.$$

Accordingly, for the number π we get:

$$\begin{aligned} \text{(a)} \quad \pi &\approx \frac{4}{\sqrt{2}} \left\{ 1 + \frac{1}{12} + \frac{3}{160} \right\} = 3.117; \\ \text{(b)} \quad \pi &\approx \frac{4}{\sqrt{2}} \left\{ 1 + \frac{1}{12} + \frac{3}{160} + \frac{5}{896} \right\} = 3.133; \\ \text{(c)} \quad \pi &\approx \frac{4}{\sqrt{2}} \left\{ 1 + \frac{1}{12} + \frac{3}{160} + \frac{5}{896} + \frac{35}{18432} \right\} = 3.138. \end{aligned}$$

Sec. 4.7

1. Take the diameter AB for the x -axis and point A for the origin. A section perpendicular to the diameter AB is a right triangle PQR with area (see Fig. 115) $S(x) = \frac{1}{2} PQ \cdot QR = \frac{1}{2} PQ^2 \tan \alpha$. But by a familiar theorem of geometry, $PQ^2 = AP \cdot PB = x(2R-x)$. Therefore $S(x) = \frac{1}{2} x(2R-x) \tan \alpha$.

$$V = \int_0^{2R} S(x) dx = \frac{2}{3} R^3 \tan \alpha. \quad 3. \quad V = 2\pi.$$

Sec. 4.8

1. $y_{\max}=2$ for $x=0$, $y_{\min}=-2$ for $x=2$. 2. There are no maxima and no minima. The curve cuts the x -axis at the point $x=1+\sqrt[3]{14} \approx 3.4$ and the y -axis at the point $y=-15$. 3. There are no maxima and no minima. The curve cuts the x -axis between the points $x=0$ and $x=-1$, the y -axis at the point $y=3$. 4. Three roots. 5. Three roots. 6. Two roots. 7. One root.

CHAPTER 5

Sec. 5.3.

1. $T \approx 1660$ years. 2. 176.5 g. 3. 53.3 g. 4. We know that $N(t) = N_0 e^{-\frac{t}{\bar{t}}}$, where N_0 is the quantity of substance at the initial time $t=0$. We are interested in the time t_1 , at which time $(100-1)\% = 99\%$ of the substance remained: $N(t_1) = \frac{99}{100} N_0$. Therefore $\frac{99}{100} N_0 = N_0 e^{-\frac{t_1}{\bar{t}}}$. From this we find $t_1 = \bar{t} \ln \frac{100}{99}$. For radium, $\bar{t} = 2400$ years, and so $t_1 = 2400 \ln \frac{100}{99} = 24$ (years). Similarly, in the other three cases we find $t_2 \approx 250$ years, $t_3 \approx 5500$ years, $t_4 \approx 11000$ years. 5. Suppose at the initial time, $t=0$, there were N_0 atoms of radium in 10^{12} atoms of rock. At the time $t=10\,000$, this amount will equal to 1. Therefore $1 = N_0 e^{-\frac{10\,000}{2400}}$ whence $N_0 = e^4 \approx 65$. In similar fashion we find that 10^6 years ago $N_0 = e^{417} \approx 10^{181}$. This is obviously absurd: 10^{12} atoms of rock contained 10^{181} atoms of radium! A still more absurd result is obtained if we compute the amount of radium 5×10^9 years ago. The absurdity of the result proves the incorrectness of the original premise that the present amount of radium may be regarded as the residue of the disintegration of radium that was present in the earth when the earth was formed.

CHAPTER 6

Sec. 6.1

1. $A(t) = -h \int_{t_0}^t v^2 dt$, A is negative since $\int_{t_0}^t v^2 dt > 0$. 2. The motion of

the body is periodic with period $T = \frac{2\pi}{\omega}$. It is required to determine the work over a half-period. Observe that during the first quarter-period the velocity is positive and so $F = -h$; in the second quarter-period, the velocity is negative and so $F = +h$. In each of these time intervals the force is a constant and so the work is equal to the product of the force by the path covered by the body in that time. For the first quarter-period, $A_1 = -hb$. For the second quarter-period, $A_2 = h(-b) = -hb$ (negative velocity!). Therefore the work during the half-period $A = A_1 + A_2 = -2hb$. 3. $A = b f_0 \omega_1$

$\times \int_0^{t_h} \sin \omega_0 t \cos \omega_1 t dt$. This integral can easily be taken in the following manner. Write down the familiar formulas

$$\sin(\omega_0 t + \omega_1 t) = \sin \omega_0 t \cos \omega_1 t + \cos \omega_0 t \sin \omega_1 t,$$

$$\sin(\omega_0 t - \omega_1 t) = \sin \omega_0 t \cos \omega_1 t - \cos \omega_0 t \sin \omega_1 t$$

Adding the right and left members, we get

$$\sin \omega_0 t \cos \omega_1 t = \frac{1}{2} [\sin(\omega_0 + \omega_1)t + \sin(\omega_0 - \omega_1)t].$$

Therefore $A = \frac{b f_0 \omega_1}{2} \int_0^{t_h} [\sin(\omega_0 + \omega_1)t + \sin(\omega_0 - \omega_1)t] dt = \frac{b f_0 \omega_1}{2} \left[\frac{1}{\omega_0 + \omega_1} + \right.$

$$+ \frac{1}{\omega_0 - \omega_1} - \frac{\cos(\omega_0 + \omega_1)t_k}{\omega_0 + \omega_1} - \frac{\cos(\omega_0 - \omega_1)t_k}{\omega_0 - \omega_1} \Big].$$
 In case $\omega = \omega_0$ we cannot employ a finite formula. But in this case, $\sin \omega_0 t \cos \omega_1 t = \frac{1}{2} \sin 2\omega_0 t$, whence

$$A = \frac{bf_0\omega_0}{2} \int_0^{t_k} \sin 2\omega_0 t \, dt = \frac{bf_0}{4} (1 - \cos 2\omega_0 t_k)$$

4. The work done by air resistance is $A_1(t) = -\frac{aS\rho g^3}{8} t^4$. The work done by the force of gravity is $A_2(t) = \frac{mg^2}{2} t^2$. For the ball: $A_1(1) = -0.00965$, $A_1(10) = -96.5$, $A_1(100) = -965 \times 10^3$, $A_2(1) = 0.177$, $A_2(10) = 1.77$, $A_2(100) = 177$. For the bullet: $A_1(1) = -1.18 \times 10^{-3}$, $A_1(10) = -11.8$, $A_1(100) = -118 \times 10^3$, $A_2(1) = 0.435$, $A_2(10) = 43.5$, $A_2(100) = 4350$ (the work is expressed in joules). 5. $A = \frac{aS\rho(v_0 - v)^2 b}{2}$. We determine the power W by the formula $W = Fv$, $W = \frac{aS\rho(v_0 - v)^2 v}{2}$. We determine for what velocity v (for a given v_0) the power will be a maximum. This requires solving the equation $\frac{dW}{dv} = 0$. We get $v_1 = v_0$ and $v_2 = \frac{v_0}{3}$. We are clearly not interested in $v = v_0$ since it makes the power vanish. The value that interests us is $v = \frac{v_0}{3}$ (the reader can carry out a full investigation with respect to the sign of $\frac{d^2W}{dv^2}$). For $v_0 = 30$ m/sec, $v = 10$ m/sec, $W_{max} = 2.6 \times 10^5$ kgf-m/sec = 3500 metric horsepower. 6. The work of the force during period is $A = c\pi f \sin \alpha$, $W = +\frac{cf\omega}{2} \sin \alpha$.

Sec. 6.3

1. For the x -axis we take the line on which the charges are located; for the origin, the point at which charge e_1 lies. Let charge e_2 be at point $x = +2a$. Equilibrium is possible only at points x where $F = -\frac{du}{dx} = 0$. Let x be the coordinate of the charge e . Then $F = \frac{e_1 e}{x^2} - \frac{4e_1 e}{(2a - x)^2}$ if $0 < x < 2a$, that is, if the charge e is located between the charges e_1 and e_2 , $F = -\frac{e_1 e}{x^2} - \frac{4e_1 e}{(2a - x)^2}$ if $x < 0$, and $F = \frac{e_1 e}{x^2} + \frac{4e_1 e}{(2a - x)^2}$ if $x > 2a$. In the first case, the equation $F = 0$ yields $x_1 = 2a/3$, $x_2 = -2a$. We discard the second root x_2 since it must be true that $x > 0$. In the cases of $x < 0$ and $x > 2a$, the equation $F = 0$ has no solution. Hence there is one position of equilibrium, $x_1 = 2a/3$. Investigate the point $x_1 = 2a/3$. To do that, compute $\frac{d^2u}{dx^2}$ at that point. The evaluation convinces us that if $e > 0$, then $\frac{d^2u}{dx^2} > 0$ and the equilibrium is stable; but if $e < 0$, then the equilibrium is unstable. 2. There is one position of equilibrium x_1 outside the charges. If the system of coor-

dinates is chosen as in Problem 1, then $x_1 = -2a$. If $ee_1 > 0$, then the equilibrium is stable; if $ee_1 < 0$, then the equilibrium is unstable.

Sec. 6.4

1. The equation of motion is $m \frac{dv}{dt} = F$. Using the fact that $v=0$ at $t=0$, we find $v = \frac{F}{m} t$. Therefore $\frac{dx}{dt} = \frac{F}{m} t$, $dx = \frac{F}{m} t dt$, whence $\int_0^x dx = \frac{F}{m} \int_0^t t dt$ since $x=0$ at $t=0$. Finally, $x = \frac{F}{2m} t^2$. 2. $x = v_0 t + \frac{F}{2m} t^2$. 3. $x = x_0 + v_0 t + \frac{F}{2m} t^2$. 4. 24.5 metres. 5. The equation of motion is $m \frac{dv}{dt} = mg$, whence we find $x = \frac{gt^2}{2}$, $t = \sqrt{\frac{2x}{g}} \approx 4.5$ sec for $x=100$ metres. 6. (a) $t=3.6$ sec; (b) $t=5.6$ sec. In each case determine the velocity at the time of touchdown. In the case of (a), $v=gt+v_0$. Let v_{td} be the velocity at touchdown, then $v_{td}=gt_{td}+v_0$, where t_{td} is the touchdown time. From the equation $v=gt+v_0$ we find $x=v_0 t + \frac{gt^2}{2}$. Let a ball fall from a height H . Then $x=H$ at $t=t_{td}$ and so $2H=2v_0 t_{td} + gt_{td}^2$ whence $t_{td} = \frac{-v_0 + \sqrt{v_0^2 + 2gH}}{g}$; for this reason, $v_{td}=gt_{td}+v_0 = \sqrt{v_0^2 + 2gH}$. In the case of (b), $v=gt-v_0$, and the rest is analogous. The terminal velocity is the same as in (a). 7. $x = \frac{k}{6m} t^3 + v_0 t$. 8. (a) $x = -\frac{f}{m\omega^2} \cos \omega t$, $T = \frac{2\pi}{\omega}$, $x_{max} = \frac{f}{m\omega^2}$, $v_{max} = \frac{f}{m\omega}$; (b) $x = \frac{f}{m\omega} t - \frac{f}{m\omega^2} \sin \omega t$. 9. Let the desired velocity be v_0 . Then the law of motion of the body is $x = x_0 + v_0(t-t_0) + \frac{F}{2m}(t-t_0)^2$. At $t=t_1$, $x=x_1$, therefore $x_1 = x_0 + v_0(t_1-t_0) + \frac{F}{2m}(t_1-t_0)^2$ whence $v_0 = \frac{x_1-x_0}{t_1-t_0} - \frac{F}{2m}(t_1-t_0)$.

Sec. 6.6

1. $K = \frac{F^2}{2m} t^2 = F(x-x_0)$. 2. $K = \frac{f^2}{2m\omega^2} \sin^2 \omega t$; $K_{max} = \frac{f^2}{2m\omega^2}$. 3. $\bar{K} = \frac{mA^2\omega^2}{2} \sin^2(\omega t + \alpha) = \frac{mA^2\omega^2}{4}$. 5. (a) $A=4.10^6$ kgf-m, $W=300$ metric horsepower; (b) $A=7.10^6$ kgf-m, $W=520$ metric horsepower. 6. Let us determine the work of each force separately. To do this, first find the velocity of the body. From the equation $m \frac{dv}{dt} = at + a(\theta-t)$ we find $v = v_0 + \frac{a\theta}{m} t$. The work done by the force F_1 is $A_1 = \int_0^\theta at \left(v_0 + \frac{a\theta}{m} t \right) dt = \frac{av_0\theta^2}{2} + \frac{a^2\theta^4}{3m}$. Similarly, we find the work A_2 done by the force F_2 . $A_2 = \frac{av_0\theta^2}{2} + \frac{a^2\theta^4}{6m}$. Form the product of the impulse by

the mean velocity:

$$I_1 = \int_0^{\theta} at \, dt = \frac{a\theta^2}{2}, \quad I_2 = \int_0^{\theta} a(\theta - t) \, dt = \frac{a\theta^2}{2}$$

$$\bar{v} = \frac{\int_0^{\theta} \left(v_0 + \frac{a\theta}{m} t \right) dt}{\theta} = v_0 + \frac{a}{2m} \theta^2$$

$$I_1 \bar{v} = \frac{a\theta^2}{2} \left(v_0 + \frac{a}{2m} \theta^2 \right), \quad I_2 \bar{v} = \frac{a\theta^2}{2} \left(v_0 + \frac{a}{2m} \theta^2 \right)$$

It is evident that $(I_1 + I_2) \bar{v} = A_1 + A_2$ although $I_1 \bar{v} \neq A_1$, $I_2 \bar{v} \neq A_2$, as was pointed out in Sec. 6.6. 7. At the start of the experiment, the mass m had a velocity v_0 (it was moving with the train) and a kinetic energy $K_1 = \frac{mv_0^2}{2}$.

After the action of the man, the velocity of the mass becomes $v_0 + v_1$, $K_2 = \frac{m(v_0 + v_1)^2}{2}$, where $v_1 = \frac{Ft}{m}$. The change in kinetic energy is $\Delta K =$

$$K_2 - K_1 = \frac{m(v_0 + v_1)^2}{2} - \frac{mv_0^2}{2}.$$

This is the work performed on the mass by the train and the man together. To find the work done by the man, note that the velocity of the mass with respect to the man riding in the train was equal to zero prior to the experiment and became equal to v_1 after the experiment. Therefore, the work done by the man is $A_1 = \frac{mv_1^2}{2} - 0 = \frac{mv_1^2}{2} =$

$\frac{F^2 t^2}{2m}$. It is now easy to determine the work done by the locomotive, A_2 : $A_2 = \Delta K - A_1 = mv_0 v_1 = v_0 F t$. This latter result can also be obtained by other

reasoning. Indeed, the work done by the locomotive is $A_2 = \int_0^t v F dt$. Since the

velocity $v = v_0$ is constant, it follows that $A_2 = v_0 \int_0^t F dt = v_0 F t$. 8. Prior to

the experiment, the velocity of the mass m and that of the man were zero. After the experiment, the mass m acquired a velocity v_1 , the man, a velocity v_2 . We find these velocities from the equations $m \frac{dv_1}{dt} = F$ and $M \frac{dv_2}{dt} = -F$ because if the man acts on a mass m with a force F , then the mass m acts on the man with a force $-F$. We find $v_1 = \frac{Ft}{m}$, $v_2 = -\frac{Ft}{M}$. The work done by the force F on the mass m and on

the man is $A = K_1 + K_2$, where $K_1 = \frac{mv_1^2}{2} = \frac{F^2 t^2}{2m}$ is the change in kinetic energy of the mass m , $K_2 = \frac{Mv_2^2}{2} = \frac{F^2 t^2}{2M}$ is the change in kinetic energy of

the man. From this we have $A = \frac{F^2 t^2 (M + m)}{2Mm}$. 9. The change in kinetic energy of mass m is $\Delta K_m = mv_0 v_1 + \frac{mv_1^2}{2}$, where $v_1 = \frac{F}{m} t$. The change in

kinetic energy of the man is $\Delta K_M = \frac{Mv_2^2}{2} + Mv_0v_2$, where $v_2 = -\frac{F}{M}t$,
 $A = \frac{F^2t^2(M+m)}{2Mm}$.

Sec. 6.7

1. Write the equation thus: $\frac{dt}{dv} = \frac{1}{\beta(v_1^2 - v^2)}$ whence [taking into account the initial condition $v(0) = v_0$] we find $t = \frac{1}{\beta} \int_{v_0}^v \frac{dv}{v_1^2 - v^2}$. To be able to perform the integration, it is necessary to write the integrand as $\frac{1}{v_1^2 - v^2} = \frac{a}{v_1 - v} + \frac{b}{v_1 + v}$ and determine the numbers a and b (see the exercises in Sec. 3.15). It is then easy to take the integral. We get $\ln \frac{v_1 + v}{v_1 - v} = \ln A + 2\beta v_1 t$, where $A = \frac{v_1 + v_0}{v_1 - v_0}$. Taking antilogs, we find $\frac{v_1 + v}{v_1 - v} = Ae^{2\beta v_1 t}$ whence $v = v_1 \frac{Ae^{2\beta v_1 t} - 1}{Ae^{2\beta v_1 t} + 1}$. For very large t , the quantity $e^{2\beta v_1 t} \gg 1$ and therefore for such t we will have $v \approx v_1$. Rewrite the solution of the equation in the form $v = v_1 \frac{A - e^{-2\beta v_1 t}}{A + e^{-2\beta v_1 t}}$. Subtracting v_1 from both sides, we get $v - v_1 = -2v_1 \frac{e^{-2\beta v_1 t}}{A + e^{-2\beta v_1 t}}$. For very large t in the denominator, we can neglect $e^{-2\beta v_1 t}$ as compared with A . For this reason, for very large t , $v - v_1 = -2v_1 \frac{e^{-2\beta v_1 t}}{A}$ or $v - v_1 = \frac{2v_1(v_0 - v_1)}{v_1 + v_0} e^{-2\beta v_1 t}$. Comparing this with (6.7-20), we find $C = \frac{2v_1(v_0 - v_1)}{v_1 + v_0}$. 2. For the case of resistance proportional to the velocity, the equation of motion takes the form $\frac{dv}{dt} = g - \frac{k}{m}v - \frac{A}{m}$, where A is the expulsive force. In this case, the velocity becomes steady at $v_1 = \frac{g}{\alpha} - \frac{A}{m\alpha}$. By the Archimedean law, $A = V\rho'g$, where V is the volume of the body, ρ' is the density of the liquid. Since $m = V\rho$ (ρ is the density of the body), it follows that $A = mg \cdot \rho'/\rho$ and $v_1 = \frac{g}{\alpha} \left(1 - \frac{\rho'}{\rho}\right)$. For $\rho' > \rho$ the body comes to the surface ($v_1 < 0$), but for $\rho' < \rho$ it submerges ($v_1 > 0$).

Sec. 6.8

2. (a) Stopping point is $x_0 = \frac{1}{4}$; (b) stopping point is $x \approx 0.95$; (c) there is no stopping point. 3. First determine the position of maximum of $u(x)$. We get $u_{max} \approx 9.5$ for $x_0 = 8/3$. Noting that the left branch of the graph goes up and the right, down, we can construct a rough graph of $u(x)$. This graph is enough to solve the problem at hand. (a) Two stopping points. These are the values of the smallest roots of the equation

$-x^3 + 4x^2 = 6$. This equation can be solved either by graph or with the aid of one of the numerical methods. We get $x_1 \approx -1.09$, $x_2 \approx 1.57$. The body oscillates between the points x_1 and x_2 . (b) One stopping point, $x \approx -2.04$. But this stopping point is to the left of the point from which the body issued at the initial time. Since the initial velocity is directed rightwards, the body will go to the right without having been at the stopping point. (c) One stopping point, $x \approx -2.04$. In this case, the body will go to the left to the stopping point and then rightwards. 4. (a) There are no stopping points. The body will go off to the right. (b) Two stopping points, $x_1 = +\sqrt{9/11}$, $x_2 = -\sqrt{9/11}$. The motion of the body is in the form of

oscillations between the points x_1 and x_2 . For (a) $t = t_0 + \int_0^x \sqrt{\frac{1+x^2}{4+3x^2}} dx$.
 For (b) $t = t_0 + \int_{0.5}^x \sqrt{\frac{20+20x^2}{9-11x^2}} dx$; for $x < x_1$, $t = t_1 - \int_{x_1}^x \sqrt{\frac{20+20x^2}{9-11x^2}} dx$

for $x_1 < x < x_2$, where t_1 is the time at which x_1 is reached, and so forth. It is to be noted that the integrals remain finite although the integrand becomes infinite at the stopping points.

Sec. 6.9

1. (a) $x = 2 \sin t$; (b) $x = \cos t$; (c) $x = \cos t + 2 \sin t$. This solution may be written as $x = C \cos(t + \alpha)$ where $C = \sqrt{5}$, $\alpha = \arctan(-2) \approx -1.11$, i.e., $x = \sqrt{5} \cos(t - 1.11)$. In all three cases, $T = 2\pi$.

Sec. 6.10

1. We assume the oscillations obey the law $x = C \cos(\omega t + \alpha)$. Then $v = -C\omega \sin(\omega t + \alpha)$. Take advantage of the relation $kC \frac{dC}{dt} = F_1 v$, where

$$F_1 = -hv |v|, \quad F_1 v = -hv^2 |v| = -hC^3 \omega^3 |\sin^3(\omega t + \alpha)|$$

Therefore $kC \frac{dC}{dt} = -hC^3 \omega^3 A$, where we put $A = |\sin^3(\omega t + \alpha)|$. Note that

$\sin(\omega t + \alpha)$ preserves sign when t varies from $t_1 = -\frac{\alpha}{\omega}$ to $t_2 = \frac{\pi - \alpha}{\omega}$ and $t_2 - t_1 = \frac{T}{2}$. Therefore

$$A = \frac{\int_{t_1}^{t_2} \sin^3(\omega t + \alpha) dt}{t_2 - t_1} = \frac{\omega}{\pi} \int_{-\frac{\alpha}{\omega}}^{\frac{\pi - \alpha}{\omega}} \sin^3(\omega t + \alpha) dt$$

Setting $\cos(\omega t + \alpha) = x$ in the last integral, we get

$$A = -\frac{1}{\pi} \int_1^{-1} (1 - x^2) dx = \frac{4}{3\pi}$$

Therefore $\frac{dC}{dt} = -\frac{hC^2 \omega^3 4}{3\pi k}$. Set $\frac{h\omega^3 4}{3\pi k} = b$, then $\frac{dC}{dt} = bC^2$ whence $\frac{dt}{dC} =$

$= -\frac{1}{bC^2}$ The solution of this equation is $t = -\frac{1}{b} \frac{C - C_0}{CC_0}$. Expressing C in

terms of t , we find $C = \frac{C_0}{1 + C_0 b t}$. Here, C_0 is used to denote the value of amplitude at the initial time, $t=0$; it is determined from the initial conditions. Observe that the same law was obtained for decay of velocity in the case of a resistance proportional to the square of the velocity [see formula (6.7-12)].

2. In this case, the work during a quarter-period is equal to $-fC$, and so the mean power is $-fC \frac{4}{T} = -fC \frac{2\omega}{\pi}$. We

get the equation $kC \frac{dC}{dt} = -\frac{2fC\omega}{\pi}$,

whence $\frac{dC}{dt} = -\frac{2f\omega}{k\pi}$. From this we have

$C = C_0 - \frac{2f\omega}{k\pi} t$. The oscillations cease

at time t_1 when $C=0$, and so $t_1 = \frac{C_0 k \pi}{2f\omega}$

(it is assumed that $t_1 \gg T$). 3. Let the pendulum at $x=0$ (position of equilibrium) have a potential energy u_0

and a kinetic energy zero. Deflect the pendulum a certain angle. Its horizontal deflection is then x (Fig. 256). In this position, the potential energy $u_1 =$

$= u_0 + mgz$, where $z = l - \sqrt{l^2 - x^2}$; the kinetic energy is equal to $\frac{m}{2} \left(\frac{dx}{dt} \right)^2$.

During the process of oscillation, the sum of the kinetic and potential energies does not change and so $u_0 + mg(l - \sqrt{l^2 - x^2}) + \frac{m}{2} \left(\frac{dx}{dt} \right)^2 = u_0$ or $\left(\frac{dx}{dt} \right)^2 =$

$= 2g(l - \sqrt{l^2 - x^2})$. Now take advantage of the fact that $x \ll l$ (small oscillations), i.e., $\frac{x}{l} \ll 1$. This makes it possible to write $\sqrt{l^2 - x^2}$ as a Maclaurin series: $\sqrt{l^2 - x^2} = l \sqrt{1 - \left(\frac{x}{l} \right)^2} \approx l \left(1 - \frac{x^2}{2l^2} \right) = l - \frac{x^2}{2l}$ (we retained two

terms of the series). The oscillation equation takes the form $\left(\frac{dx}{dt} \right)^2 = g \frac{x^2}{l}$.

Take the derivatives of both members with respect to t to get $2 \frac{dx}{dt} \cdot \frac{d^2x}{dt^2} =$

$= 2g \frac{x}{l} \frac{dx}{dt}$, whence $\frac{d^2x}{dt^2} = \frac{g}{l} x$. This is the equation of small oscillations of the pendulum.

Sec. 6.11

1. From the first equation of the system we get $C_0 \cos \alpha = x_0 - a$. Using this fact, it is easy to find from the second equation that $C_0 \sin \alpha = b \frac{\omega}{\omega_1} - \frac{v_0}{\omega_1} - \gamma \frac{x_0 - a}{\omega_1}$. Squaring these relations, we get $C_0^2 = \left(b \frac{\omega}{\omega_1} - \frac{v_0}{\omega_1} - \gamma \frac{x_0 - a}{\omega_1} \right)^2 +$

$+ (x_0 - a)^2$. Taking the root, we find C_0 ; $\tan \alpha = \frac{b \frac{\omega}{\omega_1} - \frac{v_0}{\omega_1} - \gamma \frac{x_0 - a}{\omega_1}}{x_0 - a}$.

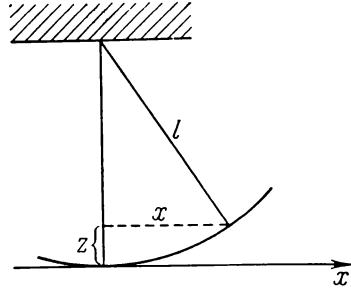


Fig. 256

Sec. 6.13

1. The problem reduces to determining z from the equation $\ln(1+z) = \frac{2z}{1+z}$. Solve this equation by graph. To do so, determine the point of intersection of the curves $y = \ln(1+z)$ and $y = \frac{2z}{1+z}$. From the graph of Fig. 257 it is evident that $z \approx 4$. Therefore $\eta_{\max} = 0.65\alpha$. 2. $r \approx 30,000$ km.

Sec. 6.14

1. For $\varphi = 30^\circ$, $x_{\max} = 565$ metres, $y_{\max} = 81.5$ metres; for $\varphi = 45^\circ$, $x_{\max} = 650$ metres, $y_{\max} = 163$ metres; for $\varphi = 60^\circ$, $x_{\max} = 565$ metres, $y_{\max} =$

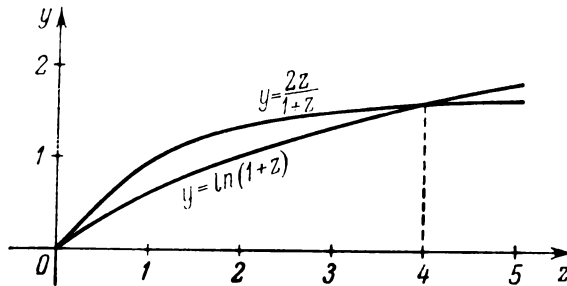


Fig. 257

$= 244$ metres. 2. The equation of the flight path is of the form $y = x \tan \varphi - x^2 \frac{g}{2v_0^2 \cos^2 \varphi}$. For a given $x = 500$ metres, we seek φ for which y becomes a maximum. This requires solving the equation $\frac{dy}{d\varphi} = 0$. We obtain $\tan \varphi = \frac{v_0^2}{gx}$. Using the identity $\frac{1}{\cos^2 \varphi} = \tan^2 \varphi + 1$, we get $\frac{1}{\cos^2 \varphi} = \frac{v_0^4 + g^2 x^2}{g^2 x^2}$. Using this, we can find, from the equation of the flight path, $y_{\max} = \frac{v_0^2}{2g} - \frac{x^2 g}{2v_0^2}$. Setting $v_0 = 80$ m/sec, $x = 500$ metres, we find $y_{\max} = 135$ metres.

Sec. 6.15

1. Putting the origin at the centre of gravity of the rod, we get

$$I_0 = \int_{-\frac{1}{2}l}^{\frac{1}{2}l} x^2 \rho dx = \rho \int_{-\frac{1}{2}l}^{\frac{1}{2}l} x^2 dx = \rho \frac{l^3}{12}$$

Since $m = \rho l$, the result can be written as $I_0 = m \frac{l^2}{12}$. 2. Put the origin at the point of contact of pieces of different density so that for the first piece $x < 0$ and for the second, $x > 0$. Then $x_C = \frac{\rho_2 l_2^2 - \rho_1 l_1^2}{2(l_1 \rho_1 + l_2 \rho_2)}$. 3. Choosing the

system of coordinates as indicated, we get $x_C = \frac{2}{3}L$, the moment of inertia about the origin is $I = \int_0^L x^2 \rho(x) dx = \frac{aL^4}{4}$. Since $m = \frac{aL^2}{2}$, it follows that $a = \frac{2m}{L^2}$ and therefore $I = \frac{mL^2}{2}$. But $I_0 = I - ml_1^2$, where $l_1 = \frac{2}{3}L$, whence $I_0 = \frac{mL^2}{18}$.

Sec. 6.16

1. (a) Denote by L the length of the pendulum. We know that $\omega = \sqrt{\frac{mgl}{I}}$ and in our case $l = \frac{2}{3}L$, $I = \frac{mL^2}{2}$ (see the exercises of Sec. 6.15). Therefore $\omega = \sqrt{\frac{4g}{3L}} = \sqrt{\frac{8g}{9l}}$, $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{3L}{4g}} \approx 5.43 \sqrt{\frac{L}{g}}$. The value of l to which a maximum frequency corresponds is determined from the formula $l_{\max} = \sqrt{\frac{I_0}{m}}$. Since in our case $I_0 = \frac{mL^2}{18}$, it follows that $l_{\max} = \frac{L}{\sqrt{18}}$ and so $\omega_{\max} = \sqrt{\frac{g}{2L} \sqrt{18}}$. Knowing $\omega_{\max}$, we find $T_{\min} = \frac{2\pi}{\omega_{\max}}$. (b) In this case, $l = \frac{1}{3}L$, $I = I_0 + ml^2 = \frac{mL^2}{18} + \frac{1}{9}mL^2 = \frac{mL^2}{6}$, $\omega = \sqrt{\frac{2g}{L}}$. The minimal period here will be the same as in case (a) for a different point of suspension and with the same value of $l_{\max}$.

CHAPTER 7

Sec. 7.3

1. $p = 1.13p_0$, $p = 1.48p_0$, $p = 3.67p_0$ where p_0 is the air pressure at ground level. 2. The pressure is given as a function of the altitude by the formula $p = p_0 e^{-\frac{gh}{b}}$, where $b = \frac{RT}{M}$. For a temperature of -40°C , $T = 273 - 40 = 233$, $b = \frac{8.3 \times 10^7 \times 233}{29.4} = 6.6 \times 10^8 \text{ cm}^2/\text{sec}^2$. In this case, $H = \frac{b}{g} = 6.6 \text{ km}$. For a temperature of $+40^\circ\text{C}$, $T = 313$, $b = 8.8 \times 10^8 \text{ cm}^2/\text{sec}^2$, $H = 8.8 \text{ km}$. 3. From the equation $\frac{dT}{dh} = -\alpha T_0$ we find $T = -\alpha T_0 h + C$. The constant C is found from the condition that $T = T_0$ at $h = 0$. We get $C = T_0$, and so $T = T_0(1 - \alpha h)$. The basic equation for determining the density is $\frac{dp}{dh} = -g\rho$. Take advantage of the Clapeyron equation $p = \rho \frac{RT}{M}$. Substituting the expression for T into this equation we find $p = \rho \frac{RT_0(1 - \alpha h)}{M}$. Set $\frac{RT_0}{M} = b_0$ and then $p = \rho b_0(1 - \alpha h)$, whence $\rho = \frac{p}{b_0(1 - \alpha h)}$. The diffe-

rential equation takes the form $\frac{dp}{dh} = -g \frac{p}{b_0(1-\alpha h)}$. Rewrite it as $\frac{dp}{p} = -\frac{g dh}{b_0(1-\alpha h)}$. Take the integrals of both sides of $\ln p = \frac{g}{b_0\alpha} \ln(1-\alpha h) + C$. Taking antilogs, we get $p = (1-\alpha h)^{\frac{g}{b_0\alpha}} e^C$. Since for $h=0$, $p=p_0$, then $e^C = p_0$ and so $p = p_0(1-\alpha h)^{\frac{g}{b_0\alpha}}$. 4. $p = p_0(1-0.037 \times 10^{-5}h)^{3.46}$, $p = 1.13p_0$, $p = 1.44p_0$, $p_0 = 2.97p_0$.

CHAPTER 8

Sec. 8.2

1. The current in the circuit decays according to the law $j = j_0 e^{-\frac{t}{RC}}$. At the time of interest, t_1 , $j = \frac{9}{10} j_0$ and so $\frac{9}{10} j_0 = j_0 e^{-\frac{t_1}{RC}}$, whence $\frac{9}{10} = e^{-\frac{t_1}{RC}}$. Taking logs, we find $\ln \frac{9}{10} = -\frac{t_1}{RC}$, whence $t_1 = RC \ln \frac{9}{10} \approx 0.105RC$. Using this formula we find that $R = 10^7$ ohms, $t_1 = 1$ sec; for $R = 10^8$ ohms, $t_1 = 10.5$ sec; for $R = 10^9$ ohms, $t_1 = 105$ sec. The time t_2 , when the current has fallen off by one half, is determined analogously: $0.5j_0 = j_0 e^{-\frac{t_2}{RC}}$, whence $t_2 = 0.693RC$. For $R = 10^7$ ohms, $t_2 = 6.93$ sec; for $R = 10^8$ ohms, $t_2 = 69.3$ sec; for $R = 10^9$ ohms, $t_2 = 693$ sec. 2. Take advantage of the formula (8.1-11) to get $\varphi_{C_1} + \varphi_R + \varphi_{C_2} = 0$, whence $\varphi_R = -(\varphi_{C_1} + \varphi_{C_2})$. The current in the circuit is everywhere the same and so $j = C_1 \frac{d\varphi_{C_1}}{dt} = C_2 \frac{d\varphi_{C_2}}{dt} = \frac{\varphi_R}{R} = -\frac{\varphi_{C_1} + \varphi_{C_2}}{R}$. We obtain the equations $\frac{d\varphi_{C_1}}{dt} = -\frac{1}{RC_1}(\varphi_{C_1} + \varphi_{C_2})$, $\frac{d\varphi_{C_2}}{dt} = -\frac{1}{RC_2}(\varphi_{C_1} + \varphi_{C_2})$. Adding these equations, we get $\frac{d(\varphi_{C_1} + \varphi_{C_2})}{dt} = \frac{\varphi_{C_1} + \varphi_{C_2}}{RC}$ where we put $C = \frac{C_1 C_2}{C_1 + C_2}$ (C is the capacitance of two capacitors C_1 and C_2 connected in series). Since $\varphi_{C_1} + \varphi_{C_2} = a$ at $t=0$, then from the last equation we find $\varphi_{C_1} + \varphi_{C_2} = a e^{-\frac{t}{RC}}$. It is clear that $C_1 \frac{d\varphi_{C_1}}{dt} - C_2 \frac{d\varphi_{C_2}}{dt} = 0$ or $\frac{d}{dt}(C_1 \varphi_{C_1} - C_2 \varphi_{C_2}) = 0$. And so $C_1 \varphi_{C_1} - C_2 \varphi_{C_2} = A$, where A is a constant. Using the initial condition $\varphi_{C_1} = a$, $\varphi_{C_2} = 0$ at $t=0$, we find $A = C_1 a$. Thus, $\varphi_{C_1} + \varphi_{C_2} = a e^{-\frac{t}{RC}}$, $C_1 \varphi_{C_1} - C_2 \varphi_{C_2} = C_1 a$. From this we find $\varphi_{C_1} = a \frac{C_1}{C_1 + C_2} \left(1 + \frac{C_2}{C_1} e^{-\frac{t}{RC}}\right)$, $\varphi_{C_2} = a \frac{C_1}{C_1 + C_2} \times \left(-1 + e^{-\frac{t}{RC}}\right)$. 3. Use the subscript 1 to denote all quantities of the circuit prior to increase of all linear dimensions, the subscript 2 to denote those same quantities after the dimensions have been increased. Then $T_1 =$

$= R_1 C_1$, $T_2 = R_2 C_2$, $C_2 = \frac{\epsilon S_2}{4\pi d_2} = \frac{\epsilon \cdot n^2 S_1}{4\pi d_1 n} = n C_1$. $R_1 = \rho \frac{l_1}{\sigma_1}$, $R_2 = \rho \frac{l_2}{\sigma_2} = \rho \frac{n l_1}{n^2 \sigma_1} = \frac{R_1}{n}$. For this reason, $T_2 = R_2 C_2 = \frac{R_1}{n} n C_1 = R_1 C_1 = T_1$. The time constant remained unchanged.

Sec. 8.8

1. $\cos \alpha = \frac{\Phi_0}{\sqrt{\Phi_0^2 + \frac{L}{C} j_0^2}}$. 2. The potential difference on the plates of

the capacitor is equal to φ . For the circuit shown in Fig. 198, $\varphi_E + \varphi_L + \varphi_C = 0$ or, noting that $\varphi_E = -E_0$, $L \frac{dj}{dt} + \varphi = E_0$. Since $j = C \frac{d\varphi}{dt}$, then $LC \frac{d^2\varphi}{dt^2} + \varphi = E_0$. Write this equation in the form $LC \frac{d^2\varphi}{dt^2} = -(\varphi - E_0)$.

Set $z = \varphi - E_0$; then the equation takes the form $LC \frac{d^2z}{dt^2} = -z$. Its solu-

tion is $z = A \cos \omega t + B \sin \omega t$, where $\omega = 1/\sqrt{LC}$. Therefore $\varphi = A \cos \omega t + B \sin \omega t + E_0$. When $t=0$, $\varphi=0$, $j=0$. Using this fact, we find $A = -E_0$, $B=0$. Finally, $\varphi = E_0(1 - \cos \omega t)$. The maximum value of φ is obtained for $\cos \omega t = -1$, i.e., when $t = \pi/\omega = T/2$ (in one half-period), $\varphi_{\max} = 2E_0$. 3. The capacitance energy is $W = C\varphi^2/2 = 4CE_0^2/2 = 2CE_0^2$. The energy released by the voltage source is $P = qE_0 = C\varphi E_0 = 2CE_0^2$.

Sec. 8.9

1. $j(t) = -\frac{\Phi_0}{L\omega} e^{-\lambda t} \sin \omega t$, where $\lambda = \frac{R}{2L}$, $\omega^2 = \frac{1}{LC} - \lambda^2$. For the three given cases we get $j(t) = -1.0025e^{-0.05t} \sin t$; $j(t) = -1.031e^{-0.25t} \sin 0.97t$; $j(t) = -1.15e^{-0.5t} \sin 0.87t$. 2. $j(t) = j_0 \left(\cos \omega t - \frac{\lambda}{\omega} \sin \omega t \right) e^{-\lambda t}$. See 1 for the formulas for λ and ω . For the specified cases, $j(t) = e^{-0.05t} (-\cos t + 0.05 \sin t)$; $j(t) = e^{-0.25t} (-\cos 0.97t + 0.26 \sin 0.97t)$; $j(t) = e^{-0.5t} \times (-\cos 0.86t + 0.58 \sin 0.86t)$. 3. If R is great, then the current flowing through the resistance is small, i.e., the current flows mainly through the inductance. Therefore, the greater R , the closer is the circuit to that of Fig. 196, where $\varphi = \Phi_0 \cos(\omega t + \alpha)$. If R is great, then in the circuit of Fig. 200 we can assume that φ is of the same form but Φ_0 is a slowly time-varying quantity. Take advantage of the relation $\frac{dP}{dt} = -\bar{h}$. But $h = Rj_1^2$, where j_1 is the current flowing through the resistance R , $j_1 = \varphi/R$. Therefore $h = \frac{\varphi^2}{R} = \frac{\Phi_0^2 \cos^2(\omega t + \alpha)}{R}$, $\bar{h} = \frac{\Phi_0^2}{2R}$. Thus, $\frac{dP}{dt} = -\frac{\Phi_0^2}{2R}$. Noting that $P = \frac{C\Phi_0^2}{2}$, we find $\frac{dP}{dt} = C\Phi_0 \frac{d\Phi_0}{dt} = -\frac{\Phi_0}{2R}$, whence $\frac{d\Phi_0}{dt} = -\frac{1}{2RC} \Phi_0$. And so $\Phi_0 = Ae^{-\frac{t}{2RC}}$; $\lambda = \frac{1}{2RC}$.

Sec. 8.10

1. $\varphi(t) = e^{-t} + te^{-t}$; $\varphi(t) = -0.03e^{-5.83t} + 1.03e^{-0.17t}$. $\varphi(t) = -0.01e^{-9.9t} + 1.01e^{-0.1t}$. 2. $\varphi(t) = e^{-t} + 2te^{-t}$; $\varphi(t) = -0.37e^{-3.73t} + 1.37e^{-0.27t}$.

CHAPTER 9

Sec. 9.2

1. Near the point $x=x_0$ we expand the function $\varphi(x)$ in a series to within second order of smallness: $\varphi(x)=\varphi(x_0)+\varphi'(x_0)(x-x_0)=\varphi'(x_0)(x-x_0)$. Thus, denoting $\varphi'(x_0)=c$, $x-x_0=y$, we get $\delta(\varphi(x))=\delta(cy)=\frac{1}{|c|}\delta(y)=\frac{1}{|\varphi'(x_0)|}\delta(x-x_0)$. 2. The function $\varphi(x)=\sin x$ vanishes for $x_0=k\pi$, $k=0, \pm 1, \dots, \pm \infty$; $|\varphi'(x_0)|=|\cos x_0|=|\cos k\pi|=1$; therefore $\delta(\sin x)=\sum_{k=-\infty}^{+\infty} \delta(x-k\pi)$ and $\int_{-\infty}^{+\infty} \psi(x) \delta(\sin x) dx = \sum_{k=-\infty}^{+\infty} \psi(k\pi)$.

Sec. 9.3

1. (a) $y'(x)=1-\delta(x-1)$. (b) *Solution:* $y(+0)=1$, $y(-0)=0$, jump of y at $x=0$: $\Delta y=1$. For $x \neq 0$, $y'(x)=-\frac{e^{1/x}}{x^2(1+e^{1/x})^2}$; therefore we finally have $y'(x)=\delta(x)-\frac{e^{1/x}}{x^2(1+e^{1/x})^2}$.

Sec. 9.5

1. Equating to zero (equilibrium!) the sum of the projections of the force on the y -axis, we obtain, for the case of small deviations ($y \ll l$): $1-k\frac{y_1}{x_1}-k\frac{y_1}{l-x_1}=0$, whence

$$y_1(x_1)=1/\left(\frac{k}{x_1}+\frac{k}{l-x_1}\right)=\frac{x_1(l-x_1)}{kl},$$

$$y(x, x_1)=\begin{cases} \left(\frac{x}{x_1}\right)/\left(\frac{k}{x_1}+\frac{k}{l-x_1}\right)=\frac{x(l-x_1)}{kl}, & 0 \leq x \leq x_1 \\ \left(\frac{l-x}{l-x_1}\right)/\left(\frac{k}{x_1}+\frac{k}{l-x_1}\right)=\frac{x_1(l-x)}{kl}, & x_1 \leq x \leq l \end{cases}$$

This function $y(x, x_1)$ is called Green's function of the problem of the string. For an arbitrarily distributed force $f(x)$, the deviation of the string is given

by the formula $y(x)=\int_0^l f(x_1)y(x, x_1)dx_1$. Observe that using Green's function

we obtained a solution for the function $y(x)$ without even knowing what equation it obeys [this equation has the form $\frac{d^2y}{dx^2}=-\frac{f(x)}{k}$, $y(0)=0$, $y(l)=0$].

2. The general solution of the equation without a compulsive force is $x(t)=c_1 \sin \omega t + c_2 \cos \omega t$, $\omega=\sqrt{k/m}$, where c_1 and c_2 are arbitrary constants. Since the delta function is nonzero only when $t=\tau$, the solution of the equation with a delta-like force and a state of rest at $t=-\infty$ is of the form

$$x(t)=\begin{cases} 0, & -\infty < t < \tau \\ c_1 \sin \omega t + c_2 \cos \omega t, & \tau < t < +\infty \end{cases}$$

The delta-like force imparts a unit impulse to the body, and so when a delta force acts on a body at rest, the body acquires an initial velocity $v_0 = \frac{\Delta p}{m} = \frac{1}{m}$. The initial position remains equal to zero. The solution of the oscillation equation at time $t = \tau$ with such initial conditions is

$$x(t, \tau) = \begin{cases} 0, & -\infty < t < \tau \\ \frac{1}{m\omega} \sin \omega(t - \tau), & \tau < t < +\infty \end{cases} \quad -$$

In other words, in the preceding formula, $c_1 = \frac{\cos \omega \tau}{m\omega}$, $c_2 = -\frac{\sin \omega \tau}{m\omega}$. The solution of the problem with an arbitrary force $f(t)$ is given by the formula

$$x(t) = \int_{-\infty}^{+\infty} f(\tau) x(t, \tau) d\tau = \int_{-\infty}^t f(\tau) \frac{1}{m\omega} \sin \omega(t - \tau) d\tau$$

Appendix

Greek Alphabet

Alpha	A, α	Eta	H, η	Nu	N, ν	Tau	T, τ
Beta	B, β	Theta	Θ, θ	Xi	Ξ, ξ	Upsilon	Υ, υ
Gamma	Γ, γ	Iota	I, ι	Omicron	O, $\omicron$	Phi	Φ, ϕ
Delta	Δ, δ	Kappa	K, κ	Pi	Π, π	Chi	χ, χ
Epsilon	E, ε	Lambda	Λ, λ	Rho	P, ρ	Psi	Ψ, ψ
Zeta	Z, ζ	Mu	M, μ	Sigma	Σ, σ	Omega	Ω, ω

Table I. Derivatives

1. $y = c$	$\frac{dy}{dx} = 0$
2. $y = x$	$\frac{dy}{dx} = 1$
3. $y = x^a$	$\frac{dy}{dx} = ax^{a-1} = a \frac{y}{x}$
4. $y = e^x$	$\frac{dy}{dx} = e^x$
5. $y = a^x$	$\frac{dy}{dx} = 2.3 \log_{10} a \cdot a^x$
6. $y = \ln x$	$\frac{dy}{dx} = \frac{1}{x}$
7. $y = \log_a x$	$\frac{dy}{dx} = \frac{0.434}{\log_{10} a} \frac{1}{x}$
8. $y = \sin x$	$\frac{dy}{dx} = \cos x$
9. $y = \cos x$	$\frac{dy}{dx} = -\sin x$
10. $y = \tan x$	$\frac{dy}{dx} = \frac{1}{\cos^2 x}$
11. $y = \cot x$	$\frac{dy}{dx} = -\frac{1}{\sin^2 x}$

$$12. y = \arcsin x \quad \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$13. y = \arccos x \quad \frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}} \quad \bullet$$

$$14. y = \arctan x \quad \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$15. y = \operatorname{arccot} x \quad \frac{dy}{dx} = -\frac{1}{1+x^2}$$

Table II. Integrals of Some Functions

1. $\int dx = x + C$
2. $\int x^a dx = \frac{x^{a+1}}{a+1} + C \quad (a \neq -1)$
3. $\int \frac{dx}{x} = \ln x + C$
4. $\int \frac{dx}{ax+b} = \frac{1}{a} \ln(ax+b) + C$
5. $\int a^x dx = \frac{a^x}{\ln a} + C$
6. $\int e^{kx} dx = \frac{1}{k} e^{kx} + C$
7. $\int x^n e^{kx} dx = \frac{1}{k} x^n e^{kx} - \frac{n}{k} \int x^{n-1} e^{kx} dx$
8. $\int \frac{dx}{1+e^{kx}} = \frac{1}{k} \ln \frac{e^{kx}}{1+e^{kx}} + C$
9. $\int e^{kx} \sin ax dx = \frac{e^{kx}}{k^2+a^2} (k \sin ax - a \cos ax) + C$
10. $\int e^{kx} \cos ax dx = \frac{e^{kx}}{k^2+a^2} (k \cos ax + a \sin ax) + C$
11. $\int \sin kx dx = -\frac{1}{k} \cos kx + C$
12. $\int \cos kx dx = \frac{1}{k} \sin kx + C$
13. $\int \frac{dx}{\sin^2 kx} = -\frac{1}{k} \cot kx + C$
14. $\int \frac{dx}{\cos^2 kx} = \frac{1}{k} \tan kx + C$
15. $\int \sin^2 kx dx = \frac{1}{2} x - \frac{1}{4k} \sin 2kx + C$
16. $\int \cos^2 kx dx = \frac{1}{2} x + \frac{1}{4k} \sin 2kx + C$

17. $\int x^n \sin kx \, dx = -\frac{x^n}{k} \cos kx + \frac{n}{k} \int x^{n-1} \cos kx \, dx$
18. $\int x^n \cos kx \, dx = \frac{x^n}{k} \sin kx - \frac{n}{k} \int x^{n-1} \sin kx \, dx$
19. $\int \sin kx \sin lx \, dx = \frac{\sin (k-l)x}{2(k-l)} - \frac{\sin (k+l)x}{2(k+l)} + C$, if
 $|k| \neq |l|$ (if $|k| = |l|$, see No. 15)
20. $\int \cos kx \cos lx \, dx = \frac{\sin (k-l)x}{2(k-l)} + \frac{\sin (k+l)x}{2(k+l)} + C$ if
 $|k| \neq |l|$ (if $|k| = |l|$, see No. 16)
21. $\int \sin kx \cos lx \, dx = -\frac{\cos (k+l)x}{2(k+l)} - \frac{\cos (k-l)x}{2(k-l)} + C$ if $|k| \neq |l|$
22. $\int \tan kx \, dx = -\frac{1}{k} \ln |\cos kx| + C$
23. $\int \cot kx \, dx = \frac{1}{k} \ln |\sin kx| + C$
24. $\int \sqrt{ax+b} \, dx = \frac{2}{3a} \sqrt{(ax+b)^3} + C$
25. $\int \frac{dx}{\sqrt{ax+b}} = \frac{2}{a} \sqrt{ax+b} + C$
26. $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C$
27. $\int x \sqrt{ax+b} \, dx = \frac{2(3ax-2b) \sqrt{(ax+b)^3}}{15a^2} + C$
28. $\int \sqrt{a^2-x^2} \, dx = \frac{1}{2} \left(x \sqrt{a^2-x^2} + a^2 \arcsin \frac{x}{a} \right) + C$
29. $\int \frac{\sqrt{a^2-x^2}}{x} \, dx = \sqrt{a^2-x^2} - a \ln \frac{a + \sqrt{a^2-x^2}}{x} + C$
30. $\int x \sqrt{x^2+m} \, dx = \frac{1}{3} \sqrt{(x^2+m)^3} + C$
31. $\int \frac{dx}{\sqrt{x^2+m}} = \ln (x + \sqrt{x^2+m}) + C$
32. $\int \frac{\sqrt{a^2+x^2}}{x} \, dx = \sqrt{a^2+x^2} - a \ln \frac{a + \sqrt{a^2+x^2}}{x} + C$
33. $\int \sqrt{x^2+m} \, dx$
 $= \frac{1}{2} [x \sqrt{x^2+m} + m \ln (x + \sqrt{x^2+m})] + C$
34. $\int \frac{\sqrt{x^2-a^2}}{x} \, dx = \sqrt{x^2-a^2} - a \arccos \frac{a}{x} + C$

$$35. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \frac{x-a}{x+a} + C$$

$$36. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \operatorname{arccot} \frac{x}{a} + C$$

$$37. \int \frac{dx}{ax^2 + bx + c} = \frac{2}{\sqrt{4ac - b^2}} \arctan \frac{2ax + b}{\sqrt{4ac - b^2}} + C \quad \text{if } 4ac - b^2 > 0$$

$$\int \frac{dx}{ax^2 + bx + c} = \frac{1}{\sqrt{b^2 - 4ac}} \ln \frac{2ax + b - \sqrt{b^2 - 4ac}}{2ax + b + \sqrt{b^2 - 4ac}} + C$$

if $4ac - b^2 < 0$

$$38. \int \frac{dx}{(ax^2 + bx + c)^n} = \frac{2ax + b}{(n-1)(4ac - b^2)(ax^2 + bx + c)^{n-1}} \\ + \frac{(2n-3)2a}{(n-1)(4ac - b^2)} \int \frac{dx}{(ax^2 + bx + c)^{n-1}}$$

$$39. \int \frac{x dx}{ax^2 + bx + c} = \frac{1}{2a} \ln(ax^2 + bx + c) - \frac{b}{2a} \int \frac{dx}{ax^2 + bx + c} \quad (\text{see No. 37})$$

$$40. \int \frac{dx}{x(ax^2 + bx + c)} = \frac{1}{2c} \ln \frac{x^2}{ax^2 + bx + c} - \frac{b}{2c} \int \frac{dx}{ax^2 + bx + c} \quad (\text{see No. 37})$$

$$41. \int \frac{dx}{x^m(ax^2 + bx + c)^n} = -\frac{1}{(m-1)cx^{m-1}(ax^2 + bx + c)^{n-1}} \\ - \frac{(2n+m-3)a}{(m-1)c} \int \frac{dx}{x^{m-2}(ax^2 + bx + c)^n} - \frac{(n+m-2)}{(m-1)c} \int \frac{dx}{x^{m-1}(ax^2 + bx + c)^n} \\ (m > 1)$$

$$42. \int \sqrt[n]{ax+b} dx = \frac{n(ax+b)}{(n+1)a} \sqrt[n]{ax+b} + C$$

$$43. \int \frac{dx}{\sqrt[n]{ax+b}} = \frac{n(ax+b)}{(n-1)a} \frac{1}{\sqrt[n]{ax+b}} + C$$

$$44. \int \ln x dx = x \ln x - x + C$$

$$45. \int (\ln x)^n dx = x (\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$46. \int \arcsin \frac{x}{a} dx = x \arcsin \frac{x}{a} + \sqrt{a^2 - x^2} + C$$

$$47. \int \arccos \frac{x}{a} dx = x \arccos \frac{x}{a} - \sqrt{a^2 - x^2} + C$$

$$48. \int \arctan \frac{x}{a} dx = x \arctan \frac{x}{a} - \frac{a}{2} \ln(a^2 + x^2) + C$$

$$49. \int \operatorname{arccot} \frac{x}{a} dx = x \operatorname{arccot} \frac{x}{a} + \frac{a}{2} \ln(a^2 + x^2) + C$$

Table III. Series Expansions

1. $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \dots$ $(-1 < x < 1)$
2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ (any x)
3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$ (any x)
4. $\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \frac{62}{2835}x^9 + \dots$ $\left(-\frac{\pi}{2} < x < \frac{\pi}{2}\right)$
5. $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ (any x)
6. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ $(-1 < x \leq 1)$
7. $\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$ $(-1 \leq x < 1)$
8. $\arcsin x = x + \frac{x^3}{2 \cdot 3} + \frac{1 \cdot 3 \cdot x^5}{2 \cdot 4 \cdot 5} + \frac{1 \cdot 3 \cdot 5 \cdot x^7}{2 \cdot 4 \cdot 6 \cdot 7} + \dots$ $(-1 < x < 1)$
9. $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ $(-1 < x < 1)$

Table IV

x	e^x	e^{-x}	x	e^x	e^{-x}
0	1.000	1.000	2.4	11.023	0.0907
0.1	1.105	0.905	2.6	13.464	0.0743
0.2	1.221	0.819	2.8	16.445	0.0608
0.3	1.350	0.741	3.0	20.086	0.0498
0.4	1.492	0.670	3.2	24.533	0.0408
0.5	1.649	0.607	3.4	29.964	0.0334
0.6	1.822	0.549	3.6	36.598	0.0273
0.7	2.014	0.497	3.8	44.701	0.0224
0.8	2.226	0.449	4.0	54.598	0.0183
0.9	2.460	0.407	4.5	90.017	0.0111
1.0	2.718	0.368	5.0	148.41	0.00674
1.1	3.004	0.333	5.5	244.69	0.00409
1.2	3.320	0.301	6.0	403.43	0.00248
1.3	3.669	0.273	6.5	665.14	0.00150
1.4	4.055	0.247	7.0	1096.6	0.000912
1.5	4.482	0.223	7.5	1808.0	0.000553
1.6	4.953	0.202	8.0	2981.0	0.000335
1.7	5.474	0.183	8.5	4914.8	0.000203
1.8	6.050	0.165	9.0	8103.1	0.000123
1.9	6.686	0.150	9.5	13360	0.000075
2.0	7.389	0.135	10.0	22026	0.000045
2.2	9.025	0.1108			

Table V

x	$\ln x$	x	$\ln x$	x	$\ln x$
1.0	0	2.2	0.788	5.0	1.609
1.1	0.0953	2.4	0.875	5.5	1.705
1.2	0.182	2.6	0.956	6.0	1.792
1.3	0.262	2.8	1.030	6.5	1.872
1.4	0.336	3.0	1.099	7.0	1.946
1.5	0.405	3.2	1.163	7.5	2.015
1.6	0.470	3.4	1.224	8.0	2.079
1.7	0.531	3.6	1.281	8.5	2.140
1.8	0.588	3.8	1.335	9.0	2.197
1.9	0.642	4.0	1.386	9.5	2.251
2.0	0.693	4.5	1.504	10.0	2.303

Table VI

x	$\sin x$	$\cos x$	$\tan x$	x	$\sin x$	$\cos x$	$\tan x$
0	0.000	1.000	0.000	3.2	-0.0584	-0.998	0.0585
0.1	0.0998	0.995	0.100	3.3	-0.158	-0.987	0.160
0.2	0.199	0.980	0.203	3.4	-0.256	-0.967	0.264
0.3	0.296	0.955	0.309	3.5	-0.351	-0.936	0.375
0.4	0.389	0.921	0.423	3.6	-0.443	-0.897	0.493
0.5	0.479	0.878	0.546	3.7	-0.530	-0.848	0.625
0.6	0.565	0.825	0.684	3.8	-0.612	-0.791	0.774
0.7	0.644	0.765	0.842	3.9	-0.688	-0.726	0.947
0.8	0.717	0.697	1.030	4.0	-0.757	-0.654	1.158
0.9	0.783	0.622	1.260	4.1	-0.818	-0.575	1.424
1.0	0.841	0.540	1.557	4.2	-0.872	-0.490	1.778
1.1	0.891	0.454	1.965	4.3	-0.916	-0.401	2.286
1.2	0.932	0.362	2.572	4.4	-0.952	-0.307	3.096
1.3	0.964	0.268	3.602	4.5	-0.978	-0.211	4.637
1.4	0.985	0.170	5.798	4.6	-0.994	-0.112	8.860
1.5	0.997	0.0707	14.101	4.7	-1.000	-0.0124	80.713
1.6	0.9996	-0.0292	-34.233	4.8	-0.996	0.0875	-11.385
1.7	0.992	-0.129	-7.697	4.9	-0.982	0.187	-5.267
1.8	0.974	-0.227	-4.286	5.0	-0.959	0.284	-3.381
1.9	0.946	-0.323	-2.927	5.1	-0.926	0.378	-2.449
2.0	0.909	-0.416	-2.185	5.2	-0.883	0.469	-1.886
2.1	0.863	-0.505	-1.710	5.3	-0.832	0.554	-1.501
2.2	0.808	-0.589	-1.374	5.4	-0.773	0.635	-1.218
2.3	0.746	-0.666	-1.119	5.5	-0.706	0.709	-0.996
2.4	0.675	-0.737	-0.916	5.6	-0.631	0.776	-0.814
2.5	0.598	-0.801	-0.747	5.7	-0.551	0.835	-0.660
2.6	0.516	-0.857	-0.602	5.8	-0.465	0.886	-0.525
2.7	0.427	-0.904	-0.473	5.9	-0.374	0.927	-0.403
2.8	0.335	-0.942	-0.356	6.0	-0.279	0.960	-0.291
2.9	0.239	-0.971	-0.246	6.1	-0.182	0.983	-0.185
3.0	0.141	-0.990	-0.143	6.2	-0.0831	0.997	-0.0834
3.1	0.0416	-0.999	-0.0416	6.3	-0.0168	1.000	0.0168

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